MBToolkit v3.4 Technical Code Documentation

Architecture, runtime flows, every callable, regression tests, build automation, and complete source-code screenshots

Documented package

`MBToolkit_v3.4_Package.zip`

Source snapshot SHA-256

`46e927e1b937fd3be024cacc2c672c5f58aabe74798babb8261367266300bfcc`

Documentation build date

26 October 2026

Verification

113/113 regression tests passed before documentation generation.

This document describes the exact source snapshot above. It does not document proposed future features, such as Modbus TCP, that are not present in this package.

The final appendix reproduces every Python and Windows batch source line as code screenshots. The explanatory chapters refer to the same line numbers.

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1. Documented source snapshot

This reference was generated from the current MBToolkit v3.4 package, without modifying the application source.

Python source lines	Documented Python + BAT lines
17,337	17,743

Package structure

```
MBToolkit_v3.4_Package.zip
├── tools/  (documentation and package builders)
├── MBToolkit_v3.4_Technical_Code_Documentation.pdf  (this document)
├── MBToolkit.bat
├── README.md
├── README.en.md
├── README.txt
└── MBToolkit v3.4/
    ├── MBToolkit.py
    ├── mb_config.py
    ├── mb_protocol.py
    ├── mb_capture.py
    ├── mb_diagnostics.py
    ├── mb_view.py
    ├── mb_slave_finder.py
    ├── mb_master.py
    ├── mb_master_ui.py
    ├── mb_slave.py
    ├── mb_slave_ui.py
    ├── mb_table_ui.py
    ├── mb_i18n.py
    ├── mb_theme.py
    ├── mb_widgets.py
    ├── mb_gui.py
    ├── tests/test_regression.py
    ├── tests/test_master.py
    ├── tests/test_slave.py
    ├── run_tests.bat
    ├── build_exe.bat
    ├── MBToolkit.ico
    ├── DEVELOPMENT_NOTES.md
    └── CHANGELOG.md
```

Source inventory

Python module	Lines	Primary responsibility
`MBToolkit.py`	29	Application launcher and top-level entry point.
`mb_config.py`	577	Central configuration, constants, user settings validation and filesystem paths.
`mb_protocol.py`	1007	Protocol engine for passive Modbus RTU parsing and session metrics.
`mb_capture.py`	902	Serial-port management and passive acquisition pipeline.
`mb_diagnostics.py`	286	Presentation-oriented diagnostics derived from captured records.
`mb_view.py`	1871	Traffic view, filtering, sorting, Frame Inspector, Bus Health updates, CSV export and TXT logging.
`mb_slave_finder.py`	1625	Active Modbus RTU Slave discovery and Device Identification.
`mb_master.py`	686	Modbus RTU master engine, frames and data formats.
`mb_master_ui.py`	1529	Master (poll) tab user interface.
`mb_slave.py`	533	Modbus RTU slave (simulator) engine.
`mb_slave_ui.py`	1412	Slave (simulator) tab user interface.
`mb_table_ui.py`	599	Window-grid code shared by the Master and Slave tabs.
`mb_i18n.py`	660	Runtime Portuguese/English translation layer.
`mb_theme.py`	692	Fixed-geometry Light/Dark theme implementation.
`mb_widgets.py`	418	Custom reusable Tkinter widgets.
`mb_gui.py`	2664	Main application composition and global UI behavior.
`test_regression.py`	787	Automated regression suite.
`test_master.py`	601	Master regression tests.
`test_slave.py`	459	Slave regression tests.

Integrity hashes

Source file	Lines	SHA-256
`MBToolkit v3.4/MBToolkit.py`	29	`44a1ab13bc860ea64ca57f2805f4ab7ab5c6e8455241973c3ae50f4e856d04e1`
`MBToolkit v3.4/mb_config.py`	577	`b6698ad9fb341dbb5a0448ef737909dc7453f0587ab4eb09d92987c7a6008578`

Source file	Lines	SHA-256
MBToolkit v3.4/mb_protocol.py	1007	8c4c41051e41f0c998692206ed0204d91f057cce5fce726b15ad2cce503c1ef2
MBToolkit v3.4/mb_capture.py	902	78c925c4aa8c9d3e1b263dc9e91f28f2c2e08ef4e62573f949e985fa6f54fac0
MBToolkit v3.4/mb_diagnostics.py	286	c67ed1df18f1d79b0feb4011a036f92cbeef338929a529a4f218acd1f78e1552
MBToolkit v3.4/mb_view.py	1871	deb7447ebdb2fb80ad32b0d4d534b9a763ca72ba062c966dddbb2c871d63c64c
MBToolkit v3.4/mb_slave_finder.py	1625	f192a6305486ea73411d920d9b6b9de19091bc612afef60ea3f5871fcd66938b
MBToolkit v3.4/mb_master.py	686	79ad42ac5c1f3a658b4252c2196ccb9829ad4cfdbcc5237052ef8a3a3423221e
MBToolkit v3.4/mb_master_ui.py	1529	9d0df865a4a1c690c8d94717316125e9dc2cf283d153dd949455884c5de9ae42
MBToolkit v3.4/mb_slave.py	533	2dbbee755b610ee797abcecc59899748fd1322e4388b7623720e5b364bda5b9d
MBToolkit v3.4/mb_slave_ui.py	1412	d8f3daaa6f69074315ad6daefe6a761a66d7ab85e4eeb1a01138d67cac8ca92a
MBToolkit v3.4/mb_table_ui.py	599	d7f10014c16f1eb74074143bcffb35705cb3a17be6731ea13f773803b096e881
MBToolkit v3.4/mb_i18n.py	660	b948e2a8a23ab71b8ebf5e65761e2025de0c832d3cea50061899f6387a5cdeeb
MBToolkit v3.4/mb_theme.py	692	7b03e5a4f4cc646a96cd1839d2a92794ea923d232ae7e50a93b5908ac4ab8d9e
MBToolkit v3.4/mb_widgets.py	418	3a732639b6813ae827290b8ca5af567158b2ee736e941c89ae3d77ea7a184d2f
MBToolkit v3.4/mb_gui.py	2664	7ab64e42fcc446f8f55c212aa4c9b102b1f367a019f34122e080ec6606842b03
MBToolkit v3.4/tests/test_regression.py	787	b5a3f7e75c0b385f8cb874fdf3b16c8e867108c59406078d966f333db4dba70c
MBToolkit v3.4/tests/test_master.py	601	d0c561dd311c1eec097e9f83527b2681d440928ff527f8e0237486ec299f6725
MBToolkit v3.4/tests/test_slave.py	459	0524c65af1da85131649a145f99868466f14f44e1293d87560edcb73d8256f9c
MBToolkit.bat	210	cae8a4af4a86a76b01b46ee66788a021c702b2143225ba946cf14294bff4bb78
MBToolkit v3.4/build_exe.bat	151	08d6ba775bd6986cdcc8a130712788b60a00c0fa0793deec076fedd01045c331

Source file	Lines	SHA-256
`MBToolkit v3.4/run_tests.bat`	45	`98c20bdbf89f4f478c4b38b7c8f5a55fae834f328edaf7d2979c8823f2236eae`

> **Regression status.** The supplied suite was executed directly from this source tree. All 113 tests passed.

2. Version history and changes in this version

Release history of the documented code base. The detailed list below applies to v3.4, the version described by every other chapter of this document.

Version	Date	Summary
v3.0	21 September 2026	Bilingual PT-PT / English interface, FC43/14 Device Identification, regression suite and first Technical Code Documentation.
v3.1	23 October 2026	Broadcast/echo-write pairing fixes, USB split-frame reassembly, in-place timeout marking, Finder attempts and idle-bus check, address-mapped read data, table-driven CRC and centralised translations.
v3.2	23 October 2026	Master (poll) tab with one window per Slave / data area, cyclic reads, writes, 16/32-bit data formats and per-window counters; the release ZIP now also carries this PDF and the documentation tools.
v3.3	23 October 2026	Renamed from MBSniffer to MBToolkit, with a new high-resolution icon and automatic migration of preferences and workspaces.
v3.4	23 October 2026	Slave (simulator) tab: several simulated Slave IDs and data tables answering a real master, with fault simulation per Slave ID; window grid code shared with the Master.

Changes in v3.4

v3.4 adds the **Slave** tab: the application answers a real master (PLC, SCADA or another Modbus Poll) as one or more Modbus RTU devices. See chapter "Slave (simulator) tab".

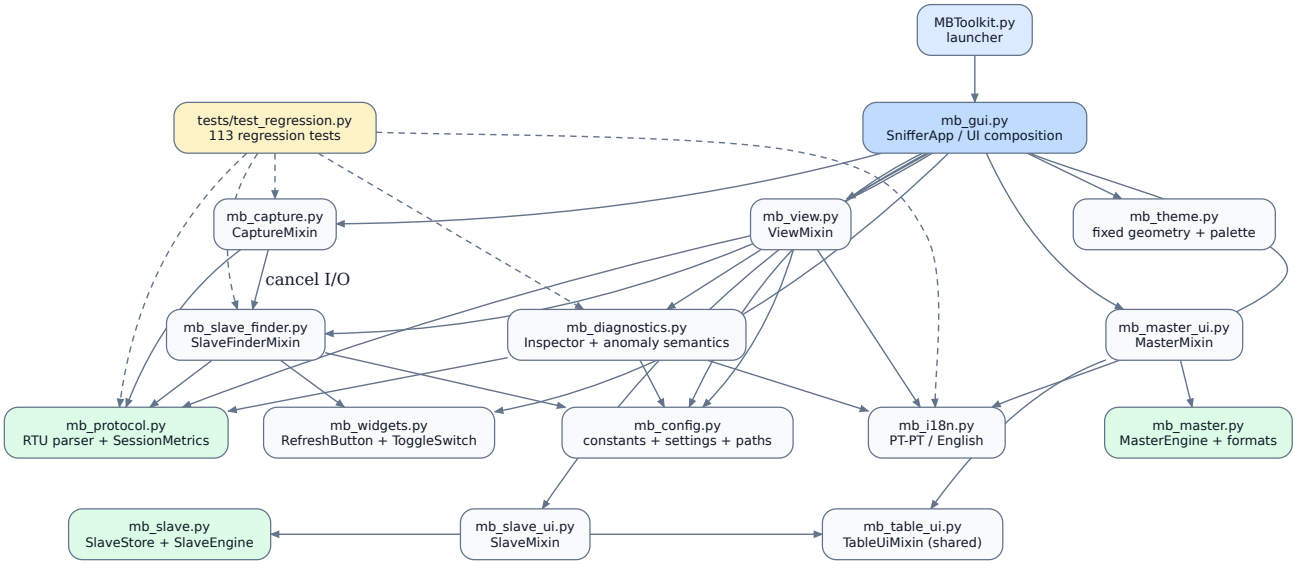
- **New modules:** `mb_slave.py` (`SlaveStore`, `handle_request()`, `SlaveEngine`; no widgets), `mb_slave_ui.py` (`SlaveMixin`, `SlaveWindow`) and `mb_table_ui.py` (`TableUiMixin`: the window header, value table, drag selection, zebra rows, row menu, scale dialog, grid reflow and Workspace menu shared by Master and Slave, plus the workspace-file helpers).
- **Master refactor:** `MasterMixin` now uses `TableUiMixin`; its behaviour is unchanged. `apply_format_to_rows()`, `validate_workspace_name()`, `workspace_path()` and `list_workspaces()` moved to `mb_table_ui.py` (which takes the folder); `mb_master_ui.py` keeps Master versions bound to the Master folder.
- **Workspace folders:** Master workspaces now live in `Workspaces\Master` and Slave workspaces in `Workspaces\Slave`; Master files left directly in `Workspaces` are moved once.
- **Requests answered:** FC01–06, FC15 and FC16; Exception 01 (function), 02 (address outside the windows) and 03 (quantity / value / byte count); broadcast writes executed on every covering Slave ID without a response; requests for other IDs ignored; the RS485 adapter echo removed.
- **Fault simulation per Slave ID:** response delay, no response, a chosen Exception, wrong CRC and a probability per request; an active fault is shown in red in the header of that Slave's windows.
- **Global busy lock:** Sniffer, Finder, Master and Slave never run at the same time.
- **Workspaces:** the Slave has its own Workspace menu (configuration only, values restart at 0) in `%APPDATA%\MBToolkit\Workspaces\Slave`; `settings.json` gains a validated `slave` section and the

"Slave" last tab.

- The regression suite grew to 113 tests, including the Master engine talking to the Slave engine over an in-memory serial link.

3. Architecture

MBToolkit uses one Tkinter root object with five behavior mixins. Protocol parsing and metrics are separated from the GUI, while the active Finder, Master and Slave have their own worker logic. This is a pragmatic modular architecture rather than a strict MVC/MVVM framework.



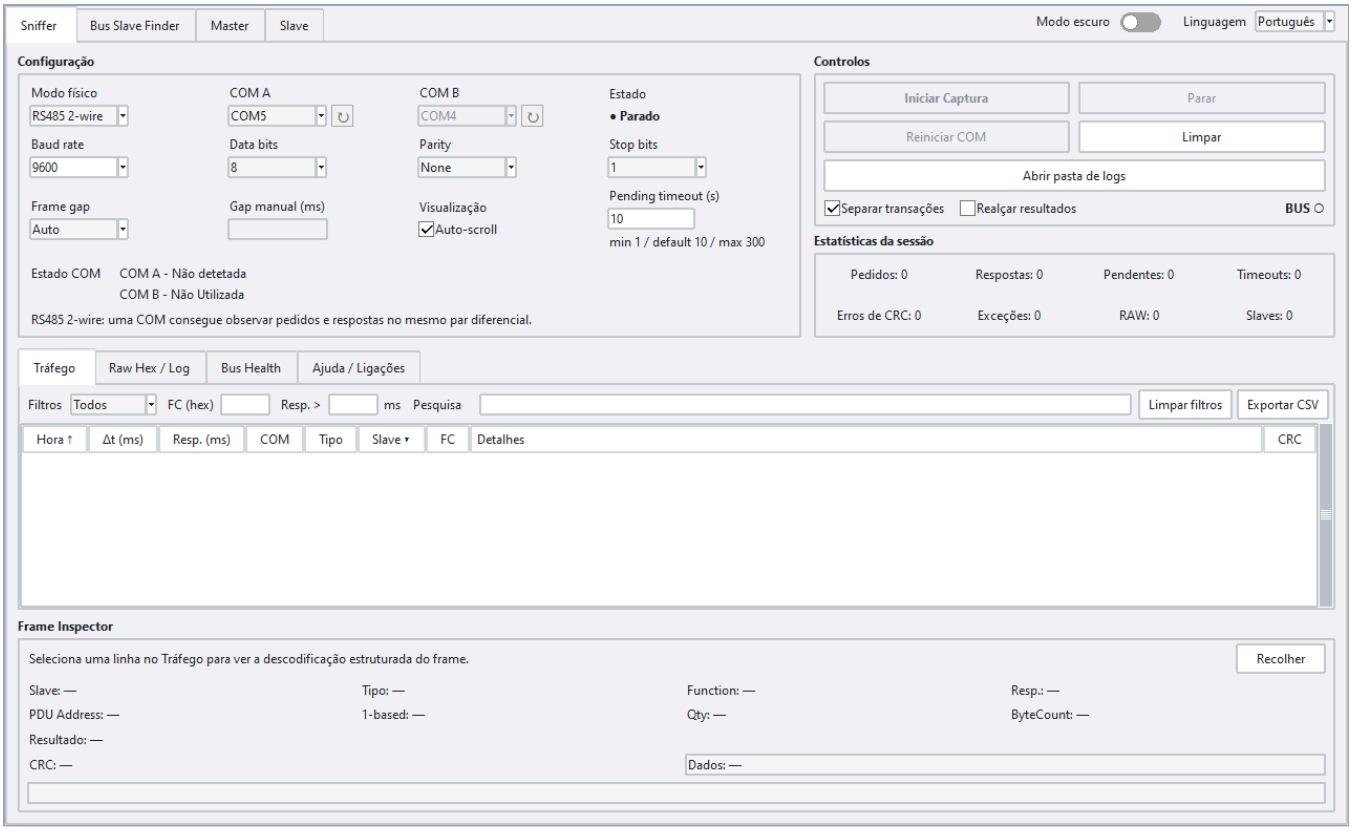
Module dependency and responsibility overview.

Composition of SnifferApp

`SnifferApp(MasterMixin, SlaveMixin, SlaveFinderMixin, CaptureMixin, ViewMixin, tk.Tk)` composes the application. `mb_gui.py` constructs and coordinates the UI; the five mixins supply the Master tab, the Slave tab, active discovery, passive capture, and presentation/logging behavior. `MasterMixin` and `SlaveMixin` both inherit `TableUiMixin` from `mb_table_ui.py`, which holds their common window-grid widgets.

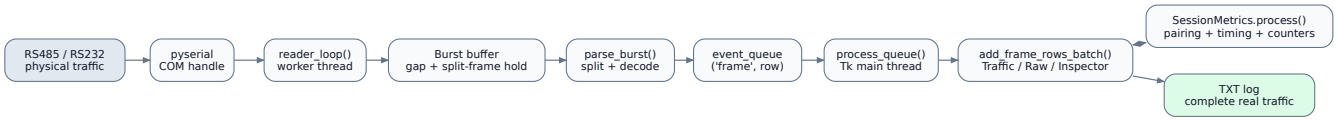
Important architectural boundaries

- **Protocol vs UI:** CRC, frame recognition, parsing and SessionMetrics are in `mb_protocol.py` and have no Tkinter dependency.
- **Workers vs Tk:** serial reader/discovery threads communicate through `event_queue`; Tk widgets are updated from the main thread.
- **Source history vs presentation:** filters and sorting rebuild the view from `_frame_history`; they do not modify the raw session record or statistics.
- **Passive vs active:** the Sniffer only reads. Bus Slave Finder intentionally transmits requests and is protected by a safety confirmation/global busy state. The Master and the Slave transmit too and share the same busy state: only one of Sniffer, Finder, Master and Slave runs at a time.
- **Appearance vs geometry:** Light/Dark changes colors; language changes text; both are designed not to rebuild widget geometry.



Current Sniffer UI generated from the documented source snapshot.

4. Passive capture pipeline



From physical bytes to UI records and disk logging.

1. **Port selection.** `selected_port_plan()` maps RS485/RS232 mode to one or two receive channels.
2. **Serial configuration.** `serial_config()` translates Baud/Data bits/Parity/Stop bits into pyserial constants.
3. **Gap calculation.** Automatic mode uses the shared Modbus t3.5 helper; manual mode validates the user-entered millisecond value.
4. **Reader workers.** One daemon thread per active receive channel reads bytes, records the arrival time of every read and closes a burst after the configured silence. If the burst ends with the start of an unfinished known frame, the reader waits up to `SNIFFER_SPLIT_FRAME_HOLD_MS` for the rest (USB latency-timer splits).
5. **Parsing.** Each burst is split/resynchronised and decoded by `parse_burst()`; each frame keeps the arrival time of the read holding its first byte.
6. **Thread boundary.** Parsed rows are placed in `event_queue`.
7. **Main-thread processing.** `process_queue()` drains events in batches and passes frame batches to `add_frame_rows_batch()`.
8. **Metrics/view/log.** The view runs SessionMetrics, updates Traffic/Raw/Inspector/Bus Health and writes complete real traffic to the lazy TXT log.

> **Threading rule.** The serial threads do not call Tkinter methods. The queue is the explicit synchronization boundary between workers and the GUI thread.

5. Frame model and parser behavior

RTU CRC and frame timing

`crc16_modbus()` implements the Modbus CRC-16 algorithm with a 256-entry lookup table (`_CRC16_TABLE`). `append_crc()` appends low byte then high byte. `modbus_rtu_interframe_gap_seconds()` provides the common t3.5 timing used by Sniffer and Finder.

Recognised frame shapes

`_candidate_prefixes_at()` recognises structural lengths for FC01/02/03/04, FC05/06/08, FC07, FC15/16, FC22, FC23 and standard Exception frames. Unknown Function Codes are deliberately handled conservatively.

Resynchronisation

The parser first tries a recognised, CRC-valid frame at offset zero. If that fails, it searches later offsets but only using recognised frame shapes. This is the performance and false-positive safeguard introduced after the earlier brute-force CRC scan.

- Known-shape prefix before a recovered valid frame + bad CRC → `CRC ERROR`.
- Unstructured bytes before a recovered frame → `RAW/UNSYNC`.
- Known-shape complete remainder + bad CRC → `CRC ERROR`.
- Unsupported-FC complete remainder with valid CRC → `UNKNOWN / OK`.
- Otherwise → `RAW/UNPARSED`.

Bursts split by USB adapters

`burst_may_continue()` re-runs the splitter on the burst and inspects only the final unvalidated segment: if it starts with a plausible Slave ID and a known Function Code but is shorter than every candidate length, more bytes are expected. The reader re-checks only when the buffer length changes, so a long noise burst is not re-parsed on every poll.

Read-response bits

FC01/FC02 responses are decoded least-significant bit first by `read_bits_from_response()`. `read_bits_details()` groups the bits per data byte, shows at most 32 of them in Traffic and trims the padding of the last byte once the matching request quantity is known.

Parser row schema

Field	Meaning
`time`	Display timestamp string.
`channel`	BUS, A→B, B→A or SIM source.
`type`	REQUEST, RESPONSE, EXCEPTION, RAW/UNSYNC, RAW/UNPARSED, UNKNOWN or REQ/RESP (FC05/06/08/22, resolved to REQUEST/RESPONSE by SessionMetrics).
`slave`, `fc`	Decoded address and hexadecimal Function Code when available.
`details`	Human-readable protocol details.
`crc`	OK, ERROR or ?.
`raw`	Original frame bytes rendered as hexadecimal.
`pdu_address`, `one_based`, `qty`	Addressing/quantity metadata when decoded.
`_frame_len`	Number of observed bytes represented by the row.
`_epoch`	Floating timestamp used for timing/pairing: arrival time of the read holding the frame's first byte.

6. Session metrics and Bus Health

`SessionMetrics` is stateful per capture/session. It records counters, pending requests, response samples and diagnostic aggregates before GUI filters are applied.

Request/response pairing

A request key is `(Slave ID, base Function Code)`. Valid REQUEST rows are queued FIFO per key. RESPONSE/EXCEPTION rows remove the oldest matching pending request and compute response time from the two epoch timestamps.

> **Pairing limitation.** Modbus RTU has no transaction identifier. The implementation therefore uses FIFO matching by Slave and Function Code. If a non-standard master pipelines multiple same-Slave/same-FC requests without waiting for responses, pairing cannot be proven uniquely from RTU headers alone.

Echo-shaped functions

FC05, FC06, FC08 and FC22 requests and their normal responses have the same length, so the parser emits `REQ/RESP`. `_resolve_echo_direction()` classifies the frame before accounting: RESPONSE when an older pending request with the same key exists (identical bytes for the exact-echo FCs 05/06/22; opposite channel in RS232 dual RX), otherwise REQUEST.

> **Echo limitation.** On a single RS485 channel, if a master repeats the same write to a Slave that never answers, the repetition is indistinguishable from an echo response and is paired as one.

Broadcast

Requests to Slave 0 are counted as requests and in `broadcast_requests`, but are never pending: by definition they receive no response and must not become timeouts.

Timeouts

Pending requests older than the configured Pending timeout are expired. The timeout is attached to the original REQUEST record through its sequence ID; MBToolkit does not invent a fake timeout frame. The view retags only the affected rows instead of rebuilding the table.

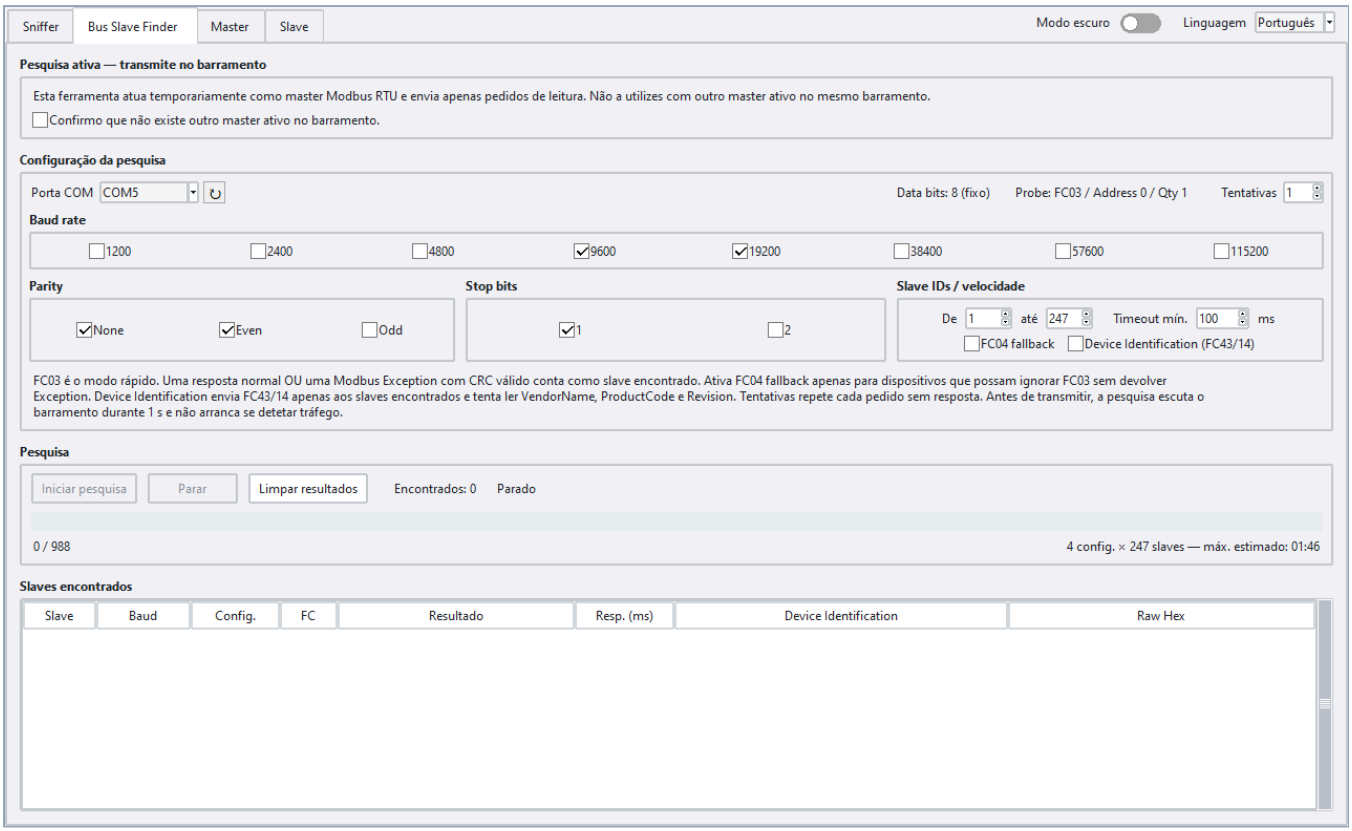
Bus Health calculations

- **Request rate:** requests divided by observed elapsed session time.
- **Response time:** average, minimum, maximum and nearest-rank P95 from valid paired responses.

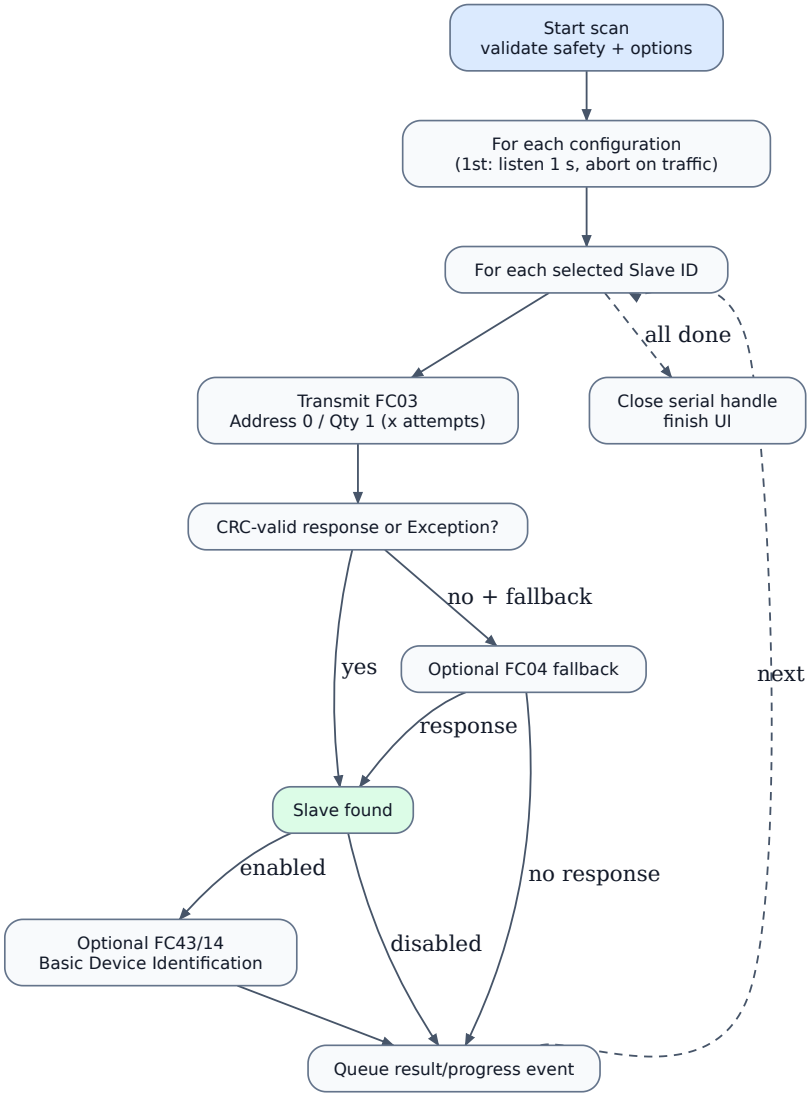
- **Slowest Slave:** highest average response time among sampled Slaves.
- **Unvalidated bytes:** bytes in explicit CRC-error frames plus RAW blocks divided by all observed bytes.
- **Timeout rate:** expired requests / observed non-broadcast requests.
- **Approximate bus utilisation:** observed bytes × configured serial bits per character / (baud × elapsed time).
- **Exceptions / Slave:** counts of Exception responses grouped by Slave ID.

> **Physical-layer boundary.** Bus utilisation and unvalidated-byte percentage are software/capture indicators. The program does not measure analog RS485 voltage, bias, reflections, ringing, common-mode level or rise/fall times.

7. Active Bus Slave Finder



Current active Finder UI generated from the documented source.



Discovery flow for each selected configuration and Slave ID.

Discovery semantics

The standard probe is FC03, Address 0, Quantity 1. A CRC-valid normal response proves the Slave exists. A CRC-valid Modbus Exception also proves the Slave exists, because the addressed device had to receive and interpret the request to generate that Exception.

FC04 fallback is optional and is attempted only after FC03 produces no accepted response. **Attempts** (1–3) repeats each unanswered probe before moving on. Device Identification is also optional and is performed only after a Slave has already been found.

Idle-bus check

Before the first transmission of a scan, `bus_activity_detected()` listens for `DISCOVERY_IDLE_CHECK_SECONDS` on the first configuration. Any byte, even one framed at another baud rate, means something else is transmitting: the worker raises an error naming the port and the first bytes, and nothing is sent. This complements the safety checkbox.

FC43/14 Device Identification

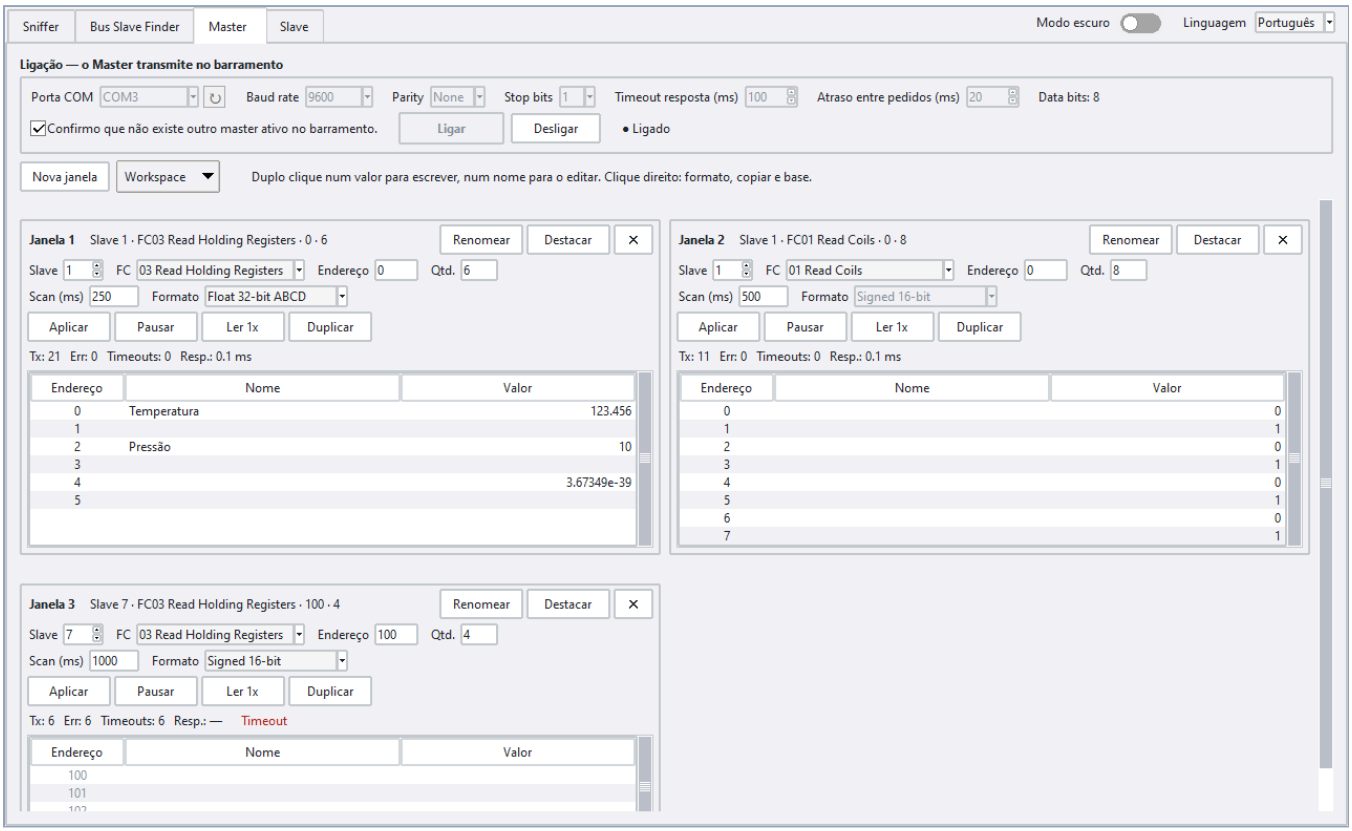
The request uses Function Code 0x2B, MEI type 0x0E and Basic Device Identification read code 0x01. The parser reads variable-length objects, validates CRC and follows `More Follows` pages up to the configured page cap. Display output focuses on object 0x00 VendorName, 0x01 ProductCode and 0x02 MajorMinorRevision.

Driver/error handling

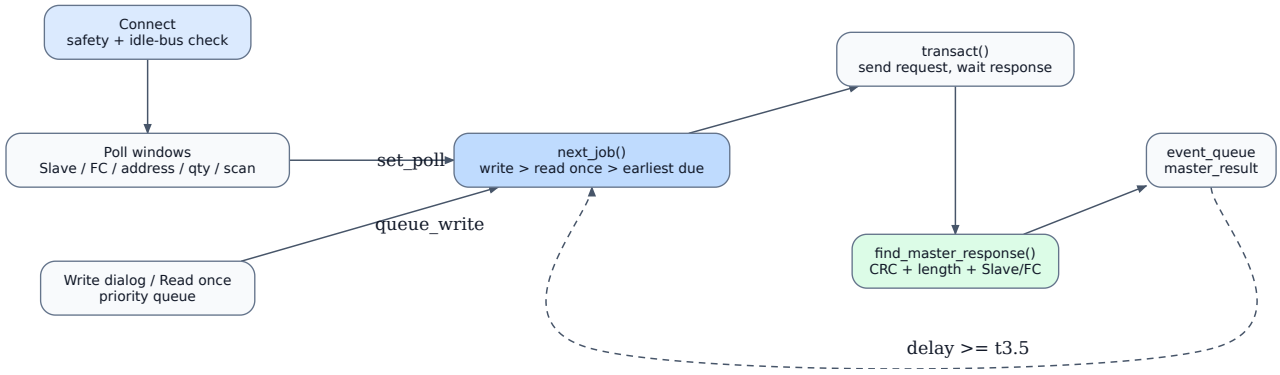
A serial write/read failure is treated as a communication/driver error, not as an ordinary "Slave did not respond" result. Write timeout recovery can cancel/clear driver buffers and retry once. Stop requests use `threading.Event` and best-effort cancel_read/cancel_write support.

> **Operational safety.** This tab actively transmits Modbus RTU requests. It must not be run on a bus that already has another active master unless the system is explicitly designed for multi-master behavior.

8. Master (poll) tab



Master tab connected to the in-memory test slave of tests/test_master.py.



Master engine: one request at a time over one serial port.

The Master tab reads and writes values as a Modbus RTU master, in the style of Modbus Poll. Each **poll window** describes one data area of one Slave; the state of a window lives in a `MasterWindow` object, independent of where its pane is shown.

Scheduling on one bus

Modbus RTU allows a single request on the bus at a time, so one `MasterEngine` thread serves every window over one serial port. Its `next_job()` returns, in this order: a queued write, a queued "Read once", or the enabled window whose next due time is earliest. A window becomes due again `scan_ms` after its previous request started (0 polls continuously). After every transaction the engine leaves the configured

delay between polls on the bus, never less than Modbus t3.5. Response time and the response timeout count from the end of the transmitted request.

Frames and responses

`build_read_request()` and `build_write_request()` build FC01–04 and FC05/06/15/16 frames within the Modbus limits (2000 bits / 125 registers per read, 1968 coils / 123 registers per write). `find_master_response()` scans the received bytes for the CRC-valid frame of the expected length from the addressed Slave, or a 5-byte exception. Read requests echoed by an RS485 adapter are skipped because their length and CRC do not fit. Write responses must echo the request (FC05/06) or its address/quantity (FC15/16). Broadcast writes (Slave 0) are not answered and are reported as sent.

Data formats

Format	Meaning
Signed / Unsigned 16-bit	One register as a two's-complement or unsigned integer.
Hex / Binary 16-bit	One register as `0xHHHH` or two binary bytes.
Signed / Unsigned / Float 32-bit	Two registers. Register n carries bytes A B and register n+1 bytes C D; the order names the value's bytes: ABCD big-endian, CDAB word swap, BADC byte swap, DCBA little-endian. The value is shown on the first register's row.
Signed / Unsigned 64-bit, Double	Four registers (bytes A B, C D, E F, G H): ABCDEFGH big-endian, GHEFCDAB registers in reverse order, BADCFEHG byte swap, HGFEDCBA little-endian. Doubles show up to 15 significant digits.

Per-address formats

The window's **Format** is the default. `MasterWindow.formats` maps the PDU address where a value starts to its own format, chosen in the right-click menu for the selected rows (drag with the mouse, or Ctrl/Shift, for several; ttk has no rubber-band selection, so `TableUiMixin._table_drag_start/move/stop` implement it, with auto-scroll past the table edges). `apply_format_to_rows()` applies 16-bit formats to every selected row and 32/64-bit formats to consecutive rows in multiples of 2/4, removing overlapping per-address formats; "Reset to window format" removes them. `register_groups()` splits the window into values; a default multi-register value cut by a per-address format, or running past the window, shows "—". The same menu copies the selected rows (also Ctrl+C) as tab-separated text and sets the address base (0 = PDU Address, 1 = manual numbering).

Scaling

`MasterWindow.scales` maps the PDU address where a value starts to a scale built by `validate_scale()`: two raw/engineering points, 0–6 decimals and a unit. `format_window_values()` shows scaled values as `e1 + (raw − r1) × (e2 − e1) / (r2 − r1)` with the unit; Hex and Binary stay raw. When writing, the typed engineering value is converted back with `raw_text_for_scaled()` (rounded for integer formats).

Open in Master

`master_definition_from_record()` takes Slave, function, address and quantity from a Traffic REQUEST or from the request matched to a RESPONSE/EXCEPTION (FC01–FC04 only). `master_open_definition()` creates the window, shows the Master tab and, when the Master is not connected, copies the port and serial parameters. The Bus Slave Finder opens a found Slave the same way from its result row (FC03/FC04, address 0, 10 registers, the baud/parity/stop it answered on).

Reconnection

The engine receives a `reopen` callable. A serial or driver failure during the session closes the port, posts `master_reconnecting` and retries every `MASTER_RECONNECT_INTERVAL_SECONDS` until the port opens again (`master_reconnected`) or the user disconnects; the values turn grey in the meantime.

Writing

Double-clicking a value in an FC01 or FC03 window opens the write dialog. `master_build_write()` validates the Slave (0–247), the address in the window's base and the values, encodes them with the format of the clicked value and builds an FC05, FC06, FC15 or FC16 request; 32/64-bit formats require FC16. Several values separated by ";" go to consecutive addresses. An accepted write queues a "Read once" of the window so the table shows the new value.

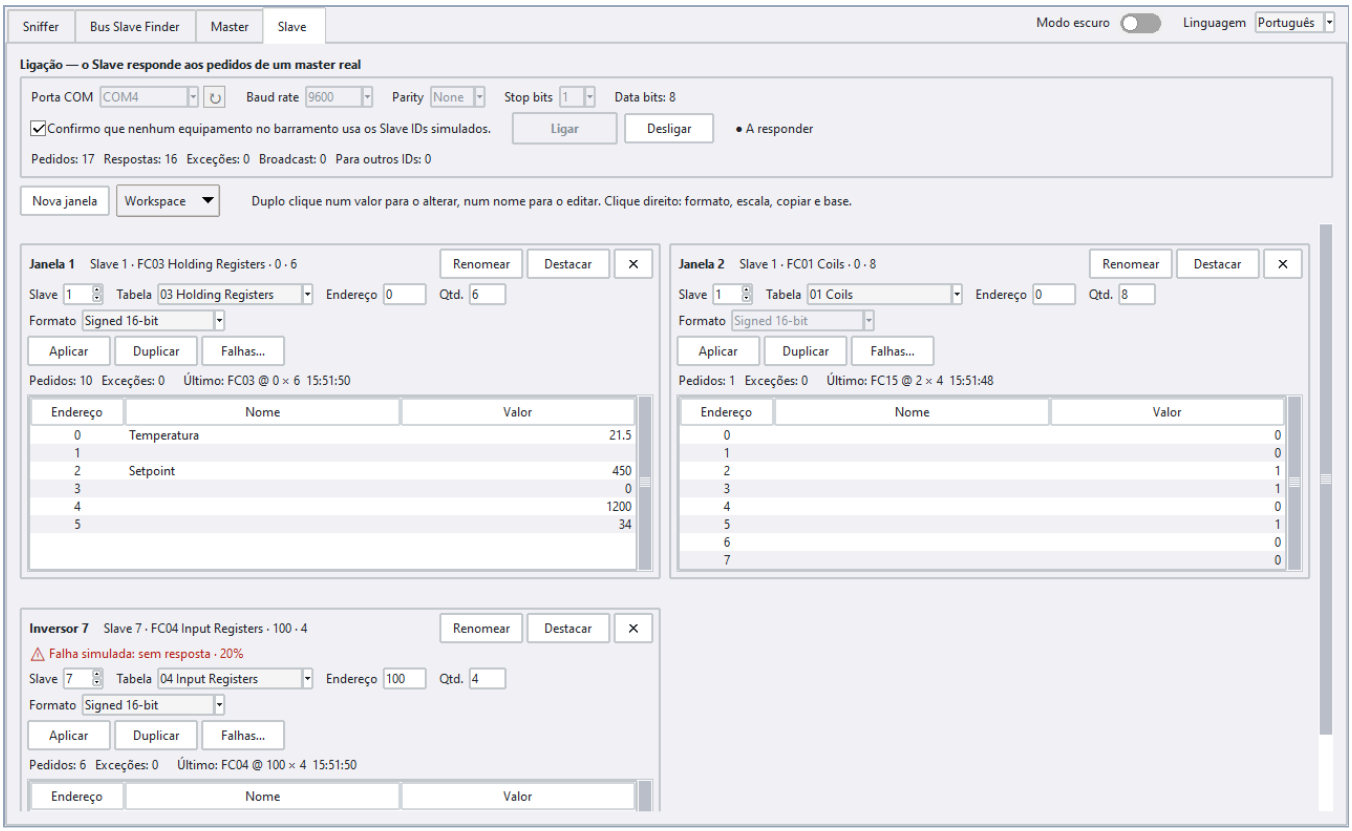
Window life cycle

- **Apply** validates the fields and updates the engine; a changed data area resets the counters. Results of a request sent with the previous definition are ignored.
- Each pane has a header with the window name (**Rename** or double-click), a summary with the Slave, the function code and name (e.g. FC03 Read Holding Registers), the address and the quantity, **Pop out** and a × close button, so closing never depends on the pane width.
- **Pop out** rebuilds the pane inside a `tk.Toplevel`; closing it or **Dock** returns it to the grid. The grid uses as many equal columns as the tab width allows, measured from the panes' requested width (which grows with Windows display scaling).
- Panes created after start-up receive the application's mouse-wheel guard tag: the wheel scrolls the window grid and never changes combobox/spinbox values.
- The **Workspace** menu saves named `.mbsw` files (JSON) with the connection parameters, every window, its name and address names in `%APPDATA%\MBToolkit\Workspaces\Master` (`get_master_workspaces_folder()`, which moves workspaces left directly in `Workspaces` by earlier versions into it once), beside `settings.json`; it lists them to open or delete. Invalid entries are skipped with a warning.
- Changes to the open workspace mark it "(modified)". `settings.json` keeps `master.last_workspace`, reopened at start-up without connecting; on exit, `master_confirm_close()` asks Yes/No/Cancel only for a saved workspace with unsaved changes.

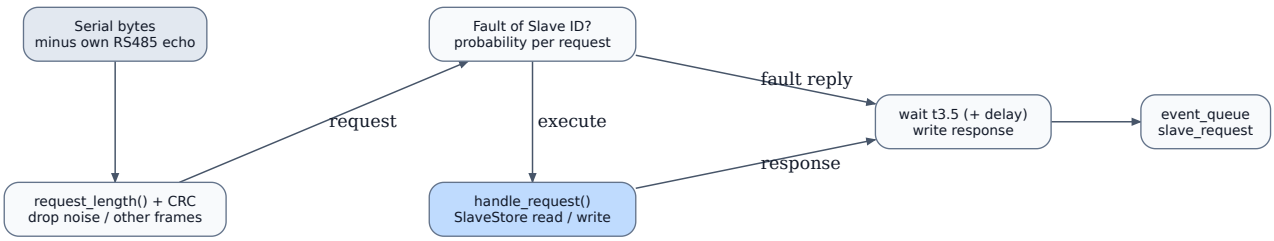
Operational safety. Like Bus Slave Finder, the Master transmits. Connecting requires the safety confirmation and an idle-bus check; while it is connected the global busy flag locks Sniffer, Simulation and Finder.

Phase boundary. Logging, conditional colours, charts, Test Center, FC22/23 and Modbus TCP are planned later phases and are not part of v3.4.

9. Slave (simulator) tab



Slave tab answering a Master engine over the in-memory link of tests/test_slave.py.



Slave engine: assemble a request, answer it from the store, apply faults.

The Slave tab is the inverse of the Master: the application behaves as one or more Modbus RTU devices, listens on a COM port and answers the requests of a real master. Each window is one data table (Coils, Discrete Inputs, Holding Registers or Input Registers) of one simulated Slave ID, defined by a start address and a quantity (up to 2000). Its state lives in a `SlaveWindow`.

Data store

`SlaveStore` keeps the values per (Slave ID, table) and address, starting at 0, plus the set of addresses defined by the windows (`set_areas()`, called whenever a window is added, applied or closed) and the active fault of each Slave ID. It is guarded by a lock: the engine thread reads and writes while the GUI edits. Windows over the same addresses therefore show the same values.

Answering one request

`handle_request(store, frame)` executes one CRC-valid request like a real device:

- a Slave ID that is not simulated is `ignored` (no reply);
- an unsupported function gets Exception 01; a wrong length, a quantity outside the Modbus limits, an FC05 value other than FF00/0000 or a byte count that does not match gets Exception 03;
- addresses not entirely inside the defined areas get Exception 02;
- reads return the stored bits or registers; writes store the values and answer with the echo (FC05/06) or the address/quantity (FC15/16), listing the written ranges in `SlaveReply.written`;
- broadcast (Slave 0) writes are executed on every simulated Slave ID whose areas cover the addresses and are never answered.

Serial loop

`SlaveEngine` reads the port continuously. `request_length()` knows the size of FC01–06 (8 bytes) and FC15/16 (9 + byte count) requests, so a request is answered as soon as it is complete and CRC-valid. Bytes that do not form a request (other devices' responses, noise) are dropped one at a time until a request lines up; after `SLAVE_RESYNC_SILENCE_MS` of silence the rest of the buffer is resolved, and a CRC-valid frame of an unknown function gets Exception 01. Before replying the engine waits the Modbus t3.5 gap; after replying it expects the RS485 adapter echo of its own response and removes it from the received bytes. Each handled request is posted as `slave_request` with its Slave ID, table, address, quantity, status and written ranges. Serial failures reopen the port like the Master.

Fault simulation

Faults are set per Slave ID (a real device fails on all its tables) and validated by `validate_fault()`: delay (0–10000 ms), no response, a chosen Exception (01, 02, 03, 04 or 06), wrong CRC, and a probability (1–100 %) drawn for every request. When the fault triggers, no response wins over the Exception, which wins over executing the request; the wrong CRC and the delay apply to whatever is sent. Only active faults are stored; their description is shown in red in the header of every window of that Slave ID.

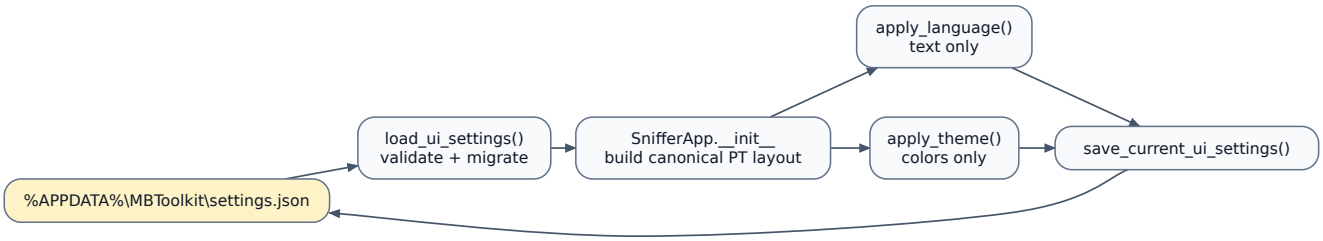
Windows and values

- Double-clicking a value edits it locally: bits toggle; registers use the value's format and scale (`slave_set_value()`). Nothing is sent on the bus.
- Master writes highlight the written rows for `SLAVE_HIGHLIGHT_SECONDS` (the `written` tag replaces the zebra tag so it is always visible); a 250 ms timer removes expired highlights.
- Each window counts the requests and exceptions that touched its addresses and shows the last request; the connection bar counts requests, responses, exceptions, broadcasts and requests for other IDs.
- The Table field and the window header show the read function code with the table name (`table_choice_label()`: "03 Holding Registers"), like the FC field of the Master.
- Formats, scales, names, Copy, address base, drag selection, Rename, Pop out and close come from `TableUiMixin`, exactly as in the Master.

- The Slave Workspace menu saves configuration only (connection, windows, names, formats, scales and faults) in `%APPDATA%\MBToolkit\Workspaces\Slave`, format `MBToolkit Slave workspace`; opening one resets the values to 0. `slave.last_workspace` is reopened at start-up and `slave_confirm_close()` asks to save a changed one on exit.

> **Operational safety.** A simulated Slave ID that also exists on the bus makes two devices answer the same request. Connecting requires a confirmation that no real device uses the simulated IDs; while listening, the global busy flag locks Sniffer, Simulation, Finder and Master. Windows and USB adapters usually add a few milliseconds to every response.

10. UI, language and theme design



Settings, language and theme lifecycle.

Canonical-language strategy

Portuguese (Portugal) is the canonical static UI language. `canonical_ui_text()` can map known English captions back to PT-PT, then `translate_text()` renders either canonical Portuguese or an explicit English translation. Technical English terms without an explicit mapping are left unchanged.

Messages with live values are stored as PT-PT templates in `PT_TO_EN` and rendered with `SnifferApp.trf(template, **values)`. Code never branches on `current_language`; a regression test enforces this.

Runtime language changes

`apply_language()` updates widget text, table headings, context menus, Help content and live status strings. Protocol bytes and parser data are not translated; only presentation fragments such as `Registos` / `Registers` are changed.

Language-independent geometry

The UI is built using the canonical Portuguese captions, then `_freeze_language_layout_geometry()` measures key dimensions and freezes them. This prevents shorter/longer English captions from moving the surrounding rectangles or controls.

Light/Dark invariant

The application uses one ttk `clam` construction. `mb_theme.py` configures geometry once and palette functions only change colors/state maps. Custom Canvas widgets receive palette updates separately.

Mouse-wheel behavior

An application bindtag lets Treeviews/Text/Listboxes/scrollbars consume wheel events normally while stopping wheel propagation elsewhere. This preserves table scrolling without allowing the Notebook to change tabs with the wheel.

11. Persistence, logs and export

User settings

On Windows, preferences are stored in `%APPDATA%\MBToolkit\settings.json`. Loading is defensive and every persisted choice/range is validated before use. Saves use a temporary file and replace operation.

Persisted data includes language, Light/Dark, last main tab, serial format, frame-gap mode/value, Pending timeout, auto-scroll, transaction separators, highlighting and Finder options (including attempts and the minimum timeout, migrated from the old 40 ms default to 100 ms) and the Master and Slave connection parameters. Master and Slave windows are not part of the settings: they are saved explicitly in workspace files. COM names, window geometry, traffic/results and the Finder safety confirmation are intentionally not persisted.

TXT capture log

Real-capture logs are lazy: the file is created only when traffic actually arrives. The disk log is complete and independent of table filters, sorting and the bounded GUI history.

CSV export

CSV exports the records currently retained in `_frame_history`, not necessarily an arbitrarily long complete session. It includes timing, channel, type, address metadata, CRC, raw bytes, timeout status and anomaly classification.

Bounded GUI history

The UI retains up to approximately `MAX_UI_FRAMES = 20000` records and prunes in chunks after a hysteresis threshold. Raw Text has a similar bounded-line policy. This prevents the GUI from growing indefinitely while keeping the capture log complete.

12. Module-by-module reference

The following chapters provide a complete callable inventory for every Python module. Each function/method is tied to its exact source-line range. The complete source itself appears as screenshots in Appendix A.

MBToolkit.py

Application launcher and top-level entry point. It sets the development simulation switch before importing the GUI, applies the Windows application identity, creates SnifferApp and enters Tkinter's main event loop.

Source lines	SHA-256
29	`44a1ab13bc860ea64ca57f2805f4ab7ab5c6e8455241973c3ae50f4e856d04e1`

Imports: `mb_config, mb_config.set_windows_app_user_model_id, mb_gui.SnifferApp`

Module constants / configuration data

Name	Definition
`debug_sim`	`0`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`main` main()	23-25	Main.	`set_windows_app_user_model_id, SnifferApp().mainloop, SnifferApp`

mb_config.py

Central configuration, constants, user settings validation and filesystem paths. It defines versioned application constants, discovery limits, GUI history limits, Modbus exception metadata, persistent settings behavior, application/log directories and PyInstaller resource lookup.

Source lines	SHA-256
577	`b6698ad9fb341dbb5a0448ef737909dc7453f0587ab4eb09d92987c7a6008578`

Imports: `ctypes, json, os, shutil, sys, pathlib.Path`

Module constants / configuration data

Name	Definition
`APP_NAME`	`'MBToolkit'`
`LEGACY_APP_NAME`	`'MBSniffer'`
`APP_VERSION`	`'3.4'`
`debug_sim`	`0`
`DISCOVERY_BAUD_RATES`	`(1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200)`
`DISCOVERY_PARITIES`	`('None', 'Even', 'Odd')`
`DISCOVERY_STOP_BITS`	`('1', '2')`
`DISCOVERY_DEFAULT_MIN_TIMEOUT_MS`	`100`
`DISCOVERY_LEGACY_DEFAULT_MIN_TIMEOUT_MS`	`40`
`DISCOVERY_MIN_TIMEOUT_MS`	`10`
`DISCOVERY_MAX_TIMEOUT_MS`	`2000`
`DISCOVERY_PROBE_ADDRESS`	`0`
`DISCOVERY_PROBE_QTY`	`1`
`DISCOVERY_WRITE_TIMEOUT_SECONDS`	`1.5`
`DISCOVERY_WRITE_RETRIES`	`1`
`DISCOVERY_RX_BUFFER_LIMIT`	`1024`
`DISCOVERY_MIN_ATTEMPTS`	`1`
`DISCOVERY_MAX_ATTEMPTS`	`3`
`DISCOVERY_DEFAULT_ATTEMPTS`	`1`

Name	Definition
DISCOVERY_IDLE_CHECK_SECONDS	1.0
UI_QUEUE_BATCH_SIZE	200
UI_QUEUE_IDLE_INTERVAL_MS	50
UI_QUEUE_BUSY_INTERVAL_MS	10
MIN_PENDING_REQUEST_TIMEOUT_SECONDS	1.0
DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS	10.0
MAX_PENDING_REQUEST_TIMEOUT_SECONDS	300.0
PENDING_REQUEST_TIMEOUT_SECONDS	DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS
MASTER_MAX_READ_BITS	2000
MASTER_MAX_READ_REGISTERS	125
MASTER_MAX_WRITE_BITS	1968
MASTER_MAX_WRITE_REGISTERS	123
MASTER_DEFAULT_RESPONSE_TIMEOUT_MS	1000
MASTER_MIN_RESPONSE_TIMEOUT_MS	10
MASTER_MAX_RESPONSE_TIMEOUT_MS	10000
MASTER_DEFAULT_DELAY_MS	20
MASTER_MIN_DELAY_MS	0
MASTER_MAX_DELAY_MS	10000
MASTER_DEFAULT_SCAN_MS	1000
MASTER_MAX_SCAN_MS	3600000
MASTER_MAX_WINDOWS	32
MASTER_RECONNECT_INTERVAL_SECONDS	2.0
MASTER_WORKSPACE_FORMAT	'MBToolkit Master workspace'
MASTER_WORKSPACE_FORMATS	(MASTER_WORKSPACE_FORMAT, 'MBSniffer Master workspace')
MASTER_WORKSPACE_VERSION	1
SLAVE_MAX_WINDOWS	32
SLAVE_MAX_QTY	2000
SLAVE_HIGHLIGHT_SECONDS	1.5

Name	Definition
`SLAVE_FAULT_MAX_DELAY_MS`	`10000`
`SLAVE_RESYNC_SILENCE_MS`	`40.0`
`SLAVE_WORKSPACE_FORMAT`	`'MBToolkit Slave workspace'`
`SLAVE_WORKSPACE_VERSION`	`1`
`SNIFFER_SPLIT_FRAME_HOLD_MS`	`40.0`
`SLOW_RESPONSE_THRESHOLD_MS`	`500.0`
`MODBUS_EXCEPTION_CODES`	`{1: ('Illegal Function', 'A função Modbus pedida não é suportada/permitida pelo dispositivo.'), 2: ('Illegal Data Address', 'O endereço ou a gama d…`
`MAX_UI_FRAMES`	`20000`
`MAX_RAW_TEXT_LINES`	`20000`
`UI_PRUNE_CHUNK_FRAMES`	`2000`
`RAW_TEXT_PRUNE_CHUNK_LINES`	`2000`
`RAW_TRANSACTION_SEPARATOR`	`'—' * 72 + '\n'`
`SIMULATION_FUNCTION_CODES`	`(1, 2, 3, 4)`
`SIMULATION_EXCEPTION_CODES`	`(1, 2, 3, 4, 6)`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`resource_path` resource_path(filename)	148-154	Return a bundled/source resource path for normal Python or PyInstaller.	`Path, Path(__file__).resolve`
`set_windows_app_user_model_id` set_windows_app_user_model_id()	156-168	Give Windows a stable application identity so the taskbar can associate the Tk window with MBToolkit instead of a generic Python process.	`ctypes.windll.shell32.SetCurrentProcessExplicitAppUserModelID`
`get_settings_file` get_settings_file()	172-193	Return the persistent per-user settings file.	`_settings_folder, folder.mkdir, folder.exists, _migrate_legacy_settings, Path.home`
`_settings_folder` _settings_folder(app_name)	196-202	Settings folder.	`os.environ.get, Path.home, Path, app_name.lower`

Callable	Lines	Responsibility	Representative direct calls
`_migrate_legacy_settings` _migrate_legacy_settings(folder)	205-217	Copy MBSniffer's settings and workspaces into the new MBToolkit folder.	`_settings_folder, legacy.is_dir, shutil.copytree`
`_validated_choice` _validated_choice(value, allowed, default)	220-221	Validated choice.	—
`_validated_bool` _validated_bool(value, default)	224-225	Validated bool.	`isinstance`
`_validated_number_string` _validated_number_string(value, minimum, maximum, default)	228-235	Validated number string.	`str, float`
`_validated_int_string` _validated_int_string(value, minimum, maximum, default)	238-245	Validated int string.	`str, int`
`load_ui_settings` load_ui_settings()	248-462	Reads settings.json defensively and validates every persisted choice/range.	`get_settings_file, isinstance, data.get, json.loads, sniffer_in.get`
`_validated_workspace_name` _validated_workspace_name(value)	465-472	Workspace name remembered in settings.json, or "" when unusable.	`value.strip, isinstance, any, len`
`save_ui_settings` save_ui_settings(settings=None, **updates)	475-499	Merges updates with validated current settings and uses a temporary file plus replace operation for atomic persistence; settings failures never prevent normal application use.	`get_settings_file, load_ui_settings, isinstance, current.update, path.parent.mkdir`
`get_application_directory` get_application_directory()	502-523	Resolves a user-facing application directory consistently for PyInstaller EXE, packaged BAT/source mode and development mode.	`getattr, (launcher_dir / 'MBToolkit.bat').is_file, Path(__file__).resolve, Path.cwd, Path(sys.executable).resolve`
`get_workspaces_folder` get_workspaces_folder()	525-533	Return/create the Master workspace folder beside settings.json:.	`folder.mkdir, get_settings_file`
`get_master_workspaces_folder` get_master_workspaces_folder()	536-555	Return/create the Master workspace folder inside the workspace folder:.	`get_workspaces_folder, folder.exists, folder.mkdir, root.glob, old.is_file`
`get_slave_workspaces_folder` get_slave_workspaces_folder()	558-566	Return/create the Slave workspace folder inside the workspace folder:.	`folder.mkdir, get_workspaces_folder`
`get_logs_folder` get_logs_folder()	569-577	Creates and returns the MBToolkit Logs directory beside the launcher/EXE.	`folder.mkdir, get_application_directory`

Detailed behavior of the key paths

`load_ui_settings` lines 248-462

Reads settings.json defensively and validates every persisted choice/range. Invalid/corrupt data falls back safely. It also migrates the older highlight_anomalies key to highlight_results and the old 40 ms Finder timeout default to 100 ms.

`save_ui_settings` lines 475-499

Merges updates with validated current settings and uses a temporary file plus replace operation for atomic persistence; settings failures never prevent normal application use.

`get_application_directory` lines 502-523

Resolves a user-facing application directory consistently for PyInstaller EXE, packaged BAT/source mode and development mode.

`get_logs_folder` lines 569-577

Creates and returns the MBToolkit Logs directory beside the launcher/EXE.

mb_protocol.py

Protocol engine for passive Modbus RTU parsing and session metrics. It implements CRC16, RTU t3.5 timing, frame-shape recognition, resynchronisation, decoding, request/response pairing, timeout accounting, echo-write and broadcast classification, split-frame detection, Bus Health metrics and debug traffic generation.

Source lines
1007

SHA-256
`8c4c41051e41f0c998692206ed0204d91f057cce5fce726b15ad2cce503c1ef2`

Imports: `math, random, time, bisect.bisect_right, collections.defaultdict, deque, datetime.datetime, mb_config.MODBUS_EXCEPTION_CODES, PENDING_REQUEST_TIMEOUT_SECONDS, SIMULATION_FUNCTION_CODES, SIMULATION_EXCEPTION_CODES, SLOW_RESPONSE_THRESHOLD_MS`

Module constants / configuration data

Name	Definition
`_CRC16_TABLE`	`_build_crc16_table()`
`KNOWN_FRAME_FUNCTION_CODES`	`(1, 2, 3, 4, 5, 6, 7, 8, 15, 16, 22, 23)`
`EXACT_ECHO_FUNCTION_CODES`	`(5, 6, 22)`
`BROADCAST_UNIT_ID`	`0`
`READ_BITS_DETAILS_LIMIT`	`32`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`_build_crc16_table` _build_crc16_table()	19-29	Build crc16 table.	`range, tuple, table.append`
`crc16_modbus` crc16_modbus(data)	36-41	Computes the standard Modbus CRC-16 value (initial 0xFFFF, reflected polynomial 0xA001) with one lookup in the precomputed 256-entry _CRC16_TABLE per byte. append_crc() stores the low CRC byte first, matching Modbus RTU wire order.	—
`crc_ok` crc_ok(frame)	43-48	Crc ok.	`crc16_modbus, len`

Callable	Lines	Responsibility	Representative direct calls
`append_crc` append_crc(payload)	50-52	Append crc.	`crc16_modbus, bytes`
`hex_bytes` hex_bytes(data)	54-55	Hex bytes.	`' '.join`
`modbus_rtu_interframe_gap_seconds` modbus_rtu_interframe_gap_seconds(baud, bits_per_char)	57-72	Implements the automatic t3.5 rule used by capture and discovery: 3.5 character times at or below 19200 bit/s, fixed 1.750 ms above 19200 bit/s.	`int, float, ValueError`
`plausible_request_qty` plausible_request_qty(fc, qty)	74-79	Plausible request qty.	—
`_plausible_unit_id` _plausible_unit_id(value)	89-94	Return True for Modbus RTU unit addresses that can appear on the wire.	`int`
`_candidate_prefixes_at` _candidate_prefixes_at(buf, start=0)	97-168	Infers only structurally plausible lengths for supported Modbus frame shapes.	`set, len, int, _plausible_unit_id, out.append`
`candidate_prefixes` candidate_prefixes(buf)	171-173	Compatibility wrapper for callers/tests that inspect offset zero.	`_candidate_prefixes_at`
`_crc_ok_at` _crc_ok_at(buf, start, length)	176-183	CRC check without copying the whole remaining capture buffer.	`crc16_modbus, len, memoryview`
`find_valid_prefix` find_valid_prefix(buf, start=0)	186-191	Find a CRC-valid, structurally recognised Modbus frame at *start*.	`_candidate_prefixes_at, _crc_ok_at, len`
`guess_invalid_whole_frame` guess_invalid_whole_frame(buf)	194-201	Classify a complete known-shape frame whose CRC is invalid.	`_candidate_prefixes_at, len, _plausible_unit_id`
`_unknown_whole_frame_is_valid` _unknown_whole_frame_is_valid(buf)	204-217	Conservative fallback for one unsupported FC occupying the whole burst.	`_candidate_prefixes_at, crc_ok, _plausible_unit_id, len`
`split_capture_buffer` split_capture_buffer(data)	220-269	Walks a burst from left to right.	`bytearray, find_valid_prefix, max, range, guess_invalid_whole_frame`
`burst_may_continue` burst_may_continue(data)	271-302	Reports whether a burst ends with the start of a known frame that has not fully arrived yet, so the reader can wait for bytes delayed by a USB adapter latency timer instead of emitting two RAW rows.	`split_capture_buffer, any, _plausible_unit_id, len, _candidate_prefixes_at`

Callable	Lines	Responsibility	Representative direct calls
`read_bits_from_response` read_bits_from_response(frame, qty=None)	305-322	Return FC01/FC02 coil/input states, least-significant bit first.	`len, range, int`
`read_bits_details` read_bits_details(frame, qty=None)	330-338	Formats FC01/FC02 response bits in address order, grouped per data byte, trimmed to the request quantity when known and capped at 32 visible bits.	`read_bits_from_response, ''.join, ' '.join, len, str`
`decode_frame` decode_frame(frame, kind_hint)	341-449	Extracts Slave, FC, PDU address, one-based address, quantity and selected payload details from a split frame.	`str(kind_hint).startswith, str, len, MODBUS_EXCEPTION_CODES.get, read_bits_details`
`format_clock_time` format_clock_time(epoch)	451-452	Format clock time.	`datetime.fromtimestamp(float(epoch)).strftime, datetime.fromtimestamp, float`
`parse_burst` parse_burst(data, timestamp, channel='', event_time=None, chunk_times=None)	455-498	Transforms split frames into the row dictionary consumed by the rest of the program.	`split_capture_buffer, time.time, float, len, decode_frame`
`class SessionMetrics`	**500-872**	**Tracks pairing, counters and bus-health diagnostics for one session.**	**Class container**
`SessionMetrics.__init__` __init__(self, pending_timeout_seconds=PENDING_REQUEST_TIMEOUT_SECONDS)	503-505	Init.	`float, self.reset`
`SessionMetrics.reset` reset(self)	507-531	Reset.	`set, defaultdict, deque`
`SessionMetrics._key` _key(self, row)	533-543	Key.	`int, row.get`
`SessionMetrics.pending_count` pending_count(self)	545-546	Pending count.	`sum, len, self.pending.values`
`SessionMetrics.pop_expired_requests` pop_expired_requests(self)	548-551	Pop expired requests.	—
`SessionMetrics.expire_pending` expire_pending(self, now=None)	553-591	Expire stale requests and retain their sequence IDs for the GUI.	`float, list, time.time, self.pending.keys, self.pending.pop`
`SessionMetrics.force_timeout` force_timeout(self, slave, fc)	593-617	Debug/simulation helper: expire the oldest pending request for one key.	`self.pending.get, pending_list.pop, dict, self.pending.pop, int`

Callable	Lines	Responsibility	Representative direct calls
`SessionMetrics._resolve_echo_direction` _resolve_echo_direction(self, row)	619-651	Classifies FC05/06/08/22 frames as REQUEST or RESPONSE from the pending queue: same key, identical bytes for exact echoes and the opposite channel in RS232 dual RX.	`self._key, self.pending.get, row.get, candidate.get`
`SessionMetrics._record_response_time` _record_response_time(self, row, response_ms)	653-664	Record response time.	`float, self.response_times_ms.append, self.response_times_by_slave[slave].append, int, row.get`
`SessionMetrics.process` process(self, row)	666-765	This is the session accounting hot path.	`float, self.expire_pending, self.pop_expired_requests, row.get, self._record_response_time`
`SessionMetrics._p95` _p95(values)	768-773	P95.	`sorted, max, min, float, len`
`SessionMetrics.health_summary` health_summary(self, *, baud=9600, data_bits=8, parity='None', stop_bits=1, now=None)	775-872	Builds the Bus Health model: request rate, response average/min/max/P95, slowest Slave, unvalidated-byte rate, timeout rate, active Slave count, approximate bus utilisation and exception distribution.	`list, self._p95, self.response_times_by_slave.items, time.time, max`
`u16_bytes` u16_bytes(value)	874-876	U16 bytes.	`bytes, int`
`corrupt_crc` corrupt_crc(frame)	878-882	Corrupt crc.	`bytearray, bytes, len`
`build_simulation_transaction` build_simulation_transaction(slave, fc, rng)	884-928	Build one syntactically valid Modbus request/response pair.	`int, rng.randint, append_crc, bytes, b''.join`
`build_random_simulation_plan` build_random_simulation_plan(rng=None)	930-952	Build exactly four simulation transactions: FC01, FC02, FC03 and FC04, one each.	`rng.sample, zip, random.Random, range, build_simulation_transaction`
`build_anomaly_simulation_plan` build_anomaly_simulation_plan(rng=None)	956-1006	Creates the deterministic categories required to preview diagnostic highlighting: normal response, slow response, Exception, bad CRC, timeout and RAW bytes.	`rng.sample, build_simulation_transaction, append_crc, corrupt_crc, random.Random`

Detailed behavior of the key paths

`crc16_modbus` **lines 36-41**

Computes the standard Modbus CRC-16 value (initial 0xFFFF, reflected polynomial 0xA001) with one lookup in the precomputed 256-entry _CRC16_TABLE per byte. append_crc() stores the low CRC byte first, matching Modbus RTU wire order.

`modbus_rtu_interframe_gap_seconds` **lines 57-72**

Implements the automatic t3.5 rule used by capture and discovery: 3.5 character times at or below 19200 bit/s, fixed 1.750 ms above 19200 bit/s.

`_candidate_prefixes_at` **lines 97-168**

Infers only structurally plausible lengths for supported Modbus frame shapes. This constrains resynchronisation and avoids the former brute-force CRC scan across arbitrary lengths.

`split_capture_buffer` **lines 220-269**

Walks a burst from left to right. A valid recognised frame is emitted immediately; otherwise the parser searches later offsets for a recognised CRC-valid frame. Bytes before the recovered frame become CRC ERROR if they exactly match a known frame shape, otherwise RAW/UNSYNC. If nothing can be recovered, the remainder becomes CRC ERROR, UNKNOWN/OK, or RAW/UNPARSED depending on its structure and CRC.

`burst_may_continue` **lines 271-302**

Reports whether a burst ends with the start of a known frame that has not fully arrived yet, so the reader can wait for bytes delayed by a USB adapter latency timer instead of emitting two RAW rows.

`read_bits_details` **lines 330-338**

Formats FC01/FC02 response bits in address order, grouped per data byte, trimmed to the request quantity when known and capped at 32 visible bits.

`decode_frame` **lines 341-449**

Extracts Slave, FC, PDU address, one-based address, quantity and selected payload details from a split frame. It handles Exceptions and common read/write Function Codes without modifying the raw bytes.

`parse_burst` **lines 455-498**

Transforms split frames into the row dictionary consumed by the rest of the program. It attaches display time, channel, decoded fields, raw hex, frame length and a floating epoch timestamp; with chunk_times each frame gets the arrival time of its own serial read.

`SessionMetrics.process` **lines 666-765**

This is the session accounting hot path. It expires old pending requests first, resolves REQ/RESP frames, updates byte/counter statistics, stores valid non-broadcast requests by (Slave, FC), FIFO-pairs responses/exceptions to pending requests, calculates response time and returns timing/pairing information to the view.

`SessionMetrics._resolve_echo_direction` **lines 619-651**

Classifies FC05/06/08/22 frames as REQUEST or RESPONSE from the pending queue: same key, identical bytes for exact echoes and the opposite channel in RS232 dual RX. Broadcast frames are always requests.

`SessionMetrics.health_summary` **lines 775-872**

Builds the Bus Health model: request rate, response average/min/max/P95, slowest Slave, unvalidated-byte rate, timeout rate, active Slave count, approximate bus utilisation and exception distribution.

`build_anomaly_simulation_plan` **lines 956-1006**

Creates the deterministic categories required to preview diagnostic highlighting: normal response, slow response, Exception, bad CRC, timeout and RAW bytes.

> **Design note.** Important design boundary: this module contains no Tkinter code and can be regression-tested independently. That separation is valuable if the project is later rebuilt for a thesis or extended with additional front ends.

mb_capture.py

Serial-port management and passive acquisition pipeline. It opens selected COM ports, starts reader threads, converts inter-frame silence into capture bursts, parses those bursts, queues GUI events, controls simulation, handles COM restart and performs orderly shutdown.

Source lines	SHA-256
902	`78c925c4aa8c9d3e1b263dc9e91f28f2c2e08ef4e62573f949e985fa6f54fac0`

Imports: `queue, random, threading, time, traceback, datetime.datetime, tkinter, tkinter.messagebox, mb_config.APP_NAME, APP_VERSION, MIN_PENDING_REQUEST_TIMEOUT_SECONDS, MAX_PENDING_REQUEST_TIMEOUT_SECONDS, SNIFFER_SPLIT_FRAME_HOLD_MS, UI_QUEUE_BATCH_SIZE, UI_QUEUE_IDLE_INTERVAL_MS, UI_QUEUE_BUSY_INTERVAL_MS, debug_sim, mb_protocol.parse_burst, burst_may_continue, build_anomaly_simulation_plan, modbus_rtu_interframe_gap_seconds, mb_slave_finder.cancel_serial_io`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class CaptureMixin`	**33-901**	**Methods for serial capture, COM management and simulation.**	**Class container**
`CaptureMixin.set_sniffer_refresh_state` set_sniffer_refresh_state(self, state)	36-41	Apply the same state to both Sniffer COM refresh buttons.	`getattr, button.configure`
`CaptureMixin.update_mode_ui` update_mode_ui(self)	43-63	Update mode ui.	`self.mode_var.get, self.port_b_combo.configure, self.update_com_status, self.mode_help_var.set, self.tr`
`CaptureMixin.update_gap_state` update_gap_state(self)	65-68	Update gap state.	`self.gap_entry.configure, self.gap_mode_var.get`
`CaptureMixin.get_serial_modules` get_serial_modules(self)	70-76	Get serial modules.	—
`CaptureMixin.refresh_ports` refresh_ports(self)	78-123	Refresh ports.	`self.get_serial_modules, hasattr, self.update_com_status, self.port_a_var.set, self.port_b_var.set`

Callable	Lines	Responsibility	Representative direct calls
`CaptureMixin.detected_ports` detected_ports(self)	125-133	Return the set of COM device names currently reported by Windows/pyserial.	`self.get_serial_modules, set, list_ports.comports`
`CaptureMixin.port_status_text` port_status_text(self, label, port, enabled=True)	135-151	Port status text.	`self.serial_by_port.get, self.detected_ports, self.tr`
`CaptureMixin.update_com_status` update_com_status(self)	153-172	Update COM status without opening/probing a port.	`self.port_a_status_var.set, self.port_b_status_var.set, hasattr, self.mode_var.get, self.port_status_text`
`CaptureMixin.poll_com_status` poll_com_status(self)	174-186	Poll com status.	`self.update_com_status, hasattr, self.update_stats_labels, self.after`
`CaptureMixin.selected_port_plan` selected_port_plan(self)	188-206	Translates physical mode into one or two receive channels.	`self.mode_var.get, self.port_a_var.get().strip, self.port_b_var.get().strip, ValueError, plan.append`
`CaptureMixin.open_selected_ports` open_selected_ports(self, serial, config)	208-221	Open the selected capture ports and return [(channel, serial_obj), ...].	`self.selected_port_plan, opened.append, self.open_serial_port, ser_obj.close`
`CaptureMixin.install_opened_ports` install_opened_ports(self, opened)	223-228	Install opened ports.	`self.update_com_status, getattr`
`CaptureMixin.start_reader_threads` start_reader_threads(self, opened, gap_ms)	230-240	Start reader threads.	`threading.Thread, self.reader_threads.append, t.start`
`CaptureMixin.restart_com_port` restart_com_port(self)	242-338	Close and reopen the selected COM port(s).	`self.get_serial_modules, self.restart_btn.configure, self.status_var.set, self.stop_event.set, list`
`CaptureMixin.set_simulation_button_state` set_simulation_button_state(self, state)	340-343	Change Simulation button state only when the optional button exists.	`self.sim_btn.configure`
`CaptureMixin.selected_pending_timeout_seconds` selected_pending_timeout_seconds(self)	345-364	Selected pending timeout seconds.	`float, ValueError, self.pending_timeout_var.get().strip().replace, self.trf, self.tr`
`CaptureMixin.calculated_gap_ms` calculated_gap_ms(self)	366-378	Returns the manual gap after validation, or computes the automatic RTU t3.5 value from baud/data/parity/stop settings using the shared protocol timing helper.	`int, float, self.gap_mode_var.get, self.baud_var.get, self.data_var.get`

Callable	Lines	Responsibility	Representative direct calls
`CaptureMixin.serial_config` serial_config(self, serial)	380-402	Serial config.	`int, self.baud_var.get, self.parity_var.get, self.stop_var.get, self.data_var.get`
`CaptureMixin.open_serial_port` open_serial_port(self, serial, port, config)	404-436	Open serial port.	`serial.Serial, ser.open, ser.reset_input_buffer`
`CaptureMixin.start_capture` start_capture(self)	438-496	Validates state and serial settings, opens the selected port(s), resets the current session, prepares lazy logging, starts reader workers and updates the GUI to capturing state.	`self.get_serial_modules, self.mode_var.get, self.port_a_var.get().strip, self.port_b_var.get().strip, self.install_opened_ports`
`CaptureMixin.reader_loop` reader_loop(self, ser, gap_ms, channel, generation)	498-575	Runs in a daemon thread per capture channel.	`max, min, bytearray, parse_burst, self.event_queue.put`
`CaptureMixin.stop_capture` stop_capture(self)	577-580	Stop capture.	`self.stop_event.set, self.status_var.set, self.tr`
`CaptureMixin.run_simulation` run_simulation(self)	582-683	Runs the internal diagnostic scenario without opening serial hardware.	`self.reset_session_metrics, self.close_log, self.status_var.set, self.start_btn.configure, self.restart_btn.configure`
`CaptureMixin.process_queue` process_queue(self)	685-833	Main-thread bridge between workers and Tkinter.	`range, flush_frames, self.metrics.expire_pending, time.time, self.metrics.pop_expired_requests`
`CaptureMixin.on_close` on_close(self)	835-901	Performs defensive shutdown: saves settings, signals all workers, cancels serial I/O, joins threads with timeouts, closes serial handles/logs and destroys the Tk root.	`self.stop_event.set, self.discovery_stop_event.set, cancel_serial_io, list, self.close_log`

Detailed behavior of the key paths

`selected_port_plan` lines 188-206

Translates physical mode into one or two receive channels. RS485 and single-RX RS232 use COM A; dual-RX RS232 requires distinct COM A and COM B and labels their directions A→B/B→A.

`calculated_gap_ms` lines 366-378

Returns the manual gap after validation, or computes the automatic RTU t3.5 value from baud/data/parity/stop settings using the shared protocol timing helper.

`start_capture` lines 438-496

Validates state and serial settings, opens the selected port(s), resets the current session, prepares lazy logging, starts reader workers and updates the GUI to capturing state.

`reader_loop` lines 498-575

Runs in a daemon thread per capture channel. It accumulates serial bytes and their read times into a burst, considers the burst complete after the configured inter-frame silence (or after the split-frame hold when the burst ends mid-frame), then calls parse_burst() and posts frame events to the GUI queue.

`run_simulation` lines 582-683

Runs the internal diagnostic scenario without opening serial hardware. It feeds the same parser/event path used by real capture, including forced timeout handling.

`process_queue` lines 685-833

Main-thread bridge between workers and Tkinter. It drains events in bounded batches, batches frame insertions for performance, handles capture/discovery status events, expires requests and reschedules itself with a faster cadence when backlog exists.

`on_close` lines 835-901

Performs defensive shutdown: saves settings, signals all workers, cancels serial I/O, joins threads with timeouts, closes serial handles/logs and destroys the Tk root.

> **Design note.** Tkinter widgets are never updated directly by reader threads. Workers communicate through event_queue; process_queue() runs on the Tk main thread.

mb_diagnostics.py

Presentation-oriented diagnostics derived from captured records. It maps Function Codes to names, recalculates/display CRC fields, extracts payloads, classifies anomalies/results and builds the structured Frame Inspector text.

Source lines	SHA-256
286	`c67ed1df18f1d79b0feb4011a036f92cbeef338929a529a4f218acd1f78e1552`

Imports: `mb_config.MODBUS_EXCEPTION_CODES, SLOW_RESPONSE_THRESHOLD_MS, mb_protocol.crc16_modbus, read_bits_from_response, mb_i18n.LANGUAGE_PORTUGUESE, translate_details, translate_text`

Module constants / configuration data

Name	Definition
`FUNCTION_NAMES`	`{1: 'Read Coils', 2: 'Read Discrete Inputs', 3: 'Read Holding Registers', 4: 'Read Input Registers', 5: 'Write Single Coil', 6: 'Write Single Regis…`
`RESULT_LABELS`	`{'success': ('✓ Sucesso', 'Verde'), 'slow': ('⚠ Resposta lenta', 'Amarelo'), 'exception': ('Exception', 'Vermelho'), 'crc': ('× CRC Error', 'Vermel…`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`raw_hex_to_bytes` raw_hex_to_bytes(raw)	26-30	Raw hex to bytes.	`bytes.fromhex, str`
`fc_number` fc_number(record)	33-41	Fc number.	`str(record.get('fc', '') or '').strip, int, record.get, str`
`function_label` function_label(record, language=LANGUAGE_PORTUGUESE)	44-49	Function label.	`fc_number, FUNCTION_NAMES.get, translate_text`
`crc_fields` crc_fields(record)	52-62	Crc fields.	`str(record.get('type', '')).startswith, raw_hex_to_bytes, crc16_modbus, record.get, len`

Callable	Lines	Responsibility	Representative direct calls
`response_payload` response_payload(record, language=LANGUAGE_PORTUGUESE)	65-127	Return (byte_count, data_hex, decoded_data) where available.	`raw_hex_to_bytes, str, record.get, len, MODBUS_EXCEPTION_CODES.get`
`anomaly_kind` anomaly_kind(record, threshold_ms=SLOW_RESPONSE_THRESHOLD_MS)	130-153	Classifies a record with explicit priority: bad CRC, timeout, Exception, RAW, then slow response.	`bool, str(record.get('type', '')).startswith, str, record.get, float`
`result_info` result_info(record, successful_request=False, language=LANGUAGE_PORTUGUESE)	168-207	Turns the anomaly/pairing state into a semantic result key, user-facing label and color name.	`anomaly_kind, str, record.get, translate_text`
`inspector_values` inspector_values(record, successful_request=False, language=LANGUAGE_PORTUGUESE)	210-264	Builds every Frame Inspector field, including recalculated CRC received/calculated values, decoded payload, timeout suffix and translated result text.	`crc_fields, response_payload, str, result_info, record.get`
`decoded_record_text` decoded_record_text(record, successful_request=False, language=LANGUAGE_PORTUGUESE)	267-286	Serialises the same Frame Inspector information into newline-separated text for the copy-decoded-frame context command.	`inspector_values, '\n'.join`

Detailed behavior of the key paths

`anomaly_kind` lines 130-153

Classifies a record with explicit priority: bad CRC, timeout, Exception, RAW, then slow response. The priority prevents conflicting colors/results.

`result_info` lines 168-207

Turns the anomaly/pairing state into a semantic result key, user-facing label and color name. Successful completed pairs are distinguished from pending requests.

`inspector_values` lines 210-264

Builds every Frame Inspector field, including recalculated CRC received/calculated values, decoded payload, timeout suffix and translated result text.

`decoded_record_text` lines 267-286

Serialises the same Frame Inspector information into newline-separated text for the copy-decoded-frame context command.

mb_view.py

Traffic view, filtering, sorting, Frame Inspector, Bus Health updates, CSV export and TXT logging. It owns the bounded in-memory frame history, converts parser rows into richer view records, pairs display state with SessionMetrics, maintains table/raw views and keeps the real capture log independent from GUI filtering.

Source lines	SHA-256
1871	`deb7447ebdb2fb80ad32b0d4d534b9a763ca72ba062c966dddbb2c871d63c64c`

Imports: `csv, math, os, subprocess, sys, time, datetime.datetime, tkinter, tkinter.filedialog, messagebox, mb_config.MAX_UI_FRAMES, MAX_RAW_TEXT_LINES, UI_PRUNE_CHUNK_FRAMES, RAW_TEXT_PRUNE_CHUNK_LINES, SLOW_RESPONSE_THRESHOLD_MS, get_application_directory, get_logs_folder, mb_diagnostics.anomaly_kind, decoded_record_text, inspector_values, mb_i18n.canonical_ui_text, translate_details, mb_protocol.read_bits_details`

Module constants / configuration data

Name	Definition
`TRAFFIC_MENU_OPEN_IN_MASTER`	`8`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class ViewMixin`	**36-1870**	**Methods for rendering, filtering, statistics and logging.**	**Class container**
`ViewMixin.reset_session_metrics` reset_session_metrics(self)	39-41	Reset session metrics.	`self.metrics.reset, self.update_stats_labels`
`ViewMixin.update_stats_labels` update_stats_labels(self)	43-108	Converts SessionMetrics into both compact session counters and Bus Health values.	`self.stat_requests_var.set, self.stat_responses_var.set, self.stat_pending_var.set, self.stat_timeouts_var.set, self.stat_crc_var.set`
`ViewMixin.pulse_bus_activity` pulse_bus_activity(self)	113-132	Pulse bus activity.	`getattr, var.set, self.after, self.after_cancel`
`ViewMixin._clear_bus_activity_pulse` _clear_bus_activity_pulse(self)	134-138	Clear bus activity pulse.	`getattr, var.set`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.schedule_advanced_filter_refresh` schedule_advanced_filter_refresh(self, *_args)	140-153	Schedules advanced filter refresh.	`getattr, self.after, self.after_cancel`
`ViewMixin.clear_advanced_filters` clear_advanced_filters(self)	155-175	Clear advanced filters.	`getattr, self.advanced_filter_type_var.set, self.advanced_filter_fc_var.set, self.advanced_filter_response_ms_var.set, self.advanced_filter…`
`ViewMixin.normalized_fc_filter` normalized_fc_filter(value)	178-190	Normalized fc filter.	`str(value or '').strip().upper, text.startswith, int, str(value or '').strip, str`
`ViewMixin.record_passes_advanced_filters` record_passes_advanced_filters(self, record)	192-253	Implements Type, FC, response-time and free-text filtering.	`getattr, canonical_ui_text, str, self.normalized_fc_filter, self.advanced_filter_response_ms_var.get().strip`
`ViewMixin.configure_anomaly_tags` configure_anomaly_tags(self, dark=None)	255-286	Configure anomaly tags.	`palette.items, hasattr, bool, self.tree.tag_configure, getattr(self, 'dark_mode_var', tk.BooleanVar(value=False)).get`
`ViewMixin.anomaly_tag_for_record` anomaly_tag_for_record(self, record)	288-325	Maps the semantic result of a record to the Treeview tag used for colour highlighting, including completed normal REQUEST/RESPONSE pairs.	`anomaly_kind, str, getattr, self.highlight_anomalies_var.get, record.get`
`ViewMixin.on_highlight_anomalies_changed` on_highlight_anomalies_changed(self)	327-334	Event handler for highlight anomalies changed.	`self.configure_anomaly_tags, self.configure_inspector_result_styles, self.render_filtered_view, self.update_frame_inspector, getattr`
`ViewMixin.mark_timed_out_records` mark_timed_out_records(self, expired, rerender=True)	336-362	Finds original REQUEST records by their sequence IDs and marks them timed_out rather than inventing artificial timeout frames.	`len, item.get, changed_records.append, canonical_ui_text, self.render_filtered_view`
`ViewMixin.refresh_record_tags` refresh_record_tags(self, records)	364-376	Re-applies the result tag to the Treeview rows of the given records and refreshes the Frame Inspector when one of them is selected.	`self.selected_record, self.anomaly_tag_for_record, self._tree_iids_by_seq.get, any, self.update_frame_inspector`
`ViewMixin.selected_record` selected_record(self)	378-382	Selected record.	`self.tree.selection, self._record_by_iid.get`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.update_frame_inspector` update_frame_inspector(self, record=None)	384-413	Builds the structured inspector values for the selected traffic record and applies result styling while preserving the raw frame.	`bool, inspector_values, self.inspector_vars.items, self.selected_raw_var.set, getattr`
`ViewMixin.toggle_frame_inspector` toggle_frame_inspector(self)	415-427	Toggle frame inspector.	`getattr, callable, self.inspector_body.grid, self.inspector_toggle_btn.configure, self.inspector_body.grid_remove`
`ViewMixin.copy_to_clipboard` copy_to_clipboard(self, text)	429-435	Copy to clipboard.	`self.clipboard_clear, self.clipboard_append, self.update_idletasks, str`
`ViewMixin.copy_selected_raw` copy_selected_raw(self)	437-440	Copy selected raw.	`self.selected_record, self.copy_to_clipboard, record.get`
`ViewMixin.copy_selected_decoded` copy_selected_decoded(self)	442-455	Copy selected decoded.	`self.selected_record, bool, self.copy_to_clipboard, decoded_record_text, record.get`
`ViewMixin.context_filter_selected_slave` context_filter_selected_slave(self)	457-470	Context filter selected slave.	`self.selected_record, str, self.slave_filter_vars.items, self.update_slave_heading, self.render_filtered_view`
`ViewMixin.context_filter_selected_fc` context_filter_selected_fc(self)	472-486	Context filter selected fc.	`self.selected_record, str, self.advanced_filter_fc_var.set, self.render_filtered_view, record.get`
`ViewMixin.context_show_selected_transaction` context_show_selected_transaction(self)	488-517	Context show selected transaction.	`self.selected_record, record.get, self.slave_filter_vars.values, self.update_slave_heading, self.advanced_filter_type_var.set`
`ViewMixin.clear_transaction_filter` clear_transaction_filter(self)	519-521	Clear transaction filter.	`self.render_filtered_view`
`ViewMixin.context_open_selected_in_master` context_open_selected_in_master(self)	523-526	Context open selected in master.	`self.selected_record, self.master_open_from_record`
`ViewMixin.show_traffic_context_menu` show_traffic_context_menu(self, event)	528-552	Show traffic context menu.	`self.tree.identify_row, self.tree.selection_set, self.tree.focus, self.on_row_select, getattr`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.export_session_csv` export_session_csv(self, path=None)	554-656	Exports the frames retained by the GUI history, including timing, decoded addressing, raw data and anomaly/timeout fields.	`list, datetime.now().strftime, filedialog.asksaveasfilename, messagebox.showinfo, open`
`ViewMixin.tree_record_column_values` tree_record_column_values(self, record)	658-669	Tree record column values.	`record.get, translate_details`
`ViewMixin.measure_tree_column_width` measure_tree_column_width(self, column, records, heading_text=None)	671-689	Measure the pixel width required by a non-wrapped Treeview column.	`self.tree_heading_font.measure, self.tree_headings.get, str, self.tree_record_column_values(record).get, max`
`ViewMixin.available_details_width` available_details_width(self)	691-709	Return the width that can be assigned to Details without intentionally forcing horizontal scrolling in the current Treeview viewport.	`int, max, self.tree.winfo_width, self.tree.column`
`ViewMixin.wrap_text_to_pixels` wrap_text_to_pixels(self, value, max_px)	711-761	Word-wrap text using the actual Tk font width.	`str, max, '\n'.join, int, text.splitlines`
`ViewMixin.details_horizontal_rule` details_horizontal_rule(self)	763-775	Return a rule that fills the Details column as far as the Treeview cell permits.	`max, int, self.tree_cell_font.measure, self.tree.column, math.ceil`
`ViewMixin.completed_pairs_for_records` completed_pairs_for_records(self, records)	777-817	Return complete visible REQUEST/RESPONSE pairs in current display order.	`pairs.sort, record.get, response.get, seq_to_index.get, min`
`ViewMixin.separator_after_sequences` separator_after_sequences(self, records)	819-836	Return logical frame seq ids that need a separator AFTER them.	`self.completed_pairs_for_records, self.separate_transactions_var.get, set, len`
`ViewMixin.tree_separator_values` tree_separator_values(self)	838-843	Tree separator values.	`self.details_horizontal_rule`
`ViewMixin.insert_tree_separator_after_seq` insert_tree_separator_after_seq(self, seq)	845-863	Insert one Details-width separator after the visual rows of seq.	`self._tree_iids_by_seq.get, self.tree.insert, self.tree.index, self.tree_separator_values`
`ViewMixin.sync_tree_transaction_separators` sync_tree_transaction_separators(self, records=None)	865-885	Synchronize standalone transaction separator rows without rebuilding normal traffic rows.	`self.separator_after_sequences, set, self.records_for_display, self._tree_separator_iid_by_seq.pop, self._raw_by_iid.pop`
`ViewMixin.detail_lines_for_record` detail_lines_for_record(self, record)	887-909	Return wrapped Details as separate visual Treeview rows.	`max, self.wrap_text_to_pixels, wrapped.splitlines, translate_details, int`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.set_tree_rowheight_for_records` set_tree_rowheight_for_records(self, records)	911-916	Treeview row height stays fixed; wrapping uses continuation rows.	`self.tree_style.configure`
`ViewMixin.update_tree_rowheight_for_record` update_tree_rowheight_for_record(self, record)	918-920	No-op by design: one logical frame may use several compact rows.	—
`ViewMixin.autofit_tree_columns` autofit_tree_columns(self, records=None)	922-965	Non-Details columns fit the longest visible value/title.	`self.measure_tree_column_width, max, self.tree.column, self.set_tree_rowheight_for_records, self.records_for_display`
`ViewMixin.expand_tree_columns_for_record` expand_tree_columns_for_record(self, record)	967-1017	Incrementally enlarge only the compact non-Details columns.	`self.tree_record_column_values, self.tree_heading_font.measure, self.tree_cell_font.measure, int, self.measure_tree_column_width`
`ViewMixin.refresh_tree_heading_width` refresh_tree_heading_width(self, column)	1019-1035	Dynamic Hora/Slave headings remain compact.	`self.records_for_display, self.measure_tree_column_width, self.tree.column`
`ViewMixin.schedule_tree_reflow` schedule_tree_reflow(self, _event=None)	1037-1051	Debounce window/Treeview resizing and then re-wrap Details to the new available width.	`self.after, self.after_cancel`
`ViewMixin.reflow_tree_after_resize` reflow_tree_after_resize(self)	1053-1057	Reflow tree after resize.	`self.render_filtered_view, hasattr`
`ViewMixin._slave_sort_key` _slave_sort_key(self, slave)	1059-1063	Slave sort key.	`int, str`
`ViewMixin.rebuild_slave_filter_menu` rebuild_slave_filter_menu(self)	1065-1092	Rebuild slave filter menu.	`self.slave_filter_menu.delete, self.slave_filter_menu.add_command, self.update_slave_heading, self.slave_filter_menu.add_separator, sorted`
`ViewMixin.update_slave_heading` update_slave_heading(self)	1094-1111	Update slave heading.	`self.tree.heading, self.refresh_tree_heading_width, self.tr, self.slave_filter_vars.items, var.get`
`ViewMixin.show_slave_filter_menu` show_slave_filter_menu(self)	1113-1126	Open the slave filter menu at the position where the Slave heading was clicked.	`self.winfo_pointerx, self.slave_filter_menu.tk_popup, self.winfo_pointery, self.slave_filter_menu.grab_release`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.register_slave_for_filter` register_slave_for_filter(self, slave)	1128-1139	Register slave for filter.	`str(slave or '').strip, tk.BooleanVar, self.rebuild_slave_filter_menu, str`
`ViewMixin.select_all_slaves` select_all_slaves(self)	1141-1146	Select all slaves.	`self.slave_filter_vars.values, self.update_slave_heading, self.render_filtered_view, var.set`
`ViewMixin.select_no_slaves` select_no_slaves(self)	1148-1153	Select no slaves.	`self.slave_filter_vars.values, self.update_slave_heading, self.render_filtered_view, var.set`
`ViewMixin.on_slave_filter_changed` on_slave_filter_changed(self)	1155-1165	Event handler for slave filter changed.	`self.update_slave_heading, self.render_filtered_view, var.get, bool, all`
`ViewMixin.selected_slave_ids` selected_slave_ids(self)	1167-1171	Selected slave ids.	`self.slave_filter_vars.items, var.get`
`ViewMixin.record_passes_slave_filter` record_passes_slave_filter(self, record)	1173-1183	Record passes slave filter.	`self.selected_slave_ids, str, record.get`
`ViewMixin.time_order_descending` time_order_descending(self)	1185-1186	Time order descending.	`canonical_ui_text(self.time_order_var.get()).startswith, canonical_ui_text, self.time_order_var.get`
`ViewMixin.on_time_order_changed` on_time_order_changed(self)	1188-1197	Event handler for time order changed.	`self.tree.heading, self.render_filtered_view, self.time_order_descending, self.tr`
`ViewMixin.toggle_time_order` toggle_time_order(self)	1199-1204	Toggle time order.	`self.time_order_descending, self.on_time_order_changed, self.time_order_var.set, self.tr`
`ViewMixin.records_for_display` records_for_display(self)	1206-1216	Applies current filtering and time-order state to the bounded source-of-truth frame history.	`self.time_order_descending, records.reverse, self.record_passes_slave_filter, self.record_passes_advanced_filters`
`ViewMixin.raw_line_for_record` raw_line_for_record(self, record)	1218-1241	Raw line for record.	`record.get, translate_details`
`ViewMixin.raw_separator_width_chars` raw_separator_width_chars(self, raw_lines=None)	1243-1263	Width of one Raw Hex separator in visible character cells.	`max, self.raw_text_font.measure, int, self.raw_text.winfo_width`
`ViewMixin.raw_transaction_separator` raw_transaction_separator(self, width_chars=None)	1265-1268	Raw transaction separator.	`max, int`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.schedule_raw_separator_reflow` schedule_raw_separator_reflow(self, _event=None)	1270-1280	Schedules raw separator reflow.	`self.after, self.after_cancel`
`ViewMixin.reflow_raw_separator_after_resize` reflow_raw_separator_after_resize(self)	1282-1285	Reflow raw separator after resize.	`self.separate_transactions_var.get, self.render_raw_text_only`
`ViewMixin.insert_tree_record` insert_tree_record(self, record, index='end', allow_separator=True)	1287-1355	Insert tree record.	`self.detail_lines_for_record, visual_rows.append, self.anomaly_tag_for_record, list, self.tree.insert`
`ViewMixin.render_raw_text_only` render_raw_text_only(self)	1357-1410	Render raw text only.	`self.records_for_display, self.separator_after_sequences, list, self.raw_separator_width_chars, self.raw_transaction_separator`
`ViewMixin.render_filtered_view` render_filtered_view(self)	1412-1472	Render filtered view.	`self.selected_record, self.tree.get_children, self._tree_history.clear, self._tree_iids_by_seq.clear, self._tree_separator_iid_by_seq.clear`
`ViewMixin.add_frame_row` add_frame_row(self, row)	1474-1476	Add frame row.	`self.add_frame_rows_batch`
`ViewMixin.add_frame_rows_batch` add_frame_rows_batch(self, rows)	1478-1676	Central view ingestion path.	`self.pulse_bus_activity, ''.join, self.update_stats_labels, self.prune_ui_history, self.metrics.process`
`ViewMixin.prune_ui_history` prune_ui_history(self)	1678-1701	Keeps memory/UI work bounded by removing old records in chunks after a hysteresis threshold.	`self._successful_request_seqs.intersection_update, self.render_filtered_view, len, self._frame_history.popleft, record.get`
`ViewMixin.on_row_select` on_row_select(self, _event=None)	1703-1705	Event handler for row select.	`self.selected_record, self.update_frame_inspector`
`ViewMixin.finish_capture_ui` finish_capture_ui(self)	1707-1720	Finish capture ui.	`self.status_var.set, self.start_btn.configure, self.stop_btn.configure, self.restart_btn.configure, self.set_simulation_button_state`
`ViewMixin.app_directory` app_directory(self)	1722-1723	App directory.	`get_application_directory`
`ViewMixin.prepare_log_session` prepare_log_session(self, prefix, started_at, header)	1725-1738	Prepare a real-capture log session without creating a file.	`self.close_log, started_at.strftime`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.ensure_log_started` ensure_log_started(self)	1740-1779	Creates the TXT capture log lazily only when real traffic actually arrives, preventing empty logs from start/stop operations with no data.	`target.open, get_logs_folder, self.log_file.write, messagebox.showerror, ''.join`
`ViewMixin.queue_log_note` queue_log_note(self, text)	1781-1789	Write a capture note to an already-open log, or keep it in memory until the first traffic frame creates the log.	`self.log_line, self.pending_log_notes.append`
`ViewMixin.log_line` log_line(self, text)	1791-1796	Log line.	`self.log_file.write`
`ViewMixin.close_log` close_log(self)	1798-1813	Close log.	`self.log_file.write, self.log_file.close, datetime.now().isoformat, datetime.now`
`ViewMixin.open_log_folder` open_log_folder(self)	1815-1825	Open log folder.	`get_logs_folder, os.startfile, messagebox.showerror, str, subprocess.Popen`
`ViewMixin.clear_view` clear_view(self)	1827-1870	Clear view.	`self.tree.get_children, self.raw_text.delete, self.selected_raw_var.set, self._tree_history.clear, self._tree_iids_by_seq.clear`

Detailed behavior of the key paths

`update_stats_labels` lines 43-108

Converts SessionMetrics into both compact session counters and Bus Health values. It recalculates health using the currently configured serial parameters so approximate bus utilisation reflects the selected format.

`record_passes_advanced_filters` lines 192-253

Implements Type, FC, response-time and free-text filtering. Filtering is display-only; it does not alter SessionMetrics or the on-disk capture log.

`anomaly_tag_for_record` lines 288-325

Maps the semantic result of a record to the Treeview tag used for colour highlighting, including completed normal REQUEST/RESPONSE pairs.

`mark_timed_out_records` **lines 336-362**

Finds original REQUEST records by their sequence IDs and marks them timed_out rather than inventing artificial timeout frames. Visible rows are retagged in place; the table is rebuilt only while the Timeouts filter is active.

`refresh_record_tags` **lines 364-376**

Re-applies the result tag to the Treeview rows of the given records and refreshes the Frame Inspector when one of them is selected.

`update_frame_inspector` **lines 384-413**

Builds the structured inspector values for the selected traffic record and applies result styling while preserving the raw frame.

`export_session_csv` **lines 554-656**

Exports the frames retained by the GUI history, including timing, decoded addressing, raw data and anomaly/timeout fields. This differs intentionally from the complete TXT capture log.

`records_for_display` **lines 1206-1216**

Applies current filtering and time-order state to the bounded source-of-truth frame history.

`add_frame_rows_batch` **lines 1478-1676**

Central view ingestion path. It assigns stable sequence IDs, runs SessionMetrics, enriches parser rows into view records (including the matched request address/quantity for read responses), tracks successful pairs, writes complete real traffic to disk, updates visible table/raw views, prunes history and auto-scrolls.

`prune_ui_history` **lines 1678-1701**

Keeps memory/UI work bounded by removing old records in chunks after a hysteresis threshold. The disk log is independent and remains complete.

`ensure_log_started` **lines 1740-1779**

Creates the TXT capture log lazily only when real traffic actually arrives, preventing empty logs from start/stop operations with no data.

Design note. There are deliberately two histories: the bounded GUI history and the complete TXT log. Sorting/filtering only changes presentation; statistics are derived before filtering.

mb_slave_finder.py

Active Modbus RTU Slave discovery and Device Identification. It checks that the bus is idle, builds safe read-only FC03/FC04 probes with configurable attempts, calculates adaptive timeouts, performs robust serial writes/reads, accepts normal or CRC-valid Exception responses as proof of a Slave, optionally reads FC43/14 Basic Device Identification and drives the Finder UI through queued events.

Source lines	SHA-256
1625	`f192a6305486ea73411d920d9b6b9de19091bc612afef60ea3f5871fcd66938b`

Imports: `threading, time, tkinter, tkinter.ttk, messagebox, mb_config.DISCOVERY_BAUD_RATES, DISCOVERY_PARITIES, DISCOVERY_STOP_BITS, DISCOVERY_DEFAULT_MIN_TIMEOUT_MS, DISCOVERY_MIN_TIMEOUT_MS, DISCOVERY_MAX_TIMEOUT_MS, DISCOVERY_PROBE_ADDRESS, DISCOVERY_PROBE_QTY, DISCOVERY_WRITE_TIMEOUT_SECONDS, DISCOVERY_WRITE_RETRIES, DISCOVERY_RX_BUFFER_LIMIT, DISCOVERY_MIN_ATTEMPTS, DISCOVERY_MAX_ATTEMPTS, DISCOVERY_DEFAULT_ATTEMPTS, DISCOVERY_IDLE_CHECK_SECONDS, MODBUS_EXCEPTION_CODES, mb_protocol.append_crc, crc_ok, hex_bytes, modbus_rtu_interframe_gap_seconds, mb_widgets.RefreshButton`

Module constants / configuration data

Name	Definition
`DEVICE_ID_FC`	`43`
`DEVICE_ID_MEI_TYPE`	`14`
`DEVICE_ID_READ_BASIC`	`1`
`DEVICE_ID_MORE_FOLLOWS`	`255`
`DEVICE_ID_BASIC_OBJECTS`	`(0, 1, 2)`
`DEVICE_ID_MAX_PAGES`	`8`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`build_discovery_probe` build_discovery_probe(slave, fc)	35-48	Constructs a read-only FC03/FC04 request for Address 0, Quantity 1 and appends the RTU CRC.	`bytes, append_crc, ValueError, int`

Callable	Lines	Responsibility	Representative direct calls
`build_device_identification_probe` build_device_identification_probe(slave, object_id=0)	57-71	Builds FC43 / MEI type 14 / Basic Device Identification requests, beginning at a selectable Object ID.	`int, append_crc, ValueError, bytes`
`_decode_device_id_value` _decode_device_id_value(value)	73-75	Decode a Modbus Device Identification ASCII object for display.	`bytes(value).decode('ascii', errors='replace').strip, bytes(value).decode, bytes`
`find_device_identification_response` find_device_identification_response(buffer, slave)	77-158	Parses FC43/14 responses and Exceptions from a receive buffer, validates CRC, decodes variable-length ASCII objects and exposes More Follows / Next Object ID for paged identification.	`bytes, int, range, max, len`
`read_device_identification_response` read_device_identification_response(ser, slave, timeout_s, stop_event)	160-182	Read until a valid FC43/14 response/exception is received or timed out.	`bytearray, find_device_identification_response, time.perf_counter, max, stop_event.is_set`
`device_identification_timeout_seconds` device_identification_timeout_seconds(baud, parity, stopbits, minimum_timeout_ms=DISCOVERY_DEFAULT_MIN_TIMEOUT_MS)	184-201	Conservative timeout for one FC43/14 response, up to one RTU frame.	`max, discovery_bits_per_char, int, float, modbus_rtu_interframe_gap_seconds`
`device_identification_request_transmit_seconds` device_identification_request_transmit_seconds(baud, parity, stopbits)	203-207	Estimated on-wire duration of the 7-byte FC43/14 request.	`float, discovery_bits_per_char, max, int`
`format_device_identification` format_device_identification(result)	209-224	Return compact Device Identification text for the Finder result table.	`any, result.get, int, objects.get, ' | '.join`
`read_basic_device_identification` read_basic_device_identification(ser, slave, baud, parity, stopbits, minimum_timeout_ms, stop_event, serial_module)	226-301	Repeats FC43/14 requests while More Follows is set, merging objects until VendorName, ProductCode and MajorMinorRevision are available or the paging limit/response terminates.	`range, stop_event.is_set, build_device_identification_probe, write_discovery_request, read_device_identification_response`
`discovery_bits_per_char` discovery_bits_per_char(parity, stopbits)	303-307	Return serial bits/character: start + 8 data + parity + stop.	`float`
`discovery_response_timeout_seconds` discovery_response_timeout_seconds(baud, parity, stopbits, minimum_timeout_ms=DISCOVERY_DEFAULT_MIN_TIMEOUT_MS)	309-330	Calculates a response deadline from serial character time, the expected 7-byte FC03/FC04 response and an inter-frame/processing margin, never below the user-configured minimum.	`max, discovery_bits_per_char, int, float, modbus_rtu_interframe_gap_seconds`

Callable	Lines	Responsibility	Representative direct calls
`discovery_request_transmit_seconds` discovery_request_transmit_seconds(baud, parity, stopbits)	332-334	Estimated on-wire duration of the fixed 8-byte discovery request.	`float, discovery_bits_per_char, max, int`
`find_discovery_response` find_discovery_response(buffer, slave, fc)	336-379	Scans received bytes for the requested Slave and Function Code.	`bytes, int, range, max, len`
`read_discovery_response` read_discovery_response(ser, slave, fc, timeout_s, stop_event)	381-407	Read non-blocking serial bytes until a valid probe response or timeout.	`bytearray, find_discovery_response, time.perf_counter, max, stop_event.is_set`
`bus_activity_detected` bus_activity_detected(ser, listen_s, stop_event)	409-427	Listens passively for the idle-check period and returns any bytes received; a non-empty result means another device is transmitting and the scan must not start.	`bytearray, bytes, time.perf_counter, max, stop_event.is_set`
`cancel_serial_io` cancel_serial_io(ser)	430-440	Best-effort cancellation of pending serial I/O.	`getattr, callable, method`
`recover_discovery_write` recover_discovery_write(ser)	442-457	Best-effort cleanup after a discovery write timeout.	`getattr, callable, ser.reset_output_buffer, ser.reset_input_buffer, cancel`
`write_discovery_request` write_discovery_request(ser, request, baud, parity, stopbits, stop_event, serial_module)	459-523	Transmits one probe without using ser.flush(), retries one write timeout after driver/buffer recovery, waits for estimated on-wire transmission time and raises actual driver errors instead of disguising them as no-response Slaves.	`bytes, getattr, discovery_bits_per_char, max, range`
`format_duration` format_duration(seconds)	525-531	Format duration.	`max, divmod, int, round, float`
`class SlaveFinderMixin`	**534-1624**	**Methods that implement the active Bus Slave Finder tab.**	**Class container**
`SlaveFinderMixin.build_discovery_tab` build_discovery_tab(self, page)	537-921	Build the full-window active Modbus RTU slave-discovery workspace.	`getattr(self, 'ui_settings', {}).get, set, page.columnconfigure, page.rowconfigure, ttk.LabelFrame`
`SlaveFinderMixin._show_discovery_context_menu` _show_discovery_context_menu(self, event)	923-932	Show discovery context menu.	`self.discovery_tree.identify_row, self.discovery_tree.selection_set, self.discovery_context_menu.entryconfigure, self.discovery_context_men…`

Callable	Lines	Responsibility	Representative direct calls
`SlaveFinderMixin.discovery_open_selected_in_master` discovery_open_selected_in_master(self)	934-950	Open the selected found Slave in a new Master window.	`self.discovery_tree.selection, self.discovery_tree.item, {'N': 'None', 'E': 'Even', 'O': 'Odd'}.get, self.master_open_definition, int`
`SlaveFinderMixin.fit_discovery_columns_to_viewport` fit_discovery_columns_to_viewport(self, _event=None)	952-981	Fit Slave Finder columns to the visible Treeview width.	`int, max, sum, len, enumerate`
`SlaveFinderMixin.on_discovery_options_changed` on_discovery_options_changed(self)	983-986	Event handler for discovery options changed.	`hasattr, self.update_discovery_estimate, self.update_discovery_start_state`
`SlaveFinderMixin.selected_discovery_configs` selected_discovery_configs(self)	988-997	Selected discovery configs.	`self.discovery_baud_vars[baud].get, self.discovery_parity_vars[p].get, self.discovery_stop_vars[s].get`
`SlaveFinderMixin.selected_discovery_slaves` selected_discovery_slaves(self)	999-1006	Selected discovery slaves.	`int, list, self.discovery_slave_start_var.get, self.discovery_slave_end_var.get, ValueError`
`SlaveFinderMixin.selected_discovery_min_timeout_ms` selected_discovery_min_timeout_ms(self)	1008-1018	Selected discovery min timeout ms.	`float, self.discovery_min_timeout_var.get, ValueError, self.trf`
`SlaveFinderMixin.selected_discovery_attempts` selected_discovery_attempts(self)	1020-1030	Selected discovery attempts.	`int, self.discovery_attempts_var.get, ValueError, self.trf`
`SlaveFinderMixin.discovery_config_label` discovery_config_label(self, parity, stopbits)	1032-1034	Discovery config label.	—
`SlaveFinderMixin.update_discovery_estimate` update_discovery_estimate(self)	1036-1093	Update discovery estimate.	`bool, self.discovery_progress_text_var.set, self.discovery_estimate_var.set, self.selected_discovery_configs, self.selected_discovery_slaves`
`SlaveFinderMixin.update_discovery_start_state` update_discovery_start_state(self)	1095-1111	Update discovery start state.	`self.discovery_start_btn.configure, hasattr, bool, self.selected_discovery_slaves, self.selected_discovery_min_timeout_ms`

Callable	Lines	Responsibility	Representative direct calls
`SlaveFinderMixin.clear_discovery_results` clear_discovery_results(self)	1113-1122	Clear discovery results.	`self.discovery_tree.get_children, self.discovery_found_var.set, self.discovery_progress_var.set, self.discovery_status_var.set, self.update…`
`SlaveFinderMixin.open_discovery_serial_port` open_discovery_serial_port(self, serial, port, baud, parity, stopbits)	1124-1162	Open discovery serial port.	`serial.Serial, int, ser.open, ser.reset_input_buffer, ser.reset_output_buffer`
`SlaveFinderMixin.start_discovery_scan` start_discovery_scan(self)	1164-1255	Validates the safety confirmation, COM selection, Slave range, attempts, selected serial configurations and global busy state.	`self.get_serial_modules, bool, self.clear_discovery_results, self.discovery_stop_event.clear, self.discovery_progress_var.set`
`SlaveFinderMixin.stop_discovery_scan` stop_discovery_scan(self)	1257-1264	Stop discovery scan.	`self.discovery_status_var.set, self.discovery_stop_event.set, self.discovery_stop_btn.configure, cancel_serial_io, self.tr`
`SlaveFinderMixin.discovery_worker` discovery_worker(self, serial, port, configs, slaves, minimum_timeout_ms, fallback_fc04, identify_devices, attempts=DISCOVERY_DEFAULT_ATTEMPTS)	1266-1545	Iterates selected serial configurations and Slave IDs.	`time.perf_counter, str, self.event_queue.put, len, self.discovery_stop_event.is_set`
`SlaveFinderMixin.apply_discovery_progress` apply_discovery_progress(self, payload)	1547-1562	Apply discovery progress.	`int, max, self.discovery_progress_var.set, self.discovery_progress_text_var.set, self.discovery_found_var.set`
`SlaveFinderMixin.add_discovery_result` add_discovery_result(self, payload)	1564-1581	Add discovery result.	`self.discovery_found_var.set, self.discovery_tree.insert, self.trf, payload.get`
`SlaveFinderMixin.finish_discovery_ui` finish_discovery_ui(self, payload)	1583-1624	Returns both Finder and Sniffer controls to idle state and presents completed/stopped/error status without leaving the application locked in a busy state.	`self.discovery_stop_event.clear, self.discovery_stop_btn.configure, self.discovery_clear_btn.configure, self.discovery_refresh_btn.configur…`

Detailed behavior of the key paths

`build_discovery_probe` lines 35-48

Constructs a read-only FC03/FC04 request for Address 0, Quantity 1 and appends the RTU CRC. Input validation restricts Slave IDs to 1..247 and FC to 03/04.

`find_discovery_response` lines 336-379

Scans received bytes for the requested Slave and Function Code. A CRC-valid normal register response or CRC-valid Modbus Exception is accepted; echoes and unrelated traffic are ignored.

`discovery_response_timeout_seconds` lines 309-330

Calculates a response deadline from serial character time, the expected 7-byte FC03/FC04 response and an inter-frame/processing margin, never below the user-configured minimum.

`bus_activity_detected` lines 409-427

Listens passively for the idle-check period and returns any bytes received; a non-empty result means another device is transmitting and the scan must not start.

`write_discovery_request` lines 459-523

Transmits one probe without using ser.flush(), retries one write timeout after driver/buffer recovery, waits for estimated on-wire transmission time and raises actual driver errors instead of disguising them as no-response Slaves.

`build_device_identification_probe` lines 57-71

Builds FC43 / MEI type 14 / Basic Device Identification requests, beginning at a selectable Object ID.

`find_device_identification_response` lines 77-158

Parses FC43/14 responses and Exceptions from a receive buffer, validates CRC, decodes variable-length ASCII objects and exposes More Follows / Next Object ID for paged identification.

`read_basic_device_identification` lines 226-301

Repeats FC43/14 requests while More Follows is set, merging objects until VendorName, ProductCode and MajorMinorRevision are available or the paging limit/response terminates.

`start_discovery_scan` lines 1164-1255

Validates the safety confirmation, COM selection, Slave range, attempts, selected serial configurations and global busy state. It disables conflicting controls and launches the discovery worker.

`discovery_worker` **lines 1266-1545**

Iterates selected serial configurations and Slave IDs. It opens one configuration at a time, checks once that the bus is idle, probes FC03 then optional FC04 with the selected attempts, optionally performs Device Identification for found Slaves, publishes progress/results, and always closes the active serial handle.

`finish_discovery_ui` **lines 1583-1624**

Returns both Finder and Sniffer controls to idle state and presents completed/stopped/error status without leaving the application locked in a busy state.

Design note. The Finder is active traffic generation, unlike the passive Sniffer. The UI requires an explicit confirmation that another master is not active and the global busy lock prevents simultaneous capture/scan inside this application.

mb_master.py

Modbus RTU master engine, frames and data formats. It builds FC01-04 read and FC05/06/15/16 write requests, matches responses and exceptions, converts registers to and from 16/32-bit formats in four byte orders and runs MasterEngine, which serves every poll window over one serial port one request at a time. It contains no widgets.

Source lines	SHA-256
686	`79ad42ac5c1f3a658b4252c2196ccb9829ad4cfdbcc5237052ef8a3a3423221e`

Imports: `math, struct, threading, time, collections.deque, dataclasses.dataclass, mb_config.MASTER_RECONNECT_INTERVAL_SECONDS, MASTER_MAX_READ_BITS, MASTER_MAX_READ_REGISTERS, MASTER_MAX_WRITE_BITS, MASTER_MAX_WRITE_REGISTERS, mb_protocol.append_crc, crc_ok, hex_bytes, modbus_rtu_interframe_gap_seconds, mb_slave_finder.bus_activity_detected, cancel_serial_io, discovery_bits_per_char, write_discovery_request`

Module constants / configuration data

Name	Definition
`READ_FUNCTION_CODES`	`(1, 2, 3, 4)`
`BIT_FUNCTION_CODES`	`(1, 2)`
`WRITABLE_READ_FCS`	`{1: (5, 15), 3: (6, 16)}`
`BYTE_ORDERS`	`('ABCD', 'CDAB', 'BADC', 'DCBA')`
`BYTE_ORDERS_64`	`('ABCDEFGH', 'GHEFCDAB', 'BADCFEHG', 'HGFEDCBA')`
`WIRE_LETTERS`	`'ABCDEFGH'`
`REGISTER_FORMATS`	`(('int16', 'Signed 16-bit', 1), ('uint16', 'Unsigned 16-bit', 1), ('hex16', 'Hex 16-bit', 1), ('bin16', 'Binary 16-bit', 1)) + tuple(((f'{kind}_{or…`
`FORMAT_LABELS`	`{key: label for key, label, _size in REGISTER_FORMATS}`
`FORMAT_KEYS_BY_LABEL`	`{label: key for key, label, _size in REGISTER_FORMATS}`
`FORMAT_SIZES`	`{key: size for key, _label, size in REGISTER_FORMATS}`
`DEFAULT_FORMAT`	`'int16'`
`_STRUCT_CODES`	`{'int32': '>i', 'uint32': '>I', 'float32': '>f', 'int64': '>q', 'uint64': '>Q', 'float64': '>d'}`

Name	Definition
`SCALE_MAX_DECIMALS`	`6`
`SCALE_MAX_UNIT_LENGTH`	`12`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`_value_bytes` _value_bytes(registers, order)	81-84	Reorder the wire bytes of 2 or 4 registers into big-endian value bytes.	`b''.join, bytes, WIRE_LETTERS.index`
`_registers_from_value_bytes` _registers_from_value_bytes(value, order)	87-91	Registers from value bytes.	`bytearray, enumerate, len, WIRE_LETTERS.index, range`
`format_float` format_float(value, digits=7)	94-99	Format float.	`math.isnan, math.isinf`
`decode_value` decode_value(registers, fmt)	102-110	Numeric value of one value (None for the Hex/Binary display formats).	`fmt.split, struct.unpack, _value_bytes`
`validate_scale` validate_scale(raw_min, raw_max, eng_min, eng_max, decimals=1, unit='')	117-133	Return a scale dict from user text/numbers; raise ValueError (PT text).	`any, int, ValueError, str(unit or '').strip, float`
`scale_value` scale_value(number, scale)	136-139	Linear map from raw units to engineering units through two points.	—
`unscale_value` unscale_value(value, scale)	142-147	Inverse of scale_value(): engineering units back to raw units.	`ValueError`
`format_scaled` format_scaled(number, scale)	150-152	Format scaled.	`scale.get, scale_value`
`format_value` format_value(registers, fmt)	155-173	Display text of one value occupying ``FORMAT_SIZES[fmt]`` registers.	`fmt.split, str, struct.unpack, format_float, _value_bytes`
`register_groups` register_groups(qty, start_address=0, default_fmt=DEFAULT_FORMAT, overrides=None)	176-205	Split a window's registers into values: [(offset, size, fmt or None)].	`overrides.get, groups.append, next, range, min`
`format_window_values` format_window_values(registers, start_address=0, default_fmt=DEFAULT_FORMAT, overrides=None, scales=None)	208-233	Return one display string per register, honouring per-address formats.	`register_groups, len, scales.get, out.extend, decode_value`

Callable	Lines	Responsibility	Representative direct calls
`raw_text_for_scaled` raw_text_for_scaled(value_text, scale, fmt)	236-245	Convert an engineering value typed by the user into raw text for *fmt*.	`unscale_value, fmt.split('_')[0].startswith, str, float, repr`
`format_register_values` format_register_values(registers, fmt)	248-250	Turns raw registers into display strings for the selected format; 32-bit values are shown on the first register of each pair and an unpaired last register shows a dash.	`format_window_values`
`_parse_int` _parse_int(text)	253-260	Parse int.	`str(text).strip().replace(' ', '').replace, text.lower, lowered.startswith, int, str(text).strip().replace`
`encode_register_value` encode_register_value(text, fmt)	263-307	Converts one user-entered value into the register(s) it occupies in the selected format and byte order, rejecting values that do not fit.	`str(text).strip, ValueError, fmt.split, kind.startswith, _registers_from_value_bytes`
`encode_coil_value` encode_coil_value(text)	310-316	Encode coil value.	`str(text).strip().lower, ValueError, str(text).strip, str`
`_u16` _u16(value)	323-324	U16.	`bytes`
`read_limit` read_limit(fc)	327-328	Read limit.	—
`validate_read_definition` validate_read_definition(slave, fc, address, qty)	331-342	Raise ValueError (PT canonical message) for an invalid poll definition.	`ValueError, int, read_limit`
`build_read_request` build_read_request(slave, fc, address, qty)	345-347	Build read request.	`validate_read_definition, append_crc, _u16, bytes, int`
`build_write_request` build_write_request(slave, fc, address, values)	350-381	Builds FC05/06/15/16 requests within the Modbus write limits; Slave 0 is accepted for broadcast.	`append_crc, int, ValueError, len, _u16`
`expected_response_length` expected_response_length(request)	384-392	Expected response length.	—
`find_master_response` find_master_response(buffer, request)	399-447	Finds the CRC-valid response of the expected length from the addressed Slave, or a Modbus exception, and decodes read values.	`bytes, expected_response_length, range, len, crc_ok`
`class PollDefinition`	**455-465**	**PollDefinition.**	**Class container**

Callable	Lines	Responsibility	Representative direct calls
`PollDefinition.request` request(self)	464-465	Request.	`build_read_request`
`class MasterEngine`	**468-686**	**Serve all poll windows over one serial port, one transaction at a time.**	**Class container**
`MasterEngine.__init__` __init__(self, ser, event_queue, *, baud, parity, stopbits, response_timeout_ms, delay_ms, serial_module=None, idle_check_seconds=0.0, reopen=None, reconnect_interval_s=MASTER_RECONNECT_INTERVAL_SECONDS)	483-508	Init.	`float, int, max, modbus_rtu_interframe_gap_seconds, threading.Lock`
`MasterEngine.set_poll` set_poll(self, definition)	511-514	Set poll.	`self._wake.set`
`MasterEngine.remove_poll` remove_poll(self, window_id)	516-519	Remove poll.	`self._polls.pop, deque`
`MasterEngine.read_once` read_once(self, window_id)	521-525	Read once.	`self._wake.set, self._once.append`
`MasterEngine.queue_write` queue_write(self, window_id, request, description='')	527-530	Queue write.	`self._wake.set, self._writes.append, bytes`
`MasterEngine.start` start(self)	532-534	Start.	`threading.Thread, self.thread.start`
`MasterEngine.stop` stop(self, join_timeout=1.0)	536-542	Stop.	`self.stop_event.set, self._wake.set, cancel_serial_io, thread.is_alive, thread.join`
`MasterEngine.next_job` next_job(self, now=None)	545-568	Chooses the next transaction: queued writes first, then "Read once" requests, then the enabled window whose next due time is earliest.	`time.perf_counter, min, self._once.popleft, self._polls.get, self._writes.popleft`
`MasterEngine.run` run(self)	571-599	Worker loop: optional idle-bus check, then one transaction at a time with the configured delay between polls, posting master_ready, master_result and master_stopped events.	`self.event_queue.put, bus_activity_detected, self.stop_event.is_set, self._serve_jobs, hex_bytes`
`MasterEngine._serve_jobs` _serve_jobs(self)	601-623	Serve writes / reads until stopped; serial errors propagate.	`self.stop_event.is_set, self.next_job, self.event_queue.put, self.stop_event.wait, self._wake.wait`
`MasterEngine._reconnect` _reconnect(self, detail)	625-640	Close the failed port and reopen it; False when stopped meanwhile.	`self.event_queue.put, self.ser.close, self.stop_event.wait, self.reopen`

Callable	Lines	Responsibility	Representative direct calls
`MasterEngine.transact` transact(self, request)	642-686	Sends one request and waits for its response until the response timeout, classifying the result as ok, exception, invalid, timeout or broadcast.	`write_discovery_request, time.perf_counter, bytearray, self.stop_event.is_set, hex_bytes`

Detailed behavior of the key paths

`format_register_values` lines 248-250

Turns raw registers into display strings for the selected format; 32-bit values are shown on the first register of each pair and an unpaired last register shows a dash.

`encode_register_value` lines 263-307

Converts one user-entered value into the register(s) it occupies in the selected format and byte order, rejecting values that do not fit.

`build_write_request` lines 350-381

Builds FC05/06/15/16 requests within the Modbus write limits; Slave 0 is accepted for broadcast.

`find_master_response` lines 399-447

Finds the CRC-valid response of the expected length from the addressed Slave, or a Modbus exception, and decodes read values. Adapter echoes of read requests are skipped.

`MasterEngine.next_job` lines 545-568

Chooses the next transaction: queued writes first, then "Read once" requests, then the enabled window whose next due time is earliest.

`MasterEngine.run` lines 571-599

Worker loop: optional idle-bus check, then one transaction at a time with the configured delay between polls, posting master_ready, master_result and master_stopped events.

`MasterEngine.transact` lines 642-686

Sends one request and waits for its response until the response timeout, classifying the result as ok, exception, invalid, timeout or broadcast.

> **Design note.** The engine owns the serial port while the Master is connected. Widgets only change definitions through set_poll(), read_once() and queue_write(); results come back through event_queue.

mb_master_ui.py

Master (poll) tab user interface. MasterMixin builds the connection bar and the grid of poll windows, validates window definitions, renders values and counters, opens the write and name dialogs, pops windows out to their own top-level windows and saves or opens workspace files.

Source lines	SHA-256
1529	`9d0df865a4a1c690c8d94717316125e9dc2cf283d153dd949455884c5de9ae42`

Imports: `json, time, tkinter, tkinter.font, tkinter.messagebox, ttk, mb_config.DISCOVERY_BAUD_RATES, DISCOVERY_IDLE_CHECK_SECONDS, MASTER_DEFAULT_DELAY_MS, MASTER_DEFAULT_RESPONSE_TIMEOUT_MS, MASTER_DEFAULT_SCAN_MS, MASTER_MAX_DELAY_MS, MASTER_MAX_RESPONSE_TIMEOUT_MS, MASTER_MAX_SCAN_MS, MASTER_MAX_WINDOWS, MASTER_MIN_DELAY_MS, MASTER_MIN_RESPONSE_TIMEOUT_MS, MASTER_WORKSPACE_FORMAT, MASTER_WORKSPACE_FORMATS, MASTER_WORKSPACE_VERSION, MODBUS_EXCEPTION_CODES, get_master_workspaces_folder, mb_diagnostics.FUNCTION_NAMES, mb_master.BIT_FUNCTION_CODES, DEFAULT_FORMAT, FORMAT_KEYS_BY_LABEL, FORMAT_LABELS, FORMAT_SIZES, READ_FUNCTION_CODES, WRITABLE_READ_FCS, MasterEngine, PollDefinition, build_write_request, encode_coil_value, encode_register_value, format_window_values, raw_text_for_scaled, register_groups, validate_read_definition, mb_table_ui.SERIAL_PARITIES, SERIAL_STOP_BITS, TABLE_MAX_TITLE_LENGTH, TABLE_PANE_MAX_ROWS, TABLE_PANE_MIN_ROWS, TABLE_PANE_MIN_WIDTH, TableUiMixin, apply_format_to_rows, parse_int_field, validate_workspace_name, validated_formats, validated_names, validated_scales, mb_table_ui.list_workspaces, mb_table_ui.workspace_path, mb_theme.theme_colors, mb_widgets.RefreshButton`

Module constants / configuration data

Name	Definition
`MASTER_PARITIES`	`SERIAL_PARITIES`
`MASTER_STOP_BITS`	`SERIAL_STOP_BITS`
`MASTER_PANE_MIN_WIDTH`	`TABLE_PANE_MIN_WIDTH`
`MASTER_PANE_MIN_ROWS`	`TABLE_PANE_MIN_ROWS`
`MASTER_PANE_MAX_ROWS`	`TABLE_PANE_MAX_ROWS`
`MASTER_OK_MESSAGE_SECONDS`	`3.0`
`MASTER_MAX_TITLE_LENGTH`	`TABLE_MAX_TITLE_LENGTH`
`MASTER_WHEEL_TAG`	`'MBToolkit.MasterWheel'`
`FC_CHOICES`	`tuple((fc_choice_label(fc) for fc in READ_FUNCTION_CODES))`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`workspace_path` workspace_path(name)	82-84	File of a Master workspace (in Workspaces\Master).	`_workspace_path, get_master_workspaces_folder`
`list_workspaces` list_workspaces()	87-89	Names of the saved Master workspaces, sorted alphabetically.	`_list_workspaces, get_master_workspaces_folder`
`fc_choice_label` fc_choice_label(fc)	92-93	Fc choice label.	—
`class MasterWindow`	**99-187**	**State of one poll window (independent of where its pane is shown).**	**Class container**
`MasterWindow.__init__` __init__(self, window_id, number, slave=1, fc=3, address=0, qty=10, scan_ms=MASTER_DEFAULT_SCAN_MS, base=0, fmt=DEFAULT_FORMAT, names=None, enabled=True, title='', formats=None, scales=None)	102-125	Init.	`dict, self.reset_counters, str(title or '').strip, int, (formats or {}).items`
`MasterWindow.reset_counters` reset_counters(self)	127-137	Reset counters.	—
`MasterWindow.definition` definition(self)	139-141	Definition.	`PollDefinition`
`MasterWindow.to_json` to_json(self)	143-152	To json.	`str, sorted, self.names.items, self.formats.items, self.scales.items`
`MasterWindow.groups` groups(self)	154-156	[(row offset, size, fmt or None)] of the values shown in this window.	`register_groups`
`MasterWindow.group_of_row` group_of_row(self, row)	158-162	Group of row.	`self.groups`
`MasterWindow.validated_kwargs` validated_kwargs(data)	165-187	Validate one workspace window entry; raise ValueError if invalid.	`validate_read_definition, int, data.get, isinstance, ValueError`
`class MasterMixin`	**190-1525**	**Master (poll) tab GUI.**	**Class container**
`MasterMixin.build_master_tab` build_master_tab(self, page)	194-341	Builds the connection bar, toolbar and scrollable grid of poll windows, restoring the saved connection parameters.	`getattr(self, 'ui_settings', {}).get, page.columnconfigure, page.rowconfigure, ttk.LabelFrame, connection.grid`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin._master_restore_workspace` _master_restore_workspace(self, name)	343-346	Master restore workspace.	`self.master_open_workspace, self.master_add_window`
`MasterMixin._master_install_wheel_tags` _master_install_wheel_tags(self, root_widget)	349-350	Master install wheel tags.	`self._table_install_wheel_tags`
`MasterMixin._master_wheel` _master_wheel(self, event)	352-353	Master wheel.	`self._table_wheel, getattr`
`MasterMixin._master_window_name` _master_window_name(self, window)	355-356	Master window name.	`self.trf`
`MasterMixin._master_window_summary` _master_window_summary(self, window)	358-360	Master window summary.	—
`MasterMixin._master_window_title` _master_window_title(self, window)	362-363	Master window title.	`self._master_window_name, self._master_window_summary`
`MasterMixin._master_counters_text` _master_counters_text(self, window)	365-368	Master counters text.	—
`MasterMixin._master_set_message` _master_set_message(self, window, template, style='normal', **values)	370-373	Master set message.	`time.monotonic, self._master_render_status`
`MasterMixin._master_render_status` _master_render_status(self, window)	375-397	Master render status.	`self._master_panes.get, self.trf, pane['counters_var'].set, pane['message_var'].set, pane['message_label'].configure`
`MasterMixin._master_build_pane` _master_build_pane(self, window, parent)	400-498	Master build pane.	`ttk.LabelFrame, frame.columnconfigure, frame.rowconfigure, self._table_build_header, ttk.Frame`
`MasterMixin._master_update_format_state` _master_update_format_state(self, window)	500-504	Master update format state.	`self._master_panes.get, pane['format_combo'].configure`
`MasterMixin._master_fill_rows` _master_fill_rows(self, window)	506-517	Master fill rows.	`self._master_panes.get, tree.delete, tree.configure, range, self._master_render_values`
`MasterMixin._master_render_values` _master_render_values(self, window)	519-537	Master render values.	`self._master_panes.get, enumerate, format_window_values, tree.item, tree.set`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin._master_schedule_reflow` _master_schedule_reflow(self, _event=None)	539-545	Master schedule reflow.	`self.after, self.after_cancel`
`MasterMixin._master_reflow` _master_reflow(self)	547-555	Arrange docked panes in as many equal columns as the width allows.	`self._table_reflow, sorted, self.master_windows.values`
`MasterMixin.master_add_window` master_add_window(self, kwargs=None)	558-576	Master add window.	`next, MasterWindow, self._master_build_pane, self._master_reflow, self._master_mark_changed`
`MasterMixin.master_duplicate_window` master_duplicate_window(self, window_id)	578-582	Master duplicate window.	`self.master_windows.get, MasterWindow.validated_kwargs, self.master_add_window, window.to_json`
`MasterMixin.master_close_window` master_close_window(self, window_id)	584-605	Master close window.	`self.master_windows.pop, self._master_panes.pop, self._master_reflow, self.master_engine.remove_poll, pane.get`
`MasterMixin.master_apply_window` master_apply_window(self, window_id)	607-644	Validates the fields of one window, updates its model, rebuilds its rows and hands the new definition to the running engine.	`self.master_windows.get, self._master_panes.get, FORMAT_KEYS_BY_LABEL.get, self._master_fill_rows, self._master_update_format_state`
`MasterMixin.master_toggle_pause` master_toggle_pause(self, window_id)	646-656	Master toggle pause.	`self.master_windows.get, self._master_render_status, self._master_mark_changed, self.master_engine.set_poll, window.definition`
`MasterMixin.master_read_once` master_read_once(self, window_id)	658-662	Master read once.	`self.master_engine.read_once, messagebox.showinfo, self.tr`
`MasterMixin.master_toggle_popout` master_toggle_popout(self, window_id)	664-692	Moves a window between the grid and its own top-level window by rebuilding its pane from the model.	`self.master_windows.get, self._master_panes.pop, pane.get, self._master_render_status, self._master_reflow`
`MasterMixin._master_on_double_click` _master_on_double_click(self, window_id, event)	695-708	Master on double click.	`self._master_panes.get, tree.identify_row, tree.selection_set, tree.identify_column, self._master_edit_name`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin._master_edit_name` _master_edit_name(self, window_id, row)	710-737	Master edit name.	`self.master_windows.get, self._table_dialog, ttk.Label(body, text=self.trf('Nome do endereço {address}', address=pdu + window.base)).grid, …`
`MasterMixin._master_edit_value` _master_edit_value(self, window_id)	739-840	Master edit value.	`self.master_windows.get, self._master_panes.get, pane['tree'].selection, int, WRITABLE_READ_FCS.get`
`MasterMixin.master_build_write` master_build_write(self, window, slave_text, address_text, values_text, fc, fmt=None, scale=None)	842-870	Validates the write dialog and encodes the values with the window format into an FC05/06/15/16 request.	`parse_int_field, build_write_request, ValueError, str(values_text).split, p.strip`
`MasterMixin._master_connection_settings` _master_connection_settings(self)	873-888	Master connection settings.	`parse_int_field, self.master_timeout_var.get, self.master_delay_var.get, self.master_port_var.get().strip, int`
`MasterMixin.master_connect` master_connect(self)	890-952	Checks the busy flag, safety confirmation and settings, opens the port, starts MasterEngine with the idle-bus check and locks the other active features.	`self.get_serial_modules, list, MasterEngine, self.master_windows.values, self._master_lock_other_features`
`MasterMixin.master_disconnect` master_disconnect(self)	954-959	Master disconnect.	`self.master_status_var.set, self.master_disconnect_btn.configure, self.master_engine.stop, self.tr`
`MasterMixin._master_lock_other_features` _master_lock_other_features(self, locked)	961-979	Master lock other features.	`self.master_connect_btn.configure, self.master_disconnect_btn.configure, self.master_refresh_btn.configure, getattr, self.start_btn.configu…`
`MasterMixin._master_finish_session` _master_finish_session(self, payload)	981-1016	Master finish session.	`self._master_lock_other_features, self.master_status_var.set, self.master_windows.values, (payload or {}).get, engine.stop`
`MasterMixin.handle_master_event` handle_master_event(self, event, payload)	1019-1039	Handle master event.	`self.master_status_var.set, self.tr, self.master_windows.values, self.trf, self._master_render_values`
`MasterMixin._master_apply_result` _master_apply_result(self, result)	1041-1090	Updates counters, values and the status line of the window that issued a transaction, ignoring results of an outdated definition.	`self.master_windows.get, result.get, self._master_render_values, self._master_set_message, self.master_engine.read_once`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin.master_workspace_data` master_workspace_data(self)	1093-1107	Master workspace data.	`self.master_port_var.get, self.master_baud_var.get, self.master_parity_var.get, self.master_stop_var.get, self.master_timeout_var.get`
`MasterMixin._master_update_workspace_label` _master_update_workspace_label(self)	1109-1117	Master update workspace label.	`getattr, self.master_workspace_var.set, self.trf`
`MasterMixin._master_mark_changed` _master_mark_changed(self)	1119-1125	Record a change to the open workspace (ignored while loading one).	`getattr, self._master_update_workspace_label`
`MasterMixin._master_build_workspace_menu` _master_build_workspace_menu(self)	1127-1134	Rebuilds the Workspace menu each time it opens, listing the saved workspaces under Open and Delete.	`self._table_fill_workspace_menu, self._master_apply_menu_palette, list_workspaces`
`MasterMixin.master_save_workspace` master_save_workspace(self, name=None)	1136-1167	Save all windows as a named workspace in the workspace folder.	`workspace_path, self._master_update_workspace_label, self._schedule_ui_settings_save, self._table_ask_text, validate_workspace_name`
`MasterMixin.master_delete_workspace` master_delete_workspace(self, name, confirm=True)	1169-1185	Master delete workspace.	`workspace_path(name).unlink, self._master_update_workspace_label, self._schedule_ui_settings_save, messagebox.askyesno, messagebox.showerror`
`MasterMixin.master_open_workspace_folder` master_open_workspace_folder(self)	1187-1188	Master open workspace folder.	`self._table_open_folder, get_master_workspaces_folder`
`MasterMixin.master_open_workspace` master_open_workspace(self, name, quiet=False)	1190-1209	Loads a named workspace from the workspace folder, validating every window and skipping invalid entries with a warning.	`self._master_load_workspace_data, open, json.load, ValueError, workspace_path`
`MasterMixin._master_load_workspace_data` _master_load_workspace_data(self, name, data)	1211-1246	Master load workspace data.	`list, data.get, self._master_update_workspace_label, self._schedule_ui_settings_save, self.master_close_window`
`MasterMixin.master_rename_window` master_rename_window(self, window_id, name=None)	1249-1266	Gives a window its own name, shown in its header and saved in the workspace; an empty name restores the default.	`self.master_windows.get, self._master_render_status, self._master_mark_changed, self._table_ask_text, name.strip`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin._master_fields_changed` _master_fields_changed(self, window, pane)	1269-1281	True when the definition fields differ from what was last applied.	`pane['slave_var'].get().strip, str, pane['fc_var'].get, fc_choice_label, pane['address_var'].get().strip`
`MasterMixin._master_update_apply_state` _master_update_apply_state(self, window_id)	1283-1293	Master update apply state.	`self.master_windows.get, self._master_panes.get, self._master_fields_changed, pane['apply_btn'].configure`
`MasterMixin._master_row_tags` _master_row_tags(index)	1295-1297	Alternate rows get a slightly darker background (zebra striping).	—
`MasterMixin._master_build_row_menu` _master_build_row_menu(self)	1300-1301	Master build row menu.	`tk.Menu`
`MasterMixin._master_selected_rows` _master_selected_rows(self, window_id)	1303-1304	Master selected rows.	`self._table_selected_rows, self._master_panes.get`
`MasterMixin._master_show_row_menu` _master_show_row_menu(self, window_id, event)	1306-1325	Master show row menu.	`self.master_windows.get, self._master_panes.get, tree.identify_row, self._table_fill_row_menu, self._master_apply_menu_palette`
`MasterMixin.master_set_row_format` master_set_row_format(self, window_id, fmt)	1327-1340	Applies a format (or the window default) to the selected rows through apply_format_to_rows() and redraws the values.	`self.master_windows.get, self._master_render_values, self._master_mark_changed, apply_format_to_rows, self._master_selected_rows`
`MasterMixin.master_definition_from_record` master_definition_from_record(self, record)	1343-1366	Read definition (slave, fc, address, qty) behind a Traffic record, or None.	`int, record.get, validate_read_definition, str`
`MasterMixin.master_open_from_record` master_open_from_record(self, record)	1368-1383	Master open from record.	`self.master_definition_from_record, self.master_open_definition, messagebox.showinfo, getattr, self.port_a_var.get().strip`
`MasterMixin.master_open_definition` master_open_definition(self, definition, port='', baud='', parity='', stopbits='')	1385-1405	Creates a Master window from a definition coming from the Sniffer or the Finder and copies the serial parameters when disconnected.	`self.master_add_window, self.main_notebook.select, str, self.master_baud_var.set, self.master_parity_var.set`
`MasterMixin._master_selected_group_starts` _master_selected_group_starts(self, window)	1407-1409	PDU addresses where the values of the selected rows start.	`self._table_selected_group_starts, self._master_panes.get`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin.master_edit_scale` master_edit_scale(self, window_id)	1411-1425	Asks for a linear scale and applies it to the values of the selected rows.	`self.master_windows.get, self._master_selected_group_starts, self._master_render_values, self._master_mark_changed, messagebox.showwarning`
`MasterMixin.master_remove_scale` master_remove_scale(self, window_id)	1427-1437	Master remove scale.	`self.master_windows.get, self._master_selected_group_starts, self._master_render_values, self._master_mark_changed, window.scales.pop`
`MasterMixin.master_copy_rows` master_copy_rows(self, window_id)	1439-1441	Copy the selected rows (all rows if none) as tab-separated text.	`self._table_copy_rows, self._master_panes.get`
`MasterMixin.master_set_base` master_set_base(self, window_id, base)	1443-1453	Master set base.	`self.master_windows.get, self._master_panes.get, pane['address_var'].set, self._master_fill_rows, self._master_render_status`
`MasterMixin.master_confirm_close` master_confirm_close(self)	1456-1473	Asks whether to save a saved workspace that has unsaved changes before the application closes; Cancel keeps it open.	`messagebox.askyesnocancel, self.tr, self.trf, self.master_save_workspace, getattr`
`MasterMixin.master_apply_language` master_apply_language(self)	1476-1489	Master apply language.	`self._master_update_workspace_label, self.master_windows.values, hasattr, self._master_panes.get, self._table_translate_headings`
`MasterMixin._master_apply_menu_palette` _master_apply_menu_palette(self)	1491-1493	Master apply menu palette.	`self._table_apply_menus_palette, getattr`
`MasterMixin.master_apply_palette` master_apply_palette(self, dark)	1495-1511	Master apply palette.	`self._master_apply_menu_palette, theme_colors, hasattr, getattr(self, '_master_panes', {}).values, getattr`
`MasterMixin.master_shutdown` master_shutdown(self)	1513-1525	Master shutdown.	`getattr, engine.stop, ser.close`
`fc_choice_label_write` fc_choice_label_write(fc)	1528-1529	Fc choice label write.	—

Detailed behavior of the key paths

`build_master_tab` lines 194-341

Builds the connection bar, toolbar and scrollable grid of poll windows, restoring the saved connection parameters.

`master_apply_window` lines 607-644

Validates the fields of one window, updates its model, rebuilds its rows and hands the new definition to the running engine.

`master_build_write` lines 842-870

Validates the write dialog and encodes the values with the window format into an FC05/06/15/16 request.

`master_connect` lines 890-952

Checks the busy flag, safety confirmation and settings, opens the port, starts MasterEngine with the idle-bus check and locks the other active features.

`_master_apply_result` lines 1041-1090

Updates counters, values and the status line of the window that issued a transaction, ignoring results of an outdated definition.

`master_toggle_popout` lines 664-692

Moves a window between the grid and its own top-level window by rebuilding its pane from the model.

`master_open_workspace` lines 1190-1209

Loads a named workspace from the workspace folder, validating every window and skipping invalid entries with a warning.

`_master_build_workspace_menu` lines 1127-1134

Rebuilds the Workspace menu each time it opens, listing the saved workspaces under Open and Delete.

`master_rename_window` lines 1249-1266

Gives a window its own name, shown in its header and saved in the workspace; an empty name restores the default.

`master_set_row_format` **lines 1327-1340**

Applies a format (or the window default) to the selected rows through apply_format_to_rows() and redraws the values.

`master_edit_scale` **lines 1411-1425**

Asks for a linear scale and applies it to the values of the selected rows.

`master_open_definition` **lines 1385-1405**

Creates a Master window from a definition coming from the Sniffer or the Finder and copies the serial parameters when disconnected.

`master_confirm_close` **lines 1456-1473**

Asks whether to save a saved workspace that has unsaved changes before the application closes; Cancel keeps it open.

> **Design note.** Window state lives in MasterWindow objects, so a pane can be destroyed and rebuilt (pop out / dock, language switch) without losing data.

mb_slave.py

Modbus RTU slave (simulator) engine. SlaveStore keeps the data tables, defined areas and faults of every simulated Slave ID; handle_request() executes one request like a real device (reads, writes, Exceptions 01/02/03, broadcast); SlaveEngine assembles requests from the serial bytes, applies the fault simulation, answers after t3.5 and removes the adapter echo. It contains no widgets.

Source lines	SHA-256
533	`2dbbee755b610ee797abcecc59899748fd1322e4388b7623720e5b364bda5b9d`

Imports: `random, threading, time, dataclasses.dataclass, field, mb_config.MASTER_MAX_READ_BITS, MASTER_MAX_READ_REGISTERS, MASTER_MAX_WRITE_BITS, MASTER_MAX_WRITE_REGISTERS, MASTER_RECONNECT_INTERVAL_SECONDS, SLAVE_FAULT_MAX_DELAY_MS, SLAVE_RESYNC_SILENCE_MS, mb_protocol.append_crc, crc_ok, modbus_rtu_interframe_gap_seconds, mb_slave_finder.cancel_serial_io, discovery_bits_per_char, write_discovery_request`

Module constants / configuration data

Name	Definition
`TABLES`	`('coils', 'discrete', 'holding', 'input')`
`TABLE_LABELS`	`{'coils': 'Coils', 'discrete': 'Discrete Inputs', 'holding': 'Holding Registers', 'input': 'Input Registers'}`
`BIT_TABLES`	`('coils', 'discrete')`
`FC_TABLES`	`{1: 'coils', 2: 'discrete', 3: 'holding', 4: 'input', 5: 'coils', 15: 'coils', 6: 'holding', 16: 'holding'}`
`SUPPORTED_FUNCTION_CODES`	`(1, 2, 3, 4, 5, 6, 15, 16)`
`WRITE_FUNCTION_CODES`	`(5, 6, 15, 16)`
`FAULT_EXCEPTION_CODES`	`(1, 2, 3, 4, 6)`
`EXCEPTION_ILLEGAL_FUNCTION`	`1`
`EXCEPTION_ILLEGAL_ADDRESS`	`2`
`EXCEPTION_ILLEGAL_VALUE`	`3`
`TABLE_READ_FC`	`{'coils': 1, 'discrete': 2, 'holding': 3, 'input': 4}`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`table_label` table_label(table)	55-56	Table label.	`TABLE_LABELS.get`
`table_choice_label` table_choice_label(table)	59-61	Table field text: read function code + name, e.g. "03 Holding Registers".	—
`default_fault` default_fault()	68-70	Default fault.	—
`validate_fault` validate_fault(data)	73-91	Return a clean fault dict from user/workspace data; raise ValueError (PT).	`isinstance, ValueError, int, bool, data.get`
`fault_active` fault_active(fault)	94-97	True when the fault settings change how the slave answers.	`bool, fault.get`
`class SlaveStore`	**104-171**	**Data tables, defined areas and faults of every simulated Slave ID.**	**Class container**
`SlaveStore.__init__` __init__(self)	113-117	Init.	`threading.RLock`
`SlaveStore.set_areas` set_areas(self, areas)	119-126	Replace the defined areas: iterable of (slave, table, address, qty).	`defined.setdefault((int(slave), table), set()).update, range, defined.setdefault, int, set`
`SlaveStore.slave_ids` slave_ids(self)	128-130	Slave ids.	—
`SlaveStore.covers` covers(self, slave, table, address, qty)	132-135	Covers.	`self._defined.get, bool, all, range`
`SlaveStore.read` read(self, slave, table, address, qty)	137-140	Read.	`self._values.get, values.get, range`
`SlaveStore.write` write(self, slave, table, address, values)	142-147	Write.	`self._values.setdefault, enumerate, int`
`SlaveStore.clear_values` clear_values(self)	149-151	Clear values.	—
`SlaveStore.set_fault` set_fault(self, slave, fault)	153-158	Set fault.	`fault_active, dict, self._faults.pop, int`
`SlaveStore.fault` fault(self, slave)	160-163	Fault.	`self._faults.get, dict`
`SlaveStore.faults` faults(self)	165-167	Faults.	`dict, self._faults.items`
`SlaveStore.clear_faults` clear_faults(self)	169-171	Clear faults.	—

Callable	Lines	Responsibility	Representative direct calls
`request_length` request_length(buffer)	178-191	Length of the request frame starting at buffer[0]; None if not known yet.	`len`
`exception_frame` exception_frame(slave, fc, code)	194-195	Exception frame.	`append_crc, bytes`
`class SlaveReply`	**199-209**	**Outcome of one request: what to send back and what it touched.**	**Class container**
`_u16` _u16(data, index)	212-213	U16.	—
`_parse_request` _parse_request(frame)	216-251	Return (table, address, qty, values or None) or raise _ExceptionReply.	`_u16, _ExceptionReply, len, range`
`class _ExceptionReply`	**254-257**	**_ExceptionReply.**	**Class container**
`_ExceptionReply.__init__` __init__(self, code)	255-257	Init.	`super().__init__, super`
`handle_request` handle_request(store, frame)	260-319	Execute one CRC-valid request frame and return a SlaveReply.	`SlaveReply, store.read, append_crc, _parse_request, store.covers`
`corrupt_crc` corrupt_crc(frame)	322-326	Same frame with a wrong CRC (fault simulation).	`bytearray, bytes`
`class SlaveEngine`	**333-533**	**Answer the requests of a real master on one serial port.**	**Class container**
`SlaveEngine.__init__` __init__(self, ser, event_queue, store, *, baud, parity, stopbits, serial_module=None, reopen=None, reconnect_interval_s=MASTER_RECONNECT_INTERVAL_SECONDS, silence_ms=SLAVE_RESYNC_SILENCE_MS, rng=None)	351-373	Init.	`int, float, discovery_bits_per_char, modbus_rtu_interframe_gap_seconds, max`
`SlaveEngine.start` start(self)	375-377	Start.	`threading.Thread, self.thread.start`
`SlaveEngine.stop` stop(self, join_timeout=1.0)	379-384	Stop.	`self.stop_event.set, cancel_serial_io, thread.is_alive, thread.join, threading.current_thread`
`SlaveEngine.run` run(self)	387-404	Run.	`self.event_queue.put, self.stop_event.is_set, self._listen, self._reconnect, type`

Callable	Lines	Responsibility	Representative direct calls
`SlaveEngine._reconnect` _reconnect(self, detail)	406-421	Reconnect.	`self._buffer.clear, self.event_queue.put, self.ser.close, self.stop_event.wait, self.reopen`
`SlaveEngine._listen` _listen(self)	423-438	Read and answer until stopped; serial errors propagate.	`time.perf_counter, self.stop_event.is_set, int, self.stop_event.wait, self._strip_echo`
`SlaveEngine.process` process(self, final)	440-465	Answer every complete request in the buffer.	`request_length, bytes, crc_ok, self.stop_event.is_set, self._answer`
`SlaveEngine._strip_echo` _strip_echo(self, data)	467-482	Remove the RS485 adapter echo of our last response from *data*.	`time.perf_counter, len`
`SlaveEngine._answer` _answer(self, frame)	484-533	Answer.	`self.event_queue.put, self.store.fault, SlaveReply, corrupt_crc, list`

> **Design note.** handle_request() is a pure function of the store and the frame, so every reply is testable without a serial port; the engine only adds framing, timing, faults and echo removal around it.

mb_slave_ui.py

Slave (simulator) tab user interface. SlaveMixin builds the connection bar and the grid of simulated tables, edits values locally, highlights master writes, shows per-window and global counters, edits the faults of each Slave ID and saves or opens Slave workspaces.

Source lines
1412

SHA-256
`d8f3daaa6f69074315ad6daefe6a761a66d7ab85e4eeb1a01138d67cac8ca92a`

Imports: `json, time, tkinter, tkinter.messagebox, ttk, mb_config.DISCOVERY_BAUD_RATES, MODBUS_EXCEPTION_CODES, SLAVE_FAULT_MAX_DELAY_MS, SLAVE_HIGHLIGHT_SECONDS, SLAVE_MAX_QTY, SLAVE_MAX_WINDOWS, SLAVE_WORKSPACE_FORMAT, SLAVE_WORKSPACE_VERSION, get_slave_workspaces_folder, mb_master.DEFAULT_FORMAT, FORMAT_KEYS_BY_LABEL, FORMAT_LABELS, FORMAT_SIZES, encode_coil_value, encode_register_value, format_window_values, raw_text_for_scaled, register_groups, mb_slave.BIT_TABLES, FAULT_EXCEPTION_CODES, TABLE_LABELS, TABLES, SlaveEngine, SlaveStore, default_fault, fault_active, TABLE_READ_FC, table_choice_label, table_label, validate_fault, mb_table_ui.SERIAL_PARITIES, SERIAL_STOP_BITS, TABLE_MAX_TITLE_LENGTH, TABLE_PANE_MAX_ROWS, TABLE_PANE_MIN_ROWS, TABLE_PANE_MIN_WIDTH, TableUiMixin, apply_format_to_rows, list_workspaces, parse_int_field, validate_workspace_name, validated_formats, validated_names, validated_scales, workspace_path, mb_theme.theme_colors, mb_widgets.RefreshButton`

Module constants / configuration data

Name	Definition
`SLAVE_WHEEL_TAG`	`'MBToolkit.SlaveWheel'`
`TABLE_CHOICES`	`tuple((table_choice_label(t) for t in TABLES))`
`TABLE_KEYS_BY_LABEL`	`{table_choice_label(t): t for t in TABLES}`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`fault_parts` fault_parts(fault)	79-90	(PT template, values) pieces describing an active fault.	`fault.get, parts.append`
`class SlaveWindow`	**93-170**	**State of one simulated data table (independent of where it is shown).**	**Class container**
`SlaveWindow.__init__` __init__(self, window_id, number, slave=1, table='holding', address=0, qty=10, base=0, fmt=DEFAULT_FORMAT, names=None, title='', formats=None, scales=None)	96-114	Init.	`dict, self.reset_counters, str(title or '').strip, int, (formats or {}).items`

Callable	Lines	Responsibility	Representative direct calls
`SlaveWindow.reset_counters` reset_counters(self)	116-119	Reset counters.	—
`SlaveWindow.bits` bits(self)	122-123	Bits.	—
`SlaveWindow.area` area(self)	125-126	Area.	—
`SlaveWindow.overlap` overlap(self, address, qty)	128-132	Row range of this window covered by [address, address + qty), or None.	`max, min`
`SlaveWindow.groups` groups(self)	134-135	Groups.	`register_groups`
`SlaveWindow.group_of_row` group_of_row(self, row)	137-141	Group of row.	`self.groups`
`SlaveWindow.to_json` to_json(self)	143-151	To json.	`str, sorted, self.names.items, self.formats.items, self.scales.items`
`SlaveWindow.validated_kwargs` validated_kwargs(data)	154-170	Validate one workspace window entry; raise ValueError if invalid.	`validate_area, data.get, isinstance, ValueError, int`
`validate_area` validate_area(slave, table, address, qty)	173-184	Raise ValueError (PT canonical message) for an invalid window area.	`ValueError, int`
`class SlaveMixin`	**187-1412**	**Slave (simulator) tab GUI.**	**Class container**
`SlaveMixin.build_slave_tab` build_slave_tab(self, page)	191-324	Build slave tab.	`getattr(self, 'ui_settings', {}).get, SlaveStore, self._slave_reset_stats, page.columnconfigure, page.rowconfigure`
`SlaveMixin._slave_restore_workspace` _slave_restore_workspace(self, name)	326-329	Slave restore workspace.	`self.slave_open_workspace, self.slave_add_window`
`SlaveMixin._slave_reset_stats` _slave_reset_stats(self)	332-334	Slave reset stats.	—
`SlaveMixin._slave_render_stats` _slave_render_stats(self)	336-340	Slave render stats.	`self.slave_stats_var.set, self.trf`
`SlaveMixin._slave_window_name` _slave_window_name(self, window)	342-343	Slave window name.	`self.trf`
`SlaveMixin._slave_window_summary` _slave_window_summary(self, window)	345-348	Slave window summary.	`table_label`

Callable	Lines	Responsibility	Representative direct calls
`SlaveMixin._slave_fault_text` _slave_fault_text(self, slave)	350-357	Slave fault text.	`self.slave_faults.get, ', '.join, self.trf, fault_active, fault_parts`
`SlaveMixin._slave_render_status` _slave_render_status(self, window)	359-383	Slave render status.	`self._slave_panes.get, pane['counters_var'].set, pane['last_var'].set, pane['name_var'].set, pane['summary_var'].set`
`SlaveMixin._slave_build_pane` _slave_build_pane(self, window, parent)	386-475	Slave build pane.	`ttk.LabelFrame, frame.columnconfigure, frame.rowconfigure, self._table_build_header, tk.StringVar`
`SlaveMixin._slave_update_format_state` _slave_update_format_state(self, window)	477-480	Slave update format state.	`self._slave_panes.get, pane['format_combo'].configure`
`SlaveMixin._slave_fill_rows` _slave_fill_rows(self, window)	482-493	Slave fill rows.	`self._slave_panes.get, tree.delete, tree.configure, range, self._slave_render_values`
`SlaveMixin._slave_render_values` _slave_render_values(self, window)	495-515	Slave render values.	`self._slave_panes.get, self.slave_store.read, time.monotonic, enumerate, format_window_values`
`SlaveMixin._slave_render_table` _slave_render_table(self, slave, table)	517-520	Slave render table.	`self.slave_windows.values, self._slave_render_values`
`SlaveMixin._slave_schedule_reflow` _slave_schedule_reflow(self, _event=None)	522-528	Slave schedule reflow.	`self.after, self.after_cancel`
`SlaveMixin._slave_reflow` _slave_reflow(self)	530-537	Slave reflow.	`self._table_reflow, sorted, self.slave_windows.values`
`SlaveMixin._slave_sync_areas` _slave_sync_areas(self)	540-542	Tell the store which addresses exist (what the windows show).	`self.slave_store.set_areas, w.area, self.slave_windows.values`
`SlaveMixin.slave_add_window` slave_add_window(self, kwargs=None)	544-561	Slave add window.	`next, SlaveWindow, self._slave_sync_areas, self._slave_build_pane, self._slave_reflow`
`SlaveMixin.slave_duplicate_window` slave_duplicate_window(self, window_id)	563-566	Slave duplicate window.	`self.slave_windows.get, self.slave_add_window, SlaveWindow.validated_kwargs, window.to_json`

Callable	Lines	Responsibility	Representative direct calls
`SlaveMixin.slave_close_window` slave_close_window(self, window_id)	568-587	Slave close window.	`self.slave_windows.pop, self._slave_sync_areas, self._slave_panes.pop, self._slave_reflow, pane.get`
`SlaveMixin._slave_fields_changed` _slave_fields_changed(self, window, pane)	589-599	Slave fields changed.	`pane['slave_var'].get().strip, str, pane['table_var'].get, table_choice_label, pane['address_var'].get().strip`
`SlaveMixin._slave_update_apply_state` _slave_update_apply_state(self, window_id)	601-610	Slave update apply state.	`self.slave_windows.get, self._slave_panes.get, self._slave_fields_changed, pane['apply_btn'].configure`
`SlaveMixin.slave_apply_window` slave_apply_window(self, window_id)	612-645	Slave apply window.	`self.slave_windows.get, self._slave_panes.get, FORMAT_KEYS_BY_LABEL.get, self._slave_fill_rows, self._slave_update_format_state`
`SlaveMixin.slave_toggle_popout` slave_toggle_popout(self, window_id)	647-674	Slave toggle popout.	`self.slave_windows.get, self._slave_panes.pop, pane.get, self._slave_render_status, self._slave_reflow`
`SlaveMixin.slave_rename_window` slave_rename_window(self, window_id, name=None)	676-693	Slave rename window.	`self.slave_windows.get, self._slave_render_status, self._slave_mark_changed, self._table_ask_text, name.strip`
`SlaveMixin._slave_on_double_click` _slave_on_double_click(self, window_id, event)	696-709	Slave on double click.	`self._slave_panes.get, tree.identify_row, tree.selection_set, tree.identify_column, self._slave_edit_name`
`SlaveMixin._slave_edit_name` _slave_edit_name(self, window_id, row)	711-730	Slave edit name.	`self.slave_windows.get, self._table_ask_text, name.strip, self._slave_panes.get, self._slave_mark_changed`
`SlaveMixin.slave_set_value` slave_set_value(self, window, row, text)	732-752	Store the value typed for *row* (first register of its value).	`self.slave_store.write, self._slave_render_table, window.group_of_row, window.scales.get, encode_register_value`
`SlaveMixin._slave_edit_value` _slave_edit_value(self, window_id)	754-812	Slave edit value.	`self.slave_windows.get, self._slave_panes.get, pane['tree'].selection, int, window.group_of_row`

Callable	Lines	Responsibility	Representative direct calls
`SlaveMixin._slave_show_row_menu` _slave_show_row_menu(self, window_id, event)	815-834	Slave show row menu.	`self.slave_windows.get, self._slave_panes.get, tree.identify_row, self._table_fill_row_menu, self._slave_apply_menu_palette`
`SlaveMixin._slave_selected_rows` _slave_selected_rows(self, window_id)	836-837	Slave selected rows.	`self._table_selected_rows, self._slave_panes.get`
`SlaveMixin.slave_set_row_format` slave_set_row_format(self, window_id, fmt)	839-851	Slave set row format.	`self.slave_windows.get, self._slave_render_values, self._slave_mark_changed, apply_format_to_rows, self._slave_selected_rows`
`SlaveMixin.slave_edit_scale` slave_edit_scale(self, window_id)	853-865	Slave edit scale.	`self.slave_windows.get, self._table_selected_group_starts, self._slave_render_values, self._slave_mark_changed, self._slave_panes.get`
`SlaveMixin.slave_remove_scale` slave_remove_scale(self, window_id)	867-877	Slave remove scale.	`self.slave_windows.get, self._table_selected_group_starts, self._slave_panes.get, self._slave_render_values, self._slave_mark_changed`
`SlaveMixin.slave_set_base` slave_set_base(self, window_id, base)	879-889	Slave set base.	`self.slave_windows.get, self._slave_panes.get, pane['address_var'].set, self._slave_fill_rows, self._slave_render_status`
`SlaveMixin.slave_set_fault` slave_set_fault(self, slave, fault)	892-901	Set (or clear, with None / an inactive fault) the fault of a Slave ID.	`fault_active, self.slave_store.set_fault, self.slave_windows.values, dict, self.slave_faults.pop`
`SlaveMixin.slave_edit_faults` slave_edit_faults(self, window_id)	903-978	Slave edit faults.	`self.slave_windows.get, self._table_dialog, ttk.Label(body, text=self.trf('Simulação de falhas do Slave {slave} (aplica-se a todas as janel…`
`SlaveMixin.slave_connect` slave_connect(self)	981-1040	Slave connect.	`self.get_serial_modules, self.slave_port_var.get().strip, list, SlaveEngine, self._slave_reset_stats`
`SlaveMixin.slave_disconnect` slave_disconnect(self)	1042-1047	Slave disconnect.	`self.slave_status_var.set, self.slave_disconnect_btn.configure, self.slave_engine.stop, self.tr`

Callable	Lines	Responsibility	Representative direct calls
`SlaveMixin._slave_lock_other_features` _slave_lock_other_features(self, locked)	1049-1066	While the Slave listens, nothing else may use a serial port.	`self.slave_connect_btn.configure, self.slave_disconnect_btn.configure, self.slave_refresh_btn.configure, self.set_sniffer_refresh_state, se…`
`SlaveMixin._slave_finish_session` _slave_finish_session(self, payload)	1068-1089	Slave finish session.	`self._slave_lock_other_features, self.slave_status_var.set, (payload or {}).get, engine.stop, self.tr`
`SlaveMixin.handle_slave_event` handle_slave_event(self, event, payload)	1092-1108	Handle slave event.	`self.slave_status_var.set, self.tr, self.trf, self._slave_finish_session, self.slave_port_var.get`
`SlaveMixin.slave_request_text` slave_request_text(self, info)	1110-1127	Short description of a request: "FC03 @ 0 × 10 → Exception 02".	`info.get, self.tr`
`SlaveMixin._slave_apply_request` _slave_apply_request(self, info)	1129-1175	Slave apply request.	`info.get, self._slave_render_stats, time.strftime, self.slave_windows.values, time.localtime`
`SlaveMixin._slave_schedule_tick` _slave_schedule_tick(self)	1177-1179	Slave schedule tick.	`self.after`
`SlaveMixin._slave_tick` _slave_tick(self)	1181-1194	Remove expired write highlights.	`time.monotonic, self.slave_windows.values, self._slave_schedule_tick, self._slave_render_values, bool`
`SlaveMixin.slave_workspace_data` slave_workspace_data(self)	1197-1213	Configuration only: the register values are not saved.	`self.slave_windows.values, self.slave_port_var.get, self.slave_baud_var.get, self.slave_parity_var.get, self.slave_stop_var.get`
`SlaveMixin._slave_update_workspace_label` _slave_update_workspace_label(self)	1215-1223	Slave update workspace label.	`self.slave_workspace_var.set, self.trf`
`SlaveMixin._slave_mark_changed` _slave_mark_changed(self)	1225-1230	Slave mark changed.	`getattr, self._slave_update_workspace_label`
`SlaveMixin._slave_build_workspace_menu` _slave_build_workspace_menu(self)	1232-1239	Slave build workspace menu.	`self._table_fill_workspace_menu, self._slave_apply_menu_palette, list_workspaces, get_slave_workspaces_folder, self._table_open_folder`

Callable	Lines	Responsibility	Representative direct calls
`SlaveMixin.slave_save_workspace` slave_save_workspace(self, name=None)	1241-1271	Slave save workspace.	`workspace_path, self._slave_update_workspace_label, self._schedule_ui_settings_save, self._table_ask_text, validate_workspace_name`
`SlaveMixin.slave_delete_workspace` slave_delete_workspace(self, name, confirm=True)	1273-1289	Slave delete workspace.	`workspace_path(name, get_slave_workspaces_folder()).unlink, self._slave_update_workspace_label, self._schedule_ui_settings_save, messagebox…`
`SlaveMixin.slave_open_workspace` slave_open_workspace(self, name, quiet=False)	1291-1307	Replace the windows and faults with a saved workspace; values restart at 0.	`self._slave_load_workspace_data, open, json.load, ValueError, workspace_path`
`SlaveMixin._slave_load_workspace_data` _slave_load_workspace_data(self, name, data)	1309-1349	Slave load workspace data.	`list, self.slave_store.clear_values, self.slave_store.clear_faults, data.get, self._slave_update_workspace_label`
`SlaveMixin.slave_confirm_close` slave_confirm_close(self)	1351-1365	Ask to save a changed Slave workspace; False cancels closing.	`messagebox.askyesnocancel, self.tr, self.trf, self.slave_save_workspace, getattr`
`SlaveMixin.slave_apply_language` slave_apply_language(self)	1368-1381	Slave apply language.	`self._slave_update_workspace_label, self._slave_render_stats, self.slave_windows.values, hasattr, self._slave_panes.get`
`SlaveMixin._slave_apply_menu_palette` _slave_apply_menu_palette(self)	1383-1385	Slave apply menu palette.	`self._table_apply_menus_palette, getattr`
`SlaveMixin.slave_apply_palette` slave_apply_palette(self, dark)	1387-1398	Slave apply palette.	`self._slave_apply_menu_palette, getattr, getattr(self, '_slave_panes', {}).values, canvas.configure, self._table_configure_tree_tags`
`SlaveMixin.slave_shutdown` slave_shutdown(self)	1400-1412	Slave shutdown.	`getattr, engine.stop, ser.close`

mb_table_ui.py

Window-grid code shared by the Master and Slave tabs. TableUiMixin builds the window header, the value table with drag selection, zebra rows and frame, the right-click menu, the scale and text dialogs, the grid reflow and the Workspace menu; module helpers validate workspace names and list workspace files.

Source lines	SHA-256
599	`d7f10014c16f1eb74074143bcffb35705cb3a17be6731ea13f773803b096e881`

Imports: `os, subprocess, sys, tkinter, tkinter.messagebox, ttk, mb_master.FORMAT_SIZES, REGISTER_FORMATS, validate_scale, mb_theme.theme_colors`

Module constants / configuration data

Name	Definition
`TABLE_PANE_MIN_WIDTH`	`600`
`TABLE_PANE_MIN_ROWS`	`3`
`TABLE_PANE_MAX_ROWS`	`15`
`TABLE_STRIPE_LIGHT`	`'#f2f4f6'`
`TABLE_STRIPE_DARK`	`'#262d34'`
`TABLE_WRITTEN_LIGHT`	`'#fff1b8'`
`TABLE_WRITTEN_DARK`	`'#5a4b12'`
`TABLE_MAX_TITLE_LENGTH`	`40`
`SERIAL_PARITIES`	`('None', 'Even', 'Odd')`
`SERIAL_STOP_BITS`	`('1', '2')`
`WORKSPACE_SUFFIX`	`'.mbsw'`
`INVALID_FILENAME_CHARS`	`'<>:"/\\|?*'`
`RESERVED_FILENAMES`	`{'CON', 'PRN', 'AUX', 'NUL'} | {f'COM{i}' for i in range(1, 10)} | {f'LPT{i}' for i in range(1, 10)}`
`WHEEL_GUARD_TAG`	`'MBToolkit.MouseWheelGuard'`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`parse_int_field` parse_int_field(text, minimum, maximum, message)	43-50	Parse int field.	`int, ValueError, str(text).strip, str`
`validate_workspace_name` validate_workspace_name(name)	53-63	Return a cleaned workspace name usable as a Windows file name.	`str(name or '').strip().rstrip, ValueError, any, name.split('.')[0].upper, str(name or '').strip`
`workspace_path` workspace_path(name, folder)	66-67	Workspace path.	`validate_workspace_name`
`list_workspaces` list_workspaces(folder)	70-76	Names of the saved workspaces in *folder*, sorted alphabetically.	`folder.glob, sorted, f.is_file`
`apply_format_to_rows` apply_format_to_rows(window, rows, fmt)	79-110	Set the per-address format of the selected rows of *window*.	`sorted, list, set, ValueError, window.formats.items`
`validated_names` validated_names(names)	113-115	Validated names.	`int, str, isinstance, names.items, str(k).isdigit`
`validated_formats` validated_formats(formats)	118-120	Validated formats.	`int, isinstance, formats.items, str(k).isdigit, str`
`validated_scales` validated_scales(scales)	123-135	Scales of a workspace window entry; invalid ones are dropped.	`isinstance, scales.items, validate_scale, item.get, int`
`class TableUiMixin`	**138-599**	**Widget code shared by the Master and Slave window grids.**	**Class container**
`TableUiMixin._table_dialog` _table_dialog(self, title)	142-150	Table dialog.	`tk.Toplevel, dialog.title, dialog.transient, dialog.resizable, self._table_apply_toplevel_palette`
`TableUiMixin._table_ask_text` _table_ask_text(self, title, caption, initial)	152-175	Small modal text prompt; returns the text or None when cancelled.	`self._table_dialog, ttk.Label(body, text=caption).grid, tk.StringVar, ttk.Entry, entry.grid`
`TableUiMixin._table_ask_scale` _table_ask_scale(self, window, starts)	177-245	Ask for a linear scale and set it on the values starting at *starts*.	`self._table_dialog, ttk.Label(body, text=self.trf('Escala de {count} valor(es) selecionado(s):', count=len(starts))).grid, ttk.Label(body, …`
`TableUiMixin._table_build_header` _table_build_header(self, frame, pane, rename, popout, close)	248-273	Window name + summary on the left; Rename / Pop out / × on the right.	`ttk.Frame, header.grid, header.columnconfigure, tk.StringVar, ttk.Label`

Callable	Lines	Responsibility	Representative direct calls
`TableUiMixin._table_build_tree` _table_build_tree(self, frame, pane, row, on_double, on_return, on_menu)	275-306	Value table (address / name / value) with a thin 1 px frame.	`tk.Frame, table.grid, table.columnconfigure, table.rowconfigure, ttk.Treeview`
`TableUiMixin._table_translate_captions` _table_translate_captions(self, *containers)	308-316	Captions are written in canonical PT; show them in the current language.	`widget.winfo_children, child.cget, child.configure, child.keys, self.tr`
`TableUiMixin._table_translate_headings` _table_translate_headings(self, pane)	318-323	Table translate headings.	`pane['tree'].heading, self.tr`
`TableUiMixin._table_row_tags` _table_row_tags(index)	326-328	Alternate rows get a slightly darker background (zebra striping).	—
`TableUiMixin._table_configure_tree_tags` _table_configure_tree_tags(self, tree)	330-346	Table configure tree tags.	`bool, theme_colors, getattr(self, 'dark_mode_var', tk.BooleanVar(value=False)).get, tree.tag_configure, isinstance`
`TableUiMixin._table_reflow` _table_reflow(self, canvas, grid, grid_item, frames, empty_label, uniform)	349-368	Arrange docked pane *frames* in as many equal columns as the width allows.	`max, canvas.itemconfigure, range, enumerate, canvas.winfo_width`
`TableUiMixin._table_build_scroll_grid` _table_build_scroll_grid(self, page, row)	370-385	Scrollable area that holds the window grid; returns (canvas, grid, item).	`ttk.Frame, holder.grid, holder.columnconfigure, holder.rowconfigure, tk.Canvas`
`TableUiMixin._table_install_wheel_tags` _table_install_wheel_tags(self, root_widget, wheel_tag)	387-401	Give new widgets the global wheel guard; plain panels scroll the grid.	`stack.pop, stack.extend, tuple, isinstance, widget.winfo_children`
`TableUiMixin._table_wheel` _table_wheel(canvas, event)	404-420	Table wheel.	`canvas.yview, canvas.yview_scroll, getattr`
`TableUiMixin._table_drag_start` _table_drag_start(self, pane, event)	427-436	Table drag start.	`tree.identify_row, bool, tuple, tree.selection, tree.identify_region`
`TableUiMixin._table_drag_move` _table_drag_move(self, pane, event)	438-444	Table drag move.	`pane.get, self._table_drag_update`
`TableUiMixin._table_drag_update` _table_drag_update(self, pane)	446-465	Table drag update.	`pane.get, tree.winfo_height, tree.identify_row, tree.get_children, sorted`
`TableUiMixin._table_drag_stop` _table_drag_stop(self, pane)	467-474	Table drag stop.	`pane.get, drag.get, self.after_cancel`

Callable	Lines	Responsibility	Representative direct calls
`TableUiMixin._table_selected_rows` _table_selected_rows(pane)	478-481	Table selected rows.	`sorted, int, pane['tree'].selection, iid.startswith`
`TableUiMixin._table_selected_group_starts` _table_selected_group_starts(self, window, pane)	483-491	PDU addresses where the values of the selected rows start.	`self._table_selected_rows, window.group_of_row, starts.append`
`TableUiMixin._table_copy_rows` _table_copy_rows(self, pane)	493-502	Copy the selected rows (all rows if none) as tab-separated text.	`list, '\t'.join, self.clipboard_clear, self.clipboard_append, tree.selection`
`TableUiMixin._table_fill_row_menu` _table_fill_row_menu(self, menu, pane, window, registers, *, set_format, edit_scale, remove_scale, copy, set_base)	504-535	Right-click menu: Formato ▸, Repor, Escala..., Remover escala, Copiar, Base ▸.	`menu.delete, menu.winfo_children, tk.Menu, format_menu.add_separator, format_menu.add_cascade`
`TableUiMixin._table_popup` _table_popup(menu, event)	538-542	Table popup.	`menu.tk_popup, menu.grab_release`
`TableUiMixin._table_fill_workspace_menu` _table_fill_workspace_menu(self, menu, names, *, save, open_, delete, folder)	545-563	Workspace menu with the saved workspaces listed directly in it.	`menu.delete, menu.winfo_children, menu.add_command, tk.Menu, menu.add_cascade`
`TableUiMixin._table_open_folder` _table_open_folder(self, folder)	565-574	Table open folder.	`os.startfile, messagebox.showerror, str, subprocess.Popen, self.tr`
`TableUiMixin._table_apply_toplevel_palette` _table_apply_toplevel_palette(self, top)	577-582	Table apply toplevel palette.	`bool, getattr(self, 'dark_mode_var', tk.BooleanVar(value=False)).get, top.configure, getattr, tk.BooleanVar`
`TableUiMixin._table_apply_menus_palette` _table_apply_menus_palette(self, *menus)	584-599	Colour menus and every submenu below them (classic Tk widgets).	`bool, theme_colors, getattr(self, 'dark_mode_var', tk.BooleanVar(value=False)).get, stack.pop, stack.extend`

> **Design note.** A pane is a plain dict of widgets and a window a model object, so the same widget code serves MasterWindow and SlaveWindow.

mb_i18n.py

Runtime Portuguese/English translation layer. Portuguese (Portugal) is the canonical UI text. Static translations, runtime-status patterns, protocol-detail fragments and the complete English Help content are kept separate from protocol logic so switching language affects presentation only.

Source lines
660

SHA-256
`b948e2a8a23ab71b8ebf5e65761e2025de0c832d3cea50061899f6387a5cdeeb`

Imports: `mb_config.DISCOVERY_IDLE_CHECK_SECONDS, SNIFFER_SPLIT_FRAME_HOLD_MS`

Module constants / configuration data

Name	Definition
`LANGUAGE_PORTUGUESE`	`'Português'`
`LANGUAGE_ENGLISH`	`'English'`
`LANGUAGE_CHOICES`	`(LANGUAGE_PORTUGUESE, LANGUAGE_ENGLISH)`
`PT_TO_EN`	`{'Linguagem': 'Language', 'Modo escuro': 'Dark mode', 'Configuração': 'Configuration', 'Modo físico': 'Physical mode', 'RS232 dual RX: COM A escuta…`
`EN_TO_PT`	`{english: portuguese for portuguese, english in PT_TO_EN.items()}`
`EXCEPTION_DESCRIPTION_EN`	`{1: 'The requested Modbus function is not supported or permitted by the device.', 2: 'The requested address or address range is not valid for this …`
`DETAIL_FRAGMENTS_PT_TO_EN`	`(('Registos (PDU Address=valor):', 'Registers (PDU Address=value):'), ('Bits (PDU Address=valor):', 'Bits (PDU Address=value):'), ('Registos=', 'Re…`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`normalize_language` normalize_language(value)	365-366	Normalize language.	—

Callable	Lines	Responsibility	Representative direct calls
`canonical_ui_text` canonical_ui_text(text)	369-372	Normalises a known English caption back to its canonical PT-PT form.	`str, EN_TO_PT.get`
`translate_text` translate_text(text, language)	375-380	Returns the English translation for known canonical PT-PT UI strings when English is selected; unknown/technical terms are returned unchanged.	`canonical_ui_text, normalize_language, PT_TO_EN.get`
`translate_details` translate_details(text, language)	391-400	Translates selected human-readable fragments embedded inside protocol detail strings (DETAIL_FRAGMENTS_PT_TO_EN) while preserving numbers, addresses, register values and other protocol data.	`str, normalize_language, value.replace`
`exception_description` exception_description(code, portuguese_text, language)	403-406	Exception description.	`normalize_language, EXCEPTION_DESCRIPTION_EN.get`
`english_help_blocks` english_help_blocks(max_ui_frames_text, max_raw_lines_text, min_pending, max_pending, debug_sim=False)	410-631	Returns the complete English Help / Connections document as structured blocks so Help switches language together with the rest of the UI.	`blocks.extend`
`translate_runtime_text` translate_runtime_text(text, language)	634-660	Uses regular expressions to translate live status strings that contain counters, elapsed time, selected configuration or estimated duration.	`str, translate_text, normalize_language, re.match, re.sub`

Detailed behavior of the key paths

`canonical_ui_text` lines 369-372

Normalises a known English caption back to its canonical PT-PT form. This makes language switching reversible instead of translating an already translated string repeatedly.

`translate_text` lines 375-380

Returns the English translation for known canonical PT-PT UI strings when English is selected; unknown/technical terms are returned unchanged.

`translate_details` **lines 391-400**

Translates selected human-readable fragments embedded inside protocol detail strings (DETAIL_FRAGMENTS_PT_TO_EN) while preserving numbers, addresses, register values and other protocol data.

`english_help_blocks` **lines 410-631**

Returns the complete English Help / Connections document as structured blocks so Help switches language together with the rest of the UI.

`translate_runtime_text` **lines 634-660**

Uses regular expressions to translate live status strings that contain counters, elapsed time, selected configuration or estimated duration.

> **Design note.** The translation design treats PT-PT as canonical and leaves technical English terms unchanged unless a translation is explicitly defined.

mb_theme.py

Fixed-geometry Light/Dark theme implementation. It configures a single ttk clam layout once, installs stable checkbox geometry and changes only palette values at runtime. This is deliberately designed so theme switching does not move or resize controls.

Source lines	SHA-256
692	`7b03e5a4f4cc646a96cd1839d2a92794ea923d232ae7e50a93b5908ac4ab8d9e`

Imports: `tkinter, tkinter.ttk`

Module constants / configuration data

Name	Definition
`BOOTSTRAP_ACTIVE`	`False`
`THEME_ENGINE`	`'clam'`
`LIGHT`	`{'root': '#f3f4f6', 'panel': '#f3f4f6', 'panel_alt': '#ffffff', 'input': '#ffffff', 'input_readonly': '#f5f6f7', 'text': '#1f2328', 'muted': '#5963…`
`DARK`	`{'root': '#171a1f', 'panel': '#171a1f', 'panel_alt': '#1f242a', 'input': '#20262c', 'input_readonly': '#252c33', 'text': '#e8eaed', 'muted': '#aab2…`
`_CHECK_IMAGES`	`None`
`_CHECK_ELEMENT`	`'MBS.Check.indicator'`
`_FIXED_GEOMETRY_READY`	`False`
`_INTERP_ID`	`None`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`theme_colors` theme_colors(root=None, dark=False)	70-71	Theme colors.	`dict`
`_ensure_engine` _ensure_engine(style)	74-95	Select/configure the fixed engine once per Tk interpreter.	`id, _configure_fixed_geometry, style.theme_use, style.theme_names`

Callable	Lines	Responsibility	Representative direct calls
`_configure_fixed_geometry` _configure_fixed_geometry(style)	98-296	Defines ttk element layouts, paddings, borders, row heights and widget metrics once.	`style.configure, style.map, _install_fixed_checkbutton_element, style.layout`
`_paint_checkbox_image` _paint_checkbox_image(image, *, fill, border, check=None)	299-314	Repaint one fixed-size checkbox image without changing its dimensions.	`image.put`
`_install_fixed_checkbutton_element` _install_fixed_checkbutton_element(style)	317-367	Creates the custom checkbox indicator element/images used to keep checkbox dimensions stable across palette changes.	`style.layout, repr, tk.PhotoImage, style.element_names, style.element_create`
`_repaint_checkboxes` _repaint_checkboxes(colors, dark)	370-408	Repaint checkboxes.	`_paint_checkbox_image`
`_apply_palette` _apply_palette(root, style, colors, dark)	411-666	Changes ttk colors/state maps and classic Tk colors without changing layout definitions, supporting the invariant that theme switching changes appearance but not geometry.	`style.configure, style.map, _repaint_checkboxes, root.configure`
`apply_theme` apply_theme(root, dark=False, style=None)	669-677	Change palette only.	`_ensure_engine, _apply_palette, ttk.Style, bool`
`apply_light_theme` apply_light_theme(root, style=None)	680-681	Apply light theme.	`apply_theme`
`apply_dark_theme` apply_dark_theme(root, style=None)	684-685	Apply dark theme.	`apply_theme`
`apply_modern_theme` apply_modern_theme(root, dark=False)	688-692	Public startup helper that selects the fixed clam engine and applies either the Light or Dark palette.	`ttk.Style, _ensure_engine, _apply_palette, bool`

Detailed behavior of the key paths

`_configure_fixed_geometry` lines 98-296

Defines ttk element layouts, paddings, borders, row heights and widget metrics once. These values are intentionally independent of Light/Dark selection.

`_install_fixed_checkbutton_element` lines 317-367

Creates the custom checkbox indicator element/images used to keep checkbox dimensions stable across palette changes.

`_apply_palette` lines 411-666

Changes ttk colors/state maps and classic Tk colors without changing layout definitions, supporting the invariant that theme switching changes appearance but not geometry.

`apply_modern_theme` lines 688-692

Public startup helper that selects the fixed clam engine and applies either the Light or Dark palette.

Design note. The theme module is geometry-sensitive: changing colors must not rebuild ttk layouts. This is why theme engine selection and fixed geometry are separated from palette application.

mb_widgets.py

Custom reusable Tkinter widgets. RefreshButton draws a stable square refresh control on a Canvas; ToggleSwitch draws the Light/Dark switch. Both implement their own state, hover/press behavior and palette updates.

Source lines	SHA-256
418	`3a732639b6813ae827290b8ca5af567158b2ee736e941c89ae3d77ea7a184d2f`

Imports: `tkinter`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class RefreshButton`	**7-250**	**Canvas-based square button that draws the ↻ glyph and recentres it from its actual rendered bounding box.**	**Class container**
`RefreshButton.__init__` __init__(self, master, command, size=26, cursor='hand2', **kwargs)	20-53	Init.	`super().__init__, self.bind, self._redraw, self._schedule_final_redraw, super`
`RefreshButton._system_color` _system_color(widget, name, fallback)	56-61	System color.	`widget.winfo_rgb`
`RefreshButton._schedule_final_redraw` _schedule_final_redraw(self)	63-77	Redraw after Tk has settled requested widget dimensions.	`self.after, self.after_cancel`
`RefreshButton._final_redraw` _final_redraw(self)	79-84	Final redraw.	`self._redraw`
`RefreshButton.configure` configure(self, cnf=None, **kwargs)	86-97	Configure.	`kwargs.pop, super().configure, str, self._redraw, self._schedule_final_redraw`
`RefreshButton.cget` cget(self, key)	101-104	Cget.	`super().cget, super`
`RefreshButton.invoke` invoke(self)	106-109	Invoke.	`callable, self._command`
`RefreshButton.set_dark_mode` set_dark_mode(self, enabled)	111-114	Set dark mode.	`bool, self._redraw, self._schedule_final_redraw`
`RefreshButton._palette` _palette(self)	116-146	Palette.	`self._system_color`

Callable	Lines	Responsibility	Representative direct calls
`RefreshButton._center_item_bbox` _center_item_bbox(self, item, width, height)	148-181	Center the visible Tk bbox of one Canvas item on the square.	`range, self.bbox, self.move, round, abs`
`RefreshButton._redraw` _redraw(self, _event=None)	183-220	Redraw.	`self._palette, self.delete, self.create_rectangle, self.create_text, self._center_item_bbox`
`RefreshButton._on_enter` _on_enter(self, _event)	222-225	Event handler for enter.	`self._redraw`
`RefreshButton._on_leave` _on_leave(self, _event)	227-230	Event handler for leave.	`self._redraw`
`RefreshButton._on_press` _on_press(self, _event)	232-235	Event handler for press.	`self._redraw`
`RefreshButton._on_release` _on_release(self, event)	237-250	Event handler for release.	`self._redraw, callable, self._command, self.winfo_width, self.winfo_height`
`class ToggleSwitch`	**254-418**	**Canvas-based boolean switch bound to a Tk variable.**	**Class container**
`ToggleSwitch.__init__` __init__(self, master, variable, command=None, width=42, height=22, cursor='hand2', **kwargs)	257-292	Init.	`super().__init__, self.bind, self.variable.trace_add, self._redraw, self._schedule_final_redraw`
`ToggleSwitch._schedule_final_redraw` _schedule_final_redraw(self)	294-300	Schedules final redraw.	`self.after, self.after_cancel`
`ToggleSwitch._final_redraw` _final_redraw(self)	302-307	Final redraw.	`self._redraw`
`ToggleSwitch.set_dark_mode` set_dark_mode(self, enabled)	309-312	Set dark mode.	`bool, self._redraw, self._schedule_final_redraw`
`ToggleSwitch._parent_bg` _parent_bg(self)	314-321	Parent bg.	`self.winfo_rgb`
`ToggleSwitch._press` _press(self, _event=None)	323-326	Press.	—
`ToggleSwitch._toggle` _toggle(self, _event=None)	328-332	Toggle.	`self.variable.set, callable, self.command, bool, self.variable.get`
`ToggleSwitch._enter` _enter(self, _event=None)	334-336	Enter.	`self._redraw`
`ToggleSwitch._leave` _leave(self, _event=None)	338-340	Leave.	`self._redraw`

Callable	Lines	Responsibility	Representative direct calls
`ToggleSwitch._redraw` _redraw(self, _event=None)	342-418	Redraw.	`bool, self.configure, self.delete, self.create_rectangle, self.create_oval`

Detailed behavior of the key paths

`RefreshButton` lines 7-250

Canvas-based square button that draws the ↻ glyph and recentres it from its actual rendered bounding box. It supports normal/hover/pressed/disabled states and palette changes.

`ToggleSwitch` lines 254-418

Canvas-based boolean switch bound to a Tk variable. It toggles the variable, invokes the callback and redraws track/knob colors while preserving fixed dimensions.

mb_gui.py

Main application composition and global UI behavior. SnifferApp combines MasterMixin, SlaveFinderMixin, CaptureMixin and ViewMixin with tk.Tk. It creates all widgets, initializes state, freezes language-independent geometry, performs runtime translation, manages theme/language persistence, prevents mouse-wheel tab switching and keeps Bus Health content responsive.

Source lines
2664

SHA-256
`7ab64e42fcc446f8f55c212aa4c9b102b1f367a019f34122e080ec6606842b03`

Imports: `queue, threading, collections.deque, tkinter, tkinter.ttk, tkinter.font, mb_config.APP_NAME, APP_VERSION, DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS, DISCOVERY_IDLE_CHECK_SECONDS, MAX_PENDING_REQUEST_TIMEOUT_SECONDS, MAX_RAW_TEXT_LINES, MAX_UI_FRAMES, MIN_PENDING_REQUEST_TIMEOUT_SECONDS, SLOW_RESPONSE_THRESHOLD_MS, SNIFFER_SPLIT_FRAME_HOLD_MS, debug_sim, load_ui_settings, resource_path, save_ui_settings, mb_protocol.SessionMetrics, mb_slave_finder.SlaveFinderMixin, mb_master_ui.MasterMixin, mb_slave_ui.SlaveMixin, mb_capture.CaptureMixin, mb_view.TRAFFIC_MENU_OPEN_IN_MASTER, ViewMixin, mb_widgets.RefreshButton, ToggleSwitch, mb_theme.apply_modern_theme, apply_theme, theme_colors, mb_i18n.LANGUAGE_CHOICES, LANGUAGE_ENGLISH, LANGUAGE_PORTUGUESE, canonical_ui_text, english_help_blocks, normalize_language, translate_text, translate_runtime_text`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class SnifferApp`	**48-2663**	**SnifferApp.**	**Class container**
`SnifferApp.__init__` __init__(self)	49-154	Creates the root window, loads persisted preferences before widget construction, initializes capture/discovery/log/history state, constructs the UI, freezes language geometry, applies language, installs event guards/persistence and schedules queue/status polling.	`super().__init__, self.title, load_ui_settings, normalize_language, bool`
`SnifferApp._freeze_language_layout_geometry` _freeze_language_layout_geometry(self)	157-217	Measures the canonical Portuguese layout and freezes key container dimensions/minimums so changing to English changes text only, not panel/control geometry.	`getattr, self.update_idletasks, enumerate, max, overlay.place_configure`

Callable	Lines	Responsibility	Representative direct calls
`SnifferApp._guard_mouse_wheel` _guard_mouse_wheel(self, event)	219-235	Allows normal wheel scrolling only on truly scrollable widgets and consumes wheel events elsewhere so a ttk Notebook cannot change tabs from mouse-wheel input.	`getattr, isinstance`
`SnifferApp._install_mouse_wheel_guard` _install_mouse_wheel_guard(self)	237-271	Prepend one application bindtag to every existing widget.	`stack.pop, self.bind_class, tuple, stack.extend, widget.bindtags`
`SnifferApp._apply_classic_widget_palette` _apply_classic_widget_palette(self, dark)	273-328	Apply palette to the small set of classic Tk widgets.	`theme_colors, hasattr, self.raw_text.configure, self.help_text.configure, getattr`
`SnifferApp.configure_inspector_result_styles` configure_inspector_result_styles(self)	330-376	Match the Inspector Result foreground to the Traffic result palette.	`theme_colors, self.ui_style.configure, hasattr, bool, tag_map.items`
`SnifferApp.tr` tr(self, text)	378-380	Translate one canonical UI string to the currently selected language.	`translate_text`
`SnifferApp.trf` trf(self, text, **values)	382-384	Translate a format template and substitute values afterwards.	`self.tr(text).format, self.tr`
`SnifferApp._capture_portuguese_help` _capture_portuguese_help(self)	386-404	Store the original PT-PT Help text and tag ranges exactly as built.	`hasattr, self.help_text.get, self.help_text.tag_names, list, self.help_text.tag_ranges`
`SnifferApp._render_help_language` _render_help_language(self)	406-434	Render Help in English or restore the original PT-PT text byte-for-byte.	`hasattr, self.help_text.configure, self.help_text.delete, self.help_text.see, f'{MAX_UI_FRAMES:_}'.replace`
`SnifferApp._translate_widget_tree` _translate_widget_tree(self)	436-468	Translate static widget captions without changing widget geometry.	`stack.pop, isinstance, stack.extend, widget.keys, widget.winfo_children`
`SnifferApp._translate_traffic_context_menu` _translate_traffic_context_menu(self)	470-487	Translate traffic context menu.	`getattr, menu.entryconfigure, self.tr`
`SnifferApp._apply_table_language` _apply_table_language(self)	489-541	Refresh headings and filter values that are not normal widget captions.	`hasattr, canonical_ui_text, self.advanced_filter_type_combo.configure, self.advanced_filter_type_var.set, self.time_order_var.set`

Callable	Lines	Responsibility	Representative direct calls
`SnifferApp.apply_language` apply_language(self, save=True)	543-605	Performs runtime translation of static widgets, tables, context menus, help content and live status strings, then re-applies saved/fixed geometry and optionally persists the selected language.	`getattr, self._translate_widget_tree, self._apply_table_language, self._translate_traffic_context_menu, self._render_help_language`
`SnifferApp.on_language_changed` on_language_changed(self, _event=None)	607-608	Event handler for language changed.	`self.apply_language`
`SnifferApp.toggle_dark_mode` toggle_dark_mode(self)	610-627	Applies the alternate palette using the already-created ttk geometry and updates custom Canvas widgets; the control structure itself is not rebuilt.	`bool, apply_theme, self._apply_classic_widget_palette, self.save_current_ui_settings, self.dark_mode_var.get`
`SnifferApp._restore_saved_main_tab` _restore_saved_main_tab(self)	629-636	Restore saved main tab.	`self.ui_settings.get, self.main_notebook.select, pages.get`
`SnifferApp._current_main_tab_name` _current_main_tab_name(self)	638-649	Current main tab name.	`self.main_notebook.select, str`
`SnifferApp._schedule_ui_settings_save` _schedule_ui_settings_save(self, *_args)	651-668	Schedules ui settings save.	`getattr, self.after, self.after_cancel`
`SnifferApp.save_current_ui_settings` save_current_ui_settings(self)	670-751	Collects validated user-facing preferences from both Sniffer and Finder and writes them to the shared settings file.	`getattr, save_ui_settings, self.after_cancel, int, bool`
`SnifferApp._install_ui_settings_persistence` _install_ui_settings_persistence(self)	753-797	Debounced persistence for all roaming-safe user preferences.	`self.main_notebook.bind, variable.trace_add, mapping.values, self._schedule_ui_settings_save`
`SnifferApp._schedule_bus_health_layout_sync` _schedule_bus_health_layout_sync(self, _event=None)	799-813	Debounce Bus Health reflow after notebook/window/Inspector resizing.	`getattr, self.after_idle, self.after_cancel`
`SnifferApp._initialize_bus_health_layout` _initialize_bus_health_layout(self)	815-837	Measure the natural full Bus Health height before adaptive reflow.	`getattr, self._sync_bus_health_layout, self.update_idletasks, card.winfo_reqheight, max`
`SnifferApp._apply_bus_health_layout` _apply_bus_health_layout(self, compact)	839-982	Re-grids Bus Health labels/values into equal-height rows.	`getattr, bool, range, label_widget.grid_forget, value_widget.grid_forget`

Callable	Lines	Responsibility	Representative direct calls
`SnifferApp._sync_bus_health_layout` _sync_bus_health_layout(self)	984-1037	Chooses full versus compact Bus Health arrangement from real available height with hysteresis, then reapplies wrapping/column behavior on width changes.	`getattr, health_frame.winfo_height, max, bool, health_frame.winfo_exists`
`SnifferApp._build_ui` _build_ui(self)	1039-2663	Constructs all top-level tabs and controls.	`self.ui_settings.get, self.columnconfigure, self.rowconfigure, ttk.Notebook, self.main_notebook.grid`

Detailed behavior of the key paths

`SnifferApp.__init__` lines 49-154

Creates the root window, loads persisted preferences before widget construction, initializes capture/discovery/log/history state, constructs the UI, freezes language geometry, applies language, installs event guards/persistence and schedules queue/status polling.

`_freeze_language_layout_geometry` lines 157-217

Measures the canonical Portuguese layout and freezes key container dimensions/minimums so changing to English changes text only, not panel/control geometry.

`_guard_mouse_wheel` lines 219-235

Allows normal wheel scrolling only on truly scrollable widgets and consumes wheel events elsewhere so a ttk Notebook cannot change tabs from mouse-wheel input.

`apply_language` lines 543-605

Performs runtime translation of static widgets, tables, context menus, help content and live status strings, then re-applies saved/fixed geometry and optionally persists the selected language.

`toggle_dark_mode` lines 610-627

Applies the alternate palette using the already-created ttk geometry and updates custom Canvas widgets; the control structure itself is not rebuilt.

`save_current_ui_settings` lines 670-751

Collects validated user-facing preferences from both Sniffer and Finder and writes them to the shared settings file. COM port names, window geometry and safety confirmation are intentionally excluded.

`_apply_bus_health_layout` **lines 839-982**

Re-grids Bus Health labels/values into equal-height rows. Full mode uses one metric pair per row; compact mode uses two pairs per row and controlled wrapping to keep content inside each rectangle.

`_sync_bus_health_layout` **lines 984-1037**

Chooses full versus compact Bus Health arrangement from real available height with hysteresis, then reapplies wrapping/column behavior on width changes.

`_build_ui` **lines 1039-2663**

Constructs all top-level tabs and controls. It is intentionally a large declarative layout method: global language/theme overlay, Sniffer configuration/actions/statistics, Traffic/Raw/Bus Health/Help notebook, filters/context menu, Frame Inspector and the Bus Slave Finder page.

> **Design note.** SnifferApp uses mixins to keep protocol capture, Finder, Master, Slave and view behavior in separate files while sharing one Tk root and common state.

test_regression.py

Automated regression suite. The tests protect RTU timing, parser/resynchronisation behavior, pairing/timeouts, broadcast and echo-write classification, split-frame reassembly, in-place timeout marking, Slave Finder attempts, idle-bus check and FC43/14 logic, plus reversible Portuguese/English translation. Fake serial classes keep the tests deterministic and hardware-independent.

Source lines	SHA-256
787	`b5a3f7e75c0b385f8cb874fdf3b16c8e867108c59406078d966f333db4dba70c`

Imports: `json, queue, random, tempfile, threading, time, unittest, collections.deque, pathlib.Path, mb_config, mb_slave_finder, mb_capture.CaptureMixin, mb_protocol.SessionMetrics, append_crc, burst_may_continue, crc16_modbus, crc_ok, modbus_rtu_interframe_gap_seconds, parse_burst, read_bits_details, split_capture_buffer, mb_slave_finder.SlaveFinderMixin, mb_view.ViewMixin, mb_diagnostics.inspector_values, mb_i18n.LANGUAGE_ENGLISH, LANGUAGE_PORTUGUESE, english_help_blocks, normalize_language, translate_details, translate_runtime_text, translate_text, mb_slave_finder.build_device_identification_probe, build_discovery_probe, device_identification_timeout_seconds, find_device_identification_response, find_discovery_response, format_device_identification, read_basic_device_identification`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`with_bad_crc` with_bad_crc(frame)	49-52	With bad crc.	`bytearray, bytes`
`device_id_response` device_id_response(slave=1, objects=None, *, more_follows=False, next_object_id=0, conformity=1)	55-83	Device id response.	`bytearray, objects.items, append_crc, bytes, payload.extend`
`class FakeSerial`	**86-116**	**Minimal non-blocking serial fake for active Device ID regression tests.**	**Class container**
`FakeSerial.__init__` __init__(self, response_by_object)	89-92	Init.	`dict, bytearray`
`FakeSerial.in_waiting` in_waiting(self)	95-96	In waiting.	`len`
`FakeSerial.reset_input_buffer` reset_input_buffer(self)	98-99	Reset input buffer.	`self.rx.clear`

Callable	Lines	Responsibility	Representative direct calls
`FakeSerial.reset_output_buffer` reset_output_buffer(self)	101-102	Reset output buffer.	—
`FakeSerial.write` write(self, data)	104-110	Write.	`bytes, self.writes.append, self.response_by_object.get, self.rx.extend, len`
`FakeSerial.read` read(self, size)	112-116	Read.	`min, bytes, int, len`
`class FakeSerialModule`	**119-124**	**FakeSerialModule.**	**Class container**
`class StaticVar`	**127-132**	**StaticVar.**	**Class container**
`StaticVar.__init__` __init__(self, value)	128-129	Init.	—
`StaticVar.get` get(self)	131-132	Get.	—
`class DummyCapture`	**135-144**	**DummyCapture.**	**Class container**
`DummyCapture.__init__` __init__(self, *, mode='Auto', baud='9600', data='8', parity='None', stop='1', manual='2.0')	138-144	Init.	`StaticVar`
`class TimingRegressionTests`	**147-179**	**TimingRegressionTests.**	**Class container**
`TimingRegressionTests.test_t35_uses_character_time_at_or_below_19200` test_t35_uses_character_time_at_or_below_19200(self)	148-158	Regression test: t35 uses character time at or below 19200.	`self.assertAlmostEqual, modbus_rtu_interframe_gap_seconds`
`TimingRegressionTests.test_t35_is_fixed_above_19200` test_t35_is_fixed_above_19200(self)	160-167	Regression test: t35 is fixed above 19200.	`self.subTest, self.assertAlmostEqual, modbus_rtu_interframe_gap_seconds`
`TimingRegressionTests.test_sniffer_auto_gap_uses_fixed_1750_us_above_19200` test_sniffer_auto_gap_uses_fixed_1750_us_above_19200(self)	169-171	Regression test: sniffer auto gap uses fixed 1750 us above 19200.	`DummyCapture, self.assertAlmostEqual, capture.calculated_gap_ms`
`TimingRegressionTests.test_sniffer_manual_gap_is_unchanged` test_sniffer_manual_gap_is_unchanged(self)	173-175	Regression test: sniffer manual gap is unchanged.	`DummyCapture, self.assertAlmostEqual, capture.calculated_gap_ms`
`TimingRegressionTests.test_device_id_timeout_covers_a_full_rtu_frame` test_device_id_timeout_covers_a_full_rtu_frame(self)	177-179	Regression test: device id timeout covers a full rtu frame.	`device_identification_timeout_seconds, self.assertGreaterEqual`
`class ParserRegressionTests`	**182-284**	**ParserRegressionTests.**	**Class container**

Callable	Lines	Responsibility	Representative direct calls
`ParserRegressionTests.test_crc_known_request` test_crc_known_request(self)	183-186	Regression test: crc known request.	`build_discovery_probe, self.assertTrue, self.assertEqual, crc_ok, bytes.fromhex`
`ParserRegressionTests.test_invalid_frame_before_valid_frame_stays_crc_error` test_invalid_frame_before_valid_frame_stays_crc_error(self)	188-193	Regression test: invalid frame before valid frame stays crc error.	`with_bad_crc, build_discovery_probe, split_capture_buffer, self.assertEqual`
`ParserRegressionTests.test_long_unsynchronised_noise_recovers_next_valid_frame` test_long_unsynchronised_noise_recovers_next_valid_frame(self)	195-201	Regression test: long unsynchronised noise recovers next valid frame.	`build_discovery_probe, split_capture_buffer, self.assertEqual, self.assertTrue, bytes`
`ParserRegressionTests.test_supported_function_code_shapes` test_supported_function_code_shapes(self)	203-235	Regression test: supported function code shapes.	`frames.items, append_crc, bytes.fromhex, self.subTest, split_capture_buffer`
`ParserRegressionTests.test_random_noise_never_crashes_and_preserves_all_bytes` test_random_noise_never_crashes_and_preserves_all_bytes(self)	237-244	Regression test: random noise never crashes and preserves all bytes.	`random.Random, range, rng.randint, bytes, split_capture_buffer`
`ParserRegressionTests.test_request_response_pairing_and_response_time` test_request_response_pairing_and_response_time(self)	246-269	Regression test: request response pairing and response time.	`SessionMetrics, metrics.process, self.assertEqual, self.assertAlmostEqual, parse_burst`
`ParserRegressionTests.test_timeout_regression` test_timeout_regression(self)	271-284	Regression test: timeout regression.	`SessionMetrics, metrics.process, self.assertEqual, metrics.pop_expired_requests, parse_burst`
`class SlaveFinderRegressionTests`	**287-369**	**SlaveFinderRegressionTests.**	**Class container**
`SlaveFinderRegressionTests.test_normal_discovery_response` test_normal_discovery_response(self)	288-292	Regression test: normal discovery response.	`append_crc, find_discovery_response, self.assertIsNotNone, self.assertEqual, bytes.fromhex`
`SlaveFinderRegressionTests.test_discovery_exception_counts_as_response` test_discovery_exception_counts_as_response(self)	294-299	Regression test: discovery exception counts as response.	`append_crc, find_discovery_response, self.assertIsNotNone, self.assertEqual, bytes.fromhex`
`SlaveFinderRegressionTests.test_device_identification_request` test_device_identification_request(self)	301-305	Regression test: device identification request.	`build_device_identification_probe, self.assertEqual, self.assertTrue, bytes.fromhex, len`

Callable	Lines	Responsibility	Representative direct calls
`SlaveFinderRegressionTests.test_device_identification_response_parser` test_device_identification_response_parser(self)	307-318	Regression test: device identification response parser.	`device_id_response, find_device_identification_response, self.assertIsNotNone, self.assertEqual, format_device_identification`
`SlaveFinderRegressionTests.test_device_identification_parser_ignores_request_echo` test_device_identification_parser_ignores_request_echo(self)	320-325	Regression test: device identification parser ignores request echo.	`build_device_identification_probe, device_id_response, find_device_identification_response, self.assertIsNotNone, self.assertEqual`
`SlaveFinderRegressionTests.test_device_identification_exception` test_device_identification_exception(self)	327-333	Regression test: device identification exception.	`append_crc, find_device_identification_response, self.assertIsNotNone, self.assertEqual, self.assertIn`
`SlaveFinderRegressionTests.test_device_identification_rejects_bad_crc` test_device_identification_rejects_bad_crc(self)	335-337	Regression test: device identification rejects bad crc.	`with_bad_crc, self.assertIsNone, device_id_response, find_device_identification_response`
`SlaveFinderRegressionTests.test_device_identification_follows_more_follows` test_device_identification_follows_more_follows(self)	339-369	Regression test: device identification follows more follows.	`device_id_response, FakeSerial, read_basic_device_identification, self.assertIsNotNone, self.assertEqual`
`class InternationalizationRegressionTests`	**372-422**	**InternationalizationRegressionTests.**	**Class container**
`InternationalizationRegressionTests.test_language_normalization_defaults_to_portuguese` test_language_normalization_defaults_to_portuguese(self)	373-375	Regression test: language normalization defaults to portuguese.	`self.assertEqual, normalize_language`
`InternationalizationRegressionTests.test_static_ui_translation_is_reversible` test_static_ui_translation_is_reversible(self)	377-383	Regression test: static ui translation is reversible.	`self.assertEqual, translate_text`
`InternationalizationRegressionTests.test_dynamic_status_translation_is_reversible` test_dynamic_status_translation_is_reversible(self)	385-389	Regression test: dynamic status translation is reversible.	`self.assertEqual, translate_runtime_text`

Callable	Lines	Responsibility	Representative direct calls
`InternationalizationRegressionTests.test_register_details_follow_language` test_register_details_follow_language(self)	391-399	Regression test: register details follow language.	`self.assertEqual, translate_details`
`InternationalizationRegressionTests.test_inspector_english_labels_and_crc` test_inspector_english_labels_and_crc(self)	401-411	Regression test: inspector english labels and crc.	`append_crc, row.update, inspector_values, self.assertTrue, self.assertIn`
`InternationalizationRegressionTests.test_english_help_contains_no_portuguese_diacritics` test_english_help_contains_no_portuguese_diacritics(self)	413-422	Regression test: english help contains no portuguese diacritics.	`''.join, self.assertIn, self.assertNotIn, text.casefold, english_help_blocks`
`bitwise_crc16` bitwise_crc16(data)	425-432	Reference CRC used up to v3.0, kept to verify the table-driven CRC.	`range`
`process_frames` process_frames(metrics, frames)	435-442	Run (bytes, channel, epoch) frames through SessionMetrics.	`enumerate, parse_burst, out.append, metrics.process`
`class ProtocolV31RegressionTests`	**445-538**	**ProtocolV31RegressionTests.**	**Class container**
`ProtocolV31RegressionTests.test_crc_table_matches_bitwise_reference` test_crc_table_matches_bitwise_reference(self)	446-450	Regression test: crc table matches bitwise reference.	`random.Random, bytes, self.assertEqual, crc16_modbus, bitwise_crc16`
`ProtocolV31RegressionTests.test_broadcast_request_never_times_out` test_broadcast_request_never_times_out(self)	452-461	Regression test: broadcast request never times out.	`append_crc, SessionMetrics, process_frames, self.assertEqual, metrics.expire_pending`
`ProtocolV31RegressionTests.test_fc06_echo_pair_is_request_then_response` test_fc06_echo_pair_is_request_then_response(self)	463-473	Regression test: fc06 echo pair is request then response.	`append_crc, SessionMetrics, process_frames, self.assertEqual, self.assertAlmostEqual`
`ProtocolV31RegressionTests.test_unanswered_fc05_write_times_out` test_unanswered_fc05_write_times_out(self)	475-480	Regression test: unanswered fc05 write times out.	`append_crc, SessionMetrics, process_frames, self.assertEqual, bytes.fromhex`
`ProtocolV31RegressionTests.test_different_fc06_writes_are_both_requests` test_different_fc06_writes_are_both_requests(self)	482-491	Regression test: different fc06 writes are both requests.	`append_crc, SessionMetrics, process_frames, self.assertEqual, bytes.fromhex`

Callable	Lines	Responsibility	Representative direct calls
`ProtocolV31RegressionTests.test_dual_rx_echo_must_come_from_opposite_channel` test_dual_rx_echo_must_come_from_opposite_channel(self)	493-504	Regression test: dual rx echo must come from opposite channel.	`append_crc, SessionMetrics, process_frames, self.assertEqual, bytes.fromhex`
`ProtocolV31RegressionTests.test_split_frame_is_held_until_complete` test_split_frame_is_held_until_complete(self)	506-511	Regression test: split frame is held until complete.	`append_crc, self.assertTrue, self.assertFalse, burst_may_continue, bytes.fromhex`
`ProtocolV31RegressionTests.test_merged_frames_keep_their_chunk_arrival_times` test_merged_frames_keep_their_chunk_arrival_times(self)	513-527	Regression test: merged frames keep their chunk arrival times.	`append_crc, parse_burst, self.assertEqual, self.assertAlmostEqual, SessionMetrics`
`ProtocolV31RegressionTests.test_read_bits_details_use_request_qty` test_read_bits_details_use_request_qty(self)	529-538	Regression test: read bits details use request qty.	`append_crc, self.assertEqual, bytes.fromhex, read_bits_details`
`class _ChunkedSerial`	**541-568**	**Delivers byte chunks at fixed delays, like a USB latency timer.**	**Class container**
`_ChunkedSerial.__init__` __init__(self, chunks, stop_event)	544-547	Init.	`time.perf_counter, list`
`_ChunkedSerial._due` _due(self)	549-555	Due.	`time.perf_counter, self.stop_event.set`
`_ChunkedSerial.in_waiting` in_waiting(self)	558-559	In waiting.	`len, self._due`
`_ChunkedSerial.read` read(self, size)	561-565	Read.	`self._due, self.chunks.pop`
`_ChunkedSerial.close` close(self)	567-568	Close.	—
`class _ReaderStub`	**571-574**	**_ReaderStub.**	**Class container**
`_ReaderStub.__init__` __init__(self)	572-574	Init.	`threading.Event, queue.Queue`
`class ReaderV31RegressionTests`	**577-591**	**ReaderV31RegressionTests.**	**Class container**
`ReaderV31RegressionTests.test_frame_split_by_usb_latency_is_reassembled` test_frame_split_by_usb_latency_is_reassembled(self)	578-591	Regression test: frame split by usb latency is reassembled.	`append_crc, _ReaderStub, _ChunkedSerial, reader.reader_loop, self.assertEqual`
`class _FakeVar`	**594-599**	**_FakeVar.**	**Class container**

Callable	Lines	Responsibility	Representative direct calls
`_FakeVar.__init__` __init__(self, value)	595-596	Init.	—
`_FakeVar.get` get(self)	598-599	Get.	—
`class _FakeTree`	**602-607**	**_FakeTree.**	**Class container**
`_FakeTree.__init__` __init__(self)	603-604	Init.	—
`_FakeTree.item` item(self, iid, tags=())	606-607	Item.	—
`class _TimeoutView`	**610-627**	**_TimeoutView.**	**Class container**
`_TimeoutView.__init__` __init__(self, type_filter)	611-621	Init.	`_FakeTree, _FakeVar, set, deque`
`_TimeoutView.render_filtered_view` render_filtered_view(self)	623-624	Render filtered view.	—
`_TimeoutView.selected_record` selected_record(self)	626-627	Selected record.	—
`class ViewV31RegressionTests`	**630-657**	**ViewV31RegressionTests.**	**Class container**
`ViewV31RegressionTests.test_timeout_retags_rows_without_full_rebuild` test_timeout_retags_rows_without_full_rebuild(self)	631-636	Regression test: timeout retags rows without full rebuild.	`_TimeoutView, self.assertEqual, view.mark_timed_out_records`
`ViewV31RegressionTests.test_timeouts_filter_still_rebuilds_view` test_timeouts_filter_still_rebuilds_view(self)	638-641	Regression test: timeouts filter still rebuilds view.	`_TimeoutView, view.mark_timed_out_records, self.assertEqual`
`ViewV31RegressionTests.test_inspector_shows_address_of_each_register` test_inspector_shows_address_of_each_register(self)	643-650	Regression test: inspector shows address of each register.	`append_crc, inspector_values, self.assertIn, bytes.fromhex, parse_burst`
`ViewV31RegressionTests.test_inspector_shows_address_of_each_bit` test_inspector_shows_address_of_each_bit(self)	652-657	Regression test: inspector shows address of each bit.	`append_crc, row.update, inspector_values, self.assertIn, bytes.fromhex`
`class _FakeFinderSerial`	**660-685**	**_FakeFinderSerial.**	**Class container**
`_FakeFinderSerial.__init__` __init__(self, incoming=b'')	661-663	Init.	`bytearray`

Callable	Lines	Responsibility	Representative direct calls
`_FakeFinderSerial.in_waiting` in_waiting(self)	666-667	In waiting.	`len`
`_FakeFinderSerial.read` read(self, size)	669-672	Read.	`bytes`
`_FakeFinderSerial.write` write(self, data)	674-676	Write.	`self.writes.append, len, bytes`
`_FakeFinderSerial.reset_input_buffer` reset_input_buffer(self)	678-679	Reset input buffer.	—
`_FakeFinderSerial.reset_output_buffer` reset_output_buffer(self)	681-682	Reset output buffer.	—
`_FakeFinderSerial.close` close(self)	684-685	Close.	—
`class _FinderStub`	**688-714**	**_FinderStub.**	**Class container**
`_FinderStub.__init__` __init__(self, fake_serial)	691-695	Init.	`queue.Queue, threading.Event`
`_FinderStub.tr` tr(self, text)	697-698	Tr.	`translate_text`
`_FinderStub.trf` trf(self, text, **values)	700-701	Trf.	`self.tr(text).format, self.tr`
`_FinderStub.open_discovery_serial_port` open_discovery_serial_port(self, serial, port, baud, parity, stopbits)	703-704	Open discovery serial port.	—
`_FinderStub.run_worker` run_worker(self, slaves, attempts)	706-714	Run worker.	`self.discovery_worker, self.event_queue.empty, events.append, self.event_queue.get`
`class FinderV31RegressionTests`	**717-763**	**FinderV31RegressionTests.**	**Class container**
`FinderV31RegressionTests.setUp` setUp(self)	718-720	SetUp.	—
`FinderV31RegressionTests.tearDown` tearDown(self)	722-723	TearDown.	—
`FinderV31RegressionTests.test_each_silent_slave_gets_all_attempts` test_each_silent_slave_gets_all_attempts(self)	725-730	Regression test: each silent slave gets all attempts.	`_FinderStub, stub.run_worker, self.assertIsNone, self.assertEqual, _FakeFinderSerial`

Callable	Lines	Responsibility	Representative direct calls
`FinderV31RegressionTests.test_traffic_before_scan_blocks_transmission` test_traffic_before_scan_blocks_transmission(self)	732-737	Regression test: traffic before scan blocks transmission.	`_FinderStub, stub.run_worker, self.assertIn, self.assertEqual, _FakeFinderSerial`
`FinderV31RegressionTests.test_legacy_default_timeout_migrates` test_legacy_default_timeout_migrates(self)	739-763	Regression test: legacy default timeout migrates.	`tempfile.TemporaryDirectory, Path, path.write_text, self.assertEqual, json.dumps`
`class SourceHygieneV31Tests`	**766-783**	**SourceHygieneV31Tests.**	**Class container**
`SourceHygieneV31Tests.test_no_inline_language_branches` test_no_inline_language_branches(self)	767-773	Regression test: no inline language branches.	`package.glob, source.read_text, self.assertNotIn, Path(__file__).resolve, Path`
`SourceHygieneV31Tests.test_message_templates_translate` test_message_templates_translate(self)	775-783	Regression test: message templates translate.	`self.assertEqual, translate_text('Encontrados: {count}', LANGUAGE_ENGLISH).format, translate_text`

test_master.py

Master regression tests. An in-memory Modbus slave behind a fake serial port answers reads and writes, so the tests cover data formats, frames from the Modbus specification, response matching, scheduling, writes, broadcast, timeouts, the idle-bus check and workspace validation without hardware.

Source lines
601

SHA-256
`d0c561dd311c1eec097e9f83527b2681d440928ff527f8e0237486ec299f6725`

Imports: `json, os, queue, struct, tempfile, time, unittest, pathlib.Path, mb_config, mb_master_ui, mb_master.BYTE_ORDERS, BYTE_ORDERS_64, format_value, format_window_values, raw_text_for_scaled, register_groups, validate_scale, MasterEngine, PollDefinition, build_read_request, build_write_request, encode_coil_value, encode_register_value, find_master_response, format_register_values, mb_master_ui.MasterMixin, MasterWindow, apply_format_to_rows, mb_protocol.append_crc, crc_ok`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class FakeSlaveSerial`	**33-115**	**In-memory Modbus RTU slave behind a fake serial port.**	**Class container**
`FakeSlaveSerial.__init__` __init__(self, slave_id=1, *, echo=False, bus_traffic=b'', exception_for=None)	41-48	Init.	`bytearray, range`
`FakeSlaveSerial.in_waiting` in_waiting(self)	51-52	In waiting.	`len`
`FakeSlaveSerial.read` read(self, size)	54-57	Read.	`bytes`
`FakeSlaveSerial.reset_input_buffer` reset_input_buffer(self)	59-60	Reset input buffer.	`self.rx.clear`
`FakeSlaveSerial.reset_output_buffer` reset_output_buffer(self)	62-63	Reset output buffer.	—
`FakeSlaveSerial.cancel_read` cancel_read(self)	65-66	Cancel read.	—
`FakeSlaveSerial.cancel_write` cancel_write(self)	68-69	Cancel write.	—

Callable	Lines	Responsibility	Representative direct calls
`FakeSlaveSerial.close` close(self)	71-72	Close.	—
`FakeSlaveSerial.write` write(self, data)	74-83	Write.	`bytes, self.requests.append, len, self.rx.extend, crc_ok`
`FakeSlaveSerial._answer` _answer(self, req)	85-115	Answer.	`append_crc, b''.join, bytearray, range, bytes`
`drain` drain(event_queue)	118-122	Drain.	`event_queue.empty, events.append, event_queue.get`
`make_engine` make_engine(serial, **overrides)	125-129	Make engine.	`dict, options.update, MasterEngine, queue.Queue`
`class MasterFormatTests`	**132-168**	**MasterFormatTests.**	**Class container**
`MasterFormatTests.test_32_bit_formats_round_trip_in_every_byte_order` test_32_bit_formats_round_trip_in_every_byte_order(self)	133-140	Regression test: 32 bit formats round trip in every byte order.	`encode_register_value, self.assertEqual, len, format_register_values`
`MasterFormatTests.test_float_byte_orders_match_wire_layout` test_float_byte_orders_match_wire_layout(self)	142-148	Regression test: float byte orders match wire layout.	`struct.pack, self.assertEqual, encode_register_value, bytes.fromhex`
`MasterFormatTests.test_16_bit_formats` test_16_bit_formats(self)	150-157	Regression test: 16 bit formats.	`self.assertEqual, format_register_values, encode_register_value`
`MasterFormatTests.test_values_outside_the_format_are_rejected` test_values_outside_the_format_are_rejected(self)	159-165	Regression test: values outside the format are rejected.	`self.assertRaises, encode_coil_value, encode_register_value`
`MasterFormatTests.test_unpaired_register_in_32_bit_format` test_unpaired_register_in_32_bit_format(self)	167-168	Regression test: unpaired register in 32 bit format.	`self.assertEqual, format_register_values`
`class MasterFrameTests`	**171-216**	**MasterFrameTests.**	**Class container**
`MasterFrameTests.test_write_frames_match_modbus_specification_examples` test_write_frames_match_modbus_specification_examples(self)	172-182	Regression test: write frames match modbus specification examples.	`self.assertEqual, bytes.fromhex, build_write_request`
`MasterFrameTests.test_invalid_definitions_are_rejected` test_invalid_definitions_are_rejected(self)	184-190	Regression test: invalid definitions are rejected.	`self.assertRaises, build_write_request, build_read_request`

Callable	Lines	Responsibility	Representative direct calls
`MasterFrameTests.test_read_response_found_after_adapter_echo` test_read_response_found_after_adapter_echo(self)	192-197	Regression test: read response found after adapter echo.	`build_read_request, append_crc, find_master_response, self.assertEqual, bytes.fromhex`
`MasterFrameTests.test_coil_response_is_trimmed_to_quantity` test_coil_response_is_trimmed_to_quantity(self)	199-202	Regression test: coil response is trimmed to quantity.	`build_read_request, find_master_response, self.assertEqual, append_crc, bytes.fromhex`
`MasterFrameTests.test_exception_and_bad_crc` test_exception_and_bad_crc(self)	204-210	Regression test: exception and bad crc.	`build_read_request, append_crc, self.assertEqual, bytearray, self.assertIsNone`
`MasterFrameTests.test_write_response_must_match_request` test_write_response_must_match_request(self)	212-216	Regression test: write response must match request.	`build_write_request, self.assertIsNotNone, append_crc, self.assertIsNone, find_master_response`
`class MasterEngineTests`	**219-288**	**MasterEngineTests.**	**Class container**
`MasterEngineTests.run_engine` run_engine(self, engine, seconds)	220-224	Run engine.	`engine.start, time.sleep, engine.stop, drain`
`MasterEngineTests.test_scheduler_serves_writes_then_read_once_then_due_polls` test_scheduler_serves_writes_then_read_once_then_due_polls(self)	226-238	Regression test: scheduler serves writes then read once then due polls.	`make_engine, engine.set_poll, engine.read_once, engine.queue_write, self.assertEqual`
`MasterEngineTests.test_fast_and_slow_windows_share_one_bus` test_fast_and_slow_windows_share_one_bus(self)	240-254	Regression test: fast and slow windows share one bus.	`FakeSlaveSerial, make_engine, engine.set_poll, self.run_engine, self.assertGreater`
`MasterEngineTests.test_write_is_executed_and_reported` test_write_is_executed_and_reported(self)	256-262	Regression test: write is executed and reported.	`FakeSlaveSerial, make_engine, engine.queue_write, self.assertEqual, build_write_request`
`MasterEngineTests.test_broadcast_write_does_not_wait_for_a_response` test_broadcast_write_does_not_wait_for_a_response(self)	264-272	Regression test: broadcast write does not wait for a response.	`FakeSlaveSerial, make_engine, engine.queue_write, time.perf_counter, engine.transact`
`MasterEngineTests.test_timeout_exception_and_invalid_results` test_timeout_exception_and_invalid_results(self)	274-279	Regression test: timeout exception and invalid results.	`make_engine, self.assertEqual, engine.transact, FakeSlaveSerial, build_read_request`
`MasterEngineTests.test_traffic_on_the_bus_blocks_the_master` test_traffic_on_the_bus_blocks_the_master(self)	281-288	Regression test: traffic on the bus blocks the master.	`FakeSlaveSerial, make_engine, engine.set_poll, self.run_engine, self.assertEqual`

Callable	Lines	Responsibility	Representative direct calls
`class MasterWindowTests`	**291-308**	**MasterWindowTests.**	**Class container**
`MasterWindowTests.test_workspace_entry_round_trip_and_validation` test_workspace_entry_round_trip_and_validation(self)	292-300	Regression test: workspace entry round trip and validation.	`MasterWindow, MasterWindow.validated_kwargs, self.assertEqual, window.to_json, self.assertRaises`
`MasterWindowTests.test_write_dialog_builds_fc16_for_float_values` test_write_dialog_builds_fc16_for_float_values(self)	302-308	Regression test: write dialog builds fc16 for float values.	`MasterWindow, MasterMixin.master_build_write, self.assertEqual, bytes.fromhex, self.assertRaises`
`class MasterWorkspaceFolderTests`	**311-385**	**MasterWorkspaceFolderTests.**	**Class container**
`MasterWorkspaceFolderTests.setUp` setUp(self)	312-315	SetUp.	`os.environ.get, tempfile.TemporaryDirectory`
`MasterWorkspaceFolderTests.tearDown` tearDown(self)	317-322	TearDown.	`self._folder.cleanup, os.environ.pop`
`MasterWorkspaceFolderTests.test_workspaces_live_beside_the_settings` test_workspaces_live_beside_the_settings(self)	324-331	Regression test: workspaces live beside the settings.	`mb_config.get_workspaces_folder, self.assertEqual, self.assertTrue, mb_config.get_master_workspaces_folder, folder.is_dir`
`MasterWorkspaceFolderTests.test_loose_master_workspaces_move_into_the_master_folder` test_loose_master_workspaces_move_into_the_master_folder(self)	333-344	Regression test: loose master workspaces move into the master folder.	`mb_config.get_workspaces_folder, (root / 'Linha 1.mbsw').write_text, (root / 'notas.txt').write_text, self.assertEqual, mb_config.get_maste…`
`MasterWorkspaceFolderTests.test_saved_workspaces_are_listed_by_name` test_saved_workspaces_are_listed_by_name(self)	346-351	Regression test: saved workspaces are listed by name.	`(mb_config.get_master_workspaces_folder() / 'notas.txt').write_text, self.assertEqual, mb_master_ui.workspace_path(name).write_text, mb_mas…`
`MasterWorkspaceFolderTests.test_workspace_names_must_be_valid_file_names` test_workspace_names_must_be_valid_file_names(self)	353-357	Regression test: workspace names must be valid file names.	`self.assertEqual, mb_master_ui.validate_workspace_name, self.assertRaises`
`MasterWorkspaceFolderTests.test_mbsniffer_settings_and_workspaces_are_migrated_once` test_mbsniffer_settings_and_workspaces_are_migrated_once(self)	359-375	Regression test: mbsniffer settings and workspaces are migrated once.	`(legacy / 'Workspaces').mkdir, (legacy / 'settings.json').write_text, (legacy / 'Workspaces' / 'Linha 1.mbsw').write_text, mb_config.get_se…`

Callable	Lines	Responsibility	Representative direct calls
`MasterWorkspaceFolderTests.test_legacy_workspace_format_is_accepted` test_legacy_workspace_format_is_accepted(self)	377-379	Regression test: legacy workspace format is accepted.	`self.assertIn, self.assertEqual`
`MasterWorkspaceFolderTests.test_window_title_is_saved_in_the_workspace` test_window_title_is_saved_in_the_workspace(self)	381-385	Regression test: window title is saved in the workspace.	`MasterWindow, self.assertEqual, MasterWindow.validated_kwargs, window.to_json`
`class MasterPerAddressFormatTests`	**388-445**	**MasterPerAddressFormatTests.**	**Class container**
`MasterPerAddressFormatTests.test_64_bit_formats_round_trip_in_every_byte_order` test_64_bit_formats_round_trip_in_every_byte_order(self)	389-396	Regression test: 64 bit formats round trip in every byte order.	`encode_register_value, self.assertEqual, len, format_value`
`MasterPerAddressFormatTests.test_double_byte_orders_match_wire_layout` test_double_byte_orders_match_wire_layout(self)	398-403	Regression test: double byte orders match wire layout.	`self.assertEqual, encode_register_value`
`MasterPerAddressFormatTests.test_per_address_formats_override_the_window_format` test_per_address_formats_override_the_window_format(self)	405-410	Regression test: per address formats override the window format.	`format_window_values, self.assertEqual, register_groups`
`MasterPerAddressFormatTests.test_default_multi_register_values_cut_by_an_override_show_a_dash` test_default_multi_register_values_cut_by_an_override_show_a_dash(self)	412-414	Regression test: default multi register values cut by an override show a dash.	`format_window_values, self.assertEqual`
`MasterPerAddressFormatTests.test_selection_rules_for_multi_register_formats` test_selection_rules_for_multi_register_formats(self)	416-431	Regression test: selection rules for multi register formats.	`MasterWindow, apply_format_to_rows, self.assertEqual, self.assertNotIn, self.assertRaises`
`MasterPerAddressFormatTests.test_formats_are_saved_and_invalid_ones_dropped` test_formats_are_saved_and_invalid_ones_dropped(self)	433-439	Regression test: formats are saved and invalid ones dropped.	`MasterWindow, MasterWindow.validated_kwargs, self.assertEqual, window.to_json`
`MasterPerAddressFormatTests.test_write_uses_the_format_of_the_clicked_value` test_write_uses_the_format_of_the_clicked_value(self)	441-445	Regression test: write uses the format of the clicked value.	`MasterWindow, MasterMixin.master_build_write, self.assertEqual, bytes.fromhex`
`class MasterScaleTests`	**448-487**	**MasterScaleTests.**	**Class container**

Callable	Lines	Responsibility	Representative direct calls
`MasterScaleTests.test_scaled_values_are_shown_in_engineering_units` test_scaled_values_are_shown_in_engineering_units(self)	451-455	Regression test: scaled values are shown in engineering units.	`format_window_values, self.assertEqual`
`MasterScaleTests.test_scale_applies_to_multi_register_values` test_scale_applies_to_multi_register_values(self)	457-461	Regression test: scale applies to multi register values.	`encode_register_value, validate_scale, self.assertEqual, format_window_values`
`MasterScaleTests.test_written_engineering_values_are_converted_to_raw` test_written_engineering_values_are_converted_to_raw(self)	463-470	Regression test: written engineering values are converted to raw.	`self.assertEqual, MasterWindow, MasterMixin.master_build_write, raw_text_for_scaled, bytes.fromhex`
`MasterScaleTests.test_invalid_scales_are_rejected` test_invalid_scales_are_rejected(self)	472-478	Regression test: invalid scales are rejected.	`self.assertRaises, raw_text_for_scaled, validate_scale`
`MasterScaleTests.test_scales_are_saved_and_invalid_ones_dropped` test_scales_are_saved_and_invalid_ones_dropped(self)	480-487	Regression test: scales are saved and invalid ones dropped.	`MasterWindow, MasterWindow.validated_kwargs, self.assertEqual, window.to_json, list`
`class MasterOpenFromTrafficTests`	**490-512**	**MasterOpenFromTrafficTests.**	**Class container**
`MasterOpenFromTrafficTests.definition` definition(self, record)	491-492	Definition.	`MasterMixin.master_definition_from_record`
`MasterOpenFromTrafficTests.test_request_and_matched_response_give_the_read_definition` test_request_and_matched_response_give_the_read_definition(self)	494-503	Regression test: request and matched response give the read definition.	`self.assertEqual, self.definition`
`MasterOpenFromTrafficTests.test_frames_without_a_read_definition_are_refused` test_frames_without_a_read_definition_are_refused(self)	505-512	Regression test: frames without a read definition are refused.	`self.assertIsNone, self.definition`
`class FlakySlaveSerial`	**515-530**	**Fake slave whose port fails (like an unplugged USB adapter).**	**Class container**
`FlakySlaveSerial.__init__` __init__(self, failures, **kwargs)	518-521	Init.	`super().__init__, super`
`FlakySlaveSerial.write` write(self, data)	523-527	Write.	`super().write, OSError, super`

Callable	Lines	Responsibility	Representative direct calls
`FlakySlaveSerial.close` close(self)	529-530	Close.	—
`class MasterReconnectTests`	**533-580**	**MasterReconnectTests.**	**Class container**
`MasterReconnectTests.test_engine_reopens_the_port_after_a_failure` test_engine_reopens_the_port_after_a_failure(self)	534-552	Regression test: engine reopens the port after a failure.	`FlakySlaveSerial, FakeSlaveSerial, make_engine, engine.set_poll, engine.start`
`MasterReconnectTests.test_engine_keeps_retrying_until_stopped` test_engine_keeps_retrying_until_stopped(self)	554-570	Regression test: engine keeps retrying until stopped.	`make_engine, engine.set_poll, engine.start, time.sleep, engine.stop`
`MasterReconnectTests.test_without_reopen_a_failure_ends_the_session` test_without_reopen_a_failure_ends_the_session(self)	572-580	Regression test: without reopen a failure ends the session.	`make_engine, engine.set_poll, engine.start, time.sleep, engine.stop`
`class MasterSettingsTests`	**583-597**	**MasterSettingsTests.**	**Class container**
`MasterSettingsTests.test_last_workspace_is_validated` test_last_workspace_is_validated(self)	584-597	Regression test: last workspace is validated.	`tempfile.TemporaryDirectory, Path, path.write_text, self.assertEqual, json.dumps`

test_slave.py

Slave regression tests. Requests and replies are checked against the store, and the real Master and Slave engines talk over an in-memory serial link, with adapter echo, foreign traffic, split and garbled requests and every fault type.

Source lines	SHA-256
459	`0524c65af1da85131649a145f99868466f14f44e1293d87560edcb73d8256f9c`

Imports: `os, queue, random, tempfile, threading, time, unittest, mb_master.MasterEngine, PollDefinition, build_read_request, build_write_request, find_master_response, mb_protocol.append_crc, crc_ok, mb_slave.SlaveEngine, SlaveStore, corrupt_crc, fault_active, handle_request, request_length, validate_fault, mb_config, mb_i18n.LANGUAGE_ENGLISH, translate_text, mb_slave_ui.SlaveWindow, fault_parts, validate_area, mb_table_ui.list_workspaces, workspace_path`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.4 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class BusEnd`	**32-90**	**One end of an in-memory serial link: what one end writes, the other reads.**	**Class container**
`BusEnd.__init__` __init__(self, echo=False)	39-44	Init.	`bytearray, threading.Lock`
`BusEnd.pair` pair(echo_a=False, echo_b=False)	47-50	Pair.	`BusEnd`
`BusEnd.in_waiting` in_waiting(self)	53-55	In waiting.	`len`
`BusEnd.read` read(self, size)	57-61	Read.	`bytes`
`BusEnd.write` write(self, data)	63-70	Write.	`bytes, self.written.append, len, self.inject, self.peer.inject`
`BusEnd.inject` inject(self, data)	72-74	Inject.	`self.rx.extend`
`BusEnd.reset_input_buffer` reset_input_buffer(self)	76-78	Reset input buffer.	`self.rx.clear`
`BusEnd.reset_output_buffer` reset_output_buffer(self)	80-81	Reset output buffer.	—

Callable	Lines	Responsibility	Representative direct calls
`BusEnd.cancel_read` cancel_read(self)	83-84	Cancel read.	—
`BusEnd.cancel_write` cancel_write(self)	86-87	Cancel write.	—
`BusEnd.close` close(self)	89-90	Close.	—
`make_store` make_store()	93-101	Make store.	`SlaveStore, store.set_areas, store.write, range`
`drain` drain(event_queue)	104-108	Drain.	`event_queue.empty, events.append, event_queue.get`
`wait_for` wait_for(predicate, timeout=2.0)	111-117	Wait for.	`predicate, time.monotonic, time.sleep`
`class SlaveRequestTests`	**120-204**	**SlaveRequestTests.**	**Class container**
`SlaveRequestTests.test_reads_answer_from_the_tables` test_reads_answer_from_the_tables(self)	121-135	Regression test: reads answer from the tables.	`make_store, handle_request, self.assertEqual, find_master_response, build_read_request`
`SlaveRequestTests.test_writes_change_the_tables_and_answer_like_a_device` test_writes_change_the_tables_and_answer_like_a_device(self)	137-153	Regression test: writes change the tables and answer like a device.	`make_store, build_write_request, handle_request, self.assertEqual, store.read`
`SlaveRequestTests.test_exceptions_like_a_real_device` test_exceptions_like_a_real_device(self)	155-170	Regression test: exceptions like a real device.	`make_store, handle_request, self.assertEqual, build_read_request, build_write_request`
`SlaveRequestTests.test_other_slave_ids_are_ignored` test_other_slave_ids_are_ignored(self)	172-175	Regression test: other slave ids are ignored.	`handle_request, self.assertEqual, self.assertIsNone, make_store, build_read_request`
`SlaveRequestTests.test_broadcast_writes_every_covering_slave_without_answering` test_broadcast_writes_every_covering_slave_without_answering(self)	177-189	Regression test: broadcast writes every covering slave without answering.	`make_store, handle_request, self.assertEqual, self.assertIsNone, build_write_request`
`SlaveRequestTests.test_request_length` test_request_length(self)	191-196	Regression test: request length.	`self.assertIsNone, self.assertEqual, request_length, build_read_request, build_write_request`
`SlaveRequestTests.test_window_areas_define_the_slave` test_window_areas_define_the_slave(self)	198-204	Regression test: window areas define the slave.	`SlaveWindow, SlaveStore, store.set_areas, self.assertEqual, self.assertTrue`

Callable	Lines	Responsibility	Representative direct calls
`class SlaveFaultTests`	**207-230**	**SlaveFaultTests.**	**Class container**
`SlaveFaultTests.test_fault_validation` test_fault_validation(self)	208-217	Regression test: fault validation.	`self.assertFalse, validate_fault, self.assertTrue, self.assertEqual, fault_active`
`SlaveFaultTests.test_inactive_faults_are_not_stored` test_inactive_faults_are_not_stored(self)	219-226	Regression test: inactive faults are not stored.	`make_store, store.set_fault, self.assertIsNone, self.assertTrue, self.assertEqual`
`SlaveFaultTests.test_corrupt_crc` test_corrupt_crc(self)	228-230	Regression test: corrupt crc.	`build_read_request, self.assertFalse, crc_ok, corrupt_crc`
`class SlaveEngineTests`	**233-381**	**The Master engine talking to the Slave engine over an in-memory bus.**	**Class container**
`SlaveEngineTests.setUp` setUp(self)	236-242	SetUp.	`make_store, BusEnd.pair, queue.Queue`
`SlaveEngineTests.tearDown` tearDown(self)	244-247	TearDown.	`engine.stop`
`SlaveEngineTests.start_slave` start_slave(self, **options)	249-252	Start slave.	`SlaveEngine, self.slave.start`
`SlaveEngineTests.start_master` start_master(self, timeout_ms=200)	254-258	Start master.	`MasterEngine, self.master.start`
`SlaveEngineTests.master_results` master_results(self, count, timeout=3.0)	260-270	Master results.	`time.monotonic, len, self.master_events.get, results.append`
`SlaveEngineTests.test_master_reads_and_writes_every_function` test_master_reads_and_writes_every_function(self)	272-292	Regression test: master reads and writes every function.	`self.start_slave, self.start_master, enumerate, self.assertEqual, self.assertIn`
`SlaveEngineTests.test_exception_reaches_the_master` test_exception_reaches_the_master(self)	294-299	Regression test: exception reaches the master.	`self.start_slave, self.start_master, self.master.set_poll, self.assertEqual, PollDefinition`
`SlaveEngineTests.test_adapter_echo_and_foreign_traffic_are_ignored` test_adapter_echo_and_foreign_traffic_are_ignored(self)	301-318	Regression test: adapter echo and foreign traffic are ignored.	`self.start_slave, self.slave_end.inject, self.start_master, self.master.set_poll, self.master_results`
`SlaveEngineTests.test_split_request_is_assembled` test_split_request_is_assembled(self)	320-328	Regression test: split request is assembled.	`self.start_slave, build_read_request, self.slave_end.inject, time.sleep, self.assertTrue`

Callable	Lines	Responsibility	Representative direct calls
`SlaveEngineTests.test_garbage_before_a_request_is_dropped` test_garbage_before_a_request_is_dropped(self)	330-335	Regression test: garbage before a request is dropped.	`self.start_slave, build_read_request, self.slave_end.inject, self.assertTrue, self.assertEqual`
`SlaveEngineTests.test_unknown_function_gets_exception_01_after_silence` test_unknown_function_gets_exception_01_after_silence(self)	337-341	Regression test: unknown function gets exception 01 after silence.	`self.start_slave, self.slave_end.inject, self.assertTrue, self.assertEqual, append_crc`
`SlaveEngineTests.test_faults` test_faults(self)	343-368	Regression test: faults.	`self.start_slave, self.start_master, self.master.set_poll, self.store.set_fault, self.master.read_once`
`SlaveEngineTests.test_fault_probability` test_fault_probability(self)	370-381	Regression test: fault probability.	`make_store, store.set_fault, SlaveEngine, range, statuses.count`
`class SlaveWindowTests`	**384-418**	**SlaveWindowTests.**	**Class container**
`SlaveWindowTests.test_workspace_entry_round_trip_and_validation` test_workspace_entry_round_trip_and_validation(self)	385-396	Regression test: workspace entry round trip and validation.	`SlaveWindow, SlaveWindow.validated_kwargs, self.assertEqual, validate_area, window.to_json`
`SlaveWindowTests.test_table_field_shows_the_read_function_code` test_table_field_shows_the_read_function_code(self)	398-404	Regression test: table field shows the read function code.	`self.assertEqual, table_choice_label`
`SlaveWindowTests.test_window_overlap` test_window_overlap(self)	406-411	Regression test: window overlap.	`SlaveWindow, self.assertEqual, self.assertIsNone, window.overlap`
`SlaveWindowTests.test_fault_description_translates` test_fault_description_translates(self)	413-418	Regression test: fault description translates.	`validate_fault, self.assertEqual, translate_text(template, LANGUAGE_ENGLISH).format, translate_text, fault_parts`
`class SlaveSettingsTests`	**421-455**	**SlaveSettingsTests.**	**Class container**
`SlaveSettingsTests.setUp` setUp(self)	422-425	SetUp.	`os.environ.get, tempfile.TemporaryDirectory`
`SlaveSettingsTests.tearDown` tearDown(self)	427-432	TearDown.	`self._folder.cleanup, os.environ.pop`
`SlaveSettingsTests.test_slave_workspaces_have_their_own_folder` test_slave_workspaces_have_their_own_folder(self)	434-440	Regression test: slave workspaces have their own folder.	`mb_config.get_slave_workspaces_folder, self.assertEqual, workspace_path('Banco', folder).write_text, mb_config.get_workspaces_folder, list_…`

Callable	Lines	Responsibility	Representative direct calls
`SlaveSettingsTests.test_slave_settings_are_validated` test_slave_settings_are_validated(self)	442-455	Regression test: slave settings are validated.	`mb_config.save_ui_settings`, `mb_config.load_ui_settings`, `self.assertEqual`

13. Regression-test suite

The documented package was executed with `python -m unittest discover -s tests -p "test_*.py" -v`. All 113 tests passed.

> Verification result: **113 tests run - 113 passed - 0 failures/errors**.

TimingRegressionTests

Protects RTU timing and automatic/manual gap calculations, including the fixed 1.750 ms t3.5 rule above 19200 bit/s and FC43 response timing.

Test	Lines	What it protects
`test_t35_uses_character_time_at_or_below_19200`	148-158	Regression test: t35 uses character time at or below 19200.
`test_t35_is_fixed_above_19200`	160-167	Regression test: t35 is fixed above 19200.
`test_sniffer_auto_gap_uses_fixed_1750_us_above_19200`	169-171	Regression test: sniffer auto gap uses fixed 1750 us above 19200.
`test_sniffer_manual_gap_is_unchanged`	173-175	Regression test: sniffer manual gap is unchanged.
`test_device_id_timeout_covers_a_full_rtu_frame`	177-179	Regression test: device id timeout covers a full rtu frame.

ParserRegressionTests

Protects CRC, resynchronisation, supported frame shapes, random-noise robustness, request/response pairing and timeout behavior.

Test	Lines	What it protects
`test_crc_known_request`	183-186	Regression test: crc known request.
`test_invalid_frame_before_valid_frame_stays_crc_error`	188-193	Regression test: invalid frame before valid frame stays crc error.
`test_long_unsynchronised_noise_recovers_next_valid_frame`	195-201	Regression test: long unsynchronised noise recovers next valid frame.
`test_supported_function_code_shapes`	203-235	Regression test: supported function code shapes.

Test	Lines	What it protects
`test_random_noise_never_crashes_and_preserves_all_bytes`	237-244	Regression test: random noise never crashes and preserves all bytes.
`test_request_response_pairing_and_response_time`	246-269	Regression test: request response pairing and response time.
`test_timeout_regression`	271-284	Regression test: timeout regression.

SlaveFinderRegressionTests

Protects FC03/FC04 discovery and FC43/14 Device Identification parsing, CRC validation, Exceptions, echo rejection and More Follows paging.

Test	Lines	What it protects
`test_normal_discovery_response`	288-292	Regression test: normal discovery response.
`test_discovery_exception_counts_as_response`	294-299	Regression test: discovery exception counts as response.
`test_device_identification_request`	301-305	Regression test: device identification request.
`test_device_identification_response_parser`	307-318	Regression test: device identification response parser.
`test_device_identification_parser_ignores_request_echo`	320-325	Regression test: device identification parser ignores request echo.
`test_device_identification_exception`	327-333	Regression test: device identification exception.
`test_device_identification_rejects_bad_crc`	335-337	Regression test: device identification rejects bad crc.
`test_device_identification_follows_more_follows`	339-369	Regression test: device identification follows more follows.

InternationalizationRegressionTests

Protects language normalization, reversible static/runtime translation, translated register details, Frame Inspector English text and English Help content.

Test	Lines	What it protects
`test_language_normalization_defaults_to_portuguese`	373-375	Regression test: language normalization defaults to portuguese.

Test	Lines	What it protects
`test_static_ui_translation_is_reversible`	377-383	Regression test: static ui translation is reversible.
`test_dynamic_status_translation_is_reversible`	385-389	Regression test: dynamic status translation is reversible.
`test_register_details_follow_language`	391-399	Regression test: register details follow language.
`test_inspector_english_labels_and_crc`	401-411	Regression test: inspector english labels and crc.
`test_english_help_contains_no_portuguese_diacritics`	413-422	Regression test: english help contains no portuguese diacritics.

ProtocolV31RegressionTests

Protects the v3.1 protocol fixes: table CRC against the bit-wise reference, broadcast accounting, FC05/06 echo pairing, dual-RX direction, split-burst detection, per-read timestamps and FC01/02 bit formatting.

Test	Lines	What it protects
`test_crc_table_matches_bitwise_reference`	446-450	Regression test: crc table matches bitwise reference.
`test_broadcast_request_never_times_out`	452-461	Regression test: broadcast request never times out.
`test_fc06_echo_pair_is_request_then_response`	463-473	Regression test: fc06 echo pair is request then response.
`test_unanswered_fc05_write_times_out`	475-480	Regression test: unanswered fc05 write times out.
`test_different_fc06_writes_are_both_requests`	482-491	Regression test: different fc06 writes are both requests.
`test_dual_rx_echo_must_come_from_opposite_channel`	493-504	Regression test: dual rx echo must come from opposite channel.
`test_split_frame_is_held_until_complete`	506-511	Regression test: split frame is held until complete.
`test_merged_frames_keep_their_chunk_arrival_times`	513-527	Regression test: merged frames keep their chunk arrival times.
`test_read_bits_details_use_request_qty`	529-538	Regression test: read bits details use request qty.

ReaderV31RegressionTests

Feeds a frame split by a simulated 16 ms USB latency into the real reader_loop() and requires one CRC-valid RESPONSE.

Test	Lines	What it protects
`test_frame_split_by_usb_latency_is_reassembled`	578-591	Regression test: frame split by usb latency is reassembled.

ViewV31RegressionTests

Protects in-place timeout retagging, the Timeouts-filter rebuild and the address-mapped register/bit data in the Frame Inspector.

Test	Lines	What it protects
`test_timeout_retags_rows_without_full_rebuild`	631-636	Regression test: timeout retags rows without full rebuild.
`test_timeouts_filter_still_rebuilds_view`	638-641	Regression test: timeouts filter still rebuilds view.
`test_inspector_shows_address_of_each_register`	643-650	Regression test: inspector shows address of each register.
`test_inspector_shows_address_of_each_bit`	652-657	Regression test: inspector shows address of each bit.

FinderV31RegressionTests

Runs the real discovery worker against fake serial ports to check attempts, the idle-bus abort and the 40 to 100 ms settings migration.

Test	Lines	What it protects
`test_each_silent_slave_gets_all_attempts`	725-730	Regression test: each silent slave gets all attempts.
`test_traffic_before_scan_blocks_transmission`	732-737	Regression test: traffic before scan blocks transmission.
`test_legacy_default_timeout_migrates`	739-763	Regression test: legacy default timeout migrates.

SourceHygieneV31Tests

Rejects inline language branches in the package and checks that message templates translate.

Test	Lines	What it protects
`test_no_inline_language_branches`	767-773	Regression test: no inline language branches.
`test_message_templates_translate`	775-783	Regression test: message templates translate.

MasterFormatTests

Protects the 16/32-bit data formats, the four byte orders and the rejection of values that do not fit.

Test	Lines	What it protects
`test_32_bit_formats_round_trip_in_every_byte_order`	133-140	Regression test: 32 bit formats round trip in every byte order.
`test_float_byte_orders_match_wire_layout`	142-148	Regression test: float byte orders match wire layout.
`test_16_bit_formats`	150-157	Regression test: 16 bit formats.
`test_values_outside_the_format_are_rejected`	159-165	Regression test: values outside the format are rejected.
`test_unpaired_register_in_32_bit_format`	167-168	Regression test: unpaired register in 32 bit format.

MasterFrameTests

Checks write frames against the Modbus specification examples, definition limits, adapter echoes, exceptions, bad CRC and write response matching.

Test	Lines	What it protects
`test_write_frames_match_modbus_specification_examples`	172-182	Regression test: write frames match modbus specification examples.
`test_invalid_definitions_are_rejected`	184-190	Regression test: invalid definitions are rejected.
`test_read_response_found_after_adapter_echo`	192-197	Regression test: read response found after adapter echo.
`test_coil_response_is_trimmed_to_quantity`	199-202	Regression test: coil response is trimmed to quantity.
`test_exception_and_bad_crc`	204-210	Regression test: exception and bad crc.
`test_write_response_must_match_request`	212-216	Regression test: write response must match request.

MasterEngineTests

Runs MasterEngine against an in-memory slave: scheduling order, fast and slow windows on one bus, writes, broadcast, timeouts, exceptions and the idle-bus abort.

Test	Lines	What it protects
`test_scheduler_serves_writes_then_read_once_then_due_polls`	226-238	Regression test: scheduler serves writes then read once then due polls.

Test	Lines	What it protects
`test_fast_and_slow_windows_share_one_bus`	240-254	Regression test: fast and slow windows share one bus.
`test_write_is_executed_and_reported`	256-262	Regression test: write is executed and reported.
`test_broadcast_write_does_not_wait_for_a_response`	264-272	Regression test: broadcast write does not wait for a response.
`test_timeout_exception_and_invalid_results`	274-279	Regression test: timeout exception and invalid results.
`test_traffic_on_the_bus_blocks_the_master`	281-288	Regression test: traffic on the bus blocks the master.

MasterWindowTests

Protects workspace entry validation and the FC16 float encoding of the write dialog.

Test	Lines	What it protects
`test_workspace_entry_round_trip_and_validation`	292-300	Regression test: workspace entry round trip and validation.
`test_write_dialog_builds_fc16_for_float_values`	302-308	Regression test: write dialog builds fc16 for float values.

MasterWorkspaceFolderTests

Checks that workspaces live beside settings.json, are listed alphabetically and use valid Windows file names, and that window names are saved.

Test	Lines	What it protects
`test_workspaces_live_beside_the_settings`	324-331	Regression test: workspaces live beside the settings.
`test_loose_master_workspaces_move_into_the_master_folder`	333-344	Regression test: loose master workspaces move into the master folder.
`test_saved_workspaces_are_listed_by_name`	346-351	Regression test: saved workspaces are listed by name.
`test_workspace_names_must_be_valid_file_names`	353-357	Regression test: workspace names must be valid file names.
`test_mbsniffer_settings_and_workspaces_are_migrated_once`	359-375	Regression test: mbsniffer settings and workspaces are migrated once.
`test_legacy_workspace_format_is_accepted`	377-379	Regression test: legacy workspace format is accepted.

Test	Lines	What it protects
`test_window_title_is_saved_in_the_workspace`	381-385	Regression test: window title is saved in the workspace.

MasterPerAddressFormatTests

Protects the 64-bit formats and byte orders, per-address formats, value grouping, the selection rules and the write encoding of the clicked value.

Test	Lines	What it protects
`test_64_bit_formats_round_trip_in_every_byte_order`	389-396	Regression test: 64 bit formats round trip in every byte order.
`test_double_byte_orders_match_wire_layout`	398-403	Regression test: double byte orders match wire layout.
`test_per_address_formats_override_the_window_format`	405-410	Regression test: per address formats override the window format.
`test_default_multi_register_values_cut_by_an_override_show_a_dash`	412-414	Regression test: default multi register values cut by an override show a dash.
`test_selection_rules_for_multi_register_formats`	416-431	Regression test: selection rules for multi register formats.
`test_formats_are_saved_and_invalid_ones_dropped`	433-439	Regression test: formats are saved and invalid ones dropped.
`test_write_uses_the_format_of_the_clicked_value`	441-445	Regression test: write uses the format of the clicked value.

MasterScaleTests

Protects scaled display, multi-register scaling, the conversion of written engineering values and scale validation.

Test	Lines	What it protects
`test_scaled_values_are_shown_in_engineering_units`	451-455	Regression test: scaled values are shown in engineering units.
`test_scale_applies_to_multi_register_values`	457-461	Regression test: scale applies to multi register values.
`test_written_engineering_values_are_converted_to_raw`	463-470	Regression test: written engineering values are converted to raw.
`test_invalid_scales_are_rejected`	472-478	Regression test: invalid scales are rejected.

Test	Lines	What it protects
`test_scales_are_saved_and_invalid_ones_dropped`	480-487	Regression test: scales are saved and invalid ones dropped.

MasterOpenFromTrafficTests

Checks which Traffic records can open a Master window and with which definition.

Test	Lines	What it protects
`test_request_and_matched_response_give_the_read_definition`	494-503	Regression test: request and matched response give the read definition.
`test_frames_without_a_read_definition_are_refused`	505-512	Regression test: frames without a read definition are refused.

MasterReconnectTests

Simulates an unplugged adapter: the engine reopens the port and resumes, keeps retrying until stopped, or ends without reopen.

Test	Lines	What it protects
`test_engine_reopens_the_port_after_a_failure`	534-552	Regression test: engine reopens the port after a failure.
`test_engine_keeps_retrying_until_stopped`	554-570	Regression test: engine keeps retrying until stopped.
`test_without_reopen_a_failure_ends_the_session`	572-580	Regression test: without reopen a failure ends the session.

MasterSettingsTests

Validates the last workspace name stored in settings.json.

Test	Lines	What it protects
`test_last_workspace_is_validated`	584-597	Regression test: last workspace is validated.

SlaveRequestTests

Checks reads and writes of every table, Exceptions 01/02/03, ignored IDs, broadcast writes and request lengths.

Test	Lines	What it protects
`test_reads_answer_from_the_tables`	121-135	Regression test: reads answer from the tables.

Test	Lines	What it protects
`test_writes_change_the_tables_and_answer_like_a_device`	137-153	Regression test: writes change the tables and answer like a device.
`test_exceptions_like_a_real_device`	155-170	Regression test: exceptions like a real device.
`test_other_slave_ids_are_ignored`	172-175	Regression test: other slave ids are ignored.
`test_broadcast_writes_every_covering_slave_without_answering`	177-189	Regression test: broadcast writes every covering slave without answering.
`test_request_length`	191-196	Regression test: request length.
`test_window_areas_define_the_slave`	198-204	Regression test: window areas define the slave.

SlaveFaultTests

Validates fault settings, keeps only active faults and corrupts the CRC.

Test	Lines	What it protects
`test_fault_validation`	208-217	Regression test: fault validation.
`test_inactive_faults_are_not_stored`	219-226	Regression test: inactive faults are not stored.
`test_corrupt_crc`	228-230	Regression test: corrupt crc.

SlaveEngineTests

Runs the Master engine against the Slave engine over an in-memory link: every function, exceptions, adapter echo, foreign traffic, split and garbled requests, unknown functions and each fault, including the probability.

Test	Lines	What it protects
`test_master_reads_and_writes_every_function`	272-292	Regression test: master reads and writes every function.
`test_exception_reaches_the_master`	294-299	Regression test: exception reaches the master.
`test_adapter_echo_and_foreign_traffic_are_ignored`	301-318	Regression test: adapter echo and foreign traffic are ignored.
`test_split_request_is_assembled`	320-328	Regression test: split request is assembled.
`test_garbage_before_a_request_is_dropped`	330-335	Regression test: garbage before a request is dropped.

Test	Lines	What it protects
`test_unknown_function_gets_exception_01_after_silence`	337-341	Regression test: unknown function gets exception 01 after silence.
`test_faults`	343-368	Regression test: faults.
`test_fault_probability`	370-381	Regression test: fault probability.

SlaveWindowTests

Protects Slave workspace entries, area validation, window overlap and the translated fault description.

Test	Lines	What it protects
`test_workspace_entry_round_trip_and_validation`	385-396	Regression test: workspace entry round trip and validation.
`test_table_field_shows_the_read_function_code`	398-404	Regression test: table field shows the read function code.
`test_window_overlap`	406-411	Regression test: window overlap.
`test_fault_description_translates`	413-418	Regression test: fault description translates.

SlaveSettingsTests

Checks the Slave workspace folder and the validated slave section of settings.json.

Test	Lines	What it protects
`test_slave_workspaces_have_their_own_folder`	434-440	Regression test: slave workspaces have their own folder.
`test_slave_settings_are_validated`	442-455	Regression test: slave settings are validated.

14. Build and launch automation

The package contains three Windows batch files. Their complete source is reproduced in Appendix A.

MBToolkit.bat

Windows batch automation included in the package.

Lines	SHA-256
210	`cae8a4af4a86a76b01b46ee66788a021c702b2143225ba946cf14294bff4bb78`

- Sets the expected versioned source folder and verifies that MBToolkit.py exists.
- Detects Python 3 through py -3, python on PATH, or standard Python 3.13 install locations.
- If Python is absent and WinGet exists, attempts a silent per-user Python 3.13 installation.
- Verifies tkinter; if missing, attempts a Python repair/reinstall.
- Verifies pyserial and installs it with pip/ensurepip when required.
- Prefers pythonw.exe so the GUI opens without a console window, with python as fallback.

MBToolkit v3.4/build_exe.bat

Windows batch automation included in the package.

Lines	SHA-256
151	`08d6ba775bd6986cdcc8a130712788b60a00c0fa0793deec076fedd01045c331`

- Normalizes application/distribution paths and finds Python 3.
- Installs/upgrades pyserial and PyInstaller.
- Runs the complete regression suite before building; any failure cancels the build.
- Cleans prior build/spec/output artifacts.
- Creates a one-file, windowed MBToolkit.exe with the icon embedded when available.

MBToolkit v3.4/run_tests.bat

Windows batch automation included in the package.

Lines	SHA-256
45	`98c20bdbf89f4f478c4b38b7c8f5a55fae834f328edaf7d2979c8823f2236eae`

- Finds a Python 3 interpreter.
- Runs unittest discovery for tests/test_*.py in verbose mode.
- Returns a non-zero exit code and explicit warning when tests fail, so a release should not proceed.

15. Known boundaries and extension points

What the current v3.4 does not implement

- No Modbus TCP transport.
- Master tab: no data logging, conditional colours, charts, Test Center, FC22/23 or OLE-style automation.
- Slave tab is phase 1: no FC08/FC22/FC23/FC43, no automatically changing values (counters, ramps), no traffic view of its own and no Modbus TCP server.
- No passive Ethernet packet capture.
- No analog RS485 electrical measurement.

Current parser scope

Known frame-shape recognition covers the Function Codes encoded in `_candidate_prefixes_at()`. An unsupported FC can only be accepted conservatively as `UNKNOWN / OK` when it occupies the complete remaining burst and has a valid CRC. This intentionally favors resynchronisation reliability over aggressive guessing.

Natural refactoring seams

- **Transport abstraction:** `mb_protocol.py` is already mostly transport-independent above RTU framing, making it the natural starting point for a future shared Modbus PDU core.
- **Active client engine:** the Finder already contains robust serial request/write/read/timeout primitives that could inform a future general Master engine, but the thesis-quality design should separate those primitives from the Finder UI.
- **Server simulator:** `handle_request()` is transport-independent; a Modbus TCP server would only need a new loop around it.
- **Testing:** the existing hardware-independent regression suite is a strong base; future transport/client/server work should extend the same pattern.

> **Documentation intent.** These are architectural extension points inferred from the current code structure, not features claimed to exist in v3.4.

Appendix A - Complete source-code screenshots

Every Python and batch source file in the package is reproduced below as syntax-highlighted screenshots with original line numbers. Very long source lines are visually wrapped inside the screenshot for readability; this does not alter the source file. The SHA-256 values in the module chapters identify the exact files represented.

MBToolkit v3.4/MBToolkit.py

29 source lines.

MBToolkit v3.4/MBToolkit.py — lines 1-29

```python
#!/usr/bin/env python3
# MBToolkit v3.4
"""MBToolkit launcher."""

# Simulação:
#   0 = simulação inativa (o botão não aparece no GUI)
#   1 = simulação ativa  (o botão aparece no GUI)
#
# Mantido neste ficheiro para preservar o mesmo ponto de configuração usado
# antes da modularização.
debug_sim = 0

import mb_config

# Apply the launcher switch before importing GUI/capture modules, which read
# the value during import.
mb_config.debug_sim = debug_sim

from mb_config import set_windows_app_user_model_id
from mb_gui import SnifferApp


def main():
    set_windows_app_user_model_id()
    SnifferApp().mainloop()


if __name__ == "__main__":
    main()
```

MBToolkit v3.4/mb_config.py

577 source lines.

MBToolkit v3.4/mb_config.py — lines 1-40

```python
#!/usr/bin/env python3
"""MBToolkit application configuration, constants and filesystem helpers."""

import ctypes
import json
import os
import shutil
import sys
from pathlib import Path

APP_NAME = "MBToolkit"
# Product name up to v3.2. Its settings folder is migrated on first start.
LEGACY_APP_NAME = "MBSniffer"
APP_VERSION = "3.4"

# Simulação:
#   0 = simulação inativa (o botão não aparece no GUI)
#   1 = simulação ativa  (o botão aparece no GUI)
debug_sim = 0

# Active Modbus RTU slave discovery.
DISCOVERY_BAUD_RATES = (1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200)
DISCOVERY_PARITIES = ("None", "Even", "Odd")
DISCOVERY_STOP_BITS = ("1", "2")
DISCOVERY_DEFAULT_MIN_TIMEOUT_MS = 100
# v3.0 default; settings saved by v3.0 with this value migrate to the new default.
DISCOVERY_LEGACY_DEFAULT_MIN_TIMEOUT_MS = 40
DISCOVERY_MIN_TIMEOUT_MS = 10
DISCOVERY_MAX_TIMEOUT_MS = 2000
DISCOVERY_PROBE_ADDRESS = 0
DISCOVERY_PROBE_QTY = 1
DISCOVERY_WRITE_TIMEOUT_SECONDS = 1.5
DISCOVERY_WRITE_RETRIES = 1
DISCOVERY_RX_BUFFER_LIMIT = 1024
# Total probe attempts per FC when a Slave does not answer.
DISCOVERY_MIN_ATTEMPTS = 1
DISCOVERY_MAX_ATTEMPTS = 3
DISCOVERY_DEFAULT_ATTEMPTS = 1
# Passive listen before the first transmission; any byte aborts the scan.
DISCOVERY_IDLE_CHECK_SECONDS = 1.0
```

MBToolkit v3.4/mb_config.py — lines 41-80

```
41
42 # GUI queue processing limits.
43 UI_QUEUE_BATCH_SIZE = 200
44 UI_QUEUE_IDLE_INTERVAL_MS = 50
45 UI_QUEUE_BUSY_INTERVAL_MS = 10
46
47 # Pending-request timeout.
48 MIN_PENDING_REQUEST_TIMEOUT_SECONDS = 1.0
49 DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS = 10.0
50 MAX_PENDING_REQUEST_TIMEOUT_SECONDS = 300.0
51 PENDING_REQUEST_TIMEOUT_SECONDS = DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS
52
53 # Master (poll) tab. Modbus limits per request; RTU response timeout and the
54 # silent gap between two requests are user settings.
55 MASTER_MAX_READ_BITS = 2000
56 MASTER_MAX_READ_REGISTERS = 125
57 MASTER_MAX_WRITE_BITS = 1968
58 MASTER_MAX_WRITE_REGISTERS = 123
59 MASTER_DEFAULT_RESPONSE_TIMEOUT_MS = 1000
60 MASTER_MIN_RESPONSE_TIMEOUT_MS = 10
61 MASTER_MAX_RESPONSE_TIMEOUT_MS = 10000
62 MASTER_DEFAULT_DELAY_MS = 20
63 MASTER_MIN_DELAY_MS = 0
64 MASTER_MAX_DELAY_MS = 10000
65 MASTER_DEFAULT_SCAN_MS = 1000
66 MASTER_MAX_SCAN_MS = 3600000
67 MASTER_MAX_WINDOWS = 32
68 # When the serial port fails (e.g. USB adapter unplugged), the Master keeps
69 # trying to reopen it at this interval until it succeeds or is disconnected.
70 MASTER_RECONNECT_INTERVAL_SECONDS = 2.0
71 MASTER_WORKSPACE_FORMAT = "MBToolkit Master workspace"
72 # Workspaces saved by MBSniffer v3.2 use this format name and still open.
73 MASTER_WORKSPACE_FORMATS = (MASTER_WORKSPACE_FORMAT, "MBSniffer Master workspace")
74 MASTER_WORKSPACE_VERSION = 1
75
76 # Slave (simulator) tab. Each window is one data table of one simulated Slave
77 # ID; requests are answered from those tables.
78 SLAVE_MAX_WINDOWS = 32
79 SLAVE_MAX_QTY = 2000
80 # A master write is shown highlighted in the table for this long.
```

MBToolkit v3.4/mb_config.py — lines 81-120

```
SLAVE_HIGHLIGHT_SECONDS = 1.5
SLAVE_FAULT_MAX_DELAY_MS = 10000
# Bytes that do not form a request (other devices' responses, noise) are
# discarded once the bus has been silent this long. Long enough for the
# latency-timer chunks of USB-serial adapters (FTDI default 16 ms).
SLAVE_RESYNC_SILENCE_MS = 40.0
SLAVE_WORKSPACE_FORMAT = "MBToolkit Slave workspace"
SLAVE_WORKSPACE_VERSION = 1

# Sniffer burst assembly: extra wait when a burst ends with the start of a
# recognised frame. USB-serial adapters deliver bytes in latency-timer
# chunks (FTDI default 16 ms), which can split one RTU frame in two.
SNIFFER_SPLIT_FRAME_HOLD_MS = 40.0

# Diagnostic thresholds.
SLOW_RESPONSE_THRESHOLD_MS = 500.0

# Standard Modbus exception response codes.
MODBUS_EXCEPTION_CODES = {
    0x01: (
        "Illegal Function",
        "A função Modbus pedida não é suportada/permitida pelo dispositivo."
    ),
    0x02: (
        "Illegal Data Address",
        "O endereço ou a gama de endereços pedida não é válida neste dispositivo."
    ),
    0x03: (
        "Illegal Data Value",
        "Um valor, quantidade ou campo do pedido é inválido para esta função."
    ),
    0x04: (
        "Server Device Failure",
        "O dispositivo encontrou um erro interno ao tentar executar o pedido."
    ),
    0x05: (
        "Acknowledge",
        "O pedido foi aceite, mas necessita de mais tempo para ser processado."
    ),
    0x06: (
```

MBToolkit v3.4/mb_config.py — lines 121-160

```
        "Server Device Busy",
        "O dispositivo está ocupado e não consegue processar o pedido neste momento."
    ),
    0x08: (
        "Memory Parity Error",
        "Foi detetado um erro de paridade ao aceder à memória do dispositivo."
    ),
    0x0A: (
        "Gateway Path Unavailable",
        "O gateway não conseguiu estabelecer o caminho para o dispositivo de destino."
    ),
    0x0B: (
        "Gateway Target Device Failed to Respond",
        "O gateway não recebeu resposta do dispositivo de destino."
    ),
}

# Bounded GUI history. The disk log remains complete.
MAX_UI_FRAMES = 20000
MAX_RAW_TEXT_LINES = 20000
UI_PRUNE_CHUNK_FRAMES = 2000
RAW_TEXT_PRUNE_CHUNK_LINES = 2000
RAW_TRANSACTION_SEPARATOR = "—" * 72 + "\n"

SIMULATION_FUNCTION_CODES = (1, 2, 3, 4)
SIMULATION_EXCEPTION_CODES = (1, 2, 3, 4, 6)

def resource_path(filename):
    """Return a bundled/source resource path for normal Python or PyInstaller."""
    try:
        base = Path(sys._MEIPASS)
    except (AttributeError, TypeError):
        base = Path(__file__).resolve().parent
    return base / filename

def set_windows_app_user_model_id():
    """
    Give Windows a stable application identity so the taskbar can associate
    the Tk window with MBToolkit instead of a generic Python process.
    """
```

MBToolkit v3.4/mb_config.py — lines 161-200

```
    if os.name != "nt":
        return
    try:
        ctypes.windll.shell32.SetCurrentProcessExplicitAppUserModelID(
            f"{APP_NAME}.{APP_VERSION}"
        )
    except Exception:
        pass



def get_settings_file() -> Path:
    """
    Return the persistent per-user settings file.

    Windows:
        %APPDATA%\\MBToolkit\\settings.json

    Fallback:
        ~/.mbtoolkit/settings.json

    This location is independent from the EXE/BAT folder, so the selected
    Light/Dark mode survives application updates and also works when the
    application folder is read-only.
    """
    try:
        folder = _settings_folder(APP_NAME)
        if not folder.exists():
            _migrate_legacy_settings(folder)
        folder.mkdir(parents=True, exist_ok=True)
        return folder / "settings.json"
    except Exception:
        return Path.home() / ".mbtoolkit_settings.json"


def _settings_folder(app_name):
    if os.name == "nt":
        roaming_app_data = os.environ.get("APPDATA")
        if roaming_app_data:
            return Path(roaming_app_data) / app_name
```

MBToolkit v3.4/mb_config.py — lines 201-240

```
201         return Path.home() / "AppData" / "Roaming" / app_name
202     return Path.home() / f".{app_name.lower()}"
203 
204 
205 def _migrate_legacy_settings(folder):
206     """Copy MBSniffer's settings and workspaces into the new MBToolkit folder.
207 
208     Runs once, when the MBToolkit folder does not exist yet. The old folder
209     is left untouched. Failures are ignored: settings are convenience state.
210     """
211     legacy = _settings_folder(LEGACY_APP_NAME)
212     if not legacy.is_dir():
213         return
214     try:
215         shutil.copytree(legacy, folder)
216     except Exception:
217         pass
218 
219 
220 def _validated_choice(value, allowed, default):
221     return value if value in allowed else default
222 
223 
224 def _validated_bool(value, default):
225     return value if isinstance(value, bool) else default
226 
227 
228 def _validated_number_string(value, minimum, maximum, default):
229     try:
230         number = float(value)
231         if minimum <= number <= maximum:
232             return f"{number:g}"
233     except (TypeError, ValueError):
234         pass
235     return str(default)
236 
237 
238 def _validated_int_string(value, minimum, maximum, default):
239     try:
240         number = int(value)
```

MBToolkit v3.4/mb_config.py — lines 241-280

```
241         if minimum <= number <= maximum:
242             return str(number)
243     except (TypeError, ValueError):
244         pass
245     return str(default)
246
247
248 def load_ui_settings() -> dict:
249     """
250     Load validated user preferences from the single Roaming settings file.
251
252     Machine-specific state such as COM-port names and window geometry is
253     deliberately excluded.
254     """
255     path = get_settings_file()
256     try:
257         data = json.loads(path.read_text(encoding="utf-8"))
258     except Exception:
259         return {}
260
261     if not isinstance(data, dict):
262         return {}
263
264     result = {}
265
266     if isinstance(data.get("dark_mode"), bool):
267         result["dark_mode"] = data["dark_mode"]
268
269     if data.get("language") in ("Português", "English"):
270         result["language"] = data["language"]
271
272     if data.get("last_main_tab") in ("Sniffer", "Bus Slave Finder", "Master", "Slave"):
273         result["last_main_tab"] = data["last_main_tab"]
274
275     sniffer_in = data.get("sniffer")
276     if isinstance(sniffer_in, dict):
277         # v2.9 migration: the UI option was renamed from "Realçar anomalias"
278         # to "Realçar resultados". Accept the old key once so existing user
279         # preferences are not lost, but expose/save only the new name.
280         highlight_results_value = sniffer_in.get(
```

MBToolkit v3.4/mb_config.py — lines 281-320

```
            "highlight_results",
            sniffer_in.get("highlight_anomalies"),
        )

        result["sniffer"] = {
            "physical_mode": _validated_choice(
                sniffer_in.get("physical_mode"),
                ("RS485 2-wire", "RS232 single RX", "RS232 dual RX"),
                "RS485 2-wire",
            ),
            "baud": _validated_choice(
                str(sniffer_in.get("baud", "")),
                tuple(str(v) for v in DISCOVERY_BAUD_RATES),
                "9600",
            ),
            "data_bits": _validated_choice(
                str(sniffer_in.get("data_bits", "")),
                ("5", "6", "7", "8"),
                "8",
            ),
            "parity": _validated_choice(
                sniffer_in.get("parity"),
                ("None", "Even", "Odd", "Mark", "Space"),
                "None",
            ),
            "stop_bits": _validated_choice(
                str(sniffer_in.get("stop_bits", "")),
                ("1", "1.5", "2"),
                "1",
            ),
            "frame_gap_mode": _validated_choice(
                sniffer_in.get("frame_gap_mode"),
                ("Auto", "Manual"),
                "Auto",
            ),
            "frame_gap_ms": str(sniffer_in.get("frame_gap_ms", "")),
            "pending_timeout_s": _validated_number_string(
                sniffer_in.get("pending_timeout_s"),
                MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
                MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
```

MBToolkit v3.4/mb_config.py — lines 321-360

```
                DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS,
            ),
            "auto_scroll": _validated_bool(
                sniffer_in.get("auto_scroll"),
                True,
            ),
            "separate_transactions": _validated_bool(
                sniffer_in.get("separate_transactions"),
                True,
            ),
            "highlight_results": _validated_bool(
                highlight_results_value,
                False,
            ),
        }

    finder_in = data.get("bus_slave_finder")
    if isinstance(finder_in, dict):
        saved_bauds = finder_in.get("baud_rates")
        if isinstance(saved_bauds, list):
            baud_rates = [
                int(v) for v in saved_bauds
                if isinstance(v, (int, float, str))
                and str(v).isdigit()
                and int(v) in DISCOVERY_BAUD_RATES
            ]
        else:
            baud_rates = [9600, 19200]

        saved_parities = finder_in.get("parities")
        if isinstance(saved_parities, list):
            parities = [
                v for v in saved_parities
                if v in DISCOVERY_PARITIES
            ]
        else:
            parities = ["None", "Even"]

        saved_stops = finder_in.get("stop_bits")
        if isinstance(saved_stops, list):
```

MBToolkit v3.4/mb_config.py — lines 361-400

```
            stop_bits = [
                str(v) for v in saved_stops
                if str(v) in DISCOVERY_STOP_BITS
            ]
        else:
            stop_bits = ["1"]

        min_timeout_ms = _validated_int_string(
            finder_in.get("min_timeout_ms"),
            DISCOVERY_MIN_TIMEOUT_MS,
            DISCOVERY_MAX_TIMEOUT_MS,
            DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
        )
        # Settings written before v3.1 have no "attempts" key. Their 40 ms
        # value is the old automatic default, not a user choice.
        if (
            "attempts" not in finder_in
            and min_timeout_ms == str(DISCOVERY_LEGACY_DEFAULT_MIN_TIMEOUT_MS)
        ):
            min_timeout_ms = str(DISCOVERY_DEFAULT_MIN_TIMEOUT_MS)

        result["bus_slave_finder"] = {
            "baud_rates": baud_rates,
            "parities": parities,
            "stop_bits": stop_bits,
            "slave_start": _validated_int_string(
                finder_in.get("slave_start"),
                1,
                247,
                1,
            ),
            "slave_end": _validated_int_string(
                finder_in.get("slave_end"),
                1,
                247,
                247,
            ),
            "min_timeout_ms": min_timeout_ms,
            "attempts": _validated_int_string(
                finder_in.get("attempts"),
```

MBToolkit v3.4/mb_config.py — lines 401-440

```
                DISCOVERY_MIN_ATTEMPTS,
                DISCOVERY_MAX_ATTEMPTS,
                DISCOVERY_DEFAULT_ATTEMPTS,
            ),
            "fc04_fallback": _validated_bool(
                finder_in.get("fc04_fallback"),
                False,
            ),
            "device_identification": _validated_bool(
                finder_in.get("device_identification"),
                False,
            ),
        }

    master_in = data.get("master")
    if isinstance(master_in, dict):
        result["master"] = {
            "baud": _validated_choice(
                str(master_in.get("baud", "")),
                tuple(str(v) for v in DISCOVERY_BAUD_RATES),
                "9600",
            ),
            "parity": _validated_choice(
                master_in.get("parity"), ("None", "Even", "Odd"), "None"
            ),
            "stop_bits": _validated_choice(
                str(master_in.get("stop_bits", "")), ("1", "2"), "1"
            ),
            "response_timeout_ms": _validated_int_string(
                master_in.get("response_timeout_ms"),
                MASTER_MIN_RESPONSE_TIMEOUT_MS,
                MASTER_MAX_RESPONSE_TIMEOUT_MS,
                MASTER_DEFAULT_RESPONSE_TIMEOUT_MS,
            ),
            "delay_ms": _validated_int_string(
                master_in.get("delay_ms"),
                MASTER_MIN_DELAY_MS,
                MASTER_MAX_DELAY_MS,
                MASTER_DEFAULT_DELAY_MS,
            ),
```

MBToolkit v3.4/mb_config.py — lines 441-480

```
            # Workspace reopened at start-up (name of a file in Workspaces).
            "last_workspace": _validated_workspace_name(master_in.get("last_workspace")),
        }

    slave_in = data.get("slave")
    if isinstance(slave_in, dict):
        result["slave"] = {
            "baud": _validated_choice(
                str(slave_in.get("baud", "")),
                tuple(str(v) for v in DISCOVERY_BAUD_RATES),
                "9600",
            ),
            "parity": _validated_choice(
                slave_in.get("parity"), ("None", "Even", "Odd"), "None"
            ),
            "stop_bits": _validated_choice(
                str(slave_in.get("stop_bits", "")), ("1", "2"), "1"
            ),
            "last_workspace": _validated_workspace_name(slave_in.get("last_workspace")),
        }

    return result


def _validated_workspace_name(value):
    """Workspace name remembered in settings.json, or "" when unusable."""
    if not isinstance(value, str):
        return ""
    value = value.strip()
    if not 0 < len(value) <= 60 or any(ch in '<>:"/\\|?*' for ch in value):
        return ""
    return value


def save_ui_settings(settings=None, **updates) -> None:
    """
    Persist UI preferences atomically in the single Roaming settings file.

    Partial updates merge with the existing file. This lets the Dark-mode
    toggle update immediately without discarding the other remembered options.
```

MBToolkit v3.4/mb_config.py — lines 481-520

```
    """
    path = get_settings_file()

    current = load_ui_settings()
    if isinstance(settings, dict):
        current.update(settings)
    current.update(updates)

    try:
        path.parent.mkdir(parents=True, exist_ok=True)
        tmp = path.with_suffix(path.suffix + ".tmp")
        tmp.write_text(
            json.dumps(current, indent=2, sort_keys=True),
            encoding="utf-8",
        )
        tmp.replace(path)
    except Exception:
        # Remembered UI state must never break application use.
        pass


def get_application_directory() -> Path:
    """
    Return the user-facing application directory.

    - PyInstaller EXE: directory containing MBToolkit.exe.
    - Packaged Python/BAT mode: directory containing the root MBToolkit.bat.
      The Python sources live inside the versioned child folder, e.g. ``MBToolkit v3.4``.
    - Development Python mode: directory containing this module.
    """
    try:
        if getattr(sys, "frozen", False):
            return Path(sys.executable).resolve().parent

        source_dir = Path(__file__).resolve().parent
        launcher_dir = source_dir.parent

        if (launcher_dir / "MBToolkit.bat").is_file():
            return launcher_dir

```

MBToolkit v3.4/mb_config.py — lines 521-560

```
        return source_dir
    except Exception:
        return Path.cwd()

def get_workspaces_folder() -> Path:
    """
    Return/create the Master workspace folder beside settings.json:

        %APPDATA%\\MBToolkit\\Workspaces
    """
    folder = get_settings_file().parent / "Workspaces"
    folder.mkdir(parents=True, exist_ok=True)
    return folder


def get_master_workspaces_folder() -> Path:
    """
    Return/create the Master workspace folder inside the workspace folder:

        %APPDATA%\\MBToolkit\\Workspaces\\Master

    Up to v3.4 the Master workspaces were kept directly in Workspaces. The
    first time this folder is created, those files are moved into it.
    """
    root = get_workspaces_folder()
    folder = root / "Master"
    if not folder.exists():
        folder.mkdir(parents=True, exist_ok=True)
        for old in root.glob("*.mbsw"):
            try:
                if old.is_file() and not (folder / old.name).exists():
                    shutil.move(str(old), str(folder / old.name))
            except OSError:
                pass
    return folder


def get_slave_workspaces_folder() -> Path:
    """
    Return/create the Slave workspace folder inside the workspace folder:
```

MBToolkit v3.4/mb_config.py — lines 561-577

```
561
562         %APPDATA%\\MBToolkit\\Workspaces\\Slave
563     """
564     folder = get_workspaces_folder() / "Slave"
565     folder.mkdir(parents=True, exist_ok=True)
566     return folder
567
568
569 def get_logs_folder() -> Path:
570     """
571     Return/create the log directory beside MBToolkit.bat or MBToolkit.exe:
572
573         <launcher/exe directory>/MBToolkit Logs
574     """
575     folder = get_application_directory() / "MBToolkit Logs"
576     folder.mkdir(parents=True, exist_ok=True)
577     return folder
```

MBToolkit v3.4/mb_protocol.py

1007 source lines.

MBToolkit v3.4/mb_protocol.py — lines 1-40

```python
#!/usr/bin/env python3
"""Modbus RTU parsing, CRC, transaction metrics and simulation helpers."""

import math
import random
import time
from bisect import bisect_right
from collections import defaultdict, deque
from datetime import datetime

from mb_config import (
    MODBUS_EXCEPTION_CODES,
    PENDING_REQUEST_TIMEOUT_SECONDS,
    SIMULATION_FUNCTION_CODES,
    SIMULATION_EXCEPTION_CODES,
    SLOW_RESPONSE_THRESHOLD_MS,
)

def _build_crc16_table():
    table = []
    for value in range(256):
        crc = value
        for _ in range(8):
            if crc & 1:
                crc = (crc >> 1) ^ 0xA001
            else:
                crc >>= 1
        table.append(crc)
    return tuple(table)

# Table-driven CRC: one lookup per byte instead of eight shift/XOR steps. The
# resynchronisation scan computes CRCs at many offsets, so this matters on
# noisy buses.
_CRC16_TABLE = _build_crc16_table()

def crc16_modbus(data: bytes) -> int:
    crc = 0xFFFF
    table = _CRC16_TABLE
    for b in data:
        crc = (crc >> 8) ^ table[(crc ^ b) & 0xFF]
```

MBToolkit v3.4/mb_protocol.py — lines 41-80

```
41      return crc
42
43  def crc_ok(frame: bytes) -> bool:
44      if len(frame) < 4:
45          return False
46      expected = crc16_modbus(frame[:-2])
47      received = frame[-2] | (frame[-1] << 8)
48      return expected == received
49
50  def append_crc(payload: bytes) -> bytes:
51      crc = crc16_modbus(payload)
52      return payload + bytes((crc & 0xFF, (crc >> 8) & 0xFF))
53
54  def hex_bytes(data: bytes) -> str:
55      return " ".join(f"{b:02X}" for b in data)
56
57  def modbus_rtu_interframe_gap_seconds(baud: int, bits_per_char: float) -> float:
58      """Return the recommended Modbus RTU t3.5 inter-frame gap.
59
60      Modbus Serial Line V1.02 requires the character-time timers to be
61      respected at baud rates up to and including 19200 bit/s. Above 19200
62      bit/s it recommends a fixed t3.5 value of 1.750 ms.
63      """
64      baud = int(baud)
65      bits_per_char = float(bits_per_char)
66      if baud <= 0:
67          raise ValueError("Baud rate must be greater than zero")
68      if bits_per_char <= 0:
69          raise ValueError("Bits per character must be greater than zero")
70      if baud > 19200:
71          return 0.001750
72      return 3.5 * bits_per_char / float(baud)
73
74  def plausible_request_qty(fc: int, qty: int) -> bool:
75      if fc in (1, 2):
76          return 1 <= qty <= 2000
77      if fc in (3, 4):
78          return 1 <= qty <= 125
79      return True
80
```

MBToolkit v3.4/mb_protocol.py — lines 81-120

```
# Function Codes whose RTU frame length the parser can derive.
KNOWN_FRAME_FUNCTION_CODES = (1, 2, 3, 4, 5, 6, 7, 8, 15, 16, 22, 23)

# Function Codes whose normal response is an exact copy of the request.
EXACT_ECHO_FUNCTION_CODES = (5, 6, 22)

BROADCAST_UNIT_ID = 0

def _plausible_unit_id(value: int) -> bool:
    """Return True for Modbus RTU unit addresses that can appear on the wire."""
    try:
        return 0 <= int(value) <= 247
    except (TypeError, ValueError):
        return False


def _candidate_prefixes_at(buf, start=0):
    """Return structurally plausible known-frame lengths at *start*."""
    remaining = len(buf) - int(start)
    if remaining < 2:
        return []

    unit = buf[start]
    if not _plausible_unit_id(unit):
        return []

    fc = buf[start + 1]
    out = []

    if fc & 0x80:
        # Exception responses use FC | 0x80 and have a fixed five-byte length.
        base_fc = fc & 0x7F
        if base_fc in KNOWN_FRAME_FUNCTION_CODES:
            out.append((5, "EXCEPTION"))
        return out

    if fc in (1, 2, 3, 4):
        if remaining >= 6:
            qty = (buf[start + 4] << 8) | buf[start + 5]
            if plausible_request_qty(fc, qty):
```

MBToolkit v3.4/mb_protocol.py — lines 121-160

```
121                 out.append((8, "REQUEST"))
122
123         if remaining >= 3:
124             byte_count = buf[start + 2]
125             resp_len = 5 + byte_count
126             if fc in (3, 4):
127                 if byte_count > 0 and byte_count % 2 == 0:
128                     out.append((resp_len, "RESPONSE"))
129             elif byte_count > 0:
130                 out.append((resp_len, "RESPONSE"))
131
132     elif fc in (5, 6, 8):
133         out.append((8, "REQ/RESP"))
134
135     elif fc == 7:
136         # FC07 request: Slave + FC + CRC = 4 bytes.
137         # FC07 response: Slave + FC + Status + CRC = 5 bytes.
138         if remaining >= 5:
139             out.append((5, "RESPONSE"))
140         out.append((4, "REQUEST"))
141
142     elif fc in (15, 16):
143         out.append((8, "RESPONSE"))
144         if remaining >= 7:
145             byte_count = buf[start + 6]
146             if byte_count > 0:
147                 out.append((9 + byte_count, "REQUEST"))
148
149     elif fc == 22:
150         out.append((10, "REQ/RESP"))
151
152     elif fc == 23:
153         if remaining >= 3:
154             bc = buf[start + 2]
155             if bc > 0 and bc % 2 == 0:
156                 out.append((5 + bc, "RESPONSE"))
157         if remaining >= 11:
158             bc = buf[start + 10]
159             if bc > 0:
160                 out.append((13 + bc, "REQUEST"))
```

MBToolkit v3.4/mb_protocol.py — lines 161-200

```
161
162     cleaned = []
163     seen = set()
164     for length, kind in out:
165         if 4 <= length <= 260 and (length, kind) not in seen:
166             seen.add((length, kind))
167             cleaned.append((length, kind))
168     return cleaned
169
170
171 def candidate_prefixes(buf: bytes):
172     """Compatibility wrapper for callers/tests that inspect offset zero."""
173     return _candidate_prefixes_at(buf, 0)
174
175
176 def _crc_ok_at(buf, start: int, length: int) -> bool:
177     """CRC check without copying the whole remaining capture buffer."""
178     if length < 4 or start < 0 or start + length > len(buf):
179         return False
180     end = start + length
181     expected = crc16_modbus(memoryview(buf)[start:end - 2])
182     received = buf[end - 2] | (buf[end - 1] << 8)
183     return expected == received
184
185
186 def find_valid_prefix(buf: bytes, start=0):
187     """Find a CRC-valid, structurally recognised Modbus frame at *start*."""
188     for length, kind in _candidate_prefixes_at(buf, start):
189         if start + length <= len(buf) and _crc_ok_at(buf, start, length):
190             return length, kind
191     return None
192
193
194 def guess_invalid_whole_frame(buf: bytes):
195     """Classify a complete known-shape frame whose CRC is invalid."""
196     if len(buf) < 4 or not _plausible_unit_id(buf[0]):
197         return None
198     for length, kind in _candidate_prefixes_at(buf, 0):
199         if len(buf) == length:
200             return kind
```

MBToolkit v3.4/mb_protocol.py — lines 201-240

```
201     return None
202 
203 
204 def _unknown_whole_frame_is_valid(buf: bytes) -> bool:
205     """Conservative fallback for one unsupported FC occupying the whole burst.
206 
207     Unknown frames cannot be length-validated from their FC, so CRC-only
208     recognition is deliberately restricted to the complete remaining burst.
209     It is never used while scanning arbitrary offsets for resynchronisation.
210     """
211     if not 4 <= len(buf) <= 260:
212         return False
213     if not _plausible_unit_id(buf[0]):
214         return False
215     if _candidate_prefixes_at(buf, 0):
216         return False
217     return crc_ok(buf)
218 
219 
220 def split_capture_buffer(data: bytes):
221     results = []
222     buf = bytearray(data)
223 
224     while buf:
225         match = find_valid_prefix(buf, 0)
226         if match:
227             length, kind = match
228             results.append((bytes(buf[:length]), kind, "OK"))
229             del buf[:length]
230             continue
231 
232         # Search the full remaining burst, but only for structurally recognised
233         # Modbus frames. The former generic CRC scan tried up to 256 candidate
234         # lengths at every offset and became very slow on long noise bursts.
235         # Restricting resynchronisation to known frame shapes makes the scan
236         # effectively linear in the burst size and strongly reduces accidental
237         # CRC-only "phantom" frames in random noise.
238         found_offset = None
239         search_limit = max(len(buf) - 4, 0)
240         for offset in range(1, search_limit + 1):
```

MBToolkit v3.4/mb_protocol.py — lines 241-280

```
            if find_valid_prefix(buf, offset):
                found_offset = offset
                break

        if found_offset is not None:
            prefix = bytes(buf[:found_offset])

            # If the bytes immediately before the recovered frame are exactly a
            # recognised Modbus frame shape, preserve the useful CRC diagnosis
            # instead of hiding it as RAW/UNSYNC. Otherwise they are genuinely
            # unsynchronised bytes and remain RAW.
            guessed = guess_invalid_whole_frame(prefix)
            if guessed:
                results.append((prefix, guessed, "ERROR"))
            else:
                results.append((prefix, "RAW/UNSYNC", "UNKNOWN"))
            del buf[:found_offset]
            continue

        guessed = guess_invalid_whole_frame(bytes(buf))
        if guessed:
            results.append((bytes(buf), guessed, "ERROR"))
        elif _unknown_whole_frame_is_valid(bytes(buf)):
            results.append((bytes(buf), "UNKNOWN", "OK"))
        else:
            results.append((bytes(buf), "RAW/UNPARSED", "UNKNOWN"))
        break

    return results

def burst_may_continue(data: bytes) -> bool:
    """Return True when *data* ends with the start of an unfinished frame.

    USB-serial adapters deliver received bytes in chunks, so a silence longer
    than t3.5 can occur in the middle of one RTU frame. The reader uses this
    to wait a little longer before closing such a burst. Only the final
    unvalidated segment is inspected, and only at its first byte.
    """
    segments = split_capture_buffer(data)
    if not segments:
```

MBToolkit v3.4/mb_protocol.py — lines 281-320

```
281         return False
282
283     tail, _kind, state = segments[-1]
284     if state == "OK":
285         return False
286
287     if not _plausible_unit_id(tail[0]):
288         return False
289     if len(tail) == 1:
290         return True
291
292     fc = tail[1]
293     if (fc & 0x7F) not in KNOWN_FRAME_FUNCTION_CODES:
294         return False
295     if len(tail) < 8 and not fc & 0x80:
296         # Too short for the request/response length fields to be known.
297         return True
298
299     return any(
300         length > len(tail)
301         for length, _kind in _candidate_prefixes_at(tail, 0)
302     )
303
304
305 def read_bits_from_response(frame: bytes, qty=None):
306     """Return FC01/FC02 coil/input states, least-significant bit first.
307
308     Without the matching request quantity the padding bits of the last data
309     byte cannot be distinguished, so every transmitted bit is returned.
310     """
311     if len(frame) < 5:
312         return []
313     byte_count = frame[2]
314     data = frame[3:3 + byte_count]
315     bits = [(byte >> bit) & 1 for byte in data for bit in range(8)]
316     try:
317         qty = int(qty) if qty is not None else None
318     except (TypeError, ValueError):
319         qty = None
320     if qty is not None and 0 < qty <= len(bits):
```

MBToolkit v3.4/mb_protocol.py — lines 321-360

```
        bits = bits[:qty]
    return bits


# FC01/FC02 can return up to 2000 bits; the Traffic row shows only the first
# ones (the Frame Inspector lists them all with their addresses).
READ_BITS_DETAILS_LIMIT = 32


def read_bits_details(frame: bytes, qty=None) -> str:
    """Compact Details text: bits in address order, grouped per data byte."""
    byte_count = frame[2] if len(frame) >= 3 else 0
    bits = read_bits_from_response(frame, qty)
    shown = "".join(str(bit) for bit in bits[:READ_BITS_DETAILS_LIMIT])
    grouped = " ".join(shown[i:i + 8] for i in range(0, len(shown), 8))
    if len(bits) > READ_BITS_DETAILS_LIMIT:
        grouped += f" …(+{len(bits) - READ_BITS_DETAILS_LIMIT})"
    return f"ByteCount={byte_count}  Bits={grouped}"


def decode_frame(frame: bytes, kind_hint: str):
    info = {
        "kind": kind_hint,
        "slave": "",
        "fc": "",
        "details": "",
        "pdu_address": None,
        "one_based": None,
        "qty": None,
    }

    if str(kind_hint).startswith("RAW"):
        info["details"] = f"{len(frame)} raw bytes"
        return info

    if len(frame) < 2:
        info["details"] = "Raw bytes"
        return info

    slave = frame[0]
```

MBToolkit v3.4/mb_protocol.py — lines 361-400

```
    fc = frame[1]
    info["slave"] = str(slave)
    info["fc"] = f"{fc:02X}"

    if fc & 0x80:
        info["kind"] = "EXCEPTION"
        code = frame[2] if len(frame) >= 3 else None

        if code is None:
            info["details"] = "Exception"
            return info

        exception_info = MODBUS_EXCEPTION_CODES.get(code)
        if exception_info is not None:
            name, _description = exception_info
            info["details"] = f"Exception 0x{code:02X} — {name}"
        else:
            info["details"] = f"Exception 0x{code:02X} — Unknown/Reserved Exception"
        return info

    if fc in (1, 2, 3, 4):
        if kind_hint == "REQUEST" and len(frame) >= 8:
            start_addr = (frame[2] << 8) | frame[3]
            qty = (frame[4] << 8) | frame[5]
            info["pdu_address"] = start_addr
            info["one_based"] = start_addr + 1
            info["qty"] = qty
            info["details"] = (
                f"PDU Address={start_addr}  1-based={start_addr + 1}  Qty={qty}"
            )
            return info

        if kind_hint == "RESPONSE" and len(frame) >= 5:
            bc = frame[2]
            details = f"ByteCount={bc}"
            if fc in (3, 4) and bc % 2 == 0 and len(frame) >= 5 + bc:
                data = frame[3:3+bc]
                regs = [(data[i] << 8) | data[i+1] for i in range(0, len(data), 2)]
                details += f"  Registos={regs}"
            elif fc in (1, 2) and len(frame) >= 5 + bc:
```

MBToolkit v3.4/mb_protocol.py — lines 401-440

```
401                 details = read_bits_details(frame)
402             info["details"] = details
403             return info
404
405     if fc in (5, 6, 8) and len(frame) >= 8:
406         addr = (frame[2] << 8) | frame[3]
407         value = (frame[4] << 8) | frame[5]
408         info["pdu_address"] = addr
409         info["one_based"] = addr + 1
410         info["details"] = (
411             f"PDU Address={addr}  1-based={addr + 1}  Value=0x{value:04X}"
412         )
413         return info
414
415     if fc == 7:
416         if kind_hint == "REQUEST":
417             info["details"] = "Read Exception Status"
418         elif kind_hint == "RESPONSE" and len(frame) >= 5:
419             info["details"] = f"Status=0x{frame[2]:02X}"
420         else:
421             info["details"] = "Read Exception Status"
422         return info
423
424     if fc in (15, 16):
425         if kind_hint == "REQUEST" and len(frame) >= 9:
426             start_addr = (frame[2] << 8) | frame[3]
427             qty = (frame[4] << 8) | frame[5]
428             bc = frame[6]
429             info["pdu_address"] = start_addr
430             info["one_based"] = start_addr + 1
431             info["qty"] = qty
432             info["details"] = (
433                 f"PDU Address={start_addr}  1-based={start_addr + 1}  "
434                 f"Qty={qty}  ByteCount={bc}"
435             )
436             return info
437         if len(frame) >= 8:
438             start_addr = (frame[2] << 8) | frame[3]
439             qty = (frame[4] << 8) | frame[5]
440             info["pdu_address"] = start_addr
```

MBToolkit v3.4/mb_protocol.py — lines 441-480

```
            info["one_based"] = start_addr + 1
            info["qty"] = qty
            info["details"] = (
                f"PDU Address={start_addr}  1-based={start_addr + 1}  Qty={qty}"
            )
            return info

    info["details"] = f"{len(frame)} bytes"
    return info

def format_clock_time(epoch: float) -> str:
    return datetime.fromtimestamp(float(epoch)).strftime("%H:%M:%S.%f")[:-3]


def parse_burst(data: bytes, timestamp: str, channel: str = "", event_time=None,
                chunk_times=None):
    """Split one received burst into decoded rows.

    *chunk_times* is an optional ascending list of ``(byte_offset, epoch)``
    pairs recording when each serial read arrived. When given, every frame
    gets the arrival time of the chunk holding its first byte instead of the
    burst start time, so frames merged into one burst keep distinct times.
    """
    rows = []
    if event_time is None:
        event_time = time.time()

    chunk_offsets = [int(o) for o, _t in chunk_times] if chunk_times else []
    offset = 0

    for frame, kind, state in split_capture_buffer(data):
        frame_time = float(event_time)
        frame_timestamp = timestamp
        if chunk_offsets:
            index = bisect_right(chunk_offsets, offset) - 1
            if index > 0:
                frame_time = float(chunk_times[index][1])
                frame_timestamp = format_clock_time(frame_time)
        offset += len(frame)

```

MBToolkit v3.4/mb_protocol.py — lines 481-520

```
        info = decode_frame(frame, kind)
        crc_status = "OK" if state == "OK" else ("ERROR" if state == "ERROR" else "?")
        rows.append({
            "time": frame_timestamp,
            "channel": channel,
            "type": info["kind"],
            "slave": info["slave"],
            "fc": info["fc"],
            "details": info["details"],
            "crc": crc_status,
            "raw": hex_bytes(frame),
            "pdu_address": info["pdu_address"],
            "one_based": info["one_based"],
            "qty": info["qty"],
            "_frame_len": len(frame),
            "_epoch": frame_time,
        })
    return rows

class SessionMetrics:
    """Tracks pairing, counters and bus-health diagnostics for one session."""

    def __init__(self, pending_timeout_seconds=PENDING_REQUEST_TIMEOUT_SECONDS):
        self.pending_timeout_seconds = float(pending_timeout_seconds)
        self.reset()

    def reset(self):
        self.first_event_time = None
        self.last_event_time = None
        self.pending = {}
        self.requests = 0
        # Broadcast requests (Slave 0) never receive a response, so they are
        # never pending and are excluded from the timeout-rate denominator.
        self.broadcast_requests = 0
        self.responses = 0
        self.timeouts = 0
        self.crc_errors = 0
        self.exceptions = 0
        self.raw = 0

```

MBToolkit v3.4/mb_protocol.py — lines 521-560

```
        # Bus Health.
        self.total_bytes = 0
        self.nonvalidated_bytes = 0
        self.active_slaves = set()
        self.exception_by_slave = defaultdict(int)
        self.response_times_ms = deque(maxlen=10000)
        self.response_times_by_slave = defaultdict(lambda: deque(maxlen=2000))

        # Timeout events are exposed to the view so the original request row can
        # be marked/highlighted without inventing a fake serial frame.
        self._expired_requests = []

    def _key(self, row):
        try:
            slave = int(row.get("slave", ""))
            fc = int(row.get("fc", ""), 16)
        except (ValueError, TypeError):
            return None

        if row.get("type") == "EXCEPTION":
            fc &= 0x7F

        return (slave, fc)

    def pending_count(self):
        return sum(len(v) for v in self.pending.values())

    def pop_expired_requests(self):
        items = self._expired_requests
        self._expired_requests = []
        return items

    def expire_pending(self, now=None):
        """
        Expire stale requests and retain their sequence IDs for the GUI.

        Returns the number of newly expired requests, preserving the original
        public behavior used by CaptureMixin.
        """
        if now is None:
```

MBToolkit v3.4/mb_protocol.py — lines 561-600

```
            now = time.time()
        now = float(now)

        expired = 0
        cutoff = now - self.pending_timeout_seconds

        for key in list(self.pending.keys()):
            pending_list = self.pending[key]

            keep_from = 0
            for item in pending_list:
                if item["time"] <= cutoff:
                    expired += 1
                    keep_from += 1
                    event = dict(item)
                    event["slave"] = key[0]
                    event["fc"] = key[1]
                    self._expired_requests.append(event)
                else:
                    break

            if keep_from:
                del pending_list[:keep_from]

            if not pending_list:
                self.pending.pop(key, None)

        if expired:
            self.timeouts += expired

        return expired

    def force_timeout(self, slave, fc):
        """
        Debug/simulation helper: expire the oldest pending request for one key.

        It uses the same pending bookkeeping and returns the expired request
        descriptor so the GUI marks the real REQUEST row as timed out.
        """
        try:
```

MBToolkit v3.4/mb_protocol.py — lines 601-640

```
601             key = (int(slave), int(fc))
602         except (TypeError, ValueError):
603             return []
604 
605         pending_list = self.pending.get(key, [])
606         if not pending_list:
607             return []
608 
609         item = pending_list.pop(0)
610         if not pending_list:
611             self.pending.pop(key, None)
612 
613         event = dict(item)
614         event["slave"] = key[0]
615         event["fc"] = key[1]
616         self.timeouts += 1
617         return [event]
618 
619     def _resolve_echo_direction(self, row):
620         """Classify a REQ/RESP frame (FC05/06/08/22) as REQUEST or RESPONSE.
621 
622         These requests and their normal responses have the same shape, so the
623         parser cannot tell them apart. A frame is the RESPONSE when an older
624         pending request with the same (Slave, FC) exists; for exact-echo FCs
625         the bytes must also be identical. In RS232 dual RX a response must
626         arrive on the opposite channel to its request.
627         """
628         if row.get("crc") != "OK":
629             return row.get("type", "")
630 
631         key = self._key(row)
632         if key is None:
633             return row.get("type", "")
634         if key[0] == BROADCAST_UNIT_ID:
635             return "REQUEST"
636 
637         pending_list = self.pending.get(key)
638         if pending_list:
639             candidate = pending_list[0]
640             channel = row.get("channel", "")
```

MBToolkit v3.4/mb_protocol.py — lines 641-680

```
641             same_direction = (
642                 channel in ("A→B", "B→A")
643                 and candidate.get("channel") == channel
644             )
645             same_bytes = candidate.get("raw") == row.get("raw")
646             if not same_direction and (
647                 key[1] not in EXACT_ECHO_FUNCTION_CODES or same_bytes
648             ):
649                 return "RESPONSE"
650 
651         return "REQUEST"
652 
653     def _record_response_time(self, row, response_ms):
654         if response_ms is None or row.get("crc") != "OK":
655             return
656 
657         response_ms = float(response_ms)
658         self.response_times_ms.append(response_ms)
659 
660         try:
661             slave = int(row.get("slave", ""))
662         except (TypeError, ValueError):
663             return
664         self.response_times_by_slave[slave].append(response_ms)
665 
666     def process(self, row):
667         event_time = float(row.get("_epoch", time.time()))
668 
669         # Expire stale requests BEFORE pairing the current frame.
670         self.expire_pending(event_time)
671         expired_requests = self.pop_expired_requests()
672 
673         if self.first_event_time is None:
674             self.first_event_time = event_time
675 
676         delta_ms = None
677         if self.last_event_time is not None and event_time >= self.last_event_time:
678             delta_ms = (event_time - self.last_event_time) * 1000.0
679         if self.last_event_time is None or event_time >= self.last_event_time:
680             self.last_event_time = event_time
```

MBToolkit v3.4/mb_protocol.py — lines 681-720

```
681
682         if row.get("type") == "REQ/RESP":
683             row["type"] = self._resolve_echo_direction(row)
684
685         row_type = row.get("type", "")
686         crc = row.get("crc", "")
687
688         try:
689             frame_len = max(0, int(row.get("_frame_len", 0)))
690         except (TypeError, ValueError):
691             frame_len = 0
692         self.total_bytes += frame_len
693
694         try:
695             slave_num = int(row.get("slave", ""))
696             if 1 <= slave_num <= 247 and not str(row_type).startswith("RAW"):
697                 self.active_slaves.add(slave_num)
698         except (TypeError, ValueError):
699             slave_num = None
700
701         if crc == "ERROR":
702             self.crc_errors += 1
703
704         # RAW bytes cannot honestly be called CRC errors: there may be no
705         # complete frame on which to validate a CRC. For Bus Health we track
706         # them together with explicit CRC failures as traffic that could not be
707         # validated as a correct Modbus RTU frame.
708         if crc == "ERROR" or str(row_type).startswith("RAW"):
709             self.nonvalidated_bytes += frame_len
710
711         if row_type == "REQUEST":
712             self.requests += 1
713             key = self._key(row)
714             if key is not None and key[0] == BROADCAST_UNIT_ID:
715                 self.broadcast_requests += 1
716             elif key is not None and crc == "OK":
717                 self.pending.setdefault(key, []).append({
718                     "time": event_time,
719                     "seq": row.get("_view_seq"),
720                     "pdu_address": row.get("pdu_address"),
```

MBToolkit v3.4/mb_protocol.py — lines 721-760

```
                    "one_based": row.get("one_based"),
                    "qty": row.get("qty"),
                    "channel": row.get("channel", ""),
                    "raw": row.get("raw", ""),
                })

        elif row_type == "RESPONSE":
            self.responses += 1

        elif row_type == "EXCEPTION":
            self.responses += 1
            self.exceptions += 1
            if slave_num is not None:
                self.exception_by_slave[slave_num] += 1

        elif str(row_type).startswith("RAW"):
            self.raw += 1

        response_ms = None
        matched_request = None

        if row_type in ("RESPONSE", "EXCEPTION"):
            key = self._key(row)
            if key is not None:
                pending_list = self.pending.get(key, [])
                if pending_list:
                    matched_request = pending_list.pop(0)
                    if not pending_list:
                        self.pending.pop(key, None)

                    if event_time >= matched_request["time"]:
                        response_ms = (
                            event_time - matched_request["time"]
                        ) * 1000.0

        self._record_response_time(row, response_ms)

        return {
            "delta_ms": delta_ms,
            "response_ms": response_ms,
```

MBToolkit v3.4/mb_protocol.py — lines 761-800

```
            "matched_request": matched_request,
            "expired_requests": expired_requests,
            "pending": self.pending_count(),
            "timeouts": self.timeouts,
        }

    @staticmethod
    def _p95(values):
        if not values:
            return None
        ordered = sorted(float(v) for v in values)
        index = max(0, min(len(ordered) - 1, math.ceil(0.95 * len(ordered)) - 1))
        return ordered[index]

    def health_summary(
        self,
        *,
        baud=9600,
        data_bits=8,
        parity="None",
        stop_bits=1,
        now=None,
    ):
        """Return compact diagnostic metrics for the Bus Health view."""
        if now is None:
            now = time.time()

        if self.first_event_time is None:
            elapsed = 0.0
        else:
            end = max(
                float(now),
                float(self.last_event_time or self.first_event_time),
            )
            elapsed = max(0.0, end - float(self.first_event_time))

        responses = list(self.response_times_ms)
        response_avg = (
            sum(responses) / len(responses)
            if responses else None
```

MBToolkit v3.4/mb_protocol.py — lines 801-840

```
        )
        response_min = min(responses) if responses else None
        response_max = max(responses) if responses else None
        response_p95 = self._p95(responses)

        slowest_slave = None
        slowest_avg = None
        for slave, samples in self.response_times_by_slave.items():
            if not samples:
                continue
            avg = sum(samples) / len(samples)
            if slowest_avg is None or avg > slowest_avg:
                slowest_slave = slave
                slowest_avg = avg

        request_rate = (
            self.requests / elapsed
            if elapsed > 0.0 else 0.0
        )

        nonvalidated_rate = (
            100.0 * self.nonvalidated_bytes / self.total_bytes
            if self.total_bytes else 0.0
        )
        answerable_requests = self.requests - self.broadcast_requests
        timeout_rate = (
            100.0 * self.timeouts / answerable_requests
            if answerable_requests > 0 else 0.0
        )

        try:
            baud = max(1.0, float(baud))
        except (TypeError, ValueError):
            baud = 9600.0
        try:
            data_bits = float(data_bits)
        except (TypeError, ValueError):
            data_bits = 8.0
        try:
            stop_bits = float(stop_bits)
```

MBToolkit v3.4/mb_protocol.py — lines 841-880

```
        except (TypeError, ValueError):
            stop_bits = 1.0

        parity_bits = 0.0 if str(parity) == "None" else 1.0
        bits_per_char = 1.0 + data_bits + parity_bits + stop_bits
        bus_load = (
            100.0 * (self.total_bytes * bits_per_char) / (baud * elapsed)
            if elapsed > 0.0 else 0.0
        )

        exception_summary = ", ".join(
            f"S{slave}:{count}"
            for slave, count in sorted(self.exception_by_slave.items())
        ) or "-"

        return {
            "elapsed_s": elapsed,
            "request_rate": request_rate,
            "response_avg_ms": response_avg,
            "response_min_ms": response_min,
            "response_max_ms": response_max,
            "response_p95_ms": response_p95,
            "nonvalidated_rate_pct": nonvalidated_rate,
            "timeout_rate_pct": timeout_rate,
            "active_slaves": len(self.active_slaves),
            "bus_load_pct": bus_load,
            "slowest_slave": slowest_slave,
            "slowest_avg_ms": slowest_avg,
            "exception_summary": exception_summary,
            "total_bytes": self.total_bytes,
            "nonvalidated_bytes": self.nonvalidated_bytes,
        }

def u16_bytes(value):
    value = int(value) & 0xFFFF
    return bytes(((value >> 8) & 0xFF, value & 0xFF))

def corrupt_crc(frame):
    damaged = bytearray(frame)
    if len(damaged) >= 2:
```

MBToolkit v3.4/mb_protocol.py — lines 881-920

```
881         damaged[-1] ^= 0xFF
882     return bytes(damaged)
883 
884 def build_simulation_transaction(slave, fc, rng):
885     """
886     Build one syntactically valid Modbus request/response pair.
887 
888     Simulation rules:
889     - FC01..FC04 only.
890     - Address random 1..32.
891     - Qty random 1..8.
892     """
893     slave = int(slave)
894     fc = int(fc)
895     start_addr = rng.randint(1, 32)
896     qty = rng.randint(1, 8)
897 
898     if fc in (1, 2):
899         request = append_crc(
900             bytes((slave, fc)) + u16_bytes(start_addr) + u16_bytes(qty)
901         )
902 
903         byte_count = (qty + 7) // 8
904         data = bytes(rng.randrange(256) for _ in range(byte_count))
905         response = append_crc(bytes((slave, fc, byte_count)) + data)
906 
907     elif fc in (3, 4):
908         request = append_crc(
909             bytes((slave, fc)) + u16_bytes(start_addr) + u16_bytes(qty)
910         )
911 
912         values = [rng.randint(0, 65535) for _ in range(qty)]
913         payload = b"".join(u16_bytes(value) for value in values)
914         response = append_crc(
915             bytes((slave, fc, len(payload))) + payload
916         )
917 
918     else:
919         raise ValueError(f"Unsupported simulation FC: {fc}")
920 
```

MBToolkit v3.4/mb_protocol.py — lines 921-960

```python
    return {
        "slave": slave,
        "fc": fc,
        "address": start_addr,
        "qty": qty,
        "request": request,
        "response": response,
    }

def build_random_simulation_plan(rng=None):
    """
    Build exactly four simulation transactions:
    FC01, FC02, FC03 and FC04, one each.

    Four unique random Slave IDs are used, one per transaction.
    """
    if rng is None:
        rng = random.Random()

    slaves = rng.sample(range(1, 248), 4)

    plan = []
    for slave, fc in zip(slaves, SIMULATION_FUNCTION_CODES):
        tx = build_simulation_transaction(slave, fc, rng)
        tx["outcome"] = "response"
        tx["reply"] = tx["response"]
        plan.append(tx)

    return {
        "slaves": slaves,
        "transactions": plan,
    }



def build_anomaly_simulation_plan(rng=None):
    """
    Build a deterministic set of anomaly categories for UI colour preview.

    Used whenever debug simulation is executed. "Realçar resultados" controls
```

MBToolkit v3.4/mb_protocol.py — lines 961-1000

```
    only whether these rows receive colours; it does not change the generated
    scenario.

    It demonstrates:
      - normal response
      - slow response
      - Modbus exception
      - CRC error
      - timeout
      - RAW/UNPARSED bytes
    """
    if rng is None:
        rng = random.Random()

    slaves = rng.sample(range(1, 248), 5)

    normal = build_simulation_transaction(slaves[0], 1, rng)
    normal["outcome"] = "response"
    normal["reply"] = normal["response"]

    slow = build_simulation_transaction(slaves[1], 2, rng)
    slow["outcome"] = "slow"
    slow["reply"] = slow["response"]

    exception = build_simulation_transaction(slaves[2], 3, rng)
    exception["outcome"] = "exception"
    exception["reply"] = append_crc(
        bytes((slaves[2], 0x83, 0x02))
    )

    crc_error = build_simulation_transaction(slaves[3], 4, rng)
    crc_error["outcome"] = "crc"
    crc_error["reply"] = corrupt_crc(crc_error["response"])

    timeout = build_simulation_transaction(slaves[4], 3, rng)
    timeout["outcome"] = "timeout"
    timeout["reply"] = None

    return {
        "slaves": slaves,
```

MBToolkit v3.4/mb_protocol.py — lines 1001-1007

```
        "transactions": [normal, slow, exception, crc_error, timeout],
        # Four bytes are deliberately too short to be accepted as the FC=0xFF
        # exception frame candidate, producing a visible RAW/UNPARSED row.
        "raw_noise": bytes((0x00, 0x00, 0x00, 0x00)),
        "slow_delay_seconds": (SLOW_RESPONSE_THRESHOLD_MS / 1000.0) + 0.15,
    }

```

MBToolkit v3.4/mb_capture.py

902 source lines.

MBToolkit v3.4/mb_capture.py — lines 1-40

```python
#!/usr/bin/env python3
"""Serial-port management, passive capture, simulation and event processing."""

import queue
import random
import threading
import time
import traceback
from datetime import datetime

import tkinter as tk
from tkinter import messagebox

from mb_config import (
    APP_NAME,
    APP_VERSION,
    MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
    MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
    SNIFFER_SPLIT_FRAME_HOLD_MS,
    UI_QUEUE_BATCH_SIZE,
    UI_QUEUE_IDLE_INTERVAL_MS,
    UI_QUEUE_BUSY_INTERVAL_MS,
    debug_sim,
)
from mb_protocol import (
    parse_burst,
    burst_may_continue,
    build_anomaly_simulation_plan,
    modbus_rtu_interframe_gap_seconds,
)
from mb_slave_finder import cancel_serial_io

class CaptureMixin:
    """Methods for serial capture, COM management and simulation."""

    def set_sniffer_refresh_state(self, state):
        """Apply the same state to both Sniffer COM refresh buttons."""
        for name in ("refresh_a_btn", "refresh_b_btn"):
            button = getattr(self, name, None)
            if button is not None:
```

MBToolkit v3.4/mb_capture.py — lines 41-80

```
41                 button.configure(state=state)
42
43     def update_mode_ui(self):
44         mode = self.mode_var.get()
45         dual = mode == "RS232 dual RX"
46
47         state = "readonly" if dual else "disabled"
48         self.port_b_combo.configure(state=state)
49
50         if dual:
51             self.mode_help_var.set(
52                 self.tr("RS232 dual RX: COM A escuta uma direção; COM B escuta a direção oposta.")
53             )
54         elif mode == "RS232 single RX":
55             self.mode_help_var.set(
56                 self.tr("RS232 single RX: uma COM escuta apenas uma direção da ligação RS232.")
57             )
58         else:
59             self.mode_help_var.set(
60                 self.tr("RS485 2-wire: uma COM consegue observar pedidos e respostas no mesmo par diferencial.")
61             )
62
63         self.update_com_status()
64
65     def update_gap_state(self):
66         self.gap_entry.configure(
67             state="normal" if self.gap_mode_var.get() == "Manual" else "disabled"
68         )
69
70     def get_serial_modules(self):
71         try:
72             import serial
73             from serial.tools import list_ports
74             return serial, list_ports
75         except ImportError:
76             return None, None
77
78     def refresh_ports(self):
79         _, list_ports = self.get_serial_modules()
80         ports = [p.device for p in list_ports.comports()] if list_ports else []
```

MBToolkit v3.4/mb_capture.py — lines 81-120

```
81
82          self.port_a_combo["values"] = ports
83          self.port_b_combo["values"] = ports
84
85          if ports:
86              if self.port_a_var.get() not in ports:
87                  self.port_a_var.set(ports[0])
88              if len(ports) > 1:
89                  if self.port_b_var.get() not in ports or self.port_b_var.get() == self.port_a_var.get():
90                      self.port_b_var.set(ports[1])
91              elif not self.port_b_var.get():
92                  self.port_b_var.set("")
93          else:
94              self.port_a_var.set("")
95              self.port_b_var.set("")
96
97          if hasattr(self, "discovery_port_combo"):
98              self.discovery_port_combo["values"] = ports
99              current_discovery = self.discovery_port_var.get().strip()
100             if ports:
101                 if current_discovery not in ports:
102                     self.discovery_port_var.set(ports[0])
103             else:
104                 self.discovery_port_var.set("")
105             self.update_discovery_start_state()
106
107         if hasattr(self, "master_port_combo"):
108             self.master_port_combo["values"] = ports
109             if ports:
110                 if self.master_port_var.get().strip() not in ports:
111                     self.master_port_var.set(ports[0])
112             elif not getattr(self, "master_active", False):
113                 self.master_port_var.set("")
114
115         if hasattr(self, "slave_port_combo"):
116             self.slave_port_combo["values"] = ports
117             if ports:
118                 if self.slave_port_var.get().strip() not in ports:
119                     self.slave_port_var.set(ports[0])
120             elif not getattr(self, "slave_active", False):
```

MBToolkit v3.4/mb_capture.py — lines 121-160

```
                self.slave_port_var.set("")

        self.update_com_status()

    def detected_ports(self):
        """Return the set of COM device names currently reported by Windows/pyserial."""
        _, list_ports = self.get_serial_modules()
        if list_ports is None:
            return set()
        try:
            return {p.device for p in list_ports.comports()}
        except Exception:
            return set()

    def port_status_text(self, label, port, enabled=True):
        if not enabled:
            return f"{label} - {self.tr('Não Utilizada')}"
        if not port:
            return f"{label} - {self.tr('Não selecionada')}"

        ser = self.serial_by_port.get(port)
        if ser is not None:
            try:
                if ser.is_open:
                    return f"{label} - {self.tr('Aberta')}"
            except Exception:
                pass

        if port in self.detected_ports():
            return f"{label} - {self.tr('Detetada')}"
        return f"{label} - {self.tr('Não detetada')}"

    def update_com_status(self):
        """
        Update COM status without opening/probing a port.

        Status means:
          - Aberta: currently opened by MBToolkit
          - Detetada: reported by Windows/pyserial, but not opened by MBToolkit
          - Não detetada: selected name is no longer reported by the OS
```

MBToolkit v3.4/mb_capture.py — lines 161-200

```
        """
        if not hasattr(self, "port_a_status_var"):
            return

        mode = self.mode_var.get() if hasattr(self, "mode_var") else "RS485 2-wire"
        dual = mode == "RS232 dual RX"
        self.port_a_status_var.set(
            self.port_status_text("COM A", self.port_a_var.get().strip(), True)
        )
        self.port_b_status_var.set(
            self.port_status_text("COM B", self.port_b_var.get().strip(), dual)
        )

    def poll_com_status(self):
        if self._closing:
            return
        try:
            self.update_com_status()
            if hasattr(self, "stat_requests_var"):
                self.update_stats_labels()
        finally:
            if not self._closing:
                try:
                    self.after(1000, self.poll_com_status)
                except tk.TclError:
                    pass

    def selected_port_plan(self):
        """Return [(channel, port), ...] for the currently selected physical mode."""
        mode = self.mode_var.get()
        port_a = self.port_a_var.get().strip()
        port_b = self.port_b_var.get().strip()

        if not port_a:
            raise ValueError(self.tr("Seleciona a COM A."))

        plan = [("BUS" if mode == "RS485 2-wire" else "A→B", port_a)]

        if mode == "RS232 dual RX":
            if not port_b:
```

MBToolkit v3.4/mb_capture.py — lines 201-240

```
                raise ValueError(self.tr("Seleciona também a COM B."))
            if port_a == port_b:
                raise ValueError(self.tr("COM A e COM B têm de ser diferentes."))
            plan.append(("B→A", port_b))

        return plan

    def open_selected_ports(self, serial, config):
        """Open the selected capture ports and return [(channel, serial_obj), ...]."""
        opened = []
        try:
            for channel, port in self.selected_port_plan():
                opened.append((channel, self.open_serial_port(serial, port, config)))
            return opened
        except Exception:
            for _, ser_obj in opened:
                try:
                    ser_obj.close()
                except Exception:
                    pass
            raise

    def install_opened_ports(self, opened):
        self.serial_objects = [ser_obj for _, ser_obj in opened]
        self.serial_by_port = {
            ser_obj.port: ser_obj for _, ser_obj in opened if getattr(ser_obj, "port", None)
        }
        self.update_com_status()

    def start_reader_threads(self, opened, gap_ms):
        self.reader_threads = []
        generation = self.reader_generation
        for channel, ser_obj in opened:
            t = threading.Thread(
                target=self.reader_loop,
                args=(ser_obj, gap_ms, channel, generation),
                daemon=True
            )
            self.reader_threads.append(t)
            t.start()
```

MBToolkit v3.4/mb_capture.py — lines 241-280

```
241
242     def restart_com_port(self):
243         """
244         Close and reopen the selected COM port(s).
245
246         During an active capture the capture continues in the same session/log.
247         When stopped, the selected COM port(s) are opened and closed once as a
248         restart/test of the application's serial handle.
249         """
250         if self.busy and not self.capture_active:
251             return
252         if self.restart_in_progress:
253             return
254
255         serial, _ = self.get_serial_modules()
256         if serial is None:
257             messagebox.showerror(self.tr("pyserial em falta"), self.tr("pyserial não está instalado."))
258             return
259
260         try:
261             config = self.serial_config(serial)
262             gap_ms = self.calculated_gap_ms()
263             self.selected_port_plan()
264         except Exception as exc:
265             messagebox.showerror(self.tr("Porta COM"), str(exc))
266             return
267
268         self.restart_in_progress = True
269         self.restart_btn.configure(state="disabled")
270
271         if not self.capture_active:
272             opened = []
273             try:
274                 opened = self.open_selected_ports(serial, config)
275                 self.install_opened_ports(opened)
276                 self.update_idletasks()
277             except Exception as exc:
278                 messagebox.showerror(self.tr("Reiniciar COM"), self.trf("Não foi possível reiniciar a porta COM:\n\n{error}", error=exc))
279             finally:
280                 for _, ser_obj in opened:
```

MBToolkit v3.4/mb_capture.py — lines 281-320

```
                    try:
                        ser_obj.close()
                    except Exception:
                        pass
                self.serial_objects = []
                self.serial_by_port = {}
                self.restart_in_progress = False
                self.restart_btn.configure(state="normal")
                self.refresh_ports()
            return

        # Active capture: stop only the current serial-reader generation.
        self.status_var.set(self.tr("● A reiniciar COM"))
        self.reader_generation += 1
        self.stop_event.set()

        for thread in list(self.reader_threads):
            try:
                thread.join(timeout=0.5)
            except Exception:
                pass

        for ser_obj in list(self.serial_objects):
            try:
                ser_obj.close()
            except Exception:
                pass

        self.reader_threads = []
        self.serial_objects = []
        self.serial_by_port = {}
        self.stop_event.clear()

        try:
            opened = self.open_selected_ports(serial, config)
        except Exception as exc:
            self.restart_in_progress = False
            self.capture_active = False
            self.busy = False
            self.stop_event.set()
```

MBToolkit v3.4/mb_capture.py — lines 321-360

```
321             self.finish_capture_ui()
322             self.update_com_status()
323             messagebox.showerror(
324                 self.tr("Reiniciar COM"),
325                 self.trf("Não foi possível reabrir a porta COM. A captura foi parada.\n\n{error}", error=exc)
326             )
327             return
328 
329         self.install_opened_ports(opened)
330         self.stop_event.clear()
331         self.start_reader_threads(opened, gap_ms)
332         self.queue_log_note(
333             f"# COM restart: {datetime.now().isoformat(timespec='milliseconds')}\n"
334         )
335         self.status_var.set(self.tr("● A capturar"))
336         self.restart_in_progress = False
337         self.restart_btn.configure(state="normal")
338         self.update_com_status()
339 
340     def set_simulation_button_state(self, state):
341         """Change Simulation button state only when the optional button exists."""
342         if self.sim_btn is not None:
343             self.sim_btn.configure(state=state)
344 
345     def selected_pending_timeout_seconds(self):
346         try:
347             value = float(self.pending_timeout_var.get().strip().replace(",", "."))
348         except ValueError:
349             raise ValueError(self.tr("Pending timeout inválido."))
350 
351         if not (
352             MIN_PENDING_REQUEST_TIMEOUT_SECONDS
353             <= value
354             <= MAX_PENDING_REQUEST_TIMEOUT_SECONDS
355         ):
356             raise ValueError(
357                 self.trf(
358                     "Pending timeout deve estar entre {minimum:g} e {maximum:g} segundos.",
359                     minimum=MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
360                     maximum=MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
```

MBToolkit v3.4/mb_capture.py — lines 361-400

```
                )
            )

        return value

    def calculated_gap_ms(self):
        if self.gap_mode_var.get() == "Manual":
            value = float(self.gap_ms_var.get().replace(",", "."))
            if value <= 0:
                raise ValueError(self.tr("Frame gap manual inválido."))
            return value

        baud = int(self.baud_var.get())
        data_bits = int(self.data_var.get())
        parity_bits = 0 if self.parity_var.get() == "None" else 1
        stop_bits = float(self.stop_var.get())
        bits_per_char = 1 + data_bits + parity_bits + stop_bits
        return modbus_rtu_interframe_gap_seconds(baud, bits_per_char) * 1000.0

    def serial_config(self, serial):
        byte_map = {
            5: serial.FIVEBITS, 6: serial.SIXBITS,
            7: serial.SEVENBITS, 8: serial.EIGHTBITS,
        }
        parity_map = {
            "None": serial.PARITY_NONE,
            "Even": serial.PARITY_EVEN,
            "Odd": serial.PARITY_ODD,
            "Mark": serial.PARITY_MARK,
            "Space": serial.PARITY_SPACE,
        }
        stop_map = {
            "1": serial.STOPBITS_ONE,
            "1.5": serial.STOPBITS_ONE_POINT_FIVE,
            "2": serial.STOPBITS_TWO,
        }
        return {
            "baudrate": int(self.baud_var.get()),
            "bytesize": byte_map[int(self.data_var.get())],
            "parity": parity_map[self.parity_var.get()],
```

MBToolkit v3.4/mb_capture.py — lines 401-440

```
            "stopbits": stop_map[self.stop_var.get()],
        }

    def open_serial_port(self, serial, port, config):
        ser = serial.Serial()
        ser.port = port
        ser.baudrate = config["baudrate"]
        ser.bytesize = config["bytesize"]
        ser.parity = config["parity"]
        ser.stopbits = config["stopbits"]
        ser.timeout = 0
        ser.write_timeout = 0
        ser.xonxoff = False
        ser.rtscts = False
        ser.dsrdtr = False

        try:
            ser.rts = False
            ser.dtr = False
        except Exception:
            pass

        ser.open()

        try:
            ser.rts = False
            ser.dtr = False
        except Exception:
            pass

        try:
            ser.reset_input_buffer()
        except Exception:
            pass

        return ser

    def start_capture(self):
        if self.busy:
            return
```

MBToolkit v3.4/mb_capture.py — lines 441-480

```
441
442         serial, _ = self.get_serial_modules()
443         if serial is None:
444             messagebox.showerror(
445                 self.tr("pyserial em falta"),
446                 self.tr("pyserial não está instalado.\n\nO build_exe.bat inclui-o automaticamente no EXE.")
447             )
448             return
449
450         mode = self.mode_var.get()
451         port_a = self.port_a_var.get().strip()
452         port_b = self.port_b_var.get().strip()
453
454         try:
455             gap_ms = self.calculated_gap_ms()
456             pending_timeout_s = self.selected_pending_timeout_seconds()
457             config = self.serial_config(serial)
458             opened = self.open_selected_ports(serial, config)
459         except Exception as exc:
460             messagebox.showerror(self.tr("Erro ao abrir porta"), str(exc))
461             self.update_com_status()
462             return
463
464         self.reader_generation += 1
465         self.install_opened_ports(opened)
466         self.reader_threads = []
467         self.stop_event.clear()
468         self.busy = True
469         self.capture_active = True
470         self.metrics.pending_timeout_seconds = pending_timeout_s
471         self.reset_session_metrics()
472
473         capture_started = datetime.now()
474         log_header = (
475             f"# {APP_NAME} v{APP_VERSION}\n"
476             f"# Start: {capture_started.isoformat(timespec='milliseconds')}\n"
477             f"# Mode: {mode}\n"
478             f"# COM A: {port_a}\n"
479             + (f"# COM B: {port_b}\n" if mode == "RS232 dual RX" else "")
480             + f"# Serial: baud={self.baud_var.get()} data={self.data_var.get()} "
```

MBToolkit v3.4/mb_capture.py — lines 481-520

```
481              f"parity={self.parity_var.get()} stop={self.stop_var.get()} "
482              f"frame_gap_ms={gap_ms:.3f}\n"
483            + f"# Pending timeout: {pending_timeout_s:g} s\n"
484            + "# Passive/read-only: no Serial.write() calls.\n#\n"
485        )
486        self.prepare_log_session("MBToolkit", capture_started, log_header)
487
488        self.status_var.set(self.tr("● A capturar"))
489        self.start_btn.configure(state="disabled")
490        self.stop_btn.configure(state="normal")
491        self.restart_btn.configure(state="normal")
492        self.set_simulation_button_state("disabled")
493        self.set_sniffer_refresh_state("disabled")
494
495        self.start_reader_threads(opened, gap_ms)
496        self.update_com_status()
497
498    def reader_loop(self, ser, gap_ms, channel, generation):
499        gap_s = gap_ms / 1000.0
500        hold_s = max(gap_s, SNIFFER_SPLIT_FRAME_HOLD_MS / 1000.0)
501        poll_sleep = min(max(gap_s / 8.0, 0.0002), 0.001)
502
503        buffer = bytearray()
504        buffer_wall = None
505        # (byte offset, epoch) of every read, so frames merged into one burst
506        # by USB latency keep their own arrival times.
507        chunk_times = []
508        last_rx = None
509        # Burst length already checked by burst_may_continue(); avoids
510        # re-parsing an unchanged buffer on every poll of the hold window.
511        hold_checked_len = None
512        hold_active = False
513
514        def flush_buffer():
515            ts = (
516                buffer_wall.strftime("%H:%M:%S.%f")[:-3]
517                if buffer_wall else datetime.now().strftime("%H:%M:%S.%f")[:-3]
518            )
519            event_epoch = (
520                buffer_wall.timestamp() if buffer_wall else time.time()
```

MBToolkit v3.4/mb_capture.py — lines 521-560

```
            )
            for row in parse_burst(
                bytes(buffer), ts, channel, event_epoch, chunk_times
            ):
                self.event_queue.put(("frame", row))

        try:
            while not self.stop_event.is_set():
                waiting = ser.in_waiting

                if waiting:
                    chunk = ser.read(waiting)
                    if chunk:
                        if not buffer:
                            buffer_wall = datetime.now()
                            chunk_times.append((0, buffer_wall.timestamp()))
                        else:
                            chunk_times.append((len(buffer), time.time()))
                        buffer.extend(chunk)
                        last_rx = time.perf_counter()

                elif buffer and last_rx is not None:
                    idle_s = time.perf_counter() - last_rx
                    if idle_s >= gap_s:
                        if idle_s < hold_s:
                            if hold_checked_len != len(buffer):
                                hold_checked_len = len(buffer)
                                hold_active = burst_may_continue(bytes(buffer))
                            if hold_active:
                                time.sleep(poll_sleep)
                                continue

                        flush_buffer()
                        buffer.clear()
                        chunk_times.clear()
                        buffer_wall = None
                        last_rx = None
                        hold_checked_len = None
                        hold_active = False

```

MBToolkit v3.4/mb_capture.py — lines 561-600

```
561                 time.sleep(poll_sleep)
562
563         except Exception as exc:
564             if not self.stop_event.is_set():
565                 self.event_queue.put(("error", (generation, channel, str(exc))))
566
567         finally:
568             if buffer:
569                 flush_buffer()
570
571             try:
572                 ser.close()
573             except Exception:
574                 pass
575             self.event_queue.put(("reader_stopped", (generation, channel)))
576
577     def stop_capture(self):
578         if self.capture_active:
579             self.status_var.set(self.tr("● A parar"))
580         self.stop_event.set()
581
582     def run_simulation(self):
583         if debug_sim != 1:
584             return
585
586         if self.busy:
587             return
588
589         try:
590             pending_timeout_s = self.selected_pending_timeout_seconds()
591         except Exception as exc:
592             messagebox.showerror(self.tr("Configuração inválida"), str(exc))
593             return
594
595         self.busy = True
596         self.metrics.pending_timeout_seconds = pending_timeout_s
597         self.reset_session_metrics()
598
599         # Simulation is GUI-only. Never create or write a .txt log.
600         self.close_log()
```

MBToolkit v3.4/mb_capture.py — lines 601-640

```
601
602         self.status_var.set(self.tr("● A simular"))
603         self.start_btn.configure(state="disabled")
604         self.restart_btn.configure(state="disabled")
605         self.set_simulation_button_state("disabled")
606         self.set_sniffer_refresh_state("disabled")
607
608         def worker():
609             try:
610                 # Simulation always exercises the complete diagnostic scenario.
611                 # "Realçar resultados" affects only Treeview row colouring; it
612                 # never changes which simulated frames are generated.
613                 rng = random.Random()
614                 plan = build_anomaly_simulation_plan(rng=rng)
615
616                 for tx in plan["transactions"]:
617                     time.sleep(rng.uniform(0.025, 0.085))
618
619                     request_now = datetime.now()
620                     request_ts = request_now.strftime("%H:%M:%S.%f")[:-3]
621
622                     request_channel = "SIM"
623
624                     for row in parse_burst(
625                         tx["request"], request_ts, request_channel,
626                         request_now.timestamp()
627                     ):
628                         self.event_queue.put(("frame", row))
629
630                     outcome = tx.get("outcome", "response")
631
632                     if outcome == "timeout":
633                         # Do not sleep for the user-configured Pending timeout.
634                         # The queue event expires this exact pending request
635                         # through SessionMetrics and marks the REQUEST row.
636                         self.event_queue.put((
637                             "simulation_force_timeout",
638                             (tx["slave"], tx["fc"]),
639                         ))
640                         continue
```

MBToolkit v3.4/mb_capture.py — lines 641-680

```
641
642                     if outcome == "slow":
643                         time.sleep(plan.get("slow_delay_seconds", 0.65))
644                     else:
645                         time.sleep(rng.uniform(0.018, 0.095))
646
647                     reply_now = datetime.now()
648                     reply_ts = reply_now.strftime("%H:%M:%S.%f")[:-3]
649                     reply_channel = "SIM"
650
651                     reply = tx.get("reply")
652                     if reply:
653                         for row in parse_burst(
654                             reply, reply_ts, reply_channel,
655                             reply_now.timestamp()
656                         ):
657                             self.event_queue.put(("frame", row))
658
659                 # One RAW row completes the diagnostic scenario. With anomaly
660                 # highlighting OFF it remains a normal, uncoloured RAW row;
661                 # with highlighting ON the same row receives the RAW colour.
662                 raw_now = datetime.now()
663                 raw_ts = raw_now.strftime("%H:%M:%S.%f")[:-3]
664                 raw_channel = "SIM"
665                 for row in parse_burst(
666                     plan["raw_noise"],
667                     raw_ts,
668                     raw_channel,
669                     raw_now.timestamp(),
670                 ):
671                     self.event_queue.put(("frame", row))
672
673                 self.event_queue.put(("simulation_done", None))
674
675             except Exception as exc:
676                 # Never leave the GUI permanently busy if the simulation worker
677                 # fails unexpectedly.
678                 self.event_queue.put((
679                     "simulation_error",
680                     f"{type(exc).__name__}: {exc}"
```

MBToolkit v3.4/mb_capture.py — lines 681-720

```
                ))

        threading.Thread(target=worker, daemon=True).start()

    def process_queue(self):
        if self._closing:
            return

        delay = UI_QUEUE_IDLE_INTERVAL_MS
        try:
            events = []
            for _ in range(UI_QUEUE_BATCH_SIZE):
                try:
                    events.append(self.event_queue.get_nowait())
                except queue.Empty:
                    break

            frame_batch = []

            def flush_frames():
                nonlocal frame_batch
                if frame_batch:
                    self.add_frame_rows_batch(frame_batch)
                    frame_batch = []

            for event, payload in events:
                if event == "frame":
                    frame_batch.append(payload)
                    continue

                flush_frames()

                if event == "error":
                    generation, channel, error_text = payload
                    if generation != self.reader_generation:
                        continue
                    # Stop all readers in this capture generation after any
                    # reader failure; never leave a half-alive dual-RX capture.
                    self.stop_event.set()
                    self.update_com_status()
```

MBToolkit v3.4/mb_capture.py — lines 721-760

```
                    messagebox.showerror(
                        self.tr("Erro de captura"), f"{channel}: {error_text}"
                    )

                elif event == "reader_stopped":
                    generation, _channel = payload
                    if generation != self.reader_generation:
                        continue
                    if not any(t.is_alive() for t in self.reader_threads):
                        self.finish_capture_ui()

                elif event == "simulation_force_timeout":
                    slave, fc = payload
                    expired = self.metrics.force_timeout(slave, fc)
                    if expired:
                        self.mark_timed_out_records(expired, rerender=True)
                    self.update_stats_labels()

                elif event == "simulation_done":
                    self.busy = False
                    self.status_var.set(self.tr("● Simulação concluída"))
                    self.start_btn.configure(state="normal")
                    self.stop_btn.configure(state="disabled")
                    self.restart_btn.configure(state="normal")
                    self.set_simulation_button_state("normal")
                    self.set_sniffer_refresh_state("normal")
                    self.close_log()
                    self.update_com_status()

                elif event == "simulation_error":
                    self.busy = False
                    self.status_var.set(self.tr("● Erro na simulação"))
                    self.start_btn.configure(state="normal")
                    self.stop_btn.configure(state="disabled")
                    self.restart_btn.configure(state="normal")
                    self.set_simulation_button_state("normal")
                    self.set_sniffer_refresh_state("normal")
                    self.close_log()
                    self.update_com_status()
                    messagebox.showerror(self.tr("Erro na simulação"), str(payload))
```

MBToolkit v3.4/mb_capture.py — lines 761-800

```
761
762                 elif event == "discovery_status":
763                     self.discovery_status_var.set(str(payload))
764
765                 elif event == "discovery_progress":
766                     self.apply_discovery_progress(payload)
767
768                 elif event == "discovery_found":
769                     self.add_discovery_result(payload)
770
771                 elif event == "discovery_done":
772                     self.finish_discovery_ui(payload)
773
774                 elif event.startswith("master_"):
775                     self.handle_master_event(event, payload)
776
777                 elif event.startswith("slave_"):
778                     self.handle_slave_event(event, payload)
779
780             flush_frames()
781
782             if self.metrics.expire_pending(time.time()):
783                 expired = self.metrics.pop_expired_requests()
784                 if expired:
785                     self.mark_timed_out_records(expired, rerender=True)
786                 self.update_stats_labels()
787
788             if (
789                 self.capture_active
790                 and not self.restart_in_progress
791                 and self.stop_event.is_set()
792                 and self.reader_threads
793                 and not any(t.is_alive() for t in self.reader_threads)
794                 and self.stop_btn["state"] != "disabled"
795             ):
796                 self.finish_capture_ui()
797
798             delay = (
799                 UI_QUEUE_BUSY_INTERVAL_MS
800                 if len(events) >= UI_QUEUE_BATCH_SIZE
```

MBToolkit v3.4/mb_capture.py — lines 801-840

```
                else UI_QUEUE_IDLE_INTERVAL_MS
            )

        except tk.TclError:
            if self._closing:
                return
            traceback.print_exc()
            delay = UI_QUEUE_IDLE_INTERVAL_MS

        except Exception as exc:
            # One unexpected event/UI error must not permanently kill the
            # periodic queue callback and make the application appear frozen.
            traceback.print_exc()
            if not self._process_queue_error_shown and not self._closing:
                self._process_queue_error_shown = True
                try:
                    messagebox.showerror(
                        self.tr("Erro interno"),
                        self.trf(
                            "Ocorreu um erro ao processar eventos da interface. O processamento continuará.\n\n{error}",
                            error=f"{type(exc).__name__}: {exc}",
                        )
                    )
                except Exception:
                    pass
            delay = UI_QUEUE_IDLE_INTERVAL_MS

        finally:
            if not self._closing:
                try:
                    self.after(delay, self.process_queue)
                except tk.TclError:
                    pass

    def on_close(self):
        if self._closing:
            return

        # A changed Master / Slave workspace may be saved first; "Cancel"
        # keeps the application open.
```

MBToolkit v3.4/mb_capture.py — lines 841-880

```
841         for name in ("master_confirm_close", "slave_confirm_close"):
842             confirm = getattr(self, name, None)
843             if callable(confirm) and not confirm():
844                 return
845 
846         try:
847             saver = getattr(self, "save_current_ui_settings", None)
848             if callable(saver):
849                 saver()
850         except Exception:
851             pass
852 
853         self._closing = True
854         self.busy = False
855         self.capture_active = False
856         self.stop_event.set()
857         self.discovery_stop_event.set()
858 
859         # Cancel driver I/O before joining workers so shutdown is not held up
860         # by a serial read/write timeout.
861         cancel_serial_io(self.discovery_serial)
862         for name in ("master_shutdown", "slave_shutdown"):
863             stopper = getattr(self, name, None)
864             if callable(stopper):
865                 stopper()
866 
867         if self.discovery_thread is not None:
868             try:
869                 if self.discovery_thread.is_alive():
870                     self.discovery_thread.join(timeout=0.8)
871             except Exception:
872                 pass
873 
874         if self.discovery_serial is not None:
875             try:
876                 self.discovery_serial.close()
877             except Exception:
878                 pass
879         self.discovery_serial = None
880 
```

MBToolkit v3.4/mb_capture.py — lines 881-902

```
        for ser in list(self.serial_objects):
            cancel_serial_io(ser)

        for thread in list(self.reader_threads):
            try:
                if thread.is_alive():
                    thread.join(timeout=0.5)
            except Exception:
                pass

        for ser in list(self.serial_objects):
            try:
                ser.close()
            except Exception:
                pass

        self.close_log()
        try:
            self.destroy()
        except tk.TclError:
            pass

```

MBToolkit v3.4/mb_diagnostics.py

286 source lines.

MBToolkit v3.4/mb_diagnostics.py — lines 1-40

```python
#!/usr/bin/env python3
"""Diagnostic helpers for MBToolkit traffic inspection and anomaly display."""

from mb_config import MODBUS_EXCEPTION_CODES, SLOW_RESPONSE_THRESHOLD_MS
from mb_protocol import crc16_modbus, read_bits_from_response
from mb_i18n import LANGUAGE_PORTUGUESE, translate_details, translate_text


FUNCTION_NAMES = {
    0x01: "Read Coils",
    0x02: "Read Discrete Inputs",
    0x03: "Read Holding Registers",
    0x04: "Read Input Registers",
    0x05: "Write Single Coil",
    0x06: "Write Single Register",
    0x07: "Read Exception Status",
    0x08: "Diagnostics",
    0x0F: "Write Multiple Coils",
    0x10: "Write Multiple Registers",
    0x16: "Mask Write Register",
    0x17: "Read/Write Multiple Registers",
    0x2B: "Encapsulated Interface Transport",
}


def raw_hex_to_bytes(raw):
    try:
        return bytes.fromhex(str(raw or ""))
    except (TypeError, ValueError):
        return b""


def fc_number(record):
    text = str(record.get("fc", "") or "").strip()
    try:
        value = int(text, 16)
    except (TypeError, ValueError):
        return None
    if record.get("type") == "EXCEPTION":
        value &= 0x7F
```

MBToolkit v3.4/mb_diagnostics.py — lines 41-80

```
41      return value
42
43
44  def function_label(record, language=LANGUAGE_PORTUGUESE):
45      fc = fc_number(record)
46      if fc is None:
47          return "—"
48      name = FUNCTION_NAMES.get(fc, translate_text("Desconhecida/outra", language))
49      return f"0x{fc:02X} — {name}"
50
51
52  def crc_fields(record):
53      if str(record.get("type", "")).startswith("RAW"):
54          return None, None, str(record.get("crc", "?"))
55
56      frame = raw_hex_to_bytes(record.get("raw"))
57      if len(frame) < 4:
58          return None, None, str(record.get("crc", "?"))
59
60      received = frame[-2] | (frame[-1] << 8)
61      calculated = crc16_modbus(frame[:-2])
62      return received, calculated, ("OK" if received == calculated else "ERROR")
63
64
65  def response_payload(record, language=LANGUAGE_PORTUGUESE):
66      """Return (byte_count, data_hex, decoded_data) where available."""
67      frame = raw_hex_to_bytes(record.get("raw"))
68      if len(frame) < 3:
69          return None, "", ""
70
71      kind = str(record.get("type", ""))
72      fc_raw = frame[1] if len(frame) >= 2 else 0
73      fc = fc_raw & 0x7F
74
75      if kind == "EXCEPTION":
76          code = frame[2] if len(frame) >= 3 else None
77          if code is None:
78              return None, "", "Exception"
79          info = MODBUS_EXCEPTION_CODES.get(code)
80          name = info[0] if info else "Unknown/Reserved Exception"
```

MBToolkit v3.4/mb_diagnostics.py — lines 81-120

```
        return None, "", f"0x{code:02X} — {name}"

    if kind == "RESPONSE" and fc in (1, 2, 3, 4) and len(frame) >= 5:
        byte_count = frame[2]
        data = frame[3:3 + byte_count]
        data_hex = " ".join(f"{b:02X}" for b in data)
        decoded = ""
        start = record.get("request_pdu_address")
        if fc in (3, 4) and byte_count % 2 == 0 and len(data) == byte_count:
            regs = [
                (data[i] << 8) | data[i + 1]
                for i in range(0, len(data), 2)
            ]
            if start is not None:
                # The matched request gives the address of every value.
                pairs = ", ".join(
                    f"{start + i}={value}" for i, value in enumerate(regs)
                )
                decoded = translate_details(
                    f"Registos (PDU Address=valor): {pairs}", language
                )
            else:
                decoded = translate_details(f"Registos={regs}", language)
        elif fc in (1, 2) and len(data) == byte_count:
            bits = read_bits_from_response(frame, record.get("request_qty"))
            if start is not None:
                pairs = ", ".join(
                    f"{start + i}={value}" for i, value in enumerate(bits)
                )
                decoded = translate_details(
                    f"Bits (PDU Address=valor): {pairs}", language
                )
            else:
                decoded = "Bits=" + "".join(str(bit) for bit in bits)
        return byte_count, data_hex, decoded

    if kind == "REQUEST" and fc in (15, 16) and len(frame) >= 9:
        byte_count = frame[6]
        data = frame[7:7 + byte_count]
        return byte_count, " ".join(f"{b:02X}" for b in data), ""
```

MBToolkit v3.4/mb_diagnostics.py — lines 121-160

```
121
122     if kind == "REQUEST" and fc == 23 and len(frame) >= 13:
123         byte_count = frame[10]
124         data = frame[11:11 + byte_count]
125         return byte_count, " ".join(f"{b:02X}" for b in data), ""
126
127     return record.get("byte_count"), str(record.get("data_hex", "") or ""), ""
128
129
130 def anomaly_kind(record, threshold_ms=SLOW_RESPONSE_THRESHOLD_MS):
131     """
132     Return one anomaly class or None.
133
134     Priority avoids ambiguous colouring when one frame satisfies several
135     conditions. CRC is strongest because the frame itself is invalid.
136     """
137     if str(record.get("crc", "")) == "ERROR":
138         return "crc"
139     if bool(record.get("timed_out")):
140         return "timeout"
141     if str(record.get("type", "")) == "EXCEPTION":
142         return "exception"
143     if str(record.get("type", "")).startswith("RAW"):
144         return "raw"
145
146     try:
147         response_ms = float(record.get("response_text", ""))
148     except (TypeError, ValueError):
149         response_ms = None
150
151     if response_ms is not None and response_ms >= float(threshold_ms):
152         return "slow"
153     return None
154
155
156 RESULT_LABELS = {
157     "success": ("✓ Sucesso", "Verde"),
158     "slow": ("⚠ Resposta lenta", "Amarelo"),
159     "exception": ("Exception", "Vermelho"),
160     "crc": ("× CRC Error", "Vermelho"),
```

MBToolkit v3.4/mb_diagnostics.py — lines 161-200

```
    "timeout": ("⚠ Timeout", "Laranja"),
    "raw": ("RAW", "Roxo"),
    "pending": ("Pendente / sem resultado", "Neutro"),
    "neutral": ("Sem classificação", "Neutro"),
}


def result_info(record, successful_request=False, language=LANGUAGE_PORTUGUESE):
    """
    Return (result_key, human_label, colour_name) for one traffic record.

    The semantic result is always available to the Frame Inspector. Whether
    the table/Inspector actually uses the category colour is controlled by
    "Realçar resultados".
    """
    if not record:
        return "neutral", "—", translate_text("Neutro", language)

    kind = anomaly_kind(record, SLOW_RESPONSE_THRESHOLD_MS)
    if kind:
        label, colour = RESULT_LABELS[kind]
        if kind == "raw":
            # Preserve the exact parser state in the Inspector.
            row_type = str(record.get("type", "RAW") or "RAW")
            label = row_type
        return kind, translate_text(label, language), translate_text(colour, language)

    row_type = str(record.get("type", ""))

    if (
        row_type == "RESPONSE"
        and record.get("crc") == "OK"
        and record.get("matched")
    ):
        label, colour = RESULT_LABELS["success"]
        return "success", translate_text(label, language), translate_text(colour, language)

    if row_type == "REQUEST":
        if successful_request and not record.get("timed_out"):
            label, colour = RESULT_LABELS["success"]
```

MBToolkit v3.4/mb_diagnostics.py — lines 201-240

```
            return "success", translate_text(label, language), translate_text(colour, language)

        label, colour = RESULT_LABELS["pending"]
        return "pending", translate_text(label, language), translate_text(colour, language)

    label, colour = RESULT_LABELS["neutral"]
    return "neutral", translate_text(label, language), translate_text(colour, language)


def inspector_values(record, successful_request=False, language=LANGUAGE_PORTUGUESE):
    if not record:
        return {
            "slave": "Slave: —",
            "type": f"{translate_text('Tipo', language)}: —",
            "function": "Function: —",
            "response": "Resp.: —",
            "result": f"{translate_text('Resultado', language)}: —",
            "result_key": "neutral",
            "pdu": "PDU Address: —",
            "one_based": "1-based: —",
            "qty": "Qty: —",
            "byte_count": "ByteCount: —",
            "crc": "CRC: —",
            "data": f"{translate_text('Dados', language)}: —",
            "raw": "",
        }

    received, calculated, crc_state = crc_fields(record)
    if received is None:
        crc_text = f"CRC: {crc_state or '—'}"
    else:
        crc_text = (
            (f"CRC: received 0x{received:04X} / calculated 0x{calculated:04X} — {crc_state}" if language == "English" else f"CRC: recebido 0x{received:04X} / calculado 0x{calculated:04X} — {crc_state}")
        )

    byte_count, data_hex, decoded = response_payload(record, language=language)
    payload = decoded or data_hex or str(record.get("details", "") or "—")

    response = str(record.get("response_text", "") or "")
    if response:
```

MBToolkit v3.4/mb_diagnostics.py — lines 241-280

```
        response += " ms"

    timeout_suffix = " — TIMEOUT" if record.get("timed_out") else ""
    result_key, result_label, _result_colour = result_info(
        record,
        successful_request=successful_request,
        language=language,
    )

    return {
        "slave": f"Slave: {record.get('slave') or '—'}",
        "type": f"{translate_text('Tipo', language)}: {record.get('type') or '—'}{timeout_suffix}",
        "function": f"Function: {function_label(record, language=language)}",
        "response": f"Resp.: {response or '—'}",
        "result": f"{translate_text('Resultado', language)}: {result_label}",
        "result_key": result_key,
        "pdu": f"PDU Address: {record.get('pdu_address') if record.get('pdu_address') is not None else '—'}",
        "one_based": f"1-based: {record.get('one_based') if record.get('one_based') is not None else '—'}",
        "qty": f"Qty: {record.get('qty') if record.get('qty') is not None else '—'}",
        "byte_count": f"ByteCount: {byte_count if byte_count is not None else '—'}",
        "crc": crc_text,
        "data": f"{translate_text('Dados', language)}: {translate_details(payload, language)}",
        "raw": str(record.get("raw", "") or ""),
    }


def decoded_record_text(record, successful_request=False, language=LANGUAGE_PORTUGUESE):
    values = inspector_values(
        record,
        successful_request=successful_request,
        language=language,
    )
    return "\n".join((
        values["slave"],
        values["type"],
        values["function"],
        values["response"],
        values["result"],
        values["pdu"],
        values["one_based"],
```

MBToolkit v3.4/mb_diagnostics.py — lines 281-286

```
        values["qty"],
        values["byte_count"],
        values["crc"],
        values["data"],
        f"Raw: {values['raw']}",
    ))
```

MBToolkit v3.4/mb_view.py

1871 source lines.

MBToolkit v3.4/mb_view.py — lines 1-40

```python
#!/usr/bin/env python3
"""Traffic rendering, filtering, Raw Hex view, statistics and log handling."""

import csv
import math
import os
import subprocess
import sys
import time
from datetime import datetime

import tkinter as tk
from tkinter import filedialog, messagebox

from mb_config import (
    MAX_UI_FRAMES,
    MAX_RAW_TEXT_LINES,
    UI_PRUNE_CHUNK_FRAMES,
    RAW_TEXT_PRUNE_CHUNK_LINES,
    SLOW_RESPONSE_THRESHOLD_MS,
    get_application_directory,
    get_logs_folder,
)
from mb_diagnostics import (
    anomaly_kind,
    decoded_record_text,
    inspector_values,
)
from mb_i18n import canonical_ui_text, translate_details
from mb_protocol import read_bits_details

# Index of "Abrir no Master" in the Traffic context menu (built in mb_gui.py).
TRAFFIC_MENU_OPEN_IN_MASTER = 8


class ViewMixin:
    """Methods for rendering, filtering, statistics and logging."""

    def reset_session_metrics(self):
        self.metrics.reset()
```

MBToolkit v3.4/mb_view.py — lines 41-80

```
41          self.update_stats_labels()
42
43      def update_stats_labels(self):
44          self.stat_requests_var.set(f"{self.tr('Pedidos')}: {self.metrics.requests}")
45          self.stat_responses_var.set(f"{self.tr('Respostas')}: {self.metrics.responses}")
46          self.stat_pending_var.set(f"{self.tr('Pendentes')}: {self.metrics.pending_count()}")
47          self.stat_timeouts_var.set(f"Timeouts: {self.metrics.timeouts}")
48          self.stat_crc_var.set(f"{self.tr('Erros de CRC')}: {self.metrics.crc_errors}")
49          self.stat_exceptions_var.set(f"{self.tr('Exceções')}: {self.metrics.exceptions}")
50          self.stat_raw_var.set(f"RAW: {self.metrics.raw}")
51
52          try:
53              health_now = (
54                  time.time()
55                  if getattr(self, "capture_active", False) or getattr(self, "busy", False)
56                  else self.metrics.last_event_time
57              )
58              summary = self.metrics.health_summary(
59                  baud=self.baud_var.get(),
60                  data_bits=self.data_var.get(),
61                  parity=self.parity_var.get(),
62                  stop_bits=self.stop_var.get(),
63                  now=health_now,
64              )
65          except Exception:
66              summary = self.metrics.health_summary()
67
68          def ms(value):
69              return "–" if value is None else f"{value:.1f} ms"
70
71          if hasattr(self, "stat_slaves_var"):
72              self.stat_slaves_var.set(f"Slaves: {summary['active_slaves']}")
73
74          mappings = {
75              "health_requests_var": str(self.metrics.requests),
76              "health_responses_var": str(self.metrics.responses),
77              "health_crc_count_var": str(self.metrics.crc_errors),
78              "health_timeout_count_var": str(self.metrics.timeouts),
79              "health_exception_count_var": str(self.metrics.exceptions),
80              "health_req_rate_var": f"{summary['request_rate']:.2f} req/s",
```

MBToolkit v3.4/mb_view.py — lines 81-120

```
            "health_resp_avg_var": ms(summary["response_avg_ms"]),
            "health_resp_min_var": ms(summary["response_min_ms"]),
            "health_resp_max_var": ms(summary["response_max_ms"]),
            "health_p95_var": ms(summary["response_p95_ms"]),
            "health_nonvalidated_rate_var": f"{summary['nonvalidated_rate_pct']:.2f} %",
            "health_timeout_rate_var": f"{summary['timeout_rate_pct']:.2f} %",
            "health_bus_load_var": f"{summary['bus_load_pct']:.2f} %",
            "health_slaves_var": str(summary["active_slaves"]),
            "health_total_bytes_var": str(summary["total_bytes"]),
            "health_exception_slaves_var": summary["exception_summary"],
        }

        slowest_slave = summary["slowest_slave"]
        if slowest_slave is None:
            mappings["health_slowest_var"] = "—"
        else:
            mappings["health_slowest_var"] = (
                self.trf(
                    "Slave {slave} — média {average:.1f} ms",
                    slave=slowest_slave,
                    average=summary["slowest_avg_ms"],
                )
            )

        for attr, value in mappings.items():
            var = getattr(self, attr, None)
            if var is not None:
                var.set(value)

    # ---------------------------------------------------------------------
    # Bus activity, filters, anomaly highlighting and Frame Inspector.
    # ---------------------------------------------------------------------
    def pulse_bus_activity(self):
        var = getattr(self, "bus_activity_var", None)
        if var is None:
            return

        var.set("BUS ● RX")
        previous = getattr(self, "_bus_activity_after_id", None)
        if previous is not None:
```

MBToolkit v3.4/mb_view.py — lines 121-160

```
            try:
                self.after_cancel(previous)
            except tk.TclError:
                pass

        try:
            self._bus_activity_after_id = self.after(
                180,
                self._clear_bus_activity_pulse,
            )
        except tk.TclError:
            self._bus_activity_after_id = None

    def _clear_bus_activity_pulse(self):
        self._bus_activity_after_id = None
        var = getattr(self, "bus_activity_var", None)
        if var is not None:
            var.set("BUS ○")

    def schedule_advanced_filter_refresh(self, *_args):
        previous = getattr(self, "_advanced_filter_after_id", None)
        if previous is not None:
            try:
                self.after_cancel(previous)
            except tk.TclError:
                pass
        try:
            self._advanced_filter_after_id = self.after(
                120,
                self.render_filtered_view,
            )
        except tk.TclError:
            self._advanced_filter_after_id = None

    def clear_advanced_filters(self):
        pending = getattr(self, "_advanced_filter_after_id", None)
        if pending is not None:
            try:
                self.after_cancel(pending)
            except tk.TclError:
```

MBToolkit v3.4/mb_view.py — lines 161-200

```
                pass
            self._advanced_filter_after_id = None

        self.advanced_filter_type_var.set(self.tr("Todos"))
        self.advanced_filter_fc_var.set("")
        self.advanced_filter_response_ms_var.set("")
        self.advanced_filter_text_var.set("")
        self._transaction_filter_seqs = None

        self.slave_filter_all_mode = True
        for var in self.slave_filter_vars.values():
            var.set(True)
        self.update_slave_heading()

        self.render_filtered_view()

    @staticmethod
    def normalized_fc_filter(value):
        text = str(value or "").strip().upper()
        if not text or text == "TODOS":
            return ""
        if text.startswith("0X"):
            text = text[2:]
        try:
            number = int(text, 16)
        except ValueError:
            return "__INVALID__"
        if not 0 <= number <= 0xFF:
            return "__INVALID__"
        return f"{number:02X}"

    def record_passes_advanced_filters(self, record):
        transaction_filter = getattr(self, "_transaction_filter_seqs", None)
        if transaction_filter is not None:
            if record.get("seq") not in transaction_filter:
                return False

        kind = canonical_ui_text(self.advanced_filter_type_var.get())
        row_type = str(record.get("type", ""))

```

MBToolkit v3.4/mb_view.py — lines 201-240

```
        if kind == "Requests" and row_type != "REQUEST":
            return False
        if kind == "Responses" and row_type != "RESPONSE":
            return False
        if kind == "Exceptions" and row_type != "EXCEPTION":
            return False
        if kind == "CRC errors" and record.get("crc") != "ERROR":
            return False
        if kind == "Timeouts" and not record.get("timed_out"):
            return False
        if kind == "RAW" and not row_type.startswith("RAW"):
            return False

        fc_filter = self.normalized_fc_filter(
            self.advanced_filter_fc_var.get()
        )
        if fc_filter == "__INVALID__":
            return False
        if fc_filter:
            record_fc_text = str(record.get("fc", "")).upper()
            candidates = {record_fc_text}
            if str(record.get("type", "")) == "EXCEPTION":
                try:
                    candidates.add(f"{int(record_fc_text, 16) & 0x7F:02X}")
                except ValueError:
                    pass
            if fc_filter not in candidates:
                return False

        response_limit = self.advanced_filter_response_ms_var.get().strip()
        if response_limit:
            try:
                minimum_ms = float(response_limit.replace(",", "."))
                response_ms = float(record.get("response_text", ""))
            except (TypeError, ValueError):
                return False
            if response_ms <= minimum_ms:
                return False

        query = self.advanced_filter_text_var.get().strip().casefold()
```

MBToolkit v3.4/mb_view.py — lines 241-280

```
        if query:
            haystack = " ".join((
                str(record.get("details", "")),
                str(record.get("raw", "")),
                str(record.get("slave", "")),
                str(record.get("fc", "")),
                str(record.get("type", "")),
                str(record.get("channel", "")),
            )).casefold()
            if query not in haystack:
                return False

        return True

    def configure_anomaly_tags(self, dark=None):
        if not hasattr(self, "tree"):
            return

        if dark is None:
            dark = bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get())

        if dark:
            palette = {
                "result_success": ("#20372a", "#a7e4b8"),
                "anomaly_exception": ("#3b2728", "#ffc0c5"),
                "anomaly_crc": ("#482427", "#ffb3ba"),
                "anomaly_timeout": ("#42351f", "#ffd27a"),
                "anomaly_slow": ("#3b3422", "#ffe19a"),
                "anomaly_raw": ("#30283c", "#d8c5ff"),
            }
        else:
            palette = {
                "result_success": ("#e4f4e9", "#1f6b3a"),
                "anomaly_exception": ("#ffe8e9", "#8c1d24"),
                "anomaly_crc": ("#ffdfe1", "#8c1d24"),
                "anomaly_timeout": ("#fff0cf", "#774b00"),
                "anomaly_slow": ("#fff6dd", "#6b5200"),
                "anomaly_raw": ("#f0e9ff", "#5d3d8f"),
            }

```

MBToolkit v3.4/mb_view.py — lines 281-320

```
281         for tag, (background, foreground) in palette.items():
282             self.tree.tag_configure(
283                 tag,
284                 background=background,
285                 foreground=foreground,
286             )
287 
288     def anomaly_tag_for_record(self, record):
289         """
290         Return the optional result-highlight tag for one logical traffic row.
291 
292         "Realçar resultados" is a presentation option only:
293         - diagnostic outcomes retain their existing anomaly colors;
294         - a successfully completed normal REQUEST/RESPONSE transaction is green;
295         - unmatched/pending requests remain neutral.
296         """
297         if not getattr(self, "highlight_anomalies_var", None):
298             return ""
299         if not self.highlight_anomalies_var.get():
300             return ""
301 
302         kind = anomaly_kind(record, SLOW_RESPONSE_THRESHOLD_MS)
303         if kind:
304             return f"anomaly_{kind}"
305 
306         row_type = str(record.get("type", ""))
307 
308         # A normal, CRC-valid matched RESPONSE is a successful result.
309         if (
310             row_type == "RESPONSE"
311             and record.get("crc") == "OK"
312             and record.get("matched")
313         ):
314             return "result_success"
315 
316         # Once the normal response has arrived, also mark its REQUEST so the
317         # complete successful transaction pair reads as one result.
318         if (
319             row_type == "REQUEST"
320             and record.get("seq") in self._successful_request_seqs
```

MBToolkit v3.4/mb_view.py — lines 321-360

```
            and not record.get("timed_out")
        ):
            return "result_success"

        return ""

    def on_highlight_anomalies_changed(self):
        self.configure_anomaly_tags()
        self.configure_inspector_result_styles()
        self.render_filtered_view()
        self.update_frame_inspector(self.selected_record())
        saver = getattr(self, "_schedule_ui_settings_save", None)
        if callable(saver):
            saver()

    def mark_timed_out_records(self, expired, rerender=True):
        if not expired:
            return 0

        seqs = {
            item.get("seq")
            for item in expired
            if item.get("seq") is not None
        }
        if not seqs:
            return 0

        changed_records = []
        for record in self._frame_history:
            if record.get("seq") in seqs and not record.get("timed_out"):
                record["timed_out"] = True
                changed_records.append(record)

        if changed_records and rerender:
            # The "Timeouts" filter changes which rows are visible, so only
            # then is a full rebuild needed. Otherwise a timeout changes just
            # the row colour: retag the already visible rows in place.
            if canonical_ui_text(self.advanced_filter_type_var.get()) == "Timeouts":
                self.render_filtered_view()
            else:
```

MBToolkit v3.4/mb_view.py — lines 361-400

```
361                 self.refresh_record_tags(changed_records)
362         return len(changed_records)
363
364     def refresh_record_tags(self, records):
365         """Re-apply the result tag of visible rows without rebuilding the table."""
366         for record in records:
367             row_tag = self.anomaly_tag_for_record(record)
368             for iid in self._tree_iids_by_seq.get(record.get("seq"), []):
369                 try:
370                     self.tree.item(iid, tags=((row_tag,) if row_tag else ()))
371                 except tk.TclError:
372                     pass
373
374         selected = self.selected_record()
375         if selected is not None and any(selected is record for record in records):
376             self.update_frame_inspector(selected)
377
378     def selected_record(self):
379         selection = self.tree.selection()
380         if not selection:
381             return None
382         return self._record_by_iid.get(selection[0])
383
384     def update_frame_inspector(self, record=None):
385         if not hasattr(self, "inspector_vars"):
386             return
387
388         successful_request = bool(
389             record
390             and record.get("type") == "REQUEST"
391             and record.get("seq") in self._successful_request_seqs
392         )
393         values = inspector_values(
394             record,
395             successful_request=successful_request,
396             language=self.current_language,
397         )
398         for key, var in self.inspector_vars.items():
399             if key in values:
400                 var.set(values[key])
```

MBToolkit v3.4/mb_view.py — lines 401-440

```
401         self.selected_raw_var.set(values.get("raw", ""))
402 
403         result_label = getattr(self, "inspector_result_label", None)
404         if result_label is not None:
405             style_key = values.get("result_key", "neutral")
406             if not self.highlight_anomalies_var.get():
407                 style_key = "neutral"
408             try:
409                 result_label.configure(
410                     style=f"MBS.Result.{style_key}.TLabel"
411                 )
412             except tk.TclError:
413                 pass
414 
415     def toggle_frame_inspector(self):
416         if getattr(self, "inspector_collapsed", False):
417             self.inspector_body.grid()
418             self.inspector_collapsed = False
419             self.inspector_toggle_btn.configure(text=self.tr("Recolher"))
420         else:
421             self.inspector_body.grid_remove()
422             self.inspector_collapsed = True
423             self.inspector_toggle_btn.configure(text=self.tr("Expandir"))
424 
425         scheduler = getattr(self, "_schedule_bus_health_layout_sync", None)
426         if callable(scheduler):
427             scheduler()
428 
429     def copy_to_clipboard(self, text):
430         try:
431             self.clipboard_clear()
432             self.clipboard_append(str(text))
433             self.update_idletasks()
434         except tk.TclError:
435             pass
436 
437     def copy_selected_raw(self):
438         record = self.selected_record()
439         if record:
440             self.copy_to_clipboard(record.get("raw", ""))
```

MBToolkit v3.4/mb_view.py — lines 441-480

```
441
442     def copy_selected_decoded(self):
443         record = self.selected_record()
444         if record:
445             successful_request = bool(
446                 record.get("type") == "REQUEST"
447                 and record.get("seq") in self._successful_request_seqs
448             )
449             self.copy_to_clipboard(
450                 decoded_record_text(
451                     record,
452                     successful_request=successful_request,
453                     language=self.current_language,
454                 )
455             )
456
457     def context_filter_selected_slave(self):
458         record = self.selected_record()
459         if not record:
460             return
461         slave = str(record.get("slave", "") or "")
462         if not slave or slave not in self.slave_filter_vars:
463             return
464
465         self._transaction_filter_seqs = None
466         self.slave_filter_all_mode = False
467         for key, var in self.slave_filter_vars.items():
468             var.set(key == slave)
469         self.update_slave_heading()
470         self.render_filtered_view()
471
472     def context_filter_selected_fc(self):
473         record = self.selected_record()
474         if not record:
475             return
476         fc = str(record.get("fc", "") or "")
477         if not fc:
478             return
479         if str(record.get("type", "")) == "EXCEPTION":
480             try:
```

MBToolkit v3.4/mb_view.py — lines 481-520

```
                fc = f"{int(fc, 16) & 0x7F:02X}"
            except ValueError:
                pass
        self.advanced_filter_fc_var.set(fc)
        self._transaction_filter_seqs = None
        self.render_filtered_view()

    def context_show_selected_transaction(self):
        record = self.selected_record()
        if not record:
            return

        seq = record.get("seq")
        seqs = {seq}

        if record.get("request_seq") is not None:
            seqs.add(record["request_seq"])
        else:
            for candidate in self._frame_history:
                if candidate.get("request_seq") == seq:
                    seqs.add(candidate.get("seq"))
                    break

        self.slave_filter_all_mode = True
        for var in self.slave_filter_vars.values():
            var.set(True)
        self.update_slave_heading()

        self.advanced_filter_type_var.set(self.tr("Todos"))
        self.advanced_filter_fc_var.set("")
        self.advanced_filter_response_ms_var.set("")
        self.advanced_filter_text_var.set("")

        self._transaction_filter_seqs = {
            value for value in seqs if value is not None
        }
        self.render_filtered_view()

    def clear_transaction_filter(self):
        self._transaction_filter_seqs = None
```

MBToolkit v3.4/mb_view.py — lines 521-560

```
521         self.render_filtered_view()
522
523     def context_open_selected_in_master(self):
524         record = self.selected_record()
525         if record is not None:
526             self.master_open_from_record(record)
527
528     def show_traffic_context_menu(self, event):
529         row_iid = self.tree.identify_row(event.y)
530         if not row_iid or row_iid not in self._record_by_iid:
531             return
532
533         self.tree.selection_set(row_iid)
534         self.tree.focus(row_iid)
535         self.on_row_select()
536
537         opener = getattr(self, "master_definition_from_record", None)
538         if callable(opener):
539             state = "normal" if opener(self._record_by_iid[row_iid]) else "disabled"
540             try:
541                 self.traffic_context_menu.entryconfigure(TRAFFIC_MENU_OPEN_IN_MASTER,
542                                                          state=state)
543             except tk.TclError:
544                 pass
545
546         try:
547             self.traffic_context_menu.tk_popup(event.x_root, event.y_root)
548         finally:
549             try:
550                 self.traffic_context_menu.grab_release()
551             except tk.TclError:
552                 pass
553
554     def export_session_csv(self, path=None):
555         interactive = path is None
556         records = list(self._frame_history)
557         if not records:
558             if path is None:
559                 messagebox.showinfo(
560                     self.tr("Exportar CSV"),
```

MBToolkit v3.4/mb_view.py — lines 561-600

```
561                     self.tr("Não existem frames na sessão atual para exportar."),
562                 )
563             return None
564 
565         if path is None:
566             filename = datetime.now().strftime(
567                 "MBToolkit_session_%Y%m%d_%H%M%S.csv"
568             )
569             path = filedialog.asksaveasfilename(
570                 title=self.tr("Exportar sessão para CSV"),
571                 defaultextension=".csv",
572                 initialfile=filename,
573                 filetypes=[("CSV", "*.csv"), (self.tr("Todos os ficheiros"), "*.*")],
574             )
575             if not path:
576                 return None
577 
578         fieldnames = [
579             "Time",
580             "Delta ms",
581             "Response ms",
582             "Channel",
583             "Type",
584             "Slave",
585             "FC",
586             "Details",
587             "CRC",
588             "Raw",
589             "PDU Address",
590             "1-based Address",
591             "Qty",
592             "Byte Count",
593             "Timed out",
594             "Anomaly",
595         ]
596 
597         try:
598             with open(path, "w", newline="", encoding="utf-8-sig") as handle:
599                 writer = csv.DictWriter(handle, fieldnames=fieldnames)
600                 writer.writeheader()
```

MBToolkit v3.4/mb_view.py — lines 601-640

```
601
602                 for record in records:
603                     inspector = inspector_values(record, language=self.current_language)
604                     byte_count_text = inspector["byte_count"].split(
605                         ":", 1
606                     )[-1].strip()
607                     writer.writerow({
608                         "Time": record.get("time", ""),
609                         "Delta ms": record.get("delta_text", ""),
610                         "Response ms": record.get("response_text", ""),
611                         "Channel": record.get("channel", ""),
612                         "Type": record.get("type", ""),
613                         "Slave": record.get("slave", ""),
614                         "FC": record.get("fc", ""),
615                         "Details": translate_details(record.get("details", ""), self.current_language),
616                         "CRC": record.get("crc", ""),
617                         "Raw": record.get("raw", ""),
618                         "PDU Address": (
619                             "" if record.get("pdu_address") is None
620                             else record.get("pdu_address")
621                         ),
622                         "1-based Address": (
623                             "" if record.get("one_based") is None
624                             else record.get("one_based")
625                         ),
626                         "Qty": (
627                             "" if record.get("qty") is None
628                             else record.get("qty")
629                         ),
630                         "Byte Count": (
631                             "" if byte_count_text == "—"
632                             else byte_count_text
633                         ),
634                         "Timed out": "Yes" if record.get("timed_out") else "No",
635                         "Anomaly": anomaly_kind(
636                             record,
637                             SLOW_RESPONSE_THRESHOLD_MS,
638                         ) or "",
639                     })
640
```

MBToolkit v3.4/mb_view.py — lines 641-680

```
            if interactive:
                try:
                    messagebox.showinfo(
                        self.tr("Exportar CSV"),
                        self.trf("Sessão exportada para:\n\n{path}", path=path),
                    )
                except tk.TclError:
                    pass
            return path

        except OSError as exc:
            messagebox.showerror(
                self.tr("Exportar CSV"),
                self.trf("Não foi possível guardar o CSV:\n\n{error}", error=exc),
            )
            return None

    def tree_record_column_values(self, record):
        return {
            "time": record.get("time", ""),
            "delta": record.get("delta_text", ""),
            "response": record.get("response_text", ""),
            "channel": record.get("channel", ""),
            "type": record.get("type", ""),
            "slave": record.get("slave", ""),
            "fc": record.get("fc", ""),
            "details": translate_details(record.get("details", ""), self.current_language),
            "crc": record.get("crc", ""),
        }

    def measure_tree_column_width(self, column, records, heading_text=None):
        """
        Measure the pixel width required by a non-wrapped Treeview column.

        Details is handled separately because it uses the remaining viewport
        width and wraps instead of expanding horizontally without limit.
        """
        heading = (
            heading_text
            if heading_text is not None
```

MBToolkit v3.4/mb_view.py — lines 681-720

```
681             else self.tree_headings.get(column, column)
682         )
683         max_px = self.tree_heading_font.measure(str(heading))
684 
685         for record in records:
686             value = self.tree_record_column_values(record).get(column, "")
687             max_px = max(max_px, self.tree_cell_font.measure(str(value)))
688 
689         return max_px + self.tree_column_padding_px
690 
691     def available_details_width(self):
692         """
693         Return the width that can be assigned to Details without intentionally
694         forcing horizontal scrolling in the current Treeview viewport.
695         """
696         tree_width = int(self.tree.winfo_width())
697         if tree_width <= 20:
698             tree_width = 1100  # sensible pre-layout fallback
699 
700         other_total = 0
701         for column in self.tree["columns"]:
702             if column == "details":
703                 continue
704             other_total += int(self.tree.column(column, "width"))
705 
706         return max(
707             self.tree_detail_min_width_px,
708             tree_width - other_total - self.tree_detail_border_reserve_px
709         )
710 
711     def wrap_text_to_pixels(self, value, max_px):
712         """
713         Word-wrap text using the actual Tk font width.
714 
715         Long individual tokens are split character-by-character as a fallback,
716         so a cell can never force the Details column wider than its target.
717         """
718         text = str(value or "")
719         if not text:
720             return ""
```

MBToolkit v3.4/mb_view.py — lines 721-760

```
721
722         max_px = max(40, int(max_px))
723         paragraphs = text.splitlines() or [text]
724         output_lines = []
725
726         for paragraph in paragraphs:
727             words = paragraph.split()
728             if not words:
729                 output_lines.append("")
730                 continue
731
732             line = ""
733             for word in words:
734                 candidate = word if not line else f"{line} {word}"
735                 if self.tree_cell_font.measure(candidate) <= max_px:
736                     line = candidate
737                     continue
738
739                 if line:
740                     output_lines.append(line)
741                     line = ""
742
743                 # The token itself may be wider than the available space.
744                 if self.tree_cell_font.measure(word) <= max_px:
745                     line = word
746                     continue
747
748                 chunk = ""
749                 for char in word:
750                     candidate_chunk = chunk + char
751                     if chunk and self.tree_cell_font.measure(candidate_chunk) > max_px:
752                         output_lines.append(chunk)
753                         chunk = char
754                     else:
755                         chunk = candidate_chunk
756                 line = chunk
757
758             if line:
759                 output_lines.append(line)
760
```

MBToolkit v3.4/mb_view.py — lines 761-800

```
761         return "\n".join(output_lines)
762 
763     def details_horizontal_rule(self):
764         """
765         Return a rule that fills the Details column as far as the Treeview cell
766         permits. A slight overdraw is intentional; Treeview clips it at the
767         column boundary.
768         """
769         details_width = max(
770             40,
771             int(self.tree.column("details", "width"))
772         )
773         char_px = max(1, self.tree_cell_font.measure("—"))
774         count = max(8, int(math.ceil(details_width / char_px)) + 4)
775         return "—" * count
776 
777     def completed_pairs_for_records(self, records):
778         """
779         Return complete visible REQUEST/RESPONSE pairs in current display order.
780 
781         Each matched RESPONSE stores the exact request sequence id. This allows
782         correct boundaries in both Hora ↑ and Hora ↓ without relying on row
783         adjacency alone.
784         """
785         seq_to_index = {
786             record.get("seq"): index
787             for index, record in enumerate(records)
788         }
789 
790         pairs = []
791         for response in records:
792             request_seq = response.get("request_seq")
793             response_seq = response.get("seq")
794 
795             if request_seq is None or request_seq not in seq_to_index:
796                 continue
797 
798             request_index = seq_to_index[request_seq]
799             response_index = seq_to_index.get(response_seq)
800             if response_index is None:
```

MBToolkit v3.4/mb_view.py — lines 801-840

```
                continue

            start_index = min(request_index, response_index)
            end_index = max(request_index, response_index)

            pairs.append({
                "request_seq": request_seq,
                "response_seq": response_seq,
                "start_index": start_index,
                "end_index": end_index,
                # Separator belongs after whichever logical frame is visually
                # the end of the pair.
                "end_seq": records[end_index].get("seq"),
            })

        pairs.sort(key=lambda item: item["start_index"])
        return pairs

    def separator_after_sequences(self, records):
        """
        Return logical frame seq ids that need a separator AFTER them.

        There is intentionally no separator after the final complete visible
        pair because there is nothing below it to divide.
        """
        if not self.separate_transactions_var.get():
            return set()

        pairs = self.completed_pairs_for_records(records)
        if len(pairs) < 2:
            return set()

        return {
            pair["end_seq"]
            for pair in pairs[:-1]
        }

    def tree_separator_values(self):
        return (
            "", "", "", "", "", "", "",
```

MBToolkit v3.4/mb_view.py — lines 841-880

```
            self.details_horizontal_rule(),
            ""
        )

    def insert_tree_separator_after_seq(self, seq):
        """Insert one Details-width separator after the visual rows of seq."""
        if seq in self._tree_separator_iid_by_seq:
            return

        iids = self._tree_iids_by_seq.get(seq, [])
        if not iids:
            return

        last_iid = iids[-1]
        insert_at = self.tree.index(last_iid) + 1

        iid = self.tree.insert(
            "",
            insert_at,
            values=self.tree_separator_values()
        )
        self._raw_by_iid[iid] = ""
        self._tree_separator_iid_by_seq[seq] = iid

    def sync_tree_transaction_separators(self, records=None):
        """
        Synchronize standalone transaction separator rows without rebuilding
        normal traffic rows.
        """
        if records is None:
            records = self.records_for_display()

        desired = self.separator_after_sequences(records)
        existing = set(self._tree_separator_iid_by_seq)

        # Remove boundaries that are no longer valid (filter/order changes).
        for seq in existing - desired:
            iid = self._tree_separator_iid_by_seq.pop(seq, None)
            if iid and self.tree.exists(iid):
                self.tree.delete(iid)
```

MBToolkit v3.4/mb_view.py — lines 881-920

```
881             self._raw_by_iid.pop(iid, None)
882 
883         # Add newly required boundaries.
884         for seq in desired - existing:
885             self.insert_tree_separator_after_seq(seq)
886 
887     def detail_lines_for_record(self, record):
888         """
889         Return wrapped Details as separate visual Treeview rows.
890 
891         ttk.Treeview has one global row height for the entire widget. Putting
892         newline-wrapped text in one item would therefore make every REQUEST and
893         RESPONSE row as tall as the largest Details message. Continuation rows
894         preserve wrapping while keeping normal rows compact.
895         """
896         content_width = max(
897             40,
898             int(self.tree.column("details", "width"))
899             - self.tree_column_padding_px
900         )
901         wrapped = self.wrap_text_to_pixels(
902             translate_details(record.get("details", ""), self.current_language),
903             content_width
904         )
905 
906         if not wrapped:
907             return [""]
908 
909         return wrapped.splitlines()
910 
911     def set_tree_rowheight_for_records(self, records):
912         """Treeview row height stays fixed; wrapping uses continuation rows."""
913         self.tree_style.configure(
914             self.tree_style_name,
915             rowheight=self.tree_single_line_rowheight
916         )
917 
918     def update_tree_rowheight_for_record(self, record):
919         """No-op by design: one logical frame may use several compact rows."""
920         return
```

MBToolkit v3.4/mb_view.py — lines 921-960

```
921
922     def autofit_tree_columns(self, records=None):
923         """
924         Non-Details columns fit the longest visible value/title.
925
926         Details deliberately behaves differently: it consumes the remaining
927         viewport width and wraps. This keeps the full row information visible
928         without letting long register lists create an extremely wide table.
929         """
930         if records is None:
931             records = self.records_for_display()
932
933         # First compact every non-Details column.
934         for column in self.tree["columns"]:
935             if column == "details":
936                 continue
937
938             width = self.measure_tree_column_width(column, records)
939             self.tree.column(
940                 column,
941                 width=width,
942                 minwidth=width,
943                 anchor="center",
944                 stretch=False
945             )
946
947         # Then allocate the remaining viewport to Details.
948         heading_min = self.measure_tree_column_width(
949             "details",
950             [],
951             heading_text=self.tree_headings["details"]
952         )
953         details_width = max(
954             heading_min,
955             self.available_details_width()
956         )
957         self.tree.column(
958             "details",
959             width=details_width,
960             minwidth=heading_min,
```

MBToolkit v3.4/mb_view.py — lines 961-1000

```
            anchor="w",
            stretch=True
        )

        self.set_tree_rowheight_for_records(records)

    def expand_tree_columns_for_record(self, record):
        """
        Incrementally enlarge only the compact non-Details columns.

        Returns True when another column became wider. If that happens the
        remaining Details width changes, so the caller can rebuild/re-wrap the
        visible table once for the batch.
        """
        values = self.tree_record_column_values(record)
        layout_changed = False

        for column in self.tree["columns"]:
            if column == "details":
                continue

            heading_px = self.tree_heading_font.measure(
                str(self.tree_headings.get(column, column))
            )
            value_px = self.tree_cell_font.measure(str(values.get(column, "")))
            required = max(heading_px, value_px) + self.tree_column_padding_px
            current = int(self.tree.column(column, "width"))

            if required > current:
                self.tree.column(
                    column,
                    width=required,
                    minwidth=required,
                    anchor="center",
                    stretch=False
                )
                layout_changed = True

        if layout_changed:
            heading_min = self.measure_tree_column_width(
```

MBToolkit v3.4/mb_view.py — lines 1001-1040

```
                "details",
                [],
                heading_text=self.tree_headings["details"]
            )
            details_width = max(
                heading_min,
                self.available_details_width()
            )
            self.tree.column(
                "details",
                width=details_width,
                minwidth=heading_min,
                anchor="w",
                stretch=True
            )

        return layout_changed

    def refresh_tree_heading_width(self, column):
        """
        Dynamic Hora/Slave headings remain compact. Details is intentionally
        excluded because it is viewport-driven and wrapped.
        """
        if column == "details":
            return

        records = self.records_for_display()
        width = self.measure_tree_column_width(column, records)
        self.tree.column(
            column,
            width=width,
            minwidth=width,
            anchor="center",
            stretch=False
        )

    def schedule_tree_reflow(self, _event=None):
        """
        Debounce window/Treeview resizing and then re-wrap Details to the new
        available width.
```

MBToolkit v3.4/mb_view.py — lines 1041-1080

```
        """
        if self._tree_reflow_after_id is not None:
            try:
                self.after_cancel(self._tree_reflow_after_id)
            except Exception:
                pass

        self._tree_reflow_after_id = self.after(
            120,
            self.reflow_tree_after_resize
        )

    def reflow_tree_after_resize(self):
        self._tree_reflow_after_id = None
        if not hasattr(self, "tree"):
            return
        self.render_filtered_view()

    def _slave_sort_key(self, slave):
        try:
            return (0, int(slave))
        except (TypeError, ValueError):
            return (1, str(slave))

    def rebuild_slave_filter_menu(self):
        self.slave_filter_menu.delete(0, "end")

        self.slave_filter_menu.add_command(
            label=self.tr("Todos"),
            command=self.select_all_slaves
        )
        self.slave_filter_menu.add_command(
            label=self.tr("Nenhum"),
            command=self.select_no_slaves
        )

        if self.slave_filter_vars:
            self.slave_filter_menu.add_separator()
            for slave in sorted(self.slave_filter_vars, key=self._slave_sort_key):
                self.slave_filter_menu.add_checkbutton(
```

MBToolkit v3.4/mb_view.py — lines 1081-1120

```
                    label=f"Slave {slave}",
                    variable=self.slave_filter_vars[slave],
                    command=self.on_slave_filter_changed
                )
        else:
            self.slave_filter_menu.add_separator()
            self.slave_filter_menu.add_command(
                label=self.tr("Nenhum slave detetado"),
                state="disabled"
            )

        self.update_slave_heading()

    def update_slave_heading(self):
        if self.slave_filter_all_mode:
            label = "Slave ▾"
        else:
            selected = [
                slave for slave, var in self.slave_filter_vars.items()
                if var.get()
            ]
            label = (self.tr("Slave (Filtro) ▾") if selected else self.tr("Slave (Nenhum) ▾"))

        self.tree_headings["slave"] = label
        self.tree.heading(
            "slave",
            text=label,
            anchor="center",
            command=self.show_slave_filter_menu
        )
        self.refresh_tree_heading_width("slave")

    def show_slave_filter_menu(self):
        """
        Open the slave filter menu at the position where the Slave heading
        was clicked.
        """
        try:
            x = self.winfo_pointerx()
            y = self.winfo_pointery() + 8
```

MBToolkit v3.4/mb_view.py — lines 1121-1160

```
            self.slave_filter_menu.tk_popup(x, y)
        finally:
            try:
                self.slave_filter_menu.grab_release()
            except tk.TclError:
                pass

    def register_slave_for_filter(self, slave):
        slave = str(slave or "").strip()
        if not slave:
            return

        if slave in self.slave_filter_vars:
            return

        self.slave_filter_vars[slave] = tk.BooleanVar(
            value=self.slave_filter_all_mode
        )
        self.rebuild_slave_filter_menu()

    def select_all_slaves(self):
        self.slave_filter_all_mode = True
        for var in self.slave_filter_vars.values():
            var.set(True)
        self.update_slave_heading()
        self.render_filtered_view()

    def select_no_slaves(self):
        self.slave_filter_all_mode = False
        for var in self.slave_filter_vars.values():
            var.set(False)
        self.update_slave_heading()
        self.render_filtered_view()

    def on_slave_filter_changed(self):
        if not self.slave_filter_vars:
            self.slave_filter_all_mode = True
        else:
            values = [var.get() for var in self.slave_filter_vars.values()]
            # If every currently known slave is checked, remain in "All" mode
```

MBToolkit v3.4/mb_view.py — lines 1161-1200

```
            # so future slaves are automatically displayed too.
            self.slave_filter_all_mode = bool(values) and all(values)

        self.update_slave_heading()
        self.render_filtered_view()

    def selected_slave_ids(self):
        return {
            slave for slave, var in self.slave_filter_vars.items()
            if var.get()
        }

    def record_passes_slave_filter(self, record):
        # "Todos" includes RAW/unsynchronised data where a slave ID could not
        # be identified. "Nenhum" hides absolutely all traffic.
        if self.slave_filter_all_mode:
            return True

        selected = self.selected_slave_ids()
        if not selected:
            return False

        return str(record.get("slave", "")) in selected

    def time_order_descending(self):
        return canonical_ui_text(self.time_order_var.get()).startswith("Decrescente")

    def on_time_order_changed(self):
        label = self.tr("Hora ↓") if self.time_order_descending() else self.tr("Hora ↑")
        self.tree_headings["time"] = label
        self.tree.heading(
            "time",
            text=label,
            anchor="center",
            command=self.toggle_time_order
        )
        self.render_filtered_view()

    def toggle_time_order(self):
        if self.time_order_descending():
```

MBToolkit v3.4/mb_view.py — lines 1201-1240

```
            self.time_order_var.set(self.tr("Crescente — antigo → recente"))
        else:
            self.time_order_var.set(self.tr("Decrescente — recente → antigo"))
        self.on_time_order_changed()

    def records_for_display(self):
        records = [
            record for record in self._frame_history
            if (
                self.record_passes_slave_filter(record)
                and self.record_passes_advanced_filters(record)
            )
        ]
        if self.time_order_descending():
            records.reverse()
        return records

    def raw_line_for_record(self, record):
        # Keep fixed-width timing fields so the rest of every Raw Hex line
        # starts at exactly the same horizontal position. When a value does not
        # exist, show a completely blank field instead of "-", "dt=", "resp="
        # or "ms".
        delta_field = (
            f"dt={record['delta_text']:>7}ms "
            if record.get("delta_text")
            else " " * 13
        )
        response_field = (
            f"resp={record['response_text']:>7}ms "
            if record.get("response_text")
            else " " * 15
        )

        return (
            f"[{record['time']}] "
            f"{delta_field}"
            f"{response_field}"
            f"{record['channel']:<6} {record['type']:<10} "
            f"Slave={record['slave'] or '-':<3} FC={record['fc'] or '-':<2} "
            f"CRC={record['crc']:<5} | {record['raw']} | {translate_details(record['details'], self.current_language)}\n"
```

MBToolkit v3.4/mb_view.py — lines 1241-1280

```
        )

    def raw_separator_width_chars(self, raw_lines=None):
        """
        Width of one Raw Hex separator in visible character cells.

        Raw Hex now wraps to the viewport, so the separator must span only the
        visible width. Using the longest unwrapped data line would make the
        separator wrap onto multiple lines.
        """
        try:
            # Leave a small pixel reserve for Text internal borders/padding so
            # the last separator character does not wrap onto a second line.
            widget_px = max(1, int(self.raw_text.winfo_width()) - 14)
            char_px = max(1, self.raw_text_font.measure("—"))
            viewport_chars = max(
                1,
                int(widget_px // char_px)
            )
        except Exception:
            viewport_chars = 1

        return viewport_chars

    def raw_transaction_separator(self, width_chars=None):
        if width_chars is None:
            width_chars = self._raw_separator_width_chars
        return "—" * max(1, int(width_chars)) + "\n"

    def schedule_raw_separator_reflow(self, _event=None):
        if self._raw_reflow_after_id is not None:
            try:
                self.after_cancel(self._raw_reflow_after_id)
            except Exception:
                pass

        self._raw_reflow_after_id = self.after(
            120,
            self.reflow_raw_separator_after_resize
        )
```

MBToolkit v3.4/mb_view.py — lines 1281-1320

```
1281
1282    def reflow_raw_separator_after_resize(self):
1283        self._raw_reflow_after_id = None
1284        if self.separate_transactions_var.get():
1285            self.render_raw_text_only()
1286
1287    def insert_tree_record(self, record, index="end", allow_separator=True):
1288        detail_lines = self.detail_lines_for_record(record)
1289        visual_rows = []
1290
1291        # Main row: all columns + first Details line.
1292        visual_rows.append({
1293            "kind": "main",
1294            "values": (
1295                record["time"],
1296                record["delta_text"],
1297                record["response_text"],
1298                record["channel"],
1299                record["type"],
1300                record["slave"],
1301                record["fc"],
1302                detail_lines[0],
1303                record["crc"],
1304            ),
1305            "raw": record["raw"],
1306        })
1307
1308        # Wrapped continuation lines keep only the Details text.
1309        for continuation in detail_lines[1:]:
1310            visual_rows.append({
1311                "kind": "continuation",
1312                "values": (
1313                    "", "", "", "", "", "", "",
1314                    continuation,
1315                    ""
1316                ),
1317                "raw": record["raw"],
1318            })
1319
1320        inserted_main_iid = None
```

MBToolkit v3.4/mb_view.py — lines 1321-1360

```
        inserted_iids = []

        # index=0 requires reverse insertion to preserve visual block order.
        rows_to_insert = (
            list(reversed(visual_rows))
            if index == 0
            else visual_rows
        )

        row_tag = self.anomaly_tag_for_record(record)

        for visual in rows_to_insert:
            iid = self.tree.insert(
                "",
                index,
                values=visual["values"],
                tags=((row_tag,) if row_tag else ()),
            )

            self._raw_by_iid[iid] = visual["raw"]
            self._record_by_iid[iid] = record

            if index == 0:
                inserted_iids.insert(0, iid)
                self._tree_history.appendleft((iid, True))
            else:
                inserted_iids.append(iid)
                self._tree_history.append((iid, True))

            if visual["kind"] == "main":
                inserted_main_iid = iid

        self._tree_iids_by_seq[record["seq"]] = inserted_iids
        self._ui_frame_count += 1
        return inserted_main_iid

    def render_raw_text_only(self):
        records = self.records_for_display()
        separator_after = self.separator_after_sequences(records)

```

MBToolkit v3.4/mb_view.py — lines 1361-1400

```
        line_by_seq = {
            record["seq"]: self.raw_line_for_record(record)
            for record in records
        }
        all_raw_lines = list(line_by_seq.values())
        self._raw_separator_width_chars = self.raw_separator_width_chars(
            all_raw_lines
        )
        separator_line = self.raw_transaction_separator(
            self._raw_separator_width_chars
        )

        # Respect the visual line cap, counting transaction separators as one
        # line each.
        if not self.time_order_descending():
            selected_reversed = []
            line_count = 0

            for record in reversed(records):
                cost = 1 + (1 if record["seq"] in separator_after else 0)
                if line_count + cost > MAX_RAW_TEXT_LINES:
                    break
                selected_reversed.append(record)
                line_count += cost

            raw_records = list(reversed(selected_reversed))
        else:
            selected = []
            line_count = 0

            for record in records:
                cost = 1 + (1 if record["seq"] in separator_after else 0)
                if line_count + cost > MAX_RAW_TEXT_LINES:
                    break
                selected.append(record)
                line_count += cost

            raw_records = selected

        parts = []
```

MBToolkit v3.4/mb_view.py — lines 1401-1440

```
        for record in raw_records:
            parts.append(line_by_seq[record["seq"]])
            if record["seq"] in separator_after:
                parts.append(separator_line)

        block = "".join(parts)
        self.raw_text.delete("1.0", "end")
        if block:
            self.raw_text.insert("end", block)
        self._raw_text_line_count = block.count("\n")

    def render_filtered_view(self):
        # This is intentionally called on user-driven filter/sort changes and
        # infrequent history-prune events, not once per received frame.
        selected_record = self.selected_record()
        selected_seq = (
            selected_record.get("seq")
            if selected_record is not None
            else None
        )

        for item in self.tree.get_children():
            self.tree.delete(item)

        self._raw_by_iid = {}
        self._record_by_iid = {}
        self._tree_history.clear()
        self._tree_iids_by_seq.clear()
        self._tree_separator_iid_by_seq.clear()
        self._ui_frame_count = 0
        self.selected_raw_var.set("")

        descending = self.time_order_descending()
        records = self.records_for_display()

        # Column sizing determines the wrap width, so it must happen before
        # inserting the visible rows.
        self.autofit_tree_columns(records)

        for record in records:
```

MBToolkit v3.4/mb_view.py — lines 1441-1480

```
            self.insert_tree_record(
                record,
                index="end",
                allow_separator=False
            )

        self.sync_tree_transaction_separators(records)
        self.render_raw_text_only()

        if selected_seq is not None:
            iids = self._tree_iids_by_seq.get(selected_seq, [])
            if iids:
                try:
                    self.tree.selection_set(iids[0])
                    self.tree.focus(iids[0])
                    self.update_frame_inspector(
                        self._record_by_iid.get(iids[0])
                    )
                except tk.TclError:
                    pass
            else:
                self.update_frame_inspector(None)

        if self.auto_scroll_var.get() and self._tree_history:
            if descending:
                iid = self._tree_history[0][0]
                self.tree.see(iid)
                self.raw_text.see("1.0")
            else:
                iid = self._tree_history[-1][0]
                self.tree.see(iid)
                self.raw_text.see("end")

    def add_frame_row(self, row):
        # Kept as a convenience wrapper for callers/tests.
        self.add_frame_rows_batch([row])

    def add_frame_rows_batch(self, rows):
        if not rows:
            return
```

MBToolkit v3.4/mb_view.py — lines 1481-1520

```
1481
1482        raw_lines_log = []
1483        any_visible = False
1484        tree_layout_changed = False
1485        pair_boundary_changed = False
1486        expired_requests = []
1487
1488        self.pulse_bus_activity()
1489
1490        for row in rows:
1491            seq = self._next_frame_seq
1492            self._next_frame_seq += 1
1493            row["_view_seq"] = seq
1494
1495            timing = self.metrics.process(row)
1496
1497            delta_text = (
1498                f"{timing['delta_ms']:.1f}" if timing["delta_ms"] is not None else ""
1499            )
1500            response_text = (
1501                f"{timing['response_ms']:.1f}"
1502                if timing["response_ms"] is not None
1503                else ""
1504            )
1505
1506            # Keep pairing for response-time/statistics, but do not copy the
1507            # request Address/1-based/Qty into RESPONSE/EXCEPTION details.
1508            # Those fields remain visible only on the REQUEST itself.
1509            matched = timing.get("matched_request")
1510            expired_requests.extend(
1511                timing.get("expired_requests", [])
1512            )
1513
1514            self.register_slave_for_filter(row.get("slave", ""))
1515
1516            # FC01/FC02: the request Qty tells which bits of the last data
1517            # byte are padding, so trim them from the response details.
1518            if (
1519                matched is not None
1520                and row.get("type") == "RESPONSE"
```

MBToolkit v3.4/mb_view.py — lines 1521-1560

```
                and row.get("fc") in ("01", "02")
            ):
                try:
                    row["details"] = read_bits_details(
                        bytes.fromhex(row.get("raw", "")),
                        matched.get("qty"),
                    )
                except ValueError:
                    pass

            record = {
                "seq": seq,
                "epoch": float(row.get("_epoch", time.time())),
                "time": row["time"],
                "delta_text": delta_text,
                "response_text": response_text,
                "channel": row["channel"],
                "type": row["type"],
                "slave": row["slave"],
                "fc": row["fc"],
                "details": row["details"],
                "crc": row["crc"],
                "raw": row["raw"],
                "pdu_address": row.get("pdu_address"),
                "one_based": row.get("one_based"),
                "qty": row.get("qty"),
                "frame_len": row.get("_frame_len", 0),
                "timed_out": False,
                "matched": matched is not None,
                "request_seq": (
                    matched.get("seq")
                    if matched is not None
                    else None
                ),
                # Used only by the Frame Inspector to show the address of
                # each returned value; RESPONSE Details stay response-native.
                "request_pdu_address": (
                    matched.get("pdu_address")
                    if matched is not None
                    else None
```

MBToolkit v3.4/mb_view.py — lines 1561-1600

```
                ),
                "request_qty": (
                    matched.get("qty")
                    if matched is not None
                    else None
                ),
            }
            self._frame_history.append(record)

            # Track successful normal transaction pairs independently of whether
            # highlighting is currently enabled. This lets the user toggle
            # "Realçar resultados" on/off against the exact same session.
            if (
                record.get("type") == "RESPONSE"
                and record.get("crc") == "OK"
                and record.get("matched")
                and anomaly_kind(record, SLOW_RESPONSE_THRESHOLD_MS) is None
                and record.get("request_seq") is not None
            ):
                self._successful_request_seqs.add(record["request_seq"])

                # If the REQUEST is already visible, update it immediately
                # without rebuilding the entire Treeview.
                if self.highlight_anomalies_var.get():
                    for request_iid in self._tree_iids_by_seq.get(
                        record["request_seq"], []
                    ):
                        try:
                            self.tree.item(
                                request_iid,
                                tags=("result_success",),
                            )
                        except tk.TclError:
                            pass

            # Disk log is always complete and ignores GUI filters/sorting.
            raw_lines_log.append(self.raw_line_for_record(record))

            if (
                self.record_passes_slave_filter(record)
```

MBToolkit v3.4/mb_view.py — lines 1601-1640

```
                and self.record_passes_advanced_filters(record)
            ):
                any_visible = True
                if self.expand_tree_columns_for_record(record):
                    tree_layout_changed = True

                self.update_tree_rowheight_for_record(record)

                if self.time_order_descending():
                    self.insert_tree_record(
                        record,
                        index=0,
                        allow_separator=False
                    )

                    raw_block = self.raw_line_for_record(record)
                    self.raw_text.insert("1.0", raw_block)
                    self._raw_text_line_count += raw_block.count("\n")
                else:
                    self.insert_tree_record(
                        record,
                        index="end",
                        allow_separator=False
                    )

                    raw_block = self.raw_line_for_record(record)
                    self.raw_text.insert("end", raw_block)
                    self._raw_text_line_count += raw_block.count("\n")

                if record.get("matched"):
                    pair_boundary_changed = True

        if expired_requests:
            # A new frame can itself cause older requests to cross the Pending
            # timeout. Mark those original REQUEST records in place.
            self.mark_timed_out_records(
                expired_requests,
                rerender=True,
            )

```

MBToolkit v3.4/mb_view.py — lines 1641-1680

```
        log_block = "".join(raw_lines_log)

        # Only real traffic is allowed to create a .txt log. GUI filters do not
        # change the complete saved traffic.
        if self.capture_active and log_block:
            if self.ensure_log_started():
                self.log_line(log_block)

        self.update_stats_labels()
        rebuilt = self.prune_ui_history()

        if tree_layout_changed and not rebuilt:
            self.render_filtered_view()
            rebuilt = True

        if not rebuilt and pair_boundary_changed:
            visible_records = self.records_for_display()
            self.sync_tree_transaction_separators(visible_records)
            self.render_raw_text_only()

        # Raw Hex has its own bounded line count. Rebuild it in the correct
        # chronological direction when its hysteresis threshold is reached.
        if (
            not rebuilt
            and self._raw_text_line_count
            > MAX_RAW_TEXT_LINES + RAW_TEXT_PRUNE_CHUNK_LINES
        ):
            self.render_raw_text_only()

        if self.auto_scroll_var.get() and any_visible and self._tree_history:
            if self.time_order_descending():
                self.tree.see(self._tree_history[0][0])
                self.raw_text.see("1.0")
            else:
                self.tree.see(self._tree_history[-1][0])
                self.raw_text.see("end")

    def prune_ui_history(self):
        """
        Keep the GUI source history bounded.
```

MBToolkit v3.4/mb_view.py — lines 1681-1720

```
1681
1682        The full disk log is independent and remains complete. Pruning happens
1683        by deque chunks and only triggers a full GUI rebuild occasionally.
1684        """
1685        trigger_frames = MAX_UI_FRAMES + UI_PRUNE_CHUNK_FRAMES
1686
1687        if len(self._frame_history) <= trigger_frames:
1688            return False
1689
1690        while len(self._frame_history) > MAX_UI_FRAMES:
1691            self._frame_history.popleft()
1692
1693        retained_sequences = {
1694            record.get("seq")
1695            for record in self._frame_history
1696            if record.get("seq") is not None
1697        }
1698        self._successful_request_seqs.intersection_update(retained_sequences)
1699
1700        self.render_filtered_view()
1701        return True
1702
1703    def on_row_select(self, _event=None):
1704        record = self.selected_record()
1705        self.update_frame_inspector(record)
1706
1707    def finish_capture_ui(self):
1708        self.busy = False
1709        self.capture_active = False
1710        self.status_var.set(self.tr("● Parado"))
1711        self.start_btn.configure(state="normal")
1712        self.stop_btn.configure(state="disabled")
1713        self.restart_btn.configure(state="normal")
1714        self.set_simulation_button_state("normal")
1715        self.set_sniffer_refresh_state("normal")
1716        self.serial_objects = []
1717        self.serial_by_port = {}
1718        self.reader_threads = []
1719        self.close_log()
1720        self.update_com_status()
```

MBToolkit v3.4/mb_view.py — lines 1721-1760

```
1721
1722    def app_directory(self):
1723        return get_application_directory()
1724
1725    def prepare_log_session(self, prefix, started_at, header):
1726        """
1727        Prepare a real-capture log session without creating a file.
1728
1729        The .txt file is created only after the first received frame/RAW block.
1730        """
1731        self.close_log()
1732        self.pending_log_filename = started_at.strftime(
1733            prefix + "_%Y%m%d_%H%M%S.txt"
1734        )
1735        self.pending_log_header = header
1736        self.pending_log_notes = []
1737        self.log_start_failed = False
1738        self.log_path = None
1739
1740    def ensure_log_started(self):
1741        """
1742        Create the pending real-capture log on first actual traffic.
1743
1744        Returns True when the log is open. A creation failure is reported only
1745        once for the current capture session.
1746        """
1747        if self.log_file:
1748            return True
1749
1750        if not self.pending_log_filename or self.log_start_failed:
1751            return False
1752
1753        try:
1754            target = get_logs_folder() / self.pending_log_filename
1755            self.log_file = target.open("w", encoding="utf-8", buffering=1)
1756            self.log_path = target
1757
1758            if self.pending_log_header:
1759                self.log_file.write(self.pending_log_header)
1760
```

MBToolkit v3.4/mb_view.py — lines 1761-1800

```
            if self.pending_log_notes:
                self.log_file.write("".join(self.pending_log_notes))

            self.pending_log_notes = []
            return True

        except OSError as exc:
            self.log_start_failed = True
            self.log_file = None
            self.log_path = None
            messagebox.showerror(
                self.tr("Erro ao criar log"),
                self.trf(
                    "Foi detetado tráfego, mas não foi possível criar o ficheiro de log em:\n\n{folder}\n\n{error}",
                    folder=get_logs_folder(),
                    error=exc,
                )
            )
            return False

    def queue_log_note(self, text):
        """
        Write a capture note to an already-open log, or keep it in memory until
        the first traffic frame creates the log.
        """
        if self.log_file:
            self.log_line(text)
        elif self.capture_active and self.pending_log_filename:
            self.pending_log_notes.append(text)

    def log_line(self, text):
        if self.log_file:
            try:
                self.log_file.write(text)
            except Exception:
                pass

    def close_log(self):
        if self.log_file:
            try:
```

MBToolkit v3.4/mb_view.py — lines 1801-1840

```
                self.log_file.write(
                    f"# Stop: {datetime.now().isoformat(timespec='milliseconds')}\n"
                )
                self.log_file.close()
            except Exception:
                pass

        self.log_file = None
        self.log_path = None
        self.pending_log_filename = None
        self.pending_log_header = ""
        self.pending_log_notes = []
        self.log_start_failed = False

    def open_log_folder(self):
        try:
            folder = get_logs_folder()
            if os.name == "nt":
                os.startfile(str(folder))
            elif sys.platform == "darwin":
                subprocess.Popen(["open", str(folder)])
            else:
                subprocess.Popen(["xdg-open", str(folder)])
        except Exception as exc:
            messagebox.showerror(self.tr("Abrir pasta"), str(exc))

    def clear_view(self):
        for item in self.tree.get_children():
            self.tree.delete(item)
        self.raw_text.delete("1.0", "end")
        self.selected_raw_var.set("")
        self._raw_by_iid = {}
        self._record_by_iid = {}
        self._tree_history.clear()
        self._tree_iids_by_seq.clear()
        self._tree_separator_iid_by_seq.clear()
        self._frame_history.clear()
        self._next_frame_seq = 1
        self._ui_frame_count = 0
        self._raw_text_line_count = 0
```

MBToolkit v3.4/mb_view.py — lines 1841-1871

```
        self._raw_separator_width_chars = 1

        self.slave_filter_vars.clear()
        self.slave_filter_all_mode = True
        self.rebuild_slave_filter_menu()

        self._transaction_filter_seqs = None
        self._successful_request_seqs.clear()
        if hasattr(self, "advanced_filter_type_var"):
            self.advanced_filter_type_var.set(self.tr("Todos"))
            self.advanced_filter_fc_var.set("")
            self.advanced_filter_response_ms_var.set("")
            self.advanced_filter_text_var.set("")
        self.update_frame_inspector(None)
        if hasattr(self, "bus_activity_var"):
            self.bus_activity_var.set("BUS ○")

        self.time_order_var.set(self.tr("Crescente — antigo → recente"))
        self.tree_headings["time"] = self.tr("Hora ↑")
        self.tree.heading(
            "time",
            text=self.tree_headings["time"],
            anchor="center",
            command=self.toggle_time_order
        )

        # With no traffic, widths/row height return to their default layout.
        self.autofit_tree_columns([])

        self.reset_session_metrics()

```

MBToolkit v3.4/mb_slave_finder.py

1625 source lines.

MBToolkit v3.4/mb_slave_finder.py — lines 1-40

```python
#!/usr/bin/env python3
"""Active Modbus RTU Bus Slave Finder logic and GUI mixin."""

import threading
import time
import tkinter as tk
from tkinter import ttk, messagebox

from mb_config import (
    DISCOVERY_BAUD_RATES,
    DISCOVERY_PARITIES,
    DISCOVERY_STOP_BITS,
    DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
    DISCOVERY_MIN_TIMEOUT_MS,
    DISCOVERY_MAX_TIMEOUT_MS,
    DISCOVERY_PROBE_ADDRESS,
    DISCOVERY_PROBE_QTY,
    DISCOVERY_WRITE_TIMEOUT_SECONDS,
    DISCOVERY_WRITE_RETRIES,
    DISCOVERY_RX_BUFFER_LIMIT,
    DISCOVERY_MIN_ATTEMPTS,
    DISCOVERY_MAX_ATTEMPTS,
    DISCOVERY_DEFAULT_ATTEMPTS,
    DISCOVERY_IDLE_CHECK_SECONDS,
    MODBUS_EXCEPTION_CODES,
)
from mb_protocol import (
    append_crc,
    crc_ok,
    hex_bytes,
    modbus_rtu_interframe_gap_seconds,
)
from mb_widgets import RefreshButton

def build_discovery_probe(slave: int, fc: int) -> bytes:
    """Build a read-only discovery probe: FC03/FC04, Address 0, Qty 1."""
    if not (1 <= int(slave) <= 247):
        raise ValueError("Slave ID fora do intervalo 1..247")
    if int(fc) not in (3, 4):
        raise ValueError("A descoberta ativa suporta apenas FC03/FC04")
```

MBToolkit v3.4/mb_slave_finder.py — lines 41-80

```
41
42      payload = bytes((
43          int(slave),
44          int(fc),
45          0x00, DISCOVERY_PROBE_ADDRESS & 0xFF,
46          0x00, DISCOVERY_PROBE_QTY & 0xFF,
47      ))
48      return append_crc(payload)
49
50  DEVICE_ID_FC = 0x2B
51  DEVICE_ID_MEI_TYPE = 0x0E
52  DEVICE_ID_READ_BASIC = 0x01
53  DEVICE_ID_MORE_FOLLOWS = 0xFF
54  DEVICE_ID_BASIC_OBJECTS = (0x00, 0x01, 0x02)
55  DEVICE_ID_MAX_PAGES = 8
56
57  def build_device_identification_probe(slave: int, object_id: int = 0x00) -> bytes:
58      """Build FC43/MEI 14 Basic Device Identification request."""
59      slave = int(slave)
60      object_id = int(object_id)
61      if not (1 <= slave <= 247):
62          raise ValueError("Slave ID fora do intervalo 1..247")
63      if not (0 <= object_id <= 0xFF):
64          raise ValueError("Object ID fora do intervalo 0..255")
65      return append_crc(bytes((
66          slave,
67          DEVICE_ID_FC,
68          DEVICE_ID_MEI_TYPE,
69          DEVICE_ID_READ_BASIC,
70          object_id,
71      )))
72
73  def _decode_device_id_value(value: bytes) -> str:
74      """Decode a Modbus Device Identification ASCII object for display."""
75      return bytes(value).decode("ascii", errors="replace").strip("\x00 ")
76
77  def find_device_identification_response(buffer: bytes, slave: int):
78      """Find one CRC-valid FC43/14 response or Modbus exception in *buffer*."""
79      data = bytes(buffer)
80      slave = int(slave)
```

MBToolkit v3.4/mb_slave_finder.py — lines 81-120

```
81
82      for offset in range(max(0, len(data) - 3)):
83          if data[offset] != slave or offset + 2 > len(data):
84              continue
85
86          response_fc = data[offset + 1]
87
88          # FC43 exception response: Slave + 0xAB + Exception + CRC.
89          if response_fc == (DEVICE_ID_FC | 0x80):
90              end = offset + 5
91              if end <= len(data):
92                  frame = data[offset:end]
93                  if crc_ok(frame):
94                      return {
95                          "kind": "EXCEPTION",
96                          "frame": frame,
97                          "exception_code": frame[2],
98                          "objects": {},
99                          "more_follows": False,
100                         "next_object_id": 0,
101                     }
102             continue
103
104         if response_fc != DEVICE_ID_FC:
105             continue
106
107         # Normal response fixed header before the object list:
108         # Slave, FC, MEI, ReadCode, Conformity, MoreFollows, NextObject, Count.
109         if offset + 8 > len(data):
110             continue
111         if data[offset + 2] != DEVICE_ID_MEI_TYPE:
112             continue
113         if data[offset + 3] != DEVICE_ID_READ_BASIC:
114             continue
115
116         more_follows = data[offset + 5] == DEVICE_ID_MORE_FOLLOWS
117         next_object_id = data[offset + 6]
118         object_count = data[offset + 7]
119         pos = offset + 8
120         objects = {}
```

MBToolkit v3.4/mb_slave_finder.py — lines 121-160

```
        complete = True

        for _ in range(object_count):
            if pos + 2 > len(data):
                complete = False
                break
            object_id = data[pos]
            object_len = data[pos + 1]
            pos += 2
            if pos + object_len > len(data):
                complete = False
                break
            objects[object_id] = _decode_device_id_value(
                data[pos:pos + object_len]
            )
            pos += object_len

        if not complete:
            continue

        end = pos + 2
        if end > len(data):
            continue
        frame = data[offset:end]
        if not crc_ok(frame):
            continue

        return {
            "kind": "DEVICE_ID",
            "frame": frame,
            "exception_code": None,
            "objects": objects,
            "more_follows": more_follows,
            "next_object_id": next_object_id,
            "conformity_level": data[offset + 4],
        }

    return None

def read_device_identification_response(ser, slave, timeout_s, stop_event):
```

MBToolkit v3.4/mb_slave_finder.py — lines 161-200

```
161     """Read until a valid FC43/14 response/exception is received or timed out."""
162     deadline = time.perf_counter() + max(0.001, float(timeout_s))
163     buffer = bytearray()
164 
165     while time.perf_counter() < deadline:
166         if stop_event.is_set():
167             return None
168 
169         waiting = int(getattr(ser, "in_waiting", 0) or 0)
170         if waiting:
171             chunk = ser.read(waiting)
172             if chunk:
173                 buffer.extend(chunk)
174                 if len(buffer) > DISCOVERY_RX_BUFFER_LIMIT:
175                     del buffer[:-DISCOVERY_RX_BUFFER_LIMIT]
176                 found = find_device_identification_response(buffer, slave)
177                 if found is not None:
178                     return found
179         elif stop_event.wait(0.001):
180             return None
181 
182     return find_device_identification_response(buffer, slave)
183 
184 def device_identification_timeout_seconds(
185     baud: int,
186     parity: str,
187     stopbits: str,
188     minimum_timeout_ms: float = DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
189 ) -> float:
190     """Conservative timeout for one FC43/14 response, up to one RTU frame."""
191     baud = max(1, int(baud))
192     bits_per_char = discovery_bits_per_char(parity, stopbits)
193     char_s = bits_per_char / float(baud)
194     # A Modbus RTU ADU is at most 256 bytes. Allow one full response frame,
195     # the recommended t3.5 gap and a small device-processing margin.
196     calculated = (
197         256.0 * char_s
198         + modbus_rtu_interframe_gap_seconds(baud, bits_per_char)
199         + 0.020
200     )
```

MBToolkit v3.4/mb_slave_finder.py — lines 201-240

```
    return max(float(minimum_timeout_ms) / 1000.0, calculated)

def device_identification_request_transmit_seconds(
    baud: int, parity: str, stopbits: str
) -> float:
    """Estimated on-wire duration of the 7-byte FC43/14 request."""
    return 7.0 * discovery_bits_per_char(parity, stopbits) / float(max(1, int(baud)))

def format_device_identification(result) -> str:
    """Return compact Device Identification text for the Finder result table."""
    if result is None:
        return "No response"
    if result.get("kind") == "EXCEPTION":
        code = int(result.get("exception_code", 0))
        name = MODBUS_EXCEPTION_CODES.get(
            code, ("Unknown/Reserved Exception", "")
        )[0]
        return f"Exception 0x{code:02X} – {name}"

    objects = result.get("objects", {}) or {}
    values = [objects.get(obj_id, "") for obj_id in DEVICE_ID_BASIC_OBJECTS]
    if any(values):
        return " | ".join(value or "—" for value in values)
    return "Response – no Basic Device Identification objects"

def read_basic_device_identification(
    ser, slave, baud, parity, stopbits, minimum_timeout_ms, stop_event, serial_module
):
    """Read Basic Device Identification, following More Follows when required."""
    object_id = 0x00
    merged_objects = {}
    frames = []
    conformity_level = None

    for _page in range(DEVICE_ID_MAX_PAGES):
        if stop_event.is_set():
            return None

        try:
            ser.reset_input_buffer()
```

MBToolkit v3.4/mb_slave_finder.py — lines 241-280

```
        except Exception:
            raise

        request = build_device_identification_probe(slave, object_id)
        sent = write_discovery_request(
            ser, request, baud, parity, stopbits, stop_event, serial_module
        )
        if not sent or stop_event.is_set():
            return None

        result = read_device_identification_response(
            ser,
            slave,
            device_identification_timeout_seconds(
                baud, parity, stopbits, minimum_timeout_ms
            ),
            stop_event,
        )
        if result is None or result.get("kind") == "EXCEPTION":
            return result

        frames.append(result["frame"])
        merged_objects.update(result.get("objects", {}))
        conformity_level = result.get("conformity_level")

        # Basic identification is complete as soon as VendorName, ProductCode
        # and MajorMinorRevision have all been received.
        if all(obj_id in merged_objects for obj_id in DEVICE_ID_BASIC_OBJECTS):
            return {
                "kind": "DEVICE_ID",
                "frame": b"".join(frames),
                "exception_code": None,
                "objects": merged_objects,
                "more_follows": bool(result.get("more_follows")),
                "next_object_id": int(result.get("next_object_id", 0)),
                "conformity_level": conformity_level,
            }

        if not result.get("more_follows"):
            break
```

MBToolkit v3.4/mb_slave_finder.py — lines 281-320

```
281
282         next_object_id = int(result.get("next_object_id", 0))
283         if next_object_id == object_id:
284             break
285         object_id = next_object_id
286
287         gap_s = modbus_rtu_interframe_gap_seconds(
288             baud, discovery_bits_per_char(parity, stopbits)
289         )
290         if stop_event.wait(gap_s):
291             return None
292
293     return {
294         "kind": "DEVICE_ID",
295         "frame": b"".join(frames),
296         "exception_code": None,
297         "objects": merged_objects,
298         "more_follows": False,
299         "next_object_id": object_id,
300         "conformity_level": conformity_level,
301     } if frames else None
302
303 def discovery_bits_per_char(parity: str, stopbits: str) -> float:
304     """Return serial bits/character: start + 8 data + parity + stop."""
305     parity_bits = 0 if parity == "None" else 1
306     stop_value = float(stopbits)
307     return 1.0 + 8.0 + parity_bits + stop_value
308
309 def discovery_response_timeout_seconds(
310     baud: int,
311     parity: str,
312     stopbits: str,
313     minimum_timeout_ms: float = DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
314 ) -> float:
315     """
316     Adaptive response deadline after the request has left the serial driver.
317
318     The deadline includes the on-wire duration of a normal 7-byte FC03/FC04
319     response plus processing/inter-frame margin, with a user-configurable
320     minimum.
```

MBToolkit v3.4/mb_slave_finder.py — lines 321-360

```
    """
    baud = max(1, int(baud))
    bits_per_char = discovery_bits_per_char(parity, stopbits)
    char_s = bits_per_char / float(baud)
    normal_response_s = 7.0 * char_s
    serial_margin_s = max(
        0.012, modbus_rtu_interframe_gap_seconds(baud, bits_per_char)
    )
    calculated = normal_response_s + serial_margin_s
    return max(float(minimum_timeout_ms) / 1000.0, calculated)

def discovery_request_transmit_seconds(baud: int, parity: str, stopbits: str) -> float:
    """Estimated on-wire duration of the fixed 8-byte discovery request."""
    return 8.0 * discovery_bits_per_char(parity, stopbits) / float(max(1, int(baud)))

def find_discovery_response(buffer: bytes, slave: int, fc: int):
    """
    Find a valid normal or exception response for one discovery probe.

    Any CRC-valid Modbus exception counts as proof that the slave exists.
    The scanner intentionally ignores other traffic and request echoes.
    """
    data = bytes(buffer)
    slave = int(slave)
    fc = int(fc)

    for offset in range(max(0, len(data) - 3)):
        if data[offset] != slave or offset + 2 > len(data):
            continue

        response_fc = data[offset + 1]

        if response_fc == (fc | 0x80):
            end = offset + 5
            if end <= len(data):
                frame = data[offset:end]
                if crc_ok(frame):
                    code = frame[2]
                    return {
                        "kind": "EXCEPTION",
```

MBToolkit v3.4/mb_slave_finder.py — lines 361-400

```
361                         "frame": frame,
362                         "exception_code": code,
363                     }
364
365         if response_fc == fc and offset + 3 <= len(data):
366             byte_count = data[offset + 2]
367             # FC03/04 register responses must contain an even positive byte count.
368             if byte_count > 0 and byte_count % 2 == 0 and byte_count <= 250:
369                 end = offset + 5 + byte_count
370                 if end <= len(data):
371                     frame = data[offset:end]
372                     if crc_ok(frame):
373                         return {
374                             "kind": "RESPONSE",
375                             "frame": frame,
376                             "exception_code": None,
377                         }
378
379     return None
380
381 def read_discovery_response(ser, slave, fc, timeout_s, stop_event):
382     """Read non-blocking serial bytes until a valid probe response or timeout."""
383     deadline = time.perf_counter() + max(0.001, float(timeout_s))
384     buffer = bytearray()
385
386     while time.perf_counter() < deadline:
387         if stop_event.is_set():
388             return None
389
390         # Do not hide serial-driver failures as ordinary slave timeouts.
391         waiting = int(getattr(ser, "in_waiting", 0) or 0)
392
393         if waiting:
394             chunk = ser.read(waiting)
395             if chunk:
396                 buffer.extend(chunk)
397                 if len(buffer) > DISCOVERY_RX_BUFFER_LIMIT:
398                     del buffer[:-DISCOVERY_RX_BUFFER_LIMIT]
399                 found = find_discovery_response(buffer, slave, fc)
400                 if found is not None:
```

MBToolkit v3.4/mb_slave_finder.py — lines 401-440

```
                    return found
        else:
            # Event.wait() keeps the Stop button responsive.
            if stop_event.wait(0.001):
                return None

    return find_discovery_response(buffer, slave, fc)

def bus_activity_detected(ser, listen_s, stop_event):
    """Listen passively for *listen_s* seconds; return the bytes received.

    Any byte, even one framed at another baud rate, means something else is
    transmitting, so the active scan must not start.
    """
    deadline = time.perf_counter() + max(0.0, float(listen_s))
    received = bytearray()
    while time.perf_counter() < deadline:
        if stop_event.is_set():
            break
        waiting = int(getattr(ser, "in_waiting", 0) or 0)
        if waiting:
            received.extend(ser.read(waiting) or b"")
            if received:
                break
        elif stop_event.wait(0.005):
            break
    return bytes(received)


def cancel_serial_io(ser):
    """Best-effort cancellation of pending serial I/O."""
    if ser is None:
        return
    for method_name in ("cancel_read", "cancel_write"):
        method = getattr(ser, method_name, None)
        if callable(method):
            try:
                method()
            except Exception:
                pass
```

MBToolkit v3.4/mb_slave_finder.py — lines 441-480

```
441
442 def recover_discovery_write(ser):
443     """Best-effort cleanup after a discovery write timeout."""
444     cancel = getattr(ser, "cancel_write", None)
445     if callable(cancel):
446         try:
447             cancel()
448         except Exception:
449             pass
450     try:
451         ser.reset_output_buffer()
452     except Exception:
453         pass
454     try:
455         ser.reset_input_buffer()
456     except Exception:
457         pass
458
459 def write_discovery_request(
460     ser, request, baud, parity, stopbits, stop_event, serial_module
461 ):
462     """
463     Transmit one discovery request robustly.
464
465     A write timeout is retried once after cancelling/clearing TX. ser.flush()
466     is deliberately avoided because some USB-serial drivers can block there.
467     The helper also waits for the request's estimated physical on-wire time so
468     the slave response timeout starts at a sensible point on slow baud rates.
469     """
470     request = bytes(request)
471     timeout_exc = getattr(serial_module, "SerialTimeoutException", TimeoutError)
472     serial_exc = getattr(serial_module, "SerialException", OSError)
473     attempts = DISCOVERY_WRITE_RETRIES + 1
474     last_error = None
475     bits_per_char = discovery_bits_per_char(parity, stopbits)
476     char_time_s = bits_per_char / float(max(1, int(baud)))
477     request_wire_s = len(request) * char_time_s
478     retry_silence_s = max(
479         0.050,
480         request_wire_s + modbus_rtu_interframe_gap_seconds(baud, bits_per_char),
```

MBToolkit v3.4/mb_slave_finder.py — lines 481-520

```
481     )
482
483     for attempt in range(attempts):
484         if stop_event.is_set():
485             return False
486
487         started_write = time.perf_counter()
488         try:
489             written_total = 0
490             while written_total < len(request):
491                 if stop_event.is_set():
492                     return False
493                 written = ser.write(request[written_total:])
494                 written = 0 if written is None else int(written)
495                 if written <= 0:
496                     raise timeout_exc(
497                         "A porta série não aceitou bytes para transmissão."
498                     )
499                 written_total += written
500
501             elapsed_write_s = time.perf_counter() - started_write
502             remaining_wire_s = max(0.0, request_wire_s - elapsed_write_s)
503             if remaining_wire_s > 0 and stop_event.wait(remaining_wire_s):
504                 return False
505             return True
506
507         except timeout_exc as exc:
508             last_error = exc
509             recover_discovery_write(ser)
510             if attempt + 1 >= attempts or stop_event.is_set():
511                 raise
512             # If any fragment reached the wire before the driver timed out,
513             # leave enough silence for that fragment to finish and for a full
514             # Modbus RTU frame gap before retrying.
515             if stop_event.wait(retry_silence_s):
516                 return False
517
518         except serial_exc:
519             raise
520
```

MBToolkit v3.4/mb_slave_finder.py — lines 521-560

```
521     if last_error is not None:
522         raise last_error
523     return False
524 
525 def format_duration(seconds: float) -> str:
526     total = max(0, int(round(float(seconds))))
527     hours, rem = divmod(total, 3600)
528     minutes, secs = divmod(rem, 60)
529     if hours:
530         return f"{hours:d}:{minutes:02d}:{secs:02d}"
531     return f"{minutes:02d}:{secs:02d}"
532 
533 
534 class SlaveFinderMixin:
535     """Methods that implement the active Bus Slave Finder tab."""
536 
537     def build_discovery_tab(self, page):
538         """Build the full-window active Modbus RTU slave-discovery workspace."""
539         finder_saved = getattr(self, "ui_settings", {}).get(
540             "bus_slave_finder", {}
541         )
542         if not isinstance(finder_saved, dict):
543             finder_saved = {}
544 
545         saved_bauds = set(
546             finder_saved.get("baud_rates", [9600, 19200])
547         )
548         saved_parities = set(
549             finder_saved.get("parities", ["None", "Even"])
550         )
551         saved_stops = set(
552             finder_saved.get("stop_bits", ["1"])
553         )
554 
555         page.columnconfigure(0, weight=1)
556         page.rowconfigure(3, weight=1)
557 
558         warning = ttk.LabelFrame(page, text="Pesquisa ativa — transmite no barramento")
559         warning.grid(row=0, column=0, sticky="ew", padx=12, pady=(12, 6))
560         warning.columnconfigure(0, weight=1)
```

MBToolkit v3.4/mb_slave_finder.py — lines 561-600

```
        ttk.Label(
            warning,
            text=(
                "Esta ferramenta atua temporariamente como master Modbus RTU e envia apenas "
                "pedidos de leitura. Não a utilizes com outro master ativo no mesmo barramento."
            ),
            wraplength=1100,
        ).grid(row=0, column=0, sticky="w", padx=10, pady=(8, 4))

        self.discovery_safety_var = tk.BooleanVar(value=False)
        safety_cb = ttk.Checkbutton(
            warning,
            text="Confirmo que não existe outro master ativo no barramento.",
            variable=self.discovery_safety_var,
            command=self.update_discovery_start_state,
        )
        safety_cb.grid(row=1, column=0, sticky="w", padx=10, pady=(0, 8))

        options = ttk.LabelFrame(page, text="Configuração da pesquisa")
        self._discovery_options_frame = options
        options.grid(row=1, column=0, sticky="ew", padx=12, pady=6)
        options.columnconfigure(0, weight=1)

        top_row = ttk.Frame(options)
        top_row.grid(row=0, column=0, sticky="ew", padx=8, pady=(8, 4))
        top_row.columnconfigure(3, weight=1)

        ttk.Label(top_row, text="Porta COM").grid(
            row=0, column=0, sticky="w", padx=(0, 4)
        )
        self.discovery_port_var = tk.StringVar()
        self.discovery_port_combo = ttk.Combobox(
            top_row,
            textvariable=self.discovery_port_var,
            width=13,
            state="readonly",
        )
        self.discovery_port_combo.grid(
            row=0, column=1, sticky="w", padx=(0, 0)
        )
```

MBToolkit v3.4/mb_slave_finder.py — lines 601-640

```
601         self.discovery_port_combo.bind(
602             "<<ComboboxSelected>>",
603             lambda _e: self.update_discovery_start_state(),
604         )
605 
606         self.discovery_refresh_btn = RefreshButton(
607             top_row,
608             command=self.refresh_ports,
609             size=24,
610         )
611         self.discovery_refresh_btn.grid(
612             row=0, column=2, sticky="w", padx=(4, 0)
613         )
614 
615         def _sync_discovery_refresh_square(_event=None):
616             height = self.discovery_port_combo.winfo_height()
617             if height > 1:
618                 self.discovery_refresh_btn.configure(
619                     width=height, height=height
620                 )
621 
622         self.discovery_port_combo.bind(
623             "<Configure>", _sync_discovery_refresh_square, add="+"
624         )
625 
626         ttk.Label(
627             top_row,
628             text="Data bits: 8 (fixo)",
629         ).grid(row=0, column=4, sticky="w", padx=(28, 20))
630         ttk.Label(
631             top_row,
632             text="Probe: FC03 / Address 0 / Qty 1",
633         ).grid(row=0, column=5, sticky="w")
634 
635         ttk.Label(top_row, text="Tentativas").grid(
636             row=0, column=6, sticky="w", padx=(28, 4)
637         )
638         self.discovery_attempts_var = tk.StringVar(
639             value=finder_saved.get("attempts", str(DISCOVERY_DEFAULT_ATTEMPTS))
640         )
```

MBToolkit v3.4/mb_slave_finder.py — lines 641-680

```
        self.discovery_attempts_spin = ttk.Spinbox(
            top_row,
            from_=DISCOVERY_MIN_ATTEMPTS,
            to=DISCOVERY_MAX_ATTEMPTS,
            increment=1,
            width=4,
            textvariable=self.discovery_attempts_var,
        )
        self.discovery_attempts_spin.grid(row=0, column=7, sticky="w")

        # Baud-rate checkboxes. Defaults keep a normal industrial scan quick.
        baud_frame = ttk.LabelFrame(options, text="Baud rate")
        baud_frame.grid(row=1, column=0, sticky="ew", padx=8, pady=4)
        baud_frame.rowconfigure(0, weight=1)
        self.discovery_baud_vars = {}
        for idx in range(len(DISCOVERY_BAUD_RATES)):
            baud_frame.columnconfigure(idx, weight=1, uniform="baud")
        for idx, baud in enumerate(DISCOVERY_BAUD_RATES):
            var = tk.BooleanVar(value=baud in saved_bauds)
            self.discovery_baud_vars[baud] = var
            cb = ttk.Checkbutton(baud_frame, text=str(baud), variable=var)
            cb.grid(row=0, column=idx, sticky="", padx=4, pady=(8, 7))
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())

        serial_frame = ttk.Frame(options)
        serial_frame.grid(row=2, column=0, sticky="ew", padx=8, pady=4)
        # Equal thirds independent of translated child captions.
        for column in range(3):
            serial_frame.columnconfigure(
                column, weight=1, uniform="finder_serial_columns"
            )

        parity_box = ttk.LabelFrame(serial_frame, text="Parity")
        parity_box.grid(row=0, column=0, sticky="nsew", padx=(0, 5))
        parity_box.rowconfigure(0, weight=1)
        for idx in range(len(DISCOVERY_PARITIES)):
            parity_box.columnconfigure(idx, weight=1, uniform="parity")
        self.discovery_parity_vars = {}
        for idx, parity in enumerate(DISCOVERY_PARITIES):
            var = tk.BooleanVar(value=parity in saved_parities)
```

MBToolkit v3.4/mb_slave_finder.py — lines 681-720

```
            self.discovery_parity_vars[parity] = var
            ttk.Checkbutton(parity_box, text=parity, variable=var).grid(
                row=0, column=idx, sticky="", padx=4, pady=(8, 7)
            )
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())
        stop_box = ttk.LabelFrame(serial_frame, text="Stop bits")
        stop_box.grid(row=0, column=1, sticky="nsew", padx=5)
        stop_box.rowconfigure(0, weight=1)
        for idx in range(len(DISCOVERY_STOP_BITS)):
            stop_box.columnconfigure(idx, weight=1, uniform="stop")
        self.discovery_stop_vars = {}
        for idx, stopbits in enumerate(DISCOVERY_STOP_BITS):
            var = tk.BooleanVar(value=stopbits in saved_stops)
            self.discovery_stop_vars[stopbits] = var
            ttk.Checkbutton(stop_box, text=stopbits, variable=var).grid(
                row=0, column=idx, sticky="", padx=4, pady=(8, 7)
            )
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())
        range_box = ttk.LabelFrame(serial_frame, text="Slave IDs / velocidade")
        range_box.grid(row=0, column=2, sticky="nsew", padx=(5, 0))
        range_box.columnconfigure(0, weight=1)
        range_box.rowconfigure(0, weight=1)

        # Keep the range and timeout controls on one centred line so this
        # LabelFrame stays as compact as the neighbouring Parity/Stop bits
        # boxes.  The option checkboxes keep their own second line so the long
        # Device Identification caption remains fully inside the rectangle.
        range_content = ttk.Frame(range_box)
        range_content.grid(
            row=0, column=0, sticky="nsew", padx=8, pady=(8, 7)
        )
        range_content.columnconfigure(0, weight=1)
        for row in range(2):
            range_content.rowconfigure(row, weight=1, uniform="range_rows")

        id_row = ttk.Frame(range_content)
        id_row.grid(row=0, column=0, sticky="")
        ttk.Label(id_row, text="De").grid(row=0, column=0, padx=(0, 3))
        self.discovery_slave_start_var = tk.StringVar(value=finder_saved.get("slave_start", "1"))
        self.discovery_slave_start_spin = ttk.Spinbox(
```

MBToolkit v3.4/mb_slave_finder.py — lines 721-760

```
            id_row, from_=1, to=247, width=5, textvariable=self.discovery_slave_start_var
        )
        self.discovery_slave_start_spin.grid(row=0, column=1, padx=3)
        ttk.Label(id_row, text="até").grid(row=0, column=2, padx=3)
        self.discovery_slave_end_var = tk.StringVar(value=finder_saved.get("slave_end", "247"))
        self.discovery_slave_end_spin = ttk.Spinbox(
            id_row, from_=1, to=247, width=5, textvariable=self.discovery_slave_end_var
        )
        self.discovery_slave_end_spin.grid(row=0, column=3, padx=3)

        ttk.Label(id_row, text="Timeout mín.").grid(
            row=0, column=4, padx=(14, 3)
        )
        self.discovery_min_timeout_var = tk.StringVar(
            value=finder_saved.get(
                "min_timeout_ms",
                str(DISCOVERY_DEFAULT_MIN_TIMEOUT_MS),
            )
        )
        self.discovery_min_timeout_spin = ttk.Spinbox(
            id_row,
            from_=DISCOVERY_MIN_TIMEOUT_MS,
            to=DISCOVERY_MAX_TIMEOUT_MS,
            increment=5,
            width=6,
            textvariable=self.discovery_min_timeout_var,
        )
        self.discovery_min_timeout_spin.grid(row=0, column=5, padx=3)
        ttk.Label(id_row, text="ms").grid(row=0, column=6, padx=3)

        toggles = ttk.Frame(range_content)
        # Small vertical separation from the Slave range/timeout row.
        # Keep this deliberately modest so the panel remains compact.
        toggles.grid(row=1, column=0, sticky="", pady=(3, 0))

        self.discovery_fc04_fallback_var = tk.BooleanVar(value=finder_saved.get("fc04_fallback", False))
        fallback_cb = ttk.Checkbutton(
            toggles,
            text="FC04 fallback",
            variable=self.discovery_fc04_fallback_var,
```

MBToolkit v3.4/mb_slave_finder.py — lines 761-800

```
        )
        fallback_cb.grid(row=0, column=0, sticky="w", padx=(0, 10))

        self.discovery_device_identification_var = tk.BooleanVar(
            value=finder_saved.get("device_identification", False)
        )
        device_id_cb = ttk.Checkbutton(
            toggles,
            text="Device Identification (FC43/14)",
            variable=self.discovery_device_identification_var,
        )
        device_id_cb.grid(row=0, column=1, sticky="w")

        for var in (
            self.discovery_slave_start_var,
            self.discovery_slave_end_var,
            self.discovery_min_timeout_var,
            self.discovery_attempts_var,
            self.discovery_fc04_fallback_var,
            self.discovery_device_identification_var,
        ):
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())

        ttk.Label(
            options,
            text=(
                "FC03 é o modo rápido. Uma resposta normal OU uma Modbus Exception com CRC válido "
                "conta como slave encontrado. Ativa FC04 fallback apenas para dispositivos que "
                "possam ignorar FC03 sem devolver Exception. Device Identification envia FC43/14 "
                "apenas aos slaves encontrados e tenta ler VendorName, ProductCode e Revision. "
                "Tentativas repete cada pedido sem resposta. Antes de transmitir, a pesquisa escuta "
                "o barramento durante 1 s e não arranca se detetar tráfego."
            ),
            wraplength=1120,
        ).grid(row=3, column=0, sticky="w", padx=10, pady=(4, 8))

        controls = ttk.LabelFrame(page, text="Pesquisa")
        controls.grid(row=2, column=0, sticky="ew", padx=12, pady=6)
        controls.columnconfigure(4, weight=1)

```

MBToolkit v3.4/mb_slave_finder.py — lines 801-840

```
        self.discovery_start_btn = ttk.Button(
            controls, text="Iniciar pesquisa", command=self.start_discovery_scan
        )
        self.discovery_start_btn.grid(row=0, column=0, padx=(10, 5), pady=8)
        self.discovery_stop_btn = ttk.Button(
            controls, text="Parar", command=self.stop_discovery_scan, state="disabled"
        )
        self.discovery_stop_btn.grid(row=0, column=1, padx=5, pady=8)
        self.discovery_clear_btn = ttk.Button(
            controls, text="Limpar resultados", command=self.clear_discovery_results
        )
        self.discovery_clear_btn.grid(row=0, column=2, padx=5, pady=8)

        self.discovery_found_var = tk.StringVar(value="Encontrados: 0")
        ttk.Label(controls, textvariable=self.discovery_found_var).grid(
            row=0, column=3, padx=(18, 8), pady=8, sticky="w"
        )

        self.discovery_status_var = tk.StringVar(value="Parado")
        ttk.Label(controls, textvariable=self.discovery_status_var).grid(
            row=0, column=4, padx=8, pady=8, sticky="w"
        )

        self.discovery_progress_var = tk.DoubleVar(value=0.0)
        self.discovery_progress = ttk.Progressbar(
            controls, variable=self.discovery_progress_var, maximum=100.0
        )
        self.discovery_progress.grid(
            row=1, column=0, columnspan=5, sticky="ew", padx=10, pady=(0, 5)
        )

        self.discovery_progress_text_var = tk.StringVar(value="0 / 0")
        self.discovery_estimate_var = tk.StringVar(value="")
        ttk.Label(controls, textvariable=self.discovery_progress_text_var).grid(
            row=2, column=0, columnspan=2, sticky="w", padx=10, pady=(0, 8)
        )
        ttk.Label(controls, textvariable=self.discovery_estimate_var).grid(
            row=2, column=2, columnspan=3, sticky="e", padx=10, pady=(0, 8)
        )

```

MBToolkit v3.4/mb_slave_finder.py — lines 841-880

```
        results = ttk.LabelFrame(page, text="Slaves encontrados")
        results.grid(row=3, column=0, sticky="nsew", padx=12, pady=(6, 12))
        results.columnconfigure(0, weight=1)
        results.rowconfigure(0, weight=1)

        columns = (
            "slave", "baud", "config", "fc", "result", "resp", "device_id", "raw"
        )
        self.discovery_tree = ttk.Treeview(
            results, columns=columns, show="headings", selectmode="browse"
        )
        headings = {
            "slave": "Slave",
            "baud": "Baud",
            "config": "Config.",
            "fc": "FC",
            "result": "Resultado",
            "resp": "Resp. (ms)",
            "device_id": "Device Identification",
            "raw": "Raw Hex",
        }
        # Match the Traffic table behaviour: no horizontal scrollbar. Keep
        # short fields compact and let the three text-heavy columns share the
        # remaining viewport width. Column widths are recalculated whenever
        # the Treeview is resized so the complete table remains visible.
        self.discovery_fixed_widths = {
            "slave": 70,
            "baud": 90,
            "config": 85,
            "fc": 55,
            "resp": 95,
        }
        self.discovery_flex_columns = ("result", "device_id", "raw")
        self.discovery_flex_weights = {
            "result": 0.27,
            "device_id": 0.38,
            "raw": 0.35,
        }
        self.discovery_flex_min_width = 90

```

MBToolkit v3.4/mb_slave_finder.py — lines 881-920

```
        initial_widths = {
            **self.discovery_fixed_widths,
            "result": 220,
            "device_id": 330,
            "raw": 300,
        }
        for col in columns:
            self.discovery_tree.heading(col, text=headings[col], anchor="center")
            self.discovery_tree.column(
                col,
                width=initial_widths[col],
                minwidth=(
                    self.discovery_flex_min_width
                    if col in self.discovery_flex_columns
                    else self.discovery_fixed_widths[col]
                ),
                anchor="w" if col in self.discovery_flex_columns else "center",
                stretch=False,
            )

        discovery_y = ttk.Scrollbar(
            results, orient="vertical", command=self.discovery_tree.yview
        )
        self.discovery_tree.configure(yscrollcommand=discovery_y.set)
        self.discovery_tree.grid(row=0, column=0, sticky="nsew")
        discovery_y.grid(row=0, column=1, sticky="ns")
        self.discovery_tree.bind(
            "<Configure>", self.fit_discovery_columns_to_viewport, add="+"
        )
        # Right-click / double-click a found Slave to read it in the Master tab.
        self.discovery_context_menu = tk.Menu(self, tearoff=False)
        self.discovery_context_menu.add_command(
            label="Abrir no Master", command=self.discovery_open_selected_in_master
        )
        self.discovery_tree.bind("<Button-3>", self._show_discovery_context_menu)
        self.discovery_tree.bind(
            "<Double-1>", lambda _e: self.discovery_open_selected_in_master()
        )

        self.update_discovery_estimate()
```

MBToolkit v3.4/mb_slave_finder.py — lines 921-960

```
921         self.update_discovery_start_state()
922 
923     def _show_discovery_context_menu(self, event):
924         row = self.discovery_tree.identify_row(event.y)
925         if not row:
926             return
927         self.discovery_tree.selection_set(row)
928         try:
929             self.discovery_context_menu.entryconfigure(0, label=self.tr("Abrir no Master"))
930             self.discovery_context_menu.tk_popup(event.x_root, event.y_root)
931         finally:
932             self.discovery_context_menu.grab_release()
933 
934     def discovery_open_selected_in_master(self):
935         """Open the selected found Slave in a new Master window."""
936         selection = self.discovery_tree.selection()
937         if not selection:
938             return
939         values = self.discovery_tree.item(selection[0], "values")
940         try:
941             slave, baud, config, fc = int(values[0]), str(values[1]), str(values[2]), int(values[3])
942         except (IndexError, ValueError):
943             return
944         parity = {"N": "None", "E": "Even", "O": "Odd"}.get(config[1:2], "None")
945         stopbits = config[2:] or "1"
946         self.master_open_definition(
947             {"slave": slave, "fc": fc, "address": 0, "qty": 10},
948             port=self.discovery_port_var.get().strip(),
949             baud=baud, parity=parity, stopbits=stopbits,
950         )
951 
952     def fit_discovery_columns_to_viewport(self, _event=None):
953         """Fit Slave Finder columns to the visible Treeview width."""
954         if not hasattr(self, "discovery_tree"):
955             return
956 
957         viewport_width = int(self.discovery_tree.winfo_width())
958         if viewport_width <= 1:
959             return
960 
```

MBToolkit v3.4/mb_slave_finder.py — lines 961-1000

```
        # Reserve a few pixels for Treeview borders so Tk never needs a hidden
        # horizontal overflow area. The vertical scrollbar is a sibling and is
        # therefore already excluded from winfo_width().
        usable_width = max(1, viewport_width - 4)
        fixed_total = sum(self.discovery_fixed_widths.values())
        flex_count = len(self.discovery_flex_columns)
        min_flex_total = self.discovery_flex_min_width * flex_count

        flex_total = max(min_flex_total, usable_width - fixed_total)
        allocated = 0
        for index, col in enumerate(self.discovery_flex_columns):
            if index == flex_count - 1:
                width = flex_total - allocated
            else:
                width = int(round(flex_total * self.discovery_flex_weights[col]))
                allocated += width
            width = max(self.discovery_flex_min_width, width)
            self.discovery_tree.column(col, width=width)

        for col, width in self.discovery_fixed_widths.items():
            self.discovery_tree.column(col, width=width)

    def on_discovery_options_changed(self):
        if hasattr(self, "discovery_estimate_var"):
            self.update_discovery_estimate()
            self.update_discovery_start_state()

    def selected_discovery_configs(self):
        bauds = [baud for baud in DISCOVERY_BAUD_RATES if self.discovery_baud_vars[baud].get()]
        parities = [p for p in DISCOVERY_PARITIES if self.discovery_parity_vars[p].get()]
        stops = [s for s in DISCOVERY_STOP_BITS if self.discovery_stop_vars[s].get()]
        return [
            (baud, parity, stopbits)
            for baud in bauds
            for parity in parities
            for stopbits in stops
        ]

    def selected_discovery_slaves(self):
        start = int(self.discovery_slave_start_var.get())
```

MBToolkit v3.4/mb_slave_finder.py — lines 1001-1040

```
        end = int(self.discovery_slave_end_var.get())
        if not (1 <= start <= 247 and 1 <= end <= 247):
            raise ValueError(self.tr("Os Slave IDs têm de estar entre 1 e 247."))
        if start > end:
            raise ValueError(self.tr("O Slave ID inicial não pode ser superior ao final."))
        return list(range(start, end + 1))

    def selected_discovery_min_timeout_ms(self):
        value = float(self.discovery_min_timeout_var.get())
        if not (DISCOVERY_MIN_TIMEOUT_MS <= value <= DISCOVERY_MAX_TIMEOUT_MS):
            raise ValueError(
                self.trf(
                    "O timeout mínimo tem de estar entre {minimum} e {maximum} ms.",
                    minimum=DISCOVERY_MIN_TIMEOUT_MS,
                    maximum=DISCOVERY_MAX_TIMEOUT_MS,
                )
            )
        return value

    def selected_discovery_attempts(self):
        value = int(self.discovery_attempts_var.get())
        if not (DISCOVERY_MIN_ATTEMPTS <= value <= DISCOVERY_MAX_ATTEMPTS):
            raise ValueError(
                self.trf(
                    "Tentativas tem de estar entre {minimum} e {maximum}.",
                    minimum=DISCOVERY_MIN_ATTEMPTS,
                    maximum=DISCOVERY_MAX_ATTEMPTS,
                )
            )
        return value

    def discovery_config_label(self, parity, stopbits):
        parity_letter = {"None": "N", "Even": "E", "Odd": "O"}[parity]
        return f"8{parity_letter}{stopbits}"

    def update_discovery_estimate(self):
        try:
            configs = self.selected_discovery_configs()
            slaves = self.selected_discovery_slaves()
            minimum_timeout_ms = self.selected_discovery_min_timeout_ms()
```

MBToolkit v3.4/mb_slave_finder.py — lines 1041-1080

```
            attempts = self.selected_discovery_attempts()
        except Exception:
            self.discovery_estimate_var.set(self.tr("Máx. estimado: —"))
            self.discovery_progress_text_var.set("0 / 0")
            return

        total_units = len(configs) * len(slaves)
        probes_per_unit = 2 if self.discovery_fc04_fallback_var.get() else 1
        identify_devices = bool(self.discovery_device_identification_var.get())

        seconds = 0.0
        for baud, parity, stopbits in configs:
            per_probe = (
                discovery_request_transmit_seconds(baud, parity, stopbits)
                + discovery_response_timeout_seconds(
                    baud, parity, stopbits, minimum_timeout_ms
                )
            )
            # Worst case: no Slave answers, so every probe uses all attempts.
            seconds += len(slaves) * probes_per_unit * attempts * per_probe
            if identify_devices:
                # Worst case: every probed Slave exists and therefore receives
                # one optional Basic Device Identification query.
                seconds += len(slaves) * (
                    device_identification_request_transmit_seconds(
                        baud, parity, stopbits
                    )
                    + device_identification_timeout_seconds(
                        baud, parity, stopbits, minimum_timeout_ms
                    )
                )
            seconds += 0.025  # small reopen/reconfigure allowance
        seconds += DISCOVERY_IDLE_CHECK_SECONDS if configs else 0.0

        self.discovery_progress_text_var.set(f"0 / {total_units}")
        suffix_parts = []
        if probes_per_unit == 2:
            suffix_parts.append("FC04 fallback")
        if identify_devices:
            suffix_parts.append("Device Identification")
```

MBToolkit v3.4/mb_slave_finder.py — lines 1081-1120

```
        suffix = (
            self.trf(" (com {options})", options=" + ".join(suffix_parts))
            if suffix_parts else ""
        )
        self.discovery_estimate_var.set(
            self.trf(
                "{configs} config. × {slaves} slaves — máx. estimado: {duration}",
                configs=len(configs),
                slaves=len(slaves),
                duration=format_duration(seconds),
            )
            + suffix
        )

    def update_discovery_start_state(self):
        if not hasattr(self, "discovery_start_btn"):
            return
        enabled = (
            not self.discovery_active
            and not self.busy
            and bool(self.discovery_port_var.get().strip())
            and bool(self.discovery_safety_var.get())
        )
        try:
            enabled = enabled and bool(self.selected_discovery_configs())
            self.selected_discovery_slaves()
            self.selected_discovery_min_timeout_ms()
            self.selected_discovery_attempts()
        except Exception:
            enabled = False
        self.discovery_start_btn.configure(state="normal" if enabled else "disabled")

    def clear_discovery_results(self):
        if self.discovery_active:
            return
        for iid in self.discovery_tree.get_children():
            self.discovery_tree.delete(iid)
        self.discovery_found_count = 0
        self.discovery_found_var.set(self.trf("Encontrados: {count}", count=0))
        self.discovery_progress_var.set(0.0)
```

MBToolkit v3.4/mb_slave_finder.py — lines 1121-1160

```
        self.discovery_status_var.set(self.tr("Parado"))
        self.update_discovery_estimate()

    def open_discovery_serial_port(self, serial, port, baud, parity, stopbits):
        parity_map = {
            "None": serial.PARITY_NONE,
            "Even": serial.PARITY_EVEN,
            "Odd": serial.PARITY_ODD,
        }
        stop_map = {
            "1": serial.STOPBITS_ONE,
            "2": serial.STOPBITS_TWO,
        }

        ser = serial.Serial()
        ser.port = port
        ser.baudrate = int(baud)
        ser.bytesize = serial.EIGHTBITS
        ser.parity = parity_map[parity]
        ser.stopbits = stop_map[stopbits]
        ser.timeout = 0

        ser.write_timeout = DISCOVERY_WRITE_TIMEOUT_SECONDS
        ser.xonxoff = False
        ser.rtscts = False
        ser.dsrdtr = False

        # Do not force RTS/DTR levels. Many USB-RS485 adapters manage direction
        # internally and forcing modem-control lines can interfere with some
        # adapters/drivers. Hardware flow control is disabled above.
        ser.open()

        try:
            ser.reset_input_buffer()
        except Exception:
            pass
        try:
            ser.reset_output_buffer()
        except Exception:
            pass
```

MBToolkit v3.4/mb_slave_finder.py — lines 1161-1200

```
1161
1162        return ser
1163
1164    def start_discovery_scan(self):
1165        if self.busy or self.discovery_active:
1166            return
1167
1168        serial, _ = self.get_serial_modules()
1169        if serial is None:
1170            messagebox.showerror(
1171                self.tr("pyserial em falta"),
1172                self.tr("pyserial não está instalado. O build_exe.bat inclui-o automaticamente no EXE.")
1173            )
1174            return
1175
1176        if not self.discovery_safety_var.get():
1177            messagebox.showwarning(
1178                self.tr("Pesquisa ativa"),
1179                self.tr("Confirma primeiro que não existe outro master ativo no barramento.")
1180            )
1181            return
1182
1183        try:
1184            port = self.discovery_port_var.get().strip()
1185            if not port:
1186                raise ValueError(self.tr("Seleciona uma porta COM."))
1187            configs = self.selected_discovery_configs()
1188            if not configs:
1189                raise ValueError(self.tr("Seleciona pelo menos uma configuração série."))
1190            slaves = self.selected_discovery_slaves()
1191            minimum_timeout_ms = self.selected_discovery_min_timeout_ms()
1192            attempts = self.selected_discovery_attempts()
1193        except Exception as exc:
1194            messagebox.showerror(self.tr("Configuração inválida"), str(exc))
1195            return
1196
1197        fallback_fc04 = bool(self.discovery_fc04_fallback_var.get())
1198        identify_devices = bool(self.discovery_device_identification_var.get())
1199
1200        self.clear_discovery_results()
```

MBToolkit v3.4/mb_slave_finder.py — lines 1201-1240

```
        self.discovery_active = True
        self.busy = True
        self.discovery_stop_event.clear()
        self.discovery_found_count = 0
        self.discovery_progress_var.set(0.0)
        self.discovery_status_var.set(self.tr("A iniciar pesquisa…"))

        self.discovery_start_btn.configure(state="disabled")
        self.discovery_stop_btn.configure(state="normal")
        self.discovery_clear_btn.configure(state="disabled")
        self.discovery_refresh_btn.configure(state="disabled")
        self.discovery_port_combo.configure(state="disabled")

        # Prevent passive capture/simulation from being started concurrently.
        self.start_btn.configure(state="disabled")
        self.stop_btn.configure(state="disabled")
        self.restart_btn.configure(state="disabled")
        self.set_sniffer_refresh_state("disabled")
        self.set_simulation_button_state("disabled")

        immutable_configs = list(configs)
        immutable_slaves = list(slaves)

        self.discovery_thread = threading.Thread(
            target=self.discovery_worker,
            args=(
                serial, port, immutable_configs, immutable_slaves,
                minimum_timeout_ms, fallback_fc04, identify_devices, attempts
            ),
            daemon=True,
        )
        try:
            self.discovery_thread.start()
        except Exception as exc:
            # Restore the interface if the worker cannot even be started.
            self.discovery_active = False
            self.busy = False
            self.discovery_thread = None
            self.discovery_stop_btn.configure(state="disabled")
            self.discovery_clear_btn.configure(state="normal")
```

MBToolkit v3.4/mb_slave_finder.py — lines 1241-1280

```
            self.discovery_refresh_btn.configure(state="normal")
            self.discovery_port_combo.configure(state="readonly")
            self.start_btn.configure(state="normal")
            self.restart_btn.configure(state="normal")
            self.set_sniffer_refresh_state("normal")
            self.set_simulation_button_state("normal")
            self.discovery_status_var.set(self.tr("Erro ao iniciar pesquisa"))
            self.update_discovery_start_state()
            messagebox.showerror(
                self.tr("Erro na Bus Slave Finder"),
                self.trf(
                    "Não foi possível iniciar a thread de pesquisa:\n\n{error}",
                    error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                )
            )

    def stop_discovery_scan(self):
        if self.discovery_active:
            self.discovery_status_var.set(self.tr("A parar…"))
            self.discovery_stop_event.set()
            self.discovery_stop_btn.configure(state="disabled")
            # Wake/cancel pending driver I/O without closing the handle from
            # the Tk thread while the discovery worker may still be using it.
            cancel_serial_io(self.discovery_serial)

    def discovery_worker(
        self, serial, port, configs, slaves, minimum_timeout_ms, fallback_fc04,
        identify_devices, attempts=DISCOVERY_DEFAULT_ATTEMPTS
    ):
        started = time.perf_counter()
        total_units = len(configs) * len(slaves)
        completed = 0
        found_count = 0
        fatal_error = None
        context = str(port)
        idle_checked = False

        try:
            for baud, parity, stopbits in configs:
                if self.discovery_stop_event.is_set():
```

MBToolkit v3.4/mb_slave_finder.py — lines 1281-1320

```
                    break

                config_label = self.discovery_config_label(parity, stopbits)
                context = f"{port} — {baud} / {config_label}"
                self.event_queue.put((
                    "discovery_status",
                    self.trf(
                        "A configurar {baud} / {config}…",
                        baud=baud,
                        config=config_label,
                    )
                ))

                ser = None
                try:
                    try:
                        ser = self.open_discovery_serial_port(
                            serial, port, baud, parity, stopbits
                        )
                    except Exception as exc:
                        raise RuntimeError(
                            self.trf(
                                "Não foi possível abrir {port} em {baud} / {config}: {error}",
                                port=port,
                                baud=baud,
                                config=config_label,
                                error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                            )
                        ) from exc

                    self.discovery_serial = ser
                    if self.discovery_stop_event.wait(0.025):
                        break

                    if not idle_checked:
                        # Last line of defence behind the safety checkbox:
                        # never transmit onto a bus that is already busy.
                        idle_checked = True
                        self.event_queue.put((
                            "discovery_status",
```

MBToolkit v3.4/mb_slave_finder.py — lines 1321-1360

```
                            self.tr("A verificar se o barramento está livre…"),
                        ))
                        traffic = bus_activity_detected(
                            ser,
                            DISCOVERY_IDLE_CHECK_SECONDS,
                            self.discovery_stop_event,
                        )
                        if traffic:
                            raise RuntimeError(
                                self.trf(
                                    "Foi detetado tráfego em {port} antes de transmitir "
                                    "({count} bytes: {raw}). Pode existir outro master "
                                    "ativo no barramento; a pesquisa não foi iniciada.",
                                    port=port,
                                    count=len(traffic),
                                    raw=hex_bytes(traffic[:16]),
                                )
                            )
                        if self.discovery_stop_event.is_set():
                            break
                        self.event_queue.put((
                            "discovery_status",
                            self.trf(
                                "A configurar {baud} / {config}…",
                                baud=baud,
                                config=config_label,
                            ),
                        ))

                    response_timeout_s = discovery_response_timeout_seconds(
                        baud, parity, stopbits, minimum_timeout_ms
                    )
                    char_gap_s = modbus_rtu_interframe_gap_seconds(
                        baud, discovery_bits_per_char(parity, stopbits)
                    )

                    for slave in slaves:
                        if self.discovery_stop_event.is_set():
                            break

```

MBToolkit v3.4/mb_slave_finder.py — lines 1361-1400

```
                    result = None
                    used_fc = None
                    elapsed_ms = None
                    probe_fcs = [3, 4] if fallback_fc04 else [3]

                    probe_plan = [
                        (fc, attempt)
                        for fc in probe_fcs
                        for attempt in range(1, attempts + 1)
                    ]
                    for fc, attempt in probe_plan:
                        if self.discovery_stop_event.is_set():
                            break

                        context = (
                            f"{port} — {baud} / {config_label} / "
                            f"Slave {slave} / FC{fc:02X}"
                        )
                        if attempts > 1:
                            context += f" / {attempt}/{attempts}"

                        # If this fails, the COM/driver is unhealthy. Do not
                        # disguise it as a normal no-response slave timeout.
                        try:
                            ser.reset_input_buffer()
                        except Exception as exc:
                            raise RuntimeError(
                                self.trf(
                                    "Falha ao preparar a receção em {context}: {error}",
                                    context=context,
                                    error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                )
                            ) from exc

                        request = build_discovery_probe(slave, fc)
                        started_probe = time.perf_counter()
                        try:
                            sent = write_discovery_request(
                                ser, request, baud, parity, stopbits,
                                self.discovery_stop_event, serial
```

MBToolkit v3.4/mb_slave_finder.py — lines 1401-1440

```
                                )
                            except Exception as exc:
                                raise RuntimeError(
                                    self.trf(
                                        "Falha ao transmitir em {context}: {error}",
                                        context=context,
                                        error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                    )
                                ) from exc

                            if not sent or self.discovery_stop_event.is_set():
                                break

                            try:
                                result = read_discovery_response(
                                    ser, slave, fc, response_timeout_s,
                                    self.discovery_stop_event
                                )
                            except Exception as exc:
                                raise RuntimeError(
                                    self.trf(
                                        "Falha ao receber em {context}: {error}",
                                        context=context,
                                        error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                    )
                                ) from exc

                            if result is not None:
                                elapsed_ms = (
                                    time.perf_counter() - started_probe
                                ) * 1000.0
                                used_fc = fc
                                if char_gap_s > 0:
                                    self.discovery_stop_event.wait(
                                        min(char_gap_s, 0.05)
                                    )
                                break

                        if result is not None and used_fc is not None:
                            found_count += 1
```

MBToolkit v3.4/mb_slave_finder.py — lines 1441-1480

```
                        if result["kind"] == "EXCEPTION":
                            code = result["exception_code"]
                            name = MODBUS_EXCEPTION_CODES.get(
                                code, ("Unknown/Reserved Exception", "")
                            )[0]
                            result_text = f"Exception 0x{code:02X} — {name}"
                        else:
                            result_text = "Response"

                        device_id_text = ""
                        if identify_devices and not self.discovery_stop_event.is_set():
                            context = (
                                f"{port} — {baud} / {config_label} / "
                                f"Slave {slave} / FC43/14"
                            )
                            try:
                                device_result = read_basic_device_identification(
                                    ser,
                                    slave,
                                    baud,
                                    parity,
                                    stopbits,
                                    minimum_timeout_ms,
                                    self.discovery_stop_event,
                                    serial,
                                )
                            except Exception as exc:
                                raise RuntimeError(
                                    self.trf(
                                        "Falha em Device Identification em {context}: {error}",
                                        context=context,
                                        error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                    )
                                ) from exc
                            device_id_text = format_device_identification(
                                device_result
                            )
                            if char_gap_s > 0:
                                self.discovery_stop_event.wait(
                                    min(char_gap_s, 0.05)
```

MBToolkit v3.4/mb_slave_finder.py — lines 1481-1520

```
                                    )

                            self.event_queue.put((
                                "discovery_found",
                                {
                                    "slave": slave,
                                    "baud": baud,
                                    "config": config_label,
                                    "fc": f"{used_fc:02X}",
                                    "result": result_text,
                                    "response_ms": elapsed_ms,
                                    "device_id": device_id_text,
                                    "raw": hex_bytes(result["frame"]),
                                }
                            ))

                        completed += 1
                        elapsed = time.perf_counter() - started
                        eta = (
                            max(
                                0.0,
                                (elapsed / completed) * (total_units - completed)
                            )
                            if completed > 0 else 0.0
                        )

                        self.event_queue.put((
                            "discovery_progress",
                            {
                                "completed": completed,
                                "total": total_units,
                                "baud": baud,
                                "config": config_label,
                                "slave": slave,
                                "elapsed": elapsed,
                                "eta": eta,
                                "found": found_count,
                            }
                        ))

```

MBToolkit v3.4/mb_slave_finder.py — lines 1521-1560

```
                finally:
                    if ser is not None:
                        cancel_serial_io(ser)
                        try:
                            ser.close()
                        except Exception:
                            pass
                    self.discovery_serial = None

        except Exception as exc:
            if not self.discovery_stop_event.is_set():
                fatal_error = str(exc) or f"{context}: {type(exc).__name__}"

        elapsed = time.perf_counter() - started
        self.event_queue.put((
            "discovery_done",
            {
                "stopped": self.discovery_stop_event.is_set(),
                "error": fatal_error,
                "elapsed": elapsed,
                "completed": completed,
                "total": total_units,
                "found": found_count,
            }
        ))

    def apply_discovery_progress(self, payload):
        completed = int(payload["completed"])
        total = max(1, int(payload["total"]))
        self.discovery_progress_var.set(100.0 * completed / total)
        self.discovery_progress_text_var.set(
            f"{completed} / {payload['total']} — "
            f"{payload['baud']} / {payload['config']} / Slave {payload['slave']}"
        )
        self.discovery_found_var.set(self.trf("Encontrados: {count}", count=payload["found"]))
        self.discovery_status_var.set(
            self.trf(
                "Decorrido {elapsed} — restante ~{remaining}",
                elapsed=format_duration(payload["elapsed"]),
                remaining=format_duration(payload["eta"]),
```

MBToolkit v3.4/mb_slave_finder.py — lines 1561-1600

```
            )
        )

    def add_discovery_result(self, payload):
        self.discovery_found_count += 1
        self.discovery_found_var.set(
            self.trf("Encontrados: {count}", count=self.discovery_found_count)
        )
        self.discovery_tree.insert(
            "", "end",
            values=(
                payload["slave"],
                payload["baud"],
                payload["config"],
                payload["fc"],
                payload["result"],
                f"{payload['response_ms']:.1f}" if payload.get("response_ms") is not None else "",
                payload.get("device_id", ""),
                payload["raw"],
            )
        )

    def finish_discovery_ui(self, payload):
        self.discovery_active = False
        self.busy = False
        self.discovery_stop_event.clear()
        self.discovery_serial = None
        self.discovery_thread = None

        self.discovery_stop_btn.configure(state="disabled")
        self.discovery_clear_btn.configure(state="normal")
        self.discovery_refresh_btn.configure(state="normal")
        self.discovery_port_combo.configure(state="readonly")

        self.start_btn.configure(state="normal")
        self.stop_btn.configure(state="disabled")
        self.restart_btn.configure(state="normal")
        self.set_sniffer_refresh_state("normal")
        self.set_simulation_button_state("normal")

```

MBToolkit v3.4/mb_slave_finder.py — lines 1601-1625

```
        if payload.get("error"):
            self.discovery_status_var.set(self.tr("Erro na pesquisa"))
            messagebox.showerror(self.tr("Erro na Bus Slave Finder"), payload["error"])
        elif payload.get("stopped"):
            self.discovery_status_var.set(
                self.trf(
                    "Pesquisa parada — {found} encontrados — {elapsed}",
                    found=payload["found"],
                    elapsed=format_duration(payload["elapsed"]),
                )
            )
        else:
            self.discovery_progress_var.set(100.0)
            self.discovery_status_var.set(
                self.trf(
                    "Pesquisa concluída — {found} encontrados — {elapsed}",
                    found=payload["found"],
                    elapsed=format_duration(payload["elapsed"]),
                )
            )

        self.update_discovery_estimate()
        self.update_discovery_start_state()
        self.update_com_status()

```

MBToolkit v3.4/mb_master.py

686 source lines.

MBToolkit v3.4/mb_master.py — lines 1-40

```python
#!/usr/bin/env python3
"""Modbus RTU master (poll) engine: frames, data formats and scheduling.

This module has no widgets. ``MasterEngine`` owns one open serial port and
serves every poll window over it: Modbus RTU allows a single request on the
bus at a time, so windows are polled in turn according to their scan rate,
with queued writes and "read once" requests served first. Results are
posted to the application's ``event_queue`` like the other workers.
"""

import math
import struct
import threading
import time
from collections import deque
from dataclasses import dataclass

from mb_config import (
    MASTER_RECONNECT_INTERVAL_SECONDS,
    MASTER_MAX_READ_BITS,
    MASTER_MAX_READ_REGISTERS,
    MASTER_MAX_WRITE_BITS,
    MASTER_MAX_WRITE_REGISTERS,
)
from mb_protocol import append_crc, crc_ok, hex_bytes, modbus_rtu_interframe_gap_seconds
from mb_slave_finder import (
    bus_activity_detected,
    cancel_serial_io,
    discovery_bits_per_char,
    write_discovery_request,
)

READ_FUNCTION_CODES = (1, 2, 3, 4)
BIT_FUNCTION_CODES = (1, 2)
WRITABLE_READ_FCS = {1: (5, 15), 3: (6, 16)}  # read window FC -> write FCs

# ---------------------------------------------------------------------------
# Data formats
# ---------------------------------------------------------------------------

```

MBToolkit v3.4/mb_master.py — lines 41-80

```
41 BYTE_ORDERS = ("ABCD", "CDAB", "BADC", "DCBA")
42 # 64-bit orders, in the same sequence: big-endian, word swap (registers in
43 # reverse order), byte swap inside each register, little-endian.
44 BYTE_ORDERS_64 = ("ABCDEFGH", "GHEFCDAB", "BADCFEHG", "HGFEDCBA")
45 WIRE_LETTERS = "ABCDEFGH"
46
47 # (key, label, registers per value).  Labels are technical terms and are not
48 # translated.  Byte order letters follow the wire order of the registers:
49 # register n = A B, register n+1 = C D, ...; "ABCD" / "ABCDEFGH" is big-endian.
50 REGISTER_FORMATS = (
51     ("int16", "Signed 16-bit", 1),
52     ("uint16", "Unsigned 16-bit", 1),
53     ("hex16", "Hex 16-bit", 1),
54     ("bin16", "Binary 16-bit", 1),
55 ) + tuple(
56     (f"{kind}_{order}", f"{label} {order}", 2)
57     for kind, label in (
58         ("int32", "Signed 32-bit"),
59         ("uint32", "Unsigned 32-bit"),
60         ("float32", "Float 32-bit"),
61     )
62     for order in BYTE_ORDERS
63 ) + tuple(
64     (f"{kind}_{order}", f"{label} {order}", 4)
65     for kind, label in (
66         ("int64", "Signed 64-bit"),
67         ("uint64", "Unsigned 64-bit"),
68         ("float64", "Double 64-bit"),
69     )
70     for order in BYTE_ORDERS_64
71 )
72 FORMAT_LABELS = {key: label for key, label, _size in REGISTER_FORMATS}
73 FORMAT_KEYS_BY_LABEL = {label: key for key, label, _size in REGISTER_FORMATS}
74 FORMAT_SIZES = {key: size for key, _label, size in REGISTER_FORMATS}
75 DEFAULT_FORMAT = "int16"
76
77 _STRUCT_CODES = {"int32": ">i", "uint32": ">I", "float32": ">f",
78                  "int64": ">q", "uint64": ">Q", "float64": ">d"}
79
80
```

MBToolkit v3.4/mb_master.py — lines 81-120

```
def _value_bytes(registers, order):
    """Reorder the wire bytes of 2 or 4 registers into big-endian value bytes."""
    wire = b"".join(bytes(((r >> 8) & 0xFF, r & 0xFF)) for r in registers)
    return bytes(wire[WIRE_LETTERS.index(letter)] for letter in order)


def _registers_from_value_bytes(value, order):
    wire = bytearray(len(order))
    for index, letter in enumerate(order):
        wire[WIRE_LETTERS.index(letter)] = value[index]
    return [(wire[i] << 8) | wire[i + 1] for i in range(0, len(wire), 2)]


def format_float(value, digits=7):
    if math.isnan(value):
        return "NaN"
    if math.isinf(value):
        return "-Inf" if value < 0 else "Inf"
    return f"{value:.{digits}g}"


def decode_value(registers, fmt):
    """Numeric value of one value (None for the Hex/Binary display formats)."""
    if fmt in ("hex16", "bin16"):
        return None
    if FORMAT_SIZES[fmt] == 1:
        value = registers[0] & 0xFFFF
        return value - 0x10000 if fmt == "int16" and value & 0x8000 else value
    kind, order = fmt.split("_")
    return struct.unpack(_STRUCT_CODES[kind], _value_bytes(registers, order))[0]


SCALE_MAX_DECIMALS = 6
SCALE_MAX_UNIT_LENGTH = 12


def validate_scale(raw_min, raw_max, eng_min, eng_max, decimals=1, unit=""):
    """Return a scale dict from user text/numbers; raise ValueError (PT text)."""
    try:
        values = [float(str(v).strip().replace(",", ".")) for v in
```

MBToolkit v3.4/mb_master.py — lines 121-160

```
                  (raw_min, raw_max, eng_min, eng_max)]
        decimals = int(str(decimals).strip())
    except ValueError:
        raise ValueError("A escala tem de ter números válidos.") from None
    if any(math.isnan(v) or math.isinf(v) for v in values):
        raise ValueError("A escala tem de ter números válidos.")
    if values[0] == values[1]:
        raise ValueError("O bruto mínimo e o bruto máximo têm de ser diferentes.")
    if not 0 <= decimals <= SCALE_MAX_DECIMALS:
        raise ValueError("As casas decimais têm de estar entre 0 e 6.")
    unit = str(unit or "").strip()[:SCALE_MAX_UNIT_LENGTH]
    return {"raw": [values[0], values[1]], "eng": [values[2], values[3]],
            "decimals": decimals, "unit": unit}


def scale_value(number, scale):
    """Linear map from raw units to engineering units through two points."""
    (r1, r2), (e1, e2) = scale["raw"], scale["eng"]
    return e1 + (number - r1) * (e2 - e1) / (r2 - r1)


def unscale_value(value, scale):
    """Inverse of scale_value(): engineering units back to raw units."""
    (r1, r2), (e1, e2) = scale["raw"], scale["eng"]
    if e1 == e2:
        raise ValueError("Esta escala não permite escrever: os valores de engenharia são iguais.")
    return r1 + (value - e1) * (r2 - r1) / (e2 - e1)


def format_scaled(number, scale):
    text = f"{scale_value(number, scale):.{scale['decimals']}f}"
    return f"{text} {scale['unit']}" if scale.get("unit") else text


def format_value(registers, fmt):
    """Display text of one value occupying ``FORMAT_SIZES[fmt]`` registers."""
    if FORMAT_SIZES[fmt] == 1:
        value = registers[0] & 0xFFFF
        if fmt == "int16":
            return str(value - 0x10000 if value & 0x8000 else value)
```

MBToolkit v3.4/mb_master.py — lines 161-200

```
161         if fmt == "uint16":
162             return str(value)
163         if fmt == "hex16":
164             return f"0x{value:04X}"
165         return f"{value >> 8:08b} {value & 0xFF:08b}"
166 
167     kind, order = fmt.split("_")
168     number = struct.unpack(_STRUCT_CODES[kind], _value_bytes(registers, order))[0]
169     if kind == "float32":
170         return format_float(number, 7)
171     if kind == "float64":
172         return format_float(number, 15)
173     return str(number)
174 
175 
176 def register_groups(qty, start_address=0, default_fmt=DEFAULT_FORMAT, overrides=None):
177     """Split a window's registers into values: [(offset, size, fmt or None)].
178 
179     *overrides* maps a PDU address to the format of the value starting there
180     (per-address formats). Other registers use *default_fmt*. A value that
181     would run past the window, or a default multi-register value cut by an
182     override, becomes ``(offset, size, None)`` and is shown as "—".
183     """
184     overrides = overrides or {}
185     default_fmt = default_fmt if default_fmt in FORMAT_SIZES else DEFAULT_FORMAT
186     groups = []
187     offset = 0
188     while offset < qty:
189         fmt = overrides.get(start_address + offset)
190         if fmt not in FORMAT_SIZES:
191             fmt = default_fmt
192             size = FORMAT_SIZES[fmt]
193             cut = next((j for j in range(offset + 1, min(offset + size, qty))
194                         if start_address + j in overrides), None)
195             if cut is not None:
196                 groups.append((offset, cut - offset, None))
197                 offset = cut
198                 continue
199         size = FORMAT_SIZES[fmt]
200         if offset + size > qty:
```

MBToolkit v3.4/mb_master.py — lines 201-240

```
            groups.append((offset, qty - offset, None))
            break
        groups.append((offset, size, fmt))
        offset += size
    return groups


def format_window_values(registers, start_address=0, default_fmt=DEFAULT_FORMAT,
                         overrides=None, scales=None):
    """Return one display string per register, honouring per-address formats.

    A multi-register value is shown on the row of its first register; the
    other rows of the group are empty. Incomplete groups show "—". *scales*
    maps the PDU address where a value starts to a scale (validate_scale());
    scaled values are shown in engineering units (Hex/Binary stay raw).
    """
    scales = scales or {}
    out = []
    for offset, size, fmt in register_groups(len(registers), start_address,
                                             default_fmt, overrides):
        if fmt is None:
            out.extend(["—"] + [""] * (size - 1))
            continue
        group = registers[offset:offset + size]
        scale = scales.get(start_address + offset)
        number = decode_value(group, fmt) if scale else None
        if scale and number is not None and not (isinstance(number, float)
                                                 and (math.isnan(number) or math.isinf(number))):
            out.append(format_scaled(number, scale))
        else:
            out.append(format_value(group, fmt))
        out.extend([""] * (size - 1))
    return out


def raw_text_for_scaled(value_text, scale, fmt):
    """Convert an engineering value typed by the user into raw text for *fmt*."""
    try:
        value = float(str(value_text).strip().replace(",", "."))
    except ValueError:
```

MBToolkit v3.4/mb_master.py — lines 241-280

```
241         raise ValueError("Valor fora do formato selecionado.") from None
242     raw = unscale_value(value, scale)
243     if fmt.split("_")[0].startswith("float"):
244         return repr(raw)
245     return str(int(round(raw)))
246
247
248 def format_register_values(registers, fmt):
249     """Display strings for registers that all use one format."""
250     return format_window_values(registers, 0, fmt, None)
251
252
253 def _parse_int(text):
254     text = str(text).strip().replace(" ", "").replace("_", "")
255     lowered = text.lower()
256     if lowered.startswith(("0x", "-0x")):
257         return int(text, 16)
258     if lowered.startswith(("0b", "-0b")):
259         return int(text, 2)
260     return int(text, 10)
261
262
263 def encode_register_value(text, fmt):
264     """Convert one user-entered value to the registers it occupies.
265
266     Raises ValueError with a message key (PT canonical text) when the value
267     does not fit the format.
268     """
269     fmt = fmt if fmt in FORMAT_SIZES else DEFAULT_FORMAT
270     text = str(text).strip()
271     if not text:
272         raise ValueError("Valor vazio.")
273     try:
274         if fmt == "hex16":
275             value = int(text[2:] if text.lower().startswith("0x") else text, 16)
276             if not 0 <= value <= 0xFFFF:
277                 raise ValueError
278             return [value]
279         if fmt == "bin16":
280             value = int(text.replace(" ", "")[2:] if text.lower().startswith("0b")
```

MBToolkit v3.4/mb_master.py — lines 281-320

```
                        else text.replace(" ", ""), 2)
            if not 0 <= value <= 0xFFFF:
                raise ValueError
            return [value]
        if fmt == "int16":
            value = _parse_int(text)
            if not -0x8000 <= value <= 0x7FFF:
                raise ValueError
            return [value & 0xFFFF]
        if fmt == "uint16":
            value = _parse_int(text)
            if not 0 <= value <= 0xFFFF:
                raise ValueError
            return [value]

        kind, order = fmt.split("_")
        code = _STRUCT_CODES[kind]
        if kind.startswith("float"):
            number = float(text.replace(",", "."))
            raw = struct.pack(code, number)
            if math.isinf(struct.unpack(code, raw)[0]) and not math.isinf(number):
                raise ValueError
        else:
            raw = struct.pack(code, _parse_int(text))
        return _registers_from_value_bytes(raw, order)
    except (ValueError, OverflowError, struct.error):
        raise ValueError("Valor fora do formato selecionado.") from None


def encode_coil_value(text):
    value = str(text).strip().lower()
    if value in ("1", "on", "true"):
        return 1
    if value in ("0", "off", "false"):
        return 0
    raise ValueError("Uma coil só aceita 0 ou 1.")


# ---------------------------------------------------------------------------
# Request frames
```

MBToolkit v3.4/mb_master.py — lines 321-360

```
321 # ---------------------------------------------------------------------------
322
323 def _u16(value):
324     return bytes(((value >> 8) & 0xFF, value & 0xFF))
325
326
327 def read_limit(fc):
328     return MASTER_MAX_READ_BITS if fc in BIT_FUNCTION_CODES else MASTER_MAX_READ_REGISTERS
329
330
331 def validate_read_definition(slave, fc, address, qty):
332     """Raise ValueError (PT canonical message) for an invalid poll definition."""
333     if not 1 <= int(slave) <= 247:
334         raise ValueError("O Slave ID de leitura tem de estar entre 1 e 247.")
335     if int(fc) not in READ_FUNCTION_CODES:
336         raise ValueError("Função de leitura não suportada.")
337     if not 0 <= int(address) <= 0xFFFF:
338         raise ValueError("O endereço tem de estar entre 0 e 65535.")
339     if not 1 <= int(qty) <= read_limit(int(fc)):
340         raise ValueError("Quantidade fora do limite desta função.")
341     if int(address) + int(qty) > 0x10000:
342         raise ValueError("O intervalo de endereços ultrapassa 65535.")
343
344
345 def build_read_request(slave, fc, address, qty):
346     validate_read_definition(slave, fc, address, qty)
347     return append_crc(bytes((int(slave), int(fc))) + _u16(int(address)) + _u16(int(qty)))
348
349
350 def build_write_request(slave, fc, address, values):
351     """Build an FC05/06/15/16 request. *values* are 0/1 bits or 16-bit registers."""
352     slave, fc, address = int(slave), int(fc), int(address)
353     values = [int(v) for v in values]
354     if not 0 <= slave <= 247:
355         raise ValueError("O Slave ID de escrita tem de estar entre 0 (broadcast) e 247.")
356     if not 0 <= address <= 0xFFFF or address + max(1, len(values)) > 0x10000:
357         raise ValueError("O endereço tem de estar entre 0 e 65535.")
358     if fc == 5:
359         if len(values) != 1:
360             raise ValueError("FC05 escreve exatamente uma coil.")
```

MBToolkit v3.4/mb_master.py — lines 361-400

```
        payload = _u16(address) + (b"\xFF\x00" if values[0] else b"\x00\x00")
    elif fc == 6:
        if len(values) != 1:
            raise ValueError("FC06 escreve exatamente um registo.")
        payload = _u16(address) + _u16(values[0] & 0xFFFF)
    elif fc == 15:
        if not 1 <= len(values) <= MASTER_MAX_WRITE_BITS:
            raise ValueError("Quantidade fora do limite desta função.")
        data = bytearray((len(values) + 7) // 8)
        for index, bit in enumerate(values):
            if bit:
                data[index // 8] |= 1 << (index % 8)
        payload = _u16(address) + _u16(len(values)) + bytes((len(data),)) + bytes(data)
    elif fc == 16:
        if not 1 <= len(values) <= MASTER_MAX_WRITE_REGISTERS:
            raise ValueError("Quantidade fora do limite desta função.")
        data = b"".join(_u16(v & 0xFFFF) for v in values)
        payload = _u16(address) + _u16(len(values)) + bytes((len(data),)) + data
    else:
        raise ValueError("Função de escrita não suportada.")
    return append_crc(bytes((slave, fc)) + payload)


def expected_response_length(request):
    fc = request[1]
    if fc in BIT_FUNCTION_CODES:
        qty = (request[4] << 8) | request[5]
        return 5 + (qty + 7) // 8
    if fc in (3, 4):
        qty = (request[4] << 8) | request[5]
        return 5 + 2 * qty
    return 8  # FC05/06/15/16


# ---------------------------------------------------------------------------
# Response matching
# ---------------------------------------------------------------------------

def find_master_response(buffer, request):
    """Find the response to *request* in the received bytes.
```

MBToolkit v3.4/mb_master.py — lines 401-440

```
401
402     Returns a dict with ``kind`` RESPONSE or EXCEPTION, the ``frame`` and,
403     for reads, the decoded ``values`` (bits or registers); None when no
404     complete, CRC-valid, matching frame is present yet. A request echoed by
405     an RS485 adapter is skipped for reads because its length and CRC do not
406     fit the expected response. For FC05/06 a slave reply is itself an exact
407     echo, so an adapter echo cannot be distinguished from it.
408     """
409     data = bytes(buffer)
410     slave, fc = request[0], request[1]
411     length = expected_response_length(request)
412
413     for offset in range(len(data)):
414         if data[offset] != slave or offset + 2 > len(data):
415             continue
416         response_fc = data[offset + 1]
417
418         if response_fc == (fc | 0x80) and offset + 5 <= len(data):
419             frame = data[offset:offset + 5]
420             if crc_ok(frame):
421                 return {"kind": "EXCEPTION", "frame": frame, "exception_code": frame[2],
422                         "values": None}
423
424         if response_fc != fc or offset + length > len(data):
425             continue
426         frame = data[offset:offset + length]
427         if not crc_ok(frame):
428             continue
429
430         if fc in READ_FUNCTION_CODES:
431             if frame[2] != length - 5:
432                 continue
433             payload = frame[3:-2]
434             qty = (request[4] << 8) | request[5]
435             if fc in BIT_FUNCTION_CODES:
436                 values = [(payload[i // 8] >> (i % 8)) & 1 for i in range(qty)]
437             else:
438                 values = [(payload[i] << 8) | payload[i + 1] for i in range(0, len(payload), 2)]
439             return {"kind": "RESPONSE", "frame": frame, "exception_code": None, "values": values}
440
```

MBToolkit v3.4/mb_master.py — lines 441-480

```
441         if fc in (5, 6) and frame != request:
442             continue
443         if fc in (15, 16) and frame[2:6] != request[2:6]:
444             continue
445         return {"kind": "RESPONSE", "frame": frame, "exception_code": None, "values": None}
446 
447     return None
448 
449 
450 # ---------------------------------------------------------------------------
451 # Poll definitions and engine
452 # ---------------------------------------------------------------------------
453 
454 @dataclass
455 class PollDefinition:
456     window_id: int
457     slave: int
458     fc: int
459     address: int
460     qty: int
461     scan_ms: int
462     enabled: bool = True
463 
464     def request(self):
465         return build_read_request(self.slave, self.fc, self.address, self.qty)
466 
467 
468 class MasterEngine:
469     """Serve all poll windows over one serial port, one transaction at a time.
470 
471     Order of service: queued writes, then "read once" requests, then the
472     enabled poll whose next due time is earliest. A poll becomes due again
473     ``scan_ms`` after its previous transaction started (0 = continuously).
474     After every transaction the bus is left silent for at least the
475     configured delay between polls, and never less than Modbus t3.5.
476 
477     With *reopen* (a callable returning a new open serial object), a serial or
478     driver failure does not end the session: the engine closes the port, posts
479     ``master_reconnecting`` and retries every MASTER_RECONNECT_INTERVAL_SECONDS
480     until the port opens again (``master_reconnected``) or the engine stops.
```

MBToolkit v3.4/mb_master.py — lines 481-520

```
481     """
482
483     def __init__(self, ser, event_queue, *, baud, parity, stopbits,
484                  response_timeout_ms, delay_ms, serial_module=None,
485                  idle_check_seconds=0.0, reopen=None,
486                  reconnect_interval_s=MASTER_RECONNECT_INTERVAL_SECONDS):
487         self.ser = ser
488         self.reopen = reopen
489         self.reconnect_interval_s = float(reconnect_interval_s)
490         self.event_queue = event_queue
491         self.baud = int(baud)
492         self.parity = parity
493         self.stopbits = stopbits
494         self.response_timeout_s = max(0.001, float(response_timeout_ms) / 1000.0)
495         gap_s = modbus_rtu_interframe_gap_seconds(
496             self.baud, discovery_bits_per_char(parity, stopbits)
497         )
498         self.delay_s = max(gap_s, float(delay_ms) / 1000.0)
499         self.serial_module = serial_module
500         self.idle_check_seconds = float(idle_check_seconds)
501
502         self._lock = threading.Lock()
503         self._polls = {}           # window_id -> [PollDefinition, next_due]
504         self._writes = deque()     # (window_id, request, description)
505         self._once = deque()       # window_id
506         self._wake = threading.Event()
507         self.stop_event = threading.Event()
508         self.thread = None
509
510     # -- control (any thread) -----------------------------------------------
511     def set_poll(self, definition):
512         with self._lock:
513             self._polls[definition.window_id] = [definition, 0.0]
514         self._wake.set()
515
516     def remove_poll(self, window_id):
517         with self._lock:
518             self._polls.pop(window_id, None)
519             self._once = deque(w for w in self._once if w != window_id)
520
```

MBToolkit v3.4/mb_master.py — lines 521-560

```
    def read_once(self, window_id):
        with self._lock:
            if window_id in self._polls and window_id not in self._once:
                self._once.append(window_id)
        self._wake.set()

    def queue_write(self, window_id, request, description=""):
        with self._lock:
            self._writes.append((window_id, bytes(request), description))
        self._wake.set()

    def start(self):
        self.thread = threading.Thread(target=self.run, daemon=True)
        self.thread.start()

    def stop(self, join_timeout=1.0):
        self.stop_event.set()
        self._wake.set()
        cancel_serial_io(self.ser)
        thread = self.thread
        if thread is not None and thread.is_alive() and thread is not threading.current_thread():
            thread.join(timeout=join_timeout)

    # -- scheduling ---------------------------------------------------------
    def next_job(self, now=None):
        """Return ('write', item), ('read', definition, once) or ('wait', seconds)."""
        now = time.perf_counter() if now is None else now
        with self._lock:
            if self._writes:
                return ("write", self._writes.popleft())
            while self._once:
                window_id = self._once.popleft()
                entry = self._polls.get(window_id)
                if entry is not None:
                    return ("read", entry[0], True)
            due = [
                (entry[1], window_id)
                for window_id, entry in self._polls.items()
                if entry[0].enabled
            ]
```

MBToolkit v3.4/mb_master.py — lines 561-600

```
            if not due:
                return ("wait", 0.5)
            next_due, window_id = min(due)
            if next_due > now:
                return ("wait", min(0.5, next_due - now))
            entry = self._polls[window_id]
            entry[1] = now + max(0, entry[0].scan_ms) / 1000.0
            return ("read", entry[0], False)

    # -- worker ------------------------------------------------------------
    def run(self):
        error = None
        try:
            if self.idle_check_seconds > 0:
                traffic = bus_activity_detected(
                    self.ser, self.idle_check_seconds, self.stop_event
                )
                if traffic:
                    error = ("traffic", hex_bytes(traffic[:16]), len(traffic))
                    return
            self.event_queue.put(("master_ready", None))

            while not self.stop_event.is_set():
                try:
                    self._serve_jobs()
                except Exception as exc:  # serial/driver failure
                    if self.stop_event.is_set():
                        break
                    detail = f"{type(exc).__name__}: {exc}"
                    if self.reopen is None:
                        error = ("serial", detail, 0)
                        break
                    if not self._reconnect(detail):
                        break
        except Exception as exc:  # failure during the idle check
            if not self.stop_event.is_set():
                error = ("serial", f"{type(exc).__name__}: {exc}", 0)
        finally:
            self.event_queue.put(("master_stopped", {"error": error}))

```

MBToolkit v3.4/mb_master.py — lines 601-640

```
601     def _serve_jobs(self):
602         """Serve writes / reads until stopped; serial errors propagate."""
603         while not self.stop_event.is_set():
604             job = self.next_job()
605             if job[0] == "wait":
606                 self._wake.wait(job[1])
607                 self._wake.clear()
608                 continue
609 
610             if job[0] == "write":
611                 window_id, request, description = job[1]
612                 result = self.transact(request)
613                 result.update(kind="write", window_id=window_id, description=description)
614             else:
615                 definition, once = job[1], job[2]
616                 result = self.transact(definition.request())
617                 result.update(kind="read", window_id=definition.window_id, once=once,
618                               fc=definition.fc, address=definition.address)
619             if self.stop_event.is_set():
620                 return
621             self.event_queue.put(("master_result", result))
622             if self.stop_event.wait(self.delay_s):
623                 return
624 
625     def _reconnect(self, detail):
626         """Close the failed port and reopen it; False when stopped meanwhile."""
627         try:
628             self.ser.close()
629         except Exception:
630             pass
631         self.event_queue.put(("master_reconnecting", {"error": detail}))
632         while not self.stop_event.wait(self.reconnect_interval_s):
633             try:
634                 ser = self.reopen()
635             except Exception:
636                 continue
637             self.ser = ser
638             self.event_queue.put(("master_reconnected", None))
639             return True
640         return False
```

MBToolkit v3.4/mb_master.py — lines 641-680

```
641
642     def transact(self, request):
643         """Send one request and wait for its response (no wait for broadcast)."""
644         try:
645             self.ser.reset_input_buffer()
646         except Exception:
647             pass
648         sent = write_discovery_request(
649             self.ser, request, self.baud, self.parity, self.stopbits,
650             self.stop_event, self.serial_module,
651         )
652         # write_discovery_request() returns once the request has left the
653         # wire, so response time and timeout count from the end of the request.
654         started = time.perf_counter()
655         result = {"tx": hex_bytes(request), "rx": "", "status": "stopped",
656                   "values": None, "exception_code": None, "response_ms": None}
657         if not sent:
658             return result
659         if request[0] == 0:
660             result["status"] = "broadcast"
661             return result
662
663         deadline = time.perf_counter() + self.response_timeout_s
664         buffer = bytearray()
665         while time.perf_counter() < deadline and not self.stop_event.is_set():
666             waiting = int(getattr(self.ser, "in_waiting", 0) or 0)
667             if waiting:
668                 buffer.extend(self.ser.read(waiting) or b"")
669                 found = find_master_response(buffer, request)
670                 if found is not None:
671                     result.update(
672                         status="exception" if found["kind"] == "EXCEPTION" else "ok",
673                         values=found["values"],
674                         exception_code=found["exception_code"],
675                         rx=hex_bytes(found["frame"]),
676                         response_ms=(time.perf_counter() - started) * 1000.0,
677                     )
678                     return result
679             elif self.stop_event.wait(0.001):
680                 break
```

MBToolkit v3.4/mb_master.py — lines 681-686

```
681
682         if self.stop_event.is_set():
683             return result
684         result["rx"] = hex_bytes(buffer)
685         result["status"] = "invalid" if buffer else "timeout"
686         return result
```

MBToolkit v3.4/mb_master_ui.py

1529 source lines.

MBToolkit v3.4/mb_master_ui.py — lines 1-40

```
#!/usr/bin/env python3
"""Master (poll) tab: one window per Slave / data area, polled over one COM.

The protocol and scheduling live in ``mb_master.py``; this mixin owns the
widgets. Each poll window has a model (``MasterWindow``) and a pane that can
sit in the tab's grid or be popped out into its own top-level window.
"""

import json
import time
import tkinter as tk
from tkinter import font as tkfont
from tkinter import messagebox, ttk

from mb_config import (
    DISCOVERY_BAUD_RATES,
    DISCOVERY_IDLE_CHECK_SECONDS,
    MASTER_DEFAULT_DELAY_MS,
    MASTER_DEFAULT_RESPONSE_TIMEOUT_MS,
    MASTER_DEFAULT_SCAN_MS,
    MASTER_MAX_DELAY_MS,
    MASTER_MAX_RESPONSE_TIMEOUT_MS,
    MASTER_MAX_SCAN_MS,
    MASTER_MAX_WINDOWS,
    MASTER_MIN_DELAY_MS,
    MASTER_MIN_RESPONSE_TIMEOUT_MS,
    MASTER_WORKSPACE_FORMAT,
    MASTER_WORKSPACE_FORMATS,
    MASTER_WORKSPACE_VERSION,
    MODBUS_EXCEPTION_CODES,
    get_master_workspaces_folder,
)
from mb_diagnostics import FUNCTION_NAMES
from mb_master import (
    BIT_FUNCTION_CODES,
    DEFAULT_FORMAT,
    FORMAT_KEYS_BY_LABEL,
    FORMAT_LABELS,
    FORMAT_SIZES,
    READ_FUNCTION_CODES,
```

MBToolkit v3.4/mb_master_ui.py — lines 41-80

```
41      WRITABLE_READ_FCS,
42      MasterEngine,
43      PollDefinition,
44      build_write_request,
45      encode_coil_value,
46      encode_register_value,
47      format_window_values,
48      raw_text_for_scaled,
49      register_groups,
50      validate_read_definition,
51  )
52  from mb_table_ui import (
53      SERIAL_PARITIES,
54      SERIAL_STOP_BITS,
55      TABLE_MAX_TITLE_LENGTH,
56      TABLE_PANE_MAX_ROWS,
57      TABLE_PANE_MIN_ROWS,
58      TABLE_PANE_MIN_WIDTH,
59      TableUiMixin,
60      apply_format_to_rows,
61      parse_int_field,
62      validate_workspace_name,
63      validated_formats,
64      validated_names,
65      validated_scales,
66  )
67  from mb_table_ui import list_workspaces as _list_workspaces
68  from mb_table_ui import workspace_path as _workspace_path
69  from mb_theme import theme_colors
70  from mb_widgets import RefreshButton
71
72  MASTER_PARITIES = SERIAL_PARITIES
73  MASTER_STOP_BITS = SERIAL_STOP_BITS
74  MASTER_PANE_MIN_WIDTH = TABLE_PANE_MIN_WIDTH
75  MASTER_PANE_MIN_ROWS = TABLE_PANE_MIN_ROWS
76  MASTER_PANE_MAX_ROWS = TABLE_PANE_MAX_ROWS
77  MASTER_OK_MESSAGE_SECONDS = 3.0
78  MASTER_MAX_TITLE_LENGTH = TABLE_MAX_TITLE_LENGTH
79  MASTER_WHEEL_TAG = "MBToolkit.MasterWheel"
80
```

MBToolkit v3.4/mb_master_ui.py — lines 81-120

```
 81
 82 def workspace_path(name):
 83     """File of a Master workspace (in Workspaces\\Master)."""
 84     return _workspace_path(name, get_master_workspaces_folder())
 85
 86
 87 def list_workspaces():
 88     """Names of the saved Master workspaces, sorted alphabetically."""
 89     return _list_workspaces(get_master_workspaces_folder())
 90
 91
 92 def fc_choice_label(fc):
 93     return f"{fc:02d} {FUNCTION_NAMES[fc]}"
 94
 95
 96 FC_CHOICES = tuple(fc_choice_label(fc) for fc in READ_FUNCTION_CODES)
 97
 98
 99 class MasterWindow:
100     """State of one poll window (independent of where its pane is shown)."""
101
102     def __init__(self, window_id, number, slave=1, fc=3, address=0, qty=10,
103                  scan_ms=MASTER_DEFAULT_SCAN_MS, base=0, fmt=DEFAULT_FORMAT,
104                  names=None, enabled=True, title="", formats=None, scales=None):
105         self.window_id = window_id
106         self.number = number
107         # User-chosen window name; empty means the default "Janela N".
108         self.title = str(title or "").strip()[:MASTER_MAX_TITLE_LENGTH]
109         self.slave = slave
110         self.fc = fc
111         self.address = address
112         self.qty = qty
113         self.scan_ms = scan_ms
114         self.base = base
115         self.fmt = fmt
116         self.names = dict(names or {})  # PDU address -> label
117         # PDU address -> format of the value that starts there (per-address
118         # formats chosen with the right-click menu); others use self.fmt.
119         self.formats = {int(a): f for a, f in (formats or {}).items() if f in FORMAT_SIZES}
120         # PDU address -> scale (raw/engineering points, decimals, unit) of the
```

MBToolkit v3.4/mb_master_ui.py — lines 121-160

```
        # value that starts there.
        self.scales = {int(a): dict(s) for a, s in (scales or {}).items()}
        self.enabled = enabled
        self.popped = False
        self.reset_counters()

    def reset_counters(self):
        self.tx = 0
        self.errors = 0
        self.timeouts = 0
        self.response_ms = None
        self.values = None
        # True when the latest read failed: the shown values are out of date.
        self.stale = False
        # (canonical PT template, values, style) shown on the status line.
        self.message = ("", {}, "normal")
        self.message_at = 0.0

    def definition(self):
        return PollDefinition(self.window_id, self.slave, self.fc, self.address,
                              self.qty, self.scan_ms, self.enabled)

    def to_json(self):
        return {
            "title": self.title,
            "slave": self.slave, "fc": self.fc, "address": self.address,
            "qty": self.qty, "scan_ms": self.scan_ms, "base": self.base,
            "format": self.fmt, "enabled": self.enabled,
            "names": {str(address): name for address, name in sorted(self.names.items())},
            "formats": {str(address): fmt for address, fmt in sorted(self.formats.items())},
            "scales": {str(address): scale for address, scale in sorted(self.scales.items())},
        }

    def groups(self):
        """[(row offset, size, fmt or None)] of the values shown in this window."""
        return register_groups(self.qty, self.address, self.fmt, self.formats)

    def group_of_row(self, row):
        for offset, size, fmt in self.groups():
            if offset <= row < offset + size:
```

MBToolkit v3.4/mb_master_ui.py — lines 161-200

```
                return offset, size, fmt
        return row, 1, None

    @staticmethod
    def validated_kwargs(data):
        """Validate one workspace window entry; raise ValueError if invalid."""
        if not isinstance(data, dict):
            raise ValueError("Janela inválida.")
        slave, fc = int(data.get("slave", 1)), int(data.get("fc", 3))
        address, qty = int(data.get("address", 0)), int(data.get("qty", 10))
        validate_read_definition(slave, fc, address, qty)
        scan_ms = int(data.get("scan_ms", MASTER_DEFAULT_SCAN_MS))
        if not 0 <= scan_ms <= MASTER_MAX_SCAN_MS:
            raise ValueError("Scan rate fora do intervalo.")
        fmt = data.get("format", DEFAULT_FORMAT)
        names = data.get("names", {})
        formats = data.get("formats", {})
        return {
            "title": str(data.get("title", "") or "").strip()[:MASTER_MAX_TITLE_LENGTH],
            "slave": slave, "fc": fc, "address": address, "qty": qty,
            "scan_ms": scan_ms, "base": 1 if data.get("base") == 1 else 0,
            "fmt": fmt if fmt in FORMAT_SIZES else DEFAULT_FORMAT,
            "enabled": bool(data.get("enabled", True)),
            "names": validated_names(names),
            "formats": validated_formats(formats),
            "scales": validated_scales(data.get("scales", {})),
        }


class MasterMixin(TableUiMixin):
    """Master (poll) tab GUI."""

    # -- construction ---------------------------------------------------------
    def build_master_tab(self, page):
        saved = getattr(self, "ui_settings", {}).get("master", {})
        if not isinstance(saved, dict):
            saved = {}

        self.master_windows = {}
        self._master_panes = {}
```

MBToolkit v3.4/mb_master_ui.py — lines 201-240

```
        self._master_next_id = 1
        self.master_engine = None
        self.master_serial = None
        self.master_active = False
        self._master_reflow_after_id = None
        # Workspace state: name of the open workspace, whether it changed since
        # it was opened/saved, and a guard that suppresses change tracking
        # while a workspace is being loaded.
        self.master_workspace_dirty = False
        self._master_loading = False

        page.columnconfigure(0, weight=1)
        page.rowconfigure(2, weight=1)

        connection = ttk.LabelFrame(page, text="Ligação — o Master transmite no barramento")
        connection.grid(row=0, column=0, sticky="ew", padx=12, pady=(12, 6))
        self._master_connection_frame = connection
        row = ttk.Frame(connection)
        row.grid(row=0, column=0, sticky="w", padx=8, pady=(8, 4))

        ttk.Label(row, text="Porta COM").grid(row=0, column=0, padx=(0, 4))
        self.master_port_var = tk.StringVar()
        self.master_port_combo = ttk.Combobox(
            row, textvariable=self.master_port_var, width=13, state="readonly"
        )
        self.master_port_combo.grid(row=0, column=1)
        self.master_refresh_btn = RefreshButton(row, command=self.refresh_ports, size=24)
        self.master_refresh_btn.grid(row=0, column=2, padx=(4, 18))

        def _sync_refresh_square(_event=None):
            height = self.master_port_combo.winfo_height()
            if height > 1:
                self.master_refresh_btn.configure(width=height, height=height)

        self.master_port_combo.bind("<Configure>", _sync_refresh_square, add="+")

        self.master_baud_var = tk.StringVar(value=str(saved.get("baud", "9600")))
        self.master_parity_var = tk.StringVar(value=saved.get("parity", "None"))
        self.master_stop_var = tk.StringVar(value=str(saved.get("stop_bits", "1")))
        self.master_timeout_var = tk.StringVar(
```

MBToolkit v3.4/mb_master_ui.py — lines 241-280

```
            value=str(saved.get("response_timeout_ms", MASTER_DEFAULT_RESPONSE_TIMEOUT_MS))
        )
        self.master_delay_var = tk.StringVar(
            value=str(saved.get("delay_ms", MASTER_DEFAULT_DELAY_MS))
        )
        fields = (
            ("Baud rate", ttk.Combobox(row, textvariable=self.master_baud_var, width=8,
                                       state="readonly",
                                       values=[str(b) for b in DISCOVERY_BAUD_RATES])),
            ("Parity", ttk.Combobox(row, textvariable=self.master_parity_var, width=6,
                                    state="readonly", values=MASTER_PARITIES)),
            ("Stop bits", ttk.Combobox(row, textvariable=self.master_stop_var, width=3,
                                       state="readonly", values=MASTER_STOP_BITS)),
            ("Timeout resposta (ms)", ttk.Spinbox(
                row, textvariable=self.master_timeout_var, width=6, increment=50,
                from_=MASTER_MIN_RESPONSE_TIMEOUT_MS, to=MASTER_MAX_RESPONSE_TIMEOUT_MS)),
            ("Atraso entre pedidos (ms)", ttk.Spinbox(
                row, textvariable=self.master_delay_var, width=6, increment=5,
                from_=MASTER_MIN_DELAY_MS, to=MASTER_MAX_DELAY_MS)),
        )
        column = 3
        self._master_setting_widgets = [self.master_port_combo]
        for caption, widget in fields:
            ttk.Label(row, text=caption).grid(row=0, column=column, padx=(0, 4))
            widget.grid(row=0, column=column + 1, padx=(0, 16))
            self._master_setting_widgets.append(widget)
            column += 2
        ttk.Label(row, text="Data bits: 8").grid(row=0, column=column)

        controls = ttk.Frame(connection)
        controls.grid(row=1, column=0, sticky="w", padx=8, pady=(0, 8))
        self.master_safety_var = tk.BooleanVar(value=False)
        ttk.Checkbutton(
            controls,
            text="Confirmo que não existe outro master ativo no barramento.",
            variable=self.master_safety_var,
        ).grid(row=0, column=0, padx=(0, 18))
        self.master_connect_btn = ttk.Button(
            controls, text="Ligar", command=self.master_connect, style="MBS.Primary.TButton"
        )
```

MBToolkit v3.4/mb_master_ui.py — lines 281-320

```
        self.master_connect_btn.grid(row=0, column=1, padx=(0, 6))
        self.master_disconnect_btn = ttk.Button(
            controls, text="Desligar", command=self.master_disconnect, state="disabled"
        )
        self.master_disconnect_btn.grid(row=0, column=2, padx=(0, 18))
        self.master_status_var = tk.StringVar(value=self.tr("Desligado"))
        ttk.Label(controls, textvariable=self.master_status_var).grid(row=0, column=3)

        toolbar = ttk.Frame(page)
        toolbar.grid(row=1, column=0, sticky="ew", padx=12, pady=(0, 6))
        toolbar.columnconfigure(3, weight=1)
        ttk.Button(toolbar, text="Nova janela", command=self.master_add_window).grid(
            row=0, column=0, padx=(0, 6))
        # Workspace menu: the saved workspaces are listed directly in it.
        self.master_workspace_btn = ttk.Menubutton(toolbar, text="Workspace")
        self.master_workspace_btn.grid(row=0, column=1, padx=(0, 6))
        self.master_workspace_menu = tk.Menu(
            self.master_workspace_btn, tearoff=False,
            postcommand=self._master_build_workspace_menu,
        )
        self.master_workspace_btn.configure(menu=self.master_workspace_menu)
        self.master_workspace_var = tk.StringVar(value="")
        ttk.Label(toolbar, textvariable=self.master_workspace_var).grid(
            row=0, column=2, padx=(0, 6))
        self.master_current_workspace = ""
        self._master_update_workspace_label()
        title_font = tkfont.nametofont("TkDefaultFont").copy()
        title_font.configure(weight="bold")
        self._master_title_font = title_font
        self.ui_style.configure("MBS.Master.Title.TLabel", font=title_font)
        ttk.Label(
            toolbar,
            text=("Duplo clique num valor para escrever, num nome para o editar. "
                  "Clique direito: formato, copiar e base."),
        ).grid(row=0, column=3, sticky="w", padx=(12, 0))

        self.master_canvas, self.master_grid, self._master_grid_item = \
            self._table_build_scroll_grid(page, 2)
        self.master_canvas.bind("<Configure>", self._master_schedule_reflow, add="+")
        self.master_empty_label = ttk.Label(
```

MBToolkit v3.4/mb_master_ui.py — lines 321-360

```
            self.master_grid,
            text="Sem janelas. Usa \"Nova janela\" para começar a ler um Slave.",
        )

        for sequence in ("<MouseWheel>", "<Button-4>", "<Button-5>"):
            self.bind_class(MASTER_WHEEL_TAG, sequence, self._master_wheel)

        for variable in (self.master_baud_var, self.master_parity_var, self.master_stop_var,
                         self.master_timeout_var, self.master_delay_var):
            variable.trace_add("write", lambda *_a: self._schedule_ui_settings_save())

        self._master_build_row_menu()

        # Reopen the workspace used last time; it is loaded once the window is
        # complete so the COM port list is already known. Never connects.
        last = saved.get("last_workspace", "")
        if last and last in list_workspaces():
            self.after(10, lambda: self._master_restore_workspace(last))
        else:
            self.master_add_window()
            self.master_workspace_dirty = False

    def _master_restore_workspace(self, name):
        if not self.master_open_workspace(name, quiet=True):
            self.master_add_window()
            self.master_workspace_dirty = False

    # -- helpers ---------------------------------------------------------
    def _master_install_wheel_tags(self, root_widget):
        self._table_install_wheel_tags(root_widget, MASTER_WHEEL_TAG)

    def _master_wheel(self, event):
        return self._table_wheel(getattr(self, "master_canvas", None), event)

    def _master_window_name(self, window):
        return window.title or self.trf("Janela {number}", number=window.number)

    def _master_window_summary(self, window):
        return (f"Slave {window.slave} · FC{window.fc:02d} {FUNCTION_NAMES[window.fc]} · "
                f"{window.address + window.base} · {window.qty}")
```

MBToolkit v3.4/mb_master_ui.py — lines 361-400

```
361
362     def _master_window_title(self, window):
363         return f"{self._master_window_name(window)} — {self._master_window_summary(window)}"
364
365     def _master_counters_text(self, window):
366         response = f"{window.response_ms:.1f} ms" if window.response_ms is not None else "—"
367         return (f"Tx: {window.tx}   Err: {window.errors}   "
368                 f"Timeouts: {window.timeouts}   Resp.: {response}")
369
370     def _master_set_message(self, window, template, style="normal", **values):
371         window.message = (template, values, style)
372         window.message_at = time.monotonic()
373         self._master_render_status(window)
374
375     def _master_render_status(self, window):
376         pane = self._master_panes.get(window.window_id)
377         if pane is None:
378             return
379         template, values, style = window.message
380         if not window.enabled and style != "error":
381             template, values, style = "Pausada", {}, "normal"
382         text = self.trf(template, **values) if template else ""
383         try:
384             pane["counters_var"].set(self._master_counters_text(window))
385             pane["message_var"].set(text)
386             pane["message_label"].configure(style={
387                 "error": "MBS.Master.Error.TLabel",
388                 "ok": "MBS.Master.Ok.TLabel",
389             }.get(style, "TLabel"))
390             pane["name_var"].set(self._master_window_name(window))
391             pane["summary_var"].set(self._master_window_summary(window))
392             pane["pause_btn"].configure(text=self.tr("Retomar" if not window.enabled else "Pausar"))
393             pane["popout_btn"].configure(text=self.tr("Ancorar" if window.popped else "Destacar"))
394             if window.popped and pane.get("toplevel") is not None:
395                 pane["toplevel"].title(f"MBToolkit — {self._master_window_title(window)}")
396         except tk.TclError:
397             pass
398
399     # -- panes ---------------------------------------------------------------
400     def _master_build_pane(self, window, parent):
```

MBToolkit v3.4/mb_master_ui.py — lines 401-440

```
        frame = ttk.LabelFrame(parent, text="")
        frame.columnconfigure(0, weight=1)
        frame.rowconfigure(5, weight=1)
        pane = {"frame": frame}
        window_id = window.window_id

        header_buttons = self._table_build_header(
            frame, pane,
            rename=lambda: self.master_rename_window(window_id),
            popout=lambda: self.master_toggle_popout(window_id),
            close=lambda: self.master_close_window(window_id),
        )

        definition = ttk.Frame(frame)
        definition.grid(row=1, column=0, sticky="w", padx=6, pady=(2, 2))
        pane["slave_var"] = tk.StringVar(value=str(window.slave))
        pane["fc_var"] = tk.StringVar(value=fc_choice_label(window.fc))
        pane["address_var"] = tk.StringVar(value=str(window.address + window.base))
        pane["qty_var"] = tk.StringVar(value=str(window.qty))
        pane["scan_var"] = tk.StringVar(value=str(window.scan_ms))
        pane["format_var"] = tk.StringVar(value=FORMAT_LABELS[window.fmt])

        layout = (
            ("Slave", ttk.Spinbox(definition, from_=1, to=247, width=4,
                                  textvariable=pane["slave_var"])),
            ("FC", ttk.Combobox(definition, values=FC_CHOICES, width=24, state="readonly",
                                textvariable=pane["fc_var"])),
            ("Endereço", ttk.Entry(definition, width=6, textvariable=pane["address_var"])),
            ("Qtd.", ttk.Entry(definition, width=5, textvariable=pane["qty_var"])),
        )
        for index, (caption, widget) in enumerate(layout):
            ttk.Label(definition, text=caption).grid(row=0, column=index * 2, padx=(0, 3))
            widget.grid(row=0, column=index * 2 + 1, padx=(0, 10))

        definition2 = ttk.Frame(frame)
        definition2.grid(row=2, column=0, sticky="w", padx=6, pady=2)
        format_combo = ttk.Combobox(
            definition2, values=[FORMAT_LABELS[k] for k in FORMAT_LABELS], width=20,
            state="readonly", textvariable=pane["format_var"],
        )
```

MBToolkit v3.4/mb_master_ui.py — lines 441-480

```
        layout2 = (
            ("Scan (ms)", ttk.Entry(definition2, width=7, textvariable=pane["scan_var"])),
            ("Formato", format_combo),
        )
        for index, (caption, widget) in enumerate(layout2):
            ttk.Label(definition2, text=caption).grid(row=0, column=index * 2, padx=(0, 3))
            widget.grid(row=0, column=index * 2 + 1, padx=(0, 10))
        pane["format_combo"] = format_combo

        buttons = ttk.Frame(frame)
        buttons.grid(row=3, column=0, sticky="w", padx=6, pady=(2, 2))
        actions = (
            ("Aplicar", lambda: self.master_apply_window(window_id)),
            ("Pausar", lambda: self.master_toggle_pause(window_id)),
            ("Ler 1x", lambda: self.master_read_once(window_id)),
            ("Duplicar", lambda: self.master_duplicate_window(window_id)),
        )
        for index, (caption, command) in enumerate(actions):
            button = ttk.Button(buttons, text=caption, command=command, width=10)
            button.grid(row=0, column=index, padx=(0, 3))
            if caption == "Pausar":
                pane["pause_btn"] = button
            elif caption == "Aplicar":
                pane["apply_btn"] = button

        status = ttk.Frame(frame)
        status.grid(row=4, column=0, sticky="ew", padx=6, pady=(2, 2))
        status.columnconfigure(1, weight=1)
        pane["counters_var"] = tk.StringVar()
        pane["message_var"] = tk.StringVar()
        ttk.Label(status, textvariable=pane["counters_var"]).grid(row=0, column=0, sticky="w")
        pane["message_label"] = ttk.Label(status, textvariable=pane["message_var"])
        pane["message_label"].grid(row=0, column=1, sticky="w", padx=(14, 0))

        self._table_build_tree(
            frame, pane, 5,
            on_double=lambda e: self._master_on_double_click(window_id, e),
            on_return=lambda: self._master_edit_value(window_id),
            on_menu=lambda e: self._master_show_row_menu(window_id, e),
        )
```

MBToolkit v3.4/mb_master_ui.py — lines 481-520

```
481
482         for widget in (definition, definition2):
483             for child in widget.winfo_children():
484                 if isinstance(child, (ttk.Entry, ttk.Spinbox)):
485                     child.bind("<Return>", lambda _e: self.master_apply_window(window_id))
486
487         self._table_translate_captions(definition, definition2, buttons, header_buttons)
488
489         self._master_panes[window_id] = pane
490         for name in ("slave_var", "fc_var", "address_var", "qty_var", "scan_var", "format_var"):
491             pane[name].trace_add("write",
492                                  lambda *_a: self._master_update_apply_state(window_id))
493         self._master_fill_rows(window)
494         self._master_update_format_state(window)
495         self._master_render_status(window)
496         self._master_update_apply_state(window_id)
497         self._master_install_wheel_tags(frame)
498         return frame
499
500     def _master_update_format_state(self, window):
501         pane = self._master_panes.get(window.window_id)
502         if pane is not None:
503             state = "disabled" if window.fc in BIT_FUNCTION_CODES else "readonly"
504             pane["format_combo"].configure(state=state)
505
506     def _master_fill_rows(self, window):
507         pane = self._master_panes.get(window.window_id)
508         if pane is None:
509             return
510         tree = pane["tree"]
511         tree.delete(*tree.get_children())
512         tree.configure(height=max(MASTER_PANE_MIN_ROWS, min(MASTER_PANE_MAX_ROWS, window.qty)))
513         for index in range(window.qty):
514             pdu = window.address + index
515             tree.insert("", "end", iid=f"r{index}", tags=self._table_row_tags(index),
516                         values=(pdu + window.base, window.names.get(pdu, ""), ""))
517         self._master_render_values(window)
518
519     def _master_render_values(self, window):
520         pane = self._master_panes.get(window.window_id)
```

MBToolkit v3.4/mb_master_ui.py — lines 521-560

```
        if pane is None:
            return
        tree = pane["tree"]
        stale = ("stale",) if window.stale else ()
        if window.values is None:
            texts = [""] * window.qty
        elif window.fc in BIT_FUNCTION_CODES:
            texts = [str(v) for v in window.values]
        else:
            texts = format_window_values(window.values, window.address, window.fmt,
                                         window.formats, window.scales)
        for index, text in enumerate(texts[:window.qty]):
            try:
                tree.item(f"r{index}", tags=self._table_row_tags(index) + stale)
                tree.set(f"r{index}", "value", text)
            except tk.TclError:
                pass

    def _master_schedule_reflow(self, _event=None):
        if self._master_reflow_after_id is not None:
            try:
                self.after_cancel(self._master_reflow_after_id)
            except tk.TclError:
                pass
        self._master_reflow_after_id = self.after(80, self._master_reflow)

    def _master_reflow(self):
        """Arrange docked panes in as many equal columns as the width allows."""
        self._master_reflow_after_id = None
        docked = [w for w in sorted(self.master_windows.values(), key=lambda w: w.number)
                  if not w.popped]
        frames = [self._master_panes[w.window_id]["frame"] for w in docked
                  if w.window_id in self._master_panes]
        self._table_reflow(self.master_canvas, self.master_grid, self._master_grid_item,
                           frames, self.master_empty_label, "master_panes")

    # -- window actions ------------------------------------------------------
    def master_add_window(self, kwargs=None):
        if len(self.master_windows) >= MASTER_MAX_WINDOWS:
            messagebox.showwarning(
```

MBToolkit v3.4/mb_master_ui.py — lines 561-600

```
                self.tr("Master"),
                self.trf("Máximo de {count} janelas.", count=MASTER_MAX_WINDOWS),
            )
            return None
        window_id = self._master_next_id
        self._master_next_id += 1
        numbers = {w.number for w in self.master_windows.values()}
        number = next(n for n in range(1, MASTER_MAX_WINDOWS + 2) if n not in numbers)
        window = MasterWindow(window_id, number, **(kwargs or {}))
        self.master_windows[window_id] = window
        self._master_build_pane(window, self.master_grid)
        self._master_reflow()
        if self.master_engine is not None:
            self.master_engine.set_poll(window.definition())
        self._master_mark_changed()
        return window

    def master_duplicate_window(self, window_id):
        window = self.master_windows.get(window_id)
        if window is not None:
            data = MasterWindow.validated_kwargs(window.to_json())
            self.master_add_window(data)

    def master_close_window(self, window_id):
        window = self.master_windows.pop(window_id, None)
        if window is None:
            return
        if self.master_engine is not None:
            self.master_engine.remove_poll(window_id)
        pane = self._master_panes.pop(window_id, None)
        if pane is not None:
            top = pane.get("toplevel")
            try:
                (top if top is not None else pane["frame"]).destroy()
            except tk.TclError:
                pass
        self._master_reflow()
        if not self.master_windows and not self._master_loading:
            # Closing every window leaves no workspace open.
            self.master_current_workspace = ""
```

MBToolkit v3.4/mb_master_ui.py — lines 601-640

```
            self.master_workspace_dirty = False
            self._master_update_workspace_label()
            self._schedule_ui_settings_save()
        else:
            self._master_mark_changed()

    def master_apply_window(self, window_id):
        window = self.master_windows.get(window_id)
        pane = self._master_panes.get(window_id)
        if window is None or pane is None:
            return False
        try:
            base = window.base
            slave = parse_int_field(pane["slave_var"].get(), 1, 247,
                                    "O Slave ID de leitura tem de estar entre 1 e 247.")
            fc = int(pane["fc_var"].get()[:2])
            address = parse_int_field(pane["address_var"].get(), base, 0xFFFF + base,
                                      "O endereço tem de estar entre 0 e 65535.") - base
            qty = parse_int_field(pane["qty_var"].get(), 1, 2000,
                                  "Quantidade fora do limite desta função.")
            scan_ms = parse_int_field(pane["scan_var"].get(), 0, MASTER_MAX_SCAN_MS,
                                      "Scan rate fora do intervalo.")
            validate_read_definition(slave, fc, address, qty)
        except ValueError as exc:
            messagebox.showerror(self.tr("Janela inválida"), self.tr(str(exc)))
            return False

        fmt = FORMAT_KEYS_BY_LABEL.get(pane["format_var"].get(), DEFAULT_FORMAT)
        changed_area = (slave, fc, address, qty) != (window.slave, window.fc,
                                                     window.address, window.qty)
        changed = changed_area or (scan_ms, fmt) != (window.scan_ms, window.fmt)
        window.slave, window.fc, window.address, window.qty = slave, fc, address, qty
        window.scan_ms, window.fmt = scan_ms, fmt
        if changed_area:
            window.reset_counters()
        self._master_fill_rows(window)
        self._master_update_format_state(window)
        self._master_render_status(window)
        self._master_update_apply_state(window_id)
        if self.master_engine is not None:
```

MBToolkit v3.4/mb_master_ui.py — lines 641-680

```
            self.master_engine.set_poll(window.definition())
        if changed:
            self._master_mark_changed()
        return True

    def master_toggle_pause(self, window_id):
        window = self.master_windows.get(window_id)
        if window is None:
            return
        window.enabled = not window.enabled
        if window.enabled and window.message[0] == "Pausada":
            window.message = ("", {}, "normal")
        self._master_render_status(window)
        if self.master_engine is not None:
            self.master_engine.set_poll(window.definition())
        self._master_mark_changed()

    def master_read_once(self, window_id):
        if self.master_engine is None:
            messagebox.showinfo(self.tr("Master"), self.tr("Liga primeiro o Master."))
            return
        self.master_engine.read_once(window_id)

    def master_toggle_popout(self, window_id):
        window = self.master_windows.get(window_id)
        pane = self._master_panes.pop(window_id, None)
        if window is None or pane is None:
            return
        old_top = pane.get("toplevel")
        try:
            pane["frame"].destroy()
            if old_top is not None:
                old_top.destroy()
        except tk.TclError:
            pass

        window.popped = not window.popped
        if window.popped:
            top = tk.Toplevel(self)
            top.geometry("660x420")
```

MBToolkit v3.4/mb_master_ui.py — lines 681-720

```
            top.minsize(MASTER_PANE_MIN_WIDTH, 300)
            top.columnconfigure(0, weight=1)
            top.rowconfigure(0, weight=1)
            top.protocol("WM_DELETE_WINDOW", lambda: self.master_toggle_popout(window_id))
            frame = self._master_build_pane(window, top)
            frame.grid(row=0, column=0, sticky="nsew", padx=8, pady=8)
            self._master_panes[window_id]["toplevel"] = top
            self._table_apply_toplevel_palette(top)
        else:
            self._master_build_pane(window, self.master_grid)
        self._master_render_status(window)
        self._master_reflow()

    # -- value / name editing ------------------------------------------------
    def _master_on_double_click(self, window_id, event):
        pane = self._master_panes.get(window_id)
        if pane is None:
            return "break"
        tree = pane["tree"]
        row = tree.identify_row(event.y)
        if not row:
            return "break"
        tree.selection_set(row)
        if tree.identify_column(event.x) == "#2":
            self._master_edit_name(window_id, row)
        else:
            self._master_edit_value(window_id)
        return "break"

    def _master_edit_name(self, window_id, row):
        window = self.master_windows.get(window_id)
        if window is None:
            return
        pdu = window.address + int(row[1:])
        dialog, body = self._table_dialog(self.tr("Nome"))
        ttk.Label(body, text=self.trf("Nome do endereço {address}",
                                      address=pdu + window.base)).grid(row=0, column=0, sticky="w")
        name_var = tk.StringVar(value=window.names.get(pdu, ""))
        entry = ttk.Entry(body, textvariable=name_var, width=36)
        entry.grid(row=1, column=0, pady=(4, 8))
```

MBToolkit v3.4/mb_master_ui.py — lines 721-760

```
721
722         def save(_event=None):
723             name = name_var.get().strip()
724             if name:
725                 window.names[pdu] = name
726             else:
727                 window.names.pop(pdu, None)
728             pane = self._master_panes.get(window_id)
729             if pane is not None:
730                 pane["tree"].set(row, "name", name)
731             self._master_mark_changed()
732             dialog.destroy()
733
734         ttk.Button(body, text=self.tr("Guardar"), command=save).grid(row=2, column=0, sticky="e")
735         entry.bind("<Return>", save)
736         entry.focus_set()
737         dialog.grab_set()
738
739     def _master_edit_value(self, window_id):
740         window = self.master_windows.get(window_id)
741         pane = self._master_panes.get(window_id)
742         if window is None or pane is None:
743             return
744         selection = pane["tree"].selection()
745         if not selection:
746             return
747         index = int(selection[0][1:])
748         write_fcs = WRITABLE_READ_FCS.get(window.fc)
749         if write_fcs is None:
750             messagebox.showinfo(
751                 self.tr("Escrita"),
752                 self.tr("FC02 e FC04 são só de leitura: abre uma janela FC01 ou FC03 para escrever."),
753             )
754             return
755         bits = window.fc in BIT_FUNCTION_CODES
756         if bits:
757             size, fmt = 1, None
758         else:
759             # Multi-register values start on the first register of their group.
760             index, size, fmt = window.group_of_row(index)
```

MBToolkit v3.4/mb_master_ui.py — lines 761-800

```
            if fmt is None:
                messagebox.showinfo(
                    self.tr("Escrita"),
                    self.tr("Este endereço não tem um valor completo no formato atual."),
                )
                return
        current = ""
        try:
            current = pane["tree"].set(f"r{index}", "value")
        except tk.TclError:
            pass
        if not bits and window.scales.get(window.address + index, {}).get("unit"):
            current = current.rsplit(" ", 1)[0]

        dialog, body = self._table_dialog(self.tr("Escrever"))
        slave_var = tk.StringVar(value=str(window.slave))
        address_var = tk.StringVar(value=str(window.address + index + window.base))
        value_var = tk.StringVar(value=current)
        single_fc, multiple_fc = write_fcs
        fc_var = tk.IntVar(value=single_fc if size == 1 else multiple_fc)

        rows = (
            ("Slave (0 = broadcast)", ttk.Spinbox(body, from_=0, to=247, width=6,
                                                  textvariable=slave_var)),
            ("Endereço", ttk.Entry(body, width=8, textvariable=address_var)),
            ("Valor(es)", ttk.Entry(body, width=30, textvariable=value_var)),
        )
        for row_index, (caption, widget) in enumerate(rows):
            ttk.Label(body, text=self.tr(caption)).grid(row=row_index, column=0, sticky="w",
                                                        pady=2, padx=(0, 8))
            widget.grid(row=row_index, column=1, sticky="w", pady=2)
        functions = ttk.Frame(body)
        functions.grid(row=3, column=0, columnspan=2, sticky="w", pady=(6, 2))
        single = ttk.Radiobutton(functions, text=fc_choice_label_write(single_fc),
                                 variable=fc_var, value=single_fc)
        single.grid(row=0, column=0, sticky="w")
        if size != 1:
            single.configure(state="disabled")
        ttk.Radiobutton(functions, text=fc_choice_label_write(multiple_fc),
                        variable=fc_var, value=multiple_fc).grid(row=1, column=0, sticky="w")
```

MBToolkit v3.4/mb_master_ui.py — lines 801-840

```
        scale = None if bits else window.scales.get(window.address + index)
        if scale is not None and fmt in ("hex16", "bin16"):
            scale = None
        format_text = "0 / 1" if bits else FORMAT_LABELS[fmt]
        if scale is not None:
            format_text += " · " + self.trf("valores em unidades da escala {unit}",
                                            unit=scale.get("unit") or "")
        ttk.Label(
            body,
            text=self.trf("Formato: {fmt}. Vários valores (FC15/FC16): separar com ;",
                          fmt=format_text),
        ).grid(row=4, column=0, columnspan=2, sticky="w", pady=(6, 8))
        message_var = tk.StringVar()
        ttk.Label(body, textvariable=message_var, style="MBS.Master.Error.TLabel").grid(
            row=5, column=0, columnspan=2, sticky="w")

        def send(_event=None):
            try:
                request, description = self.master_build_write(
                    window, slave_var.get(), address_var.get(), value_var.get(),
                    fc_var.get(), fmt, scale,
                )
            except ValueError as exc:
                message_var.set(self.tr(str(exc)))
                return
            if self.master_engine is None:
                message_var.set(self.tr("Liga primeiro o Master."))
                return
            self.master_engine.queue_write(window.window_id, request, description)
            dialog.destroy()

        buttons = ttk.Frame(body)
        buttons.grid(row=6, column=0, columnspan=2, sticky="e", pady=(8, 0))
        ttk.Button(buttons, text=self.tr("Enviar"), command=send,
                   style="MBS.Primary.TButton").grid(row=0, column=0, padx=(0, 6))
        ttk.Button(buttons, text=self.tr("Cancelar"), command=dialog.destroy).grid(row=0, column=1)
        rows[2][1].bind("<Return>", send)
        rows[2][1].focus_set()
        rows[2][1].select_range(0, "end")
        dialog.grab_set()
```

MBToolkit v3.4/mb_master_ui.py — lines 841-880

```
841
842     def master_build_write(self, window, slave_text, address_text, values_text, fc,
843                            fmt=None, scale=None):
844         """Validate the write dialog fields and return (request, description).
845
846         *fmt* is the format of the clicked value (per-address or window format).
847         With *scale*, the typed values are engineering units and are converted
848         back to raw values before encoding.
849         """
850         fmt = fmt or window.fmt
851         slave = parse_int_field(slave_text, 0, 247,
852                                 "O Slave ID de escrita tem de estar entre 0 (broadcast) e 247.")
853         address = parse_int_field(address_text, window.base, 0xFFFF + window.base,
854                                   "O endereço tem de estar entre 0 e 65535.") - window.base
855         parts = [p for p in str(values_text).split(";") if p.strip()]
856         if not parts:
857             raise ValueError("Valor vazio.")
858         if window.fc in BIT_FUNCTION_CODES:
859             values = [encode_coil_value(p) for p in parts]
860         else:
861             values = []
862             for part in parts:
863                 if scale is not None:
864                     part = raw_text_for_scaled(part, scale, fmt)
865                 values.extend(encode_register_value(part, fmt))
866         if fc in (5, 6) and len(values) != 1:
867             raise ValueError("FC05/FC06 escrevem um único valor; usa FC15/FC16 para vários.")
868         request = build_write_request(slave, fc, address, values)
869         description = f"FC{fc:02d} @ {address + window.base} × {len(values)}"
870         return request, description
871
872     # -- connection ---------------------------------------------------------
873     def _master_connection_settings(self):
874         timeout = parse_int_field(
875             self.master_timeout_var.get(), MASTER_MIN_RESPONSE_TIMEOUT_MS,
876             MASTER_MAX_RESPONSE_TIMEOUT_MS,
877             "O timeout de resposta tem de estar entre 10 e 10000 ms.")
878         delay = parse_int_field(
879             self.master_delay_var.get(), MASTER_MIN_DELAY_MS, MASTER_MAX_DELAY_MS,
880             "O atraso entre pedidos tem de estar entre 0 e 10000 ms.")
```

MBToolkit v3.4/mb_master_ui.py — lines 881-920

```
        return {
            "port": self.master_port_var.get().strip(),
            "baud": int(self.master_baud_var.get()),
            "parity": self.master_parity_var.get(),
            "stopbits": self.master_stop_var.get(),
            "response_timeout_ms": timeout,
            "delay_ms": delay,
        }

    def master_connect(self):
        if self.master_active:
            return
        if self.busy:
            messagebox.showwarning(
                self.tr("Master"),
                self.tr("Para a captura, a pesquisa ou o Slave em curso antes de ligar o Master."),
            )
            return
        if not self.master_safety_var.get():
            messagebox.showwarning(
                self.tr("Master"),
                self.tr("Confirma primeiro que não existe outro master ativo no barramento."),
            )
            return
        serial, _ = self.get_serial_modules()
        if serial is None:
            messagebox.showerror(self.tr("pyserial em falta"), self.tr("pyserial não está instalado."))
            return
        try:
            settings = self._master_connection_settings()
            if not settings["port"]:
                raise ValueError("Seleciona uma porta COM.")
        except ValueError as exc:
            messagebox.showerror(self.tr("Configuração inválida"), self.tr(str(exc)))
            return
        for window_id in list(self.master_windows):
            if not self.master_apply_window(window_id):
                return
        try:
            ser = self.open_discovery_serial_port(
```

MBToolkit v3.4/mb_master_ui.py — lines 921-960

```
                serial, settings["port"], settings["baud"], settings["parity"],
                settings["stopbits"],
            )
        except Exception as exc:
            messagebox.showerror(
                self.tr("Master"),
                self.trf("Não foi possível abrir {port}: {error}",
                         port=settings["port"], error=f"{type(exc).__name__}: {exc}"),
            )
            return

        self.master_serial = ser
        self.master_engine = MasterEngine(
            ser, self.event_queue, baud=settings["baud"], parity=settings["parity"],
            stopbits=settings["stopbits"],
            response_timeout_ms=settings["response_timeout_ms"],
            delay_ms=settings["delay_ms"], serial_module=serial,
            idle_check_seconds=DISCOVERY_IDLE_CHECK_SECONDS,
            reopen=lambda: self.open_discovery_serial_port(
                serial, settings["port"], settings["baud"], settings["parity"],
                settings["stopbits"],
            ),
        )
        for window in self.master_windows.values():
            window.reset_counters()
            self._master_render_status(window)
            self.master_engine.set_poll(window.definition())
        self.master_active = True
        self.busy = True
        self._master_lock_other_features(True)
        self.master_status_var.set(self.tr("A verificar se o barramento está livre…"))
        self.master_engine.start()

    def master_disconnect(self):
        if self.master_engine is None:
            return
        self.master_status_var.set(self.tr("A desligar…"))
        self.master_disconnect_btn.configure(state="disabled")
        self.master_engine.stop()

```

MBToolkit v3.4/mb_master_ui.py — lines 961-1000

```
961     def _master_lock_other_features(self, locked):
962         state = "disabled" if locked else "normal"
963         self.master_connect_btn.configure(state=state)
964         self.master_disconnect_btn.configure(state="normal" if locked else "disabled")
965         for widget in self._master_setting_widgets:
966             try:
967                 widget.configure(state="disabled" if locked else "readonly"
968                                  if isinstance(widget, ttk.Combobox) else "normal")
969             except tk.TclError:
970                 pass
971         self.master_refresh_btn.configure(state=state)
972         slave_button = getattr(self, "slave_connect_btn", None)
973         if slave_button is not None:
974             slave_button.configure(state=state)
975         self.start_btn.configure(state=state)
976         self.restart_btn.configure(state=state)
977         self.set_sniffer_refresh_state(state)
978         self.set_simulation_button_state(state)
979         self.update_discovery_start_state()
980 
981     def _master_finish_session(self, payload):
982         if not self.master_active:
983             return  # already finished (a late master_stopped event)
984         engine = self.master_engine
985         self.master_engine = None
986         if engine is not None:
987             engine.stop(join_timeout=0.5)
988             self.master_serial = engine.ser
989         if self.master_serial is not None:
990             try:
991                 self.master_serial.close()
992             except Exception:
993                 pass
994         self.master_serial = None
995         self.master_active = False
996         self.busy = False
997         self._master_lock_other_features(False)
998         self.master_status_var.set(self.tr("Desligado"))
999         for window in self.master_windows.values():
1000            if window.values is not None and not window.stale:
```

MBToolkit v3.4/mb_master_ui.py — lines 1001-1040

```
                window.stale = True
                self._master_render_values(window)

        error = (payload or {}).get("error")
        if error:
            kind, detail, count = error
            if kind == "traffic":
                text = self.trf(
                    "Foi detetado tráfego em {port} antes de transmitir "
                    "({count} bytes: {raw}). Pode existir outro master "
                    "ativo no barramento; a pesquisa não foi iniciada.",
                    port=self.master_port_var.get(), count=count, raw=detail,
                )
            else:
                text = self.trf("Erro na porta série: {error}", error=detail)
            messagebox.showerror(self.tr("Master"), text)

    # -- worker events ---------------------------------------------------------
    def handle_master_event(self, event, payload):
        if event == "master_ready":
            self.master_status_var.set(self.tr("● Ligado"))
        elif event == "master_reconnecting":
            self.master_status_var.set(self.trf(
                "A religar a {port}… (porta falhou: {error})",
                port=self.master_port_var.get(), error=(payload or {}).get("error", ""),
            ))
            for window in self.master_windows.values():
                if window.values is not None and not window.stale:
                    window.stale = True
                    self._master_render_values(window)
        elif event == "master_reconnected":
            engine = self.master_engine
            if engine is not None:
                self.master_serial = engine.ser
            self.master_status_var.set(self.tr("● Ligado"))
        elif event == "master_stopped":
            self._master_finish_session(payload)
        elif event == "master_result":
            self._master_apply_result(payload)

```

MBToolkit v3.4/mb_master_ui.py — lines 1041-1080

```
    def _master_apply_result(self, result):
        window = self.master_windows.get(result.get("window_id"))
        if window is None:
            return
        status = result["status"]
        if status == "stopped":
            return
        window.tx += 1
        if result.get("response_ms") is not None:
            window.response_ms = result["response_ms"]

        if result["kind"] == "read":
            if (result.get("fc"), result.get("address")) != (window.fc, window.address):
                return  # definition changed while the request was on the bus
            if status == "ok":
                window.values = result["values"]
                window.stale = False
                self._master_render_values(window)
                # A successful read clears errors; a write confirmation stays
                # visible for a few seconds so the user can see it.
                if (window.message[2] != "ok"
                        or time.monotonic() - window.message_at > MASTER_OK_MESSAGE_SECONDS):
                    self._master_set_message(window, "")
                return
        elif status == "ok":
            self._master_set_message(window, "Escrita OK: {what}", "ok",
                                     what=result.get("description", ""))
            if self.master_engine is not None:
                self.master_engine.read_once(window.window_id)
            return
        elif status == "broadcast":
            self._master_set_message(window, "Escrita broadcast enviada: {what}", "ok",
                                     what=result.get("description", ""))
            return

        window.errors += 1
        if result["kind"] == "read" and not window.stale:
            window.stale = True
            self._master_render_values(window)
        if status == "timeout":
```

MBToolkit v3.4/mb_master_ui.py — lines 1081-1120

```
            window.timeouts += 1
            self._master_set_message(window, "Timeout", "error")
        elif status == "exception":
            code = result.get("exception_code") or 0
            name = MODBUS_EXCEPTION_CODES.get(code, ("Unknown/Reserved Exception", ""))[0]
            self._master_set_message(window, "Exception 0x{code:02X} — {name}", "error",
                                     code=code, name=name)
        else:
            self._master_set_message(window, "Resposta inválida (CRC ou formato): {rx}",
                                     "error", rx=result.get("rx", "")[:47])

    # -- workspace ------------------------------------------------------------
    def master_workspace_data(self):
        return {
            "format": MASTER_WORKSPACE_FORMAT,
            "version": MASTER_WORKSPACE_VERSION,
            "connection": {
                "port": self.master_port_var.get(),
                "baud": self.master_baud_var.get(),
                "parity": self.master_parity_var.get(),
                "stop_bits": self.master_stop_var.get(),
                "response_timeout_ms": self.master_timeout_var.get(),
                "delay_ms": self.master_delay_var.get(),
            },
            "windows": [w.to_json() for w in sorted(self.master_windows.values(),
                                                    key=lambda w: w.number)],
        }

    def _master_update_workspace_label(self):
        name = getattr(self, "master_current_workspace", "")
        if not name:
            text = ""
        elif getattr(self, "master_workspace_dirty", False):
            text = self.trf("Workspace atual: {name} (alterado)", name=name)
        else:
            text = self.trf("Workspace atual: {name}", name=name)
        self.master_workspace_var.set(text)

    def _master_mark_changed(self):
        """Record a change to the open workspace (ignored while loading one)."""
```

MBToolkit v3.4/mb_master_ui.py — lines 1121-1160

```
        if getattr(self, "_master_loading", False):
            return
        if self.master_current_workspace and not self.master_workspace_dirty:
            self.master_workspace_dirty = True
            self._master_update_workspace_label()

    def _master_build_workspace_menu(self):
        """Rebuild the Workspace menu from the files in the workspace folder."""
        self._table_fill_workspace_menu(
            self.master_workspace_menu, list_workspaces(),
            save=self.master_save_workspace, open_=self.master_open_workspace,
            delete=self.master_delete_workspace, folder=self.master_open_workspace_folder,
        )
        self._master_apply_menu_palette()

    def master_save_workspace(self, name=None):
        """Save all windows as a named workspace in the workspace folder."""
        if name is None:
            name = self._table_ask_text(
                self.tr("Guardar workspace"), self.tr("Nome do workspace"),
                self.master_current_workspace or self.tr("Workspace 1"),
            )
            if name is None:
                return False
        try:
            name = validate_workspace_name(name)
        except ValueError as exc:
            messagebox.showerror(self.tr("Guardar workspace"), self.tr(str(exc)))
            return False
        path = workspace_path(name)
        if path.exists() and name != self.master_current_workspace:
            if not messagebox.askyesno(
                self.tr("Guardar workspace"),
                self.trf("Já existe um workspace chamado \"{name}\". Substituir?", name=name),
            ):
                return False
        try:
            with open(path, "w", encoding="utf-8") as handle:
                json.dump(self.master_workspace_data(), handle, indent=2, ensure_ascii=False)
        except OSError as exc:
```

MBToolkit v3.4/mb_master_ui.py — lines 1161-1200

```
            messagebox.showerror(self.tr("Guardar workspace"), str(exc))
            return False
        self.master_current_workspace = name
        self.master_workspace_dirty = False
        self._master_update_workspace_label()
        self._schedule_ui_settings_save()
        return True

    def master_delete_workspace(self, name, confirm=True):
        if confirm and not messagebox.askyesno(
            self.tr("Apagar workspace"),
            self.trf("Apagar o workspace \"{name}\"?", name=name),
        ):
            return False
        try:
            workspace_path(name).unlink()
        except OSError as exc:
            messagebox.showerror(self.tr("Apagar workspace"), str(exc))
            return False
        if name == self.master_current_workspace:
            self.master_current_workspace = ""
            self.master_workspace_dirty = False
            self._master_update_workspace_label()
            self._schedule_ui_settings_save()
        return True

    def master_open_workspace_folder(self):
        self._table_open_folder(get_master_workspaces_folder())

    def master_open_workspace(self, name, quiet=False):
        """Replace the current windows with a saved workspace (by name).

        *quiet* (start-up restore) suppresses the error dialog when the file
        cannot be read.
        """
        try:
            with open(workspace_path(name), encoding="utf-8") as handle:
                data = json.load(handle)
            if not isinstance(data, dict) or data.get("format") not in MASTER_WORKSPACE_FORMATS:
                raise ValueError(self.tr("O ficheiro não é um workspace do MBToolkit."))
```

MBToolkit v3.4/mb_master_ui.py — lines 1201-1240

```
        except (OSError, ValueError) as exc:
            if not quiet:
                messagebox.showerror(self.tr("Abrir workspace"), str(exc))
            return False
        self._master_loading = True
        try:
            return self._master_load_workspace_data(name, data)
        finally:
            self._master_loading = False

    def _master_load_workspace_data(self, name, data):
        entries, skipped = [], 0
        for item in data.get("windows", [])[:MASTER_MAX_WINDOWS]:
            try:
                entries.append(MasterWindow.validated_kwargs(item))
            except (ValueError, TypeError):
                skipped += 1

        for window_id in list(self.master_windows):
            self.master_close_window(window_id)
        connection = data.get("connection", {})
        if isinstance(connection, dict) and not self.master_active:
            if str(connection.get("baud")) in [str(b) for b in DISCOVERY_BAUD_RATES]:
                self.master_baud_var.set(str(connection["baud"]))
            if connection.get("parity") in MASTER_PARITIES:
                self.master_parity_var.set(connection["parity"])
            if str(connection.get("stop_bits")) in MASTER_STOP_BITS:
                self.master_stop_var.set(str(connection["stop_bits"]))
            for key, var in (("response_timeout_ms", self.master_timeout_var),
                             ("delay_ms", self.master_delay_var)):
                if str(connection.get(key, "")).isdigit():
                    var.set(str(connection[key]))
            if connection.get("port") in tuple(self.master_port_combo["values"]):
                self.master_port_var.set(connection["port"])
        for kwargs in entries:
            self.master_add_window(kwargs)
        self.master_current_workspace = name
        self.master_workspace_dirty = False
        self._master_update_workspace_label()
        self._schedule_ui_settings_save()
```

MBToolkit v3.4/mb_master_ui.py — lines 1241-1280

```
        if skipped:
            messagebox.showwarning(
                self.tr("Abrir workspace"),
                self.trf("{count} janela(s) inválida(s) foram ignoradas.", count=skipped),
            )
        return True

    # -- window rename ------------------------------------------------------------
    def master_rename_window(self, window_id, name=None):
        window = self.master_windows.get(window_id)
        if window is None:
            return
        if name is None:
            name = self._table_ask_text(
                self.tr("Renomear janela"),
                self.tr("Nome da janela (vazio = nome por defeito)"),
                self._master_window_name(window),
            )
            if name is None:
                return
        name = name.strip()[:MASTER_MAX_TITLE_LENGTH]
        if name == self.trf("Janela {number}", number=window.number):
            name = ""
        window.title = name
        self._master_render_status(window)
        self._master_mark_changed()

    # -- apply highlight / table tags ---------------------------------------------
    def _master_fields_changed(self, window, pane):
        """True when the definition fields differ from what was last applied."""
        try:
            return (
                pane["slave_var"].get().strip() != str(window.slave)
                or pane["fc_var"].get() != fc_choice_label(window.fc)
                or pane["address_var"].get().strip() != str(window.address + window.base)
                or pane["qty_var"].get().strip() != str(window.qty)
                or pane["scan_var"].get().strip() != str(window.scan_ms)
                or FORMAT_KEYS_BY_LABEL.get(pane["format_var"].get()) != window.fmt
            )
        except tk.TclError:
```

MBToolkit v3.4/mb_master_ui.py — lines 1281-1320

```
            return False

    def _master_update_apply_state(self, window_id):
        window = self.master_windows.get(window_id)
        pane = self._master_panes.get(window_id)
        if window is None or pane is None or "apply_btn" not in pane:
            return
        style = ("MBS.Primary.TButton" if self._master_fields_changed(window, pane)
                 else "TButton")
        try:
            pane["apply_btn"].configure(style=style)
        except tk.TclError:
            pass

    def _master_row_tags(index):
        """Alternate rows get a slightly darker background (zebra striping)."""
        return ("odd",) if index % 2 else ()

    # -- right-click menu: per-address format, copy, base ---------------------------
    def _master_build_row_menu(self):
        self._master_row_menu = tk.Menu(self, tearoff=False)

    def _master_selected_rows(self, window_id):
        return self._table_selected_rows(self._master_panes.get(window_id))

    def _master_show_row_menu(self, window_id, event):
        window = self.master_windows.get(window_id)
        pane = self._master_panes.get(window_id)
        if window is None or pane is None:
            return "break"
        tree = pane["tree"]
        row = tree.identify_row(event.y)
        if row and row not in tree.selection():
            tree.selection_set(row)
        self._table_fill_row_menu(
            self._master_row_menu, pane, window, window.fc not in BIT_FUNCTION_CODES,
            set_format=lambda fmt: self.master_set_row_format(window_id, fmt),
            edit_scale=lambda: self.master_edit_scale(window_id),
            remove_scale=lambda: self.master_remove_scale(window_id),
            copy=lambda: self.master_copy_rows(window_id),
```

MBToolkit v3.4/mb_master_ui.py — lines 1321-1360

```
            set_base=lambda base: self.master_set_base(window_id, base),
        )
        self._master_apply_menu_palette()
        self._table_popup(self._master_row_menu, event)
        return "break"

    def master_set_row_format(self, window_id, fmt):
        """Apply *fmt* (None = window format) to the selected rows."""
        window = self.master_windows.get(window_id)
        if window is None:
            return False
        try:
            apply_format_to_rows(window, self._master_selected_rows(window_id), fmt)
        except ValueError as exc:
            size = FORMAT_SIZES.get(fmt, 1)
            messagebox.showwarning(self.tr("Formato"), self.trf(str(exc), size=size))
            return False
        self._master_render_values(window)
        self._master_mark_changed()
        return True

    # -- opening windows from Sniffer / Bus Slave Finder ------------------------------
    def master_definition_from_record(self, record):
        """Read definition (slave, fc, address, qty) behind a Traffic record, or None.

        A REQUEST carries its own address/quantity; a matched RESPONSE or
        EXCEPTION carries the ones of its request. Only FC01–FC04 qualify.
        """
        try:
            slave = int(record.get("slave", ""))
            fc = int(str(record.get("fc", "")), 16) & 0x7F
        except (TypeError, ValueError):
            return None
        if fc not in READ_FUNCTION_CODES or not 1 <= slave <= 247:
            return None
        if record.get("type") == "REQUEST":
            address, qty = record.get("pdu_address"), record.get("qty")
        else:
            address, qty = record.get("request_pdu_address"), record.get("request_qty")
        if address is None or qty is None:
```

MBToolkit v3.4/mb_master_ui.py — lines 1361-1400

```
            return None
        try:
            validate_read_definition(slave, fc, address, qty)
        except ValueError:
            return None
        return {"slave": slave, "fc": fc, "address": int(address), "qty": int(qty)}

    def master_open_from_record(self, record):
        definition = self.master_definition_from_record(record)
        if definition is None:
            messagebox.showinfo(
                self.tr("Abrir no Master"),
                self.tr("Esta trama não tem um pedido de leitura FC01–FC04 com endereço "
                        "conhecido."),
            )
            return None
        port = ""
        if not getattr(self, "capture_active", False):
            port = self.port_a_var.get().strip()
        return self.master_open_definition(
            definition, port=port, baud=self.baud_var.get(),
            parity=self.parity_var.get(), stopbits=self.stop_var.get(),
        )

    def master_open_definition(self, definition, port="", baud="", parity="", stopbits=""):
        """Add a Master window and show the Master tab.

        The serial settings (and port) are copied to the Master connection only
        when it is not connected and they are valid for the Master.
        """
        if not self.master_active:
            if str(baud) in [str(b) for b in DISCOVERY_BAUD_RATES]:
                self.master_baud_var.set(str(baud))
            if parity in MASTER_PARITIES:
                self.master_parity_var.set(parity)
            if str(stopbits) in MASTER_STOP_BITS:
                self.master_stop_var.set(str(stopbits))
            if port and port in tuple(self.master_port_combo["values"]):
                self.master_port_var.set(port)
        window = self.master_add_window(definition)
```

MBToolkit v3.4/mb_master_ui.py — lines 1401-1440

```
        try:
            self.main_notebook.select(self.master_page)
        except tk.TclError:
            pass
        return window

    def _master_selected_group_starts(self, window):
        """PDU addresses where the values of the selected rows start."""
        return self._table_selected_group_starts(window, self._master_panes.get(window.window_id))

    def master_edit_scale(self, window_id):
        """Ask for a linear scale and apply it to the selected values."""
        window = self.master_windows.get(window_id)
        if window is None:
            return False
        starts = self._master_selected_group_starts(window)
        if not starts:
            messagebox.showwarning(self.tr("Escala"),
                                   self.tr("Seleciona pelo menos um endereço."))
            return False
        if not self._table_ask_scale(window, starts):
            return False
        self._master_render_values(window)
        self._master_mark_changed()
        return True

    def master_remove_scale(self, window_id):
        window = self.master_windows.get(window_id)
        if window is None:
            return False
        removed = False
        for address in self._master_selected_group_starts(window):
            removed = window.scales.pop(address, None) is not None or removed
        if removed:
            self._master_render_values(window)
            self._master_mark_changed()
        return removed

    def master_copy_rows(self, window_id):
        """Copy the selected rows (all rows if none) as tab-separated text."""
```

MBToolkit v3.4/mb_master_ui.py — lines 1441-1480

```
1441        self._table_copy_rows(self._master_panes.get(window_id))
1442
1443    def master_set_base(self, window_id, base):
1444        window = self.master_windows.get(window_id)
1445        pane = self._master_panes.get(window_id)
1446        if window is None or pane is None or window.base == base:
1447            return
1448        window.base = base
1449        pane["address_var"].set(str(window.address + window.base))
1450        self._master_fill_rows(window)
1451        self._master_render_status(window)
1452        self._master_update_apply_state(window_id)
1453        self._master_mark_changed()
1454
1455    # -- closing the application ---------------------------------------------------
1456    def master_confirm_close(self):
1457        """Ask to save a changed workspace; False cancels closing the application.
1458
1459        Plain windows that were never saved as a workspace close silently.
1460        """
1461        if not (getattr(self, "master_current_workspace", "")
1462                and getattr(self, "master_workspace_dirty", False)):
1463            return True
1464        answer = messagebox.askyesnocancel(
1465            self.tr("Workspace"),
1466            self.trf("Guardar alterações ao workspace \"{name}\"?",
1467                     name=self.master_current_workspace),
1468        )
1469        if answer is None:
1470            return False
1471        if answer:
1472            return self.master_save_workspace(self.master_current_workspace)
1473        return True
1474
1475    # -- language / palette / shutdown ---------------------------------------------
1476    def master_apply_language(self):
1477        if not hasattr(self, "master_windows"):
1478            return
1479        self._master_update_workspace_label()
1480        for window in self.master_windows.values():
```

MBToolkit v3.4/mb_master_ui.py — lines 1481-1520

```
            pane = self._master_panes.get(window.window_id)
            if pane is None:
                continue
            self._table_translate_headings(pane)
            self._master_render_status(window)
        if not self.master_active:
            self.master_status_var.set(self.tr("Desligado"))
        elif self.master_engine is not None:
            self.master_status_var.set(self.tr("● Ligado"))

    def _master_apply_menu_palette(self):
        self._table_apply_menus_palette(getattr(self, "master_workspace_menu", None),
                                        getattr(self, "_master_row_menu", None))

    def master_apply_palette(self, dark):
        self._master_apply_menu_palette()
        colors = theme_colors(self, dark)
        if hasattr(self, "master_canvas"):
            self.master_canvas.configure(background=colors["root"])
        if hasattr(self, "ui_style"):
            self.ui_style.configure("MBS.Master.Error.TLabel",
                                    foreground="#ff8a80" if dark else "#b42318")
            self.ui_style.configure("MBS.Master.Ok.TLabel",
                                    foreground="#a7e4b8" if dark else "#1f6b3a")
        for pane in getattr(self, "_master_panes", {}).values():
            if pane.get("toplevel") is not None:
                self._table_apply_toplevel_palette(pane["toplevel"])
            self._table_configure_tree_tags(pane["tree"])
        button = getattr(self, "master_refresh_btn", None)
        if button is not None:
            button.set_dark_mode(dark)

    def master_shutdown(self):
        engine = getattr(self, "master_engine", None)
        if engine is not None:
            engine.stop(join_timeout=0.8)
            self.master_serial = engine.ser
        ser = getattr(self, "master_serial", None)
        if ser is not None:
            try:
```

MBToolkit v3.4/mb_master_ui.py — lines 1521-1529

```
                ser.close()
            except Exception:
                pass
        self.master_engine = None
        self.master_serial = None


def fc_choice_label_write(fc):
    return f"FC{fc:02d} {FUNCTION_NAMES[fc]}"
```

MBToolkit v3.4/mb_slave.py

533 source lines.

MBToolkit v3.4/mb_slave.py — lines 1-40

```python
#!/usr/bin/env python3
"""Modbus RTU slave (simulator) engine: data tables, replies and the serial loop.

This module has no widgets. ``SlaveStore`` holds the data tables of every
simulated Slave ID, the address areas the windows define and the fault
settings of each Slave ID. ``handle_request()`` executes one request frame
against the store and builds the reply, the way a real device would.
``SlaveEngine`` listens on one serial port, assembles the request frames,
answers them and posts what it did to the application's ``event_queue``.
"""

import random
import threading
import time
from dataclasses import dataclass, field

from mb_config import (
    MASTER_MAX_READ_BITS,
    MASTER_MAX_READ_REGISTERS,
    MASTER_MAX_WRITE_BITS,
    MASTER_MAX_WRITE_REGISTERS,
    MASTER_RECONNECT_INTERVAL_SECONDS,
    SLAVE_FAULT_MAX_DELAY_MS,
    SLAVE_RESYNC_SILENCE_MS,
)
from mb_protocol import append_crc, crc_ok, modbus_rtu_interframe_gap_seconds
from mb_slave_finder import cancel_serial_io, discovery_bits_per_char, write_discovery_request

# Data tables of a Modbus device. Labels are technical terms (not translated).
TABLES = ("coils", "discrete", "holding", "input")
TABLE_LABELS = {
    "coils": "Coils",
    "discrete": "Discrete Inputs",
    "holding": "Holding Registers",
    "input": "Input Registers",
}
BIT_TABLES = ("coils", "discrete")
# Table read by FC01–FC04 and written by FC05/06/15/16.
FC_TABLES = {1: "coils", 2: "discrete", 3: "holding", 4: "input",
             5: "coils", 15: "coils", 6: "holding", 16: "holding"}
```

MBToolkit v3.4/mb_slave.py — lines 41-80

```
41 SUPPORTED_FUNCTION_CODES = (1, 2, 3, 4, 5, 6, 15, 16)
42 WRITE_FUNCTION_CODES = (5, 6, 15, 16)
43 # Exception codes offered by the fault simulation.
44 FAULT_EXCEPTION_CODES = (1, 2, 3, 4, 6)
45
46 EXCEPTION_ILLEGAL_FUNCTION = 0x01
47 EXCEPTION_ILLEGAL_ADDRESS = 0x02
48 EXCEPTION_ILLEGAL_VALUE = 0x03
49
50
51 # Function code that reads each table (shown with the table name).
52 TABLE_READ_FC = {"coils": 1, "discrete": 2, "holding": 3, "input": 4}
53
54
55 def table_label(table):
56     return TABLE_LABELS.get(table, table)
57
58
59 def table_choice_label(table):
60     """Table field text: read function code + name, e.g. "03 Holding Registers"."""
61     return f"{TABLE_READ_FC[table]:02d} {TABLE_LABELS[table]}"
62
63
64 # ---------------------------------------------------------------------------
65 # Fault simulation
66 # ---------------------------------------------------------------------------
67
68 def default_fault():
69     return {"delay_ms": 0, "no_response": False, "exception": 0, "bad_crc": False,
70             "probability": 100}
71
72
73 def validate_fault(data):
74     """Return a clean fault dict from user/workspace data; raise ValueError (PT)."""
75     if not isinstance(data, dict):
76         raise ValueError("Falha inválida.")
77     try:
78         delay_ms = int(str(data.get("delay_ms", 0)).strip() or 0)
79         probability = int(str(data.get("probability", 100)).strip() or 0)
80         exception = int(data.get("exception", 0) or 0)
```

MBToolkit v3.4/mb_slave.py — lines 81-120

```
    except ValueError:
        raise ValueError("A falha tem de ter números válidos.") from None
    if not 0 <= delay_ms <= SLAVE_FAULT_MAX_DELAY_MS:
        raise ValueError("O atraso tem de estar entre 0 e 10000 ms.")
    if not 1 <= probability <= 100:
        raise ValueError("A probabilidade tem de estar entre 1 e 100 %.")
    if exception and exception not in FAULT_EXCEPTION_CODES:
        raise ValueError("Código de exceção não suportado.")
    return {"delay_ms": delay_ms, "no_response": bool(data.get("no_response")),
            "exception": exception, "bad_crc": bool(data.get("bad_crc")),
            "probability": probability}


def fault_active(fault):
    """True when the fault settings change how the slave answers."""
    return bool(fault) and bool(fault.get("delay_ms") or fault.get("no_response")
                                or fault.get("exception") or fault.get("bad_crc"))


# ---------------------------------------------------------------------------
# Data store
# ---------------------------------------------------------------------------

class SlaveStore:
    """Data tables, defined areas and faults of every simulated Slave ID.

    Values live per (Slave ID, table) and address and start at 0. A request
    is valid only for addresses inside an area defined by a window; windows
    over the same addresses share the values. Thread-safe: the engine thread
    reads and writes while the GUI edits.
    """

    def __init__(self):
        self._lock = threading.RLock()
        self._values = {}    # (slave, table) -> {address: value}
        self._defined = {}   # (slave, table) -> set of addresses
        self._faults = {}    # slave -> fault dict

    def set_areas(self, areas):
        """Replace the defined areas: iterable of (slave, table, address, qty)."""
```

MBToolkit v3.4/mb_slave.py — lines 121-160

```
        defined = {}
        for slave, table, address, qty in areas:
            defined.setdefault((int(slave), table), set()).update(
                range(int(address), int(address) + int(qty)))
        with self._lock:
            self._defined = defined

    def slave_ids(self):
        with self._lock:
            return {slave for slave, _table in self._defined}

    def covers(self, slave, table, address, qty):
        with self._lock:
            defined = self._defined.get((slave, table))
            return bool(defined) and all(a in defined for a in range(address, address + qty))

    def read(self, slave, table, address, qty):
        with self._lock:
            values = self._values.get((slave, table), {})
            return [values.get(a, 0) for a in range(address, address + qty)]

    def write(self, slave, table, address, values):
        mask = 1 if table in BIT_TABLES else 0xFFFF
        with self._lock:
            target = self._values.setdefault((slave, table), {})
            for offset, value in enumerate(values):
                target[address + offset] = int(value) & mask

    def clear_values(self):
        with self._lock:
            self._values = {}

    def set_fault(self, slave, fault):
        with self._lock:
            if fault_active(fault):
                self._faults[int(slave)] = dict(fault)
            else:
                self._faults.pop(int(slave), None)

    def fault(self, slave):
```

MBToolkit v3.4/mb_slave.py — lines 161-200

```
        with self._lock:
            fault = self._faults.get(slave)
            return dict(fault) if fault else None

    def faults(self):
        with self._lock:
            return {slave: dict(fault) for slave, fault in self._faults.items()}

    def clear_faults(self):
        with self._lock:
            self._faults = {}


# ---------------------------------------------------------------------------
# Request handling
# ---------------------------------------------------------------------------

def request_length(buffer):
    """Length of the request frame starting at buffer[0]; None if not known yet.

    FC01–FC06 requests are 8 bytes; FC15/FC16 carry a byte count in byte 6.
    Unknown function codes have no known length (None).
    """
    if len(buffer) < 2:
        return None
    fc = buffer[1]
    if fc in (1, 2, 3, 4, 5, 6):
        return 8
    if fc in (15, 16):
        return 9 + buffer[6] if len(buffer) >= 7 else None
    return None


def exception_frame(slave, fc, code):
    return append_crc(bytes((slave, fc | 0x80, code)))


@dataclass
class SlaveReply:
    """Outcome of one request: what to send back and what it touched."""
```

MBToolkit v3.4/mb_slave.py — lines 201-240

```
    slave: int
    fc: int
    status: str                 # ok, exception, broadcast, ignored
    table: str = None
    address: int = None
    qty: int = None
    exception_code: int = None
    response: bytes = None
    written: list = field(default_factory=list)   # [(slave, table, address, qty)]


def _u16(data, index):
    return (data[index] << 8) | data[index + 1]


def _parse_request(frame):
    """Return (table, address, qty, values or None) or raise _ExceptionReply."""
    fc = frame[1]
    if fc not in SUPPORTED_FUNCTION_CODES:
        raise _ExceptionReply(EXCEPTION_ILLEGAL_FUNCTION)
    table = FC_TABLES[fc]
    if fc in (15, 16):
        if len(frame) < 9 or len(frame) != 9 + frame[6]:
            raise _ExceptionReply(EXCEPTION_ILLEGAL_VALUE)
    elif len(frame) != 8:
        raise _ExceptionReply(EXCEPTION_ILLEGAL_VALUE)
    address = _u16(frame, 2)

    if fc in (1, 2, 3, 4):
        qty = _u16(frame, 4)
        limit = MASTER_MAX_READ_BITS if fc in (1, 2) else MASTER_MAX_READ_REGISTERS
        if not 1 <= qty <= limit:
            raise _ExceptionReply(EXCEPTION_ILLEGAL_VALUE)
        return table, address, qty, None
    if fc == 5:
        raw = _u16(frame, 4)
        if raw not in (0xFF00, 0x0000):
            raise _ExceptionReply(EXCEPTION_ILLEGAL_VALUE)
        return table, address, 1, [1 if raw else 0]
    if fc == 6:
```

MBToolkit v3.4/mb_slave.py — lines 241-280

```
241         return table, address, 1, [_u16(frame, 4)]
242
243     qty, byte_count = _u16(frame, 4), frame[6]
244     data = frame[7:-2]
245     if fc == 15:
246         if not 1 <= qty <= MASTER_MAX_WRITE_BITS or byte_count != (qty + 7) // 8:
247             raise _ExceptionReply(EXCEPTION_ILLEGAL_VALUE)
248         return table, address, qty, [(data[i // 8] >> (i % 8)) & 1 for i in range(qty)]
249     if not 1 <= qty <= MASTER_MAX_WRITE_REGISTERS or byte_count != 2 * qty:
250         raise _ExceptionReply(EXCEPTION_ILLEGAL_VALUE)
251     return table, address, qty, [_u16(data, 2 * i) for i in range(qty)]
252
253
254 class _ExceptionReply(Exception):
255     def __init__(self, code):
256         super().__init__(code)
257         self.code = code
258
259
260 def handle_request(store, frame):
261     """Execute one CRC-valid request frame and return a SlaveReply.
262
263     Requests for Slave IDs that are not simulated are ``ignored``. Broadcast
264     (Slave 0) writes are executed on every simulated Slave ID whose areas
265     cover the addresses, and never answered. Errors answer like a real
266     device: Exception 01 (function), 02 (address outside the defined areas)
267     or 03 (quantity / value / byte count).
268     """
269     slave, fc = frame[0], frame[1] & 0x7F
270     reply = SlaveReply(slave=slave, fc=fc, status="ignored")
271     if slave != 0 and slave not in store.slave_ids():
272         return reply
273
274     try:
275         table, address, qty, values = _parse_request(frame)
276         reply.table, reply.address, reply.qty = table, address, qty
277         if address + qty > 0x10000:
278             raise _ExceptionReply(EXCEPTION_ILLEGAL_ADDRESS)
279     except _ExceptionReply as exc:
280         if slave == 0:
```

MBToolkit v3.4/mb_slave.py — lines 281-320

```
            reply.status = "broadcast"
            return reply
        reply.status = "exception"
        reply.exception_code = exc.code
        reply.response = exception_frame(slave, fc, exc.code)
        return reply

    if slave == 0:
        reply.status = "broadcast"
        if values is not None:
            for target in sorted(store.slave_ids()):
                if store.covers(target, table, address, qty):
                    store.write(target, table, address, values)
                    reply.written.append((target, table, address, qty))
        return reply

    if not store.covers(slave, table, address, qty):
        reply.status = "exception"
        reply.exception_code = EXCEPTION_ILLEGAL_ADDRESS
        reply.response = exception_frame(slave, fc, EXCEPTION_ILLEGAL_ADDRESS)
        return reply

    reply.status = "ok"
    if values is not None:
        store.write(slave, table, address, values)
        reply.written.append((slave, table, address, qty))
        reply.response = bytes(frame) if fc in (5, 6) else append_crc(bytes(frame[:6]))
        return reply

    current = store.read(slave, table, address, qty)
    if fc in (1, 2):
        data = bytearray((qty + 7) // 8)
        for index, bit in enumerate(current):
            if bit:
                data[index // 8] |= 1 << (index % 8)
    else:
        data = b"".join(bytes(((v >> 8) & 0xFF, v & 0xFF)) for v in current)
    reply.response = append_crc(bytes((slave, fc, len(data))) + bytes(data))
    return reply

```

MBToolkit v3.4/mb_slave.py — lines 321-360

```
321
322 def corrupt_crc(frame):
323     """Same frame with a wrong CRC (fault simulation)."""
324     frame = bytearray(frame)
325     frame[-1] ^= 0xFF
326     return bytes(frame)
327
328
329 # ---------------------------------------------------------------------------
330 # Engine
331 # ---------------------------------------------------------------------------
332
333 class SlaveEngine:
334     """Answer the requests of a real master on one serial port.
335
336     Request frames are taken from the received bytes by their length (FC01–06
337     and FC15/16 have a known size), so a request is answered as soon as it is
338     complete. Bytes that do not form a CRC-valid request (other devices'
339     responses, noise, unknown function codes) are discarded one at a time
340     until a request lines up, at the latest after SLAVE_RESYNC_SILENCE_MS of
341     silence. The echo of our own response from an RS485 adapter is removed.
342
343     Before answering, the engine waits the Modbus t3.5 gap, and the fault
344     simulation of the Slave ID (if any) may delay, drop, replace by an
345     exception or corrupt the response.
346
347     With *reopen*, a serial failure does not end the session: the engine posts
348     ``slave_reconnecting`` and reopens the port like the Master does.
349     """
350
351     def __init__(self, ser, event_queue, store, *, baud, parity, stopbits,
352                  serial_module=None, reopen=None,
353                  reconnect_interval_s=MASTER_RECONNECT_INTERVAL_SECONDS,
354                  silence_ms=SLAVE_RESYNC_SILENCE_MS, rng=None):
355         self.ser = ser
356         self.event_queue = event_queue
357         self.store = store
358         self.baud = int(baud)
359         self.parity = parity
360         self.stopbits = stopbits
```

MBToolkit v3.4/mb_slave.py — lines 361-400

```
        self.serial_module = serial_module
        self.reopen = reopen
        self.reconnect_interval_s = float(reconnect_interval_s)
        bits_per_char = discovery_bits_per_char(parity, stopbits)
        self.char_s = bits_per_char / float(max(1, self.baud))
        self.gap_s = modbus_rtu_interframe_gap_seconds(self.baud, bits_per_char)
        self.silence_s = max(self.gap_s, float(silence_ms) / 1000.0)
        self.rng = rng or random.Random()
        self.stop_event = threading.Event()
        self.thread = None
        self._buffer = bytearray()
        self._echo = b""
        self._echo_deadline = 0.0

    def start(self):
        self.thread = threading.Thread(target=self.run, daemon=True)
        self.thread.start()

    def stop(self, join_timeout=1.0):
        self.stop_event.set()
        cancel_serial_io(self.ser)
        thread = self.thread
        if thread is not None and thread.is_alive() and thread is not threading.current_thread():
            thread.join(timeout=join_timeout)

    # -- worker ------------------------------------------------------------
    def run(self):
        error = None
        try:
            self.event_queue.put(("slave_ready", None))
            while not self.stop_event.is_set():
                try:
                    self._listen()
                except Exception as exc:  # serial/driver failure
                    if self.stop_event.is_set():
                        break
                    detail = f"{type(exc).__name__}: {exc}"
                    if self.reopen is None:
                        error = detail
                        break
```

MBToolkit v3.4/mb_slave.py — lines 401-440

```
                    if not self._reconnect(detail):
                        break
        finally:
            self.event_queue.put(("slave_stopped", {"error": error}))

    def _reconnect(self, detail):
        try:
            self.ser.close()
        except Exception:
            pass
        self._buffer.clear()
        self.event_queue.put(("slave_reconnecting", {"error": detail}))
        while not self.stop_event.wait(self.reconnect_interval_s):
            try:
                ser = self.reopen()
            except Exception:
                continue
            self.ser = ser
            self.event_queue.put(("slave_reconnected", None))
            return True
        return False

    def _listen(self):
        """Read and answer until stopped; serial errors propagate."""
        last_rx = time.perf_counter()
        while not self.stop_event.is_set():
            waiting = int(getattr(self.ser, "in_waiting", 0) or 0)
            if waiting:
                data = self._strip_echo(self.ser.read(waiting) or b"")
                last_rx = time.perf_counter()
                if data:
                    self._buffer.extend(data)
                    self.process(final=False)
                continue
            if self._buffer and time.perf_counter() - last_rx >= self.silence_s:
                self.process(final=True)
            if self.stop_event.wait(0.001):
                return

    def process(self, final):
```

MBToolkit v3.4/mb_slave.py — lines 441-480

```
        """Answer every complete request in the buffer.
        
        *final* (the bus went silent) also resolves what is left: a frame of
        unknown length is answered if the whole rest is CRC-valid, anything
        else is dropped byte by byte.
        """
        buffer = self._buffer
        while buffer and not self.stop_event.is_set():
            length = request_length(buffer)
            if length is None or len(buffer) < length:
                if not final:
                    return
                if length is None and len(buffer) >= 4 and crc_ok(bytes(buffer)):
                    frame = bytes(buffer)
                    buffer.clear()
                    self._answer(frame)
                    return
                del buffer[0]
                continue
            frame = bytes(buffer[:length])
            if crc_ok(frame):
                del buffer[:length]
                self._answer(frame)
            else:
                del buffer[0]

    def _strip_echo(self, data):
        """Remove the RS485 adapter echo of our last response from *data*."""
        if not self._echo:
            return data
        if time.perf_counter() > self._echo_deadline:
            self._echo = b""
            return data
        matched = 0
        while (matched < len(data) and matched < len(self._echo)
               and data[matched] == self._echo[matched]):
            matched += 1
        if matched == len(data) or matched == len(self._echo):
            self._echo = self._echo[matched:]
            return data[matched:]
```

MBToolkit v3.4/mb_slave.py — lines 481-520

```
        self._echo = b""
        return data

    def _answer(self, frame):
        slave = frame[0]
        fault = None
        if slave != 0 and slave in self.store.slave_ids():
            fault = self.store.fault(slave)
            if fault and self.rng.random() * 100.0 >= fault["probability"]:
                fault = None

        if fault and fault["no_response"]:
            reply = SlaveReply(slave=slave, fc=frame[1] & 0x7F, status="no_response")
        elif fault and fault["exception"]:
            code = fault["exception"]
            reply = SlaveReply(slave=slave, fc=frame[1] & 0x7F, status="exception",
                               exception_code=code,
                               response=exception_frame(slave, frame[1] & 0x7F, code))
        else:
            reply = handle_request(self.store, frame)
        if reply.table is None and reply.fc in FC_TABLES and len(frame) >= 6:
            # Fault replies and ignored requests still name the addresses, so
            # the windows over them can count the request.
            reply.table, reply.address = FC_TABLES[reply.fc], _u16(frame, 2)
            reply.qty = 1 if reply.fc in (5, 6) else _u16(frame, 4)

        response = reply.response
        if response is not None and fault and fault["bad_crc"]:
            response = corrupt_crc(response)
        info = {
            "slave": reply.slave, "fc": reply.fc, "table": reply.table,
            "address": reply.address, "qty": reply.qty, "status": reply.status,
            "exception_code": reply.exception_code, "written": list(reply.written),
            "fault": bool(fault), "time": time.time(),
        }
        if response is not None:
            delay_s = self.gap_s + (fault["delay_ms"] / 1000.0 if fault else 0.0)
            if self.stop_event.wait(delay_s):
                return
            # A request that arrived meanwhile belongs to another exchange; a
```

MBToolkit v3.4/mb_slave.py — lines 521-533

```
            # real slave would not answer over it, but answering late is the
            # best a PC can do. Anything received before our reply is dropped.
            self._buffer.clear()
            sent = write_discovery_request(
                self.ser, response, self.baud, self.parity, self.stopbits,
                self.stop_event, self.serial_module,
            )
            if not sent:
                return
            self._echo = bytes(response)
            self._echo_deadline = time.perf_counter() + len(response) * self.char_s + 0.2
            info["response"] = len(response)
        self.event_queue.put(("slave_request", info))
```

MBToolkit v3.4/mb_slave_ui.py

1412 source lines.

MBToolkit v3.4/mb_slave_ui.py — lines 1-40

```python
#!/usr/bin/env python3
"""Slave (simulator) tab: the application answers a real master as Modbus slaves.

The data tables, request handling and serial loop live in ``mb_slave.py``;
this mixin owns the widgets. Each window shows one data table (Coils,
Discrete Inputs, Holding or Input Registers) of one simulated Slave ID; the
windows define which addresses exist. Values start at 0, are edited by
double-click and change when the master writes. Fault simulation is set per
Slave ID. The window grid, value table and Workspace menu are shared with the
Master tab (``mb_table_ui.py``).
"""

import json
import time
import tkinter as tk
from tkinter import messagebox, ttk

from mb_config import (
    DISCOVERY_BAUD_RATES,
    MODBUS_EXCEPTION_CODES,
    SLAVE_FAULT_MAX_DELAY_MS,
    SLAVE_HIGHLIGHT_SECONDS,
    SLAVE_MAX_QTY,
    SLAVE_MAX_WINDOWS,
    SLAVE_WORKSPACE_FORMAT,
    SLAVE_WORKSPACE_VERSION,
    get_slave_workspaces_folder,
)
from mb_master import (
    DEFAULT_FORMAT,
    FORMAT_KEYS_BY_LABEL,
    FORMAT_LABELS,
    FORMAT_SIZES,
    encode_coil_value,
    encode_register_value,
    format_window_values,
    raw_text_for_scaled,
    register_groups,
)
from mb_slave import (
```

MBToolkit v3.4/mb_slave_ui.py — lines 41-80

```
    BIT_TABLES,
    FAULT_EXCEPTION_CODES,
    TABLE_LABELS,
    TABLES,
    SlaveEngine,
    SlaveStore,
    default_fault,
    fault_active,
    TABLE_READ_FC,
    table_choice_label,
    table_label,
    validate_fault,
)
from mb_table_ui import (
    SERIAL_PARITIES,
    SERIAL_STOP_BITS,
    TABLE_MAX_TITLE_LENGTH,
    TABLE_PANE_MAX_ROWS,
    TABLE_PANE_MIN_ROWS,
    TABLE_PANE_MIN_WIDTH,
    TableUiMixin,
    apply_format_to_rows,
    list_workspaces,
    parse_int_field,
    validate_workspace_name,
    validated_formats,
    validated_names,
    validated_scales,
    workspace_path,
)
from mb_theme import theme_colors
from mb_widgets import RefreshButton

SLAVE_WHEEL_TAG = "MBToolkit.SlaveWheel"
TABLE_CHOICES = tuple(table_choice_label(t) for t in TABLES)
TABLE_KEYS_BY_LABEL = {table_choice_label(t): t for t in TABLES}


def fault_parts(fault):
    """(PT template, values) pieces describing an active fault."""
```

MBToolkit v3.4/mb_slave_ui.py — lines 81-120

```
    parts = []
    if fault.get("no_response"):
        parts.append(("sem resposta", {}))
    if fault.get("exception"):
        parts.append(("Exception {code:02X}", {"code": fault["exception"]}))
    if fault.get("bad_crc"):
        parts.append(("CRC errado", {}))
    if fault.get("delay_ms"):
        parts.append(("atraso {ms} ms", {"ms": fault["delay_ms"]}))
    return parts


class SlaveWindow:
    """State of one simulated data table (independent of where it is shown)."""

    def __init__(self, window_id, number, slave=1, table="holding", address=0, qty=10,
                 base=0, fmt=DEFAULT_FORMAT, names=None, title="", formats=None, scales=None):
        self.window_id = window_id
        self.number = number
        # User-chosen window name; empty means the default "Janela N".
        self.title = str(title or "").strip()[:TABLE_MAX_TITLE_LENGTH]
        self.slave = slave
        self.table = table
        self.address = address
        self.qty = qty
        self.base = base
        self.fmt = fmt
        self.names = dict(names or {})  # PDU address -> label
        self.formats = {int(a): f for a, f in (formats or {}).items() if f in FORMAT_SIZES}
        self.scales = {int(a): dict(s) for a, s in (scales or {}).items()}
        self.popped = False
        # Row index -> time.monotonic() until which it is shown as written.
        self.highlight = {}
        self.reset_counters()

    def reset_counters(self):
        self.requests = 0
        self.exceptions = 0
        self.last = ""

```

MBToolkit v3.4/mb_slave_ui.py — lines 121-160

```
    @property
    def bits(self):
        return self.table in BIT_TABLES

    def area(self):
        return (self.slave, self.table, self.address, self.qty)

    def overlap(self, address, qty):
        """Row range of this window covered by [address, address + qty), or None."""
        first = max(self.address, address)
        last = min(self.address + self.qty, address + qty)
        return (first - self.address, last - self.address) if first < last else None

    def groups(self):
        return register_groups(self.qty, self.address, self.fmt, self.formats)

    def group_of_row(self, row):
        for offset, size, fmt in self.groups():
            if offset <= row < offset + size:
                return offset, size, fmt
        return row, 1, None

    def to_json(self):
        return {
            "title": self.title, "slave": self.slave, "table": self.table,
            "address": self.address, "qty": self.qty, "base": self.base,
            "format": self.fmt,
            "names": {str(a): n for a, n in sorted(self.names.items())},
            "formats": {str(a): f for a, f in sorted(self.formats.items())},
            "scales": {str(a): s for a, s in sorted(self.scales.items())},
        }

    @staticmethod
    def validated_kwargs(data):
        """Validate one workspace window entry; raise ValueError if invalid."""
        if not isinstance(data, dict):
            raise ValueError("Janela inválida.")
        slave, table = int(data.get("slave", 1)), data.get("table", "holding")
        address, qty = int(data.get("address", 0)), int(data.get("qty", 10))
        validate_area(slave, table, address, qty)
```

MBToolkit v3.4/mb_slave_ui.py — lines 161-200

```
        fmt = data.get("format", DEFAULT_FORMAT)
        return {
            "title": str(data.get("title", "") or "").strip()[:TABLE_MAX_TITLE_LENGTH],
            "slave": slave, "table": table, "address": address, "qty": qty,
            "base": 1 if data.get("base") == 1 else 0,
            "fmt": fmt if fmt in FORMAT_SIZES else DEFAULT_FORMAT,
            "names": validated_names(data.get("names", {})),
            "formats": validated_formats(data.get("formats", {})),
            "scales": validated_scales(data.get("scales", {})),
        }


def validate_area(slave, table, address, qty):
    """Raise ValueError (PT canonical message) for an invalid window area."""
    if not 1 <= int(slave) <= 247:
        raise ValueError("O Slave ID tem de estar entre 1 e 247.")
    if table not in TABLES:
        raise ValueError("Tabela inválida.")
    if not 0 <= int(address) <= 0xFFFF:
        raise ValueError("O endereço tem de estar entre 0 e 65535.")
    if not 1 <= int(qty) <= SLAVE_MAX_QTY:
        raise ValueError("A quantidade tem de estar entre 1 e 2000.")
    if int(address) + int(qty) > 0x10000:
        raise ValueError("O intervalo de endereços ultrapassa 65535.")


class SlaveMixin(TableUiMixin):
    """Slave (simulator) tab GUI."""

    # -- construction ---------------------------------------------------------
    def build_slave_tab(self, page):
        saved = getattr(self, "ui_settings", {}).get("slave", {})
        if not isinstance(saved, dict):
            saved = {}

        self.slave_windows = {}
        self._slave_panes = {}
        self._slave_next_id = 1
        self.slave_store = SlaveStore()
        # Slave ID -> fault settings (active ones only).
```

MBToolkit v3.4/mb_slave_ui.py — lines 201-240

```
        self.slave_faults = {}
        self.slave_engine = None
        self.slave_serial = None
        self.slave_active = False
        self._slave_reflow_after_id = None
        self._slave_tick_after_id = None
        self.slave_workspace_dirty = False
        self._slave_loading = False
        self.slave_current_workspace = ""
        self._slave_reset_stats()

        page.columnconfigure(0, weight=1)
        page.rowconfigure(2, weight=1)

        connection = ttk.LabelFrame(
            page, text="Ligação — o Slave responde aos pedidos de um master real")
        connection.grid(row=0, column=0, sticky="ew", padx=12, pady=(12, 6))
        row = ttk.Frame(connection)
        row.grid(row=0, column=0, sticky="w", padx=8, pady=(8, 4))

        ttk.Label(row, text="Porta COM").grid(row=0, column=0, padx=(0, 4))
        self.slave_port_var = tk.StringVar()
        self.slave_port_combo = ttk.Combobox(
            row, textvariable=self.slave_port_var, width=13, state="readonly"
        )
        self.slave_port_combo.grid(row=0, column=1)
        self.slave_refresh_btn = RefreshButton(row, command=self.refresh_ports, size=24)
        self.slave_refresh_btn.grid(row=0, column=2, padx=(4, 18))

        def _sync_refresh_square(_event=None):
            height = self.slave_port_combo.winfo_height()
            if height > 1:
                self.slave_refresh_btn.configure(width=height, height=height)

        self.slave_port_combo.bind("<Configure>", _sync_refresh_square, add="+")

        self.slave_baud_var = tk.StringVar(value=str(saved.get("baud", "9600")))
        self.slave_parity_var = tk.StringVar(value=saved.get("parity", "None"))
        self.slave_stop_var = tk.StringVar(value=str(saved.get("stop_bits", "1")))
        fields = (
```

MBToolkit v3.4/mb_slave_ui.py — lines 241-280

```
            ("Baud rate", ttk.Combobox(row, textvariable=self.slave_baud_var, width=8,
                                       state="readonly",
                                       values=[str(b) for b in DISCOVERY_BAUD_RATES])),
            ("Parity", ttk.Combobox(row, textvariable=self.slave_parity_var, width=6,
                                    state="readonly", values=SERIAL_PARITIES)),
            ("Stop bits", ttk.Combobox(row, textvariable=self.slave_stop_var, width=3,
                                       state="readonly", values=SERIAL_STOP_BITS)),
        )
        column = 3
        self._slave_setting_widgets = [self.slave_port_combo]
        for caption, widget in fields:
            ttk.Label(row, text=caption).grid(row=0, column=column, padx=(0, 4))
            widget.grid(row=0, column=column + 1, padx=(0, 16))
            self._slave_setting_widgets.append(widget)
            column += 2
        ttk.Label(row, text="Data bits: 8").grid(row=0, column=column)

        controls = ttk.Frame(connection)
        controls.grid(row=1, column=0, sticky="w", padx=8, pady=(0, 4))
        self.slave_safety_var = tk.BooleanVar(value=False)
        ttk.Checkbutton(
            controls,
            text="Confirmo que nenhum equipamento no barramento usa os Slave IDs simulados.",
            variable=self.slave_safety_var,
        ).grid(row=0, column=0, padx=(0, 18))
        self.slave_connect_btn = ttk.Button(
            controls, text="Ligar", command=self.slave_connect, style="MBS.Primary.TButton"
        )
        self.slave_connect_btn.grid(row=0, column=1, padx=(0, 6))
        self.slave_disconnect_btn = ttk.Button(
            controls, text="Desligar", command=self.slave_disconnect, state="disabled"
        )
        self.slave_disconnect_btn.grid(row=0, column=2, padx=(0, 18))
        self.slave_status_var = tk.StringVar(value=self.tr("Desligado"))
        ttk.Label(controls, textvariable=self.slave_status_var).grid(row=0, column=3)
        self.slave_stats_var = tk.StringVar()
        ttk.Label(connection, textvariable=self.slave_stats_var).grid(
            row=2, column=0, sticky="w", padx=8, pady=(0, 8))
        self._slave_render_stats()

```

MBToolkit v3.4/mb_slave_ui.py — lines 281-320

```
        toolbar = ttk.Frame(page)
        toolbar.grid(row=1, column=0, sticky="ew", padx=12, pady=(0, 6))
        toolbar.columnconfigure(3, weight=1)
        ttk.Button(toolbar, text="Nova janela", command=self.slave_add_window).grid(
            row=0, column=0, padx=(0, 6))
        self.slave_workspace_btn = ttk.Menubutton(toolbar, text="Workspace")
        self.slave_workspace_btn.grid(row=0, column=1, padx=(0, 6))
        self.slave_workspace_menu = tk.Menu(
            self.slave_workspace_btn, tearoff=False,
            postcommand=self._slave_build_workspace_menu,
        )
        self.slave_workspace_btn.configure(menu=self.slave_workspace_menu)
        self.slave_workspace_var = tk.StringVar(value="")
        ttk.Label(toolbar, textvariable=self.slave_workspace_var).grid(
            row=0, column=2, padx=(0, 6))
        ttk.Label(
            toolbar,
            text=("Duplo clique num valor para o alterar, num nome para o editar. "
                  "Clique direito: formato, escala, copiar e base."),
        ).grid(row=0, column=3, sticky="w", padx=(12, 0))

        self.slave_canvas, self.slave_grid, self._slave_grid_item = \
            self._table_build_scroll_grid(page, 2)
        self.slave_canvas.bind("<Configure>", self._slave_schedule_reflow, add="+")
        self.slave_empty_label = ttk.Label(
            self.slave_grid,
            text="Sem janelas. Usa \"Nova janela\" para simular um Slave.",
        )
        for sequence in ("<MouseWheel>", "<Button-4>", "<Button-5>"):
            self.bind_class(SLAVE_WHEEL_TAG, sequence,
                            lambda e: self._table_wheel(self.slave_canvas, e))

        for variable in (self.slave_baud_var, self.slave_parity_var, self.slave_stop_var):
            variable.trace_add("write", lambda *_a: self._schedule_ui_settings_save())

        self._slave_row_menu = tk.Menu(self, tearoff=False)

        # Reopen the workspace used last time (never starts listening).
        last = saved.get("last_workspace", "")
        if last and last in list_workspaces(get_slave_workspaces_folder()):
```

MBToolkit v3.4/mb_slave_ui.py — lines 321-360

```
            self.after(10, lambda: self._slave_restore_workspace(last))
        else:
            self.slave_add_window()
            self.slave_workspace_dirty = False

    def _slave_restore_workspace(self, name):
        if not self.slave_open_workspace(name, quiet=True):
            self.slave_add_window()
            self.slave_workspace_dirty = False

    # -- texts -------------------------------------------------------------
    def _slave_reset_stats(self):
        self.slave_stats = {"requests": 0, "responses": 0, "exceptions": 0,
                            "broadcasts": 0, "others": 0}

    def _slave_render_stats(self):
        stats = self.slave_stats
        self.slave_stats_var.set(self.trf(
            "Pedidos: {requests}   Respostas: {responses}   Exceções: {exceptions}   "
            "Broadcast: {broadcasts}   Para outros IDs: {others}", **stats))

    def _slave_window_name(self, window):
        return window.title or self.trf("Janela {number}", number=window.number)

    def _slave_window_summary(self, window):
        return (f"Slave {window.slave} · FC{TABLE_READ_FC[window.table]:02d} "
                f"{table_label(window.table)} · "
                f"{window.address + window.base} · {window.qty}")

    def _slave_fault_text(self, slave):
        fault = self.slave_faults.get(slave)
        if not fault_active(fault):
            return ""
        text = ", ".join(self.trf(template, **values) for template, values in fault_parts(fault))
        if fault["probability"] < 100:
            text += f" · {fault['probability']}%"
        return self.trf("⚠ Falha simulada: {fault}", fault=text)

    def _slave_render_status(self, window):
        pane = self._slave_panes.get(window.window_id)
```

MBToolkit v3.4/mb_slave_ui.py — lines 361-400

```
        if pane is None:
            return
        try:
            pane["counters_var"].set(self.trf(
                "Pedidos: {requests}   Exceções: {exceptions}",
                requests=window.requests, exceptions=window.exceptions))
            pane["last_var"].set(self.trf("Último: {request}", request=window.last)
                                 if window.last else "")
            pane["name_var"].set(self._slave_window_name(window))
            pane["summary_var"].set(self._slave_window_summary(window))
            fault = self._slave_fault_text(window.slave)
            pane["fault_var"].set(fault)
            if fault:
                pane["fault_label"].grid()
            else:
                pane["fault_label"].grid_remove()
            pane["popout_btn"].configure(text=self.tr("Ancorar" if window.popped else "Destacar"))
            if window.popped and pane.get("toplevel") is not None:
                pane["toplevel"].title(
                    f"MBToolkit — {self._slave_window_name(window)} — "
                    f"{self._slave_window_summary(window)}")
        except tk.TclError:
            pass

    # -- panes ---------------------------------------------------------------
    def _slave_build_pane(self, window, parent):
        frame = ttk.LabelFrame(parent, text="")
        frame.columnconfigure(0, weight=1)
        frame.rowconfigure(5, weight=1)
        pane = {"frame": frame}
        window_id = window.window_id

        header_buttons = self._table_build_header(
            frame, pane,
            rename=lambda: self.slave_rename_window(window_id),
            popout=lambda: self.slave_toggle_popout(window_id),
            close=lambda: self.slave_close_window(window_id),
        )
        # A fault of this Slave ID is shown under the name, never hidden.
        pane["fault_var"] = tk.StringVar()
```

MBToolkit v3.4/mb_slave_ui.py — lines 401-440

```
401         pane["fault_label"] = ttk.Label(pane["header"], textvariable=pane["fault_var"],
402                                         style="MBS.Master.Error.TLabel")
403         pane["fault_label"].grid(row=1, column=0, columnspan=3, sticky="w")
404 
405         definition = ttk.Frame(frame)
406         definition.grid(row=1, column=0, sticky="w", padx=6, pady=(2, 2))
407         pane["slave_var"] = tk.StringVar(value=str(window.slave))
408         pane["table_var"] = tk.StringVar(value=table_choice_label(window.table))
409         pane["address_var"] = tk.StringVar(value=str(window.address + window.base))
410         pane["qty_var"] = tk.StringVar(value=str(window.qty))
411         pane["format_var"] = tk.StringVar(value=FORMAT_LABELS[window.fmt])
412         layout = (
413             ("Slave", ttk.Spinbox(definition, from_=1, to=247, width=4,
414                                   textvariable=pane["slave_var"])),
415             ("Tabela", ttk.Combobox(definition, values=TABLE_CHOICES, width=22,
416                                     state="readonly", textvariable=pane["table_var"])),
417             ("Endereço", ttk.Entry(definition, width=6, textvariable=pane["address_var"])),
418             ("Qtd.", ttk.Entry(definition, width=5, textvariable=pane["qty_var"])),
419         )
420         for index, (caption, widget) in enumerate(layout):
421             ttk.Label(definition, text=caption).grid(row=0, column=index * 2, padx=(0, 3))
422             widget.grid(row=0, column=index * 2 + 1, padx=(0, 10))
423 
424         definition2 = ttk.Frame(frame)
425         definition2.grid(row=2, column=0, sticky="w", padx=6, pady=2)
426         pane["format_combo"] = ttk.Combobox(
427             definition2, values=[FORMAT_LABELS[k] for k in FORMAT_LABELS], width=20,
428             state="readonly", textvariable=pane["format_var"],
429         )
430         ttk.Label(definition2, text="Formato").grid(row=0, column=0, padx=(0, 3))
431         pane["format_combo"].grid(row=0, column=1, padx=(0, 10))
432 
433         buttons = ttk.Frame(frame)
434         buttons.grid(row=3, column=0, sticky="w", padx=6, pady=(2, 2))
435         actions = (
436             ("Aplicar", lambda: self.slave_apply_window(window_id)),
437             ("Duplicar", lambda: self.slave_duplicate_window(window_id)),
438             ("Falhas…", lambda: self.slave_edit_faults(window_id)),
439         )
440         for index, (caption, command) in enumerate(actions):
```

MBToolkit v3.4/mb_slave_ui.py — lines 441-480

```
            button = ttk.Button(buttons, text=caption, command=command, width=10)
            button.grid(row=0, column=index, padx=(0, 3))
            if caption == "Aplicar":
                pane["apply_btn"] = button

        status = ttk.Frame(frame)
        status.grid(row=4, column=0, sticky="ew", padx=6, pady=(2, 2))
        status.columnconfigure(1, weight=1)
        pane["counters_var"] = tk.StringVar()
        pane["last_var"] = tk.StringVar()
        ttk.Label(status, textvariable=pane["counters_var"]).grid(row=0, column=0, sticky="w")
        ttk.Label(status, textvariable=pane["last_var"]).grid(
            row=0, column=1, sticky="w", padx=(14, 0))

        self._table_build_tree(
            frame, pane, 5,
            on_double=lambda e: self._slave_on_double_click(window_id, e),
            on_return=lambda: self._slave_edit_value(window_id),
            on_menu=lambda e: self._slave_show_row_menu(window_id, e),
        )
        for widget in (definition, definition2):
            for child in widget.winfo_children():
                if isinstance(child, (ttk.Entry, ttk.Spinbox)):
                    child.bind("<Return>", lambda _e: self.slave_apply_window(window_id))
        self._table_translate_captions(definition, definition2, buttons, header_buttons)

        self._slave_panes[window_id] = pane
        for name in ("slave_var", "table_var", "address_var", "qty_var", "format_var"):
            pane[name].trace_add("write", lambda *_a: self._slave_update_apply_state(window_id))
        self._slave_fill_rows(window)
        self._slave_update_format_state(window)
        self._slave_render_status(window)
        self._slave_update_apply_state(window_id)
        self._table_install_wheel_tags(frame, SLAVE_WHEEL_TAG)
        return frame

    def _slave_update_format_state(self, window):
        pane = self._slave_panes.get(window.window_id)
        if pane is not None:
            pane["format_combo"].configure(state="disabled" if window.bits else "readonly")
```

MBToolkit v3.4/mb_slave_ui.py — lines 481-520

```
481
482     def _slave_fill_rows(self, window):
483         pane = self._slave_panes.get(window.window_id)
484         if pane is None:
485             return
486         tree = pane["tree"]
487         tree.delete(*tree.get_children())
488         tree.configure(height=max(TABLE_PANE_MIN_ROWS, min(TABLE_PANE_MAX_ROWS, window.qty)))
489         for index in range(window.qty):
490             pdu = window.address + index
491             tree.insert("", "end", iid=f"r{index}", tags=self._table_row_tags(index),
492                         values=(pdu + window.base, window.names.get(pdu, ""), ""))
493         self._slave_render_values(window)
494
495     def _slave_render_values(self, window):
496         pane = self._slave_panes.get(window.window_id)
497         if pane is None:
498             return
499         tree = pane["tree"]
500         values = self.slave_store.read(window.slave, window.table, window.address, window.qty)
501         if window.bits:
502             texts = [str(v) for v in values]
503         else:
504             texts = format_window_values(values, window.address, window.fmt,
505                                          window.formats, window.scales)
506         now = time.monotonic()
507         for index, text in enumerate(texts[:window.qty]):
508             # The zebra tag would win over "written", so a written row has only it.
509             tags = (("written",) if window.highlight.get(index, 0) > now
510                     else self._table_row_tags(index))
511             try:
512                 tree.item(f"r{index}", tags=tags)
513                 tree.set(f"r{index}", "value", text)
514             except tk.TclError:
515                 pass
516
517     def _slave_render_table(self, slave, table):
518         for window in self.slave_windows.values():
519             if (window.slave, window.table) == (slave, table):
520                 self._slave_render_values(window)
```

MBToolkit v3.4/mb_slave_ui.py — lines 521-560

```
521
522     def _slave_schedule_reflow(self, _event=None):
523         if self._slave_reflow_after_id is not None:
524             try:
525                 self.after_cancel(self._slave_reflow_after_id)
526             except tk.TclError:
527                 pass
528         self._slave_reflow_after_id = self.after(80, self._slave_reflow)
529
530     def _slave_reflow(self):
531         self._slave_reflow_after_id = None
532         docked = [w for w in sorted(self.slave_windows.values(), key=lambda w: w.number)
533                   if not w.popped]
534         frames = [self._slave_panes[w.window_id]["frame"] for w in docked
535                   if w.window_id in self._slave_panes]
536         self._table_reflow(self.slave_canvas, self.slave_grid, self._slave_grid_item,
537                            frames, self.slave_empty_label, "slave_panes")
538
539     # -- window actions ------------------------------------------------------
540     def _slave_sync_areas(self):
541         """Tell the store which addresses exist (what the windows show)."""
542         self.slave_store.set_areas(w.area() for w in self.slave_windows.values())
543
544     def slave_add_window(self, kwargs=None):
545         if len(self.slave_windows) >= SLAVE_MAX_WINDOWS:
546             messagebox.showwarning(
547                 self.tr("Slave"),
548                 self.trf("Máximo de {count} janelas.", count=SLAVE_MAX_WINDOWS),
549             )
550             return None
551         window_id = self._slave_next_id
552         self._slave_next_id += 1
553         numbers = {w.number for w in self.slave_windows.values()}
554         number = next(n for n in range(1, SLAVE_MAX_WINDOWS + 2) if n not in numbers)
555         window = SlaveWindow(window_id, number, **(kwargs or {}))
556         self.slave_windows[window_id] = window
557         self._slave_sync_areas()
558         self._slave_build_pane(window, self.slave_grid)
559         self._slave_reflow()
560         self._slave_mark_changed()
```

MBToolkit v3.4/mb_slave_ui.py — lines 561-600

```
561         return window
562 
563     def slave_duplicate_window(self, window_id):
564         window = self.slave_windows.get(window_id)
565         if window is not None:
566             self.slave_add_window(SlaveWindow.validated_kwargs(window.to_json()))
567 
568     def slave_close_window(self, window_id):
569         window = self.slave_windows.pop(window_id, None)
570         if window is None:
571             return
572         self._slave_sync_areas()
573         pane = self._slave_panes.pop(window_id, None)
574         if pane is not None:
575             top = pane.get("toplevel")
576             try:
577                 (top if top is not None else pane["frame"]).destroy()
578             except tk.TclError:
579                 pass
580         self._slave_reflow()
581         if not self.slave_windows and not self._slave_loading:
582             self.slave_current_workspace = ""
583             self.slave_workspace_dirty = False
584             self._slave_update_workspace_label()
585             self._schedule_ui_settings_save()
586         else:
587             self._slave_mark_changed()
588 
589     def _slave_fields_changed(self, window, pane):
590         try:
591             return (
592                 pane["slave_var"].get().strip() != str(window.slave)
593                 or pane["table_var"].get() != table_choice_label(window.table)
594                 or pane["address_var"].get().strip() != str(window.address + window.base)
595                 or pane["qty_var"].get().strip() != str(window.qty)
596                 or FORMAT_KEYS_BY_LABEL.get(pane["format_var"].get()) != window.fmt
597             )
598         except tk.TclError:
599             return False
600 
```

MBToolkit v3.4/mb_slave_ui.py — lines 601-640

```
601     def _slave_update_apply_state(self, window_id):
602         window = self.slave_windows.get(window_id)
603         pane = self._slave_panes.get(window_id)
604         if window is None or pane is None or "apply_btn" not in pane:
605             return
606         style = "MBS.Primary.TButton" if self._slave_fields_changed(window, pane) else "TButton"
607         try:
608             pane["apply_btn"].configure(style=style)
609         except tk.TclError:
610             pass
611 
612     def slave_apply_window(self, window_id):
613         window = self.slave_windows.get(window_id)
614         pane = self._slave_panes.get(window_id)
615         if window is None or pane is None:
616             return False
617         try:
618             base = window.base
619             slave = parse_int_field(pane["slave_var"].get(), 1, 247,
620                                     "O Slave ID tem de estar entre 1 e 247.")
621             table = TABLE_KEYS_BY_LABEL.get(pane["table_var"].get(), window.table)
622             address = parse_int_field(pane["address_var"].get(), base, 0xFFFF + base,
623                                       "O endereço tem de estar entre 0 e 65535.") - base
624             qty = parse_int_field(pane["qty_var"].get(), 1, SLAVE_MAX_QTY,
625                                   "A quantidade tem de estar entre 1 e 2000.")
626             validate_area(slave, table, address, qty)
627         except ValueError as exc:
628             messagebox.showerror(self.tr("Janela inválida"), self.tr(str(exc)))
629             return False
630         fmt = FORMAT_KEYS_BY_LABEL.get(pane["format_var"].get(), DEFAULT_FORMAT)
631         changed_area = (slave, table, address, qty) != window.area()
632         changed = changed_area or fmt != window.fmt
633         window.slave, window.table, window.address, window.qty = slave, table, address, qty
634         window.fmt = fmt
635         if changed_area:
636             window.reset_counters()
637             window.highlight = {}
638             self._slave_sync_areas()
639         self._slave_fill_rows(window)
640         self._slave_update_format_state(window)
```

MBToolkit v3.4/mb_slave_ui.py — lines 641-680

```
641         self._slave_render_status(window)
642         self._slave_update_apply_state(window_id)
643         if changed:
644             self._slave_mark_changed()
645         return True
646 
647     def slave_toggle_popout(self, window_id):
648         window = self.slave_windows.get(window_id)
649         pane = self._slave_panes.pop(window_id, None)
650         if window is None or pane is None:
651             return
652         old_top = pane.get("toplevel")
653         try:
654             pane["frame"].destroy()
655             if old_top is not None:
656                 old_top.destroy()
657         except tk.TclError:
658             pass
659         window.popped = not window.popped
660         if window.popped:
661             top = tk.Toplevel(self)
662             top.geometry("660x420")
663             top.minsize(TABLE_PANE_MIN_WIDTH, 300)
664             top.columnconfigure(0, weight=1)
665             top.rowconfigure(0, weight=1)
666             top.protocol("WM_DELETE_WINDOW", lambda: self.slave_toggle_popout(window_id))
667             frame = self._slave_build_pane(window, top)
668             frame.grid(row=0, column=0, sticky="nsew", padx=8, pady=8)
669             self._slave_panes[window_id]["toplevel"] = top
670             self._table_apply_toplevel_palette(top)
671         else:
672             self._slave_build_pane(window, self.slave_grid)
673         self._slave_render_status(window)
674         self._slave_reflow()
675 
676     def slave_rename_window(self, window_id, name=None):
677         window = self.slave_windows.get(window_id)
678         if window is None:
679             return
680         if name is None:
```

MBToolkit v3.4/mb_slave_ui.py — lines 681-720

```
            name = self._table_ask_text(
                self.tr("Renomear janela"),
                self.tr("Nome da janela (vazio = nome por defeito)"),
                self._slave_window_name(window),
            )
            if name is None:
                return
        name = name.strip()[:TABLE_MAX_TITLE_LENGTH]
        if name == self.trf("Janela {number}", number=window.number):
            name = ""
        window.title = name
        self._slave_render_status(window)
        self._slave_mark_changed()

    # -- values and names ------------------------------------------------------
    def _slave_on_double_click(self, window_id, event):
        pane = self._slave_panes.get(window_id)
        if pane is None:
            return "break"
        tree = pane["tree"]
        row = tree.identify_row(event.y)
        if not row:
            return "break"
        tree.selection_set(row)
        if tree.identify_column(event.x) == "#2":
            self._slave_edit_name(window_id, row)
        else:
            self._slave_edit_value(window_id)
        return "break"

    def _slave_edit_name(self, window_id, row):
        window = self.slave_windows.get(window_id)
        if window is None:
            return
        pdu = window.address + int(row[1:])
        name = self._table_ask_text(
            self.tr("Nome"), self.trf("Nome do endereço {address}", address=pdu + window.base),
            window.names.get(pdu, ""),
        )
        if name is None:
```

MBToolkit v3.4/mb_slave_ui.py — lines 721-760

```
            return
        name = name.strip()
        if name:
            window.names[pdu] = name
        else:
            window.names.pop(pdu, None)
        pane = self._slave_panes.get(window_id)
        if pane is not None:
            pane["tree"].set(row, "name", name)
        self._slave_mark_changed()

    def slave_set_value(self, window, row, text):
        """Store the value typed for *row* (first register of its value).

        Coils / discrete inputs accept 0 or 1; registers use the format (and
        scale) of the value. Raises ValueError (PT canonical message).
        """
        address = window.address + row
        if window.bits:
            values = [encode_coil_value(text)]
        else:
            offset, size, fmt = window.group_of_row(row)
            if fmt is None:
                raise ValueError("Este endereço não tem um valor completo no formato atual.")
            address = window.address + offset
            scale = window.scales.get(address)
            if scale is not None and fmt not in ("hex16", "bin16"):
                text = raw_text_for_scaled(text, scale, fmt)
            values = encode_register_value(text, fmt)
        self.slave_store.write(window.slave, window.table, address, values)
        self._slave_render_table(window.slave, window.table)
        return values

    def _slave_edit_value(self, window_id):
        window = self.slave_windows.get(window_id)
        pane = self._slave_panes.get(window_id)
        if window is None or pane is None:
            return
        selection = pane["tree"].selection()
        if not selection:
```

MBToolkit v3.4/mb_slave_ui.py — lines 761-800

```
            return
        row = int(selection[0][1:])
        if window.bits:
            # A bit is simply toggled.
            current = self.slave_store.read(window.slave, window.table, window.address + row, 1)
            self.slave_set_value(window, row, "0" if current[0] else "1")
            return
        offset, _size, fmt = window.group_of_row(row)
        if fmt is None:
            messagebox.showinfo(self.tr("Valor"), self.tr(
                "Este endereço não tem um valor completo no formato atual."))
            return
        current = pane["tree"].set(f"r{offset}", "value")
        scale = window.scales.get(window.address + offset)
        if scale is not None and fmt in ("hex16", "bin16"):
            scale = None
        if scale is not None and scale.get("unit"):
            current = current.rsplit(" ", 1)[0]

        dialog, body = self._table_dialog(self.tr("Valor"))
        format_text = FORMAT_LABELS[fmt]
        if scale is not None:
            format_text += " · " + self.trf("valores em unidades da escala {unit}",
                                            unit=scale.get("unit") or "")
        ttk.Label(body, text=self.trf("Valor do endereço {address} ({fmt})",
                                      address=window.address + offset + window.base,
                                      fmt=format_text)).grid(row=0, column=0, sticky="w")
        value_var = tk.StringVar(value=current)
        entry = ttk.Entry(body, textvariable=value_var, width=30)
        entry.grid(row=1, column=0, sticky="w", pady=(4, 4))
        message_var = tk.StringVar()
        ttk.Label(body, textvariable=message_var, style="MBS.Master.Error.TLabel").grid(
            row=2, column=0, sticky="w")

        def accept(_event=None):
            try:
                self.slave_set_value(window, offset, value_var.get())
            except ValueError as exc:
                message_var.set(self.tr(str(exc)))
                return
```

MBToolkit v3.4/mb_slave_ui.py — lines 801-840

```
            dialog.destroy()

        buttons = ttk.Frame(body)
        buttons.grid(row=3, column=0, sticky="e", pady=(8, 0))
        ttk.Button(buttons, text=self.tr("Guardar"), command=accept,
                   style="MBS.Primary.TButton").grid(row=0, column=0, padx=(0, 6))
        ttk.Button(buttons, text=self.tr("Cancelar"), command=dialog.destroy).grid(row=0, column=1)
        entry.bind("<Return>", accept)
        entry.bind("<Escape>", lambda _e: dialog.destroy())
        entry.focus_set()
        entry.select_range(0, "end")
        dialog.grab_set()

    # -- right-click menu ------------------------------------------------------
    def _slave_show_row_menu(self, window_id, event):
        window = self.slave_windows.get(window_id)
        pane = self._slave_panes.get(window_id)
        if window is None or pane is None:
            return "break"
        tree = pane["tree"]
        row = tree.identify_row(event.y)
        if row and row not in tree.selection():
            tree.selection_set(row)
        self._table_fill_row_menu(
            self._slave_row_menu, pane, window, not window.bits,
            set_format=lambda fmt: self.slave_set_row_format(window_id, fmt),
            edit_scale=lambda: self.slave_edit_scale(window_id),
            remove_scale=lambda: self.slave_remove_scale(window_id),
            copy=lambda: self._table_copy_rows(self._slave_panes.get(window_id)),
            set_base=lambda base: self.slave_set_base(window_id, base),
        )
        self._slave_apply_menu_palette()
        self._table_popup(self._slave_row_menu, event)
        return "break"

    def _slave_selected_rows(self, window_id):
        return self._table_selected_rows(self._slave_panes.get(window_id))

    def slave_set_row_format(self, window_id, fmt):
        window = self.slave_windows.get(window_id)
```

MBToolkit v3.4/mb_slave_ui.py — lines 841-880

```
        if window is None:
            return False
        try:
            apply_format_to_rows(window, self._slave_selected_rows(window_id), fmt)
        except ValueError as exc:
            messagebox.showwarning(self.tr("Formato"),
                                   self.trf(str(exc), size=FORMAT_SIZES.get(fmt, 1)))
            return False
        self._slave_render_values(window)
        self._slave_mark_changed()
        return True

    def slave_edit_scale(self, window_id):
        window = self.slave_windows.get(window_id)
        if window is None:
            return False
        starts = self._table_selected_group_starts(window, self._slave_panes.get(window_id))
        if not starts:
            messagebox.showwarning(self.tr("Escala"), self.tr("Seleciona pelo menos um endereço."))
            return False
        if not self._table_ask_scale(window, starts):
            return False
        self._slave_render_values(window)
        self._slave_mark_changed()
        return True

    def slave_remove_scale(self, window_id):
        window = self.slave_windows.get(window_id)
        if window is None:
            return False
        removed = False
        for address in self._table_selected_group_starts(window, self._slave_panes.get(window_id)):
            removed = window.scales.pop(address, None) is not None or removed
        if removed:
            self._slave_render_values(window)
            self._slave_mark_changed()
        return removed

    def slave_set_base(self, window_id, base):
        window = self.slave_windows.get(window_id)
```

MBToolkit v3.4/mb_slave_ui.py — lines 881-920

```
        pane = self._slave_panes.get(window_id)
        if window is None or pane is None or window.base == base:
            return
        window.base = base
        pane["address_var"].set(str(window.address + window.base))
        self._slave_fill_rows(window)
        self._slave_render_status(window)
        self._slave_update_apply_state(window_id)
        self._slave_mark_changed()

    # -- fault simulation ------------------------------------------------------
    def slave_set_fault(self, slave, fault):
        """Set (or clear, with None / an inactive fault) the fault of a Slave ID."""
        if fault_active(fault):
            self.slave_faults[slave] = dict(fault)
        else:
            self.slave_faults.pop(slave, None)
        self.slave_store.set_fault(slave, self.slave_faults.get(slave))
        for window in self.slave_windows.values():
            if window.slave == slave:
                self._slave_render_status(window)

    def slave_edit_faults(self, window_id):
        window = self.slave_windows.get(window_id)
        if window is None:
            return
        slave = window.slave
        current = self.slave_faults.get(slave) or default_fault()
        dialog, body = self._table_dialog(self.trf("Falhas do Slave {slave}", slave=slave))
        ttk.Label(body, text=self.trf(
            "Simulação de falhas do Slave {slave} (aplica-se a todas as janelas deste ID).",
            slave=slave)).grid(row=0, column=0, columnspan=2, sticky="w", pady=(0, 8))

        delay_var = tk.StringVar(value=str(current["delay_ms"]))
        silent_var = tk.BooleanVar(value=current["no_response"])
        exception_labels = [self.tr("Nenhuma")] + [
            f"{code:02X} {MODBUS_EXCEPTION_CODES.get(code, ('', ''))[0]}"
            for code in FAULT_EXCEPTION_CODES]
        exception_var = tk.StringVar(value=exception_labels[
            (FAULT_EXCEPTION_CODES.index(current["exception"]) + 1)
```

MBToolkit v3.4/mb_slave_ui.py — lines 921-960

```
            if current["exception"] in FAULT_EXCEPTION_CODES else 0])
        crc_var = tk.BooleanVar(value=current["bad_crc"])
        probability_var = tk.StringVar(value=str(current["probability"]))

        ttk.Label(body, text=self.tr("Atraso na resposta (ms)")).grid(
            row=1, column=0, sticky="w", padx=(0, 8), pady=2)
        ttk.Spinbox(body, from_=0, to=SLAVE_FAULT_MAX_DELAY_MS, increment=50, width=8,
                    textvariable=delay_var).grid(row=1, column=1, sticky="w", pady=2)
        ttk.Checkbutton(body, text=self.tr("Não responder"), variable=silent_var).grid(
            row=2, column=0, columnspan=2, sticky="w", pady=2)
        ttk.Label(body, text=self.tr("Responder com Exception")).grid(
            row=3, column=0, sticky="w", padx=(0, 8), pady=2)
        ttk.Combobox(body, values=exception_labels, textvariable=exception_var, width=28,
                     state="readonly").grid(row=3, column=1, sticky="w", pady=2)
        ttk.Checkbutton(body, text=self.tr("CRC errado nas respostas"), variable=crc_var).grid(
            row=4, column=0, columnspan=2, sticky="w", pady=2)
        ttk.Label(body, text=self.tr("Probabilidade (%)")).grid(
            row=5, column=0, sticky="w", padx=(0, 8), pady=2)
        ttk.Spinbox(body, from_=1, to=100, increment=5, width=8,
                    textvariable=probability_var).grid(row=5, column=1, sticky="w", pady=2)
        ttk.Label(body, text=self.tr(
            "A probabilidade decide, em cada pedido, se a falha acontece.")).grid(
            row=6, column=0, columnspan=2, sticky="w", pady=(6, 2))
        message_var = tk.StringVar()
        ttk.Label(body, textvariable=message_var, style="MBS.Master.Error.TLabel").grid(
            row=7, column=0, columnspan=2, sticky="w")

        def accept(_event=None):
            index = exception_labels.index(exception_var.get())
            try:
                fault = validate_fault({
                    "delay_ms": delay_var.get(), "no_response": silent_var.get(),
                    "exception": FAULT_EXCEPTION_CODES[index - 1] if index else 0,
                    "bad_crc": crc_var.get(), "probability": probability_var.get(),
                })
            except ValueError as exc:
                message_var.set(self.tr(str(exc)))
                return
            self.slave_set_fault(slave, fault)
            self._slave_mark_changed()
```

MBToolkit v3.4/mb_slave_ui.py — lines 961-1000

```
            dialog.destroy()

        def clear():
            self.slave_set_fault(slave, None)
            self._slave_mark_changed()
            dialog.destroy()

        buttons = ttk.Frame(body)
        buttons.grid(row=8, column=0, columnspan=2, sticky="e", pady=(8, 0))
        ttk.Button(buttons, text=self.tr("Aplicar"), command=accept,
                   style="MBS.Primary.TButton").grid(row=0, column=0, padx=(0, 6))
        ttk.Button(buttons, text=self.tr("Sem falhas"), command=clear).grid(
            row=0, column=1, padx=(0, 6))
        ttk.Button(buttons, text=self.tr("Cancelar"), command=dialog.destroy).grid(
            row=0, column=2)
        dialog.bind("<Return>", accept)
        dialog.bind("<Escape>", lambda _e: dialog.destroy())
        dialog.grab_set()

    # -- connection ---------------------------------------------------------
    def slave_connect(self):
        if self.slave_active:
            return
        if self.busy:
            messagebox.showwarning(
                self.tr("Slave"),
                self.tr("Para a captura, a pesquisa ou o Master em curso antes de ligar o Slave."),
            )
            return
        if not self.slave_safety_var.get():
            messagebox.showwarning(
                self.tr("Slave"),
                self.tr("Confirma primeiro que nenhum equipamento no barramento usa os "
                        "Slave IDs simulados."),
            )
            return
        serial, _ = self.get_serial_modules()
        if serial is None:
            messagebox.showerror(self.tr("pyserial em falta"), self.tr("pyserial não está instalado."))
            return
```

MBToolkit v3.4/mb_slave_ui.py — lines 1001-1040

```
        port = self.slave_port_var.get().strip()
        if not port:
            messagebox.showerror(self.tr("Configuração inválida"),
                                 self.tr("Seleciona uma porta COM."))
            return
        for window_id in list(self.slave_windows):
            if not self.slave_apply_window(window_id):
                return
        if not self.slave_windows:
            messagebox.showwarning(self.tr("Slave"),
                                   self.tr("Cria primeiro uma janela com o Slave a simular."))
            return
        baud, parity, stopbits = (int(self.slave_baud_var.get()), self.slave_parity_var.get(),
                                  self.slave_stop_var.get())
        try:
            ser = self.open_discovery_serial_port(serial, port, baud, parity, stopbits)
        except Exception as exc:
            messagebox.showerror(
                self.tr("Slave"),
                self.trf("Não foi possível abrir {port}: {error}",
                         port=port, error=f"{type(exc).__name__}: {exc}"),
            )
            return

        self.slave_serial = ser
        self.slave_engine = SlaveEngine(
            ser, self.event_queue, self.slave_store, baud=baud, parity=parity,
            stopbits=stopbits, serial_module=serial,
            reopen=lambda: self.open_discovery_serial_port(serial, port, baud, parity, stopbits),
        )
        self._slave_reset_stats()
        self._slave_render_stats()
        for window in self.slave_windows.values():
            window.reset_counters()
            self._slave_render_status(window)
        self.slave_active = True
        self.busy = True
        self._slave_lock_other_features(True)
        self.slave_status_var.set(self.tr("A abrir a porta…"))
        self.slave_engine.start()
```

MBToolkit v3.4/mb_slave_ui.py — lines 1041-1080

```
1041
1042    def slave_disconnect(self):
1043        if self.slave_engine is None:
1044            return
1045        self.slave_status_var.set(self.tr("A desligar…"))
1046        self.slave_disconnect_btn.configure(state="disabled")
1047        self.slave_engine.stop()
1048
1049    def _slave_lock_other_features(self, locked):
1050        """While the Slave listens, nothing else may use a serial port."""
1051        state = "disabled" if locked else "normal"
1052        self.slave_connect_btn.configure(state=state)
1053        self.slave_disconnect_btn.configure(state="normal" if locked else "disabled")
1054        for widget in self._slave_setting_widgets:
1055            try:
1056                widget.configure(state="disabled" if locked else "readonly")
1057            except tk.TclError:
1058                pass
1059        self.slave_refresh_btn.configure(state=state)
1060        for name in ("start_btn", "restart_btn", "master_connect_btn"):
1061            widget = getattr(self, name, None)
1062            if widget is not None:
1063                widget.configure(state=state)
1064        self.set_sniffer_refresh_state(state)
1065        self.set_simulation_button_state(state)
1066        self.update_discovery_start_state()
1067
1068    def _slave_finish_session(self, payload):
1069        if not self.slave_active:
1070            return  # already finished (a late slave_stopped event)
1071        engine = self.slave_engine
1072        self.slave_engine = None
1073        if engine is not None:
1074            engine.stop(join_timeout=0.5)
1075            self.slave_serial = engine.ser
1076        if self.slave_serial is not None:
1077            try:
1078                self.slave_serial.close()
1079            except Exception:
1080                pass
```

MBToolkit v3.4/mb_slave_ui.py — lines 1081-1120

```
        self.slave_serial = None
        self.slave_active = False
        self.busy = False
        self._slave_lock_other_features(False)
        self.slave_status_var.set(self.tr("Desligado"))
        error = (payload or {}).get("error")
        if error:
            messagebox.showerror(self.tr("Slave"),
                                 self.trf("Erro na porta série: {error}", error=error))

    # -- worker events ---------------------------------------------------------
    def handle_slave_event(self, event, payload):
        if event == "slave_ready":
            self.slave_status_var.set(self.tr("● A responder"))
        elif event == "slave_reconnecting":
            self.slave_status_var.set(self.trf(
                "A religar a {port}… (porta falhou: {error})",
                port=self.slave_port_var.get(), error=(payload or {}).get("error", ""),
            ))
        elif event == "slave_reconnected":
            engine = self.slave_engine
            if engine is not None:
                self.slave_serial = engine.ser
            self.slave_status_var.set(self.tr("● A responder"))
        elif event == "slave_stopped":
            self._slave_finish_session(payload)
        elif event == "slave_request":
            self._slave_apply_request(payload)

    def slave_request_text(self, info):
        """Short description of a request: "FC03 @ 0 × 10 → Exception 02"."""
        text = f"FC{info['fc']:02d}"
        if info.get("address") is not None:
            text += f" @ {info['address']}"
            if info.get("qty"):
                text += f" × {info['qty']}"
        status = info.get("status")
        if status == "exception":
            code = info.get("exception_code") or 0
            text += f" → Exception {code:02X}"
```

MBToolkit v3.4/mb_slave_ui.py — lines 1121-1160

```
        elif status == "no_response":
            text += " → " + self.tr("sem resposta")
        elif status == "broadcast":
            text += " → broadcast"
        if info.get("fault"):
            text += " ⚠"
        return text

    def _slave_apply_request(self, info):
        stats = self.slave_stats
        status = info.get("status")
        if status == "ignored":
            stats["others"] += 1
            self._slave_render_stats()
            return
        stats["requests"] += 1
        if status == "broadcast":
            stats["broadcasts"] += 1
        elif status == "exception":
            stats["exceptions"] += 1
            stats["responses"] += 1
        elif status == "ok":
            stats["responses"] += 1
        self._slave_render_stats()

        stamp = time.strftime("%H:%M:%S", time.localtime(info.get("time", time.time())))
        text = f"{self.slave_request_text(info)}  {stamp}"
        address, qty = info.get("address"), info.get("qty") or 1
        targets = {(slave, table) for slave, table, _a, _q in info.get("written", [])}
        if status != "broadcast":
            targets.add((info.get("slave"), info.get("table")))
        until = time.monotonic() + SLAVE_HIGHLIGHT_SECONDS
        touched = False
        for window in self.slave_windows.values():
            if (window.slave, window.table) not in targets or address is None:
                continue
            rows = window.overlap(address, qty)
            if rows is None:
                continue
            window.requests += 1
```

MBToolkit v3.4/mb_slave_ui.py — lines 1161-1200

```
            if status == "exception":
                window.exceptions += 1
            window.last = text
            for slave, table, w_address, w_qty in info.get("written", []):
                if (slave, table) == (window.slave, window.table):
                    written = window.overlap(w_address, w_qty)
                    if written is not None:
                        for row in range(*written):
                            window.highlight[row] = until
                        touched = True
            self._slave_render_status(window)
            if info.get("written"):
                self._slave_render_values(window)
        if touched:
            self._slave_schedule_tick()

    def _slave_schedule_tick(self):
        if self._slave_tick_after_id is None:
            self._slave_tick_after_id = self.after(250, self._slave_tick)

    def _slave_tick(self):
        """Remove expired write highlights."""
        self._slave_tick_after_id = None
        now = time.monotonic()
        pending = False
        for window in self.slave_windows.values():
            expired = [row for row, until in window.highlight.items() if until <= now]
            for row in expired:
                del window.highlight[row]
            if expired:
                self._slave_render_values(window)
            pending = pending or bool(window.highlight)
        if pending:
            self._slave_schedule_tick()

    # -- workspace ------------------------------------------------------------
    def slave_workspace_data(self):
        """Configuration only: the register values are not saved."""
        used = {w.slave for w in self.slave_windows.values()}
        return {
```

MBToolkit v3.4/mb_slave_ui.py — lines 1201-1240

```
            "format": SLAVE_WORKSPACE_FORMAT,
            "version": SLAVE_WORKSPACE_VERSION,
            "connection": {
                "port": self.slave_port_var.get(),
                "baud": self.slave_baud_var.get(),
                "parity": self.slave_parity_var.get(),
                "stop_bits": self.slave_stop_var.get(),
            },
            "windows": [w.to_json() for w in sorted(self.slave_windows.values(),
                                                    key=lambda w: w.number)],
            "faults": {str(slave): fault for slave, fault in sorted(self.slave_faults.items())
                       if slave in used},
        }

    def _slave_update_workspace_label(self):
        name = self.slave_current_workspace
        if not name:
            text = ""
        elif self.slave_workspace_dirty:
            text = self.trf("Workspace atual: {name} (alterado)", name=name)
        else:
            text = self.trf("Workspace atual: {name}", name=name)
        self.slave_workspace_var.set(text)

    def _slave_mark_changed(self):
        if getattr(self, "_slave_loading", False):
            return
        if self.slave_current_workspace and not self.slave_workspace_dirty:
            self.slave_workspace_dirty = True
            self._slave_update_workspace_label()

    def _slave_build_workspace_menu(self):
        self._table_fill_workspace_menu(
            self.slave_workspace_menu, list_workspaces(get_slave_workspaces_folder()),
            save=self.slave_save_workspace, open_=self.slave_open_workspace,
            delete=self.slave_delete_workspace,
            folder=lambda: self._table_open_folder(get_slave_workspaces_folder()),
        )
        self._slave_apply_menu_palette()

```

MBToolkit v3.4/mb_slave_ui.py — lines 1241-1280

```
1241    def slave_save_workspace(self, name=None):
1242        if name is None:
1243            name = self._table_ask_text(
1244                self.tr("Guardar workspace"), self.tr("Nome do workspace"),
1245                self.slave_current_workspace or self.tr("Workspace 1"),
1246            )
1247            if name is None:
1248                return False
1249        try:
1250            name = validate_workspace_name(name)
1251        except ValueError as exc:
1252            messagebox.showerror(self.tr("Guardar workspace"), self.tr(str(exc)))
1253            return False
1254        path = workspace_path(name, get_slave_workspaces_folder())
1255        if path.exists() and name != self.slave_current_workspace:
1256            if not messagebox.askyesno(
1257                self.tr("Guardar workspace"),
1258                self.trf("Já existe um workspace chamado \"{name}\". Substituir?", name=name),
1259            ):
1260                return False
1261        try:
1262            with open(path, "w", encoding="utf-8") as handle:
1263                json.dump(self.slave_workspace_data(), handle, indent=2, ensure_ascii=False)
1264        except OSError as exc:
1265            messagebox.showerror(self.tr("Guardar workspace"), str(exc))
1266            return False
1267        self.slave_current_workspace = name
1268        self.slave_workspace_dirty = False
1269        self._slave_update_workspace_label()
1270        self._schedule_ui_settings_save()
1271        return True
1272
1273    def slave_delete_workspace(self, name, confirm=True):
1274        if confirm and not messagebox.askyesno(
1275            self.tr("Apagar workspace"),
1276            self.trf("Apagar o workspace \"{name}\"?", name=name),
1277        ):
1278            return False
1279        try:
1280            workspace_path(name, get_slave_workspaces_folder()).unlink()
```

MBToolkit v3.4/mb_slave_ui.py — lines 1281-1320

```
        except OSError as exc:
            messagebox.showerror(self.tr("Apagar workspace"), str(exc))
            return False
        if name == self.slave_current_workspace:
            self.slave_current_workspace = ""
            self.slave_workspace_dirty = False
            self._slave_update_workspace_label()
            self._schedule_ui_settings_save()
        return True

    def slave_open_workspace(self, name, quiet=False):
        """Replace the windows and faults with a saved workspace; values restart at 0."""
        try:
            with open(workspace_path(name, get_slave_workspaces_folder()),
                      encoding="utf-8") as handle:
                data = json.load(handle)
            if not isinstance(data, dict) or data.get("format") != SLAVE_WORKSPACE_FORMAT:
                raise ValueError(self.tr("O ficheiro não é um workspace do MBToolkit."))
        except (OSError, ValueError) as exc:
            if not quiet:
                messagebox.showerror(self.tr("Abrir workspace"), str(exc))
            return False
        self._slave_loading = True
        try:
            return self._slave_load_workspace_data(name, data)
        finally:
            self._slave_loading = False

    def _slave_load_workspace_data(self, name, data):
        entries, skipped = [], 0
        for item in data.get("windows", [])[:SLAVE_MAX_WINDOWS]:
            try:
                entries.append(SlaveWindow.validated_kwargs(item))
            except (ValueError, TypeError):
                skipped += 1
        for window_id in list(self.slave_windows):
            self.slave_close_window(window_id)
        self.slave_store.clear_values()
        self.slave_store.clear_faults()
        self.slave_faults = {}
```

MBToolkit v3.4/mb_slave_ui.py — lines 1321-1360

```
        connection = data.get("connection", {})
        if isinstance(connection, dict) and not self.slave_active:
            if str(connection.get("baud")) in [str(b) for b in DISCOVERY_BAUD_RATES]:
                self.slave_baud_var.set(str(connection["baud"]))
            if connection.get("parity") in SERIAL_PARITIES:
                self.slave_parity_var.set(connection["parity"])
            if str(connection.get("stop_bits")) in SERIAL_STOP_BITS:
                self.slave_stop_var.set(str(connection["stop_bits"]))
            if connection.get("port") in tuple(self.slave_port_combo["values"]):
                self.slave_port_var.set(connection["port"])
        faults = data.get("faults", {})
        for key, item in (faults.items() if isinstance(faults, dict) else ()):
            try:
                if str(key).isdigit() and 1 <= int(key) <= 247:
                    self.slave_set_fault(int(key), validate_fault(item))
            except ValueError:
                continue
        for kwargs in entries:
            self.slave_add_window(kwargs)
        self.slave_current_workspace = name
        self.slave_workspace_dirty = False
        self._slave_update_workspace_label()
        self._schedule_ui_settings_save()
        if skipped:
            messagebox.showwarning(
                self.tr("Abrir workspace"),
                self.trf("{count} janela(s) inválida(s) foram ignoradas.", count=skipped),
            )
        return True

    def slave_confirm_close(self):
        """Ask to save a changed Slave workspace; False cancels closing."""
        if not (getattr(self, "slave_current_workspace", "")
                and getattr(self, "slave_workspace_dirty", False)):
            return True
        answer = messagebox.askyesnocancel(
            self.tr("Workspace"),
            self.trf("Guardar alterações ao workspace do Slave \"{name}\"?",
                     name=self.slave_current_workspace),
        )
```

MBToolkit v3.4/mb_slave_ui.py — lines 1361-1400

```
        if answer is None:
            return False
        if answer:
            return self.slave_save_workspace(self.slave_current_workspace)
        return True

    # -- language / palette / shutdown ------------------------------------------------
    def slave_apply_language(self):
        if not hasattr(self, "slave_windows"):
            return
        self._slave_update_workspace_label()
        self._slave_render_stats()
        for window in self.slave_windows.values():
            pane = self._slave_panes.get(window.window_id)
            if pane is not None:
                self._table_translate_headings(pane)
                self._slave_render_status(window)
        if not self.slave_active:
            self.slave_status_var.set(self.tr("Desligado"))
        elif self.slave_engine is not None:
            self.slave_status_var.set(self.tr("● A responder"))

    def _slave_apply_menu_palette(self):
        self._table_apply_menus_palette(getattr(self, "slave_workspace_menu", None),
                                        getattr(self, "_slave_row_menu", None))

    def slave_apply_palette(self, dark):
        self._slave_apply_menu_palette()
        canvas = getattr(self, "slave_canvas", None)
        if canvas is not None:
            canvas.configure(background=theme_colors(self, dark)["root"])
        for pane in getattr(self, "_slave_panes", {}).values():
            if pane.get("toplevel") is not None:
                self._table_apply_toplevel_palette(pane["toplevel"])
            self._table_configure_tree_tags(pane["tree"])
        button = getattr(self, "slave_refresh_btn", None)
        if button is not None:
            button.set_dark_mode(dark)

    def slave_shutdown(self):
```

MBToolkit v3.4/mb_slave_ui.py — lines 1401-1412

```
        engine = getattr(self, "slave_engine", None)
        if engine is not None:
            engine.stop(join_timeout=0.8)
            self.slave_serial = engine.ser
        ser = getattr(self, "slave_serial", None)
        if ser is not None:
            try:
                ser.close()
            except Exception:
                pass
        self.slave_engine = None
        self.slave_serial = None
```

MBToolkit v3.4/mb_table_ui.py

599 source lines.

MBToolkit v3.4/mb_table_ui.py — lines 1-40

```python
#!/usr/bin/env python3
"""Shared parts of the Master and Slave window grids.

Both tabs show a grid of windows; each window has a header (name, summary,
Rename / Pop out / close), a value table with per-address formats, scales,
names, copy, address base, drag selection and zebra rows, and the tab has a
Workspace menu. This module holds what is common to both: module helpers
for workspace files and formats, and ``TableUiMixin`` with the widget code.
A "pane" is the dict of widgets of one window; a "window" is its model,
with ``address``, ``qty``, ``base``, ``fmt``, ``formats``, ``scales`` and
``group_of_row()``.
"""

import os
import subprocess
import sys
import tkinter as tk
from tkinter import messagebox, ttk

from mb_master import FORMAT_SIZES, REGISTER_FORMATS, validate_scale
from mb_theme import theme_colors

TABLE_PANE_MIN_WIDTH = 600
TABLE_PANE_MIN_ROWS = 3
TABLE_PANE_MAX_ROWS = 15
# Background of every second table row: one step from the table background.
TABLE_STRIPE_LIGHT = "#f2f4f6"
TABLE_STRIPE_DARK = "#262d34"
# Rows just written by a remote master (Slave tab).
TABLE_WRITTEN_LIGHT = "#fff1b8"
TABLE_WRITTEN_DARK = "#5a4b12"
TABLE_MAX_TITLE_LENGTH = 40
SERIAL_PARITIES = ("None", "Even", "Odd")
SERIAL_STOP_BITS = ("1", "2")
WORKSPACE_SUFFIX = ".mbsw"
# Characters Windows does not allow in file names.
INVALID_FILENAME_CHARS = '<>:"/\\|?*'
RESERVED_FILENAMES = ({"CON", "PRN", "AUX", "NUL"} | {f"COM{i}" for i in range(1, 10)}
                      | {f"LPT{i}" for i in range(1, 10)})
WHEEL_GUARD_TAG = "MBToolkit.MouseWheelGuard"
```

MBToolkit v3.4/mb_table_ui.py — lines 41-80

```
41
42
43 def parse_int_field(text, minimum, maximum, message):
44     try:
45         value = int(str(text).strip())
46     except ValueError:
47         raise ValueError(message) from None
48     if not minimum <= value <= maximum:
49         raise ValueError(message)
50     return value
51
52
53 def validate_workspace_name(name):
54     """Return a cleaned workspace name usable as a Windows file name."""
55     name = str(name or "").strip().rstrip(".")
56     if not name:
57         raise ValueError("O nome do workspace não pode ficar vazio.")
58     if len(name) > 60 or any(ch in INVALID_FILENAME_CHARS for ch in name) \
59             or any(ord(ch) < 32 for ch in name):
60         raise ValueError("O nome do workspace não pode ter os caracteres < > : \" / \\ | ? *.")
61     if name.split(".")[0].upper() in RESERVED_FILENAMES:
62         raise ValueError("Esse nome está reservado pelo Windows.")
63     return name
64
65
66 def workspace_path(name, folder):
67     return folder / f"{validate_workspace_name(name)}{WORKSPACE_SUFFIX}"
68
69
70 def list_workspaces(folder):
71     """Names of the saved workspaces in *folder*, sorted alphabetically."""
72     try:
73         files = folder.glob(f"*{WORKSPACE_SUFFIX}")
74         return sorted((f.stem for f in files if f.is_file()), key=str.casefold)
75     except OSError:
76         return []
77
78
79 def apply_format_to_rows(window, rows, fmt):
80     """Set the per-address format of the selected rows of *window*.
```

MBToolkit v3.4/mb_table_ui.py — lines 81-120

```
81
82      16-bit formats apply to every selected row. Multi-register formats need
83      consecutive rows in a multiple of the format size and are applied in
84      groups (2 registers for 32-bit, 4 for 64-bit). *fmt* None restores the
85      window format. Existing per-address formats overlapping the selection are
86      removed. Raises ValueError (PT canonical message) for an invalid selection.
87      """
88      rows = sorted(set(int(r) for r in rows))
89      if not rows:
90          raise ValueError("Seleciona pelo menos um endereço.")
91      size = FORMAT_SIZES[fmt] if fmt else 1
92      if size > 1:
93          if rows != list(range(rows[0], rows[0] + len(rows))):
94              raise ValueError("Seleciona registos seguidos para este formato.")
95          if len(rows) % size:
96              raise ValueError(
97                  "Este formato ocupa {size} registos: seleciona {size} registos seguidos "
98                  "(ou um múltiplo de {size})."
99              )
100         if rows[-1] >= window.qty:
101             raise ValueError("Seleciona registos seguidos para este formato.")
102
103     covered = {window.address + r for r in rows}
104     for start, existing in list(window.formats.items()):
105         span = set(range(start, start + FORMAT_SIZES[existing]))
106         if span & covered:
107             del window.formats[start]
108     if fmt is not None:
109         for index in range(0, len(rows), size):
110             window.formats[window.address + rows[index]] = fmt
111
112
113 def validated_names(names):
114     return {int(k): str(v) for k, v in (names.items() if isinstance(names, dict) else ())
115             if str(k).isdigit() and str(v).strip()}
116
117
118 def validated_formats(formats):
119     return {int(k): v for k, v in (formats.items() if isinstance(formats, dict) else ())
120             if str(k).isdigit() and int(k) <= 0xFFFF and v in FORMAT_SIZES}
```

MBToolkit v3.4/mb_table_ui.py — lines 121-160

```
121
122
123 def validated_scales(scales):
124     """Scales of a workspace window entry; invalid ones are dropped."""
125     result = {}
126     for key, item in (scales.items() if isinstance(scales, dict) else ()):
127         try:
128             if not str(key).isdigit() or int(key) > 0xFFFF or not isinstance(item, dict):
129                 raise ValueError
130             raw, eng = item.get("raw", []), item.get("eng", [])
131             result[int(key)] = validate_scale(raw[0], raw[1], eng[0], eng[1],
132                                               item.get("decimals", 1), item.get("unit", ""))
133         except (ValueError, TypeError, IndexError):
134             continue
135     return result
136
137
138 class TableUiMixin:
139     """Widget code shared by the Master and Slave window grids."""
140
141     # -- dialogs -------------------------------------------------------------
142     def _table_dialog(self, title):
143         dialog = tk.Toplevel(self)
144         dialog.title(title)
145         dialog.transient(self)
146         dialog.resizable(False, False)
147         self._table_apply_toplevel_palette(dialog)
148         body = ttk.Frame(dialog, padding=12)
149         body.grid(row=0, column=0, sticky="nsew")
150         return dialog, body
151
152     def _table_ask_text(self, title, caption, initial):
153         """Small modal text prompt; returns the text or None when cancelled."""
154         dialog, body = self._table_dialog(title)
155         ttk.Label(body, text=caption).grid(row=0, column=0, columnspan=2, sticky="w")
156         value = tk.StringVar(value=initial)
157         entry = ttk.Entry(body, textvariable=value, width=36)
158         entry.grid(row=1, column=0, columnspan=2, pady=(4, 10), sticky="ew")
159         result = {"text": None}
160
```

MBToolkit v3.4/mb_table_ui.py — lines 161-200

```
        def accept(_event=None):
            result["text"] = value.get()
            dialog.destroy()

        ttk.Button(body, text=self.tr("OK"), command=accept,
                   style="MBS.Primary.TButton").grid(row=2, column=0, sticky="e", padx=(0, 6))
        ttk.Button(body, text=self.tr("Cancelar"), command=dialog.destroy).grid(
            row=2, column=1, sticky="e")
        entry.bind("<Return>", accept)
        entry.bind("<Escape>", lambda _e: dialog.destroy())
        entry.focus_set()
        entry.select_range(0, "end")
        dialog.grab_set()
        self.wait_window(dialog)
        return result["text"]

    def _table_ask_scale(self, window, starts):
        """Ask for a linear scale and set it on the values starting at *starts*."""
        current = window.scales.get(starts[0]) or {
            "raw": [0, 65535], "eng": [0, 100], "decimals": 1, "unit": ""}

        dialog, body = self._table_dialog(self.tr("Escala"))
        ttk.Label(body, text=self.trf("Escala de {count} valor(es) selecionado(s):",
                                      count=len(starts))).grid(
            row=0, column=0, columnspan=4, sticky="w", pady=(0, 6))

        def number_text(value):
            return f"{value:g}"

        variables = {
            "raw_min": tk.StringVar(value=number_text(current["raw"][0])),
            "raw_max": tk.StringVar(value=number_text(current["raw"][1])),
            "eng_min": tk.StringVar(value=number_text(current["eng"][0])),
            "eng_max": tk.StringVar(value=number_text(current["eng"][1])),
            "decimals": tk.StringVar(value=str(current["decimals"])),
            "unit": tk.StringVar(value=current.get("unit", "")),
        }
        layout = (
            (1, 0, "Bruto mín.", "raw_min"), (1, 2, "Bruto máx.", "raw_max"),
            (2, 0, "Eng. mín.", "eng_min"), (2, 2, "Eng. máx.", "eng_max"),
```

MBToolkit v3.4/mb_table_ui.py — lines 201-240

```
            (3, 0, "Casas decimais", "decimals"), (3, 2, "Unidade", "unit"),
        )
        first_entry = None
        for row, column, caption, key in layout:
            ttk.Label(body, text=self.tr(caption)).grid(row=row, column=column, sticky="w",
                                                        padx=(0, 6), pady=2)
            entry = ttk.Entry(body, width=10, textvariable=variables[key])
            entry.grid(row=row, column=column + 1, sticky="w", padx=(0, 14), pady=2)
            first_entry = first_entry or entry
        ttk.Label(body, text=self.tr("Valor = Eng. mín. + (bruto − Bruto mín.) × "
                                     "(Eng. máx. − Eng. mín.) / (Bruto máx. − Bruto mín.)"),
                  style="TLabel").grid(row=4, column=0, columnspan=4, sticky="w", pady=(6, 4))
        message_var = tk.StringVar()
        ttk.Label(body, textvariable=message_var, style="MBS.Master.Error.TLabel").grid(
            row=5, column=0, columnspan=4, sticky="w")
        result = {"ok": False}

        def accept(_event=None):
            try:
                scale = validate_scale(
                    variables["raw_min"].get(), variables["raw_max"].get(),
                    variables["eng_min"].get(), variables["eng_max"].get(),
                    variables["decimals"].get(), variables["unit"].get(),
                )
            except ValueError as exc:
                message_var.set(self.tr(str(exc)))
                return
            for address in starts:
                window.scales[address] = dict(scale)
            result["ok"] = True
            dialog.destroy()

        buttons = ttk.Frame(body)
        buttons.grid(row=6, column=0, columnspan=4, sticky="e", pady=(8, 0))
        ttk.Button(buttons, text=self.tr("Aplicar"), command=accept,
                   style="MBS.Primary.TButton").grid(row=0, column=0, padx=(0, 6))
        ttk.Button(buttons, text=self.tr("Cancelar"), command=dialog.destroy).grid(
            row=0, column=1)
        dialog.bind("<Return>", accept)
        dialog.bind("<Escape>", lambda _e: dialog.destroy())
```

MBToolkit v3.4/mb_table_ui.py — lines 241-280

```
        if first_entry is not None:
            first_entry.focus_set()
        dialog.grab_set()
        self.wait_window(dialog)
        return result["ok"]

    # -- pane parts ---------------------------------------------------------
    def _table_build_header(self, frame, pane, rename, popout, close):
        """Window name + summary on the left; Rename / Pop out / □ on the right.

        Closing a window never depends on the pane being wide.
        """
        header = ttk.Frame(frame)
        header.grid(row=0, column=0, sticky="ew", padx=6, pady=(4, 2))
        header.columnconfigure(1, weight=1)
        pane["name_var"] = tk.StringVar()
        pane["summary_var"] = tk.StringVar()
        name_label = ttk.Label(header, textvariable=pane["name_var"],
                               style="MBS.Master.Title.TLabel")
        name_label.grid(row=0, column=0, sticky="w")
        name_label.bind("<Double-1>", lambda _e: rename())
        ttk.Label(header, textvariable=pane["summary_var"]).grid(
            row=0, column=1, sticky="w", padx=(10, 0))
        header_buttons = ttk.Frame(header)
        header_buttons.grid(row=0, column=2, sticky="e")
        ttk.Button(header_buttons, text="Renomear", width=10, command=rename).grid(
            row=0, column=0, padx=(0, 3))
        pane["popout_btn"] = ttk.Button(header_buttons, text="Destacar", width=10,
                                        command=popout)
        pane["popout_btn"].grid(row=0, column=1, padx=(0, 3))
        ttk.Button(header_buttons, text="×", width=3, command=close).grid(row=0, column=2)
        pane["header"] = header
        return header_buttons

    def _table_build_tree(self, frame, pane, row, on_double, on_return, on_menu):
        """Value table (address / name / value) with a thin 1 px frame."""
        # Classic Tk highlight ring, coloured in _table_configure_tree_tags().
        table = tk.Frame(frame, borderwidth=0, highlightthickness=1)
        table.grid(row=row, column=0, sticky="nsew", padx=6, pady=(2, 6))
        table.columnconfigure(0, weight=1)
```

MBToolkit v3.4/mb_table_ui.py — lines 281-320

```
        table.rowconfigure(0, weight=1)
        pane["table_frame"] = table
        tree = ttk.Treeview(table, columns=("address", "name", "value"), show="headings",
                            height=TABLE_PANE_MAX_ROWS, selectmode="extended")
        for column, caption, width, anchor in (
            ("address", "Endereço", 90, "center"),
            ("name", "Nome", 200, "w"),
            ("value", "Valor", 160, "e"),
        ):
            tree.heading(column, text=self.tr(caption), anchor="center")
            tree.column(column, width=width, minwidth=60, anchor=anchor, stretch=column != "address")
        tree.grid(row=0, column=0, sticky="nsew")
        scrollbar = ttk.Scrollbar(table, orient="vertical", command=tree.yview)
        scrollbar.grid(row=0, column=1, sticky="ns")
        tree.configure(yscrollcommand=scrollbar.set)
        tree.bind("<Double-1>", on_double)
        tree.bind("<Return>", lambda _e: on_return())
        tree.bind("<Button-3>", on_menu)
        tree.bind("<ButtonPress-1>", lambda e: self._table_drag_start(pane, e), add="+")
        tree.bind("<B1-Motion>", lambda e: self._table_drag_move(pane, e))
        tree.bind("<ButtonRelease-1>", lambda _e: self._table_drag_stop(pane), add="+")
        for sequence in ("<Control-c>", "<Control-C>"):
            tree.bind(sequence, lambda _e: self._table_copy_rows(pane) or "break")
        pane["tree"] = tree
        self._table_configure_tree_tags(tree)
        return tree

    def _table_translate_captions(self, *containers):
        """Captions are written in canonical PT; show them in the current language."""
        for widget in containers:
            for child in widget.winfo_children():
                try:
                    if "text" in child.keys() and child.cget("text"):
                        child.configure(text=self.tr(child.cget("text")))
                except tk.TclError:
                    pass

    def _table_translate_headings(self, pane):
        for column, caption in (("address", "Endereço"), ("name", "Nome"), ("value", "Valor")):
            try:
```

MBToolkit v3.4/mb_table_ui.py — lines 321-360

```
321                 pane["tree"].heading(column, text=self.tr(caption))
322             except tk.TclError:
323                 pass
324
325     @staticmethod
326     def _table_row_tags(index):
327         """Alternate rows get a slightly darker background (zebra striping)."""
328         return ("odd",) if index % 2 else ()
329
330     def _table_configure_tree_tags(self, tree):
331         dark = bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get())
332         colors = theme_colors(self, dark)
333         try:
334             tree.tag_configure("stale", foreground=colors["disabled"])
335             tree.tag_configure("odd", background=TABLE_STRIPE_DARK if dark
336                                else TABLE_STRIPE_LIGHT)
337             # A written row carries only this tag (no zebra "odd" tag).
338             tree.tag_configure("written", background=TABLE_WRITTEN_DARK if dark
339                                else TABLE_WRITTEN_LIGHT)
340             frame = tree.master
341             if isinstance(frame, tk.Frame):
342                 frame.configure(highlightbackground=colors["border"],
343                                 highlightcolor=colors["border"],
344                                 background=colors["input"])
345         except tk.TclError:
346             pass
347
348     # -- grid ------------------------------------------------------------------
349     def _table_reflow(self, canvas, grid, grid_item, frames, empty_label, uniform):
350         """Arrange docked pane *frames* in as many equal columns as the width allows."""
351         width = max(1, canvas.winfo_width())
352         canvas.itemconfigure(grid_item, width=width)
353         # Use the real requested pane width (it grows with Windows display
354         # scaling) so columns never become narrower than their contents.
355         pane_width = TABLE_PANE_MIN_WIDTH
356         for frame in frames:
357             pane_width = max(pane_width, frame.winfo_reqwidth() + 8)
358         columns = max(1, width // pane_width)
359         for column in range(8):
360             grid.columnconfigure(column, weight=1 if column < columns else 0,
```

MBToolkit v3.4/mb_table_ui.py — lines 361-400

```
                                uniform=uniform if column < columns else "")
        if not frames:
            empty_label.grid(row=0, column=0, sticky="w", padx=4, pady=8)
        else:
            empty_label.grid_remove()
        for index, frame in enumerate(frames):
            frame.grid(row=index // columns, column=index % columns,
                       sticky="nsew", padx=(0, 8), pady=(0, 8))

    def _table_build_scroll_grid(self, page, row):
        """Scrollable area that holds the window grid; returns (canvas, grid, item)."""
        holder = ttk.Frame(page)
        holder.grid(row=row, column=0, sticky="nsew", padx=12, pady=(0, 12))
        holder.columnconfigure(0, weight=1)
        holder.rowconfigure(0, weight=1)
        canvas = tk.Canvas(holder, highlightthickness=0, borderwidth=0)
        canvas.grid(row=0, column=0, sticky="nsew")
        scrollbar = ttk.Scrollbar(holder, orient="vertical", command=canvas.yview)
        scrollbar.grid(row=0, column=1, sticky="ns")
        canvas.configure(yscrollcommand=scrollbar.set)
        grid = ttk.Frame(canvas)
        item = canvas.create_window((0, 0), window=grid, anchor="nw")
        grid.bind("<Configure>",
                  lambda _e: canvas.configure(scrollregion=canvas.bbox("all")))
        return canvas, grid, item

    def _table_install_wheel_tags(self, root_widget, wheel_tag):
        """Give new widgets the global wheel guard; plain panels scroll the grid."""
        stack = [root_widget]
        while stack:
            widget = stack.pop()
            try:
                stack.extend(widget.winfo_children())
                tags = tuple(t for t in widget.bindtags()
                             if t not in (WHEEL_GUARD_TAG, wheel_tag))
                if isinstance(widget, (ttk.Treeview, ttk.Scrollbar, tk.Toplevel)):
                    widget.bindtags((WHEEL_GUARD_TAG,) + tags)
                else:
                    widget.bindtags((wheel_tag, WHEEL_GUARD_TAG) + tags)
            except tk.TclError:
```

MBToolkit v3.4/mb_table_ui.py — lines 401-440

```
401                 continue
402
403     @staticmethod
404     def _table_wheel(canvas, event):
405         if canvas is None:
406             return "break"
407         try:
408             top, bottom = canvas.yview()
409             if top <= 0.0 and bottom >= 1.0:
410                 return "break"
411             if getattr(event, "num", None) == 4:
412                 step = -1
413             elif getattr(event, "num", None) == 5:
414                 step = 1
415             else:
416                 step = -1 if event.delta > 0 else 1
417             canvas.yview_scroll(step * 3, "units")
418         except tk.TclError:
419             pass
420         return "break"
421
422     # -- drag selection ---------------------------------------------------------
423     # ttk.Treeview has no rubber-band selection: pressing on a row and dragging
424     # selects every row from that row to the one under the pointer, scrolling
425     # when the pointer leaves the table. Ctrl+drag adds the block to the
426     # selection that existed before the press.
427     def _table_drag_start(self, pane, event):
428         tree = pane["tree"]
429         row = tree.identify_row(event.y)
430         if not row or tree.identify_region(event.x, event.y) != "cell":
431             pane["drag"] = None
432             return
433         ctrl = bool(event.state & 0x0004)
434         before = tuple(tree.selection())
435         pane["drag"] = {"anchor": row, "base": before if ctrl else (), "y": event.y,
436                         "after_id": None}
437
438     def _table_drag_move(self, pane, event):
439         drag = pane.get("drag")
440         if not drag:
```

MBToolkit v3.4/mb_table_ui.py — lines 441-480

```
            return None
        drag["y"] = event.y
        self._table_drag_update(pane)
        return "break"

    def _table_drag_update(self, pane):
        drag = pane.get("drag")
        if not drag:
            return
        tree = pane["tree"]
        drag["after_id"] = None
        height = tree.winfo_height()
        y = drag["y"]
        if y < 0 or y > height:
            # Pointer outside the rows: scroll and keep extending the block.
            tree.yview_scroll(-1 if y < 0 else 1, "units")
            y = 1 if y < 0 else height - 2
            drag["after_id"] = self.after(60, lambda: self._table_drag_update(pane))
        row = tree.identify_row(y)
        rows = tree.get_children()
        if not row or drag["anchor"] not in rows:
            return
        first, last = sorted((rows.index(drag["anchor"]), rows.index(row)))
        block = rows[first:last + 1]
        tree.selection_set(tuple(dict.fromkeys(drag["base"] + tuple(block))))

    def _table_drag_stop(self, pane):
        drag = pane.get("drag")
        if drag and drag.get("after_id"):
            try:
                self.after_cancel(drag["after_id"])
            except tk.TclError:
                pass
        pane["drag"] = None

    # -- selection / copy / row menu ------------------------------------------------
    @staticmethod
    def _table_selected_rows(pane):
        if pane is None:
            return []
```

MBToolkit v3.4/mb_table_ui.py — lines 481-520

```
481         return sorted(int(iid[1:]) for iid in pane["tree"].selection() if iid.startswith("r"))
482
483     def _table_selected_group_starts(self, window, pane):
484         """PDU addresses where the values of the selected rows start."""
485         starts = []
486         for row in self._table_selected_rows(pane):
487             offset, _size, fmt = window.group_of_row(row)
488             address = window.address + offset
489             if fmt is not None and address not in starts:
490                 starts.append(address)
491         return starts
492
493     def _table_copy_rows(self, pane):
494         """Copy the selected rows (all rows if none) as tab-separated text."""
495         if pane is None:
496             return
497         tree = pane["tree"]
498         rows = list(tree.selection()) or list(tree.get_children())
499         lines = ["\t".join(str(v) for v in tree.item(iid, "values")) for iid in rows]
500         if lines:
501             self.clipboard_clear()
502             self.clipboard_append("\n".join(lines))
503
504     def _table_fill_row_menu(self, menu, pane, window, registers, *, set_format,
505                              edit_scale, remove_scale, copy, set_base):
506         """Right-click menu: Formato ▸, Repor, Escala…, Remover escala, Copiar, Base ▸."""
507         menu.delete(0, "end")
508         for child in menu.winfo_children():
509             child.destroy()
510         state = "normal" if registers else "disabled"
511         format_menu = tk.Menu(menu, tearoff=False)
512         sub_menus = {2: tk.Menu(format_menu, tearoff=False),
513                      4: tk.Menu(format_menu, tearoff=False)}
514         for key, label, size in REGISTER_FORMATS:
515             target = format_menu if size == 1 else sub_menus[size]
516             target.add_command(label=label, command=lambda k=key: set_format(k))
517         format_menu.add_separator()
518         format_menu.add_cascade(label=self.tr("32 bits (2 registos)"), menu=sub_menus[2])
519         format_menu.add_cascade(label=self.tr("64 bits (4 registos)"), menu=sub_menus[4])
520         menu.add_cascade(label=self.tr("Formato"), menu=format_menu, state=state)
```

MBToolkit v3.4/mb_table_ui.py — lines 521-560

```
        menu.add_command(label=self.tr("Repor formato da janela"),
                         command=lambda: set_format(None), state=state)
        menu.add_command(label=self.tr("Escala…"), command=edit_scale, state=state)
        menu.add_command(label=self.tr("Remover escala"), command=remove_scale, state=state)
        menu.add_separator()
        menu.add_command(label=self.tr("Copiar"), accelerator="Ctrl+C", command=copy)
        menu.add_separator()
        base_menu = tk.Menu(menu, tearoff=False)
        base_var = tk.IntVar(value=window.base)
        pane["menu_base_var"] = base_var  # keep a reference while the menu lives
        base_menu.add_radiobutton(label=self.tr("0 — PDU Address"), variable=base_var, value=0,
                                  command=lambda: set_base(0))
        base_menu.add_radiobutton(label=self.tr("1 — numeração dos manuais"), variable=base_var,
                                  value=1, command=lambda: set_base(1))
        menu.add_cascade(label=self.tr("Base de endereços"), menu=base_menu)

    @staticmethod
    def _table_popup(menu, event):
        try:
            menu.tk_popup(event.x_root, event.y_root)
        finally:
            menu.grab_release()

    # -- workspace menu ------------------------------------------------------
    def _table_fill_workspace_menu(self, menu, names, *, save, open_, delete, folder):
        """Workspace menu with the saved workspaces listed directly in it."""
        menu.delete(0, "end")
        for child in menu.winfo_children():
            child.destroy()
        menu.add_command(label=self.tr("Guardar workspace…"), command=save)
        open_menu = tk.Menu(menu, tearoff=False)
        delete_menu = tk.Menu(menu, tearoff=False)
        for name in names:
            open_menu.add_command(label=name, command=lambda n=name: open_(n))
            delete_menu.add_command(label=name, command=lambda n=name: delete(n))
        if not names:
            for submenu in (open_menu, delete_menu):
                submenu.add_command(label=self.tr("(nenhum workspace guardado)"),
                                    state="disabled")
        menu.add_cascade(label=self.tr("Abrir workspace"), menu=open_menu)
```

MBToolkit v3.4/mb_table_ui.py — lines 561-599

```
        menu.add_cascade(label=self.tr("Apagar workspace"), menu=delete_menu)
        menu.add_separator()
        menu.add_command(label=self.tr("Abrir pasta dos workspaces"), command=folder)

    def _table_open_folder(self, folder):
        try:
            if os.name == "nt":
                os.startfile(str(folder))
            elif sys.platform == "darwin":
                subprocess.Popen(["open", str(folder)])
            else:
                subprocess.Popen(["xdg-open", str(folder)])
        except Exception as exc:
            messagebox.showerror(self.tr("Abrir pasta dos workspaces"), str(exc))

    # -- palette ------------------------------------------------------------------
    def _table_apply_toplevel_palette(self, top):
        dark = bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get())
        try:
            top.configure(background=theme_colors(self, dark)["root"])
        except tk.TclError:
            pass

    def _table_apply_menus_palette(self, *menus):
        """Colour menus and every submenu below them (classic Tk widgets)."""
        dark = bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get())
        colors = theme_colors(self, dark)
        stack = [m for m in menus if m is not None]
        while stack:
            widget = stack.pop()
            try:
                stack.extend(widget.winfo_children())
                widget.configure(
                    background=colors["panel"], foreground=colors["text"],
                    activebackground=colors["select"], activeforeground=colors["select_text"],
                    disabledforeground=colors["disabled"],
                )
            except tk.TclError:
                pass
```

MBToolkit v3.4/mb_i18n.py

660 source lines.

MBToolkit v3.4/mb_i18n.py — lines 1-40

```python
#!/usr/bin/env python3
"""Runtime language support for MBToolkit.

Portuguese (Portugal) is the canonical UI language.  English translations are
kept here so switching language never mutates protocol data or application
logic.
"""

from mb_config import DISCOVERY_IDLE_CHECK_SECONDS, SNIFFER_SPLIT_FRAME_HOLD_MS

LANGUAGE_PORTUGUESE = "Português"
LANGUAGE_ENGLISH = "English"
LANGUAGE_CHOICES = (LANGUAGE_PORTUGUESE, LANGUAGE_ENGLISH)

# Static UI text. Technical terms that are already English are intentionally
# left out: translate_text() returns unknown strings unchanged.
PT_TO_EN = {
    "Linguagem": "Language",
    "Modo escuro": "Dark mode",
    "Configuração": "Configuration",
    "Modo físico": "Physical mode",
    "RS232 dual RX: COM A escuta uma direção; COM B escuta a direção oposta.": "RS232 dual RX: COM A monitors one direction; COM B monitors the opposite direction.",
    "RS232 single RX: uma COM escuta apenas uma direção da ligação RS232.": "RS232 single RX: one COM port monitors only one direction of the RS232 link.",
    "RS485 2-wire: uma COM consegue observar pedidos e respostas no mesmo par diferencial.": "RS485 2-wire: one COM port can monitor requests and responses on the same differential pair.",
    "Estado": "Status",
    "Visualização": "Display",
    "Gap manual (ms)": "Manual gap (ms)",
    "Estado COM": "COM status",
    "Controlos": "Controls",
    "Iniciar Captura": "Start Capture",
    "Parar": "Stop",
    "Reiniciar COM": "Restart COM",
    "Limpar": "Clear",
    "Abrir pasta de logs": "Open logs folder",
    "Simulação": "Simulation",
    "Separar transações": "Separate transactions",
    "Realçar resultados": "Highlight results",
    "Estatísticas da sessão": "Session statistics",
    "Tráfego": "Traffic",
    "Ajuda / Ligações": "Help / Connections",
```

MBToolkit v3.4/mb_i18n.py — lines 41-80

```
    "Filtros": "Filters",
    "Todos": "All",
    "Pesquisa": "Search",
    "Limpar filtros": "Clear filters",
    "Exportar CSV": "Export CSV",
    "Hora ↑": "Time ↑",
    "Hora ↓": "Time ↓",
    "Tipo": "Type",
    "Detalhes": "Details",
    "Copiar Raw Hex": "Copy Raw Hex",
    "Copiar frame descodificado": "Copy decoded frame",
    "Filtrar por este Slave": "Filter by this Slave",
    "Filtrar por este FC": "Filter by this FC",
    "Mostrar apenas esta transação": "Show only this transaction",
    "Limpar filtro de transação": "Clear transaction filter",
    "Tempos de resposta": "Response times",
    "Qualidade": "Quality",
    "Pedidos": "Requests",
    "Pendentes": "Pending",
    "Slave (Filtro) ▾": "Slave (Filter) ▾",
    "Slave (Nenhum) ▾": "Slave (None) ▾",
    "✓ Sucesso": "✓ Success",
    "⚠ Resposta lenta": "⚠ Slow response",
    "● Parado": "● Stopped",
    "● A reiniciar COM": "● Restarting COM",
    "● A capturar": "● Capturing",
    "● A parar": "● Stopping",
    "● A simular": "● Simulating",
    "● Simulação concluída": "● Simulation complete",
    "● Erro na simulação": "● Simulation error",
    "Parado": "Stopped",
    "A iniciar pesquisa…": "Starting scan…",
    "A parar…": "Stopping…",
    "Erro ao iniciar pesquisa": "Error starting scan",
    "Respostas": "Responses",
    "Taxa de pedidos": "Request rate",
    "Slaves ativos": "Active Slaves",
    "Bytes observados": "Observed bytes",
    "Utilização do bus ~": "Bus utilization ~",
    "Média": "Average",
```

MBToolkit v3.4/mb_i18n.py — lines 81-120

```
    "Mínimo": "Minimum",
    "Máximo": "Maximum",
    "Slave mais lento": "Slowest Slave",
    "Erros de CRC": "CRC errors",
    "Bytes não validados": "Unvalidated bytes",
    "Taxa de timeout": "Timeout rate",
    "Exceções": "Exceptions",
    "Exceções / Slave": "Exceptions / Slave",
    "Seleciona uma linha no Tráfego para ver a descodificação estruturada do frame.":
        "Select a row in Traffic to view the structured frame decoding.",
    "Recolher": "Collapse",
    "Expandir": "Expand",
    "Resultado": "Result",
    "Dados": "Data",
    "Sucesso": "Success",
    "Resposta lenta": "Slow response",
    "Pendente / sem resultado": "Pending / no result",
    "Sem classificação": "Unclassified",
    "Verde": "Green",
    "Amarelo": "Yellow",
    "Vermelho": "Red",
    "Laranja": "Orange",
    "Roxo": "Purple",
    "Neutro": "Neutral",
    "Desconhecida/outra": "Unknown/other",
    "Não Utilizada": "Not Used",
    "Não selecionada": "Not selected",
    "Aberta": "Open",
    "Detetada": "Detected",
    "Não detetada": "Not detected",
    "Configuração inválida": "Invalid configuration",
    "pyserial em falta": "pyserial missing",
    "Porta COM": "COM port",
    "A porta série não aceitou bytes para transmissão.": "The serial port did not accept bytes for transmission.",
    "Erro ao abrir porta": "Error opening port",
    "Erro de captura": "Capture error",
    "Erro na simulação": "Simulation error",
    "Erro interno": "Internal error",
    "Abrir pasta": "Open folder",
    "Pesquisa ativa — transmite no barramento": "Active scan — transmits on the bus",
```

MBToolkit v3.4/mb_i18n.py — lines 121-160

```
    "Esta ferramenta atua temporariamente como master Modbus RTU e envia apenas pedidos de leitura. Não a utilizes com outro master ativo no mesmo barramento.": "This tool temporarily acts as a Modbus RTU master and sends read requests only. Do not use it while another master is active on the same bus.",
    "Confirmo que não existe outro master ativo no barramento.": "I confirm that no other master is active on the bus.",
    "FC03 é o modo rápido. Uma resposta normal OU uma Modbus Exception com CRC válido conta como slave encontrado. Ativa FC04 fallback apenas para dispositivos que possam ignorar FC03 sem devolver Exception. Device Identification envia FC43/14 apenas aos slaves encontrados e tenta ler VendorName, ProductCode e Revision. Tentativas repete cada pedido sem resposta. Antes de transmitir, a pesquisa escuta o barramento durante 1 s e não arranca se detetar tráfego.": "FC03 is the fast mode. A normal response OR a CRC-valid Modbus Exception counts as a found Slave. Enable FC04 fallback only for devices that may ignore FC03 without returning an Exception. Device Identification sends FC43/14 only to found Slaves and attempts to read VendorName, ProductCode and Revision. Attempts repeats every unanswered request. Before transmitting, the scan listens to the bus for 1 s and does not start if traffic is detected.",
    "Configuração da pesquisa": "Scan configuration",
    "Porta COM": "COM port",
    "A porta série não aceitou bytes para transmissão.": "The serial port did not accept bytes for transmission.",
    "Data bits: 8 (fixo)": "Data bits: 8 (fixed)",
    "Slave IDs / velocidade": "Slave IDs / speed",
    "De": "From",
    "até": "to",
    "Timeout mín.": "Min. timeout",
    "Iniciar pesquisa": "Start scan",
    "Limpar resultados": "Clear results",
    "Slaves encontrados": "Slaves found",
    "Encontrados: 0": "Found: 0",
    "Resultado": "Result",
    "Pesquisa ativa": "Active scan",
    "Erro ao iniciar pesquisa": "Error starting scan",
    "Erro na Bus Slave Finder": "Bus Slave Finder error",
    "Erro na pesquisa": "Scan error",
    "Nenhum": "None",
    "Nenhum slave detetado": "No Slave detected",
    "Crescente — antigo → recente": "Ascending — oldest → newest",
    "Decrescente — recente → antigo": "Descending — newest → oldest",
    "Todos os ficheiros": "All files",
    "Exportar sessão para CSV": "Export session to CSV",
    # Messages with live values (templates used with SnifferApp.trf()).
    'Seleciona a COM A.': 'Select COM A.',
    'Seleciona também a COM B.': 'Select COM B as well.',
    'COM A e COM B têm de ser diferentes.': 'COM A and COM B must be different.',
    'pyserial não está instalado.': 'pyserial is not installed.',
    'Não foi possível reiniciar a porta COM:\n\n{error}': 'Could not restart the COM port:\n\n{error}',
    'Não foi possível reabrir a porta COM. A captura foi parada.\n\n{error}': 'Could not reopen the COM port. Capture was stopped.\n\n{error}',
    'Pending timeout inválido.': 'Invalid Pending timeout.',
    'Pending timeout deve estar entre {minimum:g} e {maximum:g} segundos.': 'Pending timeout must be between {minimum:g} and {maximum:g} seconds.',
    'Frame gap manual inválido.': 'Invalid manual Frame gap.',
    'pyserial não está instalado.\n\nO build_exe.bat inclui-o automaticamente no EXE.': 'pyserial is not installed.\n\nbuild_exe.bat includes it automatically in the EXE.',
    'Ocorreu um erro ao processar eventos da interface. O processamento continuará.\n\n{error}': 'An error occurred while processing interface events. Processing will continue.\n\n{error}',
    'Slave {slave} — média {average:.1f} ms': 'Slave {slave} — average {average:.1f} ms',
    'Não existem frames na sessão atual para exportar.': 'There are no frames in the current session to export.',
```

MBToolkit v3.4/mb_i18n.py — lines 161-200

```
    'Sessão exportada para:\n\n{path}': 'Session exported to:\n\n{path}',
    'Não foi possível guardar o CSV:\n\n{error}': 'Could not save the CSV:\n\n{error}',
    'Erro ao criar log': 'Error creating log',
    'Foi detetado tráfego, mas não foi possível criar o ficheiro de log em:\n\n{folder}\n\n{error}': 'Traffic was detected, but the log file could not be created at:\n\n{folder}\n\n{error}',
    'Os Slave IDs têm de estar entre 1 e 247.': 'Slave IDs must be between 1 and 247.',
    'O Slave ID inicial não pode ser superior ao final.': 'The starting Slave ID cannot be greater than the ending Slave ID.',
    'O timeout mínimo tem de estar entre {minimum} e {maximum} ms.': 'The minimum timeout must be between {minimum} and {maximum} ms.',
    'Tentativas tem de estar entre {minimum} e {maximum}.': 'Attempts must be between {minimum} and {maximum}.',
    'Tentativas': 'Attempts',
    'Máx. estimado: —': 'Estimated max.: —',
    ' (com {options})': ' (with {options})',
    '{configs} config. × {slaves} slaves — máx. estimado: {duration}': '{configs} config. × {slaves} slaves — estimated max.: {duration}',
    'Encontrados: {count}': 'Found: {count}',
    'pyserial não está instalado. O build_exe.bat inclui-o automaticamente no EXE.': 'pyserial is not installed. build_exe.bat includes it automatically in the EXE.',
    'Confirma primeiro que não existe outro master ativo no barramento.': 'First confirm that no other master is active on the bus.',
    'Seleciona uma porta COM.': 'Select a COM port.',
    'Seleciona pelo menos uma configuração série.': 'Select at least one serial configuration.',
    'Não foi possível iniciar a thread de pesquisa:\n\n{error}': 'Could not start the scan thread:\n\n{error}',
    'A configurar {baud} / {config}…': 'Configuring {baud} / {config}…',
    'Foi detetado tráfego em {port} antes de transmitir ({count} bytes: {raw}). Pode existir outro master ativo no barramento; a pesquisa não foi iniciada.': 'Traffic was detected on {port} before transmitting ({count} bytes: {raw}). Another master may be active on the bus; the scan was not started.',
    'A verificar se o barramento está livre…': 'Checking that the bus is idle…',
    'Não foi possível abrir {port} em {baud} / {config}: {error}': 'Could not open {port} at {baud} / {config}: {error}',
    'Falha ao preparar a receção em {context}: {error}': 'Failed to prepare reception at {context}: {error}',
    'Falha ao transmitir em {context}: {error}': 'Failed to transmit at {context}: {error}',
    'Falha ao receber em {context}: {error}': 'Failed to receive at {context}: {error}',
    'Falha em Device Identification em {context}: {error}': 'Device Identification failed at {context}: {error}',
    'Decorrido {elapsed} — restante ~{remaining}': 'Elapsed {elapsed} — remaining ~{remaining}',
    'Pesquisa parada — {found} encontrados — {elapsed}': 'Scan stopped — {found} found — {elapsed}',
    'Pesquisa concluída — {found} encontrados — {elapsed}': 'Scan complete — {found} found — {elapsed}',
    # Master (poll) tab — v3.2.
    'Ligação — o Master transmite no barramento': 'Connection — the Master transmits on the bus',
    'Timeout resposta (ms)': 'Response timeout (ms)',
    'Atraso entre pedidos (ms)': 'Delay between polls (ms)',
    'Ligar': 'Connect',
    'Desligar': 'Disconnect',
    'Desligado': 'Disconnected',
    '● Ligado': '● Connected',
    'A desligar…': 'Disconnecting…',
    'Nova janela': 'New window',
    'Abrir workspace': 'Open workspace',
```

MBToolkit v3.4/mb_i18n.py — lines 201-240

```
    'Guardar workspace': 'Save workspace',
    'Sem janelas. Usa "Nova janela" para começar a ler um Slave.': 'No windows. Use "New window" to start reading a Slave.',
    'Endereço': 'Address',
    'Qtd.': 'Qty',
    'Formato': 'Format',
    'Aplicar': 'Apply',
    'Pausar': 'Pause',
    'Retomar': 'Resume',
    'Pausada': 'Paused',
    'Ler 1x': 'Read once',
    'Duplicar': 'Duplicate',
    'Destacar': 'Pop out',
    'Ancorar': 'Dock',
    'Fechar': 'Close',
    'Nome': 'Name',
    'Valor': 'Value',
    'Máximo de {count} janelas.': 'Maximum of {count} windows.',
    'Janela inválida': 'Invalid window',
    'Janela inválida.': 'Invalid window.',
    'Liga primeiro o Master.': 'Connect the Master first.',
    'Escrita': 'Write',
    'FC02 e FC04 são só de leitura: abre uma janela FC01 ou FC03 para escrever.': 'FC02 and FC04 are read-only: open an FC01 or FC03 window to write.',
    'Escrever': 'Write value',
    'Slave (0 = broadcast)': 'Slave (0 = broadcast)',
    'Valor(es)': 'Value(s)',
    'Formato: {fmt}. Vários valores (FC15/FC16): separar com ;': 'Format: {fmt}. Several values (FC15/FC16): separate with ;',
    'Enviar': 'Send',
    'Cancelar': 'Cancel',
    'Nome do endereço {address}': 'Name of address {address}',
    'Guardar': 'Save',
    'Para a captura, a pesquisa ou o Slave em curso antes de ligar o Master.': 'Stop the running capture, scan or Slave before connecting the Master.',
    'Não foi possível abrir {port}: {error}': 'Could not open {port}: {error}',
    'Erro na porta série: {error}': 'Serial port error: {error}',
    'Escrita OK: {what}': 'Write OK: {what}',
    'Escrita broadcast enviada: {what}': 'Broadcast write sent: {what}',
    'Resposta inválida (CRC ou formato): {rx}': 'Invalid response (CRC or format): {rx}',
    'O ficheiro não é um workspace do MBToolkit.': 'The file is not an MBToolkit workspace.',
    '{count} janela(s) inválida(s) foram ignoradas.': '{count} invalid window(s) were skipped.',
    'O timeout de resposta tem de estar entre 10 e 10000 ms.': 'The response timeout must be between 10 and 10000 ms.',
    'O atraso entre pedidos tem de estar entre 0 e 10000 ms.': 'The delay between polls must be between 0 and 10000 ms.',
```

MBToolkit v3.4/mb_i18n.py — lines 241-280

```
    'Scan rate fora do intervalo.': 'Scan rate out of range.',
    'Valor vazio.': 'Empty value.',
    'Valor fora do formato selecionado.': 'Value does not fit the selected format.',
    'Uma coil só aceita 0 ou 1.': 'A coil only accepts 0 or 1.',
    'O Slave ID de leitura tem de estar entre 1 e 247.': 'The read Slave ID must be between 1 and 247.',
    'Função de leitura não suportada.': 'Unsupported read function.',
    'O endereço tem de estar entre 0 e 65535.': 'The address must be between 0 and 65535.',
    'Quantidade fora do limite desta função.': 'Quantity outside the limit of this function.',
    'O intervalo de endereços ultrapassa 65535.': 'The address range goes beyond 65535.',
    'O Slave ID de escrita tem de estar entre 0 (broadcast) e 247.': 'The write Slave ID must be between 0 (broadcast) and 247.',
    'FC05 escreve exatamente uma coil.': 'FC05 writes exactly one coil.',
    'FC06 escreve exatamente um registo.': 'FC06 writes exactly one register.',
    'Função de escrita não suportada.': 'Unsupported write function.',
    'FC05/FC06 escrevem um único valor; usa FC15/FC16 para vários.': 'FC05/FC06 write a single value; use FC15/FC16 for several.',
    # Master windows and workspaces — v3.2.
    'Janela {number}': 'Window {number}',
    'Renomear': 'Rename',
    'Renomear janela': 'Rename window',
    'Nome da janela (vazio = nome por defeito)': 'Window name (empty = default name)',
    'Workspace atual: {name}': 'Current workspace: {name}',
    'Guardar workspace…': 'Save workspace…',
    'Apagar workspace': 'Delete workspace',
    '(nenhum workspace guardado)': '(no saved workspace)',
    'Abrir pasta dos workspaces': 'Open workspaces folder',
    'Nome do workspace': 'Workspace name',
    'Já existe um workspace chamado "{name}". Substituir?': 'A workspace named "{name}" already exists. Replace it?',
    'Apagar o workspace "{name}"?': 'Delete workspace "{name}"?',
    'O nome do workspace não pode ficar vazio.': 'The workspace name cannot be empty.',
    'O nome do workspace não pode ter os caracteres < > : " / \\ | ? *.': 'The workspace name cannot contain the characters < > : " / \\ | ? *.',
    'Esse nome está reservado pelo Windows.': 'That name is reserved by Windows.',
    # Master per-address formats, row menu, workspace changes — v3.2.
    'Duplo clique num valor para escrever, num nome para o editar. Clique direito: formato, copiar e base.': 'Double-click a value to write it, a name to edit it. Right-click: f
ormat, copy and base.',
    'Workspace atual: {name} (alterado)': 'Current workspace: {name} (modified)',
    'Seleciona pelo menos um endereço.': 'Select at least one address.',
    'Seleciona registos seguidos para este formato.': 'Select consecutive registers for this format.',
    'Este formato ocupa {size} registos: seleciona {size} registos seguidos (ou um múltiplo de {size}).': 'This format uses {size} registers: select {size} consecutive registers
 (or a multiple of {size}).',
    'Este endereço não tem um valor completo no formato atual.': 'This address does not hold a complete value in the current format.',
    '32 bits (2 registos)': '32-bit (2 registers)',
    '64 bits (4 registos)': '64-bit (4 registers)',
    'Repor formato da janela': 'Reset to window format',
```

MBToolkit v3.4/mb_i18n.py — lines 281-320

```
    'Copiar': 'Copy',
    '1 — numeração dos manuais': '1 — manual numbering',
    'Base de endereços': 'Address base',
    'Guardar alterações ao workspace "{name}"?': 'Save changes to workspace "{name}"?',
    # Master scale, open in Master, reconnection — v3.2.
    'Escala…': 'Scale…',
    'Remover escala': 'Remove scale',
    'Escala': 'Scale',
    'Escala de {count} valor(es) selecionado(s):': 'Scale of {count} selected value(s):',
    'Bruto mín.': 'Raw min.',
    'Bruto máx.': 'Raw max.',
    'Eng. mín.': 'Eng. min.',
    'Eng. máx.': 'Eng. max.',
    'Casas decimais': 'Decimals',
    'Unidade': 'Unit',
    'Valor = Eng. mín. + (bruto − Bruto mín.) × (Eng. máx. − Eng. mín.) / (Bruto máx. − Bruto mín.)': 'Value = Eng. min. + (raw − Raw min.) × (Eng. max. − Eng. min.) / (Raw max. − Raw min.)',
    'valores em unidades da escala {unit}': 'values in scale units {unit}',
    'A escala tem de ter números válidos.': 'The scale needs valid numbers.',
    'O bruto mínimo e o bruto máximo têm de ser diferentes.': 'Raw min. and raw max. must be different.',
    'As casas decimais têm de estar entre 0 e 6.': 'Decimals must be between 0 and 6.',
    'Esta escala não permite escrever: os valores de engenharia são iguais.': 'This scale does not allow writing: the engineering values are equal.',
    'Abrir no Master': 'Open in Master',
    'Esta trama não tem um pedido de leitura FC01–FC04 com endereço conhecido.': 'This frame has no FC01–FC04 read request with a known address.',
    'A religar a {port}… (porta falhou: {error})': 'Reconnecting to {port}… (port failed: {error})',
    # Slave (simulator) tab — v3.4.
    'Ligação — o Slave responde aos pedidos de um master real': 'Connection — the Slave answers the requests of a real master',
    'Confirmo que nenhum equipamento no barramento usa os Slave IDs simulados.': 'I confirm that no device on the bus uses the simulated Slave IDs.',
    'Confirma primeiro que nenhum equipamento no barramento usa os Slave IDs simulados.': 'First confirm that no device on the bus uses the simulated Slave IDs.',
    'Para a captura, a pesquisa ou o Master em curso antes de ligar o Slave.': 'Stop the running capture, scan or Master before connecting the Slave.',
    'Cria primeiro uma janela com o Slave a simular.': 'First create a window with the Slave to simulate.',
    'Sem janelas. Usa "Nova janela" para simular um Slave.': 'No windows. Use "New window" to simulate a Slave.',
    'Duplo clique num valor para o alterar, num nome para o editar. Clique direito: formato, escala, copiar e base.': 'Double-click a value to change it, a name to edit it. Right-click: format, scale, copy and base.',
    'Pedidos: {requests}   Respostas: {responses}   Exceções: {exceptions}   Broadcast: {broadcasts}   Para outros IDs: {others}': 'Requests: {requests}   Responses: {responses}   Exceptions: {exceptions}   Broadcast: {broadcasts}   For other IDs: {others}',
    'Pedidos: {requests}   Exceções: {exceptions}': 'Requests: {requests}   Exceptions: {exceptions}',
    'Último: {request}': 'Last: {request}',
    '● A responder': '● Answering',
    'A abrir a porta…': 'Opening the port…',
    'Tabela': 'Table',
    'Tabela inválida.': 'Invalid table.',
    'O Slave ID tem de estar entre 1 e 247.': 'The Slave ID must be between 1 and 247.',
```

MBToolkit v3.4/mb_i18n.py — lines 321-360

```
    'A quantidade tem de estar entre 1 e 2000.': 'The quantity must be between 1 and 2000.',
    'Valor do endereço {address} ({fmt})': 'Value of address {address} ({fmt})',
    'Guardar alterações ao workspace do Slave "{name}"?': 'Save changes to the Slave workspace "{name}"?',
    'Falhas…': 'Faults…',
    'Falhas do Slave {slave}': 'Slave {slave} faults',
    'Simulação de falhas do Slave {slave} (aplica-se a todas as janelas deste ID).': 'Fault simulation for Slave {slave} (applies to every window of this ID).',
    'Atraso na resposta (ms)': 'Response delay (ms)',
    'Não responder': 'Do not respond',
    'Responder com Exception': 'Respond with Exception',
    'Nenhuma': 'None',
    'CRC errado nas respostas': 'Wrong CRC in responses',
    'Probabilidade (%)': 'Probability (%)',
    'A probabilidade decide, em cada pedido, se a falha acontece.': 'The probability decides, for each request, whether the fault happens.',
    'Sem falhas': 'No faults',
    '⚠ Falha simulada: {fault}': '⚠ Simulated fault: {fault}',
    'sem resposta': 'no response',
    'CRC errado': 'wrong CRC',
    'atraso {ms} ms': 'delay {ms} ms',
    'Falha inválida.': 'Invalid fault.',
    'A falha tem de ter números válidos.': 'The fault needs valid numbers.',
    'O atraso tem de estar entre 0 e 10000 ms.': 'The delay must be between 0 and 10000 ms.',
    'A probabilidade tem de estar entre 1 e 100 %.': 'The probability must be between 1 and 100 %.',
    'Código de exceção não suportado.': 'Unsupported exception code.',
}

EN_TO_PT = {english: portuguese for portuguese, english in PT_TO_EN.items()}
# "None" is also a canonical technical serial-parity value in the Portuguese UI.
# Do not reverse-translate it to "Nenhum" when restoring Portuguese; menu entries
# that need "Nenhum" are rendered from their Portuguese canonical source text.
EN_TO_PT.pop("None", None)

EXCEPTION_DESCRIPTION_EN = {
    0x01: "The requested Modbus function is not supported or permitted by the device.",
    0x02: "The requested address or address range is not valid for this device.",
    0x03: "A value, quantity or request field is invalid for this function.",
    0x04: "The device encountered an internal error while attempting to execute the request.",
    0x05: "The request was accepted but requires more time to be processed.",
    0x06: "The device is busy and cannot process the request at this time.",
    0x08: "A parity error was detected while accessing the device memory.",
    0x0A: "The gateway could not establish a path to the target device.",
```

MBToolkit v3.4/mb_i18n.py — lines 361-400

```
    0x0B: "The gateway did not receive a response from the target device.",
}


def normalize_language(value):
    return value if value in LANGUAGE_CHOICES else LANGUAGE_PORTUGUESE


def canonical_ui_text(text):
    """Return the canonical PT-PT form of a known static UI string."""
    value = str(text)
    return EN_TO_PT.get(value, value)


def translate_text(text, language):
    """Translate a known static string; leave technical/unknown text intact."""
    canonical = canonical_ui_text(text)
    if normalize_language(language) == LANGUAGE_ENGLISH:
        return PT_TO_EN.get(canonical, canonical)
    return canonical


# Human-readable fragments embedded in protocol Details / Inspector data.
DETAIL_FRAGMENTS_PT_TO_EN = (
    ("Registos (PDU Address=valor):", "Registers (PDU Address=value):"),
    ("Bits (PDU Address=valor):", "Bits (PDU Address=value):"),
    ("Registos=", "Registers="),
)


def translate_details(text, language):
    """Translate only human-readable fragments embedded in protocol details."""
    value = str(text or "")
    if normalize_language(language) == LANGUAGE_ENGLISH:
        for portuguese, english in DETAIL_FRAGMENTS_PT_TO_EN:
            value = value.replace(portuguese, english)
    else:
        for portuguese, english in DETAIL_FRAGMENTS_PT_TO_EN:
            value = value.replace(english, portuguese)
    return value
```

MBToolkit v3.4/mb_i18n.py — lines 401-440

```
401
402
403 def exception_description(code, portuguese_text, language):
404     if normalize_language(language) == LANGUAGE_ENGLISH:
405         return EXCEPTION_DESCRIPTION_EN.get(code, portuguese_text)
406     return portuguese_text
407
408
409 # English Help content. Tags match the tags configured by mb_gui.py.
410 def english_help_blocks(max_ui_frames_text, max_raw_lines_text,
411                         min_pending, max_pending, debug_sim=False):
412     blocks = [
413         ("title", "MBToolkit — Help and connections\n"),
414         (None, "MBToolkit is designed for Modbus RTU communications diagnostics. The Sniffer tab is passive: it opens the COM port(s) for reception and does not transmit request
    s. Bus Slave Finder is an active scan tool and must only be used when no other master is active on the same bus.\n"),
415         ("heading", "Quick start — Sniffer\n"),
416         ("bullet", "1. Select the correct physical mode: RS485 2-wire, RS232 single RX or RS232 dual RX.\n"),
417         ("bullet", "2. Select the COM port(s) and serial parameters used by the communication being observed.\n"),
418         ("bullet", "3. Keep Frame gap set to Auto unless you have a specific reason to change it.\n"),
419         ("bullet", "4. Click “Start Capture” and confirm activity through BUS ● RX.\n"),
420         ("bullet", "5. Use Traffic for quick diagnostics and Raw Hex / Log to confirm the actual bytes.\n"),
421         ("bullet", "6. Select a frame to inspect its detailed analysis in Frame Inspector.\n"),
422
423         ("heading", "1. RS485 2-wire\n"),
424         (None, "In this mode, one COM port observes requests and responses on the same differential pair. The sniffer adapter must be connected in parallel and must not act as a
     master.\n"),
425         ("subheading", "Typical connection:\n"),
426         ("mono", "        Device A              Device B\n            D+ ---------------- D+\n            D- ---------------- D-\n              \\              /\n
       \\             /\n                D+          D-\n                USB-RS485 sniffer\n\n"),
427         ("bullet", "• Connect sniffer D+ in parallel with bus D+.\n"),
428         ("bullet", "• Connect sniffer D− in parallel with bus D−.\n"),
429         ("bullet", "• If a C/COM/0 V reference is available, connect it only when you know it is the correct RS485 interface reference.\n"),
430         ("warning", "• Do not add a third 120 Ω termination resistor if the bus is already terminated at both ends.\n"),
431         ("bullet", "• If the USB-RS485 adapter has selectable termination or bias, leave them disabled during passive capture unless they are intentionally required by the insta
    llation.\n"),
432         ("bullet", "• A/B naming is not universal: different manufacturers may use opposite polarities. Prefer D+/D− and the equipment documentation.\n"),
433
434         ("heading", "2. RS232 single RX\n"),
435         (None, "Observes only one RS232 direction. RS232 is full-duplex: each direction uses an independent TX line, so one RX input cannot observe both directions simultaneousl
    y.\n"),
436         ("mono", "        Device A                    Device B\n            TX ----------------------------> RX\n            \\\n              \\----> USB-RS232 sniffer R
    X (COM A)\n\n            GND ------------------------- GND\n              \\--------------------------> sniffer GND\n\n        Sniffer TX: DO NOT CONNECT\n\n"),
437         ("bullet", "• Connect sniffer RX to the TX line you want to observe.\n"),
438         ("bullet", "• Connect the common GND as well.\n"),
439         ("warning", "• Leave the TX pin of the adapter used as a sniffer physically disconnected.\n"),
440
```

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```
441         ("heading", "3. RS232 dual RX\n"),
442         (None, "Allows both RS232 directions to be observed at the same time. Two independent RX channels are required, normally two USB-RS232 adapters.\n"),
443         ("mono", "        Device A                      Device B\n             TX ----------------------------> RX\n              \\----> sniffer RX COM A   (A → B)\n\n
        RX <----------------------------- TX\n                                    \\----> sniffer RX COM B   (B → A)\n\n             GND ---------------------------- GND\n
            \\---------------------------- COM A GND\n                \\---------------------------- COM B GND\n\n        TX on both sniffers: DO NOT CONNECT\n\n"),
444         ("bullet", "• COM A is shown as A→B and COM B as B→A.\n"),
445         ("bullet", "• Both COM ports must use the same serial parameters as the observed link.\n"),
446
447         ("heading", "RS232 is not TTL\n"),
448         ("warning", "Do not connect RS232 signals directly to 3.3 V or 5 V USB-TTL/UART interfaces. Use a USB-RS232 adapter with a transceiver suitable for RS232 voltage levels.
    \n"),
449
450         ("heading", "Serial communication parameters\n"),
451         (None, "Baud rate, Data bits, Parity and Stop bits must match the actual communication. Incorrect parameters may produce apparently random bytes, frames with invalid CRC
    , or no recognizable frames at all.\n"),
452         ("bullet", "• Baud rate — speed in bit/s, for example 9600 or 19200.\n"),
453         ("bullet", "• Data bits — normally 8 for Modbus RTU.\n"),
454         ("bullet", "• Parity — None, Even, Odd, Mark or Space, according to the installation.\n"),
455         ("bullet", "• Stop bits — 1, 1.5 or 2, according to the observed configuration.\n"),
456
457         ("heading", "Frame gap\n"),
458         (None, "In Auto mode, MBToolkit uses 3.5 character times up to and including 19200 bit/s. Above 19200 bit/s it uses the fixed 1.750 ms t3.5 recommended for Modbus RTU. M
    anual mode allows another millisecond value to be used when diagnosing equipment or drivers with particular behavior.\n"),
459         (None, "USB adapters and Windows may group bytes and introduce latency. MBToolkit timestamps are therefore useful for sequence diagnostics and relative timing, but they
    do not replace a logic analyzer when high-precision timing measurements are required.\n"),
460         (None, f"When a burst ends with the start of a known frame that is still incomplete, MBToolkit waits up to {SNIFFER_SPLIT_FRAME_HOLD_MS:g} ms more before closing it. A f
    rame split in two by the USB adapter therefore does not appear as RAW. Frames merged into one burst keep the arrival time of their own block of bytes.\n"),
461
462         ("heading", "Pending timeout\n"),
463         (None, "Defines how long a CRC-valid REQUEST may remain waiting for a matchable RESPONSE/EXCEPTION. When it expires, the original REQUEST is marked as timeout and the Ti
    meout/Timeout rate counters increase.\n"),
464         (None, f"Allowed range: {min_pending:g} to {max_pending:g} s. The value should be greater than the normal maximum response time of the installation to avoid false timeou
    ts.\n"),
465         (None, "Broadcast requests (Slave 0) never have a response by definition: they are not kept pending, never count as a timeout and are excluded from the Timeout rate.\n")
    ,
466
467         ("heading", "How to read the Traffic table\n"),
468         ("bullet", "• REQUEST — Modbus RTU request recognized by the parser.\n"),
469         ("bullet", "• RESPONSE — recognized Modbus RTU response that is not an EXCEPTION.\n"),
470         ("bullet", "• FC05/FC06/FC08/FC22 — the request and the response have the same format. The frame is a RESPONSE when a pending request from the same Slave/FC exists (for
    FC05/06/22 the bytes must be identical; in RS232 dual RX it must come from the opposite channel). Otherwise it is a REQUEST.\n"),
471         ("bullet", "• FC01/FC02 responses show the bits in address order, grouped per byte. In Frame Inspector, matched read responses (FC01–FC04) show the PDU Address of every
    value.\n"),
472         ("bullet", "• EXCEPTION — Modbus response with an exception Function Code (bit 7 set). Details shows the code and normalized description.\n"),
473         ("bullet", "• CRC OK — received CRC matches the calculated CRC.\n"),
474         ("bullet", "• CRC ERROR — the structure resembles a supported frame, but the received CRC does not match the calculated CRC.\n"),
475         ("bullet", "• RAW/UNSYNC — bytes discarded during resynchronization before a structurally recognized, CRC-valid frame is found later in the same burst.\n"),
476         ("bullet", "• RAW/UNPARSED — remaining bytes that could not be interpreted or resynchronized as a valid Modbus frame. This is the state normally shown for a noise block
    with no later valid frame in the same burst.\n"),
477         (None, "RAW/UNSYNC and RAW/UNPARSED are different conditions and both may appear in the Type column. Neither represents a valid Modbus response.\n"),
478
479         ("heading", "Traffic table columns\n"),
480         ("bullet", "• Time — timestamp for the start of the processed burst/frame.\n"),
```

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```
481         ("bullet", "• Δt — interval since the previous observed frame.\n"),
482         ("bullet", "• Resp. — time between a REQUEST and its matched RESPONSE/EXCEPTION. Blank when no valid pair exists.\n"),
483         ("bullet", "• COM — capture channel: BUS in RS485, A→B/B→A in RS232 dual RX and SIM in development mode. It does not necessarily represent the Windows COM number.\n"),
484         ("bullet", "• Type — REQUEST, RESPONSE, EXCEPTION or a RAW state.\n"),
485         ("bullet", "• Slave — Modbus Unit/Slave ID when identifiable.\n"),
486         ("bullet", "• FC — hexadecimal Function Code observed in the frame.\n"),
487         ("bullet", "• Details — address, quantity, data or operation description.\n"),
488         ("bullet", "• CRC — OK, ERROR or ? when a complete Modbus frame is not available.\n"),
489 
490         ("heading", "REQUEST/RESPONSE pairing\n"),
491         (None, "Modbus RTU has no Transaction ID. MBToolkit pairs frames by (Slave ID, Function Code) in FIFO order. This is appropriate for normal RTU traffic, but can become ambiguous when requests using the same Slave and FC overlap before their corresponding responses arrive.\n"),
492         (None, "An EXCEPTION uses its base Function Code for pairing. For example, response 0x83 is associated with a pending FC03 REQUEST when a compatible request exists.\n"),
493 
494         ("heading", "PDU Address, 1-based and Qty\n"),
495         (None, "PDU Address is the address encoded in the Modbus request. 1-based is simply PDU Address + 1 and helps compare with manuals that number the first register/coil as 1. Qty is the requested quantity.\n"),
496         (None, "These fields belong to the REQUEST. A RESPONSE does not automatically repeat Address or Qty in the protocol, so MBToolkit does not invent those fields on the response row.\n"),
497 
498         ("heading", "Modbus exception codes\n"),
499         ("bullet", "• 0x01 — Illegal Function: function not supported or not permitted.\n"),
500         ("bullet", "• 0x02 — Illegal Data Address: invalid address/address range.\n"),
501         ("bullet", "• 0x03 — Illegal Data Value: invalid value, quantity or field.\n"),
502         ("bullet", "• 0x04 — Server Device Failure: device-side execution failure.\n"),
503         ("bullet", "• 0x05 — Acknowledge.\n"),
504         ("bullet", "• 0x06 — Server Device Busy.\n"),
505         ("bullet", "• 0x08 — Memory Parity Error.\n"),
506         ("bullet", "• 0x0A — Gateway Path Unavailable.\n"),
507         ("bullet", "• 0x0B — Gateway Target Device Failed to Respond.\n"),
508 
509         ("heading", "Advanced filters\n"),
510         (None, "The bar at the top of Traffic can combine several filters without changing the original session data:\n"),
511         ("bullet", "• Type — All, Requests, Responses, Exceptions, CRC errors, Timeouts or RAW.\n"),
512         ("bullet", "• FC (hex) — Function Code, with or without the 0x prefix.\n"),
513         ("bullet", "• Resp. > — only completed responses above the specified time in ms.\n"),
514         ("bullet", "• Search — searches Details, Raw Hex, Slave, FC, Type and COM/channel.\n"),
515         (None, "The Slave filter remains available in the column header and combines with the advanced filters. “Clear filters” resets all display filters. Filters do not change statistics or the capture log.\n"),
516 
517         ("heading", "Highlight results\n"),
518         (None, "This option is disabled by default to keep the table neutral. When enabled, results are differentiated by color: a completed normal transaction with a CRC-valid RESPONSE is green; CRC error, timeout, Exception, RAW and slow response retain their respective diagnostic colors.\n"),
519         (None, "In a successfully completed normal transaction, both the REQUEST and its RESPONSE are green. A still-pending REQUEST remains neutral. A slow response keeps the slow-response color even if valid so that the timing result is not hidden.\n"),
520 
```

MBToolkit v3.4/mb_i18n.py — lines 521-560

```
        ("heading", "Frame Inspector\n"),
        (None, "Select a Traffic row to analyze the frame without manually interpreting every byte. Frame Inspector shows Slave, Type, Function, response time, Result, PDU Address, 1-based address, Qty, ByteCount, CRC, decoded Data and Raw Hex.\n"),
        (None, "The CRC field shows both received and calculated values for a complete frame. Result uses the same diagnostic classification as Traffic, independently of whether Highlight results is enabled.\n"),

        ("heading", "Traffic context menu\n"),
        (None, "Right-click a Traffic row to:\n"),
        ("bullet", "• Copy Raw Hex.\n"),
        ("bullet", "• Copy the decoded frame.\n"),
        ("bullet", "• Filter by that frame's Slave.\n"),
        ("bullet", "• Filter by that frame's Function Code.\n"),
        ("bullet", "• Show only the matching REQUEST/RESPONSE transaction.\n"),
        ("bullet", "• Clear the temporary transaction filter.\n"),

        ("heading", "Bus Health\n"),
        (None, "Bus Health summarizes session quality and performance without replacing the frame table.\n"),
        ("bullet", "• Requests / Responses — session counters.\n"),
        ("bullet", "• Request rate — average number of observed requests per second.\n"),
        ("bullet", "• Active Slaves — valid Modbus IDs observed in the session.\n"),
        ("bullet", "• Observed bytes — sum of bytes delivered to the parser.\n"),
        ("bullet", "• Bus utilization ~ — estimate based on observed bytes, baud rate, data bits, parity and stop bits. It is approximate and depends on the capture point.\n"),
        ("bullet", "• Average / Minimum / Maximum — paired response times.\n"),
        ("bullet", "• P95 — 95% of measured responses were at or below this value.\n"),
        ("bullet", "• Slowest Slave — Slave with the highest average response time among observed samples.\n"),
        ("bullet", "• Unvalidated bytes — percentage of observed bytes belonging to invalid-CRC frames or RAW blocks. By itself this does not prove electrical noise; incorrect serial parameters, incomplete capture or parser-unsupported functions can also produce RAW.\n"),
        ("bullet", "• Timeout rate — percentage of REQUESTs that expired without a matched response.\n"),
        ("bullet", "• Exceptions / Slave — distribution of Exception responses by Slave ID.\n"),

        ("heading", "BUS RX indicator\n"),
        (None, "BUS ● RX pulses briefly when a batch of frames reaches the interface. At rest, BUS ○ is displayed. It is only a visual confirmation of received activity; it does not indicate electrical quality or guarantee that frames are valid.\n"),

        ("heading", "Raw Hex / Log\n"),
        (None, "Shows captured bytes in hexadecimal with context information. Long lines wrap to the available width. The view follows the same display filters used in Traffic; the on-disk TXT file, when present, keeps the complete capture and is not truncated by filters.\n"),

        ("heading", "Export CSV\n"),
        (None, "Export CSV saves the frames currently retained in the UI history using Excel-compatible UTF-8. It includes Time, Δt, Response time, Channel, Type, Slave, FC, Details, CRC, Raw, PDU Address, 1-based, Qty, ByteCount, Timeout and anomaly class.\n"),
        (None, f"The interface retains approximately the latest {max_ui_frames_text} frames. In longer sessions, the CSV therefore represents the history still retained by the GUI, while the TXT file from a real capture remains the complete on-disk record.\n"),

        ("heading", "Capture logs\n"),
        (None, "The TXT log is created only after real traffic arrives. Starting and stopping without receiving frames does not create an empty file.\n"),
        (None, "Logs folder:\n"),
```

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```
561         ("mono", "        <application folder>\\MBToolkit Logs\n"),
562         (None, "Filters, sorting, Highlight results and Frame Inspector are display-only features and do not change frames written to the log.\n"),
563 
564         ("heading", "Visible history limits\n"),
565         (None, f"To keep the application responsive, the interface retains approximately {max_ui_frames_text} frames and {max_raw_lines_text} Raw Hex lines. Pruning is performed
     in blocks. The capture TXT log, when active, is independent of this limit.\n"),
566 
567         ("heading", "COM status and ports\n"),
568         (None, "COM status distinguishes a port opened by MBToolkit, a port merely detected by Windows and a port that is no longer detected. Status is refreshed periodically.\n
    "),
569         (None, "Restart COM closes and reopens the port(s) used by Sniffer. This restarts the serial-port handle inside MBToolkit; it does not restart the Windows USB/COM driver
    .\n"),
570         (None, "The ↻ buttons beside the COM selectors refresh the available port enumeration in Sniffer and Bus Slave Finder.\n"),
571 
572         ("heading", "Bus Slave Finder — active scan\n"),
573         (None, "Bus Slave Finder transmits Modbus RTU requests, so it is not passive. Before starting, confirm that no other master is active on the same bus.\n"),
574         (None, "Select COM, baud rates, parities, stop bits and the Slave ID range. Data bits is fixed at 8. The scan tries FC03 Address 0 Qty 1; either a normal response or a C
    RC-valid Modbus Exception confirms that the Slave exists.\n"),
575         (None, "FC04 fallback is optional and increases the maximum number of attempts. The minimum timeout is adapted to baud rate. A write timeout means transmission through t
    he port/driver failed and must not be interpreted simply as “slave did not respond”.\n"),
576         (None, f"Attempts (1 to 3) repeats every unanswered request before moving on to the next Slave. Before the first transmission, the scan listens to the bus for {DISCOVERY
    _IDLE_CHECK_SECONDS:g} s; if any byte is received, it does not transmit and reports that another master may be active.\n"),
577         (None, "Device Identification (FC43/14) is optional. When enabled, it is sent only after a Slave has been found and attempts to read Basic Device Identification: VendorN
    ame, ProductCode and MajorMinorRevision. An Exception or no response to FC43/14 does not invalidate Slave discovery.\n"),
578         (None, "Finder and Sniffer share the global activity lock so the same application cannot actively scan and capture at the same time.\n"),
579 
580         ("heading", "Master — read and write as a Modbus RTU master\n"),
581         (None, f"The Master tab transmits requests: use it only when no other master is active on the bus. When connecting, the application listens to the bus for {DISCOVERY_IDL
    E_CHECK_SECONDS:g} s and does not transmit if traffic is detected. While the Master is connected, Sniffer, Bus Slave Finder and Slave are locked.\n"),
582         (None, "Each window reads one data area of a Slave: Slave ID, function (FC01 Coils, FC02 Discrete Inputs, FC03 Holding Registers or FC04 Input Registers), address, quant
    ity and scan rate. Changes take effect only after “Apply”, which is highlighted in blue while some fields are not applied yet. The table shows up to 15 rows; beyond that it scro
    lls.\n"),
583         (None, "All windows share the same COM port: Modbus RTU allows only one request at a time, so windows are served in turn according to their scan rate. Writes and “Read o
    nce” go ahead of cyclic reads. The “Delay between polls” (at least t3.5) is always left between two requests.\n"),
584         (None, "Formats: signed and unsigned 16-bit, hex and binary; signed and unsigned 32-bit and float, in byte orders ABCD (big-endian), CDAB (word swap), BADC (byte swap) a
    nd DCBA (little-endian); signed and unsigned 64-bit and double, in orders ABCDEFGH, GHEFCDAB, BADCFEHG and HGFEDCBA. 32 and 64-bit values use 2 and 4 registers and appear on the
     row of the first one.\n"),
585         (None, "The “Format” field is the window's default format. Right-clicking one or more addresses (drag with the mouse, or Ctrl/Shift, to select several) sets the format o
    f those addresses only: 32-bit formats need 2 consecutive registers and 64-bit formats 4 (or a multiple, applied in groups). “Reset to window format” returns to the default. The
     same menu has “Copy” (also Ctrl+C), which copies addresses, names and values to paste into Excel, and “Address base”: 0 shows the PDU Address; 1 numbers from 1, as in many manu
    als.\n"),
586         (None, "When a read fails (Timeout, Exception or invalid response), the values turn grey until a correct read arrives; they also turn grey when disconnecting.\n"),
587         (None, "Scale: in the right-click menu, “Scale…” converts the raw value into engineering units through two points (Raw min./max. → Eng. min./max.), with decimals and a u
    nit (for example 2000–10000 → −180–180 °). When writing, the value is typed in units and converted back to raw. “Remove scale” returns to the raw value. Scales are saved in the
    workspace.\n"),
588         (None, "“Open in Master”: in the Sniffer Traffic table (right-click an FC01–FC04 request or its matched response) and in the Bus Slave Finder results (right-click or dou
    ble-click) it creates a Master window with the same Slave, function, address and quantity, and copies the serial parameters to the Master connection (when it is not connected).\
    n"),
589         (None, "If the COM port fails while connected (for example, the USB adapter was unplugged), the Master does not stop: it tries to reopen the port every 2 s and resumes r
    eading when the port is back. To stop, click “Disconnect”.\n"),
590         (None, "Writing: double-click a value in an FC01 or FC03 window. Coils with FC05 or FC15, registers with FC06 or FC16; several values separated by “;” are written to con
    secutive addresses with FC15/FC16. Slave 0 sends a broadcast, with no response. After an accepted write, the window is read again.\n"),
591         (None, "Each window shows Tx, errors, timeouts, the last response time and the last error (Timeout, Exception or invalid response). In each window header, “Rename” chang
    es the window name (also by double-clicking the name), “Pop out” opens it outside the tab (“Dock” or closing that window returns it) and “×” closes the window.\n"),
592         (None, "The “Workspace” menu saves the connection, every window, names and formats under a name of your choice, and lists the saved workspaces so they can be opened or d
    eleted directly. The files are kept in %APPDATA%\\MBToolkit\\Workspaces\\Master, next to the preferences.\n"),
593         (None, "The last workspace opened or saved is reopened when the program starts, without connecting to the bus. While there are unsaved changes, “(modified)” appears besi
```

```
        (None, "The last workspace opened or saved is reopened when the program starts, without connecting to the bus. While there are unsaved changes, \"(modified)\" appears beside its name; closing the program in that state asks whether to save. Windows never saved in a workspace close without asking, and closing every window leaves no workspace open.\n"),
594
595         ("heading", "Slave — Modbus RTU device simulator\n"),
596         (None, "The Slave tab makes the application behave as one or more Modbus RTU devices: it listens on the COM port and answers the requests of a real master (a PLC, a SCADA or another Modbus Poll). It is used to test the master without the device. It answers FC01–FC06, FC15 and FC16.\n"),
597         (None, "Each window is one table of a simulated Slave: Slave ID, table (Coils, Discrete Inputs, Holding Registers or Input Registers), start address and quantity (up to 2000). Several Slave IDs can exist, each with several windows. Only the addresses defined in the windows exist: a request outside them gets Exception 02 (Illegal Data Address), an unsupported function Exception 01 and an invalid quantity or value Exception 03, as on a real device. Requests for other Slave IDs are ignored.\n"),
598         (None, "Values start at 0. Double-clicking a value changes it (Coils and Discrete Inputs toggle between 0 and 1); changes apply at once, without going through the bus. When the master writes (FC05/06/15/16), the written registers are highlighted in yellow for 1.5 s. Windows over the same addresses show the same values. Broadcast (Slave 0) writes are executed on every simulated Slave that has those addresses, without a response.\n"),
599         (None, "Per-address formats, scale, names, Copy, Address base, mouse selection, Rename, Pop out and × work as in the Master.\n"),
600         (None, "Fault simulation: “Faults…” sets faults for the window's Slave ID, which apply to every window of that ID: response delay (ms), do not respond, always respond with a chosen Exception (01, 02, 03, 04 or 06), wrong CRC in responses and the probability (%) of the fault happening on each request. An active fault appears in red in the header of that Slave's windows. “No faults” switches them off.\n"),
```

MBToolkit v3.4/mb_i18n.py - lines 601-640

MBToolkit v3.4/mb_i18n.py — lines 601-640

```
        (None, "The connection bar shows the requests received, responses, exceptions, broadcasts and requests for other IDs; each window shows the requests and exceptions on its addresses and the last request. The RS485 adapter echo is ignored. Before answering, the Slave waits the t3.5 silence; Windows and the USB adapter usually add a few ms, so a master with a very short timeout may get timeouts.\n"),
        (None, "Before connecting, confirm that no real device on the bus uses the same Slave IDs: if two devices answer the same request, the responses are corrupted. The Slave never runs at the same time as the Sniffer, the Bus Slave Finder or the Master. If the port fails, it tries to reopen it every 2 s, like the Master.\n"),
        (None, "The Slave “Workspace” menu saves the configuration only: connection, windows, names, formats, scales and faults. Register values are not saved and return to 0 when a workspace is opened. The files are kept in %APPDATA%\\MBToolkit\\Workspaces\\Slave; the last Slave workspace is also reopened at start-up.\n"),

        ("heading", "Saved preferences\n"),
        (None, "User preferences are stored in %APPDATA%\\MBToolkit\\settings.json. They include Light/Dark, Language, last tab, serial parameters, Frame gap, Pending timeout, Auto-scroll, Separate transactions, Highlight results, Bus Slave Finder options and Master and Slave connection parameters. Master and Slave windows are saved in workspaces, in the %APPDATA%\\MBToolkit\\Workspaces folder.\n"),
        (None, "COM ports are not saved because they are specific to each computer. The Bus Slave Finder safety confirmation is not saved either.\n"),
        (None, "Up to v3.2 the program was called MBSniffer. The first time MBToolkit starts, it copies the preferences and workspaces from %APPDATA%\\MBSniffer to %APPDATA%\\MBToolkit; the old folder is not deleted. New logs go to “MBToolkit Logs”; old ones stay in “MBSniffer Logs”.\n"),

        ("heading", "Diagnostic best practices\n"),
        ("bullet", "• Confirm wiring, polarity and serial parameters first.\n"),
        ("bullet", "• On RS485, keep the sniffer stub short and avoid changing the existing termination.\n"),
        ("bullet", "• Make sure no other program has the same COM port open.\n"),
        ("bullet", "• Treat CRC ERROR and RAW as indicators of framing, noise, incorrect parameters or incomplete capture.\n"),
        ("bullet", "• Treat Exceptions as valid protocol responses: they are not equivalent to CRC errors.\n"),
        ("bullet", "• Compare response times with normal equipment behavior, not only with a generic threshold.\n"),
        ("bullet", "• Keep the TXT log when you need later analysis of the complete capture.\n"),
    ]
    if debug_sim:
        blocks.extend([
            ("heading", "Simulation mode\n"),
            (None, "This development mode appears only when debug_sim = 1 in MBToolkit.py. It does not open serial ports and never creates TXT logs.\n"),
            (None, "Each run uses the same diagnostic scenario types: one normal response, one slow response, one Modbus Exception, one CRC error, one timeout and one RAW/UNPARSED block. This makes it possible to compare the same information directly with and without visual highlighting.\n"),
            (None, "Highlight results does not change Simulation data. When disabled, all rows use normal table colors; when enabled, the completed normal transaction is green and Exception, CRC error, timeout, slow response and RAW rows use their respective diagnostic colors.\n"),
            (None, "Frames in this scenario use the SIM channel so they are not confused with real physical capture.\n"),
        ])
    blocks.extend([
        ("heading", "Important limitation\n"),
        (None, "MBToolkit interprets bytes, CRC, Function Code, addresses, quantities, pairing and timing observed by the operating system. It does not directly measure electrical levels, ringing, reflections, bias, common-mode, rise/fall times or analog signal integrity. Those problems still require an oscilloscope or an analyzer suitable for the physical interface.\n"),
    ])
    return blocks


def translate_runtime_text(text, language):
    """Translate status strings that contain live values."""
    import re

    value = str(text or "")
    target_en = normalize_language(language) == LANGUAGE_ENGLISH

```

MBToolkit v3.4/mb_i18n.py — lines 641-660

```
    # Normalize either language to canonical PT where practical, then render.
    patterns = [
        (r"^Encontrados: (\d+)$", r"Found: \1", target_en),
        (r"^Found: (\d+)$", r"Encontrados: \1", not target_en),
        (r"^Decorrido (.+) — restante ~(.+)$", r"Elapsed \1 — remaining ~\2", target_en),
        (r"^Elapsed (.+) — remaining ~(.+)$", r"Decorrido \1 — restante ~\2", not target_en),
        (r"^A configurar (.+)…$", r"Configuring \1…", target_en),
        (r"^Configuring (.+)…$", r"A configurar \1…", not target_en),
        (r"^Pesquisa parada — (\d+) encontrados — (.+)$", r"Scan stopped — \1 found — \2", target_en),
        (r"^Scan stopped — (\d+) found — (.+)$", r"Pesquisa parada — \1 encontrados — \2", not target_en),
        (r"^Pesquisa concluída — (\d+) encontrados — (.+)$", r"Scan complete — \1 found — \2", target_en),
        (r"^Scan complete — (\d+) found — (.+)$", r"Pesquisa concluída — \1 encontrados — \2", not target_en),
        (r"^(\d+) config\. × (\d+) slaves — máx\. estimado: (.+)$", r"\1 config. × \2 slaves — estimated max.: \3", target_en),
        (r"^(\d+) config\. × (\d+) slaves — estimated max\.: (.+)$", r"\1 config. × \2 slaves — máx. estimado: \3", not target_en),
    ]
    for pattern, replacement, enabled in patterns:
        if enabled and re.match(pattern, value):
            return re.sub(pattern, replacement, value)

    return translate_text(value, language)
```

MBToolkit v3.4/mb_theme.py

692 source lines.

MBToolkit v3.4/mb_theme.py — lines 1-40

```
#!/usr/bin/env python3
"""MBToolkit fixed-geometry Light/Dark theme.

The application uses one ttk construction for its entire lifetime. Light/Dark
switching changes colours only; it never changes ttk theme engine, layouts,
padding, borders, margins, widget metrics or row heights.
"""

import tkinter as tk
from tkinter import ttk


BOOTSTRAP_ACTIVE = False
THEME_ENGINE = "clam"

LIGHT = {
    "root": "#f3f4f6",
    "panel": "#f3f4f6",
    "panel_alt": "#ffffff",
    "input": "#ffffff",
    "input_readonly": "#f5f6f7",
    "text": "#1f2328",
    "muted": "#59636e",
    "disabled": "#8b949e",
    "border": "#c3c9d0",
    "divider": "#d6dbe0",
    "button": "#ffffff",
    "button_hover": "#f0f5fa",
    "button_pressed": "#e3edf6",
    "accent": "#2f80d1",
    "accent_hover": "#3b8edb",
    "accent_pressed": "#246eB8",
    "select": "#2f80d1",
    "select_text": "#ffffff",
    "scroll_trough": "#e8ebee",
    "scroll_thumb": "#b9c2ca",
    "scroll_thumb_hover": "#929faa",
}

DARK = {
```

MBToolkit v3.4/mb_theme.py — lines 41-80

```
    "root": "#171a1f",
    "panel": "#171a1f",
    "panel_alt": "#1f242a",
    "input": "#20262c",
    "input_readonly": "#252c33",
    "text": "#e8eaed",
    "muted": "#aab2ba",
    "disabled": "#707982",
    "border": "#46525d",
    "divider": "#46525d",
    "button": "#252c33",
    "button_hover": "#303942",
    "button_pressed": "#1f252b",
    "accent": "#2f8fd8",
    "accent_hover": "#43a2e8",
    "accent_pressed": "#287bb8",
    "select": "#285f80",
    "select_text": "#ffffff",
    "scroll_trough": "#1d2329",
    "scroll_thumb": "#65717c",
    "scroll_thumb_hover": "#7d8994",
}

_CHECK_IMAGES = None
_CHECK_ELEMENT = "MBS.Check.indicator"
_FIXED_GEOMETRY_READY = False
_INTERP_ID = None


def theme_colors(root=None, dark=False):
    return dict(DARK if dark else LIGHT)


def _ensure_engine(style):
    """Select/configure the fixed engine once per Tk interpreter."""
    global _FIXED_GEOMETRY_READY, _CHECK_IMAGES, _INTERP_ID

    # A normal MBToolkit process owns one Tk root. Handling a replaced Tk
    # interpreter as well makes tests/relaunches in the same Python process
    # safe and prevents stale PhotoImage command references.
```

MBToolkit v3.4/mb_theme.py — lines 81-120

```
    current_interp = id(style.master.tk)
    if _INTERP_ID != current_interp:
        _INTERP_ID = current_interp
        _FIXED_GEOMETRY_READY = False
        _CHECK_IMAGES = None

    try:
        if THEME_ENGINE in style.theme_names() and style.theme_use() != THEME_ENGINE:
            style.theme_use(THEME_ENGINE)
    except Exception:
        pass

    if not _FIXED_GEOMETRY_READY:
        _configure_fixed_geometry(style)
        _FIXED_GEOMETRY_READY = True


def _configure_fixed_geometry(style):
    """
    Define every geometry/state rule once.

    Nothing in this function is called as part of a Light/Dark toggle after
    startup. This is what guarantees that the page construction cannot move.
    """

    # ------------------------------------------------------------------
    # LabelFrames: titles above the rectangular border in both modes.
    # ------------------------------------------------------------------
    style.configure(
        "TLabelframe",
        borderwidth=1,
        relief="solid",
        padding=0,
        labeloutside=True,
        labelmargins=(0, 0, 0, 4),
    )
    style.configure(
        "TLabelframe.Label",
        borderwidth=0,
        padding=0,
```

MBToolkit v3.4/mb_theme.py — lines 121-160

```
        font=("Segoe UI", 9, "bold"),
    )

    # ------------------------------------------------------------------
    # Notebook/tabs: fixed Vista-like top geometry.
    # ------------------------------------------------------------------
    style.configure(
        "TNotebook",
        borderwidth=1,
        padding=0,
        tabmargins=(2, 2, 2, 0),
    )
    style.configure(
        "TNotebook.Tab",
        borderwidth=1,
        padding=(12, 6),
        font=("Segoe UI", 9),
    )
    style.map(
        "TNotebook.Tab",
        expand=[("selected", (2, 2, 2, 2))],
        padding=[("selected", (12, 6))],
        relief=[
            ("selected", "flat"),
            ("active", "flat"),
            ("pressed", "flat"),
        ],
    )

    # ------------------------------------------------------------------
    # Buttons.
    # ------------------------------------------------------------------
    style.configure(
        "TButton",
        borderwidth=1,
        relief="solid",
        padding=(7, 4),
        font=("Segoe UI", 9),
    )
    style.configure(
```

MBToolkit v3.4/mb_theme.py — lines 161-200

```
        "MBS.Toolbar.TButton",
        borderwidth=1,
        relief="solid",
        padding=(9, 5),
        font=("Segoe UI", 9),
    )
    style.configure(
        "MBS.Primary.TButton",
        borderwidth=1,
        relief="solid",
        padding=(9, 5),
        font=("Segoe UI", 9, "bold"),
    )
    style.configure(
        "MBS.Compact.TButton",
        borderwidth=1,
        relief="solid",
        padding=(7, 4),
        font=("Segoe UI", 9),
    )

    # ------------------------------------------------------------------
    # Input controls.
    # ------------------------------------------------------------------
    style.configure("TEntry", borderwidth=1, relief="solid", padding=1)
    style.configure("TCombobox", borderwidth=1, relief="solid", padding=1, arrowsize=10)
    style.configure("TSpinbox", borderwidth=1, relief="solid", padding=1, arrowsize=7)

    # ------------------------------------------------------------------
    # Checkboxes: fixed 15x15 image indicator. The image pixels are recoloured
    # on toggle; the element/layout never changes.
    # ------------------------------------------------------------------
    style.configure(
        "TCheckbutton",
        borderwidth=0,
        relief="flat",
        padding=(1, 1),
        font=("Segoe UI", 9),
    )

```

MBToolkit v3.4/mb_theme.py — lines 201-240

```
    # Create the indicator element from the active clam layout once.
    _install_fixed_checkbutton_element(style)

    # ------------------------------------------------------------------
    # Treeviews.
    # ------------------------------------------------------------------
    style.configure(
        "Treeview",
        borderwidth=1,
        relief="solid",
        font=("Segoe UI", 9),
    )
    style.configure(
        "MBToolkit.Treeview",
        borderwidth=1,
        relief="solid",
        font=("Segoe UI", 9),
    )

    # Important: right + bottom border only. This reproduces Light mode's
    # vertical column divider lines without making every heading look like a
    # separate raised rectangular button.
    for heading in ("Treeview.Heading", "MBToolkit.Treeview.Heading"):
        style.configure(
            heading,
            borderwidth=(0, 0, 1, 1),
            relief="solid",
            padding=(5, 4),
            font=("Segoe UI", 9),
        )
        style.map(
            heading,
            relief=[
                ("active", "solid"),
                ("pressed", "solid"),
            ],
        )

    # Keep MBToolkit.Treeview on the same heading layout as Treeview.
    try:
```

MBToolkit v3.4/mb_theme.py — lines 241-280

```
        heading_layout = style.layout("Treeview.Heading")
        if heading_layout:
            style.layout("MBToolkit.Treeview.Heading", heading_layout)
    except Exception:
        pass

    # ------------------------------------------------------------------
    # Scrollbars and named labels.
    # ------------------------------------------------------------------
    # Minimal scrollbars: slim track + thumb only, without arrow buttons or extra controls.
    style.layout(
        "Vertical.TScrollbar",
        [(
            "Vertical.Scrollbar.trough",
            {
                "sticky": "ns",
                "children": [
                    ("Vertical.Scrollbar.thumb", {"sticky": "nswe"})
                ],
            },
        )],
    )
    style.layout(
        "Horizontal.TScrollbar",
        [(
            "Horizontal.Scrollbar.trough",
            {
                "sticky": "we",
                "children": [
                    ("Horizontal.Scrollbar.thumb", {"sticky": "nswe"})
                ],
            },
        )],
    )
    style.configure(
        "Vertical.TScrollbar",
        width=6,
        borderwidth=0,
        relief="flat",
    )
```

MBToolkit v3.4/mb_theme.py — lines 281-320

```
    style.configure(
        "Horizontal.TScrollbar",
        width=6,
        borderwidth=0,
        relief="flat",
    )

    style.configure(
        "Horizontal.TProgressbar",
        borderwidth=0,
        thickness=8,
        relief="flat",
    )

    style.configure("MBS.Stat.TLabel", font=("Segoe UI", 9))
    style.configure("MBS.Status.TLabel", font=("Segoe UI", 9, "bold"))


def _paint_checkbox_image(image, *, fill, border, check=None):
    """Repaint one fixed-size checkbox image without changing its dimensions."""
    size = 15
    image.put(border, to=(0, 0, size, size))
    image.put(fill, to=(1, 1, size - 1, size - 1))

    if check:
        # White/black check mark. Coordinates remain identical in both modes.
        points = (
            (3, 7), (4, 8), (5, 9), (6, 10),
            (7, 9), (8, 8), (9, 7), (10, 6), (11, 5), (12, 4),
        )
        for x, y in points:
            image.put(check, to=(x, y, x + 1, y + 2))
            if x + 1 < size - 1:
                image.put(check, to=(x + 1, y, x + 2, y + 1))


def _install_fixed_checkbutton_element(style):
    global _CHECK_IMAGES

    root = style.master
```

MBToolkit v3.4/mb_theme.py — lines 321-360

```
    if _CHECK_IMAGES is None:
        _CHECK_IMAGES = {
            "off": tk.PhotoImage(master=root, width=15, height=15),
            "on": tk.PhotoImage(master=root, width=15, height=15),
            "disabled_off": tk.PhotoImage(master=root, width=15, height=15),
            "disabled_on": tk.PhotoImage(master=root, width=15, height=15),
        }

    try:
        if _CHECK_ELEMENT not in style.element_names():
            style.element_create(
                _CHECK_ELEMENT,
                "image",
                _CHECK_IMAGES["off"],
                ("disabled selected", _CHECK_IMAGES["disabled_on"]),
                ("disabled", _CHECK_IMAGES["disabled_off"]),
                ("selected", _CHECK_IMAGES["on"]),
                sticky="",
            )
    except tk.TclError:
        pass

    try:
        layout = style.layout("TCheckbutton")
    except Exception:
        layout = None

    if not layout:
        return

    def replace_indicator(nodes):
        result = []
        for name, options in nodes:
            options = dict(options)
            if "children" in options:
                options["children"] = replace_indicator(options["children"])
            if name == "Checkbutton.indicator":
                name = _CHECK_ELEMENT
            result.append((name, options))
        return result
```

MBToolkit v3.4/mb_theme.py — lines 361-400

```
361
362     # Avoid repeatedly replacing an already-custom element.
363     if _CHECK_ELEMENT not in repr(layout):
364         try:
365             style.layout("TCheckbutton", replace_indicator(layout))
366         except tk.TclError:
367             pass
368
369
370 def _repaint_checkboxes(colors, dark):
371     if not _CHECK_IMAGES:
372         return
373
374     if dark:
375         face = colors["input_readonly"]
376         border = colors["disabled"]
377         check = colors["select_text"]
378         disabled_face = colors["panel_alt"]
379         disabled_check = colors["disabled"]
380     else:
381         face = colors["input"]
382         border = "#7a7a7a"
383         check = "#111111"
384         disabled_face = "#e4e4e4"
385         disabled_check = "#999999"
386
387     _paint_checkbox_image(
388         _CHECK_IMAGES["off"],
389         fill=face,
390         border=border,
391     )
392     _paint_checkbox_image(
393         _CHECK_IMAGES["on"],
394         fill=face,
395         border=border,
396         check=check,
397     )
398     _paint_checkbox_image(
399         _CHECK_IMAGES["disabled_off"],
400         fill=disabled_face,
```

MBToolkit v3.4/mb_theme.py — lines 401-440

```
401         border=colors["disabled"],
402     )
403     _paint_checkbox_image(
404         _CHECK_IMAGES["disabled_on"],
405         fill=disabled_face,
406         border=colors["disabled"],
407         check=disabled_check,
408     )
409
410
411 def _apply_palette(root, style, colors, dark):
412     """
413     Apply colours only.
414
415     No call in this function changes:
416     - theme engine
417     - layout
418     - padding
419     - border width
420     - relief
421     - margins
422     - font
423     - widget dimensions
424     """
425     c = colors
426
427     style.configure(".", background=c["root"], foreground=c["text"])
428     style.configure("TFrame", background=c["root"])
429     style.configure("TLabel", background=c["root"], foreground=c["text"])
430     style.configure(
431         "TLabelframe",
432         background=c["root"],
433         foreground=c["text"],
434         bordercolor=c["border"],
435         lightcolor=c["border"],
436         darkcolor=c["border"],
437     )
438     style.configure(
439         "TLabelframe.Label",
440         background=c["root"],
```

MBToolkit v3.4/mb_theme.py — lines 441-480

```
        foreground=c["text"],
    )

    # Notebook.
    style.configure(
        "TNotebook",
        background=c["root"],
        bordercolor=c["border"],
        lightcolor=c["border"],
        darkcolor=c["border"],
    )
    style.configure(
        "TNotebook.Tab",
        background=c["panel"],
        foreground=c["text"],
        bordercolor=c["border"],
        lightcolor=c["border"],
        darkcolor=c["border"],
    )
    style.map(
        "TNotebook.Tab",
        background=[
            ("selected", c["panel_alt"]),
            ("active", c["panel_alt"]),
        ],
        foreground=[
            ("selected", c["text"]),
            ("active", c["text"]),
        ],
    )

    # Buttons: neutral controls have clear contrast; the primary action uses
    # the same restrained accent in Light and Dark.
    for name in (
        "TButton",
        "MBS.Toolbar.TButton",
        "MBS.Compact.TButton",
    ):
        style.configure(
            name,
```

MBToolkit v3.4/mb_theme.py — lines 481-520

```
            background=c["button"],
            foreground=c["text"],
            bordercolor=c["border"],
            lightcolor=c["border"],
            darkcolor=c["border"],
            focuscolor=c["accent"],
        )
        style.map(
            name,
            background=[
                ("pressed", c["button_pressed"]),
                ("active", c["button_hover"]),
                ("disabled", c["panel"]),
            ],
            foreground=[("disabled", c["disabled"])],
        )

    style.configure(
        "MBS.Primary.TButton",
        background=c["accent"],
        foreground=c["select_text"],
        bordercolor=c["accent"],
        lightcolor=c["accent"],
        darkcolor=c["accent"],
        focuscolor=c["accent"],
    )
    style.map(
        "MBS.Primary.TButton",
        background=[
            ("pressed", c["accent_pressed"]),
            ("active", c["accent_hover"]),
            ("disabled", c["panel"]),
        ],
        foreground=[
            ("disabled", c["disabled"]),
        ],
        bordercolor=[
            ("pressed", c["accent_pressed"]),
            ("active", c["accent_hover"]),
            ("disabled", c["border"]),
```

MBToolkit v3.4/mb_theme.py — lines 521-560

```
        ],
    )

    # Entry / Spinbox.
    for name in ("TEntry", "TSpinbox"):
        style.configure(
            name,
            fieldbackground=c["input"],
            background=c["input"],
            foreground=c["text"],
            bordercolor=c["border"],
            lightcolor=c["border"],
            darkcolor=c["border"],
            insertcolor=c["text"],
        )
        style.map(
            name,
            fieldbackground=[
                ("readonly", c["input_readonly"]),
                ("disabled", c["panel"]),
            ],
            foreground=[("disabled", c["disabled"])],
        )

    # Combobox.
    style.configure(
        "TCombobox",
        fieldbackground=c["input"],
        background=c["button"],
        foreground=c["text"],
        arrowcolor=c["text"],
        bordercolor=c["border"],
        lightcolor=c["border"],
        darkcolor=c["border"],
    )
    style.map(
        "TCombobox",
        fieldbackground=[
            ("readonly", c["input_readonly"]),
            ("disabled", c["panel"]),
```

MBToolkit v3.4/mb_theme.py — lines 561-600

```
        ],
        foreground=[
            ("readonly", c["text"]),
            ("disabled", c["disabled"]),
        ],
        background=[
            ("active", c["button_hover"]),
            ("disabled", c["panel"]),
        ],
        arrowcolor=[("disabled", c["disabled"])],
    )

    # Checkbox label/background only; indicator pixels are repainted separately.
    style.configure(
        "TCheckbutton",
        background=c["root"],
        foreground=c["text"],
    )
    style.map(
        "TCheckbutton",
        background=[("active", c["root"])],
        foreground=[("disabled", c["disabled"])],
    )
    _repaint_checkboxes(c, dark)

    # Treeview body and flat heading.
    for name in ("Treeview", "MBToolkit.Treeview"):
        style.configure(
            name,
            background=c["input"],
            fieldbackground=c["input"],
            foreground=c["text"],
            bordercolor=c["border"],
            lightcolor=c["border"],
            darkcolor=c["border"],
        )
        style.map(
            name,
            background=[("selected", c["select"])],
            foreground=[("selected", c["select_text"])],
```

MBToolkit v3.4/mb_theme.py — lines 601-640

```
        )

    for name in ("Treeview.Heading", "MBToolkit.Treeview.Heading"):
        style.configure(
            name,
            background=c["input"],
            foreground=c["text"],
            bordercolor=c["divider"],
            lightcolor=c["divider"],
            darkcolor=c["divider"],
        )
        style.map(
            name,
            background=[
                ("active", c["input_readonly"]),
                ("pressed", c["input_readonly"]),
            ],
            foreground=[("active", c["text"])],
        )

    # Scrollbars: only the slim thumb should be visually apparent.
    # The required trough element remains for ttk mechanics, but is painted
    # exactly like the content background so it disappears visually.
    for name in ("Vertical.TScrollbar", "Horizontal.TScrollbar", "TScrollbar"):
        style.configure(
            name,
            background=c["scroll_thumb"],
            troughcolor=c["input"],
            bordercolor=c["input"],
            darkcolor=c["scroll_thumb"],
            lightcolor=c["scroll_thumb"],
        )
        style.map(
            name,
            background=[
                ("active", c["scroll_thumb_hover"]),
                ("pressed", c["scroll_thumb_hover"]),
            ],
            darkcolor=[
                ("active", c["scroll_thumb_hover"]),
```

MBToolkit v3.4/mb_theme.py — lines 641-680

```
                ("pressed", c["scroll_thumb_hover"]),
            ],
            lightcolor=[
                ("active", c["scroll_thumb_hover"]),
                ("pressed", c["scroll_thumb_hover"]),
            ],
        )

    style.configure(
        "Horizontal.TProgressbar",
        background=c["accent"],
        troughcolor=c["scroll_trough"],
        bordercolor=c["scroll_trough"],
        lightcolor=c["accent"],
        darkcolor=c["accent"],
    )

    style.configure("MBS.Stat.TLabel", background=c["root"], foreground=c["text"])
    style.configure("MBS.Status.TLabel", background=c["root"], foreground=c["text"])

    try:
        root.configure(background=c["root"])
    except Exception:
        pass

    return style


def apply_theme(root, dark=False, style=None):
    """
    Change palette only.

    The ttk engine and all geometry were fixed before the UI was built.
    """
    style = style or ttk.Style(root)
    _ensure_engine(style)
    return _apply_palette(root, style, DARK if dark else LIGHT, bool(dark))


def apply_light_theme(root, style=None):
```

MBToolkit v3.4/mb_theme.py — lines 681-692

```
681     return apply_theme(root, dark=False, style=style)
682 
683 
684 def apply_dark_theme(root, style=None):
685     return apply_theme(root, dark=True, style=style)
686 
687 
688 def apply_modern_theme(root, dark=False):
689     """Initial application setup with the same engine used for both modes."""
690     style = ttk.Style(root)
691     _ensure_engine(style)
692     return _apply_palette(root, style, DARK if dark else LIGHT, bool(dark))
```

MBToolkit v3.4/mb_widgets.py

418 source lines.

MBToolkit v3.4/mb_widgets.py — lines 1-40

```python
#!/usr/bin/env python3
"""Small reusable Tk widgets for MBToolkit."""

import tkinter as tk


class RefreshButton(tk.Canvas):
    # The fixed clam UI gives the refresh control stable dimensions, so the ↻
    # can now use its measured rendered-bbox centre without manual offsets.
    OPTICAL_OFFSET_X = 0
    OPTICAL_OFFSET_Y = 0

    """
    Square refresh button using the Unicode ↻ glyph.

    The glyph is centered from its actual rendered Tk bounding box. No manual
    X/Y optical offset is applied.
    """

    def __init__(
        self,
        master,
        command,
        size=26,
        cursor="hand2",
        **kwargs,
    ):
        self._command = command
        self._hover = False
        self._pressed = False
        self._state = "normal"
        self._dark_mode = False
        self._final_redraw_after_id = None

        super().__init__(
            master,
            width=size,
            height=size,
            bd=0,
            highlightthickness=0,
```

MBToolkit v3.4/mb_widgets.py — lines 41-80

```
            relief="flat",
            cursor=cursor,
            takefocus=0,
            **kwargs,
        )

        self.bind("<Configure>", self._redraw, add="+")
        self.bind("<Enter>", self._on_enter, add="+")
        self.bind("<Leave>", self._on_leave, add="+")
        self.bind("<ButtonPress-1>", self._on_press, add="+")
        self.bind("<ButtonRelease-1>", self._on_release, add="+")
        self._redraw()
        self._schedule_final_redraw()

    @staticmethod
    def _system_color(widget, name, fallback):
        try:
            widget.winfo_rgb(name)
            return name
        except Exception:
            return fallback

    def _schedule_final_redraw(self):
        """
        Redraw after Tk has settled requested widget dimensions.

        Some themed comboboxes report their final height only after the first
        layout pass. A short deferred redraw prevents the Unicode glyph from
        staying centered for an earlier 24 px square after the control becomes
        26 px high.
        """
        try:
            if self._final_redraw_after_id is not None:
                self.after_cancel(self._final_redraw_after_id)
            self._final_redraw_after_id = self.after(20, self._final_redraw)
        except tk.TclError:
            self._final_redraw_after_id = None

    def _final_redraw(self):
        self._final_redraw_after_id = None
```

MBToolkit v3.4/mb_widgets.py — lines 81-120

```
        try:
            self._redraw()
        except tk.TclError:
            pass

    def configure(self, cnf=None, **kwargs):
        state = kwargs.pop("state", None)
        if state is not None:
            self._state = str(state)

        result = super().configure(cnf, **kwargs)
        try:
            self._redraw()
            self._schedule_final_redraw()
        except tk.TclError:
            pass
        return result

    config = configure

    def cget(self, key):
        if key == "state":
            return self._state
        return super().cget(key)

    def invoke(self):
        if self._state != "disabled" and callable(self._command):
            return self._command()
        return None

    def set_dark_mode(self, enabled):
        self._dark_mode = bool(enabled)
        self._redraw()
        self._schedule_final_redraw()

    def _palette(self):
        if self._dark_mode:
            face = "#2a3138"
            hover = "#36404a"
            pressed = "#20262c"
```

MBToolkit v3.4/mb_widgets.py — lines 121-160

```
            text = "#e8eaed"
            gray = "#707982"
            border = "#4a5560"
            highlight = "#2f8fd8"

            if self._state == "disabled":
                return face, gray, border
            if self._pressed:
                return pressed, text, highlight
            if self._hover:
                return hover, text, highlight
            return face, text, border

        face = self._system_color(self, "SystemButtonFace", "#f0f0f0")
        text = self._system_color(self, "SystemButtonText", "#202020")
        gray = self._system_color(self, "SystemGrayText", "#8a8a8a")
        shadow = self._system_color(self, "System3DShadow", "#adadad")
        highlight = self._system_color(self, "SystemHighlight", "#7a9fc2")

        if self._state == "disabled":
            return face, gray, shadow
        if self._pressed:
            return face, text, highlight
        if self._hover:
            return face, text, highlight
        return face, text, shadow

    def _center_item_bbox(self, item, width, height):
        """Center the visible Tk bbox of one Canvas item on the square."""
        target_x = width / 2.0
        target_y = height / 2.0

        # Tk text bboxes land on integer pixels. Repeat because a fractional
        # first move may be rounded by the Canvas implementation.
        for _ in range(4):
            bbox = self.bbox(item)
            if not bbox:
                return
            current_x = (bbox[0] + bbox[2]) / 2.0
            current_y = (bbox[1] + bbox[3]) / 2.0
```

MBToolkit v3.4/mb_widgets.py — lines 161-200

```
            dx = target_x - current_x
            dy = target_y - current_y

            if abs(dx) <= 0.5 and abs(dy) <= 0.5:
                break

            # Move by integral pixels when a full-pixel correction is needed.
            move_x = round(dx) if abs(dx) > 0.5 else 0
            move_y = round(dy) if abs(dy) > 0.5 else 0
            if move_x == 0 and move_y == 0:
                break
            self.move(item, move_x, move_y)

        # Keep the explicit offset hook at zero so all three refresh buttons
        # use the same measured centre.
        if self.OPTICAL_OFFSET_X or self.OPTICAL_OFFSET_Y:
            self.move(
                item,
                self.OPTICAL_OFFSET_X,
                self.OPTICAL_OFFSET_Y,
            )

    def _redraw(self, _event=None):
        if not self.winfo_exists():
            return

        # During a <Configure> callback winfo_width()/height() may still
        # report the previous geometry on some Tk builds. Prefer the event's
        # new dimensions so centering is calculated for the final square.
        if _event is not None and hasattr(_event, "width") and hasattr(_event, "height"):
            width = max(2, int(_event.width))
            height = max(2, int(_event.height))
        else:
            width = max(2, int(self.winfo_width()))
            height = max(2, int(self.winfo_height()))

        background, foreground, border = self._palette()

        self.delete("all")
        self.create_rectangle(
```

MBToolkit v3.4/mb_widgets.py — lines 201-240

```
            0,
            0,
            width - 1,
            height - 1,
            fill=background,
            outline=border,
            width=1,
            tags=("background",),
        )

        glyph = self.create_text(
            width / 2.0,
            height / 2.0,
            text="↻",
            fill=foreground,
            font=("Segoe UI Symbol", 13),
            anchor="center",
            tags=("glyph",),
        )
        self._center_item_bbox(glyph, width, height)

    def _on_enter(self, _event):
        if self._state != "disabled":
            self._hover = True
            self._redraw()

    def _on_leave(self, _event):
        self._hover = False
        self._pressed = False
        self._redraw()

    def _on_press(self, _event):
        if self._state != "disabled":
            self._pressed = True
            self._redraw()

    def _on_release(self, event):
        if self._state == "disabled":
            return

```

MBToolkit v3.4/mb_widgets.py — lines 241-280

```
241         was_pressed = self._pressed
242         self._pressed = False
243         self._redraw()
244 
245         inside = (
246             0 <= event.x < self.winfo_width()
247             and 0 <= event.y < self.winfo_height()
248         )
249         if was_pressed and inside and callable(self._command):
250             self._command()
251 
252 
253 
254 class ToggleSwitch(tk.Canvas):
255     """Compact on/off slider used for the Light/Dark mode control."""
256 
257     def __init__(
258         self,
259         master,
260         variable,
261         command=None,
262         width=42,
263         height=22,
264         cursor="hand2",
265         **kwargs,
266     ):
267         self.variable = variable
268         self.command = command
269         self._dark_mode = False
270         self._hover = False
271         self._final_redraw_after_id = None
272 
273         super().__init__(
274             master,
275             width=width,
276             height=height,
277             bd=0,
278             highlightthickness=0,
279             relief="flat",
280             cursor=cursor,
```

MBToolkit v3.4/mb_widgets.py — lines 281-320

```
            takefocus=0,
            **kwargs,
        )

        self.bind("<Configure>", self._redraw, add="+")
        self.bind("<ButtonPress-1>", self._press, add="+")
        self.bind("<ButtonRelease-1>", self._toggle, add="+")
        self.bind("<Enter>", self._enter, add="+")
        self.bind("<Leave>", self._leave, add="+")
        self.variable.trace_add("write", lambda *_a: self._redraw())
        self._redraw()
        self._schedule_final_redraw()

    def _schedule_final_redraw(self):
        try:
            if self._final_redraw_after_id is not None:
                self.after_cancel(self._final_redraw_after_id)
            self._final_redraw_after_id = self.after(20, self._final_redraw)
        except tk.TclError:
            self._final_redraw_after_id = None

    def _final_redraw(self):
        self._final_redraw_after_id = None
        try:
            self._redraw()
        except tk.TclError:
            pass

    def set_dark_mode(self, enabled):
        self._dark_mode = bool(enabled)
        self._redraw()
        self._schedule_final_redraw()

    def _parent_bg(self):
        if self._dark_mode:
            return "#171a1f"
        try:
            self.winfo_rgb("SystemButtonFace")
            return "SystemButtonFace"
        except Exception:
```

MBToolkit v3.4/mb_widgets.py — lines 321-360

```
321             return "#f0f0f0"
322
323     def _press(self, _event=None):
324         # Deliberately do not focus the Canvas. This removes the white focus
325         # rectangle that appeared around the switch after mouse clicks.
326         return "break"
327
328     def _toggle(self, _event=None):
329         self.variable.set(not bool(self.variable.get()))
330         if callable(self.command):
331             self.command()
332         return "break"
333
334     def _enter(self, _event=None):
335         self._hover = True
336         self._redraw()
337
338     def _leave(self, _event=None):
339         self._hover = False
340         self._redraw()
341
342     def _redraw(self, _event=None):
343         if not self.winfo_exists():
344             return
345
346         # Use the final dimensions delivered by <Configure>. Without this, the
347         # first draw can happen while Tk still reports the initial 1x1 Canvas,
348         # which made the switch look small until the first mouse hover.
349         if _event is not None and hasattr(_event, "width") and hasattr(_event, "height"):
350             width = max(22, int(_event.width))
351             height = max(14, int(_event.height))
352         else:
353             width = max(22, int(self.winfo_width()))
354             height = max(14, int(self.winfo_height()))
355
356         on = bool(self.variable.get())
357
358         self.configure(background=self._parent_bg())
359         self.delete("all")
360
```

MBToolkit v3.4/mb_widgets.py — lines 361-400

```
        pad = 2
        radius = (height - 2 * pad) / 2.0
        x0, y0 = pad, pad
        x1, y1 = width - pad, height - pad

        if self._dark_mode:
            off_track = "#5a646e"
            on_track = "#2f8fd8"
            hover_track = "#43a2e8"
            knob = "#f4f6f8"
            outline = "#68737d"
        else:
            off_track = "#a8a8a8"
            on_track = "#2689d9"
            hover_track = "#3b9be5"
            knob = "#ffffff"
            outline = "#8f8f8f"

        track = hover_track if self._hover and on else (on_track if on else off_track)

        # Rounded track: rectangle + two circles.
        self.create_rectangle(
            x0 + radius,
            y0,
            x1 - radius,
            y1,
            fill=track,
            outline=track,
        )
        self.create_oval(
            x0,
            y0,
            x0 + 2 * radius,
            y1,
            fill=track,
            outline=track,
        )
        self.create_oval(
            x1 - 2 * radius,
            y0,
```

MBToolkit v3.4/mb_widgets.py — lines 401-418

```
            x1,
            y1,
            fill=track,
            outline=track,
        )

        knob_radius = max(4.0, radius - 2)
        knob_cx = (x1 - radius) if on else (x0 + radius)
        knob_cy = height / 2.0
        self.create_oval(
            knob_cx - knob_radius,
            knob_cy - knob_radius,
            knob_cx + knob_radius,
            knob_cy + knob_radius,
            fill=knob,
            outline=outline,
            width=1,
        )
```

MBToolkit v3.4/mb_gui.py

2664 source lines.

MBToolkit v3.4/mb_gui.py — lines 1-40

```
#!/usr/bin/env python3
"""Main Tkinter window composition for MBToolkit."""

import queue
import threading
from collections import deque

import tkinter as tk
from tkinter import ttk
import tkinter.font as tkfont

from mb_config import (
    APP_NAME,
    APP_VERSION,
    DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS,
    DISCOVERY_IDLE_CHECK_SECONDS,
    MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
    MAX_RAW_TEXT_LINES,
    MAX_UI_FRAMES,
    MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
    SLOW_RESPONSE_THRESHOLD_MS,
    SNIFFER_SPLIT_FRAME_HOLD_MS,
    debug_sim,
    load_ui_settings,
    resource_path,
    save_ui_settings,
)
from mb_protocol import SessionMetrics
from mb_slave_finder import SlaveFinderMixin
from mb_master_ui import MasterMixin
from mb_slave_ui import SlaveMixin
from mb_capture import CaptureMixin
from mb_view import TRAFFIC_MENU_OPEN_IN_MASTER, ViewMixin
from mb_widgets import RefreshButton, ToggleSwitch
from mb_theme import apply_modern_theme, apply_theme, theme_colors
from mb_i18n import (
    LANGUAGE_CHOICES,
    LANGUAGE_ENGLISH,
    LANGUAGE_PORTUGUESE,
    canonical_ui_text,
```

MBToolkit v3.4/mb_gui.py — lines 41-80

```
    english_help_blocks,
    normalize_language,
    translate_text,
    translate_runtime_text,
)


class SnifferApp(MasterMixin, SlaveMixin, SlaveFinderMixin, CaptureMixin, ViewMixin, tk.Tk):
    def __init__(self):
        super().__init__()
        self.title(f"{APP_NAME} v{APP_VERSION}")

        # Load the remembered mode before any ttk widgets are constructed.
        # Light and Dark use one fixed ttk construction; only colours differ.
        self.ui_settings = load_ui_settings()
        self.current_language = normalize_language(
            self.ui_settings.get("language", LANGUAGE_PORTUGUESE)
        )
        self.initial_dark_mode = bool(self.ui_settings.get("dark_mode", False))
        self.ui_style = apply_modern_theme(
            self,
            dark=self.initial_dark_mode,
        )

        # Explicit application/window icon. This works when launched through
        # MBToolkit.bat (Python) and when bundled as a PyInstaller EXE.
        try:
            icon_file = resource_path("MBToolkit.ico")
            if icon_file.exists():
                self.iconbitmap(default=str(icon_file))
        except Exception:
            pass

        self.geometry("1280x780")
        self.minsize(1050, 650)

        self.serial_objects = []
        self.serial_by_port = {}
        self.reader_threads = []
        self.reader_generation = 0
```

MBToolkit v3.4/mb_gui.py — lines 81-120

```
        self.stop_event = threading.Event()
        self.event_queue = queue.Queue()
        self.busy = False
        self.capture_active = False
        self.restart_in_progress = False

        # Active slave-discovery state. This is deliberately independent from
        # passive capture/simulation state, but shares the global busy lock so
        # the application can never act as scanner and sniffer at the same time.
        self.discovery_active = False
        self.discovery_stop_event = threading.Event()
        self.discovery_thread = None
        self.discovery_serial = None
        self.discovery_found_count = 0
        self._closing = False
        self._process_queue_error_shown = False

        self.log_file = None
        self.log_path = None

        # Real-capture logs are created lazily: starting/stopping a capture with
        # zero received traffic must not create an empty .txt file.
        self.pending_log_filename = None
        self.pending_log_header = ""
        self.pending_log_notes = []
        self.log_start_failed = False
        self._raw_by_iid = {}
        self._record_by_iid = {}
        self._transaction_filter_seqs = None
        self._successful_request_seqs = set()
        self._advanced_filter_after_id = None
        self._bus_activity_after_id = None

        # Bounded GUI-history bookkeeping.
        self._tree_history = deque()   # current visible Treeview rows
        self._ui_frame_count = 0
        self._raw_text_line_count = 0

        # Source-of-truth for the bounded GUI history. Filtering/sorting only
        # changes the view; it never changes statistics or the disk log.
```

MBToolkit v3.4/mb_gui.py — lines 121-160

```
121         self._frame_history = deque()
122         self._next_frame_seq = 1
123 
124         # Visible Treeview bookkeeping by logical frame sequence.
125         self._tree_iids_by_seq = {}
126         self._tree_separator_iid_by_seq = {}
127 
128         # Raw Hex separator width/reflow bookkeeping.
129         self._raw_separator_width_chars = 1
130         self._raw_reflow_after_id = None
131 
132         # Slave filter state. "All mode" also controls whether newly discovered
133         # slaves are automatically selected.
134         self.slave_filter_vars = {}
135         self.slave_filter_all_mode = True
136 
137         self.metrics = SessionMetrics()
138 
139         self._build_ui()
140         self._freeze_language_layout_geometry()
141         self._capture_portuguese_help()
142         self.apply_language(save=False)
143         self._install_mouse_wheel_guard()
144         self.rebuild_slave_filter_menu()
145         self.refresh_ports()
146         self.update_mode_ui()
147         self.update_gap_state()
148         self.update_com_status()
149         self._restore_saved_main_tab()
150         self._install_ui_settings_persistence()
151 
152         self.protocol("WM_DELETE_WINDOW", self.on_close)
153         self.after(50, self.process_queue)
154         self.after(1000, self.poll_com_status)
155 
156 
157     def _freeze_language_layout_geometry(self):
158         """Freeze language-independent container geometry using the PT layout.
159 
160         The UI is constructed from the canonical Portuguese captions. Capture
```

MBToolkit v3.4/mb_gui.py — lines 161-200

```
        the natural geometry before any runtime translation and use those
        dimensions as layout constraints. Language changes may replace text,
        but must never move or resize the surrounding controls/panels.
        """
        try:
            self.update_idletasks()
        except tk.TclError:
            return

        # The global language/theme overlay is anchored to the top-right.
        # Freeze the canonical PT grid-cell widths. English captions are shorter
        # here, so this preserves the exact existing PT positions and prevents
        # the Dark Mode switch / language combobox from sliding left or right.
        overlay = getattr(self, "theme_overlay", None)
        if overlay is not None:
            try:
                self.update_idletasks()
                canonical_column_widths = [
                    max(1, int(overlay.grid_bbox(column, 0)[2]))
                    for column in range(4)
                ]
                for column, width in enumerate(canonical_column_widths):
                    overlay.columnconfigure(column, minsize=width)
                self.update_idletasks()
                self._language_overlay_width = max(1, overlay.winfo_reqwidth())
                self._language_overlay_height = max(1, overlay.winfo_reqheight())
                overlay.place_configure(
                    width=self._language_overlay_width,
                    height=self._language_overlay_height,
                )
            except (tk.TclError, IndexError, TypeError):
                pass

        # The Finder configuration includes wrapped explanatory text. It must
        # be measured while the Finder page is actually laid out; measuring a
        # hidden Notebook page can under-estimate the PT height by one line.
        page = getattr(self, "discovery_page", None)
        options = getattr(self, "_discovery_options_frame", None)
        if page is not None and options is not None:
            try:
```

MBToolkit v3.4/mb_gui.py — lines 201-240

```
201                 selected_tab = self.main_notebook.select()
202                 self.main_notebook.select(page)
203                 self.update_idletasks()
204                 height = max(options.winfo_height(), options.winfo_reqheight(), 1)
205                 # grid row height also includes the widget's external pady.
206                 # Preserve the complete PT cell, not only the LabelFrame body.
207                 try:
208                     cell_height = max(height, int(page.grid_bbox(0, 1)[3]))
209                 except (tk.TclError, TypeError, IndexError):
210                     cell_height = height + 12
211                 self._language_discovery_options_height = height
212                 page.rowconfigure(1, minsize=cell_height)
213                 options.grid_configure(sticky="nsew")
214                 self.main_notebook.select(selected_tab)
215                 self.update_idletasks()
216             except tk.TclError:
217                 pass
218 
219     def _guard_mouse_wheel(self, event):
220         """
221         Keep wheel input local to genuinely scrollable content widgets.
222 
223         Treeviews and Text widgets retain their normal vertical wheel scrolling,
224         as do the scrollbar widgets themselves.  Everywhere else the event is
225         consumed before ttk class bindings see it; in particular, this prevents
226         Windows ttk Notebook tabs from changing when the user turns the wheel
227         over a tab or over non-scrollable page content.
228         """
229         widget = getattr(event, "widget", None)
230         if isinstance(
231             widget,
232             (ttk.Treeview, tk.Text, tk.Listbox, ttk.Scrollbar, tk.Scrollbar),
233         ):
234             return None
235         return "break"
236 
237     def _install_mouse_wheel_guard(self):
238         """
239         Prepend one application bindtag to every existing widget.
240 
```

MBToolkit v3.4/mb_gui.py — lines 241-280

```
        The UI is statically constructed before this method is called, so this
        covers the Sniffer, Bus Health, Help and Bus Slave Finder pages without
        changing individual widget behavior or notebook keyboard navigation.
        """
        tag = "MBToolkit.MouseWheelGuard"

        for sequence in (
            "<MouseWheel>",
            "<Shift-MouseWheel>",
            "<Button-4>",
            "<Button-5>",
        ):
            try:
                self.bind_class(
                    tag,
                    sequence,
                    self._guard_mouse_wheel,
                )
            except tk.TclError:
                pass

        stack = [self]
        while stack:
            widget = stack.pop()
            try:
                tags = tuple(widget.bindtags())
                if tag not in tags:
                    widget.bindtags((tag,) + tags)
                stack.extend(widget.winfo_children())
            except tk.TclError:
                continue

    def _apply_classic_widget_palette(self, dark):
        """Apply palette to the small set of classic Tk widgets."""
        colors = theme_colors(self, dark)

        if hasattr(self, "raw_text"):
            self.raw_text.configure(
                background=colors["input"],
                foreground=colors["text"],
```

MBToolkit v3.4/mb_gui.py — lines 281-320

```
                insertbackground=colors["text"],
                selectbackground=colors["select"],
                selectforeground=colors["select_text"],
            )

        if hasattr(self, "help_text"):
            self.help_text.configure(
                background=colors["input"],
                foreground=colors["text"],
                insertbackground=colors["text"],
                selectbackground=colors["select"],
                selectforeground=colors["select_text"],
            )

        for menu_name in ("slave_filter_menu", "traffic_context_menu", "discovery_context_menu"):
            menu = getattr(self, menu_name, None)
            if menu is None:
                continue
            try:
                menu.configure(
                    background=colors["panel"],
                    foreground=colors["text"],
                    activebackground=colors["select"],
                    activeforeground=colors["select_text"],
                    selectcolor=colors["text"],
                )
            except tk.TclError:
                pass

        if hasattr(self, "tree"):
            self.configure_anomaly_tags(dark)
            self.configure_inspector_result_styles()

        for name in (
            "refresh_a_btn",
            "refresh_b_btn",
            "discovery_refresh_btn",
            "dark_mode_switch",
        ):
            widget = getattr(self, name, None)
```

MBToolkit v3.4/mb_gui.py — lines 321-360

```
            setter = getattr(widget, "set_dark_mode", None)
            if callable(setter):
                setter(dark)

        if hasattr(self, "master_canvas"):
            self.master_apply_palette(dark)
        if hasattr(self, "slave_canvas"):
            self.slave_apply_palette(dark)

    def configure_inspector_result_styles(self):
        """
        Match the Inspector Result foreground to the Traffic result palette.

        Only colour values are configured here; no padding/font/layout metrics
        are changed, so Light/Dark construction remains identical.
        """
        if not hasattr(self, "ui_style"):
            return

        colors = theme_colors(
            self,
            bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get()),
        )

        tag_map = {
            "success": "result_success",
            "slow": "anomaly_slow",
            "exception": "anomaly_exception",
            "crc": "anomaly_crc",
            "timeout": "anomaly_timeout",
            "raw": "anomaly_raw",
        }

        self.ui_style.configure(
            "MBS.Result.neutral.TLabel",
            background=colors["root"],
            foreground=colors["text"],
        )
        self.ui_style.configure(
            "MBS.Result.pending.TLabel",
```

MBToolkit v3.4/mb_gui.py — lines 361-400

```
            background=colors["root"],
            foreground=colors["text"],
        )

        if hasattr(self, "tree"):
            for key, tag in tag_map.items():
                try:
                    foreground = self.tree.tag_configure(tag)["foreground"]
                except Exception:
                    foreground = colors["text"]

                self.ui_style.configure(
                    f"MBS.Result.{key}.TLabel",
                    background=colors["root"],
                    foreground=foreground,
                )

    def tr(self, text):
        """Translate one canonical UI string to the currently selected language."""
        return translate_text(text, self.current_language)

    def trf(self, text, **values):
        """Translate a format template and substitute values afterwards."""
        return self.tr(text).format(**values)

    def _capture_portuguese_help(self):
        """Store the original PT-PT Help text and tag ranges exactly as built."""
        if not hasattr(self, "help_text"):
            return
        try:
            self._help_pt_text = self.help_text.get("1.0", "end-1c")
            ranges = {}
            for tag in self.help_text.tag_names():
                if tag == "sel":
                    continue
                tag_ranges = list(self.help_text.tag_ranges(tag))
                ranges[tag] = [
                    (str(tag_ranges[i]), str(tag_ranges[i + 1]))
                    for i in range(0, len(tag_ranges), 2)
                ]
```

MBToolkit v3.4/mb_gui.py — lines 401-440

```
            self._help_pt_tag_ranges = ranges
        except tk.TclError:
            self._help_pt_text = ""
            self._help_pt_tag_ranges = {}

    def _render_help_language(self):
        """Render Help in English or restore the original PT-PT text byte-for-byte."""
        if not hasattr(self, "help_text"):
            return
        try:
            self.help_text.configure(state="normal")
            self.help_text.delete("1.0", "end")

            if self.current_language == LANGUAGE_ENGLISH:
                max_ui_frames_text = f"{MAX_UI_FRAMES:_}".replace("_", " ")
                max_raw_lines_text = f"{MAX_RAW_TEXT_LINES:_}".replace("_", " ")
                for tag, text in english_help_blocks(
                    max_ui_frames_text,
                    max_raw_lines_text,
                    MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
                    MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
                    debug_sim=(debug_sim == 1),
                ):
                    self.help_text.insert("end", text, tag)
            else:
                self.help_text.insert("1.0", getattr(self, "_help_pt_text", ""))
                for tag, ranges in getattr(self, "_help_pt_tag_ranges", {}).items():
                    for start, end in ranges:
                        self.help_text.tag_add(tag, start, end)

            self.help_text.configure(state="disabled")
            self.help_text.see("1.0")
        except tk.TclError:
            pass

    def _translate_widget_tree(self):
        """Translate static widget captions without changing widget geometry."""
        stack = [self]
        while stack:
            widget = stack.pop()
```

MBToolkit v3.4/mb_gui.py — lines 441-480

```
            try:
                stack.extend(widget.winfo_children())
            except tk.TclError:
                continue

            # Text widgets contain documents, not captions.
            if isinstance(widget, tk.Text):
                continue

            try:
                keys = widget.keys()
            except Exception:
                keys = ()
            if "text" in keys:
                try:
                    current = widget.cget("text")
                    if current:
                        widget.configure(text=self.tr(current))
                except (tk.TclError, TypeError):
                    pass

            if isinstance(widget, ttk.Notebook):
                try:
                    for tab_id in widget.tabs():
                        current = widget.tab(tab_id, "text")
                        widget.tab(tab_id, text=self.tr(current))
                except tk.TclError:
                    pass

    def _translate_traffic_context_menu(self):
        menu = getattr(self, "traffic_context_menu", None)
        if menu is None:
            return
        labels = (
            (0, "Copiar Raw Hex"),
            (1, "Copiar frame descodificado"),
            (3, "Filtrar por este Slave"),
            (4, "Filtrar por este FC"),
            (5, "Mostrar apenas esta transação"),
            (6, "Limpar filtro de transação"),
```

MBToolkit v3.4/mb_gui.py — lines 481-520

```
            (TRAFFIC_MENU_OPEN_IN_MASTER, "Abrir no Master"),
        )
        for index, canonical in labels:
            try:
                menu.entryconfigure(index, label=self.tr(canonical))
            except tk.TclError:
                pass

    def _apply_table_language(self):
        """Refresh headings and filter values that are not normal widget captions."""
        if hasattr(self, "advanced_filter_type_combo"):
            current = canonical_ui_text(self.advanced_filter_type_var.get())
            canonical_values = (
                "Todos", "Requests", "Responses", "Exceptions",
                "CRC errors", "Timeouts", "RAW",
            )
            self.advanced_filter_type_combo.configure(
                values=tuple(self.tr(v) for v in canonical_values)
            )
            self.advanced_filter_type_var.set(self.tr(current))

        if hasattr(self, "time_order_var"):
            canonical_order = canonical_ui_text(self.time_order_var.get())
            self.time_order_var.set(self.tr(canonical_order))

        if hasattr(self, "tree") and hasattr(self, "tree_headings"):
            canonical_headings = {
                "time": "Hora ↓" if self.time_order_descending() else "Hora ↑",
                "delta": "Δt (ms)",
                "response": "Resp. (ms)",
                "channel": "COM",
                "type": "Tipo",
                "slave": canonical_ui_text(self.tree_headings.get("slave", "Slave ▾")),
                "fc": "FC",
                "details": "Detalhes",
                "crc": "CRC",
            }
            for col, canonical in canonical_headings.items():
                translated = self.tr(canonical)
                self.tree_headings[col] = translated
```

MBToolkit v3.4/mb_gui.py — lines 521-560

```
                try:
                    self.tree.heading(col, text=translated)
                except tk.TclError:
                    pass

        if hasattr(self, "discovery_tree"):
            canonical = {
                "slave": "Slave",
                "baud": "Baud",
                "config": "Config.",
                "fc": "FC",
                "result": "Resultado",
                "resp": "Resp. (ms)",
                "device_id": "Device Identification",
                "raw": "Raw Hex",
            }
            for col, value in canonical.items():
                try:
                    self.discovery_tree.heading(col, text=self.tr(value))
                except tk.TclError:
                    pass

    def apply_language(self, save=True):
        """Apply the selected language immediately to the complete UI."""
        selected = getattr(self, "language_var", None)
        if selected is not None:
            self.current_language = normalize_language(selected.get())
            if selected.get() != self.current_language:
                selected.set(self.current_language)

        self._translate_widget_tree()
        self._apply_table_language()
        self._translate_traffic_context_menu()
        self._render_help_language()

        # Rebuild dynamic captions/values in the selected language.
        try:
            self.update_mode_ui()
            self.update_com_status()
            self.update_stats_labels()
```

MBToolkit v3.4/mb_gui.py — lines 561-600

```
            self.rebuild_slave_filter_menu()
            self.update_slave_heading()
            self.update_frame_inspector(self.selected_record())
            self.update_discovery_estimate()
            self.master_apply_language()
            self.slave_apply_language()
        except (AttributeError, tk.TclError):
            pass

        # Translate idle/current status variables without altering state.
        for name in ("status_var", "discovery_status_var", "discovery_found_var"):
            var = getattr(self, name, None)
            if var is not None:
                try:
                    var.set(translate_runtime_text(var.get(), self.current_language))
                except tk.TclError:
                    pass

        # Refresh translated row contents without allowing the language change
        # itself to alter table/card geometry.  Column widths are restored after
        # rendering; normal viewport resizing and new traffic may still use the
        # existing responsive/autofit behaviour.
        traffic_widths = {}
        discovery_widths = {}
        try:
            if hasattr(self, "tree"):
                traffic_widths = {
                    col: int(self.tree.column(col, "width"))
                    for col in self.tree["columns"]
                }
            if hasattr(self, "discovery_tree"):
                discovery_widths = {
                    col: int(self.discovery_tree.column(col, "width"))
                    for col in self.discovery_tree["columns"]
                }
            self.render_filtered_view()
            for col, width in traffic_widths.items():
                self.tree.column(col, width=width)
            for col, width in discovery_widths.items():
                self.discovery_tree.column(col, width=width)
```

MBToolkit v3.4/mb_gui.py — lines 601-640

```
601         except (AttributeError, tk.TclError):
602             pass
603 
604         if save:
605             self.save_current_ui_settings()
606 
607     def on_language_changed(self, _event=None):
608         self.apply_language(save=True)
609 
610     def toggle_dark_mode(self):
611         """
612         Switch Light/Dark colours only.
613 
614         No ttk theme replacement, layout change, padding change or row-height
615         change occurs here. The page construction is immutable after startup.
616         """
617         dark = bool(self.dark_mode_var.get())
618 
619         self.ui_style = apply_theme(
620             self,
621             dark=dark,
622             style=self.ui_style,
623         )
624         self._apply_classic_widget_palette(dark)
625 
626         # Remember Dark mode together with the rest of the user's preferences.
627         self.save_current_ui_settings()
628 
629     def _restore_saved_main_tab(self):
630         saved = self.ui_settings.get("last_main_tab", "Sniffer")
631         pages = {"Bus Slave Finder": self.discovery_page, "Master": self.master_page,
632                  "Slave": self.slave_page}
633         try:
634             self.main_notebook.select(pages.get(saved, self.sniffer_page))
635         except tk.TclError:
636             pass
637 
638     def _current_main_tab_name(self):
639         try:
640             selected = self.main_notebook.select()
```

MBToolkit v3.4/mb_gui.py — lines 641-680

```
            if selected == str(self.discovery_page):
                return "Bus Slave Finder"
            if selected == str(self.master_page):
                return "Master"
            if selected == str(self.slave_page):
                return "Slave"
            return "Sniffer"
        except Exception:
            return "Sniffer"

    def _schedule_ui_settings_save(self, *_args):
        if getattr(self, "_closing", False):
            return

        previous = getattr(self, "_settings_save_after_id", None)
        if previous is not None:
            try:
                self.after_cancel(previous)
            except tk.TclError:
                pass

        try:
            self._settings_save_after_id = self.after(
                300,
                self.save_current_ui_settings,
            )
        except tk.TclError:
            self._settings_save_after_id = None

    def save_current_ui_settings(self):
        """Persist all roaming-safe user preferences in one settings.json."""
        pending = getattr(self, "_settings_save_after_id", None)
        if pending is not None:
            try:
                self.after_cancel(pending)
            except tk.TclError:
                pass
        self._settings_save_after_id = None

        try:
```

MBToolkit v3.4/mb_gui.py — lines 681-720

```
            finder_bauds = [
                int(baud)
                for baud, var in self.discovery_baud_vars.items()
                if bool(var.get())
            ]
            finder_parities = [
                parity
                for parity, var in self.discovery_parity_vars.items()
                if bool(var.get())
            ]
            finder_stops = [
                stop
                for stop, var in self.discovery_stop_vars.items()
                if bool(var.get())
            ]

            payload = {
                "dark_mode": bool(self.dark_mode_var.get()),
                "language": normalize_language(self.current_language),
                "last_main_tab": self._current_main_tab_name(),
                "sniffer": {
                    "physical_mode": self.mode_var.get(),
                    "baud": self.baud_var.get(),
                    "data_bits": self.data_var.get(),
                    "parity": self.parity_var.get(),
                    "stop_bits": self.stop_var.get(),
                    "frame_gap_mode": self.gap_mode_var.get(),
                    "frame_gap_ms": self.gap_ms_var.get(),
                    "pending_timeout_s": self.pending_timeout_var.get(),
                    "auto_scroll": bool(self.auto_scroll_var.get()),
                    "separate_transactions": bool(
                        self.separate_transactions_var.get()
                    ),
                    "highlight_results": bool(
                        self.highlight_anomalies_var.get()
                    ),
                },
                "bus_slave_finder": {
                    "baud_rates": finder_bauds,
                    "parities": finder_parities,
```

MBToolkit v3.4/mb_gui.py — lines 721-760

```
                    "stop_bits": finder_stops,
                    "slave_start": self.discovery_slave_start_var.get(),
                    "slave_end": self.discovery_slave_end_var.get(),
                    "min_timeout_ms": self.discovery_min_timeout_var.get(),
                    "attempts": self.discovery_attempts_var.get(),
                    "fc04_fallback": bool(
                        self.discovery_fc04_fallback_var.get()
                    ),
                    "device_identification": bool(
                        self.discovery_device_identification_var.get()
                    ),
                },
                "master": {
                    "baud": self.master_baud_var.get(),
                    "parity": self.master_parity_var.get(),
                    "stop_bits": self.master_stop_var.get(),
                    "response_timeout_ms": self.master_timeout_var.get(),
                    "delay_ms": self.master_delay_var.get(),
                    "last_workspace": self.master_current_workspace,
                },
                "slave": {
                    "baud": self.slave_baud_var.get(),
                    "parity": self.slave_parity_var.get(),
                    "stop_bits": self.slave_stop_var.get(),
                    "last_workspace": self.slave_current_workspace,
                },
            }
            save_ui_settings(settings=payload)
        except Exception:
            # Persistence is convenience state only.
            pass

    def _install_ui_settings_persistence(self):
        """Debounced persistence for all roaming-safe user preferences."""
        self._settings_save_after_id = None

        variables = (
            self.mode_var,
            self.baud_var,
            self.data_var,
```

MBToolkit v3.4/mb_gui.py — lines 761-800

```
            self.parity_var,
            self.stop_var,
            self.gap_mode_var,
            self.gap_ms_var,
            self.pending_timeout_var,
            self.auto_scroll_var,
            self.separate_transactions_var,
            self.highlight_anomalies_var,
            self.discovery_slave_start_var,
            self.discovery_slave_end_var,
            self.discovery_min_timeout_var,
            self.discovery_fc04_fallback_var,
            self.discovery_device_identification_var,
        )

        for variable in variables:
            variable.trace_add(
                "write",
                lambda *_a: self._schedule_ui_settings_save(),
            )

        for mapping in (
            self.discovery_baud_vars,
            self.discovery_parity_vars,
            self.discovery_stop_vars,
        ):
            for variable in mapping.values():
                variable.trace_add(
                    "write",
                    lambda *_a: self._schedule_ui_settings_save(),
                )

        self.main_notebook.bind(
            "<<NotebookTabChanged>>",
            lambda _e: self._schedule_ui_settings_save(),
            add="+",
        )

    def _schedule_bus_health_layout_sync(self, _event=None):
        """Debounce Bus Health reflow after notebook/window/Inspector resizing."""
```

MBToolkit v3.4/mb_gui.py — lines 801-840

```
        previous = getattr(self, "_bus_health_layout_after_id", None)
        if previous is not None:
            try:
                self.after_cancel(previous)
            except tk.TclError:
                pass

        try:
            self._bus_health_layout_after_id = self.after_idle(
                self._sync_bus_health_layout
            )
        except tk.TclError:
            self._bus_health_layout_after_id = None

    def _initialize_bus_health_layout(self):
        """Measure the natural full Bus Health height before adaptive reflow."""
        self._bus_health_layout_after_id = None

        cards = getattr(self, "_health_cards", ())
        if not cards:
            return

        try:
            self.update_idletasks()
        except tk.TclError:
            return

        if getattr(self, "_health_full_required_card_height", None) is None:
            requested = [
                card.winfo_reqheight()
                for card in cards
                if card.winfo_exists()
            ]
            if requested:
                self._health_full_required_card_height = max(requested)

        self._sync_bus_health_layout()

    def _apply_bus_health_layout(self, compact):
        """
```

MBToolkit v3.4/mb_gui.py — lines 841-880

```
        Re-grid Bus Health metrics inside equal vertical bands.

        Full mode uses one metric/value pair per row.  Compact mode uses two
        pairs per row.  In both modes every used row receives the same weight,
        so the metrics are distributed evenly over the card height instead of
        accumulating at the top.  Compact columns use proportional uniform
        widths (label/value + label/value), which keeps both halves aligned and
        prevents one metric group from consuming the neighbouring group.
        """
        groups = getattr(self, "_health_metric_groups", ())
        if not groups:
            return

        for card, pairs in groups:
            for label_widget, value_widget in pairs:
                label_widget.grid_forget()
                value_widget.grid_forget()

            # Clear weights left by the other layout before applying the new
            # one.  Uniform groups are local to each LabelFrame.
            for column in range(4):
                card.columnconfigure(column, weight=0, minsize=0, uniform="")
            for row in range(max(6, len(pairs))):
                card.rowconfigure(row, weight=0, minsize=0, uniform="")

            if compact:
                # Size the four compact columns from their real content.
                # The two label columns absorb any spare width equally; value
                # columns keep their natural size.  This avoids the clipping
                # caused by forcing every card into one fixed percentage split.
                for column in (0, 2):
                    card.columnconfigure(column, weight=1, minsize=0, uniform="")
                for column in (1, 3):
                    card.columnconfigure(column, weight=0, minsize=0, uniform="")

                used_rows = max(1, (len(pairs) + 1) // 2)
                for row in range(used_rows):
                    card.rowconfigure(
                        row,
                        weight=1,
```

MBToolkit v3.4/mb_gui.py — lines 881-920

```
                        minsize=0,
                        uniform="health_metric_rows",
                    )

                pair_count = len(pairs)
                card_width = max(320, int(card.winfo_width()))
                # At normal widths the technical labels/values remain on one
                # line.  Near the application's minimum width labels may wrap
                # so the pair stays inside the LabelFrame instead of colliding
                # with the neighbouring pair.
                compact_label_wrap = (
                    0 if card_width >= 390 else max(80, int(card_width * 0.29))
                )

                for index, (label_widget, value_widget) in enumerate(pairs):
                    row = index // 2
                    group = index % 2

                    try:
                        label_widget.configure(wraplength=compact_label_wrap)
                        value_widget.configure(wraplength=0)
                    except tk.TclError:
                        pass

                    # Odd final metrics use the complete last row, leaving the
                    # value the whole right-hand side of the card.
                    if index == pair_count - 1 and pair_count % 2:
                        label_widget.grid(
                            row=row,
                            column=0,
                            sticky="w",
                            padx=(6, 3),
                            pady=1,
                        )
                        value_widget.grid(
                            row=row,
                            column=1,
                            columnspan=3,
                            sticky="e",
                            padx=(3, 6),
```

MBToolkit v3.4/mb_gui.py — lines 921-960

```
                            pady=1,
                        )
                        continue

                    base_column = group * 2
                    label_widget.grid(
                        row=row,
                        column=base_column,
                        sticky="w",
                        padx=((6 if base_column == 0 else 5), 3),
                        pady=1,
                    )
                    value_widget.grid(
                        row=row,
                        column=base_column + 1,
                        sticky="e",
                        padx=(3, (5 if base_column == 0 else 6)),
                        pady=1,
                    )
            else:
                # One pair per row, with every row sharing the card height
                # equally.  Proportional columns keep labels and values aligned
                # while still allowing long values to use the right-hand side.
                card.columnconfigure(0, weight=1, minsize=0, uniform="")
                card.columnconfigure(1, weight=0, minsize=0, uniform="")

                used_rows = max(1, len(pairs))
                for row in range(used_rows):
                    card.rowconfigure(
                        row,
                        weight=1,
                        minsize=0,
                        uniform="health_metric_rows",
                    )

                card_width = max(320, int(card.winfo_width()))
                full_label_wrap = (
                    0 if card_width >= 330 else max(100, int(card_width * 0.55))
                )

```

MBToolkit v3.4/mb_gui.py — lines 961-1000

```
                for row, (label_widget, value_widget) in enumerate(pairs):
                    try:
                        label_widget.configure(wraplength=full_label_wrap)
                        value_widget.configure(wraplength=0)
                    except tk.TclError:
                        pass
                    label_widget.grid(
                        row=row,
                        column=0,
                        sticky="w",
                        padx=(10, 6),
                        pady=1,
                    )
                    value_widget.grid(
                        row=row,
                        column=1,
                        sticky="e",
                        padx=(6, 10),
                        pady=1,
                    )

        self._health_layout_compact = bool(compact)

    def _sync_bus_health_layout(self):
        """
        Keep Bus Health metrics inside their rectangles at every available height.

        The natural/full card height is measured from the real widgets. If the
        Frame Inspector expands and the notebook becomes too short, the metrics
        reflow automatically. When enough vertical room returns, the original
        layout is restored. Hysteresis prevents resize oscillation.
        """
        self._bus_health_layout_after_id = None

        health_frame = getattr(self, "health_frame", None)
        cards = getattr(self, "_health_cards", ())
        full_required = getattr(
            self,
            "_health_full_required_card_height",
            None,
```

MBToolkit v3.4/mb_gui.py — lines 1001-1040

```
        )
        if health_frame is None or not cards:
            return

        try:
            if not health_frame.winfo_exists():
                return

            available_frame_height = health_frame.winfo_height()
            if available_frame_height <= 1:
                return

            # Each card has 6 px external top and bottom grid padding.
            available_card_height = max(0, available_frame_height - 12)

            if full_required is None:
                self.update_idletasks()
                full_required = max(card.winfo_reqheight() for card in cards)
                self._health_full_required_card_height = full_required

            compact = bool(getattr(self, "_health_layout_compact", False))
            if compact:
                should_compact = available_card_height < (full_required + 8)
            else:
                should_compact = available_card_height < (full_required + 2)

            if should_compact != compact:
                self._apply_bus_health_layout(should_compact)
                self.update_idletasks()
            else:
                # Width can change without crossing the vertical compact/full
                # threshold. Reapply the current layout so wrap limits and the
                # equal-width grid continue to match the real card size.
                self._apply_bus_health_layout(compact)

        except tk.TclError:
            return

    def _build_ui(self):
        sniffer_saved = self.ui_settings.get("sniffer", {})
```

MBToolkit v3.4/mb_gui.py — lines 1041-1080

```
        if not isinstance(sniffer_saved, dict):
            sniffer_saved = {}

        self.columnconfigure(0, weight=1)
        self.rowconfigure(0, weight=1)

        # Refresh controls use a custom Canvas widget whose arrow geometry is
        # centered explicitly; this avoids ttk theme/font/image alignment drift.

        self.main_notebook = ttk.Notebook(self)
        self.main_notebook.grid(row=0, column=0, sticky="nsew")

        self.sniffer_page = ttk.Frame(self.main_notebook)
        self.sniffer_page.columnconfigure(0, weight=1)
        self.sniffer_page.rowconfigure(1, weight=1)
        self.main_notebook.add(self.sniffer_page, text="Sniffer")

        self.discovery_page = ttk.Frame(self.main_notebook)
        self.discovery_page.columnconfigure(0, weight=1)
        self.discovery_page.rowconfigure(3, weight=1)
        self.main_notebook.add(self.discovery_page, text="Bus Slave Finder")

        self.master_page = ttk.Frame(self.main_notebook)
        self.main_notebook.add(self.master_page, text="Master")

        self.slave_page = ttk.Frame(self.main_notebook)
        self.main_notebook.add(self.slave_page, text="Slave")

        # Global Light/Dark control: root-level overlay in the top-right corner.
        # It is outside both tabs and therefore applies equally to Sniffer and
        # Bus Slave Finder. `place()` means it consumes no layout height.
        self.theme_overlay = ttk.Frame(self)
        self.theme_overlay.place(
            relx=1.0,
            x=-12,
            y=4,
            anchor="ne",
        )

        ttk.Label(
```

MBToolkit v3.4/mb_gui.py — lines 1081-1120

```
            self.theme_overlay,
            text="Modo escuro",
        ).grid(row=0, column=0, sticky="e", padx=(0, 6))

        self.dark_mode_var = tk.BooleanVar(value=self.initial_dark_mode)
        self.dark_mode_switch = ToggleSwitch(
            self.theme_overlay,
            variable=self.dark_mode_var,
            command=self.toggle_dark_mode,
            width=42,
            height=22,
        )
        self.dark_mode_switch.grid(row=0, column=1, sticky="e")

        self.language_label = ttk.Label(
            self.theme_overlay,
            text="Linguagem",
        )
        self.language_label.grid(
            row=0, column=2, sticky="e", padx=(18, 6)
        )
        self.language_var = tk.StringVar(value=self.current_language)
        self.language_combo = ttk.Combobox(
            self.theme_overlay,
            textvariable=self.language_var,
            values=LANGUAGE_CHOICES,
            state="readonly",
            width=10,
        )
        self.language_combo.grid(row=0, column=3, sticky="e")
        self.language_combo.bind(
            "<<ComboboxSelected>>", self.on_language_changed
        )
        self.theme_overlay.lift()

        # ================================================================
        # Compact top workspace: configuration on the left, controls/stats
        # on the right.  This reclaims vertical space for the traffic view.
        # ================================================================
        top_area = ttk.Frame(self.sniffer_page)
```

MBToolkit v3.4/mb_gui.py — lines 1121-1160

```
        top_area.grid(row=0, column=0, sticky="ew", padx=10, pady=(10, 5))
        # Keep the 3:2 workspace split independent of translated text widths.
        # Without a shared uniform group Tk first honours each child widget's
        # requested width, so changing language moves the two main rectangles.
        top_area.columnconfigure(0, weight=3, uniform="top_workspace")
        top_area.columnconfigure(1, weight=2, uniform="top_workspace")
        # The left Configuração card defines the natural top-workspace height.
        # The right stack stretches to exactly the same top/bottom limits.
        top_area.rowconfigure(0, weight=1)

        cfg = ttk.LabelFrame(top_area, text="Configuração")
        cfg.grid(row=0, column=0, sticky="nsew", padx=(0, 6))
        cfg.columnconfigure(0, weight=1)

        # Four compact equal columns.  Widgets keep the same fixed widths as
        # v2.6; only the card itself is narrower, eliminating wasted whitespace.
        config_grid = ttk.Frame(cfg)
        config_grid.grid(row=0, column=0, sticky="ew", padx=10, pady=(8, 4))
        for col in range(4):
            config_grid.columnconfigure(col, weight=1, uniform="sniffer_cfg")

        def config_cell(row, column):
            cell = ttk.Frame(config_grid)
            cell.grid(
                row=row, column=column, sticky="w",
                padx=(0 if column == 0 else 8, 8 if column < 3 else 0),
                pady=(0, 7),
            )
            return cell

        # Row 1 — physical mode / COM A / COM B / capture status
        cell = config_cell(0, 0)
        ttk.Label(cell, text="Modo físico").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.mode_var = tk.StringVar(value=sniffer_saved.get("physical_mode", "RS485 2-wire"))
        self.mode_combo = ttk.Combobox(
            cell, textvariable=self.mode_var, width=13, state="readonly",
            values=["RS485 2-wire", "RS232 single RX", "RS232 dual RX"],
        )
        self.mode_combo.grid(row=1, column=0, sticky="w")
        self.mode_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_mode_ui())
```

MBToolkit v3.4/mb_gui.py — lines 1161-1200

```
1161
1162        cell = config_cell(0, 1)
1163        ttk.Label(cell, text="COM A").grid(row=0, column=0, sticky="w", pady=(0, 2))
1164        com_a_controls = ttk.Frame(cell)
1165        com_a_controls.grid(row=1, column=0, sticky="w")
1166        self.port_a_var = tk.StringVar()
1167        self.port_a_combo = ttk.Combobox(
1168            com_a_controls, textvariable=self.port_a_var, width=13, state="readonly"
1169        )
1170        self.port_a_combo.grid(row=0, column=0, sticky="w")
1171        self.port_a_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_com_status())
1172        self.refresh_a_btn = RefreshButton(com_a_controls, command=self.refresh_ports, size=24)
1173        self.refresh_a_btn.grid(row=0, column=1, sticky="w", padx=(4, 0))
1174
1175        def _sync_sniffer_refresh_a_square(_event=None):
1176            height = self.port_a_combo.winfo_height()
1177            if height > 1:
1178                self.refresh_a_btn.configure(width=height, height=height)
1179
1180        self.port_a_combo.bind("<Configure>", _sync_sniffer_refresh_a_square, add="+")
1181
1182        cell = config_cell(0, 2)
1183        self.port_b_label = ttk.Label(cell, text="COM B")
1184        self.port_b_label.grid(row=0, column=0, sticky="w", pady=(0, 2))
1185        com_b_controls = ttk.Frame(cell)
1186        com_b_controls.grid(row=1, column=0, sticky="w")
1187        self.port_b_var = tk.StringVar()
1188        self.port_b_combo = ttk.Combobox(
1189            com_b_controls, textvariable=self.port_b_var, width=13, state="readonly"
1190        )
1191        self.port_b_combo.grid(row=0, column=0, sticky="w")
1192        self.port_b_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_com_status())
1193        self.refresh_b_btn = RefreshButton(com_b_controls, command=self.refresh_ports, size=24)
1194        self.refresh_b_btn.grid(row=0, column=1, sticky="w", padx=(4, 0))
1195
1196        def _sync_sniffer_refresh_b_square(_event=None):
1197            height = self.port_b_combo.winfo_height()
1198            if height > 1:
1199                self.refresh_b_btn.configure(width=height, height=height)
1200
```

MBToolkit v3.4/mb_gui.py — lines 1201-1240

```
        self.port_b_combo.bind("<Configure>", _sync_sniffer_refresh_b_square, add="+")

        cell = config_cell(0, 3)
        ttk.Label(cell, text="Estado").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.status_var = tk.StringVar(value="● Parado")
        ttk.Label(
            cell,
            textvariable=self.status_var,
            style="MBS.Status.TLabel",
            width=22,
            anchor="w",
        ).grid(
            row=1, column=0, sticky="w"
        )

        # Row 2 — serial format
        cell = config_cell(1, 0)
        ttk.Label(cell, text="Baud rate").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.baud_var = tk.StringVar(value=sniffer_saved.get("baud", "9600"))
        ttk.Combobox(
            cell, textvariable=self.baud_var, width=13,
            values=["1200", "2400", "4800", "9600", "19200", "38400", "57600", "115200"],
        ).grid(row=1, column=0, sticky="w")

        cell = config_cell(1, 1)
        ttk.Label(cell, text="Data bits").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.data_var = tk.StringVar(value=sniffer_saved.get("data_bits", "8"))
        ttk.Combobox(
            cell, textvariable=self.data_var, width=13, state="readonly",
            values=["5", "6", "7", "8"],
        ).grid(row=1, column=0, sticky="w")

        cell = config_cell(1, 2)
        ttk.Label(cell, text="Parity").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.parity_var = tk.StringVar(value=sniffer_saved.get("parity", "None"))
        ttk.Combobox(
            cell, textvariable=self.parity_var, width=13, state="readonly",
            values=["None", "Even", "Odd", "Mark", "Space"],
        ).grid(row=1, column=0, sticky="w")

```

MBToolkit v3.4/mb_gui.py — lines 1241-1280

```
        cell = config_cell(1, 3)
        ttk.Label(cell, text="Stop bits").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.stop_var = tk.StringVar(value=sniffer_saved.get("stop_bits", "1"))
        ttk.Combobox(
            cell, textvariable=self.stop_var, width=13, state="readonly",
            values=["1", "1.5", "2"],
        ).grid(row=1, column=0, sticky="w")

        # Row 3 — timing and display options
        cell = config_cell(2, 0)
        ttk.Label(cell, text="Frame gap").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.gap_mode_var = tk.StringVar(value=sniffer_saved.get("frame_gap_mode", "Auto"))
        self.gap_combo = ttk.Combobox(
            cell, textvariable=self.gap_mode_var, width=13, state="readonly",
            values=["Auto", "Manual"],
        )
        self.gap_combo.grid(row=1, column=0, sticky="w")
        self.gap_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_gap_state())

        cell = config_cell(2, 1)
        ttk.Label(cell, text="Gap manual (ms)").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.gap_ms_var = tk.StringVar(value=sniffer_saved.get("frame_gap_ms", ""))
        self.gap_entry = ttk.Entry(cell, textvariable=self.gap_ms_var, width=15, state="disabled")
        self.gap_entry.grid(row=1, column=0, sticky="w")

        cell = config_cell(2, 2)
        ttk.Label(cell, text="Visualização").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.auto_scroll_var = tk.BooleanVar(value=sniffer_saved.get("auto_scroll", True))
        ttk.Checkbutton(cell, text="Auto-scroll", variable=self.auto_scroll_var).grid(
            row=1, column=0, sticky="w"
        )

        cell = config_cell(2, 3)
        ttk.Label(cell, text="Pending timeout (s)").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.pending_timeout_var = tk.StringVar(value=sniffer_saved.get("pending_timeout_s", f"{DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS:g}"))
        # Direct numeric entry is cleaner here than tiny spin arrows. Range
        # validation still happens in get_pending_timeout_seconds().
        self.pending_timeout_entry = ttk.Entry(
            cell,
            textvariable=self.pending_timeout_var,
```

MBToolkit v3.4/mb_gui.py — lines 1281-1320

```
            width=13,
        )
        self.pending_timeout_entry.grid(row=1, column=0, sticky="w")
        ttk.Label(
            cell,
            text=(
                f"min {MIN_PENDING_REQUEST_TIMEOUT_SECONDS:g} / "
                f"default {DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS:g} / "
                f"max {MAX_PENDING_REQUEST_TIMEOUT_SECONDS:g}"
            ),
        ).grid(row=2, column=0, sticky="w", pady=(2, 0))

        # COM status + mode help: compact footer inside the configuration card.
        com_status = ttk.Frame(cfg)
        com_status.grid(row=1, column=0, sticky="w", padx=10, pady=(0, 3))
        ttk.Label(com_status, text="Estado COM").grid(
            row=0, column=0, rowspan=2, sticky="nw", padx=(0, 14)
        )
        self.port_a_status_var = tk.StringVar(value="COM A - —")
        self.port_b_status_var = tk.StringVar(value="COM B - —")
        ttk.Label(com_status, textvariable=self.port_a_status_var).grid(row=0, column=1, sticky="w")
        ttk.Label(com_status, textvariable=self.port_b_status_var).grid(
            row=1, column=1, sticky="w", pady=(1, 0)
        )

        self.mode_help_var = tk.StringVar()
        ttk.Label(cfg, textvariable=self.mode_help_var).grid(
            row=2, column=0, sticky="w", padx=10, pady=(2, 7)
        )

        # ------------------------------------------------------------------
        # Right-hand command/status panel.
        # ------------------------------------------------------------------
        side_panel = ttk.Frame(top_area)
        side_panel.grid(row=0, column=1, sticky="nsew", padx=(6, 0))
        side_panel.columnconfigure(0, weight=1)
        side_panel.rowconfigure(0, weight=0)
        side_panel.rowconfigure(1, weight=1)

        actions = ttk.LabelFrame(side_panel, text="Controlos")
```

MBToolkit v3.4/mb_gui.py — lines 1321-1360

```
        # Same top coordinate as Configuração; only the gap below Controlos
        # separates it from Estatísticas.
        actions.grid(row=0, column=0, sticky="ew", pady=(0, 6))
        actions.columnconfigure(0, weight=1, uniform="action_columns")
        actions.columnconfigure(1, weight=1, uniform="action_columns")

        self.start_btn = ttk.Button(
            actions, text="Iniciar Captura", command=self.start_capture,
            style="MBS.Primary.TButton"
        )
        self.start_btn.grid(row=0, column=0, sticky="ew", padx=(8, 4), pady=(7, 3))

        self.stop_btn = ttk.Button(
            actions, text="Parar", command=self.stop_capture, state="disabled",
            style="MBS.Toolbar.TButton"
        )
        self.stop_btn.grid(row=0, column=1, sticky="ew", padx=(4, 8), pady=(7, 3))

        self.restart_btn = ttk.Button(
            actions, text="Reiniciar COM", command=self.restart_com_port,
            style="MBS.Toolbar.TButton"
        )
        self.restart_btn.grid(row=1, column=0, sticky="ew", padx=(8, 4), pady=3)

        ttk.Button(
            actions, text="Limpar", command=self.clear_view,
            style="MBS.Toolbar.TButton"
        ).grid(row=1, column=1, sticky="ew", padx=(4, 8), pady=3)

        ttk.Button(
            actions, text="Abrir pasta de logs", command=self.open_log_folder,
            style="MBS.Toolbar.TButton"
        ).grid(row=2, column=0, columnspan=2, sticky="ew", padx=8, pady=3)

        self.sim_btn = None
        if debug_sim == 1:
            self.sim_btn = ttk.Button(
                actions, text="Simulação", command=self.run_simulation,
                style="MBS.Toolbar.TButton"
            )
```

MBToolkit v3.4/mb_gui.py — lines 1361-1400

```
            self.sim_btn.grid(row=3, column=0, columnspan=2, sticky="ew", padx=8, pady=3)
            options_row = 4
        else:
            options_row = 3

        options_bar = ttk.Frame(actions)
        options_bar.grid(
            row=options_row,
            column=0,
            columnspan=2,
            sticky="ew",
            padx=8,
            pady=(3, 7),
        )
        options_bar.columnconfigure(2, weight=1)

        self.separate_transactions_var = tk.BooleanVar(
            value=sniffer_saved.get("separate_transactions", True)
        )
        ttk.Checkbutton(
            options_bar,
            text="Separar transações",
            variable=self.separate_transactions_var,
            command=self.render_filtered_view,
        ).grid(row=0, column=0, sticky="w", padx=(0, 12))

        self.highlight_anomalies_var = tk.BooleanVar(
            value=sniffer_saved.get("highlight_results", False)
        )
        ttk.Checkbutton(
            options_bar,
            text="Realçar resultados",
            variable=self.highlight_anomalies_var,
            command=self.on_highlight_anomalies_changed,
        ).grid(row=0, column=1, sticky="w")

        self.bus_activity_var = tk.StringVar(value="BUS ○")
        ttk.Label(
            options_bar,
            textvariable=self.bus_activity_var,
```

MBToolkit v3.4/mb_gui.py — lines 1401-1440

```
            style="MBS.Status.TLabel",
            anchor="e",
            width=9,
        ).grid(row=0, column=2, sticky="e")

        stats = ttk.LabelFrame(side_panel, text="Estatísticas da sessão")
        # Expand vertically into all remaining space so the lower border always
        # lines up with the lower border of Configuração, even if Configuração
        # changes height in the future.
        stats.grid(row=1, column=0, sticky="nsew")
        for col in range(4):
            stats.columnconfigure(col, weight=1, uniform="stats")
        # Keep the eight counters centred in a fixed 4 x 2 matrix.  The
        # labels must not hug the LabelFrame border or shift with translation.
        for row in range(2):
            stats.rowconfigure(row, weight=1, uniform="stats_rows")

        self.stat_requests_var = tk.StringVar(value="Pedidos: 0")
        self.stat_responses_var = tk.StringVar(value="Respostas: 0")
        self.stat_pending_var = tk.StringVar(value="Pendentes: 0")
        self.stat_timeouts_var = tk.StringVar(value="Timeouts: 0")
        self.stat_crc_var = tk.StringVar(value="Erros de CRC: 0")
        self.stat_exceptions_var = tk.StringVar(value="Exceções: 0")
        self.stat_raw_var = tk.StringVar(value="RAW: 0")
        self.stat_slaves_var = tk.StringVar(value="Slaves: 0")

        stat_vars = [
            self.stat_requests_var,
            self.stat_responses_var,
            self.stat_pending_var,
            self.stat_timeouts_var,
            self.stat_crc_var,
            self.stat_exceptions_var,
            self.stat_raw_var,
            self.stat_slaves_var,
        ]
        for idx, var in enumerate(stat_vars):
            ttk.Label(
                stats,
                textvariable=var,
```

MBToolkit v3.4/mb_gui.py — lines 1441-1480

```
                style="MBS.Stat.TLabel",
                anchor="center",
            ).grid(
                row=idx // 4,
                column=idx % 4,
                sticky="",
                padx=6,
                pady=3,
            )

        # Notebook
        notebook = ttk.Notebook(self.sniffer_page)
        notebook.grid(row=1, column=0, sticky="nsew", padx=10, pady=(5, 8))

        parsed = ttk.Frame(notebook)
        parsed.columnconfigure(0, weight=1)
        parsed.rowconfigure(1, weight=1)
        notebook.add(parsed, text="Tráfego")

        filter_bar = ttk.Frame(parsed)
        filter_bar.grid(
            row=0,
            column=0,
            columnspan=2,
            sticky="ew",
            padx=4,
            pady=(4, 3),
        )
        filter_bar.columnconfigure(8, weight=1)
        # "Pesquisa" is wider than "Search". Keep its canonical PT cell width
        # reserved so the search Entry never moves when language changes.
        filter_bar.columnconfigure(7, minsize=65)

        ttk.Label(filter_bar, text="Filtros").grid(
            row=0, column=0, sticky="w", padx=(2, 6)
        )

        self.advanced_filter_type_var = tk.StringVar(value="Todos")
        type_filter = ttk.Combobox(
            filter_bar,
```

MBToolkit v3.4/mb_gui.py — lines 1481-1520

```
            textvariable=self.advanced_filter_type_var,
            state="readonly",
            width=11,
            values=(
                "Todos",
                "Requests",
                "Responses",
                "Exceptions",
                "CRC errors",
                "Timeouts",
                "RAW",
            ),
        )
        type_filter.grid(row=0, column=1, sticky="w", padx=(0, 8))
        self.advanced_filter_type_combo = type_filter
        type_filter.bind(
            "<<ComboboxSelected>>",
            lambda _e: self.render_filtered_view(),
        )

        ttk.Label(filter_bar, text="FC (hex)").grid(
            row=0, column=2, sticky="w", padx=(0, 4)
        )
        self.advanced_filter_fc_var = tk.StringVar(value="")
        fc_filter = ttk.Entry(
            filter_bar,
            textvariable=self.advanced_filter_fc_var,
            width=7,
        )
        fc_filter.grid(row=0, column=3, sticky="w", padx=(0, 8))

        ttk.Label(filter_bar, text="Resp. >").grid(
            row=0, column=4, sticky="w", padx=(0, 4)
        )
        self.advanced_filter_response_ms_var = tk.StringVar(value="")
        response_filter = ttk.Entry(
            filter_bar,
            textvariable=self.advanced_filter_response_ms_var,
            width=7,
        )
```

MBToolkit v3.4/mb_gui.py — lines 1521-1560

```
        response_filter.grid(row=0, column=5, sticky="w")
        ttk.Label(filter_bar, text="ms").grid(
            row=0, column=6, sticky="w", padx=(3, 8)
        )

        ttk.Label(filter_bar, text="Pesquisa").grid(
            row=0, column=7, sticky="w", padx=(0, 4)
        )
        self.advanced_filter_text_var = tk.StringVar(value="")
        search_filter = ttk.Entry(
            filter_bar,
            textvariable=self.advanced_filter_text_var,
            width=22,
        )
        search_filter.grid(row=0, column=8, sticky="ew")

        filter_actions = ttk.Frame(filter_bar)
        filter_actions.grid(row=0, column=9, sticky="e")
        ttk.Button(
            filter_actions,
            text="Limpar filtros",
            command=self.clear_advanced_filters,
            style="MBS.Compact.TButton",
        ).grid(row=0, column=0, padx=(6, 4))
        ttk.Button(
            filter_actions,
            text="Exportar CSV",
            command=self.export_session_csv,
            style="MBS.Compact.TButton",
        ).grid(row=0, column=1)

        for variable in (
            self.advanced_filter_fc_var,
            self.advanced_filter_response_ms_var,
            self.advanced_filter_text_var,
        ):
            variable.trace_add(
                "write",
                self.schedule_advanced_filter_refresh,
            )
```

MBToolkit v3.4/mb_gui.py — lines 1561-1600

```
1561
1562        cols = ("time", "delta", "response", "channel", "type", "slave", "fc", "details", "crc")
1563        # Dedicated style so wrapped multi-line rows do not affect unrelated
1564        # widgets. Treeview has one row height for the whole table, therefore
1565        # the row height is adjusted to the largest wrapped Details cell that
1566        # is currently visible.
1567        self.tree_style_name = "MBToolkit.Treeview"
1568        self.tree_style = ttk.Style(self)
1569        self.tree = ttk.Treeview(
1570            parsed,
1571            columns=cols,
1572            show="headings",
1573            selectmode="browse",
1574            style=self.tree_style_name
1575        )
1576
1577        self._tree_reflow_after_id = None
1578        self.tree_detail_min_width_px = 220
1579        self.tree_detail_border_reserve_px = 18
1580
1581        # Internal sort state. The user changes it only by clicking Hora.
1582        self.time_order_var = tk.StringVar(
1583            value="Crescente — antigo → recente"
1584        )
1585
1586        self.tree_headings = {
1587            "time": "Hora ↑",
1588            "delta": "Δt (ms)",
1589            "response": "Resp. (ms)",
1590            "channel": "COM",
1591            "type": "Tipo",
1592            "slave": "Slave ▾",
1593            "fc": "FC",
1594            "details": "Detalhes",
1595            "crc": "CRC",
1596        }
1597
1598        # Keep columns as compact as possible while leaving breathing room
1599        # between the longest text and the column borders.
1600        self.tree_column_padding_px = 28  # total horizontal padding
```

MBToolkit v3.4/mb_gui.py — lines 1601-1640

```
        self.tree_cell_font = tkfont.nametofont("TkDefaultFont")
        try:
            heading_font_name = ttk.Style(self).lookup("Treeview.Heading", "font")
            self.tree_heading_font = (
                tkfont.nametofont(heading_font_name)
                if heading_font_name
                else self.tree_cell_font
            )
        except Exception:
            self.tree_heading_font = self.tree_cell_font

        self.tree_font_linespace = max(
            1,
            int(self.tree_cell_font.metrics("linespace"))
        )
        self.tree_single_line_rowheight = max(
            20,
            self.tree_font_linespace + 5
        )
        self.tree_style.configure(
            self.tree_style_name,
            rowheight=self.tree_single_line_rowheight
        )

        for col in cols:
            align = "w" if col == "details" else "center"
            self.tree.heading(
                col,
                text=self.tree_headings[col],
                anchor=align
            )
            initial_width = self.measure_tree_column_width(
                col,
                [],
                heading_text=self.tree_headings[col]
            )
            self.tree.column(
                col,
                width=initial_width,
                minwidth=initial_width,
```

MBToolkit v3.4/mb_gui.py — lines 1641-1680

```
                anchor=align,
                stretch=(col == "details")
            )

        self.tree.heading(
            "time",
            text=self.tree_headings["time"],
            anchor="center",
            command=self.toggle_time_order
        )
        self.tree.heading(
            "slave",
            text=self.tree_headings["slave"],
            anchor="center",
            command=self.show_slave_filter_menu
        )

        # Popup menu opened directly from the Slave column heading.
        self.slave_filter_menu = tk.Menu(self, tearoff=False)

        # Right-click actions for the selected traffic frame.
        self.traffic_context_menu = tk.Menu(self, tearoff=False)
        self.traffic_context_menu.add_command(
            label="Copiar Raw Hex",
            command=self.copy_selected_raw,
        )
        self.traffic_context_menu.add_command(
            label="Copiar frame descodificado",
            command=self.copy_selected_decoded,
        )
        self.traffic_context_menu.add_separator()
        self.traffic_context_menu.add_command(
            label="Filtrar por este Slave",
            command=self.context_filter_selected_slave,
        )
        self.traffic_context_menu.add_command(
            label="Filtrar por este FC",
            command=self.context_filter_selected_fc,
        )
        self.traffic_context_menu.add_command(
```

MBToolkit v3.4/mb_gui.py — lines 1681-1720

```
            label="Mostrar apenas esta transação",
            command=self.context_show_selected_transaction,
        )
        self.traffic_context_menu.add_command(
            label="Limpar filtro de transação",
            command=self.clear_transaction_filter,
        )
        self.traffic_context_menu.add_separator()
        self.traffic_context_menu.add_command(
            label="Abrir no Master",
            command=self.context_open_selected_in_master,
        )

        ys = ttk.Scrollbar(parsed, orient="vertical", command=self.tree.yview)
        self.tree.configure(yscrollcommand=ys.set)
        self.tree.grid(row=1, column=0, sticky="nsew")
        ys.grid(row=1, column=1, sticky="ns")
        self.tree.bind("<<TreeviewSelect>>", self.on_row_select)
        self.tree.bind("<Button-3>", self.show_traffic_context_menu, add="+")
        self.tree.bind("<Configure>", self.schedule_tree_reflow)
        self.configure_anomaly_tags(self.initial_dark_mode)

        raw = ttk.Frame(notebook)
        raw.columnconfigure(0, weight=1)
        raw.rowconfigure(0, weight=1)
        notebook.add(raw, text="Raw Hex / Log")

        self.raw_text = tk.Text(
            raw,
            wrap="word",
            font=("Consolas", 10)
        )
        self.raw_text_font = tkfont.Font(font=self.raw_text.cget("font"))
        self.raw_text.bind("<Configure>", self.schedule_raw_separator_reflow)

        raw_y = ttk.Scrollbar(raw, orient="vertical", command=self.raw_text.yview)
        self.raw_text.configure(yscrollcommand=raw_y.set)
        self.raw_text.grid(row=0, column=0, sticky="nsew")
        raw_y.grid(row=0, column=1, sticky="ns")

```

MBToolkit v3.4/mb_gui.py — lines 1721-1760

```
        # Bus Health — session-level diagnostic metrics.
        self.health_frame = ttk.Frame(notebook)
        health_frame = self.health_frame
        health_frame.columnconfigure(0, weight=1, uniform="health")
        health_frame.columnconfigure(1, weight=1, uniform="health")
        health_frame.columnconfigure(2, weight=1, uniform="health")
        health_frame.rowconfigure(0, weight=1)
        notebook.add(health_frame, text="Bus Health")

        traffic_health = ttk.LabelFrame(
            health_frame,
            text="Tráfego",
        )
        traffic_health.grid(
            row=0, column=0, sticky="nsew", padx=(6, 3), pady=6
        )
        traffic_health.columnconfigure(1, weight=1)

        response_health = ttk.LabelFrame(
            health_frame,
            text="Tempos de resposta",
        )
        response_health.grid(
            row=0, column=1, sticky="nsew", padx=3, pady=6
        )
        response_health.columnconfigure(1, weight=1)

        quality_health = ttk.LabelFrame(
            health_frame,
            text="Qualidade",
        )
        quality_health.grid(
            row=0, column=2, sticky="nsew", padx=(3, 6), pady=6
        )
        quality_health.columnconfigure(1, weight=1)

        self.health_requests_var = tk.StringVar(value="0")
        self.health_responses_var = tk.StringVar(value="0")
        self.health_crc_count_var = tk.StringVar(value="0")
        self.health_timeout_count_var = tk.StringVar(value="0")
```

MBToolkit v3.4/mb_gui.py — lines 1761-1800

```
        self.health_exception_count_var = tk.StringVar(value="0")
        self.health_req_rate_var = tk.StringVar(value="0.00 req/s")
        self.health_resp_avg_var = tk.StringVar(value="—")
        self.health_resp_min_var = tk.StringVar(value="—")
        self.health_resp_max_var = tk.StringVar(value="—")
        self.health_p95_var = tk.StringVar(value="—")
        self.health_nonvalidated_rate_var = tk.StringVar(value="0.00 %")
        self.health_timeout_rate_var = tk.StringVar(value="0.00 %")
        self.health_bus_load_var = tk.StringVar(value="0.00 %")
        self.health_slaves_var = tk.StringVar(value="0")
        self.health_total_bytes_var = tk.StringVar(value="0")
        self.health_slowest_var = tk.StringVar(value="—")
        self.health_exception_slaves_var = tk.StringVar(value="—")

        self._health_metric_groups = []

        def health_item(parent, label, variable, *, wraplength=0):
            return (
                ttk.Label(parent, text=label),
                ttk.Label(
                    parent,
                    textvariable=variable,
                    anchor="e",
                    wraplength=wraplength,
                    justify="right",
                ),
            )

        traffic_pairs = [
            health_item(traffic_health, "Pedidos", self.health_requests_var),
            health_item(traffic_health, "Respostas", self.health_responses_var),
            health_item(traffic_health, "Taxa de pedidos", self.health_req_rate_var),
            health_item(traffic_health, "Slaves ativos", self.health_slaves_var),
            health_item(traffic_health, "Bytes observados", self.health_total_bytes_var),
            health_item(traffic_health, "Utilização do bus ~", self.health_bus_load_var),
        ]
        response_pairs = [
            health_item(response_health, "Média", self.health_resp_avg_var),
            health_item(response_health, "Mínimo", self.health_resp_min_var),
            health_item(response_health, "Máximo", self.health_resp_max_var),
```

MBToolkit v3.4/mb_gui.py — lines 1801-1840

```
            health_item(response_health, "P95", self.health_p95_var),
            health_item(response_health, "Slave mais lento", self.health_slowest_var),
        ]
        quality_pairs = [
            health_item(quality_health, "Erros de CRC", self.health_crc_count_var),
            health_item(quality_health, "Bytes não validados", self.health_nonvalidated_rate_var),
            health_item(quality_health, "Timeouts", self.health_timeout_count_var),
            health_item(quality_health, "Taxa de timeout", self.health_timeout_rate_var),
            health_item(quality_health, "Exceções", self.health_exception_count_var),
            health_item(
                quality_health,
                "Exceções / Slave",
                self.health_exception_slaves_var,
                wraplength=190,
            ),
        ]

        self._health_metric_groups = [
            (traffic_health, traffic_pairs),
            (response_health, response_pairs),
            (quality_health, quality_pairs),
        ]
        self._health_cards = (
            traffic_health,
            response_health,
            quality_health,
        )
        self._health_layout_compact = False
        self._health_full_required_card_height = None
        self._bus_health_layout_after_id = None

        # Start with the normal layout, then adapt only if the real available
        # height is insufficient.
        self._apply_bus_health_layout(False)

        health_frame.bind(
            "<Configure>",
            self._schedule_bus_health_layout_sync,
            add="+",
        )
```

MBToolkit v3.4/mb_gui.py — lines 1841-1880

```
        notebook.bind(
            "<<NotebookTabChanged>>",
            self._schedule_bus_health_layout_sync,
            add="+",
        )
        self.after_idle(self._initialize_bus_health_layout)

        # Help tab
        help_frame = ttk.Frame(notebook)
        help_frame.columnconfigure(0, weight=1)
        help_frame.rowconfigure(0, weight=1)
        notebook.add(help_frame, text="Ajuda / Ligações")

        self.help_text = tk.Text(
            help_frame,
            wrap="word",
            font=("Segoe UI", 10),
            padx=14,
            pady=12,
            spacing1=2,
            spacing3=5
        )
        help_scroll = ttk.Scrollbar(help_frame, orient="vertical", command=self.help_text.yview)
        self.help_text.configure(yscrollcommand=help_scroll.set)
        self.help_text.grid(row=0, column=0, sticky="nsew")
        help_scroll.grid(row=0, column=1, sticky="ns")

        self.help_text.tag_configure("title", font=("Segoe UI", 14, "bold"), spacing1=8, spacing3=8)
        self.help_text.tag_configure("heading", font=("Segoe UI", 11, "bold"), spacing1=10, spacing3=4)
        self.help_text.tag_configure("subheading", font=("Segoe UI", 10, "bold"), spacing1=6, spacing3=2)
        self.help_text.tag_configure("mono", font=("Consolas", 10), lmargin1=18, lmargin2=18)
        self.help_text.tag_configure("warning", font=("Segoe UI", 10, "bold"), lmargin1=12, lmargin2=12)
        self.help_text.tag_configure("bullet", lmargin1=12, lmargin2=28)

        def h(text_value, tag=None):
            self.help_text.insert("end", text_value, tag)

        max_ui_frames_text = f"{MAX_UI_FRAMES:_}".replace("_", " ")
        max_raw_lines_text = f"{MAX_RAW_TEXT_LINES:_}".replace("_", " ")

```

MBToolkit v3.4/mb_gui.py — lines 1881-1920

```
        h("MBToolkit — Ajuda e ligações\n", "title")
        h(
            "O MBToolkit foi pensado para diagnóstico de comunicações Modbus RTU. "
            "O separador Sniffer é passivo: abre a(s) porta(s) COM para receção e "
            "não transmite pedidos. O separador Bus Slave Finder é uma ferramenta "
            "ativa de pesquisa e deve ser utilizado apenas quando não existe outro "
            "master ativo no mesmo barramento.\n"
        )

        h("Início rápido — Sniffer\n", "heading")
        h("1. Escolhe o modo físico correto: RS485 2-wire, RS232 single RX ou RS232 dual RX.\n", "bullet")
        h("2. Seleciona a(s) porta(s) COM e os parâmetros série da comunicação observada.\n", "bullet")
        h("3. Mantém Frame gap em Auto, salvo quando tens uma razão concreta para o ajustar.\n", "bullet")
        h("4. Clica em “Iniciar Captura” e confirma atividade através de BUS ● RX.\n", "bullet")
        h("5. Usa Tráfego para diagnóstico rápido e Raw Hex / Log para confirmar os bytes reais.\n", "bullet")
        h("6. Seleciona um frame para obter a análise detalhada no Frame Inspector.\n", "bullet")

        h("1. RS485 2-wire\n", "heading")
        h(
            "Neste modo uma única COM observa pedidos e respostas no mesmo par diferencial. "
            "O adaptador do sniffer deve ser ligado em paralelo e não deve atuar como master.\n"
        )
        h("Ligação típica:\n", "subheading")
        h(
            "        Equipamento A          Equipamento B\n"
            "             D+ --------------- D+\n"
            "             D- --------------- D-\n"
            "               \\               /\n"
            "                \\             /\n"
            "                 D+           D-\n"
            "                  USB-RS485 sniffer\n\n",
            "mono"
        )
        h("• Liga D+ do sniffer em paralelo com D+ do barramento.\n", "bullet")
        h("• Liga D− do sniffer em paralelo com D− do barramento.\n", "bullet")
        h(
            "• Se existir C/COM/0 V de referência, liga-o apenas quando souberes que "
            "corresponde à referência correta da interface RS485.\n",
            "bullet"
        )
```

MBToolkit v3.4/mb_gui.py — lines 1921-1960

```
        h(
            "• Não acrescentes uma terceira resistência de 120 Ω se o barramento já "
            "estiver terminado nas duas extremidades.\n",
            "warning"
        )
        h(
            "• Se o adaptador USB-RS485 tiver terminação ou bias selecionáveis, deixa-os "
            "desativados durante uma captura passiva, salvo se forem intencionalmente "
            "necessários na instalação.\n",
            "bullet"
        )
        h(
            "• A/B não é uma nomenclatura universal: fabricantes diferentes podem usar "
            "polaridades opostas. Dá prioridade a D+/D− e à documentação do equipamento.\n",
            "bullet"
        )

        h("2. RS232 single RX\n", "heading")
        h(
            "Observa apenas uma direção RS232. O RS232 é full-duplex: cada direção utiliza "
            "uma linha TX independente, pelo que uma única entrada RX não consegue observar "
            "simultaneamente os dois sentidos.\n"
        )
        h(
            "        Equipamento A                    Equipamento B\n"
            "             TX ----------------------------> RX\n"
            "              \\\n"
            "               \\----> RX do USB-RS232 sniffer (COM A)\n"
            "\n"
            "             GND --------------------------- GND\n"
            "               \\---------------------------> GND sniffer\n\n"
            "        TX do sniffer: NÃO LIGAR\n\n",
            "mono"
        )
        h("• Liga RX do sniffer ao TX da direção que pretendes observar.\n", "bullet")
        h("• Liga também o GND comum.\n", "bullet")
        h("• Deixa fisicamente desligado o TX do adaptador usado como sniffer.\n", "warning")

        h("3. RS232 dual RX\n", "heading")
        h(
```

MBToolkit v3.4/mb_gui.py — lines 1961-2000

```
            "Permite observar as duas direções RS232 ao mesmo tempo. São necessários "
            "dois canais RX independentes, normalmente dois adaptadores USB-RS232.\n"
        )
        h(
            "        Equipamento A                    Equipamento B\n"
            "             TX ---------------------------> RX\n"
            "              \\----> RX sniffer COM A   (A → B)\n"
            "\n"
            "             RX <----------------------------- TX\n"
            "                                   \\----> RX sniffer COM B   (B → A)\n"
            "\n"
            "             GND --------------------------- GND\n"
            "               \\--------------------------- GND COM A\n"
            "                \\--------------------------- GND COM B\n\n"
            "        TX de ambos os sniffers: NÃO LIGAR\n\n",
            "mono"
        )
        h("• COM A é apresentada como A→B e COM B como B→A.\n", "bullet")
        h("• As duas COM devem usar os mesmos parâmetros série da ligação observada.\n", "bullet")

        h("RS232 não é TTL\n", "heading")
        h(
            "Não ligues sinais RS232 diretamente a interfaces USB-TTL/UART de 3,3 V ou "
            "5 V. Usa um adaptador USB-RS232 com transceiver apropriado aos níveis RS232.\n",
            "warning"
        )

        h("Parâmetros da comunicação série\n", "heading")
        h(
            "Baud, Data bits, Parity e Stop bits têm de coincidir com a comunicação real. "
            "Parâmetros errados podem produzir bytes aparentemente aleatórios, frames com "
            "CRC inválido ou ausência total de frames reconhecíveis.\n"
        )
        h("• Baud rate — velocidade em bit/s, por exemplo 9600 ou 19200.\n", "bullet")
        h("• Data bits — normalmente 8 em Modbus RTU.\n", "bullet")
        h("• Parity — None, Even, Odd, Mark ou Space, conforme a instalação.\n", "bullet")
        h("• Stop bits — 1, 1.5 ou 2, conforme a configuração observada.\n", "bullet")

        h("Frame gap\n", "heading")
        h(
```

MBToolkit v3.4/mb_gui.py — lines 2001-2040

```
            "Em Auto, o programa usa 3,5 tempos de carácter até 19200 bit/s, inclusive. "
            "Acima de 19200 bit/s usa o t3.5 fixo de 1,750 ms recomendado para Modbus RTU. "
            "O modo Manual permite definir outro valor em milissegundos para diagnóstico "
            "de equipamentos ou drivers com comportamento particular.\n"
        )
        h(
            "Os adaptadores USB e o Windows podem agrupar bytes e introduzir latência. "
            "Por isso, os timestamps do MBToolkit são úteis para diagnóstico de sequência "
            "e tempos relativos, mas não substituem um analisador lógico quando são "
            "necessárias medições temporais de elevada precisão.\n"
        )
        h(
            "Quando um burst termina com o início de um frame conhecido ainda incompleto, "
            f"o MBToolkit espera até {SNIFFER_SPLIT_FRAME_HOLD_MS:g} ms adicionais antes de o "
            "fechar. Assim, um frame partido em dois pelo adaptador USB não aparece como RAW. "
            "Frames juntos no mesmo burst mantêm a hora de chegada do seu próprio bloco de bytes.\n"
        )

        h("Pending timeout\n", "heading")
        h(
            "Define durante quanto tempo um REQUEST CRC-válido pode permanecer à espera de "
            "uma RESPONSE/EXCEPTION emparelhável. Ao expirar, o REQUEST original é marcado "
            "como timeout e aumenta os contadores de Timeout/Timeout rate.\n"
        )
        h(
            f"Intervalo permitido: {MIN_PENDING_REQUEST_TIMEOUT_SECONDS:g} a "
            f"{MAX_PENDING_REQUEST_TIMEOUT_SECONDS:g} s. O valor deve ser superior ao "
            "tempo de resposta máximo normal da instalação para evitar falsos timeouts.\n"
        )
        h(
            "Pedidos broadcast (Slave 0) não têm resposta por definição: não ficam pendentes, "
            "nunca contam como timeout e não entram no cálculo da Timeout rate.\n"
        )

        h("Como interpretar a tabela Tráfego\n", "heading")
        h("• REQUEST — pedido Modbus RTU reconhecido pelo parser.\n", "bullet")
        h("• RESPONSE — resposta Modbus RTU reconhecida que não é uma EXCEPTION.\n", "bullet")
        h(
            "• FC05/FC06/FC08/FC22 — o pedido e a resposta têm o mesmo formato. A trama é "
            "RESPONSE quando existe um pedido pendente do mesmo Slave/FC (em FC05/06/22 os "
```

MBToolkit v3.4/mb_gui.py — lines 2041-2080

```
            "bytes têm de ser iguais; em RS232 dual RX tem de vir do canal oposto). Caso "
            "contrário é REQUEST.\n",
            "bullet"
        )
        h(
            "• Respostas FC01/FC02 mostram os bits por ordem de endereço, agrupados por byte. "
            "No Frame Inspector, as respostas de leitura emparelhadas (FC01–FC04) mostram o "
            "PDU Address de cada valor.\n",
            "bullet"
        )
        h(
            "• EXCEPTION — resposta Modbus com Function Code de exceção (bit 7 ativo). "
            "Detalhes mostra o código e a descrição normalizada.\n",
            "bullet"
        )
        h("• CRC OK — CRC recebido coincide com o CRC calculado.\n", "bullet")
        h(
            "• CRC ERROR — a estrutura parece um frame suportado, mas o CRC recebido não "
            "coincide com o CRC calculado.\n",
            "bullet"
        )
        h(
            "• RAW/UNSYNC — bytes descartados durante uma recuperação de sincronismo antes "
            "de ser encontrado, mais à frente no mesmo burst, um frame de estrutura "
            "reconhecida com CRC válido.\n",
            "bullet"
        )
        h(
            "• RAW/UNPARSED — restante conjunto de bytes que não pôde ser interpretado nem "
            "ressincronizado como um frame Modbus válido. É este estado que aparece, por "
            "exemplo, num bloco de ruído sem um frame válido posterior no mesmo burst.\n",
            "bullet"
        )
        h(
            "RAW/UNSYNC e RAW/UNPARSED são portanto situações diferentes e ambas podem "
            "aparecer na coluna Tipo. Nenhuma delas representa uma resposta Modbus válida.\n"
        )

        h("Colunas da tabela Tráfego\n", "heading")
        h("• Hora — timestamp do início do burst/frame processado.\n", "bullet")
```

MBToolkit v3.4/mb_gui.py — lines 2081-2120

```
        h("• Δt — intervalo desde o frame anterior observada.\n", "bullet")
        h(
            "• Resp. — tempo entre um REQUEST e a RESPONSE/EXCEPTION emparelhada. "
            "Fica vazio quando não existe um par válido.\n",
            "bullet"
        )
        h("• COM — identifica o canal de captura: BUS em RS485, A→B/B→A em RS232 dual RX e SIM no modo de desenvolvimento. Não representa necessariamente o número COM do Windows.\n", "bullet")
        h("• Tipo — REQUEST, RESPONSE, EXCEPTION ou um estado RAW.\n", "bullet")
        h("• Slave — Unit/Slave ID Modbus quando identificável.\n", "bullet")
        h("• FC — Function Code hexadecimal observado no frame.\n", "bullet")
        h("• Detalhes — endereço, quantidade, dados ou descrição da operação.\n", "bullet")
        h("• CRC — OK, ERROR ou ? quando não existe um frame Modbus completo.\n", "bullet")

        h("Emparelhamento REQUEST / RESPONSE\n", "heading")
        h(
            "Modbus RTU não possui Transaction ID. O MBToolkit emparelha frames por "
            "(Slave ID, Function Code) em ordem FIFO. Isto é apropriado para tráfego RTU "
            "normal, mas pode tornar-se ambíguo se existirem pedidos sobrepostos com o mesmo "
            "Slave e FC antes das respostas correspondentes.\n"
        )
        h(
            "Uma EXCEPTION usa internamente o Function Code base para emparelhamento. "
            "Exemplo: uma resposta 0x83 é associada a um REQUEST FC03 quando existe um "
            "pedido pendente compatível.\n"
        )

        h("PDU Address, 1-based e Qty\n", "heading")
        h(
            "PDU Address é o endereço codificado no pedido Modbus. 1-based é simplesmente "
            "PDU Address + 1 e serve para comparação com manuais que numeram o primeiro "
            "registo/coil como 1. Qty é a quantidade pedida.\n"
        )
        h(
            "Estes campos pertencem ao REQUEST. Uma RESPONSE não repete automaticamente "
            "Address ou Qty no protocolo, por isso o MBToolkit não inventa esses campos na "
            "linha da resposta.\n"
        )

        h("Códigos de exceção Modbus\n", "heading")
        h("• 0x01 — Illegal Function: função não suportada ou não permitida.\n", "bullet")
```

MBToolkit v3.4/mb_gui.py — lines 2121-2160

```
        h("• 0x02 — Illegal Data Address: endereço/gama de endereços inválida.\n", "bullet")
        h("• 0x03 — Illegal Data Value: valor, quantidade ou campo inválido.\n", "bullet")
        h("• 0x04 — Server Device Failure: falha interna do dispositivo.\n", "bullet")
        h("• 0x05 — Acknowledge: pedido aceite; processamento ainda em curso.\n", "bullet")
        h("• 0x06 — Server Device Busy: dispositivo temporariamente ocupado.\n", "bullet")
        h("• 0x08 — Memory Parity Error.\n", "bullet")
        h("• 0x0A — Gateway Path Unavailable.\n", "bullet")
        h("• 0x0B — Gateway Target Device Failed to Respond.\n", "bullet")

        h("Filtros avançados\n", "heading")
        h(
            "A barra no topo de Tráfego permite combinar vários filtros sem alterar os "
            "dados originais da sessão:\n"
        )
        h("• Tipo — Todos, Requests, Responses, Exceptions, CRC errors, Timeouts ou RAW.\n", "bullet")
        h("• FC (hex) — por exemplo 03, 04, 10 ou 0x04.\n", "bullet")
        h("• Resp. > — mostra apenas respostas acima do limite indicado em ms.\n", "bullet")
        h(
            "• Pesquisa — procura texto em Detalhes, Raw Hex, Slave, FC, Tipo e COM/canal.\n",
            "bullet"
        )
        h(
            "O filtro Slave continua disponível no cabeçalho da coluna e combina-se com "
            "os filtros avançados. “Limpar filtros” repõe todos os filtros de visualização. "
            "Os filtros não alteram as estatísticas nem o conteúdo do log de captura.\n"
        )

        h("Realçar resultados\n", "heading")
        h(
            "Esta opção vem desligada por defeito para manter a tabela neutra. Quando está "
            "ativa, os resultados passam a ser diferenciados por cor: uma transação normal "
            "concluída com RESPONSE CRC-válida fica verde; CRC error, timeout, Exception, "
            "RAW e resposta lenta mantêm as respetivas cores de diagnóstico.\n"
        )
        h(
            "Numa transação normal concluída com sucesso, tanto o REQUEST como a RESPONSE "
            "correspondente ficam verdes. Um REQUEST ainda pendente permanece neutro. "
            "Uma resposta lenta continua com a cor de resposta lenta, mesmo sendo válida, "
            "para não esconder o resultado de diagnóstico.\n"
        )
```

MBToolkit v3.4/mb_gui.py — lines 2161-2200

```
        h(
            f"Uma resposta é considerada lenta a partir de "
            f"{SLOW_RESPONSE_THRESHOLD_MS:g} ms. Este limite é uma referência visual do "
            "MBToolkit; deve ser interpretado no contexto do equipamento e do ciclo de poll.\n"
        )

        h("Código de cores dos resultados\n", "heading")
        h("• Verde — Sucesso: REQUEST/RESPONSE normal, emparelhada e com CRC válido.\n", "bullet")
        h(
            f"• Amarelo — Resposta lenta: tempo de resposta >= "
            f"{SLOW_RESPONSE_THRESHOLD_MS:g} ms.\n",
            "bullet"
        )
        h("• Vermelho — CRC Error ou resposta Modbus Exception.\n", "bullet")
        h("• Laranja — Timeout: REQUEST expirou sem resposta emparelhada.\n", "bullet")
        h("• Roxo — RAW/UNSYNC ou RAW/UNPARSED: bytes sem frame Modbus válido.\n", "bullet")
        h(
            "Com “Realçar resultados” desligado, a tabela permanece neutra. O Frame "
            "Inspector continua a indicar a classificação do resultado em texto; quando "
            "o destaque está ligado, o campo Resultado também usa a respetiva cor.\n"
        )

        h("Frame Inspector\n", "heading")
        h(
            "Seleciona uma linha de Tráfego para analisar o frame sem ter de interpretar "
            "manualmente todos os bytes. O Inspector mostra Slave, Tipo, Function Code + "
            "nome, tempo de resposta, Resultado, PDU Address, 1-based, "
            "Qty, ByteCount, CRC recebido, CRC calculado, dados/registos/Exception e Raw Hex.\n"
        )
        h(
            "“Recolher” esconde o conteúdo do Inspector e devolve espaço vertical à tabela. "
            "“Expandir” volta a apresentar a análise.\n"
        )

        h("Menu de contexto do Tráfego\n", "heading")
        h("Clica com o botão direito numa frame real para:\n")
        h("• Copiar Raw Hex.\n", "bullet")
        h("• Copiar o frame descodificado.\n", "bullet")
        h("• Filtrar pelo Slave dessa frame.\n", "bullet")
        h("• Filtrar pelo Function Code dessa frame.\n", "bullet")
```

MBToolkit v3.4/mb_gui.py — lines 2201-2240

```
2201        h("• Mostrar apenas a transação REQUEST/RESPONSE correspondente.\n", "bullet")
2202        h("• Limpar o filtro temporário de transação.\n", "bullet")
2203
2204        h("Bus Health\n", "heading")
2205        h(
2206            "O separador Bus Health resume a qualidade e o desempenho da sessão sem "
2207            "substituir a tabela de frames.\n"
2208        )
2209        h("• Pedidos / Respostas — contadores da sessão.\n", "bullet")
2210        h("• Taxa de pedidos — número médio de pedidos observados por segundo.\n", "bullet")
2211        h("• Slaves ativos — IDs Modbus válidos observados na sessão.\n", "bullet")
2212        h("• Bytes observados — soma dos bytes entregues ao parser.\n", "bullet")
2213        h(
2214            "• Utilização bus ~ — estimativa com base em bytes observados, baud, data bits, "
2215            "parity e stop bits. É aproximada e depende do ponto de captura.\n",
2216            "bullet"
2217        )
2218        h("• Média / Mínimo / Máximo — tempos de resposta emparelhados.\n", "bullet")
2219        h("• P95 — 95% das respostas medidas ficaram abaixo ou iguais a este valor.\n", "bullet")
2220        h(
2221            "• Slave mais lento — Slave com maior tempo médio de resposta entre as amostras "
2222            "observadas.\n",
2223            "bullet"
2224        )
2225        h("• Bytes não validados — percentagem dos bytes observados que pertencem a frames com CRC inválido ou a blocos RAW. Não prova, por si só, a existência de ruído elétrico
    ; parâmetros série errados, captura incompleta ou funções não suportadas pelo parser também podem produzir RAW.\n", "bullet")
2226        h("• Taxa de timeout — percentagem de REQUEST que expiraram sem resposta emparelhada.\n", "bullet")
2227        h("• Exceções / Slave — distribuição das respostas Exception por Slave ID.\n", "bullet")
2228
2229        h("Indicador BUS RX\n", "heading")
2230        h(
2231            "BUS ● RX pulsa brevemente quando chega um lote de frames à interface. Em repouso "
2232            "é apresentado BUS ○. Serve apenas como confirmação visual de atividade recebida; "
2233            "não representa qualidade elétrica nem garante que os frames sejam válidos.\n"
2234        )
2235
2236        h("Raw Hex / Log\n", "heading")
2237        h(
2238            "Mostra os bytes capturados em hexadecimal com informação de contexto. As linhas "
2239            "longas fazem wrap à largura disponível. A vista acompanha os mesmos filtros "
2240            "visuais usados em Tráfego; o ficheiro TXT em disco, quando existe, mantém a "
```

MBToolkit v3.4/mb_gui.py — lines 2241-2280

```
            "captura completa e não é truncado pelos filtros.\n"
        )

        h("Exportar CSV\n", "heading")
        h(
            "Exportar CSV grava os frames atualmente retidos no histórico da interface em "
            "UTF-8 compatível com Excel. Inclui Time, Δt, Response time, Channel, Type, "
            "Slave, FC, Details, CRC, Raw, PDU Address, 1-based, Qty, ByteCount, Timeout "
            "e classe de anomalia.\n"
        )
        h(
            f"A interface mantém aproximadamente os últimos {max_ui_frames_text} frames. "
            "Por isso, em sessões maiores o CSV representa o histórico ainda retido pela GUI, "
            "enquanto o TXT de uma captura real continua a ser o registo completo em disco.\n"
        )

        h("Logs de captura\n", "heading")
        h(
            "O log TXT só é criado depois de chegar tráfego real. Iniciar e parar sem receber "
            "frames não cria um ficheiro vazio.\n"
        )
        h("Pasta dos logs:\n")
        h("        <pasta da aplicação>\\MBToolkit Logs\n", "mono")
        h(
            "Os filtros, ordenação, Realçar resultados e o Frame Inspector são apenas funções "
            "de visualização e não alteram os frames gravados no log.\n"
        )

        h("Limites do histórico visível\n", "heading")
        h(
            f"Para manter a aplicação responsiva, a interface conserva aproximadamente "
            f"{max_ui_frames_text} frames e {max_raw_lines_text} linhas de Raw Hex. "
            "A remoção é feita por blocos. O log TXT de captura, quando ativo, é independente "
            "deste limite.\n"
        )

        h("Estado e portas COM\n", "heading")
        h(
            "Estado COM distingue uma porta aberta pelo MBToolkit, uma porta apenas detetada "
            "pelo Windows e uma porta que deixou de ser detetada. O estado é atualizado "
```

MBToolkit v3.4/mb_gui.py — lines 2281-2320

```
            "periodicamente.\n"
        )
        h(
            "Reiniciar COM fecha e volta a abrir a(s) porta(s) usada(s) pelo Sniffer. "
            "Isto reinicia o handle da porta série dentro do MBToolkit; não reinicia o "
            "driver USB/COM do Windows.\n"
        )
        h(
            "Os botões ↻ ao lado das COM atualizam a enumeração de portas disponível no "
            "Sniffer e no Bus Slave Finder.\n"
        )

        h("Bus Slave Finder — pesquisa ativa\n", "heading")
        h(
            "O Bus Slave Finder transmite pedidos Modbus RTU, por isso não é uma função "
            "passiva. Antes de iniciar, confirma que não existe outro master ativo no mesmo "
            "barramento.\n"
        )
        h(
            "Seleciona COM, baud rates, paridades, stop bits e intervalo de Slave IDs. "
            "Data bits é fixo em 8. A pesquisa tenta FC03 Address 0 Qty 1; uma resposta "
            "normal ou uma Exception Modbus com CRC válido confirma que o Slave existe.\n"
        )
        h(
            "FC04 fallback é opcional e aumenta o número máximo de tentativas. O timeout "
            "mínimo é adaptado ao baud rate. Um write timeout significa falha ao transmitir "
            "pela porta/driver e não deve ser interpretado simplesmente como “slave sem resposta”.\n"
        )
        h(
            "Tentativas (1 a 3) repete cada pedido sem resposta antes de passar ao Slave "
            "seguinte. Antes da primeira transmissão, a pesquisa escuta o barramento durante "
            f"{DISCOVERY_IDLE_CHECK_SECONDS:g} s; se receber qualquer byte, não transmite e "
            "indica que pode existir outro master ativo.\n"
        )
        h(
            "Device Identification (FC43/14) é opcional. Quando ativo, só é enviado depois "
            "de um Slave ter sido encontrado e tenta ler Basic Device Identification: "
            "VendorName, ProductCode e MajorMinorRevision. Uma Exception ou ausência de "
            "resposta a FC43/14 não invalida a descoberta do Slave.\n"
        )
```

MBToolkit v3.4/mb_gui.py — lines 2321-2360

```
        h(
            "O Finder e o Sniffer partilham o bloqueio global de atividade para impedir que "
            "a mesma aplicação faça pesquisa ativa e captura ao mesmo tempo.\n"
        )

        h("Master — leitura e escrita como master Modbus RTU\n", "heading")
        h(
            "O separador Master transmite pedidos: só deve ser usado quando não existe "
            "outro master ativo no barramento. Ao ligar, a aplicação escuta o barramento "
            f"durante {DISCOVERY_IDLE_CHECK_SECONDS:g} s e não transmite se detetar tráfego. "
            "Enquanto o Master está ligado, o Sniffer, o Bus Slave Finder e o Slave ficam "
            "bloqueados.\n"
        )
        h(
            "Cada janela lê uma zona de memória de um Slave: Slave ID, função (FC01 Coils, "
            "FC02 Discrete Inputs, FC03 Holding Registers ou FC04 Input Registers), endereço, "
            "quantidade e scan rate. As alterações só valem depois de “Aplicar”, que fica "
            "destacado a azul enquanto houver campos por aplicar. A tabela mostra até 15 "
            "linhas; acima disso tem scroll.\n"
        )
        h(
            "Todas as janelas partilham a mesma porta COM: o Modbus RTU só admite um pedido "
            "de cada vez, por isso as janelas são servidas à vez conforme o scan rate. "
            "Escritas e “Ler 1x” passam à frente das leituras cíclicas. Entre dois pedidos "
            "fica sempre o “Atraso entre pedidos” (no mínimo t3.5).\n"
        )
        h(
            "Formatos: 16 bits com e sem sinal, hex e binário; 32 bits com e sem sinal e "
            "float, nas ordens de bytes ABCD (big-endian), CDAB (palavras trocadas), BADC "
            "(bytes trocados) e DCBA (little-endian); 64 bits com e sem sinal e double, nas "
            "ordens ABCDEFGH, GHEFCDAB, BADCFEHG e HGFEDCBA. Os valores de 32 e 64 bits "
            "ocupam 2 e 4 registos e aparecem na linha do primeiro.\n"
        )
        h(
            "O campo “Formato” é o formato por defeito da janela. Com clique direito num ou "
            "mais endereços (arrastar com o rato, ou Ctrl/Shift, para selecionar vários) escolhe-se o formato só "
            "desses endereços: os de 32 bits exigem 2 registos seguidos e os de 64 bits 4 "
            "(ou um múltiplo, aplicado em grupos). “Repor formato da janela” volta ao "
            "formato por defeito. O mesmo menu tem “Copiar” (também Ctrl+C), que copia "
            "endereços, nomes e valores para colar no Excel, e “Base de endereços”: 0 mostra "
```

MBToolkit v3.4/mb_gui.py — lines 2361-2400

```
            "o PDU Address; 1 numera a partir de 1, como em muitos manuais.\n"
        )
        h(
            "Quando uma leitura falha (Timeout, Exception ou resposta inválida), os valores "
            "ficam a cinzento até voltar a haver uma leitura correta; ao desligar também.\n"
        )
        h(
            "Escala: no clique direito, “Escala…” converte o valor bruto em unidades de "
            "engenharia com dois pontos (Bruto mín./máx. → Eng. mín./máx.), com casas "
            "decimais e unidade (por exemplo 2000–10000 → −180–180 °). Na escrita, o valor "
            "é escrito em unidades e convertido para bruto. “Remover escala” volta ao valor "
            "bruto. As escalas ficam guardadas no workspace.\n"
        )
        h(
            "“Abrir no Master”: no Tráfego do Sniffer (clique direito num pedido FC01–FC04 "
            "ou na resposta correspondente) e nos resultados do Bus Slave Finder (clique "
            "direito ou duplo clique) cria uma janela do Master com o mesmo Slave, função, "
            "endereço e quantidade, e copia os parâmetros série para a ligação do Master "
            "(se não estiver ligado).\n"
        )
        h(
            "Se a porta COM falhar durante a ligação (por exemplo, o adaptador USB foi "
            "desligado), o Master não para: tenta voltar a abrir a porta a cada 2 s e "
            "retoma as leituras quando a porta volta. Para parar, carregar em “Desligar”.\n"
        )
        h(
            "Escrita: duplo clique num valor de uma janela FC01 ou FC03. Coils com FC05 ou "
            "FC15, registos com FC06 ou FC16; vários valores separados por “;” são escritos "
            "em endereços consecutivos com FC15/FC16. Slave 0 envia em broadcast, sem "
            "resposta. Depois de uma escrita aceite, a janela é lida de novo.\n"
        )
        h(
            "Cada janela mostra Tx, erros, timeouts, o último tempo de resposta e o último "
            "erro (Timeout, Exception ou resposta inválida). No cabeçalho de cada janela, "
            "“Renomear” muda o nome da janela (também com duplo clique no nome), "
            "“Destacar” abre-a à parte do separador (“Ancorar” ou fechar essa janela "
            "devolve-a) e “×” fecha a janela.\n"
        )
        h(
            "O menu “Workspace” guarda a ligação, todas as janelas, os nomes e os formatos "
```

MBToolkit v3.4/mb_gui.py — lines 2401-2440

```
            "com um nome à escolha, e lista os workspaces guardados para os abrir ou apagar "
            "diretamente. Os ficheiros ficam em %APPDATA%\\MBToolkit\\Workspaces\\Master, junto das "
            "preferências.\n"
        )
        h(
            "O último workspace aberto ou guardado é reaberto ao arrancar o programa, sem "
            "ligar ao barramento. Enquanto houver alterações por guardar, aparece "
            "“(alterado)” ao lado do nome; ao fechar o programa nessa situação, pergunta se "
            "quer guardar. Janelas que nunca foram guardadas num workspace fecham sem "
            "perguntar, e fechar todas as janelas deixa de ter workspace aberto.\n"
        )

        h("Slave — simulador de equipamentos Modbus RTU\n", "heading")
        h(
            "O separador Slave faz a aplicação comportar-se como um ou mais equipamentos Modbus "
            "RTU: fica à escuta na porta COM e responde aos pedidos de um master real (um PLC, um "
            "SCADA ou outro Modbus Poll). Serve para testar o master sem o equipamento. Responde "
            "a FC01–FC06, FC15 e FC16.\n"
        )
        h(
            "Cada janela é uma tabela de um Slave simulado: Slave ID, tabela (Coils, Discrete "
            "Inputs, Holding Registers ou Input Registers), endereço inicial e quantidade (até "
            "2000). Podem existir vários Slave IDs, cada um com várias janelas. Só existem os "
            "endereços definidos nas janelas: um pedido fora deles recebe a Exception 02 (Illegal "
            "Data Address), uma função não suportada a Exception 01 e uma quantidade ou valor "
            "inválido a Exception 03, como num equipamento real. Pedidos para outros Slave IDs "
            "são ignorados.\n"
        )
        h(
            "Os valores começam a 0. Duplo clique num valor altera-o (nas Coils e Discrete Inputs "
            "alterna entre 0 e 1); as alterações valem logo, sem passar pelo barramento. Quando o "
            "master escreve (FC05/06/15/16), os registos escritos ficam destacados a amarelo "
            "durante 1,5 s. Janelas sobre os mesmos endereços mostram os mesmos valores. Pedidos "
            "broadcast (Slave 0) de escrita são executados em todos os Slaves simulados que têm "
            "esses endereços, sem resposta.\n"
        )
        h(
            "Formatos por endereço, escala, nomes, Copiar, Base de endereços, seleção com o rato, "
            "Renomear, Destacar e × funcionam como no Master.\n"
        )
```

MBToolkit v3.4/mb_gui.py — lines 2441-2480

```
        h(
            "Simulação de falhas: “Falhas…” configura falhas para o Slave ID da janela, que valem "
            "para todas as janelas desse ID: atraso na resposta (ms), não responder, responder "
            "sempre com uma Exception à escolha (01, 02, 03, 04 ou 06), CRC errado nas respostas "
            "e a probabilidade (%) de a falha acontecer em cada pedido. Uma falha ativa aparece a "
            "vermelho no cabeçalho das janelas desse Slave. “Sem falhas” desliga-as.\n"
        )
        h(
            "A barra de ligação mostra os pedidos recebidos, as respostas, as exceções, os "
            "broadcasts e os pedidos para outros IDs; cada janela mostra os pedidos e exceções "
            "sobre os seus endereços e o último pedido. O eco do adaptador RS485 é ignorado. "
            "Antes de responder, o Slave espera o silêncio t3.5; o Windows e o adaptador USB "
            "juntam normalmente alguns ms, por isso um master com timeout muito curto pode dar "
            "timeouts.\n"
        )
        h(
            "Antes de ligar, confirma que nenhum equipamento real no barramento usa os mesmos "
            "Slave IDs: se dois responderem ao mesmo pedido, as respostas corrompem-se. O Slave "
            "nunca funciona ao mesmo tempo que o Sniffer, o Bus Slave Finder ou o Master. Se a "
            "porta falhar, tenta voltar a abri-la a cada 2 s, como o Master.\n"
        )
        h(
            "O menu “Workspace” do Slave guarda só a configuração: ligação, janelas, nomes, "
            "formatos, escalas e falhas. Os valores dos registos não são guardados e voltam a 0 "
            "ao abrir um workspace. Os ficheiros ficam em %APPDATA%\\MBToolkit\\Workspaces\\Slave; o "
            "último workspace do Slave também é reaberto ao arrancar.\n"
        )

        h("Preferências guardadas\n", "heading")
        h(
            "As preferências do utilizador são guardadas em "
            "%APPDATA%\\MBToolkit\\settings.json. Incluem Light/Dark, último separador, "
            "parâmetros série, Frame gap, Pending timeout, Auto-scroll, Separar transações, "
            "Realçar resultados, opções do Bus Slave Finder e parâmetros de ligação do "
            "Master e do Slave. As janelas do Master e do Slave guardam-se em "
            "workspaces, na pasta %APPDATA%\\MBToolkit\\Workspaces.\n"
        )
        h(
            "As portas COM não são guardadas porque são específicas de cada computador. "
            "A confirmação de segurança do Bus Slave Finder também não é persistida.\n"
```

MBToolkit v3.4/mb_gui.py — lines 2481-2520

```
        )
        h(
            "Até à v3.2 o programa chamava-se MBSniffer. Na primeira vez que o MBToolkit "
            "arranca, copia as preferências e os workspaces de %APPDATA%\\MBSniffer para "
            "%APPDATA%\\MBToolkit; a pasta antiga não é apagada. Os logs novos ficam em "
            "“MBToolkit Logs”; os antigos continuam em “MBSniffer Logs”.\n"
        )

        h("Boas práticas de diagnóstico\n", "heading")
        h("• Confirma primeiro cablagem, polaridade e parâmetros série.\n", "bullet")
        h("• Em RS485, mantém a derivação do sniffer curta e evita alterar a terminação existente.\n", "bullet")
        h("• Garante que nenhum outro programa tem a mesma COM aberta.\n", "bullet")
        h("• Usa CRC ERROR e RAW como indícios de framing, ruído, parâmetros errados ou captura incompleta.\n", "bullet")
        h("• Usa Exceptions como respostas válidas do protocolo: não são equivalentes a CRC error.\n", "bullet")
        h("• Compara tempos de resposta com o comportamento normal do equipamento, não apenas com um limite genérico.\n", "bullet")
        h("• Guarda o TXT quando precisares de análise posterior da captura completa.\n", "bullet")

        if debug_sim == 1:
            h("Modo Simulação\n", "heading")
            h(
                "Este modo de desenvolvimento aparece apenas quando debug_sim = 1 em "
                "MBToolkit.py. Não abre portas série e nunca cria logs TXT.\n"
            )
            h(
                "Cada execução usa sempre o mesmo tipo de cenário de diagnóstico: uma "
                "resposta normal, uma resposta lenta, uma Modbus Exception, um CRC error, "
                "um timeout e um bloco RAW/UNPARSED. Assim é possível comparar diretamente "
                "a mesma informação com e sem destaque visual.\n"
            )
            h(
                "“Realçar resultados” não altera os dados gerados pela Simulação. Desligado, "
                "todas as linhas usam as cores normais da tabela; ligado, a transação normal "
                "concluída fica verde e as linhas de Exception, CRC error, timeout, resposta "
                "lenta e RAW recebem as respetivas cores de diagnóstico.\n"
            )
            h(
                "As frames deste cenário usam o canal SIM para não serem confundidas com uma "
                "captura física real.\n"
            )

```

MBToolkit v3.4/mb_gui.py — lines 2521-2560

```
        h("Limitação importante\n", "heading")
        h(
            "O MBToolkit interpreta bytes, CRC, Function Code, endereços, quantidades, "
            "emparelhamento e tempos observados pelo sistema operativo. Não mede diretamente "
            "níveis elétricos, ringing, reflexões, bias, common-mode, tempos de subida/descida "
            "ou integridade analógica do sinal. Para esses problemas continua a ser necessário "
            "um osciloscópio ou analisador adequado à interface física.\n"
        )

        self.help_text.configure(state="disabled")

        detail = ttk.LabelFrame(self.sniffer_page, text="Frame Inspector")
        detail.grid(row=2, column=0, sticky="ew", padx=10, pady=(0, 10))
        detail.columnconfigure(0, weight=1)

        inspector_header = ttk.Frame(detail)
        inspector_header.grid(
            row=0, column=0, sticky="ew", padx=8, pady=(4, 2)
        )
        inspector_header.columnconfigure(0, weight=1)
        ttk.Label(
            inspector_header,
            text=(
                "Seleciona uma linha no Tráfego para ver a descodificação "
                "estruturada do frame."
            ),
        ).grid(row=0, column=0, sticky="w")

        self.inspector_collapsed = False
        self.inspector_toggle_btn = ttk.Button(
            inspector_header,
            text="Recolher",
            command=self.toggle_frame_inspector,
            style="MBS.Compact.TButton",
        )
        self.inspector_toggle_btn.grid(row=0, column=1, sticky="e")

        self.inspector_body = ttk.Frame(detail)
        self.inspector_body.grid(
            row=1, column=0, sticky="ew", padx=8, pady=(2, 7)
```

MBToolkit v3.4/mb_gui.py — lines 2561-2600

```
        )
        for col in range(4):
            self.inspector_body.columnconfigure(
                col,
                weight=1,
                uniform="inspector",
            )

        self.inspector_vars = {
            "slave": tk.StringVar(value="Slave: —"),
            "type": tk.StringVar(value="Tipo: —"),
            "function": tk.StringVar(value="Function: —"),
            "response": tk.StringVar(value="Resp.: —"),
            "result": tk.StringVar(value="Resultado: —"),
            "pdu": tk.StringVar(value="PDU Address: —"),
            "one_based": tk.StringVar(value="1-based: —"),
            "qty": tk.StringVar(value="Qty: —"),
            "byte_count": tk.StringVar(value="ByteCount: —"),
            "crc": tk.StringVar(value="CRC: —"),
            "data": tk.StringVar(value="Dados: —"),
        }

        inspector_grid = (
            ("slave", 0, 0),
            ("type", 0, 1),
            ("function", 0, 2),
            ("response", 0, 3),
            ("pdu", 1, 0),
            ("one_based", 1, 1),
            ("qty", 1, 2),
            ("byte_count", 1, 3),
        )
        for key, row, col in inspector_grid:
            ttk.Label(
                self.inspector_body,
                textvariable=self.inspector_vars[key],
                anchor="w",
            ).grid(
                row=row,
                column=col,
```

MBToolkit v3.4/mb_gui.py — lines 2601-2640

```
                sticky="ew",
                padx=(0 if col == 0 else 8, 8 if col < 3 else 0),
                pady=2,
            )

        self.inspector_result_label = ttk.Label(
            self.inspector_body,
            textvariable=self.inspector_vars["result"],
            anchor="w",
            style="MBS.Result.neutral.TLabel",
        )
        self.inspector_result_label.grid(
            row=2,
            column=0,
            columnspan=4,
            sticky="ew",
            pady=(3, 2),
        )

        ttk.Label(
            self.inspector_body,
            textvariable=self.inspector_vars["crc"],
            anchor="w",
        ).grid(
            row=3, column=0, columnspan=2, sticky="ew", pady=2
        )
        # Data/Register payloads can be very long. A readonly Entry keeps the
        # inspector compact instead of letting content change requested width.
        ttk.Entry(
            self.inspector_body,
            textvariable=self.inspector_vars["data"],
            state="readonly",
        ).grid(
            row=3,
            column=2,
            columnspan=2,
            sticky="ew",
            padx=(8, 0),
            pady=2,
        )
```

MBToolkit v3.4/mb_gui.py — lines 2641-2664

```
2641
2642        self.selected_raw_var = tk.StringVar()
2643        ttk.Entry(
2644            self.inspector_body,
2645            textvariable=self.selected_raw_var,
2646            state="readonly",
2647        ).grid(
2648            row=4,
2649            column=0,
2650            columnspan=4,
2651            sticky="ew",
2652            pady=(4, 0),
2653        )
2654
2655        self.build_discovery_tab(self.discovery_page)
2656        self.build_master_tab(self.master_page)
2657        self.build_slave_tab(self.slave_page)
2658        self.update_stats_labels()
2659        self.update_frame_inspector(None)
2660
2661        # Classic Tk/Canvas widgets are outside ttk.Style; apply the remembered
2662        # startup palette after every such widget has been created.
2663        self._apply_classic_widget_palette(self.initial_dark_mode)
2664
```

MBToolkit v3.4/tests/test_regression.py

787 source lines.

MBToolkit v3.4/tests/test_regression.py — lines 1-40

```
import json
import queue
import random
import tempfile
import threading
import time
import unittest
from collections import deque
from pathlib import Path

import mb_config
import mb_slave_finder
from mb_capture import CaptureMixin
from mb_protocol import (
    SessionMetrics,
    append_crc,
    burst_may_continue,
    crc16_modbus,
    crc_ok,
    modbus_rtu_interframe_gap_seconds,
    parse_burst,
    read_bits_details,
    split_capture_buffer,
)
from mb_slave_finder import SlaveFinderMixin
from mb_view import ViewMixin
from mb_diagnostics import inspector_values
from mb_i18n import (
    LANGUAGE_ENGLISH,
    LANGUAGE_PORTUGUESE,
    english_help_blocks,
    normalize_language,
    translate_details,
    translate_runtime_text,
    translate_text,
)

from mb_slave_finder import (
    build_device_identification_probe,
    build_discovery_probe,
```

MBToolkit v3.4/tests/test_regression.py — lines 41-80

```
    device_identification_timeout_seconds,
    find_device_identification_response,
    find_discovery_response,
    format_device_identification,
    read_basic_device_identification,
)


def with_bad_crc(frame: bytes) -> bytes:
    damaged = bytearray(frame)
    damaged[-1] ^= 0xFF
    return bytes(damaged)


def device_id_response(
    slave=1,
    objects=None,
    *,
    more_follows=False,
    next_object_id=0,
    conformity=0x01,
):
    if objects is None:
        objects = {
            0x00: b"Schneider Electric",
            0x01: b"TEST-PRODUCT",
            0x02: b"1.2.3",
        }
    payload = bytearray((
        slave,
        0x2B,
        0x0E,
        0x01,
        conformity,
        0xFF if more_follows else 0x00,
        next_object_id,
        len(objects),
    ))
    for object_id, value in objects.items():
        value = bytes(value)
```

MBToolkit v3.4/tests/test_regression.py — lines 81-120

```
81          payload.extend((int(object_id) & 0xFF, len(value)))
82          payload.extend(value)
83      return append_crc(bytes(payload))
84
85
86  class FakeSerial:
87      """Minimal non-blocking serial fake for active Device ID regression tests."""
88
89      def __init__(self, response_by_object):
90          self.response_by_object = dict(response_by_object)
91          self.rx = bytearray()
92          self.writes = []
93
94      @property
95      def in_waiting(self):
96          return len(self.rx)
97
98      def reset_input_buffer(self):
99          self.rx.clear()
100
101     def reset_output_buffer(self):
102         pass
103
104     def write(self, data):
105         data = bytes(data)
106         self.writes.append(data)
107         object_id = data[4] if len(data) >= 5 else 0
108         response = self.response_by_object.get(object_id, b"")
109         self.rx.extend(response)
110         return len(data)
111
112     def read(self, size):
113         size = min(int(size), len(self.rx))
114         chunk = bytes(self.rx[:size])
115         del self.rx[:size]
116         return chunk
117
118
119 class FakeSerialModule:
120     class SerialTimeoutException(TimeoutError):
```

MBToolkit v3.4/tests/test_regression.py — lines 121-160

```
121         pass
122
123     class SerialException(OSError):
124         pass
125
126
127 class StaticVar:
128     def __init__(self, value):
129         self.value = value
130
131     def get(self):
132         return self.value
133
134
135 class DummyCapture:
136     calculated_gap_ms = CaptureMixin.calculated_gap_ms
137
138     def __init__(self, *, mode="Auto", baud="9600", data="8", parity="None", stop="1", manual="2.0"):
139         self.gap_mode_var = StaticVar(mode)
140         self.baud_var = StaticVar(baud)
141         self.data_var = StaticVar(data)
142         self.parity_var = StaticVar(parity)
143         self.stop_var = StaticVar(stop)
144         self.gap_ms_var = StaticVar(manual)
145
146
147 class TimingRegressionTests(unittest.TestCase):
148     def test_t35_uses_character_time_at_or_below_19200(self):
149         self.assertAlmostEqual(
150             modbus_rtu_interframe_gap_seconds(9600, 11),
151             3.5 * 11 / 9600,
152             places=9,
153         )
154         self.assertAlmostEqual(
155             modbus_rtu_interframe_gap_seconds(19200, 11),
156             3.5 * 11 / 19200,
157             places=9,
158         )
159
160     def test_t35_is_fixed_above_19200(self):
```

MBToolkit v3.4/tests/test_regression.py — lines 161-200

```
161         for baud in (38400, 57600, 115200):
162             with self.subTest(baud=baud):
163                 self.assertAlmostEqual(
164                     modbus_rtu_interframe_gap_seconds(baud, 10),
165                     0.001750,
166                     places=9,
167                 )
168 
169     def test_sniffer_auto_gap_uses_fixed_1750_us_above_19200(self):
170         capture = DummyCapture(baud="115200", parity="None", stop="1")
171         self.assertAlmostEqual(capture.calculated_gap_ms(), 1.750, places=6)
172 
173     def test_sniffer_manual_gap_is_unchanged(self):
174         capture = DummyCapture(mode="Manual", manual="2,75")
175         self.assertAlmostEqual(capture.calculated_gap_ms(), 2.75, places=6)
176 
177     def test_device_id_timeout_covers_a_full_rtu_frame(self):
178         timeout = device_identification_timeout_seconds(9600, "None", "1", 40)
179         self.assertGreaterEqual(timeout, 256 * 10 / 9600 + 0.001750)
180 
181 
182 class ParserRegressionTests(unittest.TestCase):
183     def test_crc_known_request(self):
184         frame = build_discovery_probe(1, 3)
185         self.assertTrue(crc_ok(frame))
186         self.assertEqual(frame[:6], bytes.fromhex("01 03 00 00 00 01"))
187 
188     def test_invalid_frame_before_valid_frame_stays_crc_error(self):
189         first = with_bad_crc(build_discovery_probe(1, 3))
190         second = build_discovery_probe(2, 4)
191         parts = split_capture_buffer(first + second)
192         self.assertEqual(parts[0], (first, "REQUEST", "ERROR"))
193         self.assertEqual(parts[1], (second, "REQUEST", "OK"))
194 
195     def test_long_unsynchronised_noise_recovers_next_valid_frame(self):
196         noise = bytes([0xFF]) * 4000
197         frame = build_discovery_probe(7, 3)
198         parts = split_capture_buffer(noise + frame)
199         self.assertEqual(parts[-1], (frame, "REQUEST", "OK"))
200         self.assertTrue(parts[0][1].startswith("RAW/"))
```

MBToolkit v3.4/tests/test_regression.py — lines 201-240

```
        self.assertEqual(sum(len(part[0]) for part in parts), len(noise) + len(frame))

    def test_supported_function_code_shapes(self):
        frames = {
            "fc01_req": append_crc(bytes.fromhex("01 01 00 00 00 08")),
            "fc01_resp": append_crc(bytes.fromhex("01 01 01 5A")),
            "fc02_req": append_crc(bytes.fromhex("01 02 00 00 00 08")),
            "fc02_resp": append_crc(bytes.fromhex("01 02 01 A5")),
            "fc03_req": append_crc(bytes.fromhex("01 03 00 00 00 01")),
            "fc03_resp": append_crc(bytes.fromhex("01 03 02 00 2A")),
            "fc04_req": append_crc(bytes.fromhex("01 04 00 00 00 01")),
            "fc04_resp": append_crc(bytes.fromhex("01 04 02 00 2A")),
            "fc05": append_crc(bytes.fromhex("01 05 00 01 FF 00")),
            "fc06": append_crc(bytes.fromhex("01 06 00 01 00 2A")),
            "fc07_req": append_crc(bytes.fromhex("01 07")),
            "fc07_resp": append_crc(bytes.fromhex("01 07 AA")),
            "fc08": append_crc(bytes.fromhex("01 08 00 00 12 34")),
            "fc15_req": append_crc(bytes.fromhex("01 0F 00 00 00 08 01 5A")),
            "fc15_resp": append_crc(bytes.fromhex("01 0F 00 00 00 08")),
            "fc16_req": append_crc(bytes.fromhex("01 10 00 00 00 01 02 00 2A")),
            "fc16_resp": append_crc(bytes.fromhex("01 10 00 00 00 01")),
            "fc22": append_crc(bytes.fromhex("01 16 00 01 FF 00 00 2A")),
            "fc23_req": append_crc(bytes.fromhex(
                "01 17 00 00 00 01 00 10 00 01 02 00 2A"
            )),
            "fc23_resp": append_crc(bytes.fromhex("01 17 02 00 2A")),
            "exception": append_crc(bytes.fromhex("01 83 02")),
        }
        for name, frame in frames.items():
            with self.subTest(name=name):
                parts = split_capture_buffer(frame)
                self.assertEqual(len(parts), 1)
                self.assertEqual(parts[0][0], frame)
                self.assertEqual(parts[0][2], "OK")
                self.assertFalse(parts[0][1].startswith("RAW"))

    def test_random_noise_never_crashes_and_preserves_all_bytes(self):
        rng = random.Random(20260910)
        for _ in range(250):
            length = rng.randint(1, 512)
```

MBToolkit v3.4/tests/test_regression.py — lines 241-280

```
            data = bytes(rng.randrange(256) for _ in range(length))
            parts = split_capture_buffer(data)
            self.assertEqual(sum(len(part[0]) for part in parts), len(data))
            self.assertTrue(all(len(part[0]) > 0 for part in parts))

    def test_request_response_pairing_and_response_time(self):
        req = parse_burst(
            append_crc(bytes.fromhex("01 03 00 00 00 01")),
            "12:00:00.000",
            "BUS",
            100.000,
        )[0]
        resp = parse_burst(
            append_crc(bytes.fromhex("01 03 02 00 2A")),
            "12:00:00.125",
            "BUS",
            100.125,
        )[0]
        req["_view_seq"] = 10
        resp["_view_seq"] = 11

        metrics = SessionMetrics(pending_timeout_seconds=1.0)
        first = metrics.process(req)
        second = metrics.process(resp)

        self.assertEqual(first["pending"], 1)
        self.assertEqual(second["pending"], 0)
        self.assertEqual(second["matched_request"]["seq"], 10)
        self.assertAlmostEqual(second["response_ms"], 125.0, places=6)

    def test_timeout_regression(self):
        req = parse_burst(
            append_crc(bytes.fromhex("05 04 00 00 00 01")),
            "12:00:00.000",
            "BUS",
            10.0,
        )[0]
        req["_view_seq"] = 77
        metrics = SessionMetrics(pending_timeout_seconds=1.0)
        metrics.process(req)
```

MBToolkit v3.4/tests/test_regression.py — lines 281-320

```
281         self.assertEqual(metrics.expire_pending(11.1), 1)
282         expired = metrics.pop_expired_requests()
283         self.assertEqual(expired[0]["seq"], 77)
284         self.assertEqual(metrics.timeouts, 1)
285
286
287 class SlaveFinderRegressionTests(unittest.TestCase):
288     def test_normal_discovery_response(self):
289         response = append_crc(bytes.fromhex("04 03 02 12 34"))
290         found = find_discovery_response(response, 4, 3)
291         self.assertIsNotNone(found)
292         self.assertEqual(found["kind"], "RESPONSE")
293
294     def test_discovery_exception_counts_as_response(self):
295         response = append_crc(bytes.fromhex("04 83 02"))
296         found = find_discovery_response(response, 4, 3)
297         self.assertIsNotNone(found)
298         self.assertEqual(found["kind"], "EXCEPTION")
299         self.assertEqual(found["exception_code"], 2)
300
301     def test_device_identification_request(self):
302         request = build_device_identification_probe(4)
303         self.assertEqual(request[:5], bytes.fromhex("04 2B 0E 01 00"))
304         self.assertEqual(len(request), 7)
305         self.assertTrue(crc_ok(request))
306
307     def test_device_identification_response_parser(self):
308         response = device_id_response(slave=4)
309         found = find_device_identification_response(response, 4)
310         self.assertIsNotNone(found)
311         self.assertEqual(found["kind"], "DEVICE_ID")
312         self.assertEqual(found["objects"][0x00], "Schneider Electric")
313         self.assertEqual(found["objects"][0x01], "TEST-PRODUCT")
314         self.assertEqual(found["objects"][0x02], "1.2.3")
315         self.assertEqual(
316             format_device_identification(found),
317             "Schneider Electric | TEST-PRODUCT | 1.2.3",
318         )
319
320     def test_device_identification_parser_ignores_request_echo(self):
```

MBToolkit v3.4/tests/test_regression.py — lines 321-360

```
        request = build_device_identification_probe(4)
        response = device_id_response(slave=4)
        found = find_device_identification_response(request + response, 4)
        self.assertIsNotNone(found)
        self.assertEqual(found["objects"][0x01], "TEST-PRODUCT")

    def test_device_identification_exception(self):
        response = append_crc(bytes.fromhex("04 AB 01"))
        found = find_device_identification_response(response, 4)
        self.assertIsNotNone(found)
        self.assertEqual(found["kind"], "EXCEPTION")
        self.assertEqual(found["exception_code"], 1)
        self.assertIn("Illegal Function", format_device_identification(found))

    def test_device_identification_rejects_bad_crc(self):
        response = with_bad_crc(device_id_response(slave=4))
        self.assertIsNone(find_device_identification_response(response, 4))

    def test_device_identification_follows_more_follows(self):
        page0 = device_id_response(
            slave=4,
            objects={0x00: b"Vendor"},
            more_follows=True,
            next_object_id=0x01,
        )
        page1 = device_id_response(
            slave=4,
            objects={0x01: b"Product", 0x02: b"2.0"},
            more_follows=False,
            next_object_id=0x00,
        )
        serial = FakeSerial({0x00: page0, 0x01: page1})
        result = read_basic_device_identification(
            serial,
            4,
            115200,
            "None",
            "1",
            10,
            threading.Event(),
```

MBToolkit v3.4/tests/test_regression.py — lines 361-400

```
361             FakeSerialModule,
362         )
363         self.assertIsNotNone(result)
364         self.assertEqual(result["objects"][0x00], "Vendor")
365         self.assertEqual(result["objects"][0x01], "Product")
366         self.assertEqual(result["objects"][0x02], "2.0")
367         self.assertEqual(len(serial.writes), 2)
368         self.assertEqual(serial.writes[0][4], 0x00)
369         self.assertEqual(serial.writes[1][4], 0x01)
370
371
372 class InternationalizationRegressionTests(unittest.TestCase):
373     def test_language_normalization_defaults_to_portuguese(self):
374         self.assertEqual(normalize_language("invalid"), LANGUAGE_PORTUGUESE)
375         self.assertEqual(normalize_language(LANGUAGE_ENGLISH), LANGUAGE_ENGLISH)
376
377     def test_static_ui_translation_is_reversible(self):
378         self.assertEqual(translate_text("Configuração", LANGUAGE_ENGLISH), "Configuration")
379         self.assertEqual(translate_text("Configuration", LANGUAGE_PORTUGUESE), "Configuração")
380         self.assertEqual(translate_text("Frame gap", LANGUAGE_ENGLISH), "Frame gap")
381         # "None" is a technical parity value and must remain English in the PT UI.
382         self.assertEqual(translate_text("None", LANGUAGE_PORTUGUESE), "None")
383         self.assertEqual(translate_text("Nenhum", LANGUAGE_ENGLISH), "None")
384
385     def test_dynamic_status_translation_is_reversible(self):
386         pt = "Pesquisa concluída — 7 encontrados — 00:12"
387         en = "Scan complete — 7 found — 00:12"
388         self.assertEqual(translate_runtime_text(pt, LANGUAGE_ENGLISH), en)
389         self.assertEqual(translate_runtime_text(en, LANGUAGE_PORTUGUESE), pt)
390
391     def test_register_details_follow_language(self):
392         self.assertEqual(
393             translate_details("ByteCount=2  Registos=[42]", LANGUAGE_ENGLISH),
394             "ByteCount=2  Registers=[42]",
395         )
396         self.assertEqual(
397             translate_details("ByteCount=2  Registers=[42]", LANGUAGE_PORTUGUESE),
398             "ByteCount=2  Registos=[42]",
399         )
400
```

MBToolkit v3.4/tests/test_regression.py — lines 401-440

```
    def test_inspector_english_labels_and_crc(self):
        frame = append_crc(bytes.fromhex("01 03 02 00 2A"))
        row = parse_burst(frame, "12:00:00.000", "BUS", 1.0)[0]
        row.update({"response_text": "12.3", "matched": True})
        values = inspector_values(row, language=LANGUAGE_ENGLISH)
        self.assertTrue(values["type"].startswith("Type:"))
        self.assertTrue(values["result"].startswith("Result:"))
        self.assertTrue(values["data"].startswith("Data:"))
        self.assertIn("Registers=[42]", values["data"])
        self.assertIn("received", values["crc"])
        self.assertIn("calculated", values["crc"])

    def test_english_help_contains_no_portuguese_diacritics(self):
        text = "".join(
            block for _tag, block in english_help_blocks(
                "20 000", "20 000", 1.0, 300.0, debug_sim=True
            )
        )
        for character in "ãõçáéíóúâê":
            self.assertNotIn(character, text.casefold())
        self.assertIn("Bus Slave Finder — active scan", text)
        self.assertIn("Device Identification (FC43/14)", text)


def bitwise_crc16(data: bytes) -> int:
    """Reference CRC used up to v3.0, kept to verify the table-driven CRC."""
    crc = 0xFFFF
    for b in data:
        crc ^= b
        for _ in range(8):
            crc = (crc >> 1) ^ 0xA001 if crc & 1 else crc >> 1
    return crc


def process_frames(metrics, frames):
    """Run (bytes, channel, epoch) frames through SessionMetrics."""
    out = []
    for index, (frame, channel, epoch) in enumerate(frames):
        for row in parse_burst(frame, "12:00:00.000", channel, epoch):
            row["_view_seq"] = index
```

MBToolkit v3.4/tests/test_regression.py — lines 441-480

```
            out.append((row, metrics.process(row)))
    return out


class ProtocolV31RegressionTests(unittest.TestCase):
    def test_crc_table_matches_bitwise_reference(self):
        rng = random.Random(31)
        for length in (0, 1, 2, 7, 64, 256):
            data = bytes(rng.randrange(256) for _ in range(length))
            self.assertEqual(crc16_modbus(data), bitwise_crc16(data))

    def test_broadcast_request_never_times_out(self):
        request = append_crc(bytes.fromhex("00 10 00 00 00 01 02 00 05"))
        metrics = SessionMetrics(pending_timeout_seconds=1.0)
        process_frames(metrics, [(request, "BUS", 10.0)])
        self.assertEqual(metrics.requests, 1)
        self.assertEqual(metrics.broadcast_requests, 1)
        self.assertEqual(metrics.pending_count(), 0)
        metrics.expire_pending(20.0)
        self.assertEqual(metrics.timeouts, 0)
        self.assertEqual(metrics.health_summary(now=20.0)["timeout_rate_pct"], 0.0)

    def test_fc06_echo_pair_is_request_then_response(self):
        write = append_crc(bytes.fromhex("01 06 00 01 00 09"))
        metrics = SessionMetrics(pending_timeout_seconds=1.0)
        results = process_frames(
            metrics, [(write, "BUS", 100.0), (write, "BUS", 100.040)]
        )
        self.assertEqual(
            [row["type"] for row, _timing in results], ["REQUEST", "RESPONSE"]
        )
        self.assertEqual((metrics.requests, metrics.responses), (1, 1))
        self.assertAlmostEqual(results[1][1]["response_ms"], 40.0, places=6)

    def test_unanswered_fc05_write_times_out(self):
        write = append_crc(bytes.fromhex("02 05 00 10 FF 00"))
        metrics = SessionMetrics(pending_timeout_seconds=1.0)
        results = process_frames(metrics, [(write, "BUS", 5.0)])
        self.assertEqual(results[0][0]["type"], "REQUEST")
        self.assertEqual(metrics.expire_pending(6.5), 1)
```

MBToolkit v3.4/tests/test_regression.py — lines 481-520

```
481
482     def test_different_fc06_writes_are_both_requests(self):
483         first = append_crc(bytes.fromhex("01 06 00 01 00 09"))
484         second = append_crc(bytes.fromhex("01 06 00 02 00 0A"))
485         metrics = SessionMetrics()
486         results = process_frames(
487             metrics, [(first, "BUS", 1.0), (second, "BUS", 1.1)]
488         )
489         self.assertEqual(
490             [row["type"] for row, _timing in results], ["REQUEST", "REQUEST"]
491         )
492
493     def test_dual_rx_echo_must_come_from_opposite_channel(self):
494         write = append_crc(bytes.fromhex("01 06 00 01 00 09"))
495         metrics = SessionMetrics()
496         results = process_frames(metrics, [
497             (write, "A→B", 1.0),
498             (write, "A→B", 1.5),
499             (write, "B→A", 1.52),
500         ])
501         self.assertEqual(
502             [row["type"] for row, _timing in results],
503             ["REQUEST", "REQUEST", "RESPONSE"],
504         )
505
506     def test_split_frame_is_held_until_complete(self):
507         response = append_crc(bytes.fromhex("01 03 08") + bytes(8))
508         self.assertTrue(burst_may_continue(response[:7]))
509         self.assertTrue(burst_may_continue(response[:1]))
510         self.assertFalse(burst_may_continue(response))
511         self.assertFalse(burst_may_continue(bytes((0xFF,) * 6)))
512
513     def test_merged_frames_keep_their_chunk_arrival_times(self):
514         request = append_crc(bytes.fromhex("01 03 00 00 00 01"))
515         response = append_crc(bytes.fromhex("01 03 02 00 2A"))
516         rows = parse_burst(
517             request + response,
518             "12:00:00.000",
519             "BUS",
520             100.0,
```

MBToolkit v3.4/tests/test_regression.py — lines 521-560

```
            chunk_times=[(0, 100.0), (len(request), 100.030)],
        )
        self.assertEqual([row["type"] for row in rows], ["REQUEST", "RESPONSE"])
        self.assertAlmostEqual(rows[1]["_epoch"], 100.030, places=6)
        metrics = SessionMetrics()
        metrics.process(rows[0])
        self.assertAlmostEqual(metrics.process(rows[1])["response_ms"], 30.0, places=6)

    def test_read_bits_details_use_request_qty(self):
        response = append_crc(bytes.fromhex("01 01 02 CD 01"))
        self.assertEqual(
            read_bits_details(response),
            "ByteCount=2  Bits=10110011 10000000",
        )
        self.assertEqual(
            read_bits_details(response, qty=10),
            "ByteCount=2  Bits=10110011 10",
        )


class _ChunkedSerial:
    """Delivers byte chunks at fixed delays, like a USB latency timer."""

    def __init__(self, chunks, stop_event):
        self.started = time.perf_counter()
        self.chunks = list(chunks)  # [(delay_s, bytes)]
        self.stop_event = stop_event

    def _due(self):
        elapsed = time.perf_counter() - self.started
        if self.chunks and elapsed >= self.chunks[0][0]:
            return self.chunks[0][1]
        if not self.chunks and elapsed > 0.3:
            self.stop_event.set()
        return b""

    @property
    def in_waiting(self):
        return len(self._due())

```

MBToolkit v3.4/tests/test_regression.py — lines 561-600

```
    def read(self, size):
        data = self._due()
        if data:
            self.chunks.pop(0)
        return data

    def close(self):
        pass


class _ReaderStub(CaptureMixin):
    def __init__(self):
        self.stop_event = threading.Event()
        self.event_queue = queue.Queue()


class ReaderV31RegressionTests(unittest.TestCase):
    def test_frame_split_by_usb_latency_is_reassembled(self):
        response = append_crc(bytes.fromhex("01 03 08") + bytes(8))
        reader = _ReaderStub()
        serial = _ChunkedSerial(
            [(0.0, response[:7]), (0.016, response[7:])], reader.stop_event
        )
        # 115200 bit/s: Auto t3.5 is 1.75 ms, far shorter than the 16 ms split.
        reader.reader_loop(serial, 1.75, "BUS", 1)
        rows = []
        while not reader.event_queue.empty():
            name, payload = reader.event_queue.get()
            if name == "frame":
                rows.append(payload)
        self.assertEqual([(row["type"], row["crc"]) for row in rows], [("RESPONSE", "OK")])


class _FakeVar:
    def __init__(self, value):
        self.value = value

    def get(self):
        return self.value

```

MBToolkit v3.4/tests/test_regression.py — lines 601-640

```
601
602 class _FakeTree:
603     def __init__(self):
604         self.tags = {}
605
606     def item(self, iid, tags=()):
607         self.tags[iid] = tags
608
609
610 class _TimeoutView(ViewMixin):
611     def __init__(self, type_filter):
612         self.tree = _FakeTree()
613         self.advanced_filter_type_var = _FakeVar(type_filter)
614         self.highlight_anomalies_var = _FakeVar(True)
615         self._successful_request_seqs = set()
616         self._frame_history = deque([{
617             "seq": 5, "type": "REQUEST", "crc": "OK", "timed_out": False,
618             "response_text": "",
619         }])
620         self._tree_iids_by_seq = {5: ["I001", "I002"]}
621         self.full_renders = 0
622
623     def render_filtered_view(self):
624         self.full_renders += 1
625
626     def selected_record(self):
627         return None
628
629
630 class ViewV31RegressionTests(unittest.TestCase):
631     def test_timeout_retags_rows_without_full_rebuild(self):
632         view = _TimeoutView("Todos")
633         self.assertEqual(view.mark_timed_out_records([{"seq": 5}]), 1)
634         self.assertEqual(view.full_renders, 0)
635         self.assertEqual(view.tree.tags["I001"], ("anomaly_timeout",))
636         self.assertEqual(view.tree.tags["I002"], ("anomaly_timeout",))
637
638     def test_timeouts_filter_still_rebuilds_view(self):
639         view = _TimeoutView("Timeouts")
640         view.mark_timed_out_records([{"seq": 5}])
```

MBToolkit v3.4/tests/test_regression.py — lines 641-680

```
641         self.assertEqual(view.full_renders, 1)
642
643     def test_inspector_shows_address_of_each_register(self):
644         frame = append_crc(bytes.fromhex("01 03 04 00 2A 00 2B"))
645         row = parse_burst(frame, "12:00:00.000", "BUS", 1.0)[0]
646         row["request_pdu_address"] = 100
647         pt = inspector_values(row, language=LANGUAGE_PORTUGUESE)
648         en = inspector_values(row, language=LANGUAGE_ENGLISH)
649         self.assertIn("Registos (PDU Address=valor): 100=42, 101=43", pt["data"])
650         self.assertIn("Registers (PDU Address=value): 100=42, 101=43", en["data"])
651
652     def test_inspector_shows_address_of_each_bit(self):
653         frame = append_crc(bytes.fromhex("01 02 01 05"))
654         row = parse_burst(frame, "12:00:00.000", "BUS", 1.0)[0]
655         row.update({"request_pdu_address": 8, "request_qty": 3})
656         values = inspector_values(row, language=LANGUAGE_PORTUGUESE)
657         self.assertIn("Bits (PDU Address=valor): 8=1, 9=0, 10=1", values["data"])
658
659
660 class _FakeFinderSerial:
661     def __init__(self, incoming=b""):
662         self.incoming = bytearray(incoming)
663         self.writes = []
664
665     @property
666     def in_waiting(self):
667         return len(self.incoming)
668
669     def read(self, size):
670         data = bytes(self.incoming[:size])
671         del self.incoming[:size]
672         return data
673
674     def write(self, data):
675         self.writes.append(bytes(data))
676         return len(data)
677
678     def reset_input_buffer(self):
679         pass
680
```

MBToolkit v3.4/tests/test_regression.py — lines 681-720

```
    def reset_output_buffer(self):
        pass

    def close(self):
        pass


class _FinderStub(SlaveFinderMixin):
    current_language = LANGUAGE_PORTUGUESE

    def __init__(self, fake_serial):
        self.fake_serial = fake_serial
        self.event_queue = queue.Queue()
        self.discovery_stop_event = threading.Event()
        self.discovery_serial = None

    def tr(self, text):
        return translate_text(text, self.current_language)

    def trf(self, text, **values):
        return self.tr(text).format(**values)

    def open_discovery_serial_port(self, serial, port, baud, parity, stopbits):
        return self.fake_serial

    def run_worker(self, slaves, attempts):
        self.discovery_worker(
            None, "COM9", [(115200, "None", "1")], slaves, 10, False, False,
            attempts,
        )
        events = []
        while not self.event_queue.empty():
            events.append(self.event_queue.get())
        return events


class FinderV31RegressionTests(unittest.TestCase):
    def setUp(self):
        self._idle_check = mb_slave_finder.DISCOVERY_IDLE_CHECK_SECONDS
        mb_slave_finder.DISCOVERY_IDLE_CHECK_SECONDS = 0.05
```

MBToolkit v3.4/tests/test_regression.py — lines 721-760

```
721
722     def tearDown(self):
723         mb_slave_finder.DISCOVERY_IDLE_CHECK_SECONDS = self._idle_check
724
725     def test_each_silent_slave_gets_all_attempts(self):
726         stub = _FinderStub(_FakeFinderSerial())
727         events = stub.run_worker([1, 2], attempts=3)
728         done = [payload for name, payload in events if name == "discovery_done"][0]
729         self.assertIsNone(done["error"])
730         self.assertEqual(len(stub.fake_serial.writes), 6)
731
732     def test_traffic_before_scan_blocks_transmission(self):
733         stub = _FinderStub(_FakeFinderSerial(b"\x01\x03\x00"))
734         events = stub.run_worker([1], attempts=1)
735         done = [payload for name, payload in events if name == "discovery_done"][0]
736         self.assertIn("Foi detetado tráfego em COM9", done["error"])
737         self.assertEqual(stub.fake_serial.writes, [])
738
739     def test_legacy_default_timeout_migrates(self):
740         with tempfile.TemporaryDirectory() as folder:
741             path = Path(folder) / "settings.json"
742             original = mb_config.get_settings_file
743             mb_config.get_settings_file = lambda: path
744             try:
745                 path.write_text(
746                     json.dumps({"bus_slave_finder": {"min_timeout_ms": 40}}),
747                     encoding="utf-8",
748                 )
749                 finder = mb_config.load_ui_settings()["bus_slave_finder"]
750                 self.assertEqual(finder["min_timeout_ms"], "100")
751                 self.assertEqual(finder["attempts"], "1")
752
753                 path.write_text(
754                     json.dumps({"bus_slave_finder": {
755                         "min_timeout_ms": 40, "attempts": 2,
756                     }}),
757                     encoding="utf-8",
758                 )
759                 finder = mb_config.load_ui_settings()["bus_slave_finder"]
760                 self.assertEqual(finder["min_timeout_ms"], "40")
```

MBToolkit v3.4/tests/test_regression.py — lines 761-787

```
                self.assertEqual(finder["attempts"], "2")
            finally:
                mb_config.get_settings_file = original


class SourceHygieneV31Tests(unittest.TestCase):
    def test_no_inline_language_branches(self):
        package = Path(__file__).resolve().parents[1]
        for source in package.glob("*.py"):
            text = source.read_text(encoding="utf-8")
            self.assertNotIn(
                'current_language == "English"', text, source.name
            )

    def test_message_templates_translate(self):
        self.assertEqual(
            translate_text("Encontrados: {count}", LANGUAGE_ENGLISH).format(count=3),
            "Found: 3",
        )
        self.assertEqual(
            translate_text("A verificar se o barramento está livre…", LANGUAGE_ENGLISH),
            "Checking that the bus is idle…",
        )


if __name__ == "__main__":
    unittest.main(verbosity=2)
```

MBToolkit v3.4/tests/test_master.py

601 source lines.

MBToolkit v3.4/tests/test_master.py — lines 1-40

```python
import json
import os
import queue
import struct
import tempfile
import time
import unittest
from pathlib import Path

import mb_config
import mb_master_ui
from mb_master import (
    BYTE_ORDERS,
    BYTE_ORDERS_64,
    format_value,
    format_window_values,
    raw_text_for_scaled,
    register_groups,
    validate_scale,
    MasterEngine,
    PollDefinition,
    build_read_request,
    build_write_request,
    encode_coil_value,
    encode_register_value,
    find_master_response,
    format_register_values,
)
from mb_master_ui import MasterMixin, MasterWindow, apply_format_to_rows
from mb_protocol import append_crc, crc_ok


class FakeSlaveSerial:
    """In-memory Modbus RTU slave behind a fake serial port.

    Answers FC01/03 reads and FC05/06/15/16 writes for ``slave_id`` from its
    coil/register banks, can echo every request like some RS485 adapters, and
    can hold bytes that are already on the bus before the master transmits.
    """

```

MBToolkit v3.4/tests/test_master.py — lines 41-80

```
41    def __init__(self, slave_id=1, *, echo=False, bus_traffic=b"", exception_for=None):
42        self.slave_id = slave_id
43        self.echo = echo
44        self.exception_for = exception_for or {}
45        self.registers = {i: i * 10 for i in range(200)}
46        self.coils = {i: i % 2 for i in range(200)}
47        self.rx = bytearray(bus_traffic)
48        self.requests = []
49
50    @property
51    def in_waiting(self):
52        return len(self.rx)
53
54    def read(self, size):
55        data = bytes(self.rx[:size])
56        del self.rx[:size]
57        return data
58
59    def reset_input_buffer(self):
60        self.rx.clear()
61
62    def reset_output_buffer(self):
63        pass
64
65    def cancel_read(self):
66        pass
67
68    def cancel_write(self):
69        pass
70
71    def close(self):
72        pass
73
74    def write(self, data):
75        data = bytes(data)
76        self.requests.append(data)
77        if self.echo:
78            self.rx.extend(data)
79        if crc_ok(data) and data[0] == self.slave_id:
80            self.rx.extend(self._answer(data))
```

MBToolkit v3.4/tests/test_master.py — lines 81-120

```
        elif crc_ok(data) and data[0] == 0:
            self._answer(data)  # broadcast: execute, never answer
        return len(data)

    def _answer(self, req):
        fc = req[1]
        if fc in self.exception_for:
            return append_crc(bytes((req[0], fc | 0x80, self.exception_for[fc])))
        address = (req[2] << 8) | req[3]
        qty = (req[4] << 8) | req[5]
        if fc == 3:
            payload = b"".join(struct.pack(">H", self.registers.get(address + i, 0))
                               for i in range(qty))
            return append_crc(bytes((req[0], 3, len(payload))) + payload)
        if fc == 1:
            data = bytearray((qty + 7) // 8)
            for i in range(qty):
                if self.coils.get(address + i):
                    data[i // 8] |= 1 << (i % 8)
            return append_crc(bytes((req[0], 1, len(data))) + bytes(data))
        if fc == 6:
            self.registers[address] = qty
            return req
        if fc == 5:
            self.coils[address] = 1 if qty == 0xFF00 else 0
            return req
        if fc == 16:
            for i in range(qty):
                self.registers[address + i] = (req[7 + 2 * i] << 8) | req[8 + 2 * i]
            return append_crc(req[:6])
        if fc == 15:
            for i in range(qty):
                self.coils[address + i] = (req[7 + i // 8] >> (i % 8)) & 1
            return append_crc(req[:6])
        return append_crc(bytes((req[0], fc | 0x80, 1)))


def drain(event_queue):
    events = []
    while not event_queue.empty():
```

MBToolkit v3.4/tests/test_master.py — lines 121-160

```
        events.append(event_queue.get())
    return events


def make_engine(serial, **overrides):
    options = dict(baud=115200, parity="None", stopbits="1",
                   response_timeout_ms=30, delay_ms=0)
    options.update(overrides)
    return MasterEngine(serial, queue.Queue(), **options)


class MasterFormatTests(unittest.TestCase):
    def test_32_bit_formats_round_trip_in_every_byte_order(self):
        for order in BYTE_ORDERS:
            for kind, text in (("int32", "-123456"), ("uint32", "4000000000"),
                               ("float32", "123.456")):
                fmt = f"{kind}_{order}"
                registers = encode_register_value(text, fmt)
                self.assertEqual(len(registers), 2)
                self.assertEqual(format_register_values(registers, fmt)[0], text)

    def test_float_byte_orders_match_wire_layout(self):
        big_endian = struct.pack(">f", 1.5)  # 3F C0 00 00 = A B C D
        self.assertEqual(encode_register_value("1.5", "float32_ABCD"), [0x3FC0, 0x0000])
        self.assertEqual(encode_register_value("1.5", "float32_CDAB"), [0x0000, 0x3FC0])
        self.assertEqual(encode_register_value("1.5", "float32_BADC"), [0xC03F, 0x0000])
        self.assertEqual(encode_register_value("1.5", "float32_DCBA"), [0x0000, 0xC03F])
        self.assertEqual(big_endian, bytes.fromhex("3FC00000"))

    def test_16_bit_formats(self):
        self.assertEqual(format_register_values([0xFFFE], "int16"), ["-2"])
        self.assertEqual(format_register_values([0xFFFE], "uint16"), ["65534"])
        self.assertEqual(format_register_values([0x00AB], "hex16"), ["0x00AB"])
        self.assertEqual(format_register_values([0x0105], "bin16"), ["00000001 00000101"])
        self.assertEqual(encode_register_value("-2", "int16"), [0xFFFE])
        self.assertEqual(encode_register_value("0x1A2B", "hex16"), [0x1A2B])
        self.assertEqual(encode_register_value("0000 0001 0000 0101", "bin16"), [0x0105])

    def test_values_outside_the_format_are_rejected(self):
        for text, fmt in (("40000", "int16"), ("-1", "uint16"), ("abc", "float32_ABCD"),
```

MBToolkit v3.4/tests/test_master.py — lines 161-200

```
                        ("1e40", "float32_ABCD"), ("", "int16")):
            with self.assertRaises(ValueError):
                encode_register_value(text, fmt)
        with self.assertRaises(ValueError):
            encode_coil_value("2")

    def test_unpaired_register_in_32_bit_format(self):
        self.assertEqual(format_register_values([1, 2, 3], "uint32_ABCD"), ["65538", "", "—"])


class MasterFrameTests(unittest.TestCase):
    def test_write_frames_match_modbus_specification_examples(self):
        # Modbus Application Protocol V1.1b3 examples (CRC appended for RTU).
        self.assertEqual(build_write_request(0x11, 16, 1, [0x000A, 0x0102])[:-2],
                         bytes.fromhex("11 10 0001 0002 04 000A 0102"))
        coils = [1, 0, 1, 1, 0, 0, 1, 1, 1, 0]
        self.assertEqual(build_write_request(0x11, 15, 0x13, coils)[:-2],
                         bytes.fromhex("11 0F 0013 000A 02 CD 01"))
        self.assertEqual(build_write_request(0x11, 5, 0xAC, [1])[:-2],
                         bytes.fromhex("11 05 00AC FF00"))
        self.assertEqual(build_write_request(0x11, 6, 1, [3])[:-2],
                         bytes.fromhex("11 06 0001 0003"))

    def test_invalid_definitions_are_rejected(self):
        for args in ((0, 3, 0, 1), (1, 7, 0, 1), (1, 3, 0, 126), (1, 1, 0, 2001),
                     (1, 3, 65535, 2)):
            with self.assertRaises(ValueError):
                build_read_request(*args)
        with self.assertRaises(ValueError):
            build_write_request(1, 6, 0, [1, 2])

    def test_read_response_found_after_adapter_echo(self):
        request = build_read_request(1, 3, 0, 2)
        response = append_crc(bytes.fromhex("01 03 04 0001 0002"))
        found = find_master_response(request + response, request)
        self.assertEqual(found["kind"], "RESPONSE")
        self.assertEqual(found["values"], [1, 2])

    def test_coil_response_is_trimmed_to_quantity(self):
        request = build_read_request(1, 1, 0, 3)
```

MBToolkit v3.4/tests/test_master.py — lines 201-240

```
        found = find_master_response(append_crc(bytes.fromhex("01 01 01 FD")), request)
        self.assertEqual(found["values"], [1, 0, 1])

    def test_exception_and_bad_crc(self):
        request = build_read_request(1, 3, 0, 1)
        exception = append_crc(bytes.fromhex("01 83 02"))
        self.assertEqual(find_master_response(exception, request)["exception_code"], 2)
        damaged = bytearray(append_crc(bytes.fromhex("01 03 02 0001")))
        damaged[-1] ^= 0xFF
        self.assertIsNone(find_master_response(bytes(damaged), request))

    def test_write_response_must_match_request(self):
        request = build_write_request(1, 16, 5, [1, 2])
        self.assertIsNotNone(find_master_response(append_crc(request[:6]), request))
        other = append_crc(bytes.fromhex("01 10 0006 0002"))
        self.assertIsNone(find_master_response(other, request))


class MasterEngineTests(unittest.TestCase):
    def run_engine(self, engine, seconds):
        engine.start()
        time.sleep(seconds)
        engine.stop()
        return drain(engine.event_queue)

    def test_scheduler_serves_writes_then_read_once_then_due_polls(self):
        engine = make_engine(FakeSlaveSerial())
        engine.set_poll(PollDefinition(1, 1, 3, 0, 1, 1000))
        engine.set_poll(PollDefinition(2, 1, 3, 0, 1, 1000))
        engine.read_once(2)
        engine.queue_write(1, build_write_request(1, 6, 0, [5]))
        self.assertEqual(engine.next_job(now=10.0)[0], "write")
        self.assertEqual(engine.next_job(now=10.0)[1:], (engine._polls[2][0], True))
        first = engine.next_job(now=10.0)
        second = engine.next_job(now=10.0)
        self.assertEqual({first[1].window_id, second[1].window_id}, {1, 2})
        self.assertEqual(engine.next_job(now=10.5)[0], "wait")
        self.assertEqual(engine.next_job(now=11.01)[0], "read")

    def test_fast_and_slow_windows_share_one_bus(self):
```

MBToolkit v3.4/tests/test_master.py — lines 241-280

```
        serial = FakeSlaveSerial()
        engine = make_engine(serial)
        engine.set_poll(PollDefinition(1, 1, 3, 0, 4, 0))
        engine.set_poll(PollDefinition(2, 1, 1, 0, 8, 10000))
        events = self.run_engine(engine, 0.4)
        results = [p for name, p in events if name == "master_result"]
        fast = [r for r in results if r["window_id"] == 1]
        slow = [r for r in results if r["window_id"] == 2]
        self.assertGreater(len(fast), 3)
        self.assertEqual(len(slow), 1)
        self.assertEqual(fast[0]["values"], [0, 10, 20, 30])
        self.assertEqual(slow[0]["values"], [0, 1, 0, 1, 0, 1, 0, 1])
        self.assertEqual(events[0][0], "master_ready")
        self.assertEqual(events[-1], ("master_stopped", {"error": None}))

    def test_write_is_executed_and_reported(self):
        serial = FakeSlaveSerial(echo=True)
        engine = make_engine(serial)
        engine.queue_write(1, build_write_request(1, 16, 10, [0xABCD, 0x1234]), "FC16")
        results = [p for n, p in self.run_engine(engine, 0.15) if n == "master_result"]
        self.assertEqual(results[0]["status"], "ok")
        self.assertEqual((serial.registers[10], serial.registers[11]), (0xABCD, 0x1234))

    def test_broadcast_write_does_not_wait_for_a_response(self):
        serial = FakeSlaveSerial()
        engine = make_engine(serial, response_timeout_ms=5000)
        engine.queue_write(1, build_write_request(0, 6, 3, [77]))
        started = time.perf_counter()
        result = engine.transact(engine._writes.popleft()[1])
        self.assertLess(time.perf_counter() - started, 1.0)
        self.assertEqual(result["status"], "broadcast")
        self.assertEqual(serial.registers[3], 77)

    def test_timeout_exception_and_invalid_results(self):
        engine = make_engine(FakeSlaveSerial(slave_id=9))
        self.assertEqual(engine.transact(build_read_request(1, 3, 0, 1))["status"], "timeout")
        engine = make_engine(FakeSlaveSerial(exception_for={3: 2}))
        result = engine.transact(build_read_request(1, 3, 0, 1))
        self.assertEqual((result["status"], result["exception_code"]), ("exception", 2))

```

MBToolkit v3.4/tests/test_master.py — lines 281-320

```
281     def test_traffic_on_the_bus_blocks_the_master(self):
282         serial = FakeSlaveSerial(bus_traffic=b"\x01\x03\x00")
283         engine = make_engine(serial, idle_check_seconds=0.05)
284         engine.set_poll(PollDefinition(1, 1, 3, 0, 1, 0))
285         events = self.run_engine(engine, 0.2)
286         self.assertEqual(serial.requests, [])
287         self.assertEqual(events[-1][0], "master_stopped")
288         self.assertEqual(events[-1][1]["error"][0], "traffic")
289 
290 
291 class MasterWindowTests(unittest.TestCase):
292     def test_workspace_entry_round_trip_and_validation(self):
293         window = MasterWindow(1, 1, slave=5, fc=3, address=100, qty=4, scan_ms=250, base=1,
294                               fmt="float32_CDAB", names={100: "Temperatura"})
295         data = MasterWindow.validated_kwargs(window.to_json())
296         self.assertEqual(data["names"], {100: "Temperatura"})
297         self.assertEqual((data["slave"], data["fmt"], data["base"]), (5, "float32_CDAB", 1))
298         for bad in ({"slave": 0}, {"fc": 6}, {"qty": 200}, {"scan_ms": -1}, "x"):
299             with self.assertRaises((ValueError, TypeError)):
300                 MasterWindow.validated_kwargs(bad)
301 
302     def test_write_dialog_builds_fc16_for_float_values(self):
303         window = MasterWindow(1, 1, slave=2, fc=3, address=0, qty=10, base=1,
304                               fmt="float32_ABCD")
305         request, _ = MasterMixin.master_build_write(None, window, "2", "11", "1.5;2", 16)
306         self.assertEqual(request[:-2], bytes.fromhex("02 10 000A 0004 08 3FC0 0000 4000 0000"))
307         with self.assertRaises(ValueError):
308             MasterMixin.master_build_write(None, window, "2", "11", "1.5", 6)
309 
310 
311 class MasterWorkspaceFolderTests(unittest.TestCase):
312     def setUp(self):
313         self._appdata = os.environ.get("APPDATA")
314         self._folder = tempfile.TemporaryDirectory()
315         os.environ["APPDATA"] = self._folder.name
316 
317     def tearDown(self):
318         if self._appdata is None:
319             os.environ.pop("APPDATA", None)
320         else:
```

MBToolkit v3.4/tests/test_master.py — lines 321-360

```
            os.environ["APPDATA"] = self._appdata
        self._folder.cleanup()

    def test_workspaces_live_beside_the_settings(self):
        folder = mb_config.get_workspaces_folder()
        self.assertEqual(folder.name, "Workspaces")
        self.assertEqual(folder.parent, mb_config.get_settings_file().parent)
        self.assertTrue(folder.is_dir())
        master = mb_config.get_master_workspaces_folder()
        self.assertEqual((master.name, master.parent), ("Master", folder))
        self.assertEqual(mb_master_ui.workspace_path("Linha 1").parent, master)

    def test_loose_master_workspaces_move_into_the_master_folder(self):
        root = mb_config.get_workspaces_folder()
        (root / "Linha 1.mbsw").write_text('{"windows": []}', encoding="utf-8")
        (root / "notas.txt").write_text("x", encoding="utf-8")
        self.assertEqual(mb_master_ui.list_workspaces(), ["Linha 1"])
        master = mb_config.get_master_workspaces_folder()
        self.assertTrue((master / "Linha 1.mbsw").is_file())
        self.assertFalse((root / "Linha 1.mbsw").exists())
        self.assertTrue((root / "notas.txt").exists())
        # Only the first time: a file put in Workspaces later stays there.
        (root / "Outro.mbsw").write_text("{}", encoding="utf-8")
        self.assertEqual(mb_master_ui.list_workspaces(), ["Linha 1"])

    def test_saved_workspaces_are_listed_by_name(self):
        for name in ("Subestação B", "armazém", "Linha 1"):
            mb_master_ui.workspace_path(name).write_text("{}", encoding="utf-8")
        (mb_config.get_master_workspaces_folder() / "notas.txt").write_text("x", encoding="utf-8")
        self.assertEqual(mb_master_ui.list_workspaces(),
                         ["armazém", "Linha 1", "Subestação B"])

    def test_workspace_names_must_be_valid_file_names(self):
        self.assertEqual(mb_master_ui.validate_workspace_name("  Linha 1. "), "Linha 1")
        for bad in ("", "   ", "a/b", "x:y", "CON", "com3", "a" * 61, 'q"'):
            with self.assertRaises(ValueError):
                mb_master_ui.validate_workspace_name(bad)

    def test_mbsniffer_settings_and_workspaces_are_migrated_once(self):
        legacy = Path(self._folder.name) / "MBSniffer"
```

MBToolkit v3.4/tests/test_master.py — lines 361-400

```
361         (legacy / "Workspaces").mkdir(parents=True)
362         (legacy / "settings.json").write_text('{"language": "English"}', encoding="utf-8")
363         (legacy / "Workspaces" / "Linha 1.mbsw").write_text(
364             '{"format": "MBSniffer Master workspace", "windows": []}', encoding="utf-8")
365 
366         settings = mb_config.get_settings_file()
367         self.assertEqual(settings.parent.name, "MBToolkit")
368         self.assertEqual(mb_config.load_ui_settings()["language"], "English")
369         self.assertEqual(mb_master_ui.list_workspaces(), ["Linha 1"])
370         self.assertTrue((legacy / "settings.json").exists())  # old folder kept
371 
372         # Later changes in the old folder are not copied again.
373         (legacy / "Workspaces" / "Outro.mbsw").write_text("{}", encoding="utf-8")
374         mb_config.get_settings_file()
375         self.assertEqual(mb_master_ui.list_workspaces(), ["Linha 1"])
376 
377     def test_legacy_workspace_format_is_accepted(self):
378         self.assertIn("MBSniffer Master workspace", mb_config.MASTER_WORKSPACE_FORMATS)
379         self.assertEqual(mb_config.MASTER_WORKSPACE_FORMAT, "MBToolkit Master workspace")
380 
381     def test_window_title_is_saved_in_the_workspace(self):
382         window = MasterWindow(1, 2, title="  Inversor 3  ")
383         self.assertEqual(window.title, "Inversor 3")
384         self.assertEqual(MasterWindow.validated_kwargs(window.to_json())["title"], "Inversor 3")
385         self.assertEqual(MasterWindow.validated_kwargs({"slave": 1})["title"], "")
386 
387 
388 class MasterPerAddressFormatTests(unittest.TestCase):
389     def test_64_bit_formats_round_trip_in_every_byte_order(self):
390         for order in BYTE_ORDERS_64:
391             for kind, text in (("int64", "-1234567890123"), ("uint64", "18000000000000000000"),
392                                ("float64", "123456.789012345")):
393                 fmt = f"{kind}_{order}"
394                 registers = encode_register_value(text, fmt)
395                 self.assertEqual(len(registers), 4)
396                 self.assertEqual(format_value(registers, fmt), text)
397 
398     def test_double_byte_orders_match_wire_layout(self):
399         # 1.0 = 3FF0 0000 0000 0000 in IEEE 754 big-endian.
400         self.assertEqual(encode_register_value("1", "float64_ABCDEFGH"), [0x3FF0, 0, 0, 0])
```

MBToolkit v3.4/tests/test_master.py — lines 401-440

```
401         self.assertEqual(encode_register_value("1", "float64_GHEFCDAB"), [0, 0, 0, 0x3FF0])
402         self.assertEqual(encode_register_value("1", "float64_BADCFEHG"), [0xF03F, 0, 0, 0])
403         self.assertEqual(encode_register_value("1", "float64_HGFEDCBA"), [0, 0, 0, 0xF03F])
404
405     def test_per_address_formats_override_the_window_format(self):
406         registers = [0x42F6, 0xE979, 0xFFFF, 0x0001, 0x0002, 7]
407         texts = format_window_values(registers, 100, "int16", {100: "float32_ABCD"})
408         self.assertEqual(texts, ["123.456", "", "-1", "1", "2", "7"])
409         self.assertEqual(register_groups(6, 100, "int16", {102: "int64_ABCDEFGH"}),
410                          [(0, 1, "int16"), (1, 1, "int16"), (2, 4, "int64_ABCDEFGH")])
411
412     def test_default_multi_register_values_cut_by_an_override_show_a_dash(self):
413         texts = format_window_values([1, 2, 3, 4, 5], 0, "uint32_ABCD", {1: "int16"})
414         self.assertEqual(texts, ["—", "2", "196612", "", "—"])
415
416     def test_selection_rules_for_multi_register_formats(self):
417         window = MasterWindow(1, 1, address=100, qty=10)
418         apply_format_to_rows(window, [2, 3, 4, 5], "float32_CDAB")
419         self.assertEqual(window.formats, {102: "float32_CDAB", 104: "float32_CDAB"})
420         apply_format_to_rows(window, [4, 5, 6, 7], "float64_ABCDEFGH")
421         self.assertEqual(window.formats, {102: "float32_CDAB", 104: "float64_ABCDEFGH"})
422         apply_format_to_rows(window, [0, 1], "hex16")
423         self.assertEqual(window.formats[100], "hex16")
424         self.assertEqual(window.formats[101], "hex16")
425         apply_format_to_rows(window, [4], None)
426         self.assertNotIn(104, window.formats)
427         for rows, fmt in (([1], "float32_ABCD"), ([1, 3], "float32_ABCD"),
428                           ([0, 1, 2], "float32_ABCD"), ([0, 1], "int64_ABCDEFGH"),
429                           ([8, 9, 10, 11], "int64_ABCDEFGH"), ([], "int16")):
430             with self.assertRaises(ValueError):
431                 apply_format_to_rows(window, rows, fmt)
432
433     def test_formats_are_saved_and_invalid_ones_dropped(self):
434         window = MasterWindow(1, 1, address=0, qty=4, formats={0: "float32_ABCD"})
435         data = MasterWindow.validated_kwargs(window.to_json())
436         self.assertEqual(data["formats"], {0: "float32_ABCD"})
437         data = MasterWindow.validated_kwargs(
438             {"formats": {"0": "nonsense", "x": "int16", "70000": "int16", "2": "hex16"}})
439         self.assertEqual(data["formats"], {2: "hex16"})
440
```

MBToolkit v3.4/tests/test_master.py — lines 441-480

```
441     def test_write_uses_the_format_of_the_clicked_value(self):
442         window = MasterWindow(1, 1, slave=3, address=0, qty=8, fmt="int16")
443         request, _ = MasterMixin.master_build_write(None, window, "3", "4", "1", 16,
444                                                     "float64_ABCDEFGH")
445         self.assertEqual(request[:-2], bytes.fromhex("03 10 0004 0004 08 3FF0 0000 0000 0000"))
446 
447 
448 class MasterScaleTests(unittest.TestCase):
449     SCALE = validate_scale("2000", "10000", "-180", "180", 1, "°")
450 
451     def test_scaled_values_are_shown_in_engineering_units(self):
452         texts = format_window_values([2000, 6000, 10000, 0x00AB], 0, "int16", {3: "hex16"},
453                                      {0: self.SCALE, 1: self.SCALE, 2: self.SCALE,
454                                       3: self.SCALE})
455         self.assertEqual(texts, ["-180.0 °", "0.0 °", "180.0 °", "0x00AB"])
456 
457     def test_scale_applies_to_multi_register_values(self):
458         registers = encode_register_value("50", "float32_ABCD")
459         scale = validate_scale(0, 100, 0, 10, 2, "bar")
460         self.assertEqual(format_window_values(registers, 7, "float32_ABCD", None, {7: scale}),
461                          ["5.00 bar", ""])
462 
463     def test_written_engineering_values_are_converted_to_raw(self):
464         self.assertEqual(raw_text_for_scaled("90", self.SCALE, "int16"), "8000")
465         self.assertEqual(raw_text_for_scaled("-180", self.SCALE, "uint16"), "2000")
466         window = MasterWindow(1, 1, slave=1, address=0, qty=4,
467                               scales={0: self.SCALE})
468         request, _ = MasterMixin.master_build_write(None, window, "1", "0", "45", 6,
469                                                     "int16", self.SCALE)
470         self.assertEqual(request[:-2], bytes.fromhex("01 06 0000 1B58"))  # 7000
471 
472     def test_invalid_scales_are_rejected(self):
473         for args in (("1", "1", "0", "10"), ("a", "1", "0", "1"), ("0", "1", "0", "1", "9"),
474                      ("0", "nan", "0", "1")):
475             with self.assertRaises(ValueError):
476                 validate_scale(*args)
477         with self.assertRaises(ValueError):
478             raw_text_for_scaled("5", validate_scale(0, 10, 3, 3), "int16")
479 
480     def test_scales_are_saved_and_invalid_ones_dropped(self):
```

MBToolkit v3.4/tests/test_master.py — lines 481-520

```
        window = MasterWindow(1, 1, scales={4: self.SCALE})
        data = MasterWindow.validated_kwargs(window.to_json())
        self.assertEqual(data["scales"], {4: self.SCALE})
        data = MasterWindow.validated_kwargs({"scales": {
            "1": {"raw": [0, 0], "eng": [0, 1]}, "x": self.SCALE, "2": "bad",
            "3": {"raw": [0, 10], "eng": [0, 1], "decimals": 2, "unit": "A"}}})
        self.assertEqual(list(data["scales"]), [3])


class MasterOpenFromTrafficTests(unittest.TestCase):
    def definition(self, record):
        return MasterMixin.master_definition_from_record(None, record)

    def test_request_and_matched_response_give_the_read_definition(self):
        request = {"type": "REQUEST", "slave": "5", "fc": "03", "pdu_address": 100, "qty": 4}
        self.assertEqual(self.definition(request),
                         {"slave": 5, "fc": 3, "address": 100, "qty": 4})
        response = {"type": "RESPONSE", "slave": "5", "fc": "04",
                    "request_pdu_address": 7, "request_qty": 2}
        self.assertEqual(self.definition(response)["address"], 7)
        exception = {"type": "EXCEPTION", "slave": "5", "fc": "83",
                     "request_pdu_address": 1, "request_qty": 1}
        self.assertEqual(self.definition(exception)["fc"], 3)

    def test_frames_without_a_read_definition_are_refused(self):
        for record in (
            {"type": "REQUEST", "slave": "5", "fc": "06", "pdu_address": 1, "qty": None},
            {"type": "RESPONSE", "slave": "5", "fc": "03"},
            {"type": "RAW/UNPARSED", "slave": "", "fc": ""},
            {"type": "REQUEST", "slave": "0", "fc": "03", "pdu_address": 1, "qty": 1},
        ):
            self.assertIsNone(self.definition(record))


class FlakySlaveSerial(FakeSlaveSerial):
    """Fake slave whose port fails (like an unplugged USB adapter)."""

    def __init__(self, failures, **kwargs):
        super().__init__(**kwargs)
        self.failures = failures
```

MBToolkit v3.4/tests/test_master.py — lines 521-560

```
        self.closed = False

    def write(self, data):
        if self.failures:
            self.failures -= 1
            raise OSError("device disconnected")
        return super().write(data)

    def close(self):
        self.closed = True


class MasterReconnectTests(unittest.TestCase):
    def test_engine_reopens_the_port_after_a_failure(self):
        broken = FlakySlaveSerial(failures=1)
        fresh = FakeSlaveSerial()
        reopen_calls = []
        engine = make_engine(broken, reconnect_interval_s=0.05,
                             reopen=lambda: reopen_calls.append(1) or fresh)
        engine.set_poll(PollDefinition(1, 1, 3, 0, 2, 0))
        engine.start()
        time.sleep(0.4)
        engine.stop()
        events = drain(engine.event_queue)
        names = [name for name, _payload in events]
        self.assertIn("master_reconnecting", names)
        self.assertIn("master_reconnected", names)
        self.assertTrue(broken.closed)
        self.assertIs(engine.ser, fresh)
        after = names[names.index("master_reconnected"):]
        self.assertIn("master_result", after)
        self.assertEqual(events[-1], ("master_stopped", {"error": None}))

    def test_engine_keeps_retrying_until_stopped(self):
        attempts = []

        def reopen():
            attempts.append(1)
            raise OSError("still unplugged")

```

MBToolkit v3.4/tests/test_master.py — lines 561-600

```
        engine = make_engine(FlakySlaveSerial(failures=5), reconnect_interval_s=0.02,
                             reopen=reopen)
        engine.set_poll(PollDefinition(1, 1, 3, 0, 1, 0))
        engine.start()
        time.sleep(0.3)
        engine.stop()
        events = drain(engine.event_queue)
        self.assertGreater(len(attempts), 3)
        self.assertNotIn("master_reconnected", [n for n, _p in events])
        self.assertEqual(events[-1][0], "master_stopped")

    def test_without_reopen_a_failure_ends_the_session(self):
        engine = make_engine(FlakySlaveSerial(failures=1))
        engine.set_poll(PollDefinition(1, 1, 3, 0, 1, 0))
        engine.start()
        time.sleep(0.2)
        engine.stop()
        stopped = drain(engine.event_queue)[-1]
        self.assertEqual(stopped[0], "master_stopped")
        self.assertEqual(stopped[1]["error"][0], "serial")


class MasterSettingsTests(unittest.TestCase):
    def test_last_workspace_is_validated(self):
        with tempfile.TemporaryDirectory() as folder:
            path = Path(folder) / "settings.json"
            original = mb_config.get_settings_file
            mb_config.get_settings_file = lambda: path
            try:
                for value, expected in (("Linha 1", "Linha 1"), ("a/b", ""), (5, ""),
                                        ("x" * 61, ""), ("  Subestação  ", "Subestação")):
                    path.write_text(json.dumps({"master": {"last_workspace": value}}),
                                    encoding="utf-8")
                    self.assertEqual(
                        mb_config.load_ui_settings()["master"]["last_workspace"], expected)
            finally:
                mb_config.get_settings_file = original


if __name__ == "__main__":
```

MBToolkit v3.4/tests/test_master.py — lines 601-601

```
601     unittest.main(verbosity=2)
```

MBToolkit v3.4/tests/test_slave.py

459 source lines.

MBToolkit v3.4/tests/test_slave.py — lines 1-40

```
import os
import queue
import random
import tempfile
import threading
import time
import unittest

from mb_master import (
    MasterEngine,
    PollDefinition,
    build_read_request,
    build_write_request,
    find_master_response,
)
from mb_protocol import append_crc, crc_ok
from mb_slave import (
    SlaveEngine,
    SlaveStore,
    corrupt_crc,
    fault_active,
    handle_request,
    request_length,
    validate_fault,
)
import mb_config
from mb_i18n import LANGUAGE_ENGLISH, translate_text
from mb_slave_ui import SlaveWindow, fault_parts, validate_area
from mb_table_ui import list_workspaces, workspace_path


class BusEnd:
    """One end of an in-memory serial link: what one end writes, the other reads.

    With *echo*, every byte written also comes back to the writer, like an
    RS485 adapter that hears its own transmission.
    """

    def __init__(self, echo=False):
        self.echo = echo
```

MBToolkit v3.4/tests/test_slave.py — lines 41-80

```
        self.peer = None
        self.rx = bytearray()
        self.written = []
        self._lock = threading.Lock()

    @staticmethod
    def pair(echo_a=False, echo_b=False):
        a, b = BusEnd(echo_a), BusEnd(echo_b)
        a.peer, b.peer = b, a
        return a, b

    @property
    def in_waiting(self):
        with self._lock:
            return len(self.rx)

    def read(self, size):
        with self._lock:
            data = bytes(self.rx[:size])
            del self.rx[:size]
            return data

    def write(self, data):
        data = bytes(data)
        self.written.append(data)
        if self.echo:
            self.inject(data)
        if self.peer is not None:
            self.peer.inject(data)
        return len(data)

    def inject(self, data):
        with self._lock:
            self.rx.extend(data)

    def reset_input_buffer(self):
        with self._lock:
            self.rx.clear()

    def reset_output_buffer(self):
```

MBToolkit v3.4/tests/test_slave.py — lines 81-120

```
        pass

    def cancel_read(self):
        pass

    def cancel_write(self):
        pass

    def close(self):
        pass


def make_store():
    store = SlaveStore()
    store.set_areas([(1, "holding", 0, 20), (1, "coils", 0, 16), (1, "input", 100, 4),
                     (1, "discrete", 0, 8), (2, "holding", 0, 10)])
    store.write(1, "holding", 0, [i * 10 for i in range(20)])
    store.write(1, "input", 100, [7, 8, 9, 10])
    store.write(1, "coils", 0, [1, 0, 1, 1])
    store.write(1, "discrete", 0, [0, 1])
    return store


def drain(event_queue):
    events = []
    while not event_queue.empty():
        events.append(event_queue.get())
    return events


def wait_for(predicate, timeout=2.0):
    deadline = time.monotonic() + timeout
    while time.monotonic() < deadline:
        if predicate():
            return True
        time.sleep(0.005)
    return predicate()


class SlaveRequestTests(unittest.TestCase):
```

MBToolkit v3.4/tests/test_slave.py — lines 121-160

```
121     def test_reads_answer_from_the_tables(self):
122         store = make_store()
123         reply = handle_request(store, build_read_request(1, 3, 2, 3))
124         self.assertEqual(reply.status, "ok")
125         found = find_master_response(reply.response, build_read_request(1, 3, 2, 3))
126         self.assertEqual(found["values"], [20, 30, 40])
127         found = find_master_response(handle_request(store, build_read_request(1, 1, 0, 4)).response,
128                                      build_read_request(1, 1, 0, 4))
129         self.assertEqual(found["values"], [1, 0, 1, 1])
130         found = find_master_response(handle_request(store, build_read_request(1, 4, 100, 4)).response,
131                                      build_read_request(1, 4, 100, 4))
132         self.assertEqual(found["values"], [7, 8, 9, 10])
133         found = find_master_response(handle_request(store, build_read_request(1, 2, 0, 2)).response,
134                                      build_read_request(1, 2, 0, 2))
135         self.assertEqual(found["values"], [0, 1])
136 
137     def test_writes_change_the_tables_and_answer_like_a_device(self):
138         store = make_store()
139         request = build_write_request(1, 6, 5, [1234])
140         reply = handle_request(store, request)
141         self.assertEqual(reply.response, request)
142         self.assertEqual(reply.written, [(1, "holding", 5, 1)])
143         self.assertEqual(store.read(1, "holding", 5, 1), [1234])
144 
145         request = build_write_request(1, 16, 10, [1, 2, 3])
146         reply = handle_request(store, request)
147         self.assertEqual(reply.response, append_crc(request[:6]))
148         self.assertEqual(store.read(1, "holding", 10, 3), [1, 2, 3])
149 
150         handle_request(store, build_write_request(1, 5, 1, [1]))
151         handle_request(store, build_write_request(1, 15, 8, [1, 1, 0, 1, 0, 0, 1]))
152         self.assertEqual(store.read(1, "coils", 0, 2), [1, 1])
153         self.assertEqual(store.read(1, "coils", 8, 7), [1, 1, 0, 1, 0, 0, 1])
154 
155     def test_exceptions_like_a_real_device(self):
156         store = make_store()
157         # Address outside the defined areas, also when only partly outside.
158         for request in (build_read_request(1, 3, 20, 1), build_read_request(1, 3, 18, 4),
159                         build_read_request(1, 4, 0, 1), build_write_request(1, 6, 50, [1])):
160             reply = handle_request(store, request)
```

MBToolkit v3.4/tests/test_slave.py — lines 161-200

```
            self.assertEqual((reply.status, reply.exception_code), ("exception", 2))
            self.assertEqual(reply.response, append_crc(bytes((1, request[1] | 0x80, 2))))
        # Unsupported function.
        reply = handle_request(store, append_crc(bytes((1, 0x08, 0, 0, 0x12, 0x34))))
        self.assertEqual(reply.exception_code, 1)
        # Bad quantity and bad FC05 value.
        reply = handle_request(store, append_crc(bytes((1, 3, 0, 0, 0, 0))))
        self.assertEqual(reply.exception_code, 3)
        reply = handle_request(store, append_crc(bytes((1, 5, 0, 1, 0x12, 0x34))))
        self.assertEqual(reply.exception_code, 3)

    def test_other_slave_ids_are_ignored(self):
        reply = handle_request(make_store(), build_read_request(9, 3, 0, 1))
        self.assertEqual(reply.status, "ignored")
        self.assertIsNone(reply.response)

    def test_broadcast_writes_every_covering_slave_without_answering(self):
        store = make_store()
        reply = handle_request(store, build_write_request(0, 6, 3, [77]))
        self.assertEqual(reply.status, "broadcast")
        self.assertIsNone(reply.response)
        self.assertEqual(sorted(reply.written), [(1, "holding", 3, 1), (2, "holding", 3, 1)])
        self.assertEqual(store.read(2, "holding", 3, 1), [77])
        # Slave 2 does not cover address 15: only slave 1 is written.
        reply = handle_request(store, build_write_request(0, 6, 15, [5]))
        self.assertEqual(reply.written, [(1, "holding", 15, 1)])
        # Broadcast reads and errors are never answered.
        self.assertIsNone(handle_request(store, append_crc(bytes((0, 3, 0, 0, 0, 1)))).response)
        self.assertIsNone(handle_request(store, append_crc(bytes((0, 6, 0, 90, 0, 1)))).response)

    def test_request_length(self):
        self.assertIsNone(request_length(b"\x01"))
        self.assertEqual(request_length(build_read_request(1, 3, 0, 1)), 8)
        self.assertEqual(request_length(build_write_request(1, 16, 0, [1, 2])), 13)
        self.assertIsNone(request_length(b"\x01\x10\x00\x00\x00\x02"))
        self.assertIsNone(request_length(b"\x01\x2B\x0E\x01"))

    def test_window_areas_define_the_slave(self):
        window = SlaveWindow(1, 1, slave=4, table="input", address=10, qty=5)
        store = SlaveStore()
```

MBToolkit v3.4/tests/test_slave.py — lines 201-240

```
        store.set_areas([window.area()])
        self.assertEqual(store.slave_ids(), {4})
        self.assertTrue(store.covers(4, "input", 10, 5))
        self.assertFalse(store.covers(4, "input", 10, 6))


class SlaveFaultTests(unittest.TestCase):
    def test_fault_validation(self):
        self.assertFalse(fault_active(validate_fault({})))
        fault = validate_fault({"delay_ms": "250", "probability": "10", "exception": 6})
        self.assertTrue(fault_active(fault))
        self.assertEqual((fault["delay_ms"], fault["probability"], fault["exception"]),
                         (250, 10, 6))
        for bad in ({"delay_ms": -1}, {"delay_ms": 10001}, {"probability": 0},
                    {"probability": 101}, {"exception": 5}, {"delay_ms": "x"}):
            with self.assertRaises(ValueError):
                validate_fault(bad)

    def test_inactive_faults_are_not_stored(self):
        store = make_store()
        store.set_fault(1, validate_fault({"probability": 50}))
        self.assertIsNone(store.fault(1))
        store.set_fault(1, validate_fault({"bad_crc": True}))
        self.assertTrue(store.fault(1)["bad_crc"])
        store.set_fault(1, None)
        self.assertEqual(store.faults(), {})

    def test_corrupt_crc(self):
        frame = build_read_request(1, 3, 0, 1)
        self.assertFalse(crc_ok(corrupt_crc(frame)))


class SlaveEngineTests(unittest.TestCase):
    """The Master engine talking to the Slave engine over an in-memory bus."""

    def setUp(self):
        self.store = make_store()
        self.master_end, self.slave_end = BusEnd.pair()
        self.slave_events = queue.Queue()
        self.master_events = queue.Queue()
```

MBToolkit v3.4/tests/test_slave.py — lines 241-280

```
        self.slave = None
        self.master = None

    def tearDown(self):
        for engine in (self.master, self.slave):
            if engine is not None:
                engine.stop()

    def start_slave(self, **options):
        self.slave = SlaveEngine(self.slave_end, self.slave_events, self.store, baud=115200,
                                 parity="None", stopbits="1", **options)
        self.slave.start()

    def start_master(self, timeout_ms=200):
        self.master = MasterEngine(self.master_end, self.master_events, baud=115200,
                                   parity="None", stopbits="1",
                                   response_timeout_ms=timeout_ms, delay_ms=0)
        self.master.start()

    def master_results(self, count, timeout=3.0):
        results = []
        deadline = time.monotonic() + timeout
        while len(results) < count and time.monotonic() < deadline:
            try:
                event, payload = self.master_events.get(timeout=0.05)
            except queue.Empty:
                continue
            if event == "master_result":
                results.append(payload)
        return results

    def test_master_reads_and_writes_every_function(self):
        self.start_slave()
        self.start_master()
        for window_id, (fc, address, qty) in enumerate(
                ((3, 0, 5), (4, 100, 4), (1, 0, 4), (2, 0, 2)), start=1):
            self.master.set_poll(PollDefinition(window_id, 1, fc, address, qty, 10000))
        values = {r["window_id"]: r["values"] for r in self.master_results(4)}
        self.assertEqual(values, {1: [0, 10, 20, 30, 40], 2: [7, 8, 9, 10],
                                  3: [1, 0, 1, 1], 4: [0, 1]})
```

MBToolkit v3.4/tests/test_slave.py — lines 281-320

```
281
282         for request in (build_write_request(1, 6, 0, [999]), build_write_request(1, 16, 1, [5, 6]),
283                         build_write_request(1, 5, 1, [1]),
284                         build_write_request(1, 15, 2, [0, 0])):
285             self.master.queue_write(1, request, "w")
286         statuses = [r["status"] for r in self.master_results(4) if r["kind"] == "write"]
287         self.assertEqual(statuses, ["ok"] * 4)
288         self.assertEqual(self.store.read(1, "holding", 0, 3), [999, 5, 6])
289         self.assertEqual(self.store.read(1, "coils", 0, 4), [1, 1, 0, 0])
290         written = [p["written"] for e, p in drain(self.slave_events)
291                    if e == "slave_request" and p["written"]]
292         self.assertIn([(1, "holding", 1, 2)], written)
293
294     def test_exception_reaches_the_master(self):
295         self.start_slave()
296         self.start_master()
297         self.master.set_poll(PollDefinition(1, 1, 3, 30, 2, 10000))
298         result = self.master_results(1)[0]
299         self.assertEqual((result["status"], result["exception_code"]), ("exception", 2))
300
301     def test_adapter_echo_and_foreign_traffic_are_ignored(self):
302         self.slave_end.echo = True
303         self.start_slave()
304         # A response of another slave on the bus before the request.
305         self.slave_end.inject(append_crc(bytes((9, 3, 2, 0x12, 0x34))))
306         self.start_master()
307         self.master.set_poll(PollDefinition(1, 1, 3, 2, 2, 20))
308         results = self.master_results(5)
309         self.assertEqual(len(results), 5)
310         self.assertTrue(all(r["status"] == "ok" and r["values"] == [20, 30] for r in results))
311         answered = []
312
313         def five_answered():
314             answered.extend(p for e, p in drain(self.slave_events)
315                             if e == "slave_request" and p["status"] == "ok")
316             return len(answered) >= 5
317
318         self.assertTrue(wait_for(five_answered))
319
320     def test_split_request_is_assembled(self):
```

MBToolkit v3.4/tests/test_slave.py — lines 321-360

```
321         self.start_slave()
322         request = build_read_request(1, 3, 0, 2)
323         self.slave_end.inject(request[:3])
324         time.sleep(0.01)
325         self.slave_end.inject(request[3:])
326         self.assertTrue(wait_for(lambda: len(self.slave_end.written) == 1))
327         found = find_master_response(self.slave_end.written[0], request)
328         self.assertEqual(found["values"], [0, 10])
329 
330     def test_garbage_before_a_request_is_dropped(self):
331         self.start_slave(silence_ms=20)
332         request = build_read_request(1, 3, 0, 1)
333         self.slave_end.inject(b"\x01\x10\xFF" + request)
334         self.assertTrue(wait_for(lambda: len(self.slave_end.written) == 1))
335         self.assertEqual(find_master_response(self.slave_end.written[0], request)["values"], [0])
336 
337     def test_unknown_function_gets_exception_01_after_silence(self):
338         self.start_slave(silence_ms=10)
339         self.slave_end.inject(append_crc(bytes((1, 0x2B, 0x0E, 0x01, 0x00))))
340         self.assertTrue(wait_for(lambda: len(self.slave_end.written) == 1))
341         self.assertEqual(self.slave_end.written[0], append_crc(bytes((1, 0xAB, 1))))
342 
343     def test_faults(self):
344         self.start_slave(rng=random.Random(1))
345         self.start_master(timeout_ms=100)
346         self.master.set_poll(PollDefinition(1, 1, 3, 0, 1, 10000))
347 
348         self.store.set_fault(1, validate_fault({"no_response": True}))
349         self.master.read_once(1)
350         self.assertEqual(self.master_results(2)[-1]["status"], "timeout")
351 
352         self.store.set_fault(1, validate_fault({"exception": 6}))
353         self.master.read_once(1)
354         result = self.master_results(1)[0]
355         self.assertEqual((result["status"], result["exception_code"]), ("exception", 6))
356 
357         self.store.set_fault(1, validate_fault({"bad_crc": True}))
358         self.master.read_once(1)
359         self.assertEqual(self.master_results(1)[0]["status"], "invalid")
360 
```

MBToolkit v3.4/tests/test_slave.py — lines 361-400

```
361         self.store.set_fault(1, validate_fault({"delay_ms": 300}))
362         self.master.read_once(1)
363         self.assertEqual(self.master_results(1)[0]["status"], "timeout")
364         time.sleep(0.35)  # let the late response pass before the next request
365 
366         self.store.set_fault(1, None)
367         self.master.read_once(1)
368         self.assertEqual(self.master_results(1)[0]["status"], "ok")
369 
370     def test_fault_probability(self):
371         store = make_store()
372         store.set_fault(1, validate_fault({"no_response": True, "probability": 30}))
373         engine = SlaveEngine(BusEnd(), queue.Queue(), store, baud=115200, parity="None",
374                              stopbits="1", rng=random.Random(7))
375         engine.gap_s = 0
376         for _ in range(400):
377             engine._answer(build_read_request(1, 3, 0, 1))
378         statuses = [p["status"] for _e, p in drain(engine.event_queue)]
379         silent = statuses.count("no_response")
380         self.assertTrue(80 <= silent <= 160, silent)
381         self.assertEqual(silent + statuses.count("ok"), 400)
382 
383 
384 class SlaveWindowTests(unittest.TestCase):
385     def test_workspace_entry_round_trip_and_validation(self):
386         window = SlaveWindow(1, 1, slave=3, table="coils", address=5, qty=16, base=1,
387                              title=" Saídas ", names={5: "Bomba"})
388         data = SlaveWindow.validated_kwargs(window.to_json())
389         self.assertEqual((data["title"], data["slave"], data["table"], data["address"],
390                           data["qty"], data["base"], data["names"]),
391                          ("Saídas", 3, "coils", 5, 16, 1, {5: "Bomba"}))
392         for bad in ({"slave": 0}, {"slave": 248}, {"table": "x"}, {"qty": 0},
393                     {"qty": 2001}, {"address": 65530, "qty": 10}):
394             with self.assertRaises(ValueError):
395                 SlaveWindow.validated_kwargs(bad)
396         validate_area(1, "input", 65535, 1)
397 
398     def test_table_field_shows_the_read_function_code(self):
399         from mb_slave import table_choice_label
400         from mb_slave_ui import TABLE_CHOICES, TABLE_KEYS_BY_LABEL
```

MBToolkit v3.4/tests/test_slave.py — lines 401-440

```
        self.assertEqual(TABLE_CHOICES, ("01 Coils", "02 Discrete Inputs",
                                         "03 Holding Registers", "04 Input Registers"))
        for table in ("coils", "discrete", "holding", "input"):
            self.assertEqual(TABLE_KEYS_BY_LABEL[table_choice_label(table)], table)

    def test_window_overlap(self):
        window = SlaveWindow(1, 1, address=10, qty=5)
        self.assertEqual(window.overlap(0, 12), (0, 2))
        self.assertEqual(window.overlap(12, 100), (2, 5))
        self.assertIsNone(window.overlap(15, 3))
        self.assertIsNone(window.overlap(0, 10))

    def test_fault_description_translates(self):
        fault = validate_fault({"no_response": True, "delay_ms": 50, "exception": 4})
        parts = [translate_text(template, LANGUAGE_ENGLISH).format(**values)
                 for template, values in fault_parts(fault)]
        self.assertEqual(parts, ["no response", "Exception 04", "delay 50 ms"])
        self.assertEqual(translate_text("Falhas…", LANGUAGE_ENGLISH), "Faults…")


class SlaveSettingsTests(unittest.TestCase):
    def setUp(self):
        self._appdata = os.environ.get("APPDATA")
        self._folder = tempfile.TemporaryDirectory()
        os.environ["APPDATA"] = self._folder.name

    def tearDown(self):
        if self._appdata is None:
            os.environ.pop("APPDATA", None)
        else:
            os.environ["APPDATA"] = self._appdata
        self._folder.cleanup()

    def test_slave_workspaces_have_their_own_folder(self):
        folder = mb_config.get_slave_workspaces_folder()
        self.assertEqual(folder.parent, mb_config.get_workspaces_folder())
        workspace_path("Banco", folder).write_text("{}", encoding="utf-8")
        self.assertEqual(list_workspaces(folder), ["Banco"])
        # Not listed with the Master ones.
        self.assertEqual(list_workspaces(mb_config.get_master_workspaces_folder()), [])
```

MBToolkit v3.4/tests/test_slave.py — lines 441-459

```
441
442     def test_slave_settings_are_validated(self):
443         mb_config.save_ui_settings(settings={
444             "slave": {"baud": "19200", "parity": "Even", "stop_bits": "2",
445                       "last_workspace": "Banco"},
446             "last_main_tab": "Slave"})
447         loaded = mb_config.load_ui_settings()
448         self.assertEqual(loaded["slave"], {"baud": "19200", "parity": "Even",
449                                            "stop_bits": "2", "last_workspace": "Banco"})
450         self.assertEqual(loaded["last_main_tab"], "Slave")
451         mb_config.save_ui_settings(settings={
452             "slave": {"baud": "1", "parity": "x", "stop_bits": "3", "last_workspace": "a/b"}})
453         self.assertEqual(mb_config.load_ui_settings()["slave"],
454                          {"baud": "9600", "parity": "None", "stop_bits": "1",
455                           "last_workspace": ""})
456
457
458 if __name__ == "__main__":
459     unittest.main()
```

MBToolkit.bat

210 source lines.

MBToolkit.bat — lines 1-40

```
@echo off
setlocal EnableExtensions
title MBToolkit 3.4
set "ROOT=%~dp0"
set "APPDIR=%~dp0MBToolkit v3.4"
set "SCRIPT=%APPDIR%\MBToolkit.py"
cd /d "%APPDIR%"
set "PYRUN="
set "PYEXE="
set "PYWEXE="

if not exist "%SCRIPT%" (
    echo.
    echo ERRO: os ficheiros do MBToolkit nao foram encontrados em:
    echo   %SCRIPT%
    echo.
    pause
    exit /b 1
)

rem ============================================================
rem 1. Procurar uma instalacao funcional de Python 3
rem ============================================================
call :DETECT_PYTHON

if defined PYRUN goto PYTHON_OK

echo.
echo Python 3 nao foi encontrado.
echo A tentar instalar automaticamente Python 3.13 com WinGet...
echo.

where winget >nul 2>&1
if errorlevel 1 goto NO_WINGET

winget install --id Python.Python.3.13 --exact --scope user --silent ^
    --accept-package-agreements --accept-source-agreements

if errorlevel 1 (
    echo.
```

MBToolkit.bat — lines 41-80

```
41      echo ERRO: A instalacao automatica do Python falhou.
42      echo O WinGet pode estar bloqueado pela politica da empresa ou sem acesso a Internet.
43      echo.
44      pause
45      exit /b 1
46  )
47
48  rem O instalador pode alterar o PATH sem atualizar esta janela CMD.
49  rem Por isso voltamos a procurar tambem nos caminhos standard.
50  call :DETECT_PYTHON
51
52  if not defined PYRUN (
53      echo.
54      echo ERRO: Python foi instalado, mas nao consegui localizar python.exe.
55      echo Fecha esta janela e volta a executar MBToolkit.bat.
56      echo.
57      pause
58      exit /b 1
59  )
60
61  :PYTHON_OK
62  rem ============================================================
63  rem 2. Descobrir o python.exe real e o respetivo pythonw.exe
64  rem ============================================================
65  for /f "usebackq delims=" %%P in (`%PYRUN% -c "import sys; print(sys.executable)" 2^>nul`) do (
66      set "PYEXE=%%P"
67  )
68
69  if not defined PYEXE (
70      echo.
71      echo ERRO: Nao foi possivel determinar a instalacao do Python.
72      echo.
73      pause
74      exit /b 1
75  )
76
77  for %%P in ("%PYEXE%") do set "PYWEXE=%%~dpPpythonw.exe"
78
79  rem ============================================================
80  rem 3. Verificar tkinter
```

MBToolkit.bat — lines 81-120

```
rem ============================================================
%PYRUN% -c "import tkinter" >nul 2>&1
if not errorlevel 1 goto TK_OK

echo.
echo tkinter nao esta disponivel nesta instalacao do Python.
echo A tentar instalar/reparar Python 3.13 com WinGet...
echo.

where winget >nul 2>&1
if errorlevel 1 goto TK_ERROR

winget install --id Python.Python.3.13 --exact --scope user --silent --force ^
    --accept-package-agreements --accept-source-agreements

call :DETECT_PYTHON
if not defined PYRUN goto TK_ERROR

%PYRUN% -c "import tkinter" >nul 2>&1
if errorlevel 1 goto TK_ERROR

for /f "usebackq delims=" %%P in (`%PYRUN% -c "import sys; print(sys.executable)" 2^>nul`) do (
    set "PYEXE=%%P"
)
for %%P in ("%PYEXE%") do set "PYWEXE=%%~dpPpythonw.exe"

:TK_OK
rem ============================================================
rem 4. Verificar pyserial (obrigatorio)
rem ============================================================
%PYRUN% -c "import serial" >nul 2>&1
if not errorlevel 1 goto PYSERIAL_OK

echo.
echo pyserial nao esta instalado.
echo A instalar pyserial...
echo.

%PYRUN% -m pip --version >nul 2>&1
if errorlevel 1 (
```

MBToolkit.bat — lines 121-160

```
121     %PYRUN% -m ensurepip --upgrade
122     if errorlevel 1 goto DEPENDENCIES_ERROR
123 )
124
125 %PYRUN% -m pip install --user pyserial
126 if errorlevel 1 goto DEPENDENCIES_ERROR
127
128 %PYRUN% -c "import serial" >nul 2>&1
129 if errorlevel 1 goto DEPENDENCIES_ERROR
130
131 :PYSERIAL_OK
132 rem ============================================================
133 rem 5. Dependencias verificadas
134 rem ============================================================
135 :DEPENDENCIES_OK
136 rem ============================================================
137 rem 6. Executar o MBToolkit
138 rem ============================================================
139 if exist "%PYWEXE%" (
140     start "" "%PYWEXE%" "%SCRIPT%"
141     exit /b 0
142 )
143
144 rem Fallback caso pythonw.exe nao exista.
145 start "" %PYRUN% "%SCRIPT%"
146 exit /b 0
147
148
149 :DETECT_PYTHON
150 set "PYRUN="
151
152 rem Python Launcher for Windows
153 py -3 -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
154 if not errorlevel 1 (
155     set "PYRUN=py -3"
156     exit /b 0
157 )
158
159 rem Python no PATH
160 python -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
```

MBToolkit.bat — lines 161-200

```
161 if not errorlevel 1 (
162     set "PYRUN=python"
163     exit /b 0
164 )
165
166 rem Caminhos standard para Python 3.13 instalado por utilizador/WinGet.
167 if exist "%LocalAppData%\Programs\Python\Python313\python.exe" (
168     set PYRUN="%LocalAppData%\Programs\Python\Python313\python.exe"
169     exit /b 0
170 )
171
172 if exist "%ProgramFiles%\Python313\python.exe" (
173     set PYRUN="%ProgramFiles%\Python313\python.exe"
174     exit /b 0
175 )
176
177 if exist "%ProgramFiles(x86)%\Python313\python.exe" (
178     set PYRUN="%ProgramFiles(x86)%\Python313\python.exe"
179     exit /b 0
180 )
181
182 exit /b 0
183
184
185 :NO_WINGET
186 echo.
187 echo ERRO: Python 3 nao esta instalado e o WinGet nao esta disponivel.
188 echo Nao e possivel fazer a instalacao automatica neste PC.
189 echo Instala Python 3 manualmente e volta a executar MBToolkit.bat.
190 echo.
191 pause
192 exit /b 1
193
194
195 :TK_ERROR
196 echo.
197 echo ERRO: tkinter continua indisponivel.
198 echo E necessario usar uma instalacao normal do Python para Windows com Tcl/Tk.
199 echo.
200 pause
```

MBToolkit.bat — lines 201-210

```
exit /b 1


:DEPENDENCIES_ERROR
echo.
echo ERRO: Nao foi possivel instalar pyserial.
echo Verifica acesso a Internet, proxy/firewall e permissoes do Python.
echo.
pause
exit /b 1
```

MBToolkit v3.4/build_exe.bat

151 source lines.

MBToolkit v3.4/build_exe.bat — lines 1-40

```
@echo off
setlocal EnableExtensions
title Build MBToolkit 3.4

rem Normalizar caminhos sem barra final. Isto evita problemas de parsing
rem do Windows/Python quando um argumento quoted termina em "\".
for %%I in ("%~dp0.") do set "APPDIR=%%~fI"
for %%I in ("%APPDIR%\..") do set "DISTDIR=%%~fI"

set "SCRIPT=%APPDIR%\MBToolkit.py"
set "ICON=%APPDIR%\MBToolkit.ico"
set "WORKDIR=%APPDIR%\build"
set "SPECFILE=%APPDIR%\MBToolkit.spec"
set "PYRUN="

cd /d "%APPDIR%"

if not exist "%SCRIPT%" (
    echo.
    echo ERRO: MBToolkit.py nao foi encontrado.
    echo.
    pause
    exit /b 1
)

rem ============================================================
rem 1. Procurar Python 3
rem ============================================================
py -3 -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
if not errorlevel 1 (
    set "PYRUN=py -3"
    goto PYTHON_OK
)

python -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
if not errorlevel 1 (
    set "PYRUN=python"
    goto PYTHON_OK
)

```

MBToolkit v3.4/build_exe.bat — lines 41-80

```
41 if exist "%LocalAppData%\Programs\Python\Python313\python.exe" (
42     set PYRUN="%LocalAppData%\Programs\Python\Python313\python.exe"
43     goto PYTHON_OK
44 )
45
46 echo.
47 echo ERRO: Python 3 nao foi encontrado.
48 echo Executa primeiro MBToolkit.bat para instalar/verificar as dependencias.
49 echo.
50 pause
51 exit /b 1
52
53
54 :PYTHON_OK
55 rem ============================================================
56 rem 2. Instalar/atualizar dependencias de build
57 rem ============================================================
58 echo.
59 echo [1/4] A verificar pyserial e PyInstaller...
60 %PYRUN% -m pip install --upgrade pyserial pyinstaller
61 if errorlevel 1 (
62     echo.
63     echo ERRO ao instalar pyserial/PyInstaller.
64     echo.
65     pause
66     exit /b 1
67 )
68
69 rem ============================================================
70 rem 3. Executar testes de regressao antes do build
71 rem ============================================================
72 echo.
73 echo [2/4] A executar testes de regressao...
74 %PYRUN% -m unittest discover -s tests -p "test_*.py" -v
75 if errorlevel 1 (
76     echo.
77     echo ERRO: os testes de regressao falharam. Build cancelado.
78     echo.
79     pause
80     exit /b 1
```

MBToolkit v3.4/build_exe.bat — lines 81-120

```
)

rem ============================================================
rem 4. Limpar builds anteriores
rem ============================================================
echo.
echo [3/4] A limpar build anterior...
if exist "%WORKDIR%" rmdir /s /q "%WORKDIR%"
if exist "%SPECFILE%" del /q "%SPECFILE%"
if exist "%DISTDIR%\MBToolkit.exe" del /q "%DISTDIR%\MBToolkit.exe"

rem ============================================================
rem 5. Criar EXE
rem ============================================================
echo.
echo [4/4] A criar MBToolkit.exe...

if exist "%ICON%" (
    %PYRUN% -m PyInstaller ^
      --noconfirm ^
      --clean ^
      --onefile ^
      --windowed ^
      --name MBToolkit ^
      --distpath "%DISTDIR%" ^
      --workpath "%WORKDIR%" ^
      --icon "%ICON%" ^
      --add-data "%ICON%;." ^
      "%SCRIPT%"
) else (
    echo AVISO: MBToolkit.ico nao foi encontrado.
    %PYRUN% -m PyInstaller ^
      --noconfirm ^
      --clean ^
      --onefile ^
      --windowed ^
      --name MBToolkit ^
      --distpath "%DISTDIR%" ^
      --workpath "%WORKDIR%" ^
      "%SCRIPT%"
```

MBToolkit v3.4/build_exe.bat — lines 121-151

```
)

if errorlevel 1 (
    echo.
    echo ERRO: o build falhou.
    echo.
    pause
    exit /b 1
)

if not exist "%DISTDIR%\MBToolkit.exe" (
    echo.
    echo ERRO: o PyInstaller terminou sem criar:
    echo   %DISTDIR%\MBToolkit.exe
    echo.
    pause
    exit /b 1
)

echo.
echo ============================================================
echo BUILD CONCLUIDO
echo ============================================================
echo.
echo Executavel criado em:
echo   %DISTDIR%\MBToolkit.exe
echo.
echo O icone MBToolkit.ico foi aplicado ao EXE e incluido
echo como recurso para a janela/barra de tarefas.
echo.
pause
```

MBToolkit v3.4/run_tests.bat

45 source lines.

MBToolkit v3.4/run_tests.bat — lines 1-40

```
@echo off
setlocal EnableExtensions
title MBToolkit regression tests

for %%I in ("%~dp0.") do set "APPDIR=%%~fI"
cd /d "%APPDIR%"
set "PYRUN="

py -3 -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
if not errorlevel 1 set "PYRUN=py -3"

if not defined PYRUN (
    python -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
    if not errorlevel 1 set "PYRUN=python"
)

if not defined PYRUN (
    echo.
    echo ERRO: Python 3 nao foi encontrado.
    echo.
    pause
    exit /b 1
)

echo.
echo A executar testes de regressao do MBToolkit...
echo.
%PYRUN% -m unittest discover -s tests -p "test_*.py" -v
if errorlevel 1 (
    echo.
    echo ============================================================
    echo TESTES FALHARAM - build/release nao deve prosseguir.
    echo ============================================================
    echo.
    pause
    exit /b 1
)

echo.
echo ============================================================
```

MBToolkit v3.4/run_tests.bat — lines 41-45

```
41 echo TODOS OS TESTES PASSARAM
42 echo ============================================================
43 echo.
44 pause
45 exit /b 0
```